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Inventor(s)

Kawada; Hatsuo et al.

Cyclic compound having selective KRAS inhibitory effect on HRAS and NRAS

Abstract

Cyclic compounds that selectively inhibit KRAS were found. Moreover, cyclic compounds were found to interact with an amino acid residue specific to KRAS.

Inventors: Kawada; Hatsuo (Kamakura, JP), Takano; Koji (Kamakura, JP), Kotake; Tomoya (Gotemba, JP), Kage; Mirai (Gotemba, JP), Hashimoto; Satoshi (Gotemba, JP), Tamiya; Minoru (Kamakura, JP), Wakamiya; Yuma (Kamakura, JP), Hayashi; Ryuji (Kamakura, JP), Morita; Yuya (Kamakura, JP)

Applicant: CHUGAI SEIYAKU KABUSHIKI KAISHA (Tokyo, JP)

Family ID: 88646541

Assignee: CHUGAI SEIYAKU KABUSHIKI KAISHA (Tokyo, JP)

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References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
9409952	12/2015	Kariyuki et al.	N/A	N/A
10815489	12/2019	Ohta et al.	N/A	N/A
10829458	12/2019	Li et al.	N/A	N/A
11492369	12/2021	Nomura et al.	N/A	N/A
11542299	12/2022	Nomura et al.	N/A	N/A
11732002	12/2022	Iwaskai et al.	N/A	N/A
11787836	12/2022	Nomura et al.	N/A	N/A
11891457	12/2023	Kariyuki et al.	N/A	N/A
12071396	12/2023	Wadamoto et al.	N/A	N/A
2014/0128572	12/2013	Monnaie et al.	N/A	N/A
2015/0080549	12/2014	Kariyuki et al.	N/A	N/A
2016/0311858	12/2015	Kariyuki et al.	N/A	N/A
2018/0127761	12/2017	Ohta et al.	N/A	N/A
2019/0338050	12/2018	Nakano et al.	N/A	N/A
2019/0380958	12/2018	Tampo et al.	N/A	N/A
2020/0040372	12/2019	Tanaka et al.	N/A	N/A
2020/0131669	12/2019	Muraoka et al.	N/A	N/A
2020/0277327	12/2019	Nomura et al.	N/A	N/A
2020/0339623	12/2019	Nomura et al.	N/A	N/A
2021/0024579	12/2020	Shipman	N/A	N/A
2021/0061860	12/2020	Kariyuki et al.	N/A	N/A
2021/0087572	12/2020	Ohta et al.	N/A	N/A
2022/0017456	12/2021	Ishizawa	N/A	N/A
2022/0024972	12/2021	Iwasaki et al.	N/A	N/A
2022/0096379	12/2021	Tampo et al.	N/A	N/A
2022/0144762	12/2021	Wadamoto et al.	N/A	N/A
2022/0205009	12/2021	Shinohara et al.	N/A	N/A
2022/0411462	12/2021	Hou et al.	N/A	N/A
2023/0026641	12/2022	Nomura et al.	N/A	N/A
2023/0056969	12/2022	Kondo et al.	N/A	N/A
2023/0096766	12/2022	Muraoka et al.	N/A	N/A
2023/0108274	12/2022	Kagotani et al.	N/A	N/A
2023/0138226	12/2022	Nomura et al.	N/A	N/A

2023/0151060	12/2022	Tanada et al.	N/A	N/A
2023/0295221	12/2022	Iwasaki et al.	N/A	N/A
2023/0303619	12/2022	Iwasaki et al.	N/A	N/A
2023/0391818	12/2022	Nomura et al.	N/A	N/A
2023/0406879	12/2022	Nomura et al.	N/A	N/A
2024/0052340	12/2023	Nishimura et al.	N/A	N/A
2024/0067674	12/2023	Sekita et al.	N/A	N/A
2024/0124517	12/2023	Morita et al.	N/A	N/A
2024/0148821	12/2023	Tanada et al.	N/A	N/A
2024/0158446	12/2023	Kawada et al.	N/A	N/A
2024/0166689	12/2023	Kariyuki et al.	N/A	N/A
2024/0239842	12/2023	Hayashi et al.	N/A	N/A
2024/0366711	12/2023	Ueto et al.	N/A	N/A
2024/0376044	12/2023	Wadamoto	N/A	N/A

FOREIGN PATENT DOCUMENTS

Patent No.	Application Date	Country	CPC
2813512	12/2013	EP	N/A
3636807	12/2019	EP	N/A
3896056	12/2020	EP	N/A
4043478	12/2021	EP	N/A
4309741	12/2023	EP	N/A
4316503	12/2023	EP	N/A
WO2012122059	12/2011	WO	N/A
WO2013100132	12/2012	WO	N/A
WO2016144635	12/2015	WO	N/A
WO2016148044	12/2015	WO	N/A
WO2017150732	12/2016	WO	N/A
WO2017181061	12/2016	WO	N/A
WO2018124162	12/2017	WO	N/A
WO2018143145	12/2017	WO	N/A
WO2018217651	12/2017	WO	N/A
WO2018225851	12/2017	WO	N/A
WO2018225864	12/2017	WO	N/A
WO2019099524	12/2018	WO	N/A
WO2019117274	12/2018	WO	N/A
WO2020095983	12/2019	WO	N/A
WO2020111238	12/2019	WO	N/A
WO2020122182	12/2019	WO	N/A
WO2020138336	12/2019	WO	N/A
WO2020189540	12/2019	WO	N/A
WO-2021090855	12/2020	WO	A61K 38/12
WO2021090856	12/2020	WO	N/A
WO2021132545	12/2020	WO	N/A
WO2021132546	12/2020	WO	N/A
WO2021246471	12/2020	WO	N/A
WO2021261577	12/2020	WO	N/A
WO2022097540	12/2021	WO	N/A
WO2022138891	12/2021	WO	N/A
WO2022145444	12/2021	WO	N/A

WO2022234850	12/2021	WO	N/A
WO2022234851	12/2021	WO	N/A
WO-2022234852	12/2021	WO	A61K 38/12
WO-2022234853	12/2021	WO	A61K 38/12
WO2023127869	12/2022	WO	N/A
WO2023140329	12/2022	WO	N/A
WO2023214577	12/2022	WO	N/A

OTHER PUBLICATIONS

Bockus, A. T., et al., "Form and Function in Cyclic Peptide Natural Products: A Pharmacokinetic Perspective," *Curr Topics Med Chem.*, 13:821-836 (2013). cited by applicant

Cox, A. D., et al., "Drugging the undruggable RAS: Mission Possible?," *Nat Rev Drug Discov.*, 13(11):828-851 (2014). cited by applicant

Das, M. and Himaja, M. , "Design, Synthesis and Biological Evaluation of Linear Tetrapeptide D-Ala-L-(Gly-Val-Val) (AGVV)," *UJPB*, 01(02):21-24 (2013). cited by applicant

Eggen, I. F., et al., "A novel method for repetitive peptide synthesis in solution without isolation of intermediates," *J Pep Sci.*, 11:633-641 (2005). cited by applicant

Gracia, S. R., et al., "Synthesis of chemically modified bioactive peptides: recent advances, challenges and developments for medicinal chemistry," *Future Med Chem.*, 1(7):1289-1310 (2009). cited by applicant

Isidro-Llobet, A., et al., "Amino Acid-protecting Groups," *Chemical Reviews*, 109(6):2455-2504 (2009). cited by applicant

Lambert, J.N., et al., "The synthesis of cyclic peptides," *J Chem Soc, Perkin Trans.*, 1:471- 484 (2001). cited by applicant

Niida, A., et al., "Investigation of the Structural Requirements of K-ras(G12d) Selective Inhibitory Peptide Krpep-2d Using Alanine Scans and Cysteine Bridging," *Bioorganic & Medicinal Chemistry Letters*, 27(12):2757-2761 (2017). cited by applicant

Ostrem, J. M. L. and Shokat, K. M., "Direct small-molecule inhibitors of KRAS: from structural insights to mechanism-based design," *Nat Rev Drug Discov.*, 15(11):771-785 (2016). cited by applicant

Sakamoto, K., et al., "K-Ras(G12D)-selective inhibitory peptides generated by random peptide T7 phage display technology," *Biochem Biophys Res Comm.*, 484:605-611 (2017). cited by applicant

Samatar, A. A. and Poulikakos, P. I., "Targeting RAS-ERK signaling in cancer: promises and challenges," *Nat Rev Drug Discov.*, 13:928-942 (2014). cited by applicant

Sogabe, S., et al., "Crystal Structure of a Human K-Ras G12D Mutant in Complex with GDP and the Cyclic Inhibitory Peptide KRpep-2D," *ACS Med Chem Lett.*, 8:732-736 (2017). cited by applicant

Upadhyaya, P., et al., "Direct Ras inhibitors identified from a structurally rigidified bicyclic peptide library," *Tetrahedron*, 70:7714-7720 (2014). cited by applicant

U.S. Appl. No. 14/368,564, filed Jun. 25, 2014, Kariyuki et al., related application. cited by applicant

U.S. Appl. No. 15/166,550, filed May 27, 2016, Kariyuki et al., related application. cited by applicant

U.S. Appl. No. 15/557,532, filed Sep. 12, 2017, Ohta et al., related application. cited by applicant

U.S. Appl. No. 16/081,522, filed Jul. 8, 2019, Nakano et al., related application. cited by applicant

U.S. Appl. No. 16/471,837, filed Jun. 20, 2019, Tampo et al., related application. cited by applicant

U.S. Appl. No. 16/479,736, filed Jul. 22, 2019, Tanaka et al., related application. cited by applicant

U.S. Appl. No. 16/619,014, filed Dec. 3, 2019, Muraoka et al., related application. cited by applicant

U.S. Appl. No. 16/619,388, filed Dec. 4, 2019, Nomura et al., related application. cited by applicant

U.S. Appl. No. 16/771,335, filed Jun. 10, 2020, Nomura et al., related application. cited by applicant

U.S. Appl. No. 17/011,815, filed Sep. 3, 2020, Kariyuki et al., related application. cited by applicant

U.S. Appl. No. 17/024,944, filed Sep. 18, 2020, Ohta et al., related application. cited by applicant

U.S. Appl. No. 17/291,099, filed Jun. 3, 2021, Ishizawa, related application. cited by applicant

U.S. Appl. No. 17/297,231, filed May 26, 2021, Iwasaki et al., related application. cited by applicant

U.S. Appl. No. 17/312,296, filed Jun. 9, 2021, Muraoka et al., related application. cited by applicant

U.S. Appl. No. 17/417,822, filed Jun. 24, 2021, Shinohara et al., related application. cited by applicant

U.S. Appl. No. 17/437,535, filed Sep. 9, 2021, Wadamoto et al., related application. cited by applicant

U.S. Appl. No. 17/502,525, filed Oct. 15, 2021, Tampo et al., related application. cited by applicant

U.S. Appl. No. 17/738,283, filed May 6, 2022, Hou et al., related application. cited by applicant

U.S. Appl. No. 17/773,733, filed May 2, 2022, Tanada et al., related application. cited by applicant

U.S. Appl. No. 17/773,734, filed May 2, 2022, Nomura et al., related application. cited by applicant

U.S. Appl. No. 17/787,809, filed Jun. 21, 2022, Kagotani et al., related application. cited by applicant

U.S. Appl. No. 17/788,506, filed Jun. 23, 2022, Kondo et al., related application. cited by applicant

U.S. Appl. No. 17/928,759, filed Nov. 30, 2022, Iwasaki et al., related application. cited by applicant

U.S. Appl. No. 17/976,942, filed Oct. 31, 2022, Nomura et al., related application. cited by applicant

U.S. Appl. No. 18/010,608, filed Dec. 15, 2022, Nishimura et al., related application. cited by applicant

U.S. Appl. No. 18/034,424, filed Apr. 28, 2023, Nomura et al., related application. cited by applicant

U.S. Appl. No. 18/203,371, filed May 30, 2023, Iwasaki et al., related application. cited by applicant

U.S. Appl. No. 18/268,737, filed Jun. 21, 2023, Morita et al., related application. cited by applicant

U.S. Appl. No. 18/269,334, filed Jun. 23, 2023, Sekita et al., related application. cited by applicant

U.S. Appl. No. 18/289,071, filed Oct. 31, 2023, Hayashi et al., related application. cited by applicant

U.S. Appl. No. 18/289,392, filed Nov. 3, 2023, Ueto et al., related application. cited by applicant

U.S. Appl. No. 18/289,451, filed Nov. 3, 2023, Tanada et al., related application. cited by applicant

U.S. Appl. No. 18/289,592, filed Nov. 6, 2023, Kawada et al., related application. cited by applicant

U.S. Appl. No. 18/459,998, filed Sep. 1, 2023, Nomura et al., related application. cited by applicant

U.S. Appl. No. 18/460,300, filed Sep. 1, 2023, Kariyuki et al., related application. cited by applicant

U.S. Appl. No. 18/723,993, filed Jun. 25, 2024, Komiya et al., related application. cited by applicant

U.S. Appl. No. 18/728,922, filed Jul. 15, 2024, Sase et al., related application. cited by applicant

U.S. Appl. No. 18/773,066, filed Jul. 15, 2024, Tanada et al., related application. cited by applicant

U.S. Appl. No. 18/781,112, filed Jul. 23, 2024, Wadamoto, related application. cited by applicant

U.S. Appl. No. 18/860,859, filed Oct. 28, 2024, Kage et al., related application. cited by applicant

Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS (1) This application is a continuation of PCT Application No. PCT/JP2023/017112, filed May 2, 2023, which claims the benefit of Japanese Patent Application No. 2022-076449, filed May 6, 2022, each of which is incorporated herein by reference in its entirety.

REFERENCE TO SEQUENCE LISTING SUBMITTED ELECTRONICALLY

(1) The content of the electronically submitted sequence listing (Name: 6663_0310 Sequence_Listing.xml; Size: 3.94 KB; and Date of Creation: Sep. 9, 2024) filed with the application is incorporated herein by reference in its entirety.

TECHNICAL FIELD

(2) The present invention in one aspect relates to cyclic compounds.

BACKGROUND ART

(3) RAS is a protein belonging to the small GTPase family, and KRAS, NRAS, and HRAS are known. RAS is in an active state or an inactive state according to whether it is bound to GTP or GDP. It is activated by the exchange reaction from GDP to GTP by GEFs (guanine nucleotide exchange factors) and inactivated by the hydrolysis reaction of GTP by GAPs (GTPase-activating proteins)(NPL 1). Activated RAS induces cell proliferation, survival, and differentiation by activating various downstream signals in the MAPK pathway, PI3K/Akt pathway, RAL pathway, and such, and the constitutive activation of RAS plays an important role in the development and progression of cancer. In cancer, it is known that the RAS-RAF-MEK-ERK pathway is activated by the activation of an upstream signal of RAS, constitutive activation of RAS, and/or activating mutations of RAS (NPL 2). These activating mutations of RAS have been found in numerous cancer types. G12, G13, and Q61 are known as hot spots of RAS mutation, and G12 is frequently found mutated in KRAS and Q61 in NRAS. These mutations are also known to be associated with the prognosis of patients (NPL 3).

(4) Meanwhile, when it comes to access to a tough target, as typified by inhibition of a protein-protein interaction, medium sized molecules (having a molecular weight of 500 to 2000 g/mol) may be superior to low molecular weight compounds. Also, medium sized molecules may be superior to antibodies in that they can migrate into cells. Among biologically active medium sized molecules, peptide drugs are highly valuable molecular species, with more than 40 peptide drugs being already commercially available (NPL 4). Representative examples of such peptide drugs include cyclosporin A and polymyxin B, which are peptides containing some non-natural amino acids. A non-natural amino acid refers to an amino acid that is not naturally encoded on mRNA. It is highly interesting that non-natural amino acids are contained in naturally-occurring cyclosporin A and polymyxin B.

(5) Since the discovery of the pharmaceutical utility of naturally-occurring peptides, peptides having pharmacological activity and bioabsorbability have been attracting attention, and those having a molecular weight of about 500 to 2000 g/mol have been actively researched (NPL 5).

(6) There is a report on conditions for medium molecular weight peptides to have increased membrane permeability and metabolic stability, which may contribute to improving their biokinetics (conditions necessary for satisfying drug-likeness)(PTL 1). Moreover, as for the conditions that may contribute to improving the biokinetics of medium molecular weight peptides, conditions necessary for cyclic peptides to satisfy drug-likeness have been shown (PTL 2).

(7) Peptides that bind to RAS have been found, and the binding site between a cyclic peptide and RAS has been studied by X-ray structural analysis (NPL 6, NPL 7, and NPL 8). Also, cyclic peptides that apparently inhibit binding between RAS and SOS have been found (PTL 3). Moreover, a competition assay for binding with RAS has suggested that some cyclic peptides inhibit binding between a particular compound and RAS (PTL 4).

Citation List

Patent Literature

(8) [PTL 1] WO 2013/100132 [PTL 2] WO 2018/225864 [PTL 3] WO 2012/122059 [PTL 4] WO 2017/181061

Non-Patent Literature

(9) [NPL 1] Nat. Rev. Drug Discov. 2014 Nov.; 13 (11): 828-851. [NPL 2] Nat. Rev. Drug Discov. 2014 Dec.; 13 (12): 928-942. [NPL 3] Nat. Rev. Drug Discov. 2016 Nov.; 15 (11): 771-785. [NPL 4] Future Med. Chem. 2009, 1, 1289-1310. [NPL 5] Current Topics in Medicinal Chemistry, 2013, Vol. 13, No. 7, 821-836. [NPL 6] Biochem. Biophys. Res. Commun. 2017, 484, 605-611. [NPL 7] Bioorg. Med. Chem. Lett. 2017, 27, 2757-2761. [NPL 8] ACS Med. Chem. Lett. 2017, 8, 732-736.

SUMMARY OF INVENTION

Technical Problem

(10) The present invention relates to cyclic compounds effective for RAS-mutant cancer and non-natural amino acids and peptide compounds useful for the production thereof.

(11) PTL 1 and PTL 2 describe drug-like peptides, but do not describe a peptide having an antitumor effect on cancers including RAS-mutant cancer.

(12) PTL 3 describes the inhibition of binding between RAS and SOS, and PTL 4 describes a peptide competing with a compound that binds to RAS. However, none of these documents shows any pharmacological action, especially action on tumor cells. These documents do not describe drug-like peptides, either.

(13) NPL 1 shows the relationship between RAS and cancer in detail. This document describes molecules that bind to RAS. Although their efficacy was shown in preclinical studies, no compound was shown to be effective as a drug specifically on RAS-mutant cancer. Also, no drug-like cyclic peptide is disclosed.

(14) NPL 2 provides detailed descriptions about RAS and the RAF-MEK-ERK pathway, which is downstream of RAS. Although this document suggests the possibility of treating RAS-mutant cancer with RAF, MEK, and ERK inhibitors, it does not show any compound that directly inhibits RAS.

(15) NPL 3 describes a compound that binds to the GTP/GDP binding site of RAS and inhibits the function of RAS, and the mechanism thereof. This document describes the interaction with the GTP/GDP binding site in detail, but does not show pharmacological action, especially action on tumor cells.

(16) NPL 4 describes peptides that are used as drugs, but does not describe a drug-like peptide or a peptide useful for RAS-mutant cancer.

(17) NPL 5 describes the molecular form and pharmacokinetics of cyclic peptides, but does not describe a compound useful for RAS-mutant cancer.

(18) NPLs 6 to 8 describe peptides that bind to RAS, but their action on tumor cells is limited, and, in addition, a drug-like peptide is not described.

(19) Moreover, to the present inventors' knowledge, there is no report of a compound having sufficiently selective inhibitory action on KRAS over HRAS and NRAS.

Solution to Problem

(20) As a result of dedicated research to search for cyclic compounds having selective inhibitory action on KRAS over HRAS and NRAS, the present inventors found cyclic compounds that interact with KRAS selectively as compared to HRAS and NRAS. In addition, the inventors found that the cyclic compounds have pharmacological action of inhibiting the growth of tumor cells

having a RAS mutation.

(21) In a specific non-limiting embodiment, the present invention encompasses the following:

(22) A cyclic compound represented by formula (1):

(23) ##STR00001##

wherein L.sub.1 is a single bond; R.sub.1 is a C.sub.1 to C.sub.7 alkyl, or R.sub.1 joins together with R.sub.5 to form a divalent group, wherein a partial structure: *-CR.sub.1Q.sub.1-L.sub.1-CO—NP.sub.2-CR.sub.2Q.sub.2-CO—NP.sub.3-CR.sub.3Q.sub.3—CO—NP.sub.4-CR.sub.4Q.sub.4—CO—NP.sub.5-CR.sub.5Q.sub.5-* in the cyclic compound represented by formula (1) is represented by the following formula:

(24) ##STR00002##

wherein: X.sub.2 is —Li—CO—NP.sub.2-CR.sub.2Q.sub.2-CO—NP.sub.3-CR.sub.3Q.sub.3—CO—NP.sub.4-CR.sub.4Q.sub.4—CO—NP.sub.5—,  custom character

is a single bond or a double bond,  custom character

may take E or Z configuration, if

 custom character

is a double bond, n is 0, 1 or 2, m is 0, 1, 2, 3 or 4, represents an attachment point to an adjacent atom; P.sub.1 is a C.sub.1 to C.sub.6 alkyl; Q.sub.1 is a hydrogen atom; R.sub.2 is a C.sub.1 to C.sub.6 alkyl; P.sub.2 is a hydrogen atom; Q.sub.2 is a hydrogen atom; R.sub.3 is a hydrogen atom or R.sub.3 joins together with a carbon atom to which P.sub.3 and R.sub.3 are attached and a nitrogen atom to which P.sub.3 is attached, to form a 4 to 7-membered saturated heterocycle; P.sub.3 is a C.sub.1 to C.sub.6 alkyl or a C.sub.3 to C.sub.8 cycloalkyl except the case where R.sub.3 and P.sub.3 form a 4 to 7-membered saturated heterocycle; Q.sub.3 is a hydrogen atom; R.sub.4 joins together with P.sub.5 to form a divalent group, wherein a partial structure *-CR.sub.4Q.sub.4-CO—NP.sub.5-* in the cyclic compound represented by formula (1) is represented by the following formula:

(25) ##STR00003## P.sub.4 is a C.sub.1 to C.sub.6 alkyl; Q.sub.4 is a hydrogen atom, R.sub.5 is a benzyl optionally substituted with one or more groups selected from the group consisting of a C.sub.1 to C.sub.6 alkyl, a C.sub.1 to C.sub.6 haloalkyl, and a C.sub.3 to C.sub.8 cycloalkyl except the case where R.sub.1 and R.sub.5 form a divalent group; Q.sub.5 is a hydrogen atom, R.sub.6 is a hydrogen atom; P.sub.6 is a C.sub.1 to C.sub.6 alkyl, Q.sub.6 is a hydrogen atom, R.sub.7 is a phenethyl optionally substituted with one or more groups selected from the group consisting of a halogen atom, a C.sub.1 to C.sub.6 haloalkyl and a C.sub.1 to C.sub.6 alkoxy; P.sub.7 is a hydrogen atom, Q.sub.7 is a hydrogen atom, R.sub.8 joins together with a carbon atom to which P.sub.8 and R.sub.8 are attached and a nitrogen atom to which P.sub.8 is attached, to form a 4 to 7-membered saturated heterocycle, wherein the 4 to 7-membered saturated heterocycle is optionally substituted with a C.sub.1 to C.sub.6 alkoxy; Q.sub.8 is a hydrogen atom, R.sub.9 joins together with Q.sub.9 and a carbon atom to which R.sub.9 and Q.sub.9 are attached to form a 3 to 8-membered alicyclic ring, wherein the 3 to 8-membered alicyclic ring is optionally substituted with one or more C.sub.1 to C.sub.6 alkyls; P.sub.9 is a hydrogen atom or a C.sub.1 to C.sub.6 alkyl, R.sub.10 is a C.sub.1 to C.sub.6 alkyl or a C.sub.3 to C.sub.8 cycloalkyl; P.sub.10 is a C.sub.1 to C.sub.6 alkyl, Q.sub.10 is a hydrogen atom, and L.sub.11 is —CH.sub.2—, R.sub.11 is a di-C.sub.1 to C.sub.6 alkylaminocarbonyl or a 4 to 8-membered cyclic aminocarbonyl; P.sub.11 is a C.sub.1 to C.sub.6 alkyl, and Q.sub.11 is a hydrogen atom,

or a salt thereof, or a solvate thereof, wherein the cyclic compound represented by formula (1) is selected from the group consisting of: PP1574: (2S,8S,12R,14S,20S,23S,27S,30S,36S,38Z)-20-cyclopentyl-8-[2-[3,5-difluoro-4-(trifluoromethyl)phenyl]ethyl]-12-ethoxy-N-ethyl-27-isobutyl-N,4,19,22,26,32,35-heptamethyl-30-[(1 S)-1-methylpropyl]-3,6,9,15,18,21,25,28,31,34,42-undecaoxo-2-[[4-(trifluoromethyl)phenyl]methyl]spiro[1,4,7,10,16,19,22,26,29,32,35-undecazatricyclo[34.5.1.0.sup.10,14]dotetracont-38-ene-17,1'-cyclopentane]-23-carboxamide, PP1650: (2S,8S,12R,14S,20S,23S,27S,30S,36S,38Z)-20-cyclopentyl-8-[2-[3,5-difluoro-4-

(trifluoromethyl)phenyl]ethyl]-12-ethoxy-27-isobutyl-N,N,3',3',4,19,22,26,32,35-decamethyl-30-[(1 S)-1-methylpropyl]-3,6,9,15,18,21,25,28,31,34,42-undecaoxo-2-[[4-(trifluoromethyl)phenyl]methyl]spiro[1,4,7,10,16,19,22,26,29,32,35-undecazatricyclo[34.5.1.0.sup.10,14]dotetracont-38-ene-17,1'-cyclobutane]-23-carboxamide, PP1827: (1S,7S,11R,13S,19S,22S,26S,29S,35S,37Z,47E)-19-cyclopentyl-7-[2-[3,5-difluoro-4-(trifluoromethyl)phenyl]ethyl]-11-ethoxy-N-ethyl-N,3,18,21,25,31,34-heptamethyl-29-[(1 S)-1-methylpropyl]-2,5,8,14,17,20,24,27,30,33,55-undecaoxo-spiro[3,6,9,15,18,21,25,28,31,34,41-undecazapentacyclo[24.15.11.2.sup.43,46.1.sup.35,41.0.sup.9,13]pentapentaconta-37,43(54),44,46(53),47-pentaene-16,1'-cyclobutane]-22-carboxamide, PP1830: (1S,7S,11R,13S,19S,22S,26S,29S,35S,37Z,47E)-19-cyclopentyl-11-ethoxy-N-ethyl-7-[2-[3-methoxy-4-(trifluoromethyl)phenyl]ethyl]-N,3,3',3',18,21,25,31,34-nonamethyl-29-[(1S)-1-methylpropyl]-2,5,8,14,17,20,24,27,30,33,55-undecaoxo-spiro[3,6,9,15,18,21,25,28,31,34,41-undecazapentacyclo[24.15.11.2.sup.43,46.1.sup.35,41.0.sup.9,13]pentapentaconta-37,43(54),44,46(53),47-pentaene-16,1'-cyclobutane]-22-carboxamide, PP2093: (2S,8S,12R,14S,20S,23S,27S,30S,36S,38Z)-20-cyclopentyl-32-cyclopropyl-8-[2-[3,5-difluoro-4-(trifluoromethyl)phenyl]ethyl]-12-ethoxy-N,N,3',3',4,19,22,26,35-nonamethyl-30-[(1 S)-1-methylpropyl]-3,6,9,15,18,21,25,28,31,34,42-undecaoxo-27-propyl-2-(p-tolylmethyl)spiro[1,4,7,10,16,19,22,26,29,32,35-undecazatricyclo[34.5.1.0.sup.10,14]dotetracont-38-ene-17,1'-cyclobutane]-23-carboxamide, PP2260: (1S,7S,11R,13S,19S,22S,26S,29S,35S,37Z)-19-cyclopentyl-11-ethoxy-N-ethyl-7-[2-[3-methoxy-4-(trifluoromethyl)phenyl]ethyl]-N,3,3',3',18,21,25,31,34-nonamethyl-29-[(1 S)-1-methylpropyl]-2,5,8,14,17,20,24,27,30,33,55-undecaoxo-spiro[3,6,9,15,18,21,25,28,31,34,41-undecazapentacyclo[24.15.11.2.sup.43,46.1.sup.35,41.0.sup.9,13]pentapentaconta-37,43(54),44,46(53)-tetraene-16,1'-cyclobutane]-22-carboxamide, PP2316: (2S,8S, 12R, 14S,20S,23S,27S,30S,36S,38Z)-20-cyclopentyl-32-cyclopropyl- 12-ethoxy-8-[2-[3-methoxy-4-(trifluoromethyl)phenyl]ethyl]-N,N,4,16,19,22,26,35-octamethyl-30-[(1 S)-1-methylpropyl]-3,6,9,15,18,21,25,28,31,34,42-undecaoxo-27-propyl-2-[[4-(trifluoromethyl)phenyl]methyl]spiro[1,4,7,10,16,19,22,26,29,32,35-undecazatricyclo[34.5.1.0.sup.10'.sup.14]dotetracont-38-ene-17,1'-cyclobutane]-23-carboxamide, PP2320: (1S,4S,10S,13S,17S,20S,26S,28R,32S,38S,42Z)-20-cyclopentyl-28-ethoxy-32-[2-[3-methoxy-4-(trifluoromethyl)phenyl]ethyl]-N, N, 2,14,18,21, 24,36-octamethyl-10-[(1 S)-1-methylpropyl]-3,9,12,15,19,22,25,31,34,37,45-undecaoxo-13-propyl-38-[[4-(trifluoromethyl)phenyl]methyl]spiro[2,8,11,14,18,21,24,30,33,36,39-undecazatetracyclo[37.5.1.0.sup.4,8.0.sup.26,30]pentatetracont-42-ene-23,1'-cyclobutane]-17-carboxamide, PP2328: (2S,8S,12R,14S,20S,23S,27S,30S,36S,38Z)-20-cyclopentyl-32-cyclopropyl-12-ethoxy-8-[2-[3-methoxy-4-(trifluoromethyl)phenyl]ethyl]-N,N,4,16,19,22,26,35-octamethyl-30-[(1 S)-1-methylpropyl]-3,6,9,15,18,21,25,28,31,34,42-undecaoxo-27-propyl-2-(p-tolylmethyl)spiro[1,4,7,10,16,19,22,26,29,32,35-undecazatricyclo[34.5.1.0.sup.10,14]dotetracont-38-ene-17,1'-cyclobutane]-23-carboxamide, PP2574: (1S,7S,11R,13S,19S,22S,26S,29S,35S,37Z,47E)-19-cyclopentyl-7-[2-[3,5-difluoro-4-(trifluoromethyl)phenyl]ethyl]-11-ethoxy-3,3',3',18,21,25,31,34-octamethyl-29-[(1 S)-1-methylpropyl]-22-(pyrrolidine-1-carbonyl)spiro[3,6,9,15,18,21,25,28,31,34,41-undecazapentacyclo[24.15.10.2.sup.43,46.1.sup.35,41.0.sup.9,13]tetrapentaconta-37,43(53),44,46(52),47-pentaene-16,1'-cyclobutane]-2,5,8,14,17,20,24,27,30,33,54-undecone, PP2576: (1S,7S,11R,13S,19S,22S,26S,29S,35S,37Z,47E)-7-[2-[3,5-difluoro-4-(trifluoromethyl)phenyl]ethyl]-11-ethoxy-19-(1-ethylpropyl)-3,3',3',18,21,25,31,34-octamethyl-29-[(1 S)-1-methylpropyl]-22-(pyrrolidine-1-carbonyl)spiro[3,6,9,15,18,21,25,28,31,34,41-undecazapentacyclo[24.15.10.2.sup.43,46.1.sup.35,41.0.sup.9,13]tetrapentaconta-37,43(53),44,46(52),47-pentaene-16,1'-cyclobutane]-2,5,8,14,17,20,24,27,30,33,54-undecone, PP2583: (1S,7S,11R,13S,19S,22S,26S,29S,35S,37Z,47E)-19-cyclopentyl-31-cyclopropyl-7-[2-

[3,5-difluoro-4-(trifluoromethyl)phenyl]ethyl]-11-ethoxy-3,3',3',18,21,25,34-heptamethyl-29-[(1S)-1-methylpropyl]-22-(pyrrolidine-1-carbonyl)spiro[3,6,9,15,18,21,25,28,31,34,41-undecazapentacyclo[24.15.10.2.sup.43,46.1.sup.35,41.0.sup.9,13]tetrapentaconta-37,43(53),44,46(52),47-pentaene-16,1'-cyclobutane]-2,5,8,14,17,20,24,27,30,33,54-undecone, PP2687: (2S,8S,12R,14S,20S,23S,27S,30S,36S,38Z)-20-cyclopentyl-12-ethoxy-2-[(4-ethylphenyl)methyl]-27-isobutyl-8-[2-[3-methoxy-4-(trifluoromethyl)phenyl]ethyl]-N,N,3',3',4,19,22,26,32,35-decamethyl-30-[(1S)-1-methylpropyl]-3,6,9,15,18,21,25,28,31,34,42-undecaoxo-spiro[1,4,7,10,16,19,22,26,29,32,35-undecazatricyclo[34.5.1.0.sup.10,14]dotetracont-38-ene-17,1'-cyclobutane]-23-carboxamide, PP2691: (2S,8S,12R,14S,20S,23S,27S,30S,36S,38Z)-20-cyclopentyl-2-[(4-cyclopropylphenyl)methyl]-12-ethoxy-27-isobutyl-8-[2-[3-methoxy-4-(trifluoromethyl)phenyl]ethyl]-N,N,3',3',4,19,22,26,32,35-decamethyl-30-[(1S)-1-methylpropyl]-3,6,9,15,18,21,25,28,31,34,42-undecaoxo-spiro[1,4,7,10,16,19,22,26,29,32,35-undecazatricyclo[34.5.1.0.sup.10'.sup.14]dotetracont-38-ene-17,1'-cyclobutane]-23-carboxamide, PP2957: (1S,7S,11R,13S,19S,22S,26S,29S,35S,37Z,47E)-31-cyclopropyl-7-[2-[3,5-difluoro-4-(trifluoromethyl)phenyl]ethyl]-11-ethoxy-19-(1-ethylpropyl)-3,18,21,25,34-pentamethyl-29-[(1S)-1-methylpropyl]-22-(pyrrolidine-1-carbonyl)spiro[3,6,9,15,18,21,25,28,31,34,41-undecazapentacyclo[24.15.10.2.sup.43,46.1.sup.35,41.0.sup.9,13]tetrapentaconta-37,43(53),44,46(52),47-pentaene-16,1'-cyclobutane]-2,5,8,14,17,20,24,27,30,33,54-undecone, PP3033: (1S,7S,11R,13S,19S,22S,26S,29S,35S,37Z)-19-cyclopentyl-11-ethoxy-7-[2-[3-methoxy-4-(trifluoromethyl)phenyl]ethyl]-3,18,21,25,31,34-hexamethyl-29-[(1S)-1-methylpropyl]-22-(piperidine-1-carbonyl)spiro[3,6,9,15,18,21,25,28,31,34,41-undecazapentacyclo[24.15.11.2.sup.43,46.1.sup.35,41.0.sup.9,13]pentapentaconta-37,43(54),44,46(53)-tetraene-16,1'-cyclobutane]-2,5,8,14,17,20,24,27,30,33,55-undecone, PP3034: (1S,7S,11R,13S,19S,22S,26S,29S,35S,37Z)-19-cyclopentyl-11-ethoxy-7-[2-[3-methoxy-4-(trifluoromethyl)phenyl]ethyl]-3,3',3',18,21,25,31,34-octamethyl-29-[(1S)-1-methylpropyl]-22-(pyrrolidine-1-carbonyl)spiro[3,6,9,15,18,21,25,28,31,34,41-undecazapentacyclo[24.15.11.2.sup.43,46.1.sup.35,41.0.sup.9,13]pentapentaconta-37,43(54),44,46(53)-tetraene-16,1'-cyclobutane]-2,5,8,14,17,20,24,27,30,33,55-undecone, PP3036: (1S,7S,11R,13S,19S,22S,26S,29S,35S,37Z)-19-cyclopentyl-11-ethoxy-7-[2-[3-methoxy-4-(trifluoromethyl)phenyl]ethyl]-3,3',3',18,21,25,31,34-octamethyl-29(1S)-1-methylpropyl]-22-(piperidine-1-carbonyl)spiro[3,6,9,15,18,21,25,28,31,34,41-undecazapentacyclo[24.15.10.2.sup.43,46.1.sup.35,41.0.sup.9,13]tetrapentaconta-37,43(53),44,46(52)-tetraene-16,1'-cyclobutane]-2,5,8,14,17,20,24,27,30,33,54-undecone, PP3037: (1S,7S,11R,13S,19S,22S,26S,29S,35S,37Z)-19-cyclopentyl-11-ethoxy-7-[2-[3-methoxy-4-(trifluoromethyl)phenyl]ethyl]-3,3',3',18,21,25,31,34-octamethyl-29-[(1S)-1-methylpropyl]-22-(pyrrolidine-1-carbonyl)spiro[3,6,9,15,18,21,25,28,31,34,41-undecazapentacyclo[24.15.10.2.sup.43,46.1.sup.35,41.0.sup.9,13]tetrapentaconta-37,43(53),44,46(52)-tetraene-16,1'-cyclobutane]-2,5,8,14,17,20,24,27,30,33,54-undecone, PP3047: (1S,7S,11R,13S,19S,22S,26S,29S,35S,37Z,47E)-7-[2-(3,4-dichlorophenyl)ethyl]-11-ethoxy-19-(1-ethylpropyl)-3,18,21,25,31,34-hexamethyl-29-[(1S)-1-methylpropyl]-22-(pyrrolidine-1-carbonyl)spiro[3,6,9,15,18,21,25,28,31,34,41-undecazapentacyclo[24.15.11.2.sup.43,46.1.sup.35,41.0.sup.9,13]pentapentaconta-37,43(54),44,46(53),47-pentaene-16,1'-cyclobutane]-2,5,8,14,17,20,24,27,30,33,55-undecone, PP3093: (1S,7S,11R,13S,19S,22S,26S,29S,35S,37Z)-19-cyclopentyl-11-ethoxy-7-[2-[3-methoxy-4-(trifluoromethyl)phenyl]ethyl]-3,3',3',18,21,25,31,34-octamethyl-29-[(1S)-1-methylpropyl]-22-(piperidine-1-carbonyl)spiro[3,6,9,15,18,21,25,28,31,34,41-undecazapentacyclo[24.15.9.2.sup.43,46.1.sup.35,41.0.sup.9,13]tripentaconta-37,43(52),44,46(51)-tetraene-16,1'-cyclobutane]-2,5,8,14,17,20,24,27,30,33,53-undecone, PP3094: (1S,7S,11R,13S,19S,22S,26S,29S,35S,37Z)-19-cyclopentyl-7-[2-[3-methoxy-4-

(trifluoromethyl)phenyl]ethyl]-3,3',18,21,25,31,34-octamethyl-29-[(1 S)-1-methylpropyl]-11-propoxy-22-(pyrrolidine-1-carbonyl)spiro[3,6,9,15,18,21,25,28,31,34,41-undecazapentacyclo[24.15.9.2.sup.43,46.1.sup.35,41.0.sup.9,13]tripentaconta-37,43(52),44,46(51)-tetraene-16,1'-cyclobutane]-2,5,8,14,17,20,24,27,30,33,53-undecone, PP3095:
(2S,8S,12R,14S,20S,23S,27S,30S,36S,38Z)-20-cyclopentyl-32-cyclopropyl-12-ethoxy-8-[2-[3-methoxy-4-(trifluoromethyl)phenyl]ethyl]-4,16,19,22,26,35-hexamethyl-30-[(1S)-1-methylpropyl]-23-(piperidine-1-carbonyl)-27-propyl-2-[[4-(trifluoromethyl)phenyl]methyl]spiro[1,4,7,10,16,19,22,26,29,32,35-undecazatricyclo[34.5.1.0.sup.10,14]dotetracont-38-ene-17,1'-cyclobutane]-3,6,9,15,18,21,25,28,31,34,42-undecone, PP3096:
(2S,8S,12R,14S,20S,23S,27S,30S,36S,38Z)-20-cyclopentyl-32-cyclopropyl-12-ethoxy-8-[2-[3-methoxy-4-(trifluoromethyl)phenyl]ethyl]-4,16,19,22,26,35-hexamethyl-30-[(1S)-1-methylpropyl]-23-(piperidine-1-carbonyl)-27-propyl-2-(p-tolylmethyl)spiro[1,4,7,10,16,19,22,26,29,32,35-undecazatricyclo[34.5.1.0.sup.10,14]dotetracont-38-ene-17,1-cyclobutane]-3,6,9,15,18,21,25,28,31,34,42-undecone, PP3097:
(2S,8S,12R,14S,20S,23S,27S,30S,36S,38Z)-20-cyclopentyl-32-cyclopropyl-12-ethoxy-8-[2-[3-methoxy-4-(trifluoromethyl)phenyl]ethyl]-4,16,19,22,26,35-hexamethyl-30-[(1S)-1-methylpropyl]-27-propyl-23-(pyrrolidine-1-carbonyl)-2-[[4-(trifluoromethyl)phenyl]methyl]spiro[1,4,7,10,16,19,22,26,29,32,35-undecazatricyclo[34.5.1.0.sup.10,14]dotetracont-38-ene-17,1'-cyclobutane]-3,6,9,15,18,21,25,28,31,34,42-undecone, PP3098:
(2S,8S,12R,14S,20S,23S,27S,30S,36S,38Z)-20-cyclopentyl-32-cyclopropyl-12-ethoxy-8-[2-[3-methoxy-4-(trifluoromethyl)phenyl]ethyl]-4,16,19,22,26,35-hexamethyl-30-[(1S)-1-methylpropyl]-27-propyl-2-(p-tolylmethyl)-23-(pyrrolidine-1-carbonyl)spiro[1,4,7,10,16,19,22,26,29,32,35-undecazatricyclo[34.5.1.0.sup.10,14]dotetracont-38-ene-17,1-cyclobutane]-3,6,9,15,18,21,25,28,31,34,42-undecone, PP3099:
(1S,4S,10S,13S,17S,20S,26S,28R,32S,38S,42Z)-20-cyclopentyl-28-ethoxy-32-[2-[3-methoxy-4-(trifluoromethyl)phenyl]ethyl]-2,14,18,21,24,36-hexamethyl-10-[(1 S)-1-methylpropyl]-17-(piperidine-1-carbonyl)-13-propyl-38-[[4-(trifluoromethyl)phenyl]methyl]spiro[2,8,11,14,18,21,24,30,33,36,39-undecazatetracyclo[37.5.1.0.sup.4,8.0.sup.26,30]pentatetracont-42-ene-23,1'-cyclobutane]-3,9,12,15,19,22,25,31,34,37,45-undecone, PP3100:
(1S,4S,10S,13S,17S,20S,26S,28R,32S,38S,42Z)-20-cyclopentyl-28-ethoxy-32-[2-[3-methoxy-4-(trifluoromethyl)phenyl]ethyl]-2,14,18,21,24,36-hexamethyl-10-[(1 S)-1-methylpropyl]-17-(piperidine-1-carbonyl)-13-propyl-38-(p-tolylmethyl)spiro[2,8,11,14,18,21,24,30,33,36,39-undecazatetracyclo[37.5.1.0.sup.4,8.0.sup.26,30]pentatetracont-42-ene-23,1'-cyclobutane]-3,9,12,15,19,22,25,31,34,37,45-undecone, PP3101:
(1S,4S,10S,13S,17S,20S,26S,28R,32S,38S,42Z)-20-cyclopentyl-28-ethoxy-32-[2-[3-methoxy-4-(trifluoromethyl)phenyl]ethyl]-2,14,18,21,24,36-hexamethyl-10-[(1 S)-1-methylpropyl]-13-propyl-17-(pyrrolidine-1-carbonyl)-38-[[4-(trifluoromethyl)phenyl]methyl]spiro[2,8,11,14,18,21,24,30,33,36,39-undecazatetracyclo[37.5.1.0.sup.4,8.0.sup.26,30]pentatetracont-42-ene-23,1'-cyclobutane]-3,9,12,15,19,22,25,31,34,37,45-undecone, PP3102:
(1S,4S,10S,13S,17S,20S,26S,28R,32S,38S,42Z)-20-cyclopentyl-28-ethoxy-32-[2-[3-methoxy-4-(trifluoromethyl)phenyl]ethyl]-2,14,18,21,24,36-hexamethyl-10-[(1 S)-1-methylpropyl]-13-propyl-38-(p-tolylmethyl)-17-(pyrrolidine-1-carbonyl)spiro[2,8,11,14,18,21,24,30,33,36,39-undecazatetracyclo[37.5.1.0.sup.4,8.0.sup.26,30]pentatetracont-42-ene-23,1'-cyclobutane]-3,9,12,15,19,22,25,31,34,37,45-undecone, PP3103:
(2S,8S,12R,14S,20S,23S,27S,30S,36S,38Z)-20-cyclopentyl-12-ethoxy-8-[2-[3-methoxy-4-(trifluoromethyl)phenyl]ethyl]-4,16,19,22,26,32,35-heptamethyl-30-[(1 S)-1-methylpropyl]-23-

(piperidine-1-carbonyl)-27-propyl-2-[[4-(trifluoromethyl)phenyl]methyl]spiro[1,4,7,10,16,19,22,26,29,32,35-undecazatricyclo[34.5.1.0.sup.10,14]dotetracont-38-ene-17,1'-cyclobutane]-3,6,9,15,18,21,25,28,31,34,42-undecone, PP3104:
(2S,8S,12R,14S,20S,23S,27S,30S,36S,38Z)-20-cyclopentyl-12-ethoxy-8-[2-[3-methoxy-4-(trifluoromethyl)phenyl]ethyl]-4,16,19,22,26,32,35-heptamethyl-30-[(1S)-1-methylpropyl]-23-(piperidine-1-carbonyl)-27-propyl-2-(p-tolylmethyl)spiro[1,4,7,10,16,19,22,26,29,32,35-undecazatricyclo[34.5.1.0.sup.10,14]dotetracont-38-ene-17,1'-cyclobutane]-3,6,9,15,18,21,25,28,31,34,42-undecone, PP3105:
(2S,8S,12R,14S,20S,23S,27S,30S,36S,38Z)-20-cyclopentyl-12-ethoxy-8-[2-[3-methoxy-4-(trifluoromethyl)phenyl]ethyl]-4,16,19,22,26,32,35-heptamethyl-30-[(1 S)-1-methylpropyl]-27-propyl-23-(pyrrolidine-1-carbonyl)-2-[[4-(trifluoromethyl)phenyl]methyl]spiro[1,4,7,10,16,19,22,26,29,32,35-undecazatricyclo[34.5.1.0.sup.10,14]dotetracont-38-ene-17,1'-cyclobutane]-3,6,9,15,18,21,25,28,31,34,42-undecone, PP3106:
(2S,8S,12R,14S,20S,23S,27S,30S,36S,38Z)-20-cyclopentyl-12-ethoxy-8-[2-[3-methoxy-4-(trifluoromethyl)phenyl]ethyl]-4,16,19,22,26,32,35-heptamethyl-30-[(1S)-1-methylpropyl]-27-propyl-2-(p-tolylmethyl)-23-(pyrrolidine-1-carbonyl)spiro[1,4,7,10,16,19,22,26,29,32,35-undecazatricyclo[34.5.1.0.sup.10,14]dotetracont-38-ene-17,1'-cyclobutane]-3,6,9,15,18,21,25,28,31,34,42-undecone, PP3110:
(2S,8S,12R,14S,20S,23S,27S,30S,36S,38Z)-20-cyclopentyl-32-cyclopropyl-8-[2-[3,5-difluoro-4-(trifluoromethyl)phenyl]ethyl]-12-ethoxy-4,16,19,22, 26,35-hexamethyl-30-11(1S)- 1-methylpropyl]-23-(piperidine-1-carbonyl)-27-propyl-2-11114-(trifluoromethyl)phenyl]methyl]spiro[1,4,7,10,16,19,22,26,29,32,35-undecazatricyclo[34.5.1.0.sup.10,14]dotetracont-38-ene-17,1'-cyclobutane]-3,6,9,15,18,21,25,28,31,34,42-undecone, PP3111:
(2S,8S,12R,14S,20S,23S,27S,30S,36S,38Z)-20-cyclopentyl-32-cyclopropyl-8-[2-[3,5-difluoro-4-(trifluoromethyl)phenyl]ethyl]-12-ethoxy-4,16,19,22,26,35-hexamethyl-30-[(1 S)-1-methylpropyl]-23-(piperidine-1-carbonyl)-27-propyl-2-(p-tolylmethyl)spiro[[1,4,7,10,16,19,22,26,29,32,35-undecazatricyclo[34.5.1.0.sup.10,14]dotetracont-38-ene-17,1-cyclobutane]-3,6,9,15,18,21,25,28,31,34,42-undecone, PP3112:
(2S,8S,12R,14S,20S,23S,27S,30S,36S,38Z)-20-cyclopentyl-32-cyclopropyl-8-[2-[3,5-difluoro-4-(trifluoromethyl)phenyl]ethyl]-12-ethoxy-4,16,19,22,26,35-hexamethyl-30-[(1 S)-1-methylpropyl]-27-propyl-23-(pyrrolidine-1-carbonyl)-2-[[4-(trifluoromethyl)phenyl]methyl]spiro[1,4,7,10,16,19,22,26,29,32,35-undecazatricyclo[34.5.1.0.sup.10,14]dotetracont-38-ene-17,1'-cyclobutane]-3,6,9,15,18,21,25,28,31,34,42-undecone, PP3113:
(2S,8S,12R,14S,20S,23S,27S,30S,36S,38Z)-20-cyclopentyl-32-cyclopropyl-8-[2-[3,5-difluoro-4-(trifluoromethyl)phenyl]ethyl]-12-ethoxy-4,16,19,22,26,35-hexamethyl-30-[(1 S)-1-methylpropyl]-27-propyl-2-(p-tolylmethyl)-23-(pyrrolidine-1-carbonyl)spiro[1,4,7,10,16,19,22,26,29,32,35-undecazatric clo[34.5.1.0.sup.10,14]dotetracont-38-ene-17,1-cyclobutane]-3,6,9,15,18,21,25,28,31,34,42-undecone, PP3114:
(1S,4S,10S,13S,17S,20S,26S,28R,32S,38S,42Z)-20-cyclopentyl-32-[2-[3,5-difluoro-4-(trifluoromethyl)phenyl]ethyl]-28-ethoxy-2,14,18,21,24,36-hexamethyl-10-[(1 S)-1-methylpropyl]-17-(piperidine-1-carbonyl)-13-propyl-38-[[4-(trifluoromethyl)phenyl]methyl]spiro[2,8,11,14,18,21,24,30,33,36,39-undecazatetracyclo[37.5.1.0.sup.4,8.0.sup.26,30]pentatetracont-42-ene-23,1'-cyclobutane]-3,9,12,15,19,22,25,31,34,37,45-undecone, PP3115:
(1S,4S,10S,13S,17S,20S,26S,28R,32S,38S,42Z)-20-cyclopentyl-32-[2-[3,5-difluoro-4-(trifluoromethyl)phenyl]ethyl]-28-ethoxy-2,14,18,21,24,36-hexamethyl-10-[(1 S)-1-

methylpropyl]-17-(piperidine-1-carbonyl)-13-propyl-38-(p-
 tolylmethyl)spiro[2,8,11,14,18,21,24,30,33,36,39-
 undecazatetracyclo[37.5.1.0.sup.4,8.0.sup.26,30]pentatetracont-42-ene-23,1'-
 cyclobutane]-3,9,12,15,19,22,25,31,34,37,45-undecone, PP3116:
 (1S,4S,10S,13S,17S,20S,26S,28R,32S,38S,42Z)-20-cyclopentyl-32-[2-[3,5-difluoro-4-
 (trifluoromethyl)phenyl]ethyl]-28-ethoxy-2,14,18,21,24,36-hexamethyl-10-[(1 S)-1-
 methylpropyl]-13-propyl-17-(pyrrolidine-1-carbonyl)-38-[[4-
 (trifluoromethyl)phenyl]methyl]spiro[2,8,11,14,18,21,24,30,33,36,39-
 undecazatetracyclo[37.5.1.0.sup.4,8.0.sup.26,30]pentatetracont-42-ene-23,1'-
 cyclobutane]-3,9,12,15,19,22,25,31,34,37,45-undecone, PP3117:
 (1S,4S,10S,13S,17S,20S,26S,28R,32S,38S,42Z)-20-cyclopentyl-32-[2-[3,5-difluoro-4-
 (trifluoromethyl)phenyl]ethyl]-28-ethoxy-2,14,18,21,24,36-hexamethyl-10-[(1 S)-1-
 methylpropyl]-13-propyl-38-(p-tolylmethyl)-17-(pyrrolidine-1-
 carbonyl)spiro[2,8,11,14,18,21,24,30,33,36,39-
 undecazatetracyclo[37.5.1.0.sup.4,8.0.sup.26,30]pentatetracont-42-ene-23,1'-
 cyclobutane]-3,9,12,15,19,22,25,31,34,37,45-undecone, PP3118:
 (2S,8S,12R,14S,20S,23S,27S,30S,36S,38Z)-20-cyclopentyl-8-[2-[3,5-difluoro-4-
 (trifluoromethyl)phenyl]ethyl]-12-ethoxy-4,16,19,22,26,32,35-heptamethyl-30-[(1 S)-1-
 methylpropyl]-23-(piperidine-1-carbonyl)-27-propyl-2-[[4-
 (trifluoromethyl)phenyl]methyl]spiro[1,4,7,10,16,19,22,26,29,32,35-
 undecazatricyclo[34.5.1.0.sup.10,14]dotetracont-38-ene-17,1'-
 cyclobutane]-3,6,9,15,18,21,25,28,31,34,42-undecone, PP3119:
 (2S,8S,12R,14S,20S,23S,27S,30S,36S,38Z)-20-cyclopentyl-8-[2-[3,5-difluoro-4-
 (trifluoromethyl)phenyl]ethyl]-12-ethoxy-4,16,19,22,26,32,35-heptamethyl-30-[(1 S)-1-
 methylpropyl]-23-(piperidine-1-carbonyl)-27-propyl-2-(p-
 tolylmethyl)spiro[[1,4,7,10,16,19,22,26,29,32,35-undecazatricyclo[34.5.1.0.sup.10,14]dotetracont-
 38-ene-17,1-cyclobutane]-3,6,9,15,18,21,25,28,31,34,42-undecone, PP3120:
 (2S,8S,12R,14S,20S,23S,27S,30S,36S,38Z)-20-cyclopentyl-8-[2-[3,5-difluoro-4-
 (trifluoromethyl)phenyl]ethyl]-12-ethoxy-4,16,19,22,26,32,35-heptamethyl-30-[(1 S)-1-
 methylpropyl]-27-propyl-23-(pyrrolidine-1-carbonyl)-2-[[4-
 (trifluoromethyl)phenyl]methyl]spiro[1,4,7,10,16,19,22,26,29,32,35-
 undecazatricyclo[34.5.1.0.sup.10,14]dotetracont-38-ene-17,1'-
 cyclobutane]-3,6,9,15,18,21,25,28,31,34,42-undecone, PP3121:
 (2S,8S,12R,14S,20S,23S,27S,30S,36S,38Z)-20-cyclopentyl-8-[2-[3,5-difluoro-4-
 (trifluoromethyl)phenyl]ethyl]-12-ethoxy-4,16,19,22,26,32,35-heptamethyl-30-[(1 S)-1-
 methylpropyl]-27-propyl-2-(p-tolylmethyl)-23-(pyrrolidine-1-
 carbonyl)spiro[1,4,7,10,16,19,22,26,29,32,35-undecazatricyclo[34.5.1.0.sup.10,14]dotetracont-38-
 ene-17,1-cyclobutane]-3,6,9,15,18,21,25,28,31,34,42-undecone,
 (1S,7S,11R,13S,19S,22S,26S,29S,35S,37Z,47Z)-19-cyclopentyl-7-[2-[3,5-difluoro-4-
 (trifluoromethyl)phenyl]ethyl]-11-ethoxy-N-ethyl-N,3,18,21,25,31,34-heptamethyl-29-[(1S)-1-
 methylpropyl]-2,5,8,14,17,20,24,27,30,33,55-undeca-oxo-spiro[3,6,9,15,18,21,25,28,31,34,41-
 undecazapentacyclo[24.15.11.2.sup.43,46.1.sup.35,41.0.sup.9,13]pentapentaconta-
 37,43(54),44,46(53),47-pentaene-16,1'-cyclobutane]-22-carboxamide,
 (1S,7S,11R,13S,19S,22S,26S,29S,35S,37Z,47Z)-19-cyclopentyl-11-ethoxy-N-ethyl-7-[2-[3-
 methoxy-4-(trifluoromethyl)phenyl]ethyl]-N,3,3',3',18,21,25,31,34-nonamethyl-29-[(1 S)-1-
 methylpropyl]-2,5,8,14,17,20,24,27,30,33,55-undeca-oxo-spiro[3,6,9,15,18,21,25,28,31,34,41-
 undecazapentacyclo[24.15.11.2.sup.43,46.1.sup.35,41.0.sup.9,13]pentapentaconta-
 37,43(54),44,46(53),47-pentaene-16,1'-cyclobutane]-22-carboxamide, (1 S,7S,11 R,13
 S,19S,22S,26S,29S,35S,37Z,47Z)-19-cyclopentyl-7-[2-[3,5-difluoro-4-
 (trifluoromethyl)phenyl]ethyl]-11-ethoxy-3,3',3',18,21,25,31,34-octamethyl-29-[(1 S)-1-

methylpropyl]-22-(pyrrolidine-1-carbonyl)spiro[3,6,9,15,18,21,25,28,31,34,41-undecazapentacyclo[24.15.10.2.sup.43,46.1.sup.35,41.0.sup.9,13]tetrapentaconta-37,43(53),44,46(52),47-pentaene-16,1'-cyclobutane]-2,5,8,14,17,20,24,27,30,33,54-undecone, (1S,7S,11R,13S,19S,22S,26S,29S,35S,37Z,47Z)-7-[2-[3,5-difluoro-4-(trifluoromethyl)phenyl]ethyl]-11-ethoxy-19-(1-ethylpropyl)-3,3',3',18,21,25,31,34-octamethyl-29-[(1 S)-1-methylpropyl]-22-(pyrrolidine-1-carbonyl)spiro[3,6,9,15,18,21,25,28,31,34,41-undecazapentacyclo[24.15.10.2.sup.43,46.1.sup.35,41.0.sup.9,13]tetrapentaconta-37,43(53),44,46(52),47-pentaene-16,1'-cyclobutane]-2,5,8,14,17,20,24,27,30,33,54-undecone, (1S,7S,11R,13S,19S,22S,26S,29S,35S,37Z,47Z)-19-cyclopentyl-31-cyclopropyl-7-[2-[3,5-difluoro-4-(trifluoromethyl)phenyl]ethyl]-11-ethoxy-3,3',3',18,21,25,34-heptamethyl-29-[(1 S)-1-methylpropyl]-22-(pyrrolidine-1-carbonyl)spiro[3,6,9,15,18,21,25,28,31,34,41-undecazapentacyclo[24.15.10.2.sup.43,46.1.sup.35,41.0.sup.9,13]tetrapentaconta-37,43(53),44,46(52),47-pentaene-16,1'-cyclobutane]-2,5,8,14,17,20,24,27,30,33,54-undecone, (1S,7S,11R,13S,19S,22S,26S,29S,35S,37Z,47Z)-31-cyclopropyl-7-[2-[3,5-difluoro-4-(trifluoromethyl)phenyl]ethyl]-11-ethoxy-19-(1-ethylpropyl)-3,18,21,25,34-pentamethyl-29-[(1 S)-1-methylpropyl]-22-(pyrrolidine-1-carbonyl)spiro[3,6,9,15,18,21,25,28,31,34,41-undecazapentacyclo[24.15.10.2.sup.43,46.1.sup.35,41.0.sup.9,13]tetrapentaconta-37,43(53),44,46(52),47-pentaene-16,1'-cyclobutane]-2,5,8,14,17,20,24,27,30,33,54-undecone, and (1S,7S,11R,13S,19S,22S,26S,29S,35S,37Z,47Z)-7-[2-(3,4-dichlorophenyl)ethyl]-11-ethoxy-19-(1-ethylpropyl)-3,18,21,25,31,34-hexamethyl-29-[(1 S)-1-methylpropyl]-22-(pyrrolidine-1-carbonyl)spiro[3,6,9,15,18,21,25,28,31,34,41-undecazapentacyclo[24.15.11.2.sup.43,46.1.sup.35,41.0.sup.9,13]pentapentaconta-37,43(54),44,46(53),47-pentaene-16,1'-cyclobutane]-2,5,8,14,17,20,24,27,30,33,55-undecone.

(26) The cyclic compound or a salt thereof, or a solvate thereof according to [1], wherein the formula (1) is expressed by formula (2):

(27) ##STR00004## wherein  custom character,  custom character, n, m, P.sub.1, R.sub.2, R.sub.3, P.sub.3, P.sub.4, P.sub.6, R.sub.7, R.sub.8, P.sub.8, R.sub.9, P.sub.9, Q.sub.9, R.sub.10, P.sub.10, R.sub.11 and P.sub.11 are the same as defined in [1], the cyclic compound represented by formula (2) is selected from the group consisting of: PP1827:

(1S,7S,11R,13S,19S,22S,26S,29S,35S,37Z,47E)-19-cyclopentyl-7-[2-[3,5-difluoro-4-(trifluoromethyl)phenyl]ethyl]-11-ethoxy-N-ethyl-N,3,18,21,25,31,34-heptamethyl-29-[(1 S)-1-methylpropyl]-2,5,8,14,17,20,24,27,30,33,55-undecaoxo-spiro[3,6,9,15,18,21,25,28,31,34,41-undecazapentacyclo[24.15.11.2.sup.43,46.1.sup.35,41.0.sup.9,13]pentapentaconta-37,43(54),44,46(53),47-pentaene-16,1'-cyclobutane]-22-carboxamide, PP1830:

(1S,7S,11R,13S,19S,22S,26S,29S,35S,37Z,47E)-19-cyclopentyl-11-ethoxy-N-ethyl-7-[2-[3-methoxy-4-(trifluoromethyl)phenyl]ethyl]-N,3,3',3',18,21,25,31,34-nonamethyl-29-[(1 S)-1-methylpropyl]-2,5,8,14,17,20,24,27,30,33,55-undecaoxo-spiro[3,6,9,15,18,21,25,28,31,34,41-undecazapentacyclo[24.15.11.2.sup.43,46.1.sup.35,41.0.sup.9,13]pentapentaconta-37,43(54),44,46(53),47-pentaene-16,1'-cyclobutane]-22-carboxamide, PP2260:

(1S,7S,11R,13S,19S,22S,26S,29S,35S,37Z)-19-cyclopentyl-11-ethoxy-N-ethyl-7-[2-[3-methoxy-4-(trifluoromethyl)phenyl]ethyl]-N,3,3',3',18,21,25,31,34-nonamethyl-29-[(1 S)-1-methylpropyl]-2,5,8,14,17,20,24,27,30,33,55-undecaoxo-spiro[3,6,9,15,18,21,25,28,31,34,41-undecazapentacyclo[24.15.11.2.sup.43,46.1.sup.35,41.0.sup.9,13]pentapentaconta-37,43(54),44,46(53)-tetraene-16,1'-cyclobutane]-22-carboxamide, PP2574:

(1S,7S,11R,13S,19S,22S,26S,29S,35S,37Z,47E)-19-cyclopentyl-7-[2-[3,5-difluoro-4-(trifluoromethyl)phenyl]ethyl]-11-ethoxy-3,3',3',18,21,25,31,34-octamethyl-29-[(1 S)-1-methylpropyl]-22-(pyrrolidine-1-carbonyl)spiro[3,6,9,15,18,21,25,28,31,34,41-undecazapentacyclo[24.15.10.2.sup.43,46.1.sup.35,41.0.sup.9,13]tetrapentaconta-37,43(53),44,46(52),47-pentaene-16,1'-cyclobutane]-2,5,8,14,17,20,24,27,30,33,54-undecone, PP2576: (1S,7S,11R,13S,19S,22S,26S,29S,35S,37Z,47E)-7-[2-[3,5-difluoro-4-

(trifluoromethyl)phenyl]ethyl]-11-ethoxy-19-(1-ethylpropyl)-3,3',3',18,21,25,31,34-octamethyl-29-[(1 S)-1-methylpropyl]-22-(pyrrolidine-1-carbonyl)spiro[3,6,9,15,18,21,25,28,31,34,41-undecazapentacyclo[24.15.10.2.sup.43,46.1.sup.35,41.0.sup.9,13]tetrapentaconta-37,43(53),44,46(52),47-pentaene-16,1'-cyclobutane]-2,5,8,14,17,20,24,27,30,33,54-undecone, PP2583: (1S,7S,11R,13S,19S,22S,26S,29S,35S,37Z,47E)-19-cyclopentyl-31-cyclopropyl-7-[2-[3,5-difluoro-4-(trifluoromethyl)phenyl]ethyl]-11-ethoxy-3,3',3',18,21,25,34-heptamethyl-29-[(1S)-1-methylpropyl]-22-(pyrrolidine-1-carbonyl)spiro[3,6,9,15,18,21,25,28,31,34,41-undecazapentacyclo[24.15.10.2.sup.43,46.1.sup.35,41.0.sup.9,13]tetrapentaconta-37,43(53),44,46(52),47-pentaene-16,1'-cyclobutane]-2,5,8,14,17,20,24,27,30,33,54-undecone, PP2957: (1S,7S,11R,13S,19S,22S,26S,29S,35S,37Z,47E)-31-cyclopropyl-7-[2-[3,5-difluoro-4-(trifluoromethyl)phenyl]ethyl]-11-ethoxy-19-(1-ethylpropyl)-3,18,21,25,34-pentamethyl-29-[(1S)-1-methylpropyl]-22-(pyrrolidine-1-carbonyl)spiro[3,6,9,15,18,21,25,28,31,34,41-undecazapentacyclo[24.15.10.2.sup.43,46.1.sup.35,41.0.sup.9,13]tetrapentaconta-37,43(53),44,46(52),47-pentaene-16,1'-cyclobutane]-2,5,8,14,17,20,24,27,30,33,54-undecone, PP3033: (1S,7S,11R,13S,19S,22S,26S,29S,35S,37Z)-19-cyclopentyl-11-ethoxy-7-[2-[3-methoxy-4-(trifluoromethyl)phenyl]ethyl]-3,18,21,25,31,34-hexamethyl-29-[(1 S)-1-methylpropyl]-22-(piperidine-1-carbonyl)spiro[3,6,9,15,18,21,25,28,31,34,41-undecazapentacyclo[24.15.11.2.sup.43,46.1.sup.35,41.0.sup.9,13]pentapentaconta-37,43(54),44,46(53)-tetraene-16,1'-cyclobutane]-2,5,8,14,17,20,24,27,30,33,55-undecone, PP3034: (1S,7S,11R,13S,19S,22S,26S,29S,35S,37Z)-19-cyclopentyl-11-ethoxy-7-[2-[3-methoxy-4-(trifluoromethyl)phenyl]ethyl]-3,3',3',18,21,25,31,34-octamethyl-29-[(1S)-1-methylpropyl]-22-(pyrrolidine-1-carbonyl)spiro[3,6,9,15,18,21,25,28,31,34,41-undecazapentacyclo[24.15.11.2.sup.43,46.1.sup.35,41.0.sup.9,13]pentapentaconta-37,43(54),44,46(53)-tetraene-16,1'-cyclobutane]-2,5,8,14,17,20,24,27,30,33,55-undecone, PP3036: (1S,7S,11R,13S,19S,22S,26S,29S,35S,37Z)-19-cyclopentyl-11-ethoxy-7-[2-[3-methoxy-4-(trifluoromethyl)phenyl]ethyl]-3,3',3',18,21,25,31,34-octamethyl-294(1S)-1-methylpropyl]-22-(piperidine-1-carbonyl)spiro[3,6,9,15,18,21,25,28,31,34,41-undecazapentacyclo[24.15.10.2.sup.43,46.1.sup.35,41.0.sup.9,13]tetrapentaconta-37,43(53),44,46(52)-tetraene-16,1'-cyclobutane]-2,5,8,14,17,20,24,27,30,33,54-undecone, PP3037: (1S,7S,11R,13S,19S,22S,26S,29S,35S,37Z)-19-cyclopentyl-11-ethoxy-7-[2-[3-methoxy-4-(trifluoromethyl)phenyl]ethyl]-3,3',3',18,21,25,31,34-octamethyl-29-[(1S)-1-methylpropyl]-22-(pyrrolidine-1-carbonyl)spiro[3,6,9,15,18,21,25,28,31,34,41-undecazapentacyclo[24.15.10.2.sup.43,46.1.sup.35,41.0.sup.9,13]tetrapentaconta-37,43(53),44,46(52)-tetraene-16,1'-cyclobutane]-2,5,8,14,17,20,24,27,30,33,54-undecone, PP3047: (1S,7S,IIR,13S,19S,22S,26S,29S,35S,37Z,47E)-7-[2-(3,4-dichlorophenyl)ethyl]-11-ethoxy-19-(1-ethylpropyl)-3,18,21,25,31,34-hexamethyl-29-[(1S)-1-methylpropyl]-22-(pyrrolidine-1-carbonyl)spiro[3,6,9,15,18,21,25,28,31,34,41-undecazapentacyclo[24.15.11.2.sup.43,46.1.sup.35,41.0.sup.9,13]pentapentaconta-37,43(54),44,46(53),47-pentaene-16,1'-cyclobutane]-2,5,8,14,17,20,24,27,30,33,55-undecone, PP3093: (1S,7S,11R,13S,19S,22S,26S,29S,35S,37Z)-19-cyclopentyl-11-ethoxy-7-[2-[3-methoxy-4-(trifluoromethyl)phenyl]ethyl]-3,3',3',18,21,25,31,34-octamethyl-29-[(1S)-1-methylpropyl]-22-(piperidine-1-carbonyl)spiro[3,6,9,15,18,21,25,28,31,34,41-undecazapentacyclo[24.15.9.2.sup.43,46.1.sup.35,41.0.sup.9,13]tripentaconta-37,43(52),44,46(51)-tetraene-16,1'-cyclobutane]-2,5,8,14,17,20,24,27,30,33,53-undecone, PP3094: (1S,7S,11R,13S,19S,22S,26S,29S,35S,37Z)-19-cyclopentyl-7-[2-[3-methoxy-4-(trifluoromethyl)phenyl]ethyl]-3,3',3',18,21,25,31,34-octamethyl-29-[(1 S)-1-methylpropyl]-11-propoxy-22-(pyrrolidine-1-carbonyl)spiro[3,6,9,15,18,21,25,28,31,34,41-undecazapentacyclo[24.15.9.2.sup.43,46.1.sup.35,41.0.sup.9,13]tripentaconta-37,43(52),44,46(51)-tetraene-16,1'-cyclobutane]-2,5,8,14,17,20,24,27,30,33,53-undecone, (1S,7S,11R,13S,19S,22S,26S,29S,35S,37Z,47Z)-19-cyclopentyl-7-[2-[3,5-difluoro-4-

(trifluoromethyl)phenyl]ethyl]-11-ethoxy-N-ethyl-N,3,18,21,25,31,34-heptamethyl-29-[(1 S)-1-methylpropyl]-2,5,8,14,17,20,24,27,30,33,55-undecaoxo-spiro[3,6,9,15,18,21,25,28,31,34,41-undecazapentacyclo[24.15.11.2.sup.43,46.1.sup.35,41.0.sup.9,13]pentapentaconta-37,43(54),44,46(53),47-pentaene-16,1'-cyclobutane]-22-carboxamide, (1 S,7S,11R,13S,19S,22S,26S,29S,35S,37Z,47Z)-19-cyclopentyl-11-ethoxy-N-ethyl-7-[2-[3-methoxy-4-(trifluoromethyl)phenyl]ethyl]-N,3,3',3',18,21,25,31,34-nonamethyl-29-[(1 S)-1-methylpropyl]-2,5,8,14,17,20,24,27,30,33,55-undecaoxo-spiro[3,6,9,15,18,21,25,28,31,34,41-undecazapentacyclo[24.15.11.2.sup.43,46.1.sup.35,41.0.sup.9,13]pentapentaconta-37,43(54),44,46(53),47-pentaene-16,1'-cyclobutane]-22-carboxamide, (1 S,7S,11R,13S,19S,22S,26S,29S,35S,37Z,47Z)-19-cyclopentyl-7-[2-[3,5-difluoro-4-(trifluoromethyl)phenyl]ethyl]-11-ethoxy-3,3',3',18,21,25,31,34-octamethyl-29-[(1 S)-1-methylpropyl]-22-(pyrrolidine-1-carbonyl)spiro[3,6,9,15,18,21,25,28,31,34,41-undecazapentacyclo[24.15.10.2.sup.43,46.1.sup.35,41.0.sup.9,13]tetrapentaconta-37,43(53),44,46(52),47-pentaene-16,1'-cyclobutane]-2,5,8,14,17,20,24,27,30,33,54-undecone, (1 S,7S,11R,13S,19S,22S,26S,29S,35S,37Z,47Z)-7-[2-[3,5-difluoro-4-(trifluoromethyl)phenyl]ethyl]-11-ethoxy-19-(1-ethylpropyl)-3,3',3',18,21,25,31,34-octamethyl-29-[(1 S)-1-methylpropyl]-22-(pyrrolidine-1-carbonyl)spiro[3,6,9,15,18,21,25,28,31,34,41-undecazapentacyclo[24.15.10.2.sup.43,46.1.sup.35,41.0.sup.9,13]tetrapentaconta-37,43(53),44,46(52),47-pentaene-16,1'-cyclobutane]-2,5,8,14,17,20,24,27,30,33,54-undecone, (1 S,7S,11R,13S,19S,22S,26S,29S,35S,37Z,47Z)-19-cyclopentyl-31-cyclopropyl-7-[2-[3,5-difluoro-4-(trifluoromethyl)phenyl]ethyl]-11-ethoxy-3,3',3',18,21,25,34-heptamethyl-29-[(1 S)-1-methylpropyl]-22-(pyrrolidine-1-carbonyl)spiro[3,6,9,15,18,21,25,28,31,34,41-undecazapentacyclo[24.15.10.2.sup.43,46.1.sup.35,41.0.sup.9,13]tetrapentaconta-37,43(53),44,46(52),47-pentaene-16,1'-cyclobutane]-2,5,8,14,17,20,24,27,30,33,54-undecone, (1S,7S,11R,13S,19S,22S,26S,29S,35S,37Z,47Z)-31-cyclopropyl-7-[2-[3,5-difluoro-4-(trifluoromethyl)phenyl]ethyl]-11-ethoxy-19-(1-ethylpropyl)-3,18,21,25,34-pentamethyl-29-[(1 S)-1-methylpropyl]-22-(pyrrolidine-1-carbonyl)spiro[3,6,9,15,18,21,25,28,31,34,41-undecazapentacyclo[24.15.10.2.sup.43,46.1.sup.35,41.0.sup.9,13]tetrapentaconta-37,43(53),44,46(52),47-pentaene-16,1'-cyclobutane]-2,5,8,14,17,20,24,27,30,33,54-undecone, and (1 S,7S,11R,13 S,19S,22S,26S,29S,35 S,37Z,47Z)-7-[2-(3,4-dichlorophenyl)ethyl]-11-ethoxy-19-(1-ethylpropyl)-3,18,21,25,31,34-hexamethyl-29-[(1 S)-1-methylpropyl]-22-(pyrrolidine-1-carbonyl)spiro[3,6,9,15,18,21,25,28,31,34,41-undecazapentacyclo[24.15.11.2.sup.43,46.1.sup.35,41.0.sup.9,13]pentapentaconta-37,43(54),44,46(53),47-pentaene-16,1'-cyclobutane]-2,5,8,14,17,20,24,27,30,33,55-undecone.

(28) The cyclic compound or a salt thereof, or a solvate thereof according to [1], wherein the formula (1) is expressed by formula (3):

(29) ##STR00005## wherein, R.sub.1 is a C.sub.1 to C.sub.7 alkyl; R.sub.5 is benzyl optionally substituted with one or more groups selected from the group consisting of a C.sub.1 to C.sub.6 alkyl, a C.sub.1 to C.sub.6 haloalkyl and a C.sub.3 to C.sub.8 cycloalkyl;  custom character,  custom character, n, m, P.sub.1, R.sub.2, R.sub.3, P.sub.3, P.sub.4, P.sub.6, R.sub.7, R.sub.8, P.sub.9, R.sub.9, P.sub.9, Q.sub.9, R.sub.10, P.sub.10, R.sub.11 and P.sub.11 are the same as defined in [1], wherein the cyclic compound represented by formula (3) is selected from the group consisting of: PP1574: (2S,8S,12R,14S,20S,23S,27S,30S,36S,38Z)-20-cyclopentyl-8-[2-[3,5-difluoro-4-(trifluoromethyl)phenyl]ethyl]-12-ethoxy-N-ethyl-27-isobutyl-N,4,19,22,26,32,35-heptamethyl-30-[(1 S)-1-methylpropyl]-3,6,9,15,18,21,25,28,31,34,42-undecaoxo-2-[[4-(trifluoromethyl)phenyl]methyl]spiro[1,4,7,10,16,19,22,26,29,32,35-undecazatricyclo[34.5.1.0.sup.10,14]dotetracont-38-ene-17,1'-cyclopentane]-23-carboxamide, PP1650: (2S,8S,12R,14S,20S,23S,27S,30S,36S,38Z)-20-cyclopentyl-8-[2-[3,5-difluoro-4-(trifluoromethyl)phenyl]ethyl]-12-ethoxy-27-isobutyl-N,N,3',3',4,19,22,26,32,35-decamethyl-30-[(1 S)-1-methylpropyl]-3,6,9,15,18,21,25,28,31,34,42-undecaoxo-2-[[4-

(trifluoromethyl)phenyl)methyl]spiro[1,4,7,10,16,19,22,26,29,32,35-undecazatricyclo[34.5.1.0.sup.10,14]dotetracont-38-ene-17,1'-cyclobutane]-23-carboxamide, PP1827: (1S,7S,11R,13S,19S,22S,26S,29S,35S,37Z,47E)-19-cyclopentyl-7-[2-[3,5-difluoro-4-(trifluoromethyl)phenyl]ethyl]-11-ethoxy-N-ethyl-N,3,18,21,25,31,34-heptamethyl-29-[(1 S)-1-methylpropyl]-2,5,8,14,17,20,24,27,30,33,55-undecaoxo-spiro[3,6,9,15,18,21,25,28,31,34,41-undecazapentacyclo[24.15.11.2.sup.43,46.1.sup.35,41.0.sup.9,13]pentapentaconta-37,43(54),44,46(53),47-pentaene-16,1'-cyclobutane]-22-carboxamide, PP1830: (1S,7S,11R,13S,19S,22S,26S,29S,35S,37Z,47E)-19-cyclopentyl-11-ethoxy-N-ethyl-7-[2-[3-methoxy-4-(trifluoromethyl)phenyl]ethyl]-N,3,3',3',18,21,25,31,34-nonamethyl-29-[(1S)-1-methylpropyl]-2,5,8,14,17,20,24,27,30,33,55-undecaoxo-spiro[3,6,9,15,18,21,25,28,31,34,41-undecazapentacyclo[24.15.11.2.sup.43,46.1.sup.35,41.0.sup.9,13]pentapentaconta-37,43(54),44,46(53),47-pentaene-16,1'-cyclobutane]-22-carboxamide, PP2093: (2S,8S,12R,14S,20S,23S,27S,30S,36S,38Z)-20-cyclopentyl-32-cyclopropyl-8-[2-[3,5-difluoro-4-(trifluoromethyl)phenyl]ethyl]-12-ethoxy-N,N,3',3',4,19,22,26,35-nonamethyl-30-[(1 S)-1-methylpropyl]-3,6,9,15,18,21,25,28,31,34,42-undecaoxo-27-propyl-2-(p-tolylmethyl)spiro[[1,4,7,10,16,19,22,26,29,32,35-undecazatricyclo[34.5.1.0.sup.10,14]dotetracont-38-ene-17,1'-cyclobutane]-23-carboxamide, PP2260: (1S,7S,11R,13S,19S,22S,26S,29S,35S,37Z)-19-cyclopentyl-11-ethoxy-N-ethyl-7-[2-[3-methoxy-4-(trifluoromethyl)phenyl]ethyl]-N,3,3',3',18,21,25,31,34-nonamethyl-29-[(1 S)-1-methylpropyl]-2,5,8,14,17,20,24,27,30,33,55-undecaoxo-spiro[3,6,9,15,18,21,25,28,31,34,41-undecazapentacyclo[24.15.11.2.sup.43,46.1.sup.35,41.0.sup.9,13]pentapentaconta-37,43(54),44,46(53)-tetraene-16,1'-cyclobutane]-22-carboxamide, PP2316: (2S,8S, 12R, 14S,20S,23S,27S,30S,36S,38Z)-20-cyclopentyl-32-cyclopropyl-12-ethoxy-8-[2-[3-methoxy-4-(trifluoromethyl)phenyl]ethyl]-N,N,4,16,19,22,26,35-octamethyl-30-[(1 S)-1-methylpropyl]-3,6,9,15,18,21,25,28,31,34,42-undecaoxo-27-propyl-2-[[4-(trifluoromethyl)phenyl]methyl]spiro[1,4,7,10,16,19,22,26,29,32,35-undecazatricyclo[34.5.1.0.sup.10'.sup.14]dotetracont-38-ene-17,1'-cyclobutane]-23-carboxamide, PP2320: (1S,4S,10S,13S,17S,20S,26S,28R,32S,38S,42Z)-20-cyclopentyl-28-ethoxy-32-[2-[3-methoxy-4-(trifluoromethyl)phenyl]ethyl]-N, N, 2,14,18,21, 24,36-octamethyl-10-[(1 S)-1-methylpropyl]-3,9,12,15,19,22,25,31,34,37,45-undecaoxo-13-propyl-38-[[4-(trifluoromethyl)phenyl]methyl]spiro[2,8,11,14,18,21,24,30,33,36,39-undecazatetracyclo[37.5.1.0.sup.4,8.0.sup.26,30]pentatetracont-42-ene-23,1'-cyclobutane]-17-carboxamide, PP2328: (2S,8S,12R,14S,20S,23S,27S,30S,36S,38Z)-20-cyclopentyl-32-cyclopropyl-12-ethoxy-8-[2-[3-methoxy-4-(trifluoromethyl)phenyl]ethyl]-N,N,4,16,19,22,26,35-octamethyl-30-[(1 S)-1-methylpropyl]-3,6,9,15,18,21,25,28,31,34,42-undecaoxo-27-propyl-2-(p-tolylmethyl)spiro[[1,4,7,10,16,19,22,26,29,32,35-undecazatricyclo[34.5.1.0.sup.10,14]dotetracont-38-ene-17,1'-cyclobutane]-23-carboxamide, PP2574: (1S,7S,11R,13S,19S,22S,26S,29S,35S,37Z,47E)-19-cyclopentyl-7-[2-[3,5-difluoro-4-(trifluoromethyl)phenyl]ethyl]-11-ethoxy-3,3',3',18,21,25,31,34-octamethyl-29-[(1 S)-1-methylpropyl]-22-(pyrrolidine-1-carbonyl)spiro[3,6,9,15,18,21,25,28,31,34,41-undecazapentacyclo[24.15.10.2.sup.43,46.1.sup.35,41.0.sup.9,13]tetrapentaconta-37,43(53),44,46(52),47-pentaene-16,1'-cyclobutane]-2,5,8,14,17,20,24,27,30,33,54-undecone, PP2576: (1S,7S,11R,13S,19S,22S,26S,29S,35S,37Z,47E)-7-[2-[3,5-difluoro-4-(trifluoromethyl)phenyl]ethyl]-11-ethoxy-19-(1-ethylpropyl)-3,3',3',18,21,25,31,34-octamethyl-29-[(1 S)-1-methylpropyl]-22-(pyrrolidine-1-carbonyl)spiro[3,6,9,15,18,21,25,28,31,34,41-undecazapentacyclo[24.15.10.2.sup.43,46.1.sup.35,41.0.sup.9,13]tetrapentaconta-37,43(53),44,46(52),47-pentaene-16,1'-cyclobutane]-2,5,8,14,17,20,24,27,30,33,54-undecone, PP2583: (1S,7S,11R,13S,19S,22S,26S,29S,35S,37Z,47E)-19-cyclopentyl-31-cyclopropyl-7-[2-[3,5-difluoro-4-(trifluoromethyl)phenyl]ethyl]-11-ethoxy-3,3',3',18,21,25,34-heptamethyl-29-[(1 S)-1-methylpropyl]-22-(pyrrolidine-1-carbonyl)spiro[3,6,9,15,18,21,25,28,31,34,41-

undecapentacyclo[24.15.10.2.sup.43,46.1.sup.35,41.0.sup.9,13]tetrapentaconta-37,43(53),44,46(52),47-pentaene-16,1'-cyclobutane]-2,5,8,14,17,20,24,27,30,33,54-undecone, PP2687: (2S,8S,12R,14S,20S,23S,27S,30S,36S,38Z)-20-cyclopentyl-12-ethoxy-2-[(4-ethylphenyl)methyl]-27-isobutyl-8-[2-[3-methoxy-4-(trifluoromethyl)phenyl]ethyl]-N,N,3',3',4,19,22,26,32,35-decamethyl-30-[(1S)-1-methylpropyl]-3,6,9,15,18,21,25,28,31,34,42-undecaoxo-spiro[1,4,7,10,16,19,22,26,29,32,35-undecazatricyclo[34.5.1.0.sup.10,141]dotetracont-38-ene-17,1'-cyclobutane]-23-carboxamide, PP2691: (2S,8S,12R,14S,20S,23S,27S,30S,36S,38Z)-20-cyclopentyl-2-[(4-cyclopropylphenyl)methyl]-12-ethoxy-27-isobutyl-8-[2-[3-methoxy-4-(trifluoromethyl)phenyl]ethyl]-N,N,3',3',4,19,22,26,32,35-decamethyl-30-[(1 S)-1-methylpropyl]-3,6,9,15,18,21,25,28,31,34,42-undecaoxo-spiro[1,4,7,10,16,19,22,26,29,32,35-undecazatricyclo[34.5.1.0.sup.10'.sup.14]dotetracont-38-ene-17,1'-cyclobutane]-23-carboxamide, PP2957: (1S,7S,11R,13S,19S,22S,26S,29S,35S,37Z,47E)-31-cyclopropyl-7-[2-[3,5-difluoro-4-(trifluoromethyl)phenyl]ethyl]-11-ethoxy-19-(1-ethylpropyl)-3,18,21,25,34-pentamethyl-29-[(1 S)-1-methylpropyl]-22-(pyrrolidine-1-carbonyl)spiro[3,6,9,15,18,21,25,28,31,34,41-undecapentacyclo[24.15.10.2.sup.43,46.1.sup.35,41.0.sup.9,13]tetrapentaconta-37,43(53),44,46(52),47-pentaene-16,1'-cyclobutane]-2,5,8,14,17,20,24,27,30,33,54-undecone, PP3033: (1S,7S,11R,13S,19S,22S,26S,29S,35S,37Z)-19-cyclopentyl-11-ethoxy-7-[2-[3-methoxy-4-(trifluoromethyl)phenyl]ethyl]-3,18,21,25,31,34-hexamethyl-29-[(1 S)-1-methylpropyl]-22-(piperidine-1-carbonyl)spiro[3,6,9,15,18,21,25,28,31,34,41-undecapentacyclo[24.15.11.2.sup.43,46.1.sup.35,41.0.sup.9,13]pentapentaconta-37,43(54),44,46(53)-tetraene-16,1'-cyclobutane]-2,5,8,14,17,20,24,27,30,33,55-undecone, PP3034: (1S,7S,11R,13S,19S,22S,26S,29S,35S,37Z)-19-cyclopentyl-11-ethoxy-7-[2-[3-methoxy-4-(trifluoromethyl)phenyl]ethyl]-3,3',3',18,21,25,31,34-octamethyl-29-[(1S)-1-methylpropyl]-22-(pyrrolidine-1-carbonyl)spiro[3,6,9,15,18,21,25,28,31,34,41-undecapentacyclo[24.15.11.2.sup.43,46.1.sup.35,41.0.sup.9,13]pentapentaconta-37,43(54),44,46(53)-tetraene-16,1'-cyclobutane]-2,5,8,14,17,20,24,27,30,33,55-undecone, PP3036: (1S,7S,11R,13S,19S,22S,26S,29S,35S,37Z)-19-cyclopentyl-11-ethoxy-7-[2-[3-methoxy-4-(trifluoromethyl)phenyl]ethyl]-3,3',3',18,21,25,31,34-octamethyl-29[(1S)-1-methylpropyl]-22-(piperidine-1-carbonyl)spiro[3,6,9,15,18,21,25,28,31,34,41-undecapentacyclo[24.15.10.2.sup.43,46.1.sup.35,41.0.sup.9,13]tetrapentaconta-37,43(53),44,46(52)-tetraene-16,1'-cyclobutane]-2,5,8,14,17,20,24,27,30,33,54-undecone, PP3037: (1S,7S,11R,13S,19S,22S,26S,29S,35S,37Z)-19-cyclopentyl-11-ethoxy-7-[2-[3-methoxy-4-(trifluoromethyl)phenyl]ethyl]-3,3',3',18,21,25,31,34-octamethyl-29-[(1S)-1-methylpropyl]-22-(pyrrolidine-1-carbonyl)spiro[3,6,9,15,18,21,25,28,31,34,41-undecapentacyclo[24.15.10.2.sup.43,46.1.sup.35,41.0.sup.9,13]tetrapentaconta-37,43(53),44,46(52)-tetraene-16,1'-cyclobutane]-2,5,8,14,17,20,24,27,30,33,54-undecone, PP3047: (1S,7S,11R,13S,19S,22S,26S,29S,35S,37Z,47E)-7-[2-(3,4-dichlorophenyl)ethyl]-11-ethoxy-19-(1-ethylpropyl)-3,18,21,25,31,34-hexamethyl-29-[(1S)-1-methylpropyl]-22-(pyrrolidine-1-carbonyl)spiro[3,6,9,15,18,21,25,28,31,34,41-undecapentacyclo[24.15.11.2.sup.43,46.1.sup.35,41.0.sup.9,13]pentapentaconta-37,43(54),44,46(53),47-pentaene-16,1'-cyclobutane]-2,5,8,14,17,20,24,27,30,33,55-undecone, PP3093: (1S,7S,11R,13S,19S,22S,26S,29S,35S,37Z)-19-cyclopentyl-11-ethoxy-7-[2-[3-methoxy-4-(trifluoromethyl)phenyl]ethyl]-3,3',3',18,21,25,31,34-octamethyl-29-[(1S)-1-methylpropyl]-22-(piperidine-1-carbonyl)spiro[3,6,9,15,18,21,25,28,31,34,41-undecapentacyclo[24.15.9.2.sup.43,46.1.sup.35,41.0.sup.9,13]tripentaconta-37,43(52),44,46(51)-tetraene-16,1'-cyclobutane]-2,5,8,14,17,20,24,27,30,33,53-undecone, PP3094: (1S,7S,11R,13S,19S,22S,26S,29S,35S,37Z)-19-cyclopentyl-7-[2-[3-methoxy-4-(trifluoromethyl)phenyl]ethyl]-3,3',3',18,21,25,31,34-octamethyl-29-[(1 S)-1-methylpropyl]-11-propoxy-22-(pyrrolidine-1-carbonyl)spiro[3,6,9,15,18,21,25,28,31,34,41-

undecazapentacyclo[24.15.9.2.sup.43,46.1.sup.35,41.0.sup.9,13]tripentaconta-37,43(52),44,46(51)-tetraene-16,1'-cyclobutane]-2,5,8,14,17,20,24,27,30,33,53-undecone, PP3095:
(2S,8S,12R,14S,20S,23S,27S,30S,36S,38Z)-20-cyclopentyl-32-cyclopropyl-12-ethoxy-8-[2-[3-methoxy-4-(trifluoromethyl)phenyl]ethyl]-4,16,19,22,26,35-hexamethyl-30-[(1S)-1-methylpropyl]-23-(piperidine-1-carbonyl)-27-propyl-2-[[4-(trifluoromethyl)phenyl]methyl]spiro[1,4,7,10,16,19,22,26,29,32,35-undecazatricyclo[34.5.1.0.sup.10,14]dotetracont-38-ene-17,1'-cyclobutane]-3,6,9,15,18,21,25,28,31,34,42-undecone, PP3096:
(2S,8S,12R,14S,20S,23S,27S,30S,36S,38Z)-20-cyclopentyl-32-cyclopropyl-12-ethoxy-8-[2-[3-methoxy-4-(trifluoromethyl)phenyl]ethyl]-4,16,19,22,26,35-hexamethyl-30-[(1S)-1-methylpropyl]-23-(piperidine-1-carbonyl)-27-propyl-2-(p-tolylmethyl)spiro[[1,4,7,10,16,19,22,26,29,32,35-undecazatricyclo[34.5.1.0.sup.10,14]dotetracont-38-ene-17,1-cyclobutane]-3,6,9,15,18,21,25,28,31,34,42-undecone, PP3097:
(2S,8S,12R,14S,20S,23S,27S,30S,36S,38Z)-20-cyclopentyl-32-cyclopropyl-12-ethoxy-8-[2-[3-methoxy-4-(trifluoromethyl)phenyl]ethyl]-4,16,19,22,26,35-hexamethyl-30-[(1S)-1-methylpropyl]-27-propyl-23-(pyrrolidine-1-carbonyl)-2-[[4-(trifluoromethyl)phenyl]methyl]spiro[1,4,7,10,16,19,22,26,29,32,35-undecazatricyclo[34.5.1.0.sup.10,14]dotetracont-38-ene-17,1'-cyclobutane]-3,6,9,15,18,21,25,28,31,34,42-undecone, PP3098:
(2S,8S,12R,14S,20S,23S,27S,30S,36S,38Z)-20-cyclopentyl-32-cyclopropyl-12-ethoxy-8-[2-[3-methoxy-4-(trifluoromethyl)phenyl]ethyl]-4,16,19,22,26,35-hexamethyl-30-[(1S)-1-methylpropyl]-27-propyl-2-(p-tolylmethyl)-23-(pyrrolidine-1-carbonyl)spiro[1,4,7,10,16,19,22,26,29,32,35-undecazatricyclo[34.5.1.0.sup.10,14]dotetracont-38-ene-17,1-cyclobutane]-3,6,9,15,18,21,25,28,31,34,42-undecone, PP3099:
(1S,4S,10S,13S,17S,20S,26S,28R,32S,38S,42Z)-20-cyclopentyl-28-ethoxy-32-[2-[3-methoxy-4-(trifluoromethyl)phenyl]ethyl]-2,14,18,21,24,36-hexamethyl-10-[(1 S)-1-methylpropyl]-17-(piperidine-1-carbonyl)-13-propyl-38-[[4-(trifluoromethyl)phenyl]methyl]spiro[2,8,11,14,18,21,24,30,33,36,39-undecazatetracyclo[37.5.1.0.sup.4,8.0.sup.26,30]pentatetracont-42-ene-23,1'-cyclobutane]-3,9,12,15,19,22,25,31,34,37,45-undecone, PP3100:
(1S,4S,10S,13S,17S,20S,26S,28R,32S,38S,42Z)-20-cyclopentyl-28-ethoxy-32-[2-[3-methoxy-4-(trifluoromethyl)phenyl]ethyl]-2,14,18,21,24,36-hexamethyl-10-[(1 S)-1-methylpropyl]-17-(piperidine-1-carbonyl)-13-propyl-38-(p-tolylmethyl)spiro[2,8,11,14,18,21,24,30,33,36,39-undecazatetracyclo[37.5.1.0.sup.4,8.0.sup.26,30]pentatetracont-42-ene-23,1'-cyclobutane]-3,9,12,15,19,22,25,31,34,37,45-undecone, PP3101:
(1S,4S,10S,13S,17S,20S,26S,28R,32S,38S,42Z)-20-cyclopentyl-28-ethoxy-32-[2-[3-methoxy-4-(trifluoromethyl)phenyl]ethyl]-2,14,18,21,24,36-hexamethyl-10-[(1 S)-1-methylpropyl]-13-propyl-17-(pyrrolidine-1-carbonyl)-38-[[4-(trifluoromethyl)phenyl]methyl]spiro[2,8,11,14,18,21,24,30,33,36,39-undecazatetracyclo[37.5.1.0.sup.4,8.0.sup.26,30]pentatetracont-42-ene-23,1'-cyclobutane]-3,9,12,15,19,22,25,31,34,37,45-undecone, PP3102:
(1S,4S,10S,13S,17S,20S,26S,28R,32S,38S,42Z)-20-cyclopentyl-28-ethoxy-32-[2-[3-methoxy-4-(trifluoromethyl)phenyl]ethyl]-2,14,18,21,24,36-hexamethyl-10-[(1 S)-1-methylpropyl]-13-propyl-38-(p-tolylmethyl)-17-(pyrrolidine-1-carbonyl)spiro[2,8,11,14,18,21,24,30,33,36,39-undecazatetracyclo[37.5.1.0.sup.4,8.0.sup.26,30]pentatetracont-42-ene-23,1'-cyclobutane]-3,9,12,15,19,22,25,31,34,37,45-undecone, PP3103:
(2S,8S,12R,14S,20S,23S,27S,30S,36S,38Z)-20-cyclopentyl-12-ethoxy-8-[2-[3-methoxy-4-(trifluoromethyl)phenyl]ethyl]-4,16,19,22,26,32,35-heptamethyl-30-[(1 S)-1-methylpropyl]-23-(piperidine-1-carbonyl)-27-propyl-2-[[4-(trifluoromethyl)phenyl]methyl]spiro[1,4,7,10,16,19,22,26,29,32,35-

undecazatricyclo[34.5.1.0.sup.10,14]dotetracont-38-ene-17,1'-cyclobutane]-3,6,9,15,18,21,25,28,31,34,42-undecone, PP3104: (2S,8S,12R,14S,20S,23S,27S,30S,36S,38Z)-20-cyclopentyl-12-ethoxy-8-[2-[3-methoxy-4-(trifluoromethyl)phenyl]ethyl]-4,16,19,22,26,32,35-heptamethyl-30-[(1S)-1-methylpropyl]-23-(piperidine-1-carbonyl)-27-propyl-2-(p-tolylmethyl)spiro[1,4,7,10,16,19,22,26,29,32,35-undecazatricyclo[34.5.1.0.sup.10,14]dotetracont-38-ene-17,1'-cyclobutane]-3,6,9,15,18,21,25,28,31,34,42-undecone, PP3105: (2S,8S,12R,14S,20S,23S,27S,30S,36S,38Z)-20-cyclopentyl-12-ethoxy-8-[2-[3-methoxy-4-(trifluoromethyl)phenyl]ethyl]-4,16,19,22,26,32,35-heptamethyl-30-[(1 S)-1-methylpropyl]-27-propyl-23-(pyrrolidine-1-carbonyl)-2-[[4-(trifluoromethyl)phenyl]methyl]spiro[1,4,7,10,16,19,22,26,29,32,35-undecazatricyclo[34.5.1.0.sup.10,14]dotetracont-38-ene-17,1'-cyclobutane]-3,6,9,15,18,21,25,28,31,34,42-undecone, PP3106: (2S,8S,12R,14S,20S,23S,27S,30S,36S,38Z)-20-cyclopentyl-12-ethoxy-8-[2-[3-methoxy-4-(trifluoromethyl)phenyl]ethyl]-4,16,19,22,26,32,35-heptamethyl-30-[(1S)-1-methylpropyl]-27-propyl-2-(p-tolylmethyl)-23-(pyrrolidine-1-carbonyl)spiro[1,4,7,10,16,19,22,26,29,32,35-undecazatricyclo[34.5.1.0.sup.10,14]dotetracont-38-ene-17,1'-cyclobutane]-3,6,9,15,18,21,25,28,31,34,42-undecone, PP3110: (2S,8S,12R,14S,20S,23S,27S,30S,36S,38Z)-20-cyclopentyl-32-cyclopropyl-8-[2-[3,5-difluoro-4-(trifluoromethyl)phenyl]ethyl]-12-ethoxy-4,16,19,22,26,35-hexamethyl-30-11(1S)-1-methylpropyl]-23-(piperidine-1-carbonyl)-27-propyl-2-[1114-(trifluoromethyl)phenyl]methyl]spiro[1,4,7,10,16,19,22,26,29,32,35-undecazatricyclo[34.5.1.0.sup.10,14]dotetracont-38-ene-17,1'-cyclobutane]-3,6,9,15,18,21,25,28,31,34,42-undecone, PP3111: (2S,8S,12R,14S,20S,23S,27S,30S,36S,38Z)-20-cyclopentyl-32-cyclopropyl-8-[2-[3,5-difluoro-4-(trifluoromethyl)phenyl]ethyl]-12-ethoxy-4,16,19,22,26,35-hexamethyl-30-[(1 S)-1-methylpropyl]-23-(piperidine-1-carbonyl)-27-propyl-2-(p-tolylmethyl)spiro[[1,4,7,10,16,19,22,26,29,32,35-undecazatricyclo[34.5.1.0.sup.10,14]dotetracont-38-ene-17,1-cyclobutane]-3,6,9,15,18,21,25,28,31,34,42-undecone, PP3112: (2S,8S,12R,14S,20S,23S,27S,30S,36S,38Z)-20-cyclopentyl-32-cyclopropyl-8-[2-[3,5-difluoro-4-(trifluoromethyl)phenyl]ethyl]-12-ethoxy-4,16,19,22,26,35-hexamethyl-30-[(1 S)-1-methylpropyl]-27-propyl-23-(pyrrolidine-1-carbonyl)-2-[[4-(trifluoromethyl)phenyl]methyl]spiro[1,4,7,10,16,19,22,26,29,32,35-undecazatricyclo[34.5.1.0.sup.10,14]dotetracont-38-ene-17,1'-cyclobutane]-3,6,9,15,18,21,25,28,31,34,42-undecone, PP3113: (2S,8S,12R,14S,20S,23S,27S,30S,36S,38Z)-20-cyclopentyl-32-cyclopropyl-8-[2-[3,5-difluoro-4-(trifluoromethyl)phenyl]ethyl]-12-ethoxy-4,16,19,22,26,35-hexamethyl-30-[(1 S)-1-methylpropyl]-27-propyl-2-(p-tolylmethyl)-23-(pyrrolidine-1-carbonyl)spiro[1,4,7,10,16,19,22,26,29,32,35-undecazatricyclo[34.5.1.0.sup.10,14]dotetracont-38-ene-17,1-cyclobutane]-3,6,9,15,18,21,25,28,31,34,42-undecone, PP3114: (1S,4S,10S,13S,17S,20S,26S,28R,32S,38S,42Z)-20-cyclopentyl-32-[2-[3,5-difluoro-4-(trifluoromethyl)phenyl]ethyl]-28-ethoxy-2,14,18,21,24,36-hexamethyl-10-[(1 S)-1-methylpropyl]-17-(piperidine-1-carbonyl)-13-propyl-38-[[4-(trifluoromethyl)phenyl]methyl]spiro[2,8,11,14,18,21,24,30,33,36,39-undecazatetracyclo[37.5.1.0.sup.4,8.0.sup.26,30]pentatetracont-42-ene-23,1'-cyclobutane]-3,9,12,15,19,22,25,31,34,37,45-undecone, PP3115: (1S,4S,10S,13S,17S,20S,26S,28R,32S,38S,42Z)-20-cyclopentyl-32-[2-[3,5-difluoro-4-(trifluoromethyl)phenyl]ethyl]-28-ethoxy-2,14,18,21,24,36-hexamethyl-10-[(1 S)-1-methylpropyl]-17-(piperidine-1-carbonyl)-13-propyl-38-(p-tolylmethyl)spiro[2,8,11,14,18,21,24,30,33,36,39-

undecazatetracyclo[37.5.1.0.sup.4,8.0.sup.26,30]pentatetracont-42-ene-23,1'-cyclobutane]-3,9,12,15,19,22,25,31,34,37,45-undecone, PP3116: (1S,4S,10S,13S,17S,20S,26S,28R,32S,38S,42Z)-20-cyclopentyl-32-[2-[3,5-difluoro-4-(trifluoromethyl)phenyl]ethyl]-28-ethoxy-2,14,18,21,24,36-hexamethyl-10-[(1 S)-1-methylpropyl]-13-propyl-17-(pyrrolidine-1-carbonyl)-38-[[4-(trifluoromethyl)phenyl]methyl]spiro[2,8,11,14,18,21,24,30,33,36,39-undecazatetracyclo[37.5.1.0.sup.4,8.0.sup.26,30]pentatetracont-42-ene-23,1'-cyclobutane]-3,9,12,15,19,22,25,31,34,37,45-undecone, PP3117: (1S,4S,10S,13S,17S,20S,26S,28R,32S,38S,42Z)-20-cyclopentyl-32-[2-[3,5-difluoro-4-(trifluoromethyl)phenyl]ethyl]-28-ethoxy-2,14,18,21,24,36-hexamethyl-10-[(1 S)-1-methylpropyl]-13-propyl-38-(p-tolylmethyl)-17-(pyrrolidine-1-carbonyl)spiro[2,8,11,14,18,21,24,30,33,36,39-undecazatetracyclo[37.5.1.0.sup.4,8.0.sup.26,30]pentatetracont-42-ene-23,1'-cyclobutane]-3,9,12,15,19,22,25,31,34,37,45-undecone, PP3118: (2S,8S,12R,14S,20S,23S,27S,30S,36S,38Z)-20-cyclopentyl-8-[2-[3,5-difluoro-4-(trifluoromethyl)phenyl]ethyl]-12-ethoxy-4,16,19,22,26,32,35-heptamethyl-30-[(1 S)-1-methylpropyl]-23-(piperidine-1-carbonyl)-27-propyl-2-[[4-(trifluoromethyl)phenyl]methyl]spiro[1,4,7,10,16,19,22,26,29,32,35-undecazatricyclo[34.5.1.0.sup.10,14]dotetracont-38-ene-17,1'-cyclobutane]-3,6,9,15,18,21,25,28,31,34,42-undecone, PP3119: (2S,8S,12R,14S,20S,23S,27S,30S,36S,38Z)-20-cyclopentyl-8-[2-[3,5-difluoro-4-(trifluoromethyl)phenyl]ethyl]-12-ethoxy-4,16,19,22,26,32,35-heptamethyl-30-[(1 S)-1-methylpropyl]-23-(piperidine-1-carbonyl)-27-propyl-2-(p-tolylmethyl)spiro[[1,4,7,10,16,19,22,26,29,32,35-undecazatricyclo[34.5.1.0.sup.10,14]dotetracont-38-ene-17,1-cyclobutane]-3,6,9,15,18,21,25,28,31,34,42-undecone, PP3120: (2S,8S,12R,14S,20S,23S,27S,30S,36S,38Z)-20-cyclopentyl-8-[2-[3,5-difluoro-4-(trifluoromethyl)phenyl]ethyl]-12-ethoxy-4,16,19,22,26,32,35-heptamethyl-30-[(1 S)-1-methylpropyl]-27-propyl-23-(pyrrolidine-1-carbonyl)-2-[[4-(trifluoromethyl)phenyl]methyl]spiro[1,4,7,10,16,19,22,26,29,32,35-undecazatricyclo[34.5.1.0.sup.10,14]dotetracont-38-ene-17,1'-cyclobutane]-3,6,9,15,18,21,25,28,31,34,42-undecone, and PP3121: (2S,8S,12R,14S,20S,23S,27S,30S,36S,38Z)-20-cyclopentyl-8-[2-[3,5-difluoro-4-(trifluoromethyl)phenyl]ethyl]-12-ethoxy-4,16,19,22,26,32,35-heptamethyl-30-[(1 S)-1-methylpropyl]-27-propyl-2-(p-tolylmethyl)-23-(pyrrolidine-1-carbonyl)spiro[1,4,7,10,16,19,22,26,29,32,35-undecazatricyclo[34.5.1.0.sup.10,14]dotetracont-38-ene-17,1-cyclobutane]-3,6,9,15,18,21,25,28,31,34,42-undecone. [4-1] A cyclic compound which is PP1574: (2S,8S,12R,14S,20S,23S,27S,30S,36S,38Z)-20-cyclopentyl-8-[112-.sub.113,5-difluoro-4-(trifluoromethyl)phenyl]ethyl]-12-ethoxy-N-ethyl-27-isobutyl-N,4,19,22,26,32,35-heptamethyl-30-[(1 S)-1-methylpropyl]-3,6,9,15,18,21,25,28,31,34,42-undecaoso-2-[[4-(trifluoromethyl)phenyl]methyl]spiro[4,7,10,16,19,22,26,29,32,35-undecazatricyclo[34.5.1.0.sup.10'.sup.14]dotetracont-38-ene-17,1'-cyclopentane]-23-carboxamide, or a salt thereof, or a solvate thereof. [4-2] A cyclic compound which is PP1650: (2S,8S,12R,14S,20S,23S,27S,30S,36S,38Z)-20-cyclopentyl-8-[112-.sub.113,5-difluoro-4-(trifluoromethyl)phenyl]ethyl]-12-ethoxy-27-isobutyl-N,N,3,3,4, 19,22,26,32,35-decamethyl-30-[(1 S)-1-methylpropyl]-3,6,9,15,18,21,25,28,31,34,42-undecaoso-2-[[4-(trifluoromethyl)phenyl]methyl]spiro[1,4,7,10,16,19,22,26,29,32,35-undecazatricyclo[34.5.1.0.sup.10'.sup.14]dotetracont-38-ene-17,1'-cyclobutane]-23-carboxamide, or a salt thereof, or a solvate thereof. [4-3] A cyclic compound which is PP1827: (1S,7S,11R,13S,19S,22S,26S,29S,35S,37Z,47E)-19-cyclopentyl-7-[2-[3,5-difluoro-4-(trifluoromethyl)phenyl]ethyl]-11-ethoxy-N-ethyl-N,3,18,21,25,31,34-heptamethyl-29-[(1S)-1-

methylpropyl]-2,5,8,14,17,20,24,27,30,33,55-undecaoxo-spiro[3,6,9,15,18,21,25,28,31,34,41-undecazapentacyclo[24.15.11.2.sup.43,46.1.sup.35,41.0.sup.9,13]pentapentaconta-37,43(54),44,46(53),47-pentaene-16,1'-cyclobutane]-22-carboxamide, or a salt thereof, or a solvate thereof. [4-4] A cyclic compound which is PP1830: (1S,7S,11R,13S,19S,22S,26S,29S,35S,37Z,47E)-19-cyclopentyl-11-ethoxy-N-ethyl-7-[112-.sub.113-methoxy-4-(trifluoromethyl)phenyl]ethyl]-N,3,3,3, 18,21,25,31,34-nonamethyl-29-11(1N,3,3',3',18,21,25,31,34-nonamethyl-29-[(1S)-1-methylpropyl]-2,5,8,14,17,20,24,27,30,33,55-undecaoxo-spiro[3,6,9,15,18,21,25,28,31,34,41-undecazapentacyclo[24.15.11.2.sup.43,46.1.sup.35,41.0.sup.9,13]pentapentaconta-37,43(54),44,46(53),47-pentaene-16,1'-cyclobutane]-22-carboxamide, or a salt thereof, or a solvate thereof. [4-5] A cyclic compound which is PP2093: (2S,8S,12R,14S,20S,23S,27S,30S,36S,38Z)-20-cyclopentyl-32-cyclopropyl-8-[112-.sub.113,5-difluoro-4-(trifluoromethyl)phenyl]ethyl]-12-ethoxy-N,N,3',3',4,19,22,26,35-nonamethyl-30-[(1S)-1-methylpropyl]-3,6,9,15,18,21,25,28,31,34,42-undecaoxo-27-propyl-2-(p-tolylmethyl)spiro[1,4,7,10,16,19,22,26,29,32,35-undecazatricyclo[34.5.1.0.sup.10'.sup.14]dotetracont-38-ene-17,1'-cyclobutane]-23-carboxamide, or a salt thereof, or a solvate thereof. [4-6] A cyclic compound which is PP2260: (1S,7S,11R,13S,19S,22S,26S,29S,35S,37Z)-19-cyclopentyl-11-ethoxy-N-ethyl-7-[2-[3-methoxy-4-(trifluoromethyl)phenyl]ethyl]-N,3,3',3',18,21,25,31,34-nonamethyl-29-[(1S)-1-methylpropyl]-2,5,8,14,17,20,24,27,30,33,55-undecaoxo-spiro[3,6,9,15,18,21,25,28,31,34,41-undecazapentacyclo[24.15.11.2.sup.43,46.1.sup.35,41.0.sup.9,13]pentapentaconta-37,43(54),44,46(53)-tetraene-16,1'-cyclobutane]-22-carboxamide, or a salt thereof, or a solvate thereof. [4-7] A cyclic compound which is PP2316: (2S,8S, 12R, 14S,20S,23S,27S,30S,36S,38Z)-20-cyclopentyl-32-cyclopropyl-12-ethoxy-8-[2-[3-methoxy-4-(trifluoromethyl)phenyl]ethyl]-N,N,4,16,19,22,26,35-octamethyl-30-[(1S)-1-methylpropyl]-3,6,9,15,18,21,25,28,31,34,42-undecaoxo-27-propyl-2-[[4-(trifluoromethyl)phenyl]methyl]spiro[1,4,7,10,16,19,22,26,29,32,35-undecazatricyclo[34.5.1.0.sup.10'.sup.14]dotetracont-38-ene-17,1'-cyclobutane]-23-carboxamide, or a salt thereof, or a solvate thereof. [4-8] A cyclic compound which is PP2320: (1S,4S,10S,13S,17S,20S,26S,28R,32S,38S,42Z)-20-cyclopentyl-28-ethoxy-32-[2-[3-methoxy-4-(trifluoromethyl)phenyl]ethyl]-N,N,2,14,18,21,24,36-octamethyl-10-[(1S)-1-methylpropyl]-3,9,12,15,19,22,25,31,34,37,45-undecaoxo-13-propyl-38-[[4-(trifluoromethyl)phenyl]methyl]spiro[2,8,11,14,18,21,24,30,33,36,39-undecazatetracyclo[37.5.1.0.sup.4,8.0.sup.26,30]pentatetracont-42-ene-23,1'-cyclobutane]-17-carboxamide, or a salt thereof, or a solvate thereof. [4-9] A cyclic compound which is PP2328: (2S,8S, 12R, 14S,20S,23S,27S,30S,36S,38Z)-20-cyclopentyl-32-cyclopropyl-12-ethoxy-8-[2-[3-methoxy-4-(trifluoromethyl)phenyl]ethyl]-N,N,4,16,19,22,26,35-octamethyl-30-I1(1S)-1-methylpropyl]-3,6,9,15,18,21,25,28,31,34,42-undecaoxo-27-propyl-2-(p-tolylmethyl)spiro[1,4,7,10,16,19,22,26,29,32,35-undecazatricyclo[34.5.1.0.sup.10'.sup.14]dotetracont-38-ene-17,1'-cyclobutane]-23-carboxamide, or a salt thereof, or a solvate thereof. [4-10] A cyclic compound which is PP2574: (1S,7S,11R,13S,19S,22S,26S,29S,35S,37Z,47E)-19-cyclopentyl-7-[2-[3,5-difluoro-4-(trifluoromethyl)phenyl]ethyl]-11-ethoxy-3,3',3',18,21,25,31,34-octamethyl-29-[(1S)-1-methylpropyl]-22-(pyrrolidine-l-carbonyl)spiro[3,6,9,15,18,21,25,28,31,34,41-undecazapentacyclo[24.15.10.2.sup.43,46.1.sup.35,41.0.sup.9,13]tetrapentaconta-37,43(53),44,46(52),47-pentaene-16,1'-cyclobutane]-2,5,8,14,17,20,24,27,30,33,54-undecone, or a salt thereof, or a solvate thereof. [4-11] A cyclic compound which is PP2576: (1S,7S,11R,13S,19S,22S,26S,29S,35S,37Z,47E)-7-[2-[3,5-difluoro-4-(trifluoromethyl)phenyl]ethyl]-11-ethoxy-19-(1-ethylpropyl)-3,3',3',18,21,25,31,34-octamethyl-29-[(1S)-1-methylpropyl]-22-(pyrrolidine-l-carbonyl)spiro[3,6,9,15,18,21,25,28,31,34,41-

undecazapentacyclo[24.15.10.2.sup.43,46.1.sup.35,41.0.sup.9,13]tetrapentaconta-37,43(53),44,46(52),47-pentaene-16,1'-cyclobutane]-2,5,8,14,17,20,24,27,30,33,54-undecone, or a salt thereof, or a solvate thereof. [4-12] A cyclic compound which is PP2583: (1S,7S,11R,13S,19S,22S,26S,29S,35S,37Z,47E)-19-cyclopentyl-31-cyclopropyl-7-[2-[3,5-difluoro-4-(trifluoromethyl)phenyl]ethyl]-11-ethoxy-3,3',3',18,21,25,34-heptamethyl-29-[(1 S)-1-methylpropyl]-22-(pyrrolidine-l-carbonyl)spiro[3,6,9,15,18,21,25,28,31,34,41-undecazapentacyclo[24.15.10.2.sup.43,46.1.sup.35,41.0.sup.9,13]tetrapentaconta-37,43(53),44,46(52),47-pentaene-16,1'-cyclobutane]-2,5,8,14,17,20,24,27,30,33,54-undecone, or a salt thereof, or a solvate thereof. [4-13] A cyclic compound which is PP2687: (2S,8S,12R,14S,20S,23S,27S,30S,36S,38Z)-20-cyclopentyl-12-ethoxy-2-[(4-ethylphenyl)methyl]-27-isobutyl-8-[2-[3-methoxy-4-(trifluoromethyl)phenyl]ethyl]-N,N,3',3',4,19,22,26,32,35-decamethyl-30-[(1 S)-1-methylpropyl]-3,6,9,15,18,21,25,28,31,34,42-undecaoxo-spiro[1,4,7,10,16,19,22,26,29,32,35-undecazatricyclo[34.5.1.0.sup.10'.sup.14]dotetracont-38-ene-17,1'-cyclobutane]-23-carboxamide, or a salt thereof, or a solvate thereof. [4-14] A cyclic compound which is PP2691: (2S,8S,12R,14S,20S,23S,27S,30S,36S,38Z)-20-cyclopentyl-2-[(4-cyclopropylphenyl)methyl]-12-ethoxy-27-isobutyl-8-[112-.sub.113-methoxy-4-(trifluoromethyl)phenyl]ethyl]-N,N,3,3,4,19,22,26,32,35-decamethyl-30-i(1 S)-1-methylpropyl]-3,6,9,15,18,21,25,28,31,34,42-undecaoxo-spiro[1,4,7,10,16,19,22,26,29,32,35-undecazatricyclo[34.5.1.0.sup.10141dotetracont-38-ene-17,1'-cyclobutane]-23-carboxamide, or a salt thereof, or a solvate thereof. [4-15] A cyclic compound which is PP2957: (1S,7S,11R,13S,19S,22S,26S,29S,35S,37Z,47E)-31-cyclopropyl-7-[2-[3,5-difluoro-4-(trifluoromethyl)phenyl]ethyl]-11-ethoxy-19-(1-ethylpropyl)-3,18,21,25,34-pentamethyl-29-[(1 S)-1-methylpropyl]-22-(pyrrolidine-l-carbonyl)spiro[3,6,9,15,18,21,25,28,31,34,41-undecazapentacyclo[24.15.10.2.sup.43,46.1.sup.35,41.0.sup.9,13]tetrapentaconta-37,43(53),44,46(52),47-pentaene-16,1'-cyclobutane]-2,5,8,14,17,20,24,27,30,33,54-undecone, or a salt thereof, or a solvate thereof. [4-16] A cyclic compound which is PP3033: (1S,7S,11R,13S,19S,22S,26S,29S,35S,37Z)-19-cyclopentyl-11-ethoxy-7-[2-[3-methoxy-4-(trifluoromethyl)phenyl]ethyl]-3,18,21,25,31,34-hexamethyl-29-[(1S)-1-methylpropyl]-22-(piperidine-l-carbonyl)spiro[3,6,9,15,18,21,25,28,31,34,41-undecazapentacyclo[24.15.11.2.sup.43,46.1.sup.35,41.0.sup.9,13]pentapentaconta-37,43(54),44,46(53)-tetraene-16,1'-cyclobutane]-2,5,8,14,17,20,24,27,30,33,55-undecone, or a salt thereof, or a solvate thereof. [4-17] A cyclic compound which is PP3034: (1S,7S,11R,13S,19S,22S,26S,29S,35S,37Z)-19-cyclopentyl-11-ethoxy-7-[2-[3-methoxy-4-(trifluoromethyl)phenyl]ethyl]-3,3',3',18,21,25,31,34-octamethyl-29-[(1S)-1-methylpropyl]-22-(pyrrolidine-l-carbonyl)spiro[3,6,9,15,18,21,25,28,31,34,41-undecazapentacyclo[24.15.11.2.sup.43,46.1.sup.35,41.0.sup.9,13]pentapentaconta-37,43(54),44,46(53)-tetraene-16,1'-cyclobutane]-2,5,8,14,17,20,24,27,30,33,55-undecone, or a salt thereof, or a solvate thereof. [4-18] A cyclic compound which is PP3036: (1S,7S,11R,13S,19S,22S,26S,29S,35S,37Z)-19-cyclopentyl-11-ethoxy-7-[2-[3-methoxy-4-(trifluoromethyl)phenyl]ethyl]-3,3',3',18,21,25,31,34-octamethyl-29-[(1 S)-1-methylpropyl]-22-(piperidine-l-carbonyl)spiro[3,6,9,15,18,21,25,28,31,34,41-undecazapentacyclo[24.15.10.2.sup.43,46.1.sup.35,41.0.sup.9,13]tetrapentaconta-37,43(53),44,46(52)-tetraene-16,1'-cyclobutane]-2,5,8,14,17,20,24,27,30,33,54-undecone, or a salt thereof, or a solvate thereof. [4-19] A cyclic compound which is PP3037: (1S,7S,11R,13S,19S,22S,26S,29S,35S,37Z)-19-cyclopentyl-11-ethoxy-7-[2-[3-methoxy-4-(trifluoromethyl)phenyl]ethyl]-3,3',3',18,21,25,31,34-octamethyl-29-[(1 S)-1-methylpropyl]-22-(pyrrolidine-1-carbonyl)spiro[3,6,9,15,18,21,25,28,31,34,41-undecazapentacyclo[24.15.10.2.sup.43,46.1.sup.35,41.0.sup.9,13]tetrapentaconta-37,43(53),44,46(52)-tetraene-16,1'-cyclobutane]-2,5,8,14,17,20,24,27,30,33,54-undecone, or a salt

thereof, or a solvate thereof. [4-20] A cyclic compound which is PP3047: (1S,7S,11R,13S,19S,22S,26S,29S,35S,37Z,47E)-7-[2-(3,4-dichlorophenyl)ethyl]-11-ethoxy-19-(1-ethylpropyl)-3,18,21,25,31,34-hexamethyl-29-[(1 S)-1-methylpropyl]-22-(pyrrolidine-1-carbonyl)Spiro[3,6,9,15,18,21,25,28,31,34,41-undecazapentacyclo[24.15.11.2.sup.43,46.1.sup.35,41.0.sup.9,13]pentapentaconta-37,43(54),44,46(53),47-pentaene-16,1'-cyclobutane]-2,5,8,14,17,20,24,27,30,33,55-undecone, or a salt thereof, or a solvate thereof. [4-21] A cyclic compound which is PP3093: (1S,7S,11R,13S,19S,22S,26S,29S,35S,37Z)-19-cyclopentyl-11-ethoxy-7-[2-[3-methoxy-4-(trifluoromethyl)phenyl]ethyl]-3,3',3',18,21,25,31,34-octamethyl-29-[(1 S)-1-methylpropyl]-22-(piperidine-1-carbonyl)spiro[3,6,9,15,18,21,25,28,31,34,41-undecazapentacyclo[24.15.9.2.sup.43,46.1.sup.35,41.0.sup.9,13]tripentaconta-37,43(52),44,46(51)-tetraene-16,1'-cyclobutane]-2,5,8,14,17,20,24,27,30,33,53-undecone, or a salt thereof, or a solvate thereof. [4-22] A cyclic compound which is PP3094: (1S,7S,11R,13S,19S,22S,26S,29S,35S,37Z)-19-cyclopentyl-7-[2-[3-methoxy-4-(trifluoromethyl)phenyl]ethyl]-3,3',3',18,21,25,31,34-octamethyl-29-[(1 S)-1-methylpropyl]-11-propoxy-22-(pyrrolidine-1-carbonyl)spiro[3,6,9,15,18,21,25,28,31,34,41-undecazapentacyclo[24.15.9.2.sup.43,46.1.sup.35,41.0.sup.9,13]tripentaconta-37,43(52),44,46(51)-tetraene-16,1'-cyclobutane]-2,5,8,14,17,20,24,27,30,33,53-undecone, or a salt thereof, or a solvate thereof. [4-23] A cyclic compound which is PP3095: (2S,8S, 12R, 14S,20S,23S,27S,30S,36S,38Z)-20-cyclopentyl-32-cyclopropyl-12-ethoxy-8-[2-[3-methoxy-4-(trifluoromethyl)phenyl]ethyl]-4,16,19,22,26,35-hexamethyl-30-[(1 S)-1-methylpropyl]-23-(piperidine-1-carbonyl)-27-propyl-2-[[4-(trifluoromethyl)phenyl]methyl]spiro[1,4,7,10,16,19,22,26,29,32,35-undecazatricyclo[34.5.1.0.sup.10,14]dotetracont-38-ene-17,1'-cyclobutane]-3,6,9,15,18,21,25,28,31,34,42-undecone, or a salt thereof, or a solvate thereof. [4-24] A cyclic compound which is PP3096: (2S,8S,12R,14S,20S,23S,27S,30S,36S,38Z)-20-cyclopentyl-32-cyclopropyl-12-ethoxy-8-[2-[3-methoxy-4-(trifluoromethyl)phenyl]ethyl]-4,16,19,22,26,35-hexamethyl-30-[(1 S)-1-methylpropyl]-23-(piperidine-1-carbonyl)-27-propyl-2-(p-tolylmethyl)spiro[1,4,7,10,16,19,22,26,29,32,35-undecazatricyclo[34.5.1.0.sup.10,14]dotetracont-38-ene-17,1'-cyclobutane]-3,6,9,15,18,21,25,28,31,34,42-undecone, or a salt thereof, or a solvate thereof. [4-25] A cyclic compound which is PP3097: (2S,8S,12R,14S,20S,23S,27S,30S,36S,38Z)-20-cyclopentyl-32-cyclopropyl-12-ethoxy-8-[2-[3-methoxy-4-(trifluoromethyl)phenyl]ethyl]-4,16,19,22,26,35-hexamethyl-30-[(1 S)-1-methylpropyl]-27-propyl-23-(pyrrolidine-1-carbonyl)-2-[[4-(trifluoromethyl)phenyl]methyl]spiro[1,4,7,10,16,19,22,26,29,32,35-undecazatricyclo[34.5.1.0.sup.10,14]dotetracont-38-ene-17,1'-cyclobutane]-3,6,9,15,18,21,25,28,31,34,42-undecone, or a salt thereof, or a solvate thereof. [4-26] A cyclic compound which is PP3098: (2S,8S,12R,14S,20S,23S,27S,30S,36S,38Z)-20-cyclopentyl-32-cyclopropyl-12-ethoxy-8-[2-[3-methoxy-4-(trifluoromethyl)phenyl]ethyl]-4,16,19,22,26,35-hexamethyl-30-[(1 S)-1-methylpropyl]-27-propyl-2-(p-tolylmethyl)-23-(pyrrolidine-1-carbonyl)spiro[1,4,7,10,16,19,22,26,29,32,35-undecazatricyclo[34.5.1.0.sup.10,14]dotetracont-38-ene-17,1'-cyclobutane]-3,6,9,15,18,21,25,28,31,34,42-undecone, or a salt thereof, or a solvate thereof. [4-27] A cyclic compound which is PP3099: (1S,4S,10S,13S,17S,20S,26S,28R,32S,38S,42Z)-20-cyclopentyl-28-ethoxy-32-[112-113-methoxy-4-(trifluoromethyl)phenyl]ethyl]-2, 14,18,21,24,36-hexamethyl-10-[(1 S)-1-methylpropyl]-17-(piperidine-1-carbonyl)-13-propyl-38-[[4-(trifluoromethyl)phenyl]methyl]spiro[2,8,11,14,18,21,24,30,33,36,39-undecazatetracyclo[37.5.1.0.sup.4,8.0.sup.26,30]pentatetracont-42-ene-23,1'-cyclobutane]-3,9,12,15,19,22,25,31,34,37,45-undecone, or a salt thereof, or a solvate thereof. [4-28] A cyclic compound which is PP3100: (1S,4S,10S,13S,17S,20S,26S,28R,32S,38S,42Z)-20-

cyclopentyl-28-ethoxy-32-[112-113-methoxy-4-(trifluoromethyl)phenyl]ethyl]-2, 14,18,21,24,36-hexamethyl-10-[(1 S)-1-methylpropyl]-17-(piperidine-1-carbonyl)-13-propyl-38-(p-tolylmethyl)spiro[2,8,11,14,18,21,24,30,33,36,39-undecazatetracyclo[37.5.1.0.sup.4,8.0.sup.26,30]pentatetracont-42-ene-23,1'-cyclobutane]-3,9,12,15,19,22,25,31,34,37,45-undecone, or a salt thereof, or a solvate thereof. [4-29] A cyclic compound which is PP3101: (1S,4S,10S,13S,17S,20S,26S,28R,32S,38S,42Z)-20-cyclopentyl-28-ethoxy-32-[112-113-methoxy-4-(trifluoromethyl)phenyl]ethyl]-2, 14,18,21,24,36-hexamethyl-10-[(1 S)-1-methylpropyl]-13-propyl-17-(pyrrolidine-1-carbonyl)-38-[[4-(trifluoromethyl)phenyl]methyl]spiro[2,8,11,14,18,21,24,30,33,36,39-undecazatetracyclo[37.5.1.0.sup.4,8.0.sup.26,30]pentatetracont-42-ene-23,1'-cyclobutane]-3,9,12,15,19,22,25,31,34,37,45-undecone, or a salt thereof, or a solvate thereof. [4-30] A cyclic compound which is PP3102: (1S,4S,10S,13S,17S,20S,26S,28R,32S,38S,42Z)-20-cyclopentyl-28-ethoxy-32-[112-113-methoxy-4-(trifluoromethyl)phenyl]ethyl]-2, 14,18,21,24,36-hexamethyl-10-[(1 S)-1-methylpropyl]-13-propyl-38-(p-tolylmethyl)-17-(pyrrolidine-1-carbonyl)spiro[2,8,11,14,18,21,24,30,33,36,39-undecazatetracyclo[37.5.1.0.sup.4,8.0.sup.26,30]pentatetracont-42-ene-23,1'-cyclobutane]-3,9,12,15,19,22,25,31,34,37,45-undecone, or a salt thereof, or a solvate thereof. [4-31] A cyclic compound which is PP3103: (2S,8S,12R,14S,20S,23S,27S,30S,36S,38Z)-20-cyclopentyl-12-ethoxy-8-[2-[3-methoxy-4-(trifluoromethyl)phenyl]ethyl]-4,16,19,22,26,32,35-heptamethyl-30-[(1 S)-1-methylpropyl]-23-(piperidine-1-carbonyl)-27-propyl-2-[[4-(trifluoromethyl)phenyl]methyl]spiro[1,4,7,10,16,19,22,26,29,32,35-undecazatricyclo[34.5.1.0.sup.10,14]dotetracont-38-ene-17,1'-cyclobutane]-3,6,9,15,18,21,25,28,31,34,42-undecone, or a salt thereof, or a solvate thereof. [4-32] A cyclic compound which is PP3104: (2S,8S,12R,14S,20S,23S,27S,30S,36S,38Z)-20-cyclopentyl-12-ethoxy-8-[2-[3-methoxy-4-(trifluoromethyl)phenyl]ethyl]-4,16,19,22,26,32,35-heptamethyl-30-[(1 S)-1-methylpropyl]-23-(piperidine-1-carbonyl)-27-propyl-2-(p-tolylmethyl)spiro[[1,4,7,10,16,19,22,26,29,32,35-undecazatric clo[34.5.1.0.sup.10,14]dotetracont-38-ene-17,1'-cyclobutane]-3,6,9,15,18,21,25,28,31,34,42-undecone, or a salt thereof, or a solvate thereof. [4-33] A cyclic compound which is PP3105: (2S,8S,12R,14S,20S,23S,27S,30S,36S,38Z)-20-cyclopentyl-12-ethoxy-8-[2-[3-methoxy-4-(trifluoromethyl)phenyl]ethyl]-4,16,19,22,26,32,35-heptamethyl-30-[(1 S)-1-methylpropyl]-27-propyl-23-(pyrrolidine-1-carbonyl)-2-[[4-(trifluoromethyl)phenyl]methyl]spiro[1,4,7,10,16,19,22,26,29,32,35-undecazatricyclo[34.5.1.0.sup.10,14]dotetracont-38-ene-17,1'-cyclobutane]-3,6,9,15,18,21,25,28,31,34,42-undecone, or a salt thereof, or a solvate thereof. [4-34] A cyclic compound which is PP3106: (2S,8S,12R,14S,20S,23S,27S,30S,36S,38Z)-20-cyclopentyl-12-ethoxy-8-[2-[3-methoxy-4-(trifluoromethyl)phenyl]ethyl]-4,16,19,22,26,32,35-heptamethyl-30-[(1 S)-1-methylpropyl]-27-propyl-2-(p-tolylmethyl)-23-(pyrrolidine-1-carbonyl)spiro[1,4,7,10,16,19,22,26,29,32,35-undecazatric clo[34.5.1.0.sup.10,14]dotetracont-38-ene-17,1'-cyclobutane]-3,6,9,15,18,21,25,28,31,34,42-undecone, or a salt thereof, or a solvate thereof. [4-35] A cyclic compound which is PP3110: (2S,8S,12R,14S,20S,23S,27S,30S,36S,38Z)-20-cyclopentyl-32-cyclopropyl-8-[2-[3,5-difluoro-4-(trifluoromethyl)phenyl]ethyl]-12-ethoxy-4,16,19,22,26,35-hexamethyl-30-[(1 S)-1-methylpropyl]-23-(piperidine-1-carbonyl)-27-propyl-2-[[4-(trifluoromethyl)phenyl]methyl]spiro[1,4,7,10,16,19,22,26,29,32,35-undecazatricyclo[34.5.1.0.sup.10,14]dotetracont-38-ene-17,1'-cyclobutane]-3,6,9,15,18,21,25,28,31,34,42-undecone, or a salt thereof, or a solvate thereof. [4-36] A cyclic compound which is PP3111: (2S,8S,12R,14S,20S,23S,27S,30S,36S,38Z)-20-cyclopentyl-32-cyclopropyl-8-[2-[3,5-difluoro-4-(trifluoromethyl)phenyl]ethyl]-12-ethoxy-4,16,19,22,26,35-hexamethyl-30-[(1 S)-1-methylpropyl]-23-(piperidine-1-carbonyl)-27-propyl-2-

tolylmethyl)spiro[[1,4,7,10,16,19,22,26,29,32,35]-undecazatricyclo[34.5.1.0.sup.10,14]dotetracont-38-ene-17,1'-cyclobutane]-3,6,9,15,18,21,25,28,31,34,42-undecone, or a salt thereof, or a solvate thereof. [4-37] A cyclic compound which is PP3112:

(2S,8S,12R,14S,20S,23S,27S,30S,36S,38Z)-20-cyclopentyl-32-cyclopropyl-8-[2-[3,5-difluoro-4-(trifluoromethyl)phenyl]ethyl]-12-ethoxy-4,16,19,22,26,35-hexamethyl-30-[(1 S)-1-methylpropyl]-27-propyl-23-(pyrrolidine-1-carbonyl)-2-[[4-(trifluoromethyl)phenyl]methyl]spiro[1,4,7,10,16,19,22,26,29,32,35-undecazatricyclo[34.5.1.0.sup.10,14]dotetracont-38-ene-17,1'-cyclobutane]-3,6,9,15,18,21,25,28,31,34,42-undecone, or a salt thereof, or a solvate thereof. [4-38]

A cyclic compound which is PP3113: (2S,8S,12R,14S,20S,23S,27S,30S,36S,38Z)-20-cyclopentyl-32-cyclopropyl-8-[2-[3,5-difluoro-4-(trifluoromethyl)phenyl]ethyl]-12-ethoxy-4,16,19,22,26,35-hexamethyl-30-[(1 S)-1-methylpropyl]-27-propyl-2-(p-tolylmethyl)-23-(pyrrolidine-1-carbonyl)spiro[1,4,7,10,16,19,22,26,29,32,35-undecazatricyclo[34.5.1.0.sup.10,14]dotetracont-38-ene-17,1'-cyclobutane]-3,6,9,15,18,21,25,28,31,34,42-undecone, or a salt thereof, or a solvate thereof. [4-39] A cyclic compound which is PP3114:

(1S,4S,10S,13S,17S,20S,26S,28R,32S,38S,42Z)-20-cyclopentyl-32-[2-[3,5-difluoro-4-(trifluoromethyl)phenyl]ethyl]-28-ethoxy-2,14,18,21,24,36-hexamethyl-10-[(1 S)-1-methylpropyl]-17-(piperidine-1-carbonyl)-13-propyl-38-[[4-(trifluoromethyl)phenyl]methyl]spiro[2,8,11,14,18,21,24,30,33,36,39-undecazatetracyclo[37.5.1.0.sup.4,8.0.sup.26,30]pentatetracont-42-ene-23,1'-cyclobutane]-3,9,12,15,19,22,25,31,34,37,45-undecone, or a salt thereof, or a solvate thereof. [4-40]

A cyclic compound which is PP3115: (1S,4S,10S,13S,17S,20S,26S,28R,32S,38S,42Z)-20-cyclopentyl-32-[2-[3,5-difluoro-4-(trifluoromethyl)phenyl]ethyl]-28-ethoxy-2,14,18,21, 24,36-hexamethyl-10-[(1 S)-1-methylpropyl]-17-(piperidine-1-carbonyl)-13-propyl-38-(p-tolylmethyl)spiro[2,8,11,14,18,21,24,30,33,36,39-undecazatetracyclo[37.5.1.0.sup.4,8.0.sup.26,30]pentatetracont-42-ene-23,1'-cyclobutane]-3,9,12,15,19,22,25,31,34,37,45-undecone, or a salt thereof, or a solvate thereof. [4-41]

A cyclic compound which is PP3116: (1S,4S,10S,13S,17S,20S,26S,28R,32S,38S,42Z)-20-cyclopentyl-32-[2-[3,5-difluoro-4-(trifluoromethyl)phenyl]ethyl]-28-ethoxy-2,14,18,21, 24,36-hexamethyl-10-[(1 S)-1-methylpropyl]-13-propyl-17-(pyrrolidine-1-carbonyl)-38-[[4-(trifluoromethyl)phenyl]methyl]spiro[2,8,11,14,18,21,24,30,33,36,39-undecazatetracyclo[37.5.1.0.sup.4,8.0.sup.26,30]pentatetracont-42-ene-23,1'-cyclobutane]-3,9,12,15,19,22,25,31,34,37,45-undecone, or a salt thereof, or a solvate thereof. [4-42]

A cyclic compound which is PP3117: (1S,4S,10S,13S,17S,20S,26S,28R,32S,38S,42Z)-20-cyclopentyl-32-[2-[3,5-difluoro-4-(trifluoromethyl)phenyl]ethyl]-28-ethoxy-2,14,18,21, 24,36-hexamethyl-10-[(1 S)-1-methylpropyl]-13-propyl-38-(p-tolylmethyl)-17-(pyrrolidine-1-carbonyl)spiro[2,8,11,14,18,21,24,30,33,36,39-undecazatetracyclo[37.5.1.0.sup.4,8.0.sup.26,30]pentatetracont-42-ene-23,1'-cyclobutane]-3,9,12,15,19,22,25,31,34,37,45-undecone, or a salt thereof, or a solvate thereof. [4-43]

A cyclic compound which is PP3118: (2S,8S,12R,14S,20S,23S,27S,30S,36S,38Z)-20-cyclopentyl-8-[2-[3,5-difluoro-4-(trifluoromethyl)phenyl]ethyl]-12-ethoxy-4,16,19,22,26,32,35-heptamethyl-30-[(1 S)-1-methylpropyl]-23-(piperidine-1-carbonyl)-27-propyl-2-[[4-(trifluoromethyl)phenyl]methyl]spiro[1,4,7,10,16,19,22,26,29,32,35-undecazatricyclo[34.5.1.0.sup.10,14]dotetracont-38-ene-17,1'-cyclobutane]-3,6,9,15,18,21,25,28,31,34,42-undecone, or a salt thereof, or a solvate thereof. [4-44]

A cyclic compound which is PP3119: (2S,8S,12R,14S,20S,23S,27S,30S,36S,38Z)-20-cyclopentyl-8-[2-[3,5-difluoro-4-(trifluoromethyl)phenyl]ethyl]-12-ethoxy-4,16,19,22,26,32,35-heptamethyl-30-[(1 S)-1-methylpropyl]-23-(piperidine-1-carbonyl)-27-propyl-2-(p-tolylmethyl)spiro[[1,4,7,10,16,19,22,26,29,32,35-undecazatricyclo[34.5.1.0.sup.10,14]dotetracont-38-ene-17,1'-cyclobutane]-3,6,9,15,18,21,25,28,31,34,42-undecone, or a salt thereof, or a solvate

thereof. [4-45] A cyclic compound is PP3120: (2S,8S,12R,14S,20S,23S,27S,30S,36S,38Z)-20-cyclopentyl-8-[2-[3,5-difluoro-4-(trifluoromethyl)phenyl]ethyl]-12-ethoxy-4,16,19,22,26,32,35-heptamethyl-30-[(1 S)-1-methylpropyl]-27-propyl-23-(pyrrolidine-1-carbonyl)-2-[[4-(trifluoromethyl)phenyl]methyl]spiro[1,4,7,10,16,19,22,26,29,32,35-undecazatricyclo[34.5.1.0.sup.10,14]dotetracont-38-ene-17,1'-cyclobutane]-3,6,9,15,18,21,25,28,31,34,42-undecone, or a salt thereof, or a solvate thereof. [4-46] A cyclic compound which is PP3121: (2S,8S,12R,14S,20S,23S,27S,30S,36S,38Z)-20-cyclopentyl-8-[2-[3,5-difluoro-4-(trifluoromethyl)phenyl]ethyl]-12-ethoxy-4,16,19,22,26,32,35-heptamethyl-30-[(1 S)-1-methylpropyl]-27-propyl-2-(p-tolylmethyl)-23-(pyrrolidine-1-carbonyl)spiro[1,4,7,10,16,19,22,26,29,32,35-undecazatric clo[34.5.1.0.sup.10,14]dotetracont-38-ene-17,1'-cyclobutane]-3,6,9,15,18,21,25,28,31,34,42-undecone, or a salt thereof, or a solvate thereof. [4-47] A cyclic compound which is (1S,7S,11R,13S,19S,22S,26S,29S,35S,37Z,47Z)-19-cyclopentyl-7-[2-[3,5-difluoro-4-(trifluoromethyl)phenyl]ethyl]-11-ethoxy-N-ethyl-N,3,18,21,25,31,34-heptamethyl-29-[(1S)-1-methylpropyl]-2,5,8,14,17,20,24,27,30,33,55-undecaoxo-spiro[3,6,9,15,18,21,25,28,31,34,41-undecazapentacyclo[24.15.11.2.sup.43,46.1.sup.35,41.0.sup.9,13]pentapentaconta-37,43(54),44,46(53),47-pentaene-16,1'-cyclobutane]-22-carboxamide, or a salt thereof, or a solvate thereof. [4-48] A cyclic compound which is (1S,7S,11R,13S,19S,22S,26S,29S,35S,37Z,47Z)-19-cyclopentyl-11-ethoxy-N-ethyl-7-[112-.sub.113-methoxy-4-(trifluoromethyl)phenyl]ethyl]-N,3,3,3,18,21,25,31,34-nonamethyl-29-I1(1N,3,3',3',18,21,25,31,34-nonamethyl-29-[(1S)-1-methylpropyl]-2,5,8,14,17,20,24,27,30,33,55-undecaoxo-spiro[3,6,9,15,18,21,25,28,31,34,41-undecazapentacyclo[24.15.11.2.sup.43,46.1.sup.35,41.0.sup.9,13]pentapentaconta-37,43(54),44,46(53),47-pentaene-16,1'-cyclobutane]-22-carboxamide, or a salt thereof, or a solvate thereof. [4-49] A cyclic compound which is (1S,7S,11R,13S,19S,22S,26S,29S,35S,37Z,47Z)-19-cyclopentyl-7-[2-[3,5-difluoro-4-(trifluoromethyl)phenyl]ethyl]-11-ethoxy-3,3',3',18,21,25,31,34-octamethyl-29-[(1 S)-1-methylpropyl]-22-(pyrrolidine-1-carbonyl)spiro[3,6,9,15,18,21,25,28,31,34,41-undecazapentacyclo[24.15.10.2.sup.43,46.1.sup.35,41.0.sup.9,13]tetrapentaconta-37,43(53),44,46(52),47-pentaene-16,1'-cyclobutane]-2,5,8,14,17,20,24,27,30,33,54-undecone, or a salt thereof, or a solvate thereof. [4-50] A cyclic compound which is (1S,7S,11R,13S,19S,22S,26S,29S,35S,37Z,47Z)-7-[2-[3,5-difluoro-4-(trifluoromethyl)phenyl]ethyl]-11-ethoxy-19-(1-ethylpropyl)-3,3',3',18,21,25,31,34-octamethyl-29-[(1 S)-1-methylpropyl]-22-(pyrrolidine-1-carbonyl)spiro[3,6,9,15,18,21,25,28,31,34,41-undecazapentacyclo[24.15.10.2.sup.43,46.1.sup.35,41.0.sup.9,13]tetrapentaconta-37,43(53),44,46(52),47-pentaene-16,1'-cyclobutane]-2,5,8,14,17,20,24,27,30,33,54-undecone, or a salt thereof, or a solvate thereof. [4-51] A cyclic compound which is (1S,7S,11R,13S,19S,22S,26S,29S,35S,37Z,47Z)-19-cyclopentyl-31-cyclopropyl-7-[2-[3,5-difluoro-4-(trifluoromethyl)phenyl]ethyl]-11-ethoxy-3,3',3',18,21,25,34-heptamethyl-29-[(1 S)-1-methylpropyl]-22-(pyrrolidine-1-carbonyl)spiro[3,6,9,15,18,21,25,28,31,34,41-undecazapentacyclo[24.15.10.2.sup.43,46.1.sup.35,41.0.sup.9,13]tetrapentaconta-37,43(53),44,46(52),47-pentaene-16,1'-cyclobutane]-2,5,8,14,17,20,24,27,30,33,54-undecone, or a salt thereof, or a solvate thereof. [4-52] A cyclic compound which is (1S,7S,11R,13S,19S,22S,26S,29S,35S,37Z,47Z)-31-cyclopropyl-7-[2-[3,5-difluoro-4-(trifluoromethyl)phenyl]ethyl]-11-ethoxy-19-(1-ethylpropyl)-3,18,21,25,34-pentamethyl-29-[(1S)-1-methylpropyl]-22-(pyrrolidine-1-carbonyl)spiro[3,6,9,15,18,21,25,28,31,34,41-undecazapentacyclo[24.15.10.2.sup.43,46.1.sup.35,41.0.sup.9,13]tetrapentaconta-37,43(53),44,46(52),47-pentaene-16,1'-cyclobutane]-2,5,8,14,17,20,24,27,30,33,54-undecone, or a salt thereof, or a solvate thereof. [4-53] A cyclic compound which is (1S,7S,11R,13S,19S,22S,26S,29S,35S,37Z,47Z)-7-[2-(3,4-dichlorophenyl)ethyl]-11-ethoxy-19-(I-

ethylpropyl)-3,18,21,25,31,34-hexamethyl-29-[(1S)-1-methylpropyl]-22-(pyrrolidine-1-carbonyl)spiro[3,6,9,15,18,21,25,28,31,34,41]-undecazapentacyclo[24.15.11.2.sup.43,46.1.sup.35,41.0.sup.9,13]pentapentaconta-37,43(54),44,46(53),47-pentaene-16,1'-cyclobutane]-2,5,8,14,17,20,24,27,30,33,55-undecone

(30) The cyclic compound or a salt thereof according to any one of [1] to [4-53].

(31) The cyclic compound or a solvate thereof according to any one of [1] to [4-53].

(32) A solvate of the cyclic compound or a salt thereof according to any one of [1] to [4-53].

(33) The cyclic compound according to any one of [1] to [4-53].

(34) The cyclic compound or a salt thereof, or a solvate thereof according to any one of [1] to [4-53], which binds to KRAS.

(35) The cyclic compound or a salt thereof, or a solvate thereof according to any one of [1] to [4-53], which inhibits KRAS.

(36) The cyclic compound or a salt thereof, or a solvate thereof according to any one of [1] to [4-53], which has high selectivity for KRAS.

(37) The cyclic compound or a salt thereof, or a solvate thereof according to any one of [1] to [4-53], having KRAS selectivity (KRAS selectivity to NRAS and/or KRAS selectivity to HRAS) higher than PP1820: ((3S,9S,12S, 17S,20S,23S,27S,30S,36S)-3-[2-[3-chloro-4-(trifluoromethyl)phenyl]ethyl] 30)-cyclopentyl-23-isobutyl-9-(isopentyloxymethyl)-N,N,7,17,18,24,28,31-octamethyl-20-[(1S)-1-methylpropyl]-2,5,8, 11,16,19,22,25,29,32,35-undecaoxo-10-propyl-spiro[1,4,7, 10,15, 18,21,24,28,31,34-undecazatricyclo[34.3.0.012, 15] nonatriacontane-33,l'-cyclobutane]-27-carboxamide).

(38) The cyclic compound or a salt thereof, or a solvate thereof according to any one of [1] to [4-53], which selectively binds to KRAS.

(39) The cyclic compound or a salt thereof, or a solvate thereof according to any one of [1] to [4-53], which selectively inhibits KRAS.

(40) The cyclic compound or a salt thereof, or a solvate thereof according to any one of [1] to [4-53], which has KRAS binding activity that is 3 times or more higher than NRAS binding activity and HRAS binding activity.

(41) The cyclic compound or a salt thereof, or a solvate thereof according to [15], which has KRAS binding activity that is 5 times, 7 times, 10 times, 15 times, or 20 times or more higher than NRAS binding activity and HRAS binding activity.

(42) The cyclic compound or a salt thereof, or a solvate thereof according to any one of [1] to [4-53], which has KRAS inhibitory activity that is 3 times or more higher than NRAS inhibitory activity and HRAS inhibitory activity.

(43) The cyclic compound or a salt thereof, or a solvate thereof according to [17], which has KRAS inhibitory activity that is 5 times, 7 times, 10 times, 15 times, or 20 times or more higher than NRAS inhibitory activity and HRAS inhibitory activity.

(44) A pharmaceutical composition comprising the cyclic compound or a salt thereof, or a solvate thereof according to any one of [1] to [4-53].

(45) A pharmaceutical composition for selectively inhibiting KRAS in a subject, the composition comprising the cyclic compound or a salt thereof, or a solvate thereof according to any one of [1] to [4-53].

(46) The pharmaceutical composition according to [20], wherein KRAS inhibitory activity of the compound is 3 times or more higher than NRAS inhibitory activity and HRAS inhibitory activity of the compound.

(47) The pharmaceutical composition according to [21], wherein KRAS inhibitory activity of the compound is 5 times, 7 times, 10 times, 15 times, or 20 times or more higher than NRAS inhibitory activity and HRAS inhibitory activity of the compound.

(48) A pharmaceutical composition for treating or preventing cancer in a subject, the composition comprising an effective amount of the cyclic compound or a salt thereof, or a solvate thereof

according to any one of [1] to [4-53].

(49) The pharmaceutical composition according to [23], wherein the cancer is lung cancer.

(50) The pharmaceutical composition of any one of to [24], wherein the subject is a human.

(51) The cyclic compound or a salt thereof, or a solvate thereof according to any one of [1] to [4-53] for use in treatment or prevention of cancer in a subject.

(52) The cyclic compound or a salt thereof, or a solvate thereof according to [26], wherein the cancer is lung cancer.

(53) The cyclic compound or a salt thereof, or a solvate thereof according to any one of [1] to [4-53] for use in selectively inhibiting KRAS in a subject.

(54) The cyclic compound or a salt thereof, or a solvate thereof according to any one of to [28], wherein the subject is a human.

(55) Use of the cyclic compound or a salt thereof, or a solvate thereof according to any one of [1] to [4-53] in the manufacture of a medicament for treating or preventing cancer in a subject. The use according to [30], wherein the cancer is lung cancer.

(56) Use of the cyclic compound or a salt thereof, or a solvate thereof according to any one of [1] to [4-53] in the manufacture of a medicament for selectively inhibiting KRAS in a subject.

(57) The use according to any one of to [32], wherein the subject is a human.

(58) A method for treating or preventing cancer in a subject, the method comprising administering an effective amount of the cyclic compound or a salt thereof, or a solvate thereof according to any one of [1] to [4-53] to a subject in need thereof.

(59) The method according to [34], wherein the cancer is lung cancer.

(60) A method for selectively inhibiting KRAS in a subject, the method comprising administering an effective amount of the cyclic compound or a salt thereof, or a solvate thereof according to any one of [1] to [4-53] to a subject in need thereof.

(61) The method according to any one of to [36], wherein the subject is a human.

Effects of the Invention

(62) The present invention can provide novel cyclic compounds having selective KRAS inhibitory action.

Description

DESCRIPTION OF EMBODIMENTS

Abbreviations

(1) The abbreviations used herein are as follows. AA: Ammonium acetate Boc: tert-Butoxycarbonyl CSA: (+)-10-Camphorsulfonic acid CPME: Cyclopentyl methyl ether DAST: (Diethylamino) sulfur trifluoride DBU: 1,8-Diazabicyclo [5.4.0]-7-undecene DCM: Dichloromethane DCE: 1,2-Dichloroethane DEAD: Diethyl azodicarboxylate DEPBT: 3-(Diethoxyphosphoryloxy)-1,2,3-benzotriazin-4 (3H)-one DIAD: Diisopropyl azodicarboxylate DIC: N,N'-Diisopropylcarbodiimide DIPEA: N,N-Diisopropylethylamine DHP: 3,4-Dihydro-2H-pyran DMA: N,N-Dimethylacetamide DMAP: N,N-Dimethyl-4-aminopyridine DMF: N,N-Dimethylformamide dtbbpy: 4,4'-Di-tert-butyl-2,2'-bipyridine EDTA: Ethylenediaminetetraacetic acid FA: Formic acid Fmoc: 9-Fluorenylmethyloxycarbonyl NMP: N-Methyl-2-pyrrolidone TBME: t-Butyl methyl ether TES: Triethylsilane TFA: Trifluoroacetic acid TFE: 2,2,2-Trifluoroethanol THF: Tetrahydrofuran THP: Tetrahydropyranyl TMSCI: Chlorotrimethylsilane HFIP: 1,1,1,3,3,3-Hexafluoroisopropyl alcohol HOAt: 1-Hydroxy-7-azabenzotriazole HOBt: 1-Hydroxybenzotriazole HOOBt: 3,4-Dihydro-3-hydroxy-4-oxo-1,2,3-benzotriazine IPAC: Isopropyl acetate oxyma: Ethyl cyano (hydroxyimino)acetate PPTS: Pyridinium p-toluenesulfonate Pis: 2-Phenylisopropyl WSCI.Math.HCl, WSCDI: 1-Ethyl-3-(3-dimethylaminopropyl) carbodiimide hydrochloride TIPS: Triisopropylsilane TfOH: Trifluoromethanesulfonic acid HATU: O-(7-Aza-

1H-benzotriazole-1-yl)-N,N,N',N'-tetramethyluronium hexafluorophosphate DMSO: Dimethylsulfoxide Fmoc-C1: (9H-Fluoren-9-yl)methyl carbonochloridate Fmoc-OSu: 9-Fluorenylmethyl N-succinimidyl carbonate Ns: o-Nitrobenzenesulfonyl Trt: Triphenylmethyl 9-BBN: 9-Borabicyclo[3.3.1]nonane HMDS: 1,1,1,3,3,3-Hexamethyldisilazane LDA: Lithium diisopropylamide TMSOTf: Trimethylsilyl trifluoromethanesulfonate PPA: Polyphosphoric acid

(2) Herein, the term “about” when used in combination with a numerical value means a numerical range of +10% and -10% of that numerical value.

(3) Herein, “-” indicating a range includes values at opposite ends of the range. For example, “A-B” means a range of A or more and B or less.

(4) The unit of molecular weight herein is “g/mol” (hereinafter, the unit of molecular weight may be omitted).

(5) The use of the articles “a”, “an”, and “the” in both the description and the claims are to be construed as covering both the singular and the plural, unless otherwise indicated herein or clearly contradicted by context.

(6) (Definitions of Functional Groups and the Like)

(7) Examples of “halogen atoms” herein include F, Cl, Br, and I.

(8) “Alkyl” herein means a monovalent group derived by removing any one hydrogen atom from an aliphatic hydrocarbon, and has a subset of hydrocarbyl or hydrocarbon group structures not containing either a heteroatom (which refers to an atom other than carbon and hydrogen atoms) or an unsaturated carbon-carbon bond but containing hydrogen and carbon atoms in its backbone. The alkyl includes linear and branched alkyls. Specifically, the alkyl has 1 to 20 carbon atoms (C.sub.1-C.sub.20, hereinafter “C.sub.p-C.sub.q” means that the number of carbon atoms is p to q), and is preferably C.sub.1-C.sub.10 alkyl, and more preferably C.sub.1-C.sub.6 alkyl. Specific examples of alkyl include methyl, ethyl, n-propyl, i-propyl, n-butyl, s-butyl, t-butyl, isobutyl(2-methylpropyl), n-pentyl, s-pentyl(1-methylbutyl), t-pentyl(1,1-dimethylpropyl), neopentyl(2,2-dimethylpropyl), isopentyl(3-methylbutyl), 3-pentyl(1-ethylpropyl), 1,2-dimethylpropyl, 2-methylbutyl, n-hexyl, 1,1,2-trimethylpropyl, 1,2,2-trimethylpropyl, 1,1,2,2-tetramethylpropyl, 1,1-dimethylbutyl, 1,2-dimethylbutyl, 1,3-dimethylbutyl, 2,2-dimethylbutyl, 2,3-dimethylbutyl, 3,3-dimethylbutyl, 1-ethylbutyl, and 2-ethylbutyl.

(9) “Alkenyl” herein means a monovalent group having at least one double bond (two adjacent SP.sup.2 carbon atoms). Depending on the configuration of a double bond and a substituent (if present), the geometrical form of the double bond can be entgegen (E) or zusammen (Z) as well as cis or trans configuration. The alkenyl includes linear and branched alkenyls. The alkenyl is preferably C.sub.2-C.sub.10 alkenyl, and more preferably C.sub.2-C.sub.7 alkenyl or C.sub.2-C.sub.6 alkenyl, and specific examples include vinyl, allyl, 1-propenyl, 2-propenyl, 1-butenyl, 2-butenyl (including cis and trans forms), 3-butenyl, pentenyl, 3-methyl-2-butenyl, hexenyl, and 6-heptenyl.

(10) “Alkynyl” herein means a monovalent group having at least one triple bond (two adjacent SP carbon atoms). The alkynyl includes linear and branched alkynyls. The alkynyl is preferably C.sub.2-C.sub.10 alkynyl, and more preferably C.sub.2-C.sub.6 alkynyl, and specific examples include ethynyl, 1-propynyl, propargyl, 3-butyne, pentynyl, hexynyl, 3-phenyl-2-propynyl, 3-(2'-fluorophenyl)-2-propynyl, 2-hydroxy-2-propynyl, 3-(3-fluorophenyl)-2-propynyl, and 3-methyl-(5-phenyl)-4-pentynyl.

(11) “Cycloalkyl” herein means a saturated or partially saturated cyclic monovalent aliphatic hydrocarbon group and includes a monocyclic ring, a bicyclo ring, and a spiro ring. The cycloalkyl is preferably C.sub.3-C.sub.8 cycloalkyl, and specific examples include cyclopropyl, cyclobutyl, cyclopentyl, cyclohexyl, cycloheptyl, cyclooctyl, bicyclo[2.2.1]heptyl, and spiro[3.3] heptyl.

(12) “Aryl” herein means a monovalent aromatic hydrocarbon ring, and is preferably C₆-C₁₀ aryl. Specific examples of the aryl include phenyl and naphthyl (e.g., 1-naphthyl and 2-naphthyl). Aryl herein includes bicyclic aryl in which the aromatic hydrocarbon ring is condensed with another

saturated ring or unsaturated ring, and, for example, includes aryl having a condensed ring structure in which the aromatic hydrocarbon ring is a benzene ring and the saturated ring is a 5-, 6-, or 7-membered saturated hydrocarbon ring or saturated heterocyclic ring. Specific examples include indanyl, 1,2,3,4-tetrahydronaphthyl, and 2,3-dihydrobenzofuran.

(13) “Heterocyclyl” herein means a non-aromatic cyclic monovalent group containing 1 to 5 hetero atoms in addition to carbon atoms. The heterocyclyl may have a double and/or triple bond within the ring, a carbon atom within the ring may be oxidized to form carbonyl, and heterocyclyl may be a monocyclic ring or a condensed ring. The number of atoms constituting the ring is preferably 3 to 10 (3- to 10-membered heterocyclyl) or 4 to 10 (4- to 10-membered heterocyclyl), and more preferably 3 to 7 (3- to 7-membered heterocyclyl) or 4 to 7 (4- to 7-membered heterocyclyl). Specific examples of the heterocyclyl include azetidiny, oxiranyl, oxetanyl, azetidiny, dihydrofuryl, tetrahydrofuryl, dihydropyranyl, tetrahydropyranyl, tetrahydropyridyl, tetrahydropyrimidyl, morpholinyl, thiomorpholinyl, pyrrolidinyl, piperidinyl, piperazinyl, pyrazolidinyl, imidazolinyl, imidazolidinyl, oxazolidinyl, isoxazolidinyl, thiazolidinyl, isothiazolidinyl, 1,2-thiazinane, thiadiazolidinyl, azetidiny, oxazolidone, benzodioxanyl, benzoxazolyl, dioxolanyl, dioxanyl, tetrahydropyrrolo[1,2-c]imidazole, thietanyl, 3,6-diazabicyclo[3.1.1]heptanyl, 2,5-diazabicyclo[2.2.1]heptanyl, 3-oxa-8-azabicyclo[3.2.1]octanyl, sultam, and 2-oxaspiro[3.3] heptyl.

(14) “Protected heterocyclyl” herein means a group in which one or more functional groups, such as an amino group, contained in the above-defined “heterocyclyl” are protected with a protecting group, and is preferably 4- to 7-membered protected heterocyclyl. Specific examples of the protecting group include Boc, Fmoc, Cbz, Troc, and Alloc, and specific examples of the protected heterocyclyl include Boc-protected azetidine.

(15) “Heterocycloalkylidene” herein means a divalent group obtained by removing two hydrogen atoms from one carbon atom of the above-defined “heterocyclyl”, in which a free valence forms a part of a double bond. The heterocycloalkylidene is preferably 4- to 7-membered heterocycloalkylidene, and specific examples include tetrahydropyran-4-ylidene and azetidin-3-ylidene.

(16) “Protected heterocycloalkylidene” herein means a group in which one or more functional groups, such as an amino group, contained in the above-defined “heterocycloalkylidene” are protected with a protecting group, and is preferably 4- to 7-membered protected heterocycloalkylidene. Specific examples of the protecting group include Boc, Fmoc, Cbz, Troc, and Alloc, and specific examples of the protected heterocycloalkylidene include Boc-protected azetidin-3-ylidene.

(17) “Heteroaryl” herein means an aromatic cyclic monovalent group containing 1 to 5 heteroatoms in addition to carbon atoms. The ring may be a monocyclic ring, may be a condensed ring formed with another ring, or may be partially saturated. The number of atoms constituting the ring is preferably 5 to 10 (5- to 10-membered heteroaryl) and more preferably 5 to 7 (5- to 7-membered heteroaryl). Specific examples of the heteroaryl include furyl, thienyl, pyrrolyl, imidazolyl, pyrazolyl, thiazolyl, isothiazolyl, oxazolyl, isoxazolyl, oxadiazolyl, thiadiazolyl, triazolyl, tetrazolyl, pyridyl, pyrimidyl, pyridazinyl, pyrazinyl, triazinyl, benzofuranyl, benzothienyl, benzothiadiazolyl, benzothiazolyl, benzoxazolyl, benzoxadiazolyl, benzoimidazolyl, indolyl, isoindolyl, indazolyl, quinolyl, isoquinolyl, cinnolinyl, quinazolinyl, quinoxalinyl, benzodioxolyl, indoliziny, and imidazopyridyl.

(18) “Alkoxy” herein means an oxy group to which the above-defined “alkyl” is bonded, and is preferably C.sub.1-C.sub.6 alkoxy. Specific examples of the alkoxy include methoxy, ethoxy, 1-propoxy, 2-propoxy, n-butoxy, i-butoxy, s-butoxy, t-butoxy, pentyloxy, and 3-methylbutoxy.

(19) “Alkylthio” herein means a thiol group to which the above-defined “alkyl” is bonded, and is preferably C.sub.1-C.sub.6 alkylthio. Specific examples of alkylthio include methylthio, ethylthio, 1-propylthio, 2-propylthio, n-butylthio, i-butylthio, s-butylthio, and t-butylthio. [00] 34]

(20) “Alkenyloxy” herein means an oxy group to which the above-defined “alkenyl” is bonded, and is preferably C.sub.2-C.sub.6 alkenyloxy. Specific examples of the alkenyloxy include vinyloxy, allyloxy, 1-propenyloxy, 2-propenyloxy, 1-butenyloxy, 2-butenyloxy (including cis and trans forms), 3-butenyloxy, pentenyloxy, and hexenyloxy.

(21) “Cycloalkoxy” herein means an oxy group to which the above-defined “cycloalkyl” is bonded, and is preferably C.sub.3-C.sub.8 cycloalkoxy. Specific examples of the cycloalkoxy include cyclopropoxy, cyclobutoxy, and cyclopentyloxy.

(22) “Aryloxy” herein means an oxy group to which the above-defined “aryl” is bonded, and is preferably C.sub.6-C.sub.10 aryloxy. Specific examples of the aryloxy include phenoxy, 1-naphthyloxy, and 2-naphthyloxy.

(23) “Amino” herein means-NH₂ in a narrow sense and —NRR' in a broad sense, wherein R and R' are independently selected from hydrogen, alkyl, alkenyl, alkynyl, cycloalkyl, heterocyclyl, aryl, or heteroaryl, or R and R', together with the nitrogen atom to which they are attached, form a ring. The amino is preferably-NH₂, mono-C.sub.1-C.sub.6 alkylamino, di-C.sub.1-C.sub.6 alkylamino, 4- to 8-membered cyclic amino, or the like.

(24) “Monoalkylamino” herein means a group corresponding to the above-defined “amino” wherein R is hydrogen and R' is the above-defined “alkyl”, and is preferably mono-C.sub.1-C.sub.6 alkylamino. Specific examples of the monoalkylamino include methylamino, ethylamino, n-propylamino, i-propylamino, n-butylamino, s-butylamino, and t-butylamino.

(25) “Dialkylamino” herein means a group corresponding to the above-defined “amino” wherein R and R' are independently the above-defined “alkyl”, and is preferably di-C.sub.1-C.sub.6 alkylamino. Specific examples of the dialkylamino include dimethylamino and diethylamino.

(26) “Cyclic amino” herein means a group corresponding to the above-defined “amino” wherein R and R', together with the nitrogen atom to which they are attached, form a ring, and is preferably 4- to 8-membered cyclic amino. Specific examples of the cyclic amino include 1-azetidyl, 1-pyrrolidyl, 1-piperidyl, 1-piperazyl, 4-morpholinyl, 3-oxazolidinyl, 1,1-dioxidethiomorpholinyl-4-yl, and 3-oxa-8-azabicyclo[3.2.1]octan-8-yl.

(27) “Protected amino” herein means an amino group protected with any protecting group. Specific examples of the protected amino include amino protected with a protecting group such as Boc, Fmoc, Cbz, Troc, or Alloc.

(28) “Alkylcarbonyl” herein means a carbonyl group to which the above-defined “alkyl” is bonded, and is preferably C.sub.1-C.sub.6 alkylcarbonyl. Specific examples of alkylcarbonyl include acetyl, propionyl, and butyryl. The number of carbon atoms set forth in the above definition indicates the number of carbon atoms in the alkyl moiety. For example, “C.sub.1-C.sub.6” in “C.sub.1-C.sub.6 alkylcarbonyl” indicates that the alkyl moiety has 1 to 6 carbon atoms.

(29) “Aminocarbonyl” herein means a carbonyl group to which the above-defined “amino” is bonded, and is preferably —CONH.sub.2, mono-C.sub.1-C.sub.6 alkylaminocarbonyl, di-C.sub.1-C.sub.6 alkylaminocarbonyl, and 4- to 8-membered cyclic aminocarbonyl. Specific examples of the aminocarbonyl include —CONH.sub.2, dimethylaminocarbonyl, 1-azetidinyldicarbonyl, 1-pyrrolidinylcarbonyl, 1-piperidinylcarbonyl, 1-piperazinylcarbonyl, 4-morpholinylcarbonyl, 3-oxazolidinylcarbonyl, 1,1-dioxidethiomorpholinyl-4-ylcarbonyl, and 3-oxa-8-azabicyclo[3.2.1]octan-8-ylcarbonyl.

(30) “Alkenyloxycarbonyl” herein means a carbonyl group to which the above-defined “alkenyloxy” is bonded, and is preferably C.sub.2-C.sub.6 alkenyloxycarbonyl. Specific examples of the alkenyloxycarbonyl include vinyloxycarbonyl, allyloxycarbonyl, 1-propenyloxycarbonyl, 2-propenyloxycarbonyl, 1-butenyloxycarbonyl, 2-butenyloxycarbonyl (including cis and trans forms), 3-butenyloxycarbonyl, pentenyloxycarbonyl, and hexenyloxycarbonyl. “Alkylsulfonyl” herein means a sulfonyl group to which the above-defined “alkyl” is bonded, and is preferably C.sub.1-C.sub.6 alkylsulfonyl. Specific examples of the alkylsulfonyl include methylsulfonyl.

(31) “Hydroxyalkyl” herein means a group in which one or more hydrogens of the above-defined

“alkyl” are replaced with hydroxyl groups, and is preferably C.sub.1-C.sub.6 hydroxyalkyl. Specific examples of the hydroxyalkyl include hydroxymethyl, 1-hydroxyethyl, 2-hydroxyethyl, 2-hydroxy-2-methylpropyl, and 5-hydroxypentyl.

(32) “Haloalkyl” herein means a group in which one or more hydrogens of the above-defined “alkyl” are replaced with halogen, and is preferably C.sub.1-C.sub.6 haloalkyl, and more preferably C.sub.1-C.sub.6 fluoroalkyl. Specific examples of the haloalkyl include difluoromethyl, trifluoromethyl, 2,2-difluoroethyl, 2,2,2-trifluoroethyl, 3,3-difluoropropyl, 4,4-difluorobutyl, 5,5-difluoropentyl, and 1,1-difluoroethyl.

(33) “Cyanoalkyl” herein means a group in which one or more hydrogens of the above-defined “alkyl” are replaced with cyano, and is preferably C.sub.1-C.sub.6 cyanoalkyl. Specific examples of the cyanoalkyl include cyanomethyl and 2-cyanoethyl.

(34) “Aminoalkyl” herein means a group in which one or more hydrogens of the above-defined “alkyl” are replaced with the above-defined “amino”, and is preferably C.sub.1-C.sub.6 aminoalkyl. Specific examples of the aminoalkyl include 1-pyridylmethyl, 2-(1-piperidyl)ethyl, 3-(1-piperidyl) propyl, 4-aminobutyl, and 2-aminoethyl.

(35) “Carboxyalkyl” herein means a group in which one or more hydrogens of the above-defined “alkyl” are replaced with carboxy, and is preferably C.sub.1-C.sub.6 carboxyalkyl or C.sub.2-C.sub.6 carboxyalkyl. Specific examples of the carboxyalkyl include carboxymethyl. The number of carbon atoms set forth in the above definition indicates the number of carbon atoms in the alkyl moiety. For example, “C.sub.1-C.sub.6” in “C.sub.1-C.sub.6 carboxyalkyl” indicates that the alkyl moiety has 1 to 6 carbon atoms.

(36) “Alkenyloxycarbonylalkyl” herein means a group in which one or more hydrogens of the above-defined “alkyl” are replaced with the above-defined “alkenyloxycarbonyl”, and is preferably C.sub.2-C.sub.6 alkenyloxycarbonyl C.sub.1-C.sub.6 alkyl, and more preferably C.sub.2-C.sub.6 alkenyloxycarbonyl C.sub.1-C.sub.2 alkyl. Specific examples of the alkenyloxycarbonylalkyl include allyloxycarbonylmethyl and 2-(allyloxycarbonyl)ethyl.

(37) “Alkoxyalkyl” herein means a group in which one or more hydrogens of the above-defined “alkyl” are replaced with the above-defined “alkoxy”, and is preferably C.sub.1-C.sub.6 alkoxy C.sub.1-C.sub.6 alkyl, and more preferably C.sub.1-C.sub.6 alkoxy C.sub.1-C.sub.2 alkyl. Specific examples of the alkoxyalkyl include methoxymethyl, ethoxymethyl, 1-propoxymethyl, 2-propoxymethyl, n-butoxymethyl, i-butoxymethyl, s-butoxymethyl, t-butoxymethyl, pentyloxymethyl, 3-methylbutoxymethyl, 1-methoxyethyl, 2-methoxyethyl, 2-ethoxyethyl, 1-ethoxyethyl, and 1-n-propyloxyethyl.

(38) “Alkylthioalkyl” herein means a group in which one or more hydrogens of the above-defined “alkyl” are replaced with the above-defined “alkylthio”, and is preferably C.sub.1-C.sub.6 alkylthioC.sub.1—C.sub.6 alkyl, and more preferably C.sub.1-C.sub.6 alkylthioC.sub.1—C.sub.2 alkyl. Specific examples of alkylthioalkyl include methylthiomethyl, ethylthiomethyl, 1-propylthiomethyl, 2-propylthiomethyl, n-butylthiomethyl, i-butylthiomethyl, s-butylthiomethyl, and t-butylthiomethyl.

(39) “Alkenyloxyalkyl” herein means a group in which one or more hydrogens of the above-defined “alkyl” are replaced with the above-defined “alkenyloxy”, and is preferably C.sub.2-C.sub.6 alkenyloxyC.sub.1—C.sub.6 alkyl, and more preferably C.sub.1-C.sub.6 alkenyloxyC.sub.1—C.sub.2 alkyl. Specific examples of alkenyloxyalkyl include vinyloxymethyl and allyloxymethyl.

(40) “Cycloalkylalkyl” herein means a group in which one or more hydrogens of the above-defined “alkyl” are replaced with the above-defined “cycloalkyl”, and is preferably C.sub.3-C.sub.8 cycloalkyl C.sub.1-C.sub.6 alkyl, and more preferably C.sub.3-C.sub.6 cycloalkyl C.sub.1-C.sub.2 alkyl. Specific examples of the cycloalkylalkyl include cyclopropylmethyl, cyclobutylmethyl, cyclopentylmethyl, and cyclohexylmethyl.

(41) “Cycloalkoxylalkyl” herein means a group in which one or more hydrogens of the above-

defined “alkyl” are replaced with the above-defined “cycloalkoxy”, and is preferably C.sub.3-C.sub.8 cycloalkoxy C.sub.1-C.sub.6 alkyl, and more preferably C.sub.3-C.sub.6 cycloalkoxy C.sub.1-C.sub.2 alkyl. Specific examples of the cycloalkoxyalkyl include cyclopropoxymethyl and cyclobutoxymethyl.

(42) “Heterocyclalkyl” herein means a group in which one or more hydrogens of the above-defined “alkyl” are replaced with the above-defined “heterocycl”, and is preferably 4- to 7-membered heterocycl C.sub.1-C.sub.6 alkyl, and more preferably 4- to 7-membered heterocycl C.sub.1-C.sub.2 alkyl. Specific examples of the heterocyclalkyl include 2-(tetrahydro-2H-pyran-4-yl)ethyl and 2-(azetidin-3-yl)ethyl.

(43) “Alkylsulfonylalkyl” herein means a group in which one or more hydrogens of the above-defined “alkyl” are replaced with the above-defined “alkylsulfonyl”, and is preferably C.sub.1-C.sub.6 alkylsulfonyl C.sub.1-C.sub.6 alkyl, and more preferably C.sub.1-C.sub.6 alkylsulfonyl C.sub.1-C.sub.2 alkyl. Specific examples of the alkylsulfonylalkyl include methylsulfonylmethyl and 2-(methylsulfonyl)ethyl.

(44) “Aminocarbonylalkyl” herein means a group in which one or more hydrogens of the above-defined “alkyl” are replaced with the above-defined “aminocarbonyl”, and is preferably aminocarbonyl C.sub.1-C.sub.6 alkyl, and more preferably aminocarbonyl C.sub.1-C.sub.4 alkyl. Specific examples of the aminocarbonylalkyl include methylaminocarbonylmethyl, dimethylaminocarbonylmethyl, t-butylaminocarbonylmethyl, 1-azetidinylocarbonylmethyl, 1-pyrrolidinylcarbonylmethyl, 1-piperidinylcarbonylmethyl, 4-morpholinylcarbonylmethyl, 2-(methylaminocarbonyl)ethyl, 2-(dimethylaminocarbonyl)ethyl, 2-(1-azetidinylocarbonyl)ethyl, 2-(1-pyrrolidinylcarbonyl)ethyl, 2-(4-morpholinylcarbonyl)ethyl, 3-(dimethylaminocarbonyl) propyl, and 4-(dimethylaminocarbonyl)butyl.

(45) “Aryloxyalkyl” herein means a group in which one or more hydrogens of the above-defined “alkyl” are replaced with the above-defined “aryloxy”, and is preferably C.sub.6-C.sub.10 aryloxy C.sub.1-C.sub.6 alkyl, and more preferably C.sub.6-C.sub.10 aryloxy C.sub.1-C.sub.2 alkyl. Specific examples of the aryloxyalkyl include phenoxyethyl and 2-phenoxyethyl.

(46) “Aralkyl(arylalkyl)” herein means a group in which one or more hydrogen atoms of the above-defined “alkyl” are replaced with the above-defined “aryl”, and is preferably C.sub.7-C.sub.14 aralkyl, and more preferably C.sub.7-C.sub.10 aralkyl. Specific examples of the aralkyl include benzyl, phenethyl, and 3-phenylpropyl.

(47) “Aralkoxy” herein means an oxy group to which the above-defined “aralkyl” is bonded, and is preferably C.sub.7-C.sub.14 aralkoxy, and more preferably C.sub.7-C.sub.10 aralkoxy. Specific examples of the aralkoxy include benzyloxy, phenethyloxy, and 3-phenylpropoxy.

(48) “Aralkoxyalkyl” herein means a group in which one or more hydrogens of the above-defined “alkyl” are replaced with the above-defined “aralkoxy”, and is preferably C.sub.7-C.sub.14 aralkoxy C.sub.1-C.sub.6 alkyl, and more preferably C.sub.7-C.sub.14 aralkoxy C.sub.1-C.sub.2 alkyl. Specific examples of the aralkoxyalkyl include benzyloxymethyl and 1-(benzyloxy)ethyl.

(49) “Heteroarylalkyl” herein means a group in which one or more hydrogen atoms of the above-defined “alkyl” are replaced with the above-defined “heteroaryl”, and is preferably 5- to 10-membered heteroaryl C.sub.1-C.sub.6 alkyl, and more preferably 5- to 10-membered heteroaryl C.sub.1-C.sub.2 alkyl. Specific examples of the heteroarylalkyl include 3-thienylmethyl, 4-thiazolylmethyl, 2-pyridylmethyl, 3-pyridylmethyl, 4-pyridylmethyl, 2-(2-pyridyl)ethyl, 2-(3-pyridyl)ethyl, 2-(4-pyridyl)ethyl, 2-(6-quinolyl)ethyl, 2-(7-quinolyl)ethyl, 2-(6-indolyl)ethyl, 2-(5-indolyl)ethyl, and 2-(5-benzofuranyl)ethyl.

(50) “Heteroarylalkoxy” herein means an oxy group to which the above-defined “heteroarylalkyl” is bonded, and is preferably 5- to 10-membered heteroaryl C.sub.1-C.sub.6 alkoxy, and more preferably 5- to 10-membered heteroaryl C.sub.1-C.sub.2 alkoxy. Specific examples of the heteroarylalkoxy include 3-thienylmethoxy and 3-pyridylmethoxy.

(51) “Heteroarylalkoxyalkyl” herein means a group in which one or more hydrogens of the above-

defined “alkyl” are replaced with the above-defined “heteroarylalkoxy”, and is preferably 5- to 10-membered heteroaryl C.sub.1-C.sub.6 alkoxy C.sub.1-C.sub.6 alkyl, and more preferably 5- to 10-membered heteroaryl C.sub.1-C.sub.2 alkoxy C.sub.1-C.sub.2 alkyl. Specific examples of the heteroarylalkoxyalkyl include 3-pyridylmethoxymethyl.

(52) “Heterocycloalkylidenealkyl” herein means a group in which one or more hydrogens of the above-defined “alkyl” are replaced with the above-defined “heterocycloalkylidene”, and is preferably 4- to 7-membered heterocycloalkylidene C.sub.1-C.sub.6 alkyl, and more preferably 4- to 7-membered heterocycloalkylidene C.sub.1-C.sub.2 alkyl. Specific examples of the heterocycloalkylidenealkyl include tetrahydro-4H-pyran-4-ylidenemethyl and azetidin-3-ylidenemethyl.

(53) “Alkoxyalkenyl” herein means a group in which one or more hydrogens of the above-defined “alkenyl” are replaced with the above-defined “alkoxy”, and is preferably C.sub.1-C.sub.6 alkoxy C.sub.2-C.sub.6 alkenyl. Specific examples of the alkoxyalkenyl include (E)-4-methoxybut-2-en-1-yl.

(54) “Aminocarbonylalkenyl” herein means a group in which one or more hydrogens of the above-defined “alkenyl” are replaced with the above-defined “aminocarbonyl”, and is preferably aminocarbonyl C.sub.2-C.sub.6 alkenyl. Specific examples of the aminocarbonylalkenyl include (E)-3-(dimethylaminocarbonylcarbonyl)-prop-2-en-1-yl.

(55) “Haloalkoxy” herein means a group in which one or more hydrogens of the above-defined “alkoxy” are replaced with halogen, and is preferably C.sub.1-C.sub.6 haloalkoxy. Specific examples of the haloalkoxy include difluoromethoxy, trifluoromethoxy, 2,2-difluoroethoxy, and 2,2,2-trifluoroethoxy.

(56) “Alkylene” herein means a divalent group derived by further removing any one hydrogen atom from the above “alkyl”, and is preferably C.sub.4-C.sub.8 alkylene. Specific examples of the alkylene include —CH.sub.2—, —(CH.sub.2).sub.2—, —(CH.sub.2).sub.3—, —CH(CH.sub.3)CH.sub.2—, —C(CH.sub.3).sub.2—, —(CH.sub.2) 4—, —CH(CH.sub.3)CH.sub.2CH.sub.2—, —C(CH.sub.3).sub.2CH.sub.2—, —CH.sub.2CH(CH.sub.3)CH.sub.2—, —CH.sub.2C(CH.sub.3).sub.2—, —CH.sub.2CH.sub.2CH(CH.sub.3)—, —(CH.sub.2).sub.5—, —(CH.sub.2).sub.6—, —(CH.sub.2).sub.7—, and —(CH.sub.2).sub.8—.

(57) “Cycloalkylene” herein means a divalent group derived by further removing any one hydrogen atom from the above “cycloalkyl”, and is preferably C.sub.3-C.sub.8 cycloalkylene. Specific examples of cycloalkylene include cyclopropane-1,2-diyl, cyclobutane-1,2-diyl, cyclopentane-1,2-diyl, and cyclohexane-1,2-diyl.

(58) “Heterocyclylene” herein means a divalent group derived by removing any one hydrogen atom from the above “heterocyclyl”, and is preferably 3- to 7-membered heterocyclylene. Specific examples of heterocyclylene include oxylan-2,3-diyl, oxetan-2,3-diyl, tetrahydrofuran-2,5-diyl, and tetrahydropyran-2,6-diyl.

(59) “Alkenylene” herein means a divalent group derived by further removing any one hydrogen atom from the above “alkenyl”. Depending on the configuration of a double bond and a substituent (if present), the geometrical form of the double bond can be entgegen (E) or zusammen (Z) as well as cis or trans configuration. Alkenylene may be linear or branched, and is preferably C.sub.2-C.sub.10 alkenylene and more preferably C.sub.2-C.sub.6 alkenylene.

(60) “Alkynylene” herein means a divalent group derived by further removing any one hydrogen atom from the above “alkynyl”. Alkynylene may be linear or branched, and is preferably C.sub.2-C.sub.10 alkynylene and more preferably C.sub.2-C.sub.6 alkynylene.

(61) “Arylene” herein means a divalent group derived by further removing any one hydrogen atom from the above “aryl”. Arylene may be a monocyclic ring or a condensed ring. The number of atoms constituting the ring is not particularly limited, and is preferably 6 to 10 (C.sub.6-10 arylene). Specific examples of arylene include 1,2-phenylene, 1,3-phenylene, 1,4-phenylene, 1,2-naphthylene, 1,3-naphthylene, and 1,4-naphthylene.

(62) "Spirocycloalkyl" herein means a group formed by sharing of one carbon atom constituting a cycloalkane ring with a carbon atom present in a group to be bonded. Spirocycloalkyl is preferably C3-Cs spirocycloalkyl, and specific examples include spirocyclopropyl, spirocyclobutyl, spirocyclopentyl, spirocyclohexyl, spirocycloheptyl, and spirocyclooctyl.

(63) "Spiroheterocyclyl" herein means a group obtained by replacing one or more carbon atoms in the above "spirocycloalkyl" with heteroatoms. Heterospirocycloalkyl is preferably 4- to 10-membered spiroheterocyclyl.

(64) "Alicyclic ring" herein means a non-aromatic hydrocarbon ring. The alicyclic ring may have an unsaturated bond within the ring, and may be a polycyclic ring having two or more rings. A carbon atom constituting the ring may be oxidized to form carbonyl. The alicyclic ring is preferably a 3- to 8-membered alicyclic ring, and specific examples include a cyclopropane ring, a cyclobutane ring, a cyclopentane ring, a cyclohexane ring, a cycloheptane ring, a cyclooctane ring, and a bicyclo[2.2.1]heptane ring.

(65) "Saturated heterocyclic ring" herein means a non-aromatic heterocyclic ring containing 1 to 5 hetero atoms in addition to carbon atoms and not containing a double bond and/or a triple bond within the ring. The saturated heterocyclic ring may be a monocyclic ring, or may form a condensed ring with another ring, e.g., an aromatic ring such as a benzene ring. The saturated heterocyclic ring is preferably a 4- to 7-membered saturated heterocyclic ring, and specific examples include an azetidine ring, an oxetane ring, a tetrahydrofuran ring, a tetrahydropyran ring, a morpholine ring, a thiomorpholine ring, a pyrrolidine ring, a 4-oxopyrrolidine ring, a piperidine ring, a 4-oxopiperidine ring, a piperazine ring, a pyrazolidine ring, an imidazolidine ring, an oxazolidine ring, an isoxazolidine ring, a thiazolidine ring, an isothiazolidine ring, a thiadiazolidine ring, an oxazolidone ring, a dioxolane ring, a dioxane ring, a thietane ring, an octahydroindole ring, an indoline ring, and an azepane ring.

(66) "Peptide chain" herein refers to a peptide chain in which 1, 2, 3, 4, or more natural amino acids and/or non-natural amino acids are connected by an amide bond and/or an ester bond. The peptide chain is preferably a peptide chain comprising 1 to 4 amino acid residues, and more preferably a peptide chain consisting of 1 to 4 amino acid residues.

(67) "Optionally substituted" herein means that a group may be substituted with any substituent.

(68) "Optionally protected" herein means that a group may be protected with any protecting group.

(69) "One or more" herein means one or two or more. When "one or more" is used in a context relating to the substituent of a group, the phrase means a number encompassing one to the maximum number of substituents permitted by that group. Specific examples of "one or more" include 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, and/or a greater number.

(70) The wavy line in the structural formulae herein can mean that any stereochemistry is permitted. For example, when the asymmetric center is provided with a wavy line, the stereochemistry of the asymmetric center may be in the S configuration or in the R configuration. When a double bond is provided with a wavy line, the stereochemistry of the double bond may be in the E configuration or in the Z configuration.

(71) As used herein, a symbol represented by "PPn.sup.1n.sup.2n.sup.3n.sup.4, where n.sup.1, n.sup.2, n.sup.3 and n.sup.4 each independently represent an integer of 0 to 9", for example, PP1574 and PP1640, represents the number of a compound.

(72) The compound of the present invention can be a salt thereof, and preferably a chemically or pharmaceutically acceptable salt thereof. Also, the compound of the present invention or a salt thereof can be a solvate thereof, and preferably a chemically or pharmaceutically acceptable solvate thereof. Examples of salts of the compound of the present invention include hydrochloride: hydrobromide: hydroiodide: phosphate: phosphonate: sulfate: sulfonates such as methanesulfonate and p-toluenesulfonate: carboxylates such as acetate, citrate, malate, tartrate, succinate, and salicylate: alkali metal salts such as a sodium salt and a potassium salt: alkaline earth metal salts such as a magnesium salt and a calcium salt; and ammonium salts such as an ammonium salt, an

alkylammonium salt, a dialkylammonium salt, a trialkylammonium salt, and a tetraalkylammonium salt. These salts are produced by, for example, bringing the compound into contact with an acid or a base usable in the production of pharmaceutical products. In the present invention, a solvate of a compound refers to one molecular group formed by the compound together with a solvent, and is called a hydrate when the solvent is water. The solvate of the compound of the present invention is preferably a hydrate, and specific examples of such hydrates include mono- to deca-hydrates, preferably mono- to penta-hydrates, and more preferably mono- to tri-hydrates. The solvate of the compound of the present invention includes not only a solvate formed of a single solvent such as water, alcohol (e.g., methanol, ethanol, 1-propanol, or 2-propanol), or dimethylformamide, but also a solvate formed of a plurality of solvents.

(73) The term “amino acid” as used herein includes natural and unnatural amino acids. The term “natural amino acid” as used herein refers to Gly, Ala, Ser, Thr, Val, Leu, Ile, Phe, Tyr, Trp, His, Glu, Asp, Gln, Asn, Cys, Met, Lys, Arg, or Pro. Examples of the unnatural amino acid include, but are not particularly limited to, β -amino acids, α -amino acids, D-amino acids, N-substituted amino acids, α , α -disubstituted amino acids, amino acids having side chains that are different from those of natural amino acids, and hydroxycarboxylic acids. Amino acids herein may have any conformation. There is no particular limitation on the selection of amino acid side chain, but in addition to a hydrogen atom, it can be freely selected from, for example, an alkyl group, an alkenyl group, an alkynyl group, an aryl group, a heteroaryl group, an aralkyl group, and a cycloalkyl group. One or two non-adjacent methylene groups in such a group are optionally substituted with an oxygen atom, a carbonyl group (—CO—), or a sulfonyl group ($\text{—SO}_2\text{—}$). Each group may have a substituent, and there are no limitations on the substituent. For example, one or more substituents may be freely and independently selected from any substituents including a halogen atom, an O atom, an S atom, an N atom, a B atom, a Si atom, or a P atom. Examples include an optionally substituted alkyl group, alkenyl group, alkynyl group, aryl group, heteroaryl group, aralkyl group, and cycloalkyl group. In a non-limiting embodiment, amino acids herein may be compounds having a carboxy group and an amino group in the same molecule (even in this case, imino acids such as proline and hydroxyproline are also included in amino acids).

(74) The main chain amino group of an amino acid may be unsubstituted (an NH_2 group) or substituted (i.e., an —NHR group, where R represents alkyl, alkenyl, alkynyl, aryl, heteroaryl, aralkyl, or cycloalkyl which may have a substituent, one or two non-adjacent methylene groups in such a group may be substituted with an oxygen atom, a carbonyl group (—CO—), or a sulfonyl group ($\text{—SO}_2\text{—}$), and the carbon chain bonded to the N atom and the carbon atom at the position α may form a ring, as in proline). The R substituent is selected as the substituent in the aforementioned amino acid side chain is selected. When the main chain amino group is substituted, the R is included in the “amino acid side chain” as used herein. Such amino acids in which the main chain amino group is substituted are herein called “N-substituted amino acids.” Preferred examples of the “N-substituted amino acids” as used herein include, but are not limited to, N-alkylamino acids, N-C₁₋₆ alkylamino acids, N-C₁₋₄ alkylamino acids, and N-methylamino acids.

(75) “Amino acids” as used herein which constitute a peptide compound include all isotopes corresponding to each amino acid. The isotope of the “amino acid” refers to one having at least one atom replaced with an atom of the same atomic number (number of protons) and different mass number (total number of protons and neutrons). Examples of isotopes contained in the “amino acid” constituting the peptide compounds of the present invention include a hydrogen atom, a carbon atom, a nitrogen atom, an oxygen atom, a phosphorus atom, a sulfur atom, a fluorine atom, and a chlorine atom, which respectively include ^2H and ^3H ; ^{13}C and ^{14}C ; ^{15}N ; ^{17}O and ^{18}O ; ^{31}P and ^{32}P ; ^{35}S ; ^{18}F ; and ^{36}Cl .

(76) Substituents containing a halogen atom as used herein include a halogen-substituted alkyl group, cycloalkyl group, alkenyl group, alkynyl group, aryl group, heteroaryl group, or aralkyl

group. More specific examples include fluoroalkyl, difluoroalkyl, and trifluoroalkyl.

(77) Substituents containing an O atom include groups such as hydroxy (—OH), oxy (—OR), carbonyl (—C(=O)—R), carboxy (—CO.sub.2H), oxycarbonyl (—C(=O)—OR), carbonyloxy (—O—C(=O)—R), thiocarbonyl (—C(=O)—SR), carbonylthio (—S—C(=O)—R), aminocarbonyl (—C(=O)—NHR), carbonylamino (—NH—C(=O)—R), oxycarbonylamino (—NH—C(=O)—OR), sulfonylamino (—NH—SO.sub.2—R), aminosulfonyl (—SO.sub.2—NHR), sulfamoylamino (—NH—SO.sub.2—NHR), thiocarboxy (—C(=O)—SH), and carboxycarbonyl (—C(=O)—CO.sub.2H).

(78) Examples of oxy (—OR) include alkoxy, cycloalkoxy, alkenyloxy, alkynyloxy, aryloxy, heteroaryloxy, and aralkyloxy. Alkoxy is preferably C.sub.1-C.sub.4 alkoxy or C.sub.1-C.sub.2 alkoxy, and particularly preferably methoxy or ethoxy.

(79) Examples of carbonyl (—C(=O)—R) include formyl (—C(=O)—H), alkylcarbonyl, cycloalkylcarbonyl, alkenylcarbonyl, alkynylcarbonyl, arylcarbonyl, heteroarylcarbonyl, and aralkylcarbonyl.

(80) Examples of oxycarbonyl (—C(=O)—OR) include alkyloxycarbonyl, cycloalkyloxycarbonyl, alkenyloxycarbonyl, alkynyloxycarbonyl, aryloxycarbonyl, heteroaryloxycarbonyl, and aralkyloxycarbonyl.

(81) Examples of carbonyloxy (—O—C(=O)—R) include alkylcarbonyloxy, cycloalkylcarbonyloxy, alkenylcarbonyloxy, alkynylcarbonyloxy, arylcarbonyloxy, heteroarylcarbonyloxy, and aralkylcarbonyloxy.

(82) Examples of thiocarbonyl (—C(=O)—SR) include alkylthiocarbonyl, cycloalkylthiocarbonyl, alkenylthiocarbonyl, alkynylthiocarbonyl, arylthiocarbonyl, heteroarylthiocarbonyl, and aralkylthiocarbonyl.

(83) Examples of carbonylthio (—S—C(=O)—R) include alkylcarbonylthio, cycloalkylcarbonylthio, alkenylcarbonylthio, alkynylcarbonylthio, arylcarbonylthio, heteroarylcarbonylthio, and aralkylcarbonylthio.

(84) Examples of aminocarbonyl (—C(=O)—NHR) include alkylaminocarbonyl (such as C.sub.1-C.sub.6 or C.sub.1-C.sub.4 alkylaminocarbonyl and, in particular, ethylaminocarbonyl and methylaminocarbonyl), cycloalkylaminocarbonyl, alkenylaminocarbonyl, alkynylaminocarbonyl, arylaminocarbonyl, heteroarylaminocarbonyl, and aralkylaminocarbonyl. In addition, examples further include compounds obtained by replacing the H atom bonded to the N atom in —C(=O)—NHR with alkyl, cycloalkyl, alkenyl, alkynyl, aryl, heteroaryl, or aralkyl.

(85) Examples of carbonylamino (—NH—C(=O)—R) include alkylcarbonylamino, cycloalkylcarbonylamino, alkenylcarbonylamino, alkynylcarbonylamino, arylcarbonylamino, heteroarylcarbonylamino, and aralkylcarbonylamino. In addition, examples further include compounds obtained by replacing the H atom bonded to the N atom in —NH—C(=O)—R with alkyl, cycloalkyl, alkenyl, alkynyl, aryl, heteroaryl, or aralkyl.

(86) Examples of oxycarbonylamino (—NH—C(=O)—OR) include alkoxycarbonylamino, cycloalkoxycarbonylamino, alkenyloxycarbonylamino, alkynyloxycarbonylamino, aryloxycarbonylamino, heteroaryloxycarbonylamino, and aralkyloxycarbonylamino. In addition, examples further include compounds obtained by replacing the H atom bonded to the N atom in —NH—C(=O)—OR with alkyl, cycloalkyl, alkenyl, alkynyl, aryl, heteroaryl, or aralkyl.

(87) Examples of sulfonylamino (—NH—SO.sub.2—R) include alkylsulfonylamino, cycloalkylsulfonylamino, alkenylsulfonylamino, alkynylsulfonylamino, arylsulfonylamino, heteroarylsulfonylamino, and aralkylsulfonylamino. In addition, examples further include compounds obtained by replacing the H atom bonded to the N atom in —NH—SO.sub.2—R with alkyl, cycloalkyl, alkenyl, alkynyl, aryl, heteroaryl, or aralkyl.

(88) Examples of aminosulfonyl (—SO.sub.2—NHR) include alkylaminosulfonyl, cycloalkylaminosulfonyl, alkenylaminosulfonyl, alkynylaminosulfonyl, arylaminosulfonyl, heteroarylamino sulfonyl, and aralkylaminosulfonyl. In addition, examples further include

compounds obtained by replacing the H atom bonded to the N atom in SO.sub.2-NHR with alkyl, cycloalkyl, alkenyl, alkynyl, aryl, heteroaryl, or aralkyl.

(89) Examples of sulfamoylamino (—NH—SO.sub.2—NHR) include alkylsulfamoylamino, cycloalkylsulfamoylamino, alkenylsulfamoylamino, alkynylsulfamoylamino, arylsulfamoylamino, heteroaryl sulfamoylamino, and aralkylsulfamoylamino. Moreover, two H atoms bonded to the N atoms in —NH—SO.sub.2—NHR may be replaced with substituents independently selected from the group consisting of alkyl, cycloalkyl, alkenyl, alkynyl, aryl, heteroaryl, and aralkyl, and these two substituents may form a ring.

(90) Substituents containing an S atom include groups such as thiol (—SH), thio (—S—R), sulfinyl (—S(=O)—R), sulfonyl (—SO.sub.2—R), and sulfo (—SO.sub.3H).

(91) Examples of thio (—S—R) are selected from alkylthio, cycloalkylthio, alkenylthio, alkynylthio, arylthio, heteroarylthio, aralkylthio, and the like.

(92) Examples of sulfonyl (—SO.sub.2—R) include alkylsulfonyl, cycloalkylsulfonyl, alkenylsulfonyl, alkynylsulfonyl, arylsulfonyl, heteroaryl sulfonyl, and aralkylsulfonyl.

(93) Substituents containing an N atom include groups such as azide (—N.sub.3 , also referred to as an “azido group”), cyano (—CN), primary amino (—NH.sub.2), secondary amino (—NH—R ; also referred to as mono-substituted amino), tertiary amino (—NR(R') ; also referred to as di-substituted amino), amidino (—C(=NH)—NH.sub.2), substituted amidino (—C(=NR)—NR'R''), guanidino ($\text{—NH—C(=NH)—NH.sub.2}$), substituted guanidino ($\text{—NR—C(=NR''')—NR'R''}$), aminocarbonylamino (—NR—CO—NR'R''), pyridyl, piperidino, morpholino, and azetidiny.

(94) Examples of secondary amino (—NH—R ; mono-substituted amino) include alkylamino, cycloalkylamino, alkenylamino, alkynylamino, arylamino, heteroaryl amino, and aralkylamino. Examples of tertiary amino (—NR(R') ; di-substituted amino) include amino groups that have any two substituents each independently selected from alkyl, cycloalkyl, alkenyl, alkynyl, aryl, heteroaryl, and aralkyl, such as alkyl(aralkyl) amino, and these two substituents may form a ring. Specific examples include dialkylamino, in particular, C.sub.1-C.sub.6 dialkylamino, C.sub.1-C.sub.4 dialkylamino, dimethylamino, and diethylamino. The “C.sub.p-C.sub.q dialkylamino group” herein refers to an amino group substituted with two C.sub.p-C.sub.q alkyl groups, and the C.sub.p-C.sub.q alkyl groups may be the same or different.

(95) Examples of substituted amidino (—C(=NR)—NR'R'') include groups in which three substituents R, R', and R'' on the N atoms are each independently selected from alkyl, cycloalkyl, alkenyl, alkynyl, aryl, heteroaryl, and aralkyl, such as alkyl(aralkyl)(aryl) amidino.

(96) Examples of substituted guanidino ($\text{—NR—C(=NR'')—NR'R''}$) include groups in which R, R', R'', and R'' are each independently selected from alkyl, cycloalkyl, alkenyl, alkynyl, aryl, heteroaryl, and aralkyl, and groups in which these substituents form a ring.

(97) Examples of aminocarbonylamino (—NR—CO—NR'R'') include groups in which R, R', and R'' are each independently selected from a hydrogen atom, alkyl, cycloalkyl, alkenyl, alkynyl, aryl, heteroaryl, and aralkyl, or groups in which these substituents form a ring.

(98) Herein, a “peptide residue” or an “amino acid residue” constituting the peptide compound may be simply referred to as a “peptide” or an “amino acid”, respectively.

(99) In the present invention, the meaning of the term “and/or” includes any combination attained by suitably combining “and” and “or”. Specifically, for example, “A, B, and/or C” includes the following 7 variations: (i) A, (ii) B, (iii) C, (iv) A and B, (v) A and C, (vi) B and C, (vii) A, B, and C.

(100) In an embodiment, the present invention relates to a cyclic compound represented by formula (1) below or a salt thereof, or a solvate thereof.

(101) ##STR00006##

(102) In the cyclic compound of formula (1), the ring is composed of 11 amino acid residues. Herein, the amino acid residue having P.sub.1, Q.sub.1, R.sub.1, and L.sub.1 in the formula may be referred to as core 1, the amino acid residue having P.sub.2, Q.sub.2, and R.sub.2 as core 2, the

amino acid residue having P.sub.3, Q.sub.3, and R.sub.3 as core 3, the amino acid residue having P.sub.4, Q.sub.4, and R.sub.4 as core 4, the amino acid residue having P.sub.5, Q.sub.5, and R.sub.5 as core 5, the amino acid residue having P.sub.6, Q.sub.6, and R.sub.6 as core 6, the amino acid residue having P.sub.7, Q.sub.7, and R.sub.7 as core 7, the amino acid residue having P.sub.8, Q.sub.8, and R.sub.8 as core 8, the amino acid residue having P.sub.9, Q.sub.9, and R.sub.9 as core 9, the amino acid residue having P.sub.10, Q.sub.10, and R.sub.10 as core 10, and the amino acid residue having P.sub.11, Q.sub.1, R.sub.11, and L.sub.11 as core 11.

(103) In an embodiment, in formula (1), L.sub.1 is a single bond.

(104) In an embodiment, in formula (1), R.sub.1 is a C.sub.1 to C.sub.7 alkyl, and preferably, a 2-methylpropyl or a n-propyl.

(105) In an embodiment, R.sub.1 joins together with R.sub.5 to form a divalent group, wherein a partial structure: *-CR.sub.1Q.sub.1-L.sub.1-CO—NP.sub.2-CR.sub.2Q.sub.2-CO—NP.sub.3-CR.sub.3Q.sub.3—CO—NP.sub.4-CR.sub.4Q.sub.4—CO—NP.sub.5-CR.sub.5Q.sub.5—* in the cyclic compound represented by formula (1) is represented by the following formula:

(106) ##STR00007## wherein: X.sub.2 is—L.sub.1-CO—NP.sub.2-CR.sub.2Q.sub.2-CO—NP.sub.3-CR.sub.3Q.sub.3—CO—NP.sub.4-CR.sub.4Q.sub.4—CO—NP.sub.5—,

 custom character is a single bond or a double bond, n is 0, 1 or 2, m is 0, 1, 2, 3 or 4, * represents an attachment point to an adjacent atom; wherein  custom character may take E or Z configuration, if  custom character is a double bond.

(107) The formula is preferably a formula:

(108) ##STR00008##

(109) In an embodiment, P.sub.1 is a C.sub.1 to C.sub.6 alkyl, and preferably methyl.

(110) In an embodiment, Q.sub.1 is a hydrogen atom.

(111) The amino acid residue of core 1 is, for example, MeLeu or MeNva except the case where a side chain (R.sub.1) of core 1 and a side chain (R.sub.5) of core 5 join together to form a divalent group.

(112) If a side chain (R.sub.1) of core 1 and a side chain (R.sub.5) of core 5 join together to form a divalent group, the group at the position corresponding to R.sub.1 of MeAhpe (2), MeAocte (2) or MeAhxe (2) and the group at the position corresponding to side chain (R.sub.5) of core 5 can be linked by use of a method described, for example, in the “common process” later described.

(113) In an embodiment, in formula (1), R.sub.2 is a C.sub.1 to C.sub.6 alkyl, and preferably 1-methylpropyl.

(114) In an embodiment, P.sub.2 is a hydrogen atom.

(115) In an embodiment, Q.sub.2 is a hydrogen atom.

(116) The amino acid residue of core 2 is, for example, Ile.

(117) In an embodiment, in formula (1), R.sub.3 is a hydrogen atom.

(118) In an embodiment, R.sub.3 joins together with a carbon atom to which P.sub.3 and R.sub.3 are attached and a nitrogen atom to which P.sub.3 is attached, to form a 4 to 7-membered saturated heterocycle. The 4 to 7-membered saturated heterocycle is, for example, a pyrrolidine ring.

(119) In an embodiment, P.sub.3 is a C.sub.1 to C.sub.6 alkyl or a C.sub.3 to C.sub.8 cycloalkyl, and preferably a methyl or a cyclopropyl.

(120) In an embodiment, Q.sub.3 is a hydrogen atom.

(121) The amino acid residue of core 3 is, for example, MeGly, Pro or cPrGly.

(122) In an embodiment, R.sub.4 joins together with P.sub.5 to form a divalent group, wherein a partial structure: *-CR.sub.4Q.sub.4—CO—NP.sub.5—* of a cyclic compound represented by formula (1) is expressed by the following formula:

(123) ##STR00009##

(124) In an embodiment, P.sub.4 is a C.sub.1 to C.sub.6 alkyl, and preferably methyl.

(125) In an embodiment, Q.sub.4 is a hydrogen atom.

(126) If a side chain (R.sub.4) of core 4 and N substituent (P.sub.5) of core 5 join together to form a

divalent group, preferably, a group at the position corresponding to R.sub.4 of MeAlgly and a group at the position corresponding to N substituent (P.sub.5) of core 5 can be linked by use of a method described, for example, in the “common process” later described.

(127) In an embodiment, in formula (1), R.sub.5 is a benzyl optionally substituted with one or more groups independently selected from the group consisting of a C.sub.1 to C.sub.6 alkyl, a C.sub.1 to C.sub.6 haloalkyl, and a C.sub.3 to C.sub.8 cycloalkyl, and preferably, 4-cyclopropylbenzyl, 4-(trifluoromethyl)benzyl, 4-methylbenzyl or 4-ethylbenzyl.

(128) In an embodiment, in formula (1), R.sub.5 joins together with R.sub.1 to form a divalent group. The details of this group are the same as defined in the above.

(129) In an embodiment, P.sub.5 joins together with R.sub.4 to form a divalent group. The details of this group are the same as defined in the above.

(130) In an embodiment, Q.sub.5 is a hydrogen atom.

(131) If a side chain (R.sub.5) of core 5 and a side chain (R.sub.1) of core 1 join together to form a divalent group, and a N substituent (P.sub.5) of core 5 and side chain (R.sub.4) of core 4 join together to form a divalent group, more specifically, R.sub.5 of ButenylPhe (4—CH=CH.sub.2) and a group at the position corresponding to P.sub.5, and a side chain (R.sub.1) of core 1 and a group at the position corresponding to a side chain (R.sub.4) of core 4, can be linked respectively, by use of a method described, for example, in the “common process” later described.

(132) In the case where a side chain (R.sub.5) of core 5 and a side chain (R.sub.1) of core 1 do not form a divalent group, and a N substituent (P.sub.5) of core 5 and a side chain (R.sub.4) of core 4 form a divalent group, more specifically, a group at the position corresponding to P.sub.5 of ButenylPhe (4-Et), ButenylPhe (4-cPr) or AllylPhe (4—CF₃) and a group at the position corresponding to side chain (R.sub.4) of core 4 can be linked by use of a method described, for example, in the “common process” later described.

(133) In an embodiment, in formula (1), R.sub.6 is a hydrogen atom.

(134) In an embodiment, P.sub.6 is a C.sub.1 to C.sub.6 alkyl, and preferably methyl.

(135) In an embodiment, Q.sub.6 is a hydrogen atom.

(136) The amino acid residue of core 6 is, for example, MeGly.

(137) In an embodiment, in formula (1), phenethyl optionally substituted with one or more groups independently selected from the group consisting of halogen, a C.sub.1 to C.sub.6 haloalkyl and a C.sub.1 to C.sub.6 alkoxy, is preferably, 3,5-difluoro-4-(trifluoromethyl) phenethyl, 3,4-dichlorophenethyl or 3-methoxy-4-(trifluoromethyl) phenethyl.

(138) In an embodiment, P.sub.7 is a hydrogen atom.

(139) In an embodiment, Q.sub.7 is a hydrogen atom.

(140) The amino acid residue of core 7 is, for example, Hph(4—CF₃-35-F₂), Hph(34-C₁₂) or Hph(4—CF₃-3-OMe).

(141) In an embodiment, in formula (1), R& joins together with a carbon atom to which P.sub.8 and R.sub.8 are attached and a nitrogen atom to which P& is attached, to form a 4 to 7-membered saturated heterocycle. The 4 to 7-membered saturated heterocycle is optionally substituted with a C.sub.1 to C.sub.6 alkoxy, and preferably ethoxy or n-propoxy. The 4 to 7-membered saturated heterocycle is, for example, a pyrrolidine ring.

(142) In an embodiment, Q.sub.8 is a hydrogen atom.

(143) The amino acid residue of core 8 is, for example, Hyp (Et) or Hyp (nPr).

(144) In an embodiment, in formula (1), R.sub.9 joins together with Q.sub.9 and a carbon atom to which R.sub.9 and Q.sub.9 are attached to form a 3 to 8-membered alicyclic ring. The 3 to 8-membered alicyclic ring is optionally substituted with one or more C.sub.1 to C.sub.6 alkyls, for example, two methyl groups. The 3 to 8-membered alicyclic ring is, for example, a cyclobutane ring or a cyclopentane ring.

(145) In an embodiment, P.sub.9 is a hydrogen atom or a C.sub.1 to C.sub.6 alkyl, and preferably a hydrogen atom or a methyl.

(146) The amino acid residue of core 9 is, for example, cLeu, cVal (3-Me2) or Mec Val.

(147) In an embodiment, in formula (1), R.sub.10 is a C.sub.1 to C.sub.6 alkyl or a C.sub.3 to C.sub.8 cycloalkyl, and preferably, a pentan-3-yl or cyclopentyl.

(148) In an embodiment, P.sub.10 is a C.sub.1 to C.sub.6 alkyl, and preferably methyl.

(149) In an embodiment, Q.sub.10 is a hydrogen atom.

(150) The amino acid residue of core 10 is, for example, MeGly (cPent) or MeNva (3-Et).

(151) In an embodiment, in formula (1), L.sub.11 is —CH.sub.2—.

(152) In an embodiment, in formula (1), R.sub.11 is a di-C.sub.1 to C.sub.6 alkylaminocarbonyl or a 4 to 8-membered cyclic aminocarbonyl, and preferably, a dimethylaminocarbonyl, a N-ethyl-N-methylaminocarbonyl, a pyrrolidinylcarbonyl or a piperidinylcarbonyl.

(153) In an embodiment, P.sub.11 is a C.sub.1 to C.sub.6 alkyl, and preferably, a methyl.

(154) In an embodiment, Q.sub.11 is a hydrogen atom.

(155) In an embodiment, the compound of the present invention can be one or more compounds selected from the following: PP1574: (2S,8S,12R,14S,20S,23S,27S,30S,36S,38Z)-20-cyclopentyl-8-[2-[3,5-difluoro-4-(trifluoromethyl)phenyl]ethyl]-12-ethoxy-N-ethyl-27-isobutyl-N,4,19,22,26,32,35-heptamethyl-30-[(1 S)-1-methylpropyl]-3,6,9,15,18,21,25,28,31,34,42-undecaoxo-2-[[4-(trifluoromethyl)phenyl]methyl]spiro[1,4,7,10,16,19,22,26,29,32,35-undecazatricyclo[34.5.1.0.sup.10,14]dotetracont-38-ene-17,1'-cyclopentane]-23-carboxamide, PP1650: (2S,8S,12R,14S,20S,23S,27S,30S,36S,38Z)-20-cyclopentyl-8-[2-[3,5-difluoro-4-(trifluoromethyl)phenyl]ethyl]-12-ethoxy-27-isobutyl-N,N,3',3',4,19,22,26,32,35-decamethyl-30-[(1 S)-1-methylpropyl]-3,6,9,15,18,21,25,28,31,34,42-undecaoxo-2-[[4-(trifluoromethyl)phenyl]methyl]spiro[1,4,7,10,16,19,22,26,29,32,35-undecazatricyclo[34.5.1.0.sup.10'.sup.14]dotetracont-38-ene-17,1'-cyclobutane]-23-carboxamide, PP1827: (1S,7S,IIR,13S,19S,22S,26S,29S,35S,37Z,47E)-19-cyclopentyl-7-[2-[3,5-difluoro-4-(trifluoromethyl)phenyl]ethyl]-11-ethoxy-N-ethyl-N,3,18,21,25,31,34-heptamethyl-29-[(1 S)-1-methylpropyl]-2,5,8,14,17,20,24,27,30,33,55-undecaoxo-spiro[3,6,9,15,18,21,25,28,31,34,41-undecazapentacyclo[24.15.11.2.sup.43,46.1.sup.35,41.0.sup.9,13]pentapentaconta-37,43(54),44,46(53),47-pentaene-16,1'-cyclobutane]-22-carboxamide, PP1830: (1S,7S,IIR,13S,19S,22S,26S,29S,35S,37Z,47E)-19-cyclopentyl-11-ethoxy-N-ethyl-7-[2-[3-methoxy-4-(trifluoromethyl)phenyl]ethyl]-N,3,3',3',18,21,25,31,34-nonamethyl-29-[(1 S)-1-methylpropyl]-2,5,8,14,17,20,24,27,30,33,55-undecaoxo-spiro[3,6,9,15,18,21,25,28,31,34,41-undecazapentacyclo[24.15.11.2.sup.43,46.1.sup.35,41.0.sup.9,13]pentapentaconta-37,43(54),44,46(53),47-pentaene-16,1'-cyclobutane]-22-carboxamide, PP2093: (2S,8S,12R,14S,20S,23S,27S,30S,36S,38Z)-20-cyclopentyl-32-cyclopropyl-8-[2-[3,5-difluoro-4-(trifluoromethyl)phenyl]ethyl]-12-ethoxy-N,N,3',3',4,19,22,26,35-nonamethyl-30-[(1 S)-1-methylpropyl]-3,6,9,15,18,21,25,28,31,34,42-undecaoxo-27-propyl-2-(p-tolylmethyl)spiro[[1,4,7,10,16,19,22,26,29,32,35-undecazatricyclo[34.5.1.0.sup.10,14]dotetracont-38-ene-17,1'-cyclobutane]-23-carboxamide, PP2260: (1S,7S,11R,13S,19S,22S,26S,29S,35S,37Z)-19-cyclopentyl-11-ethoxy-N-ethyl-7-[2-[3-methoxy-4-(trifluoromethyl)phenyl]ethyl]-N,3,3',3',18,21,25,31,34-nonamethyl-29-[(1 S)-1-methylpropyl]-2,5,8,14,17,20,24,27,30,33,55-undecaoxo-spiro[3,6,9,15,18,21,25,28,31,34,41-undecazapentacyclo[24.15.11.2.sup.43,46.1.sup.35,41.0.sup.9,13]pentapentaconta-37,43(54),44,46(53)-tetraene-16,1'-cyclobutane]-22-carboxamide, PP2316: (2S,8S, 12R, 14S,20S,23S,27S,30S,36S,38Z)-20-cyclopentyl-32-cyclopropyl-12-ethoxy-8-[2-[3-methoxy-4-(trifluoromethyl)phenyl]ethyl]-N,N,4,16,19,22,26,35-octamethyl-30-[(1 S)-1-methylpropyl]-3,6,9,15,18,21,25,28,31,34,42-undecaoxo-27-propyl-2-[[4-(trifluoromethyl)phenyl]methyl]spiro[1,4,7,10,16,19,22,26,29,32,35-undecazatricyclo[34.5.1.0.sup.10,14]dotetracont-38-ene-17,1'-cyclobutane]-23-carboxamide, PP2320: (1S,4S,10S,13S,17S,20S,26S,28R,32S,38S,42Z)-20-cyclopentyl-28-ethoxy-32-[2-[3-methoxy-4-(trifluoromethyl)phenyl]ethyl]-N, N, 2,14,18,21, 24,36-octamethyl-10-[(1 S)-1-

methylpropyl]-3,9,12,15,19,22,25,31,34,37,45-undecaoxo-13-propyl-38-[[4-(trifluoromethyl)phenyl]methyl]spiro[2,8,11,14,18,21,24,30,33,36,39-undecazatetracyclo[37.5.1.0.sup.4,8.0.sup.26,30]pentatetracont-42-ene-23,1'-cyclobutane]-17-carboxamide, PP2328: (2S,8S,12R,14S,20S,23S,27S,30S,36S,38Z)-20-cyclopentyl-32-cyclopropyl-12-ethoxy-8-[2-[3-methoxy-4-(trifluoromethyl)phenyl]ethyl]-N,N,4,16,19,22,26,35-octamethyl-30-[(1 S)-1-methylpropyl]-3,6,9,15,18,21,25,28,31,34,42-undecaoxo-27-propyl-2-(p-tolylmethyl)spiro[[1,4,7,10,16,19,22,26,29,32,35-undecazatricyclo[34.5.1.0.sup.10,14]dotetracont-38-ene-17,1'-cyclobutane]-23-carboxamide, PP2574: (1S,7S,11R,13S,19S,22S,26S,29S,35S,37Z,47E)-19-cyclopentyl-7-[2-[3,5-difluoro-4-(trifluoromethyl)phenyl]ethyl]-11-ethoxy-3,3',3',18,21,25,31,34-octamethyl-29-[(1 S)-1-methylpropyl]-22-(pyrrolidine-1-carbonyl)spiro[3,6,9,15,18,21,25,28,31,34,41-undecazapentacyclo[24.15.10.2.sup.43,46.1.sup.35,41.0.sup.9,13]tetrapentaconta-37,43(53),44,46(52),47-pentaene-16,1'-cyclobutane]-2,5,8,14,17,20,24,27,30,33,54-undecone, PP2576: (1S,7S,11R,13S,19S,22S,26S,29S,35S,37Z,47E)-7-[2-[3,5-difluoro-4-(trifluoromethyl)phenyl]ethyl]-11-ethoxy-19-(1-ethylpropyl)-3,3',3',18,21,25,31,34-octamethyl-29-[(1 S)-1-methylpropyl]-22-(pyrrolidine-1-carbonyl)spiro[3,6,9,15,18,21,25,28,31,34,41-undecazapentacyclo[24.15.10.2.sup.43,46.1.sup.35,41.0.sup.9,13]tetrapentaconta-37,43(53),44,46(52),47-pentaene-16,1'-cyclobutane]-2,5,8,14,17,20,24,27,30,33,54-undecone, PP2583: (1S,7S,11R,13S,19S,22S,26S,29S,35S,37Z,47E)-19-cyclopentyl-31-cyclopropyl-7-[2-[3,5-difluoro-4-(trifluoromethyl)phenyl]ethyl]-11-ethoxy-3,3',3',18,21,25,34-heptamethyl-29-[(1 S)-1-methylpropyl]-22-(pyrrolidine-1-carbonyl)spiro[3,6,9,15,18,21,25,28,31,34,41-undecazapentacyclo[24.15.10.2.sup.43,46.1.sup.35,41.0.sup.9,13]tetrapentaconta-37,43(53),44,46(52),47-pentaene-16,1'-cyclobutane]-2,5,8,14,17,20,24,27,30,33,54-undecone, PP2687: (2S,8S,12R,14S,20S,23S,27S,30S,36S,38Z)-20-cyclopentyl-12-ethoxy-2-[(4-ethylphenyl)methyl]-27-isobutyl-8-[2-[3-methoxy-4-(trifluoromethyl)phenyl]ethyl]-N,N,3',3',4,19,22,26,32,35-decamethyl-30-[(1S)-1-methylpropyl]-3,6,9,15,18,21,25,28,31,34,42-undecaoxo-spiro[1,4,7,10,16,19,22,26,29,32,35-undecazatricyclo[34.5.1.0.sup.10,14]dotetracont-38-ene-17,1'-cyclobutane]-23-carboxamide, PP2691: (2S,8S,12R,14S,20S,23S,27S,30S,36S,38Z)-20-cyclopentyl-2-[(4-cyclopropylphenyl)methyl]-12-ethoxy-27-isobutyl-8-[2-[3-methoxy-4-(trifluoromethyl)phenyl]ethyl]-N,N,3',3',4,19,22,26,32,35-decamethyl-30-[(1 S)-1-methylpropyl]-3,6,9,15,18,21,25,28,31,34,42-undecaoxo-spiro[1,4,7,10,16,19,22,26,29,32,35-undecazatricyclo[34.5.1.0.sup.10'.sup.14]dotetracont-38-ene-17,1'-cyclobutane]-23-carboxamide, PP2957: (1S,7S,11R,13S,19S,22S,26S,29S,35S,37Z,47E)-31-cyclopropyl-7-[2-[3,5-difluoro-4-(trifluoromethyl)phenyl]ethyl]-11-ethoxy-19-(1-ethylpropyl)-3,18,21,25,34-pentamethyl-29-[(1 S)-1-methylpropyl]-22-(pyrrolidine-1-carbonyl)spiro[3,6,9,15,18,21,25,28,31,34,41-undecazapentacyclo[24.15.10.2.sup.43,46.1.sup.35,41.0.sup.9,13]tetrapentaconta-37,43(53),44,46(52),47-pentaene-16,1'-cyclobutane]-2,5,8,14,17,20,24,27,30,33,54-undecone, PP3033: (1S,7S,11R,13S,19S,22S,26S,29S,35S,37Z)-19-cyclopentyl-11-ethoxy-7-[2-[3-methoxy-4-(trifluoromethyl)phenyl]ethyl]-3,18,21,25,31,34-hexamethyl-29-[(1 S)-1-methylpropyl]-22-(piperidine-1-carbonyl)spiro[3,6,9,15,18,21,25,28,31,34,41-undecazapentacyclo[24.15.11.2.sup.43,46.1.sup.35,41.0.sup.9,13]pentapentaconta-37,43(54),44,46(53)-tetraene-16,1'-cyclobutane]-2,5,8,14,17,20,24,27,30,33,55-undecone, PP3034: (1S,7S,11R,13S,19S,22S,26S,29S,35S,37Z)-19-cyclopentyl-11-ethoxy-7-[2-[3-methoxy-4-(trifluoromethyl)phenyl]ethyl]-3,3',3',18,21,25,31,34-octamethyl-29-[(1S)-1-methylpropyl]-22-(pyrrolidine-1-carbonyl)spiro[3,6,9,15,18,21,25,28,31,34,41-undecazapentacyclo[24.15.11.2.sup.43,46.1.sup.35,41.0.sup.9,13]pentapentaconta-37,43(54),44,46(53)-tetraene-16,1'-cyclobutane]-2,5,8,14,17,20,24,27,30,33,55-undecone, PP3036: (1S,7S,11R,13S,19S,22S,26S,29S,35S,37Z)-19-cyclopentyl-11-ethoxy-7-[2-[3-methoxy-4-(trifluoromethyl)phenyl]ethyl]-3,3',3',18,21,25,31,34-octamethyl-29-[(1S)-1-methylpropyl]-22-

(piperidine-1-carbonyl)spiro[3,6,9,15,18,21,25,28,31,34,41-undecazapentacyclo[24.15.10.2.sup.43,46.1.sup.35,41.0.sup.9,13]tetrapentaconta-37,43(53),44,46(52)-tetraene-16,1'-cyclobutane]-2,5,8,14,17,20,24,27,30,33,54-undecone, PP3037: (1S,7S,11R,13S,19S,22S,26S,29S,35S,37Z)-19-cyclopentyl-11-ethoxy-7-[2-[3-methoxy-4-(trifluoromethyl)phenyl]ethyl]-3,3',3',18,21,25,31,34-octamethyl-29-[(1S)-1-methylpropyl]-22-(pyrrolidine-1-carbonyl)spiro[3,6,9,15,18,21,25,28,31,34,41-undecazapentacyclo[24.15.10.2.sup.43,46.1.sup.35,41.0.sup.9,13]tetrapentaconta-37,43(53),44,46(52)-tetraene-16,1'-cyclobutane]-2,5,8,14,17,20,24,27,30,33,54-undecone, PP3047: (1S,7S,11R,13S,19S,22S,26S,29S,35S,37Z,47E)-7-[2-(3,4-dichlorophenyl)ethyl]-11-ethoxy-19-(1-ethylpropyl)-3,18,21,25,31,34-hexamethyl-29-[(1S)-1-methylpropyl]-22-(pyrrolidine-1-carbonyl)spiro[3,6,9,15,18,21,25,28,31,34,41-undecazapentacyclo[24.15.11.2.sup.43,46.1.sup.35,41.0.sup.9,13]pentapentaconta-37,43(54),44,46(53),47-pentaene-16,1'-cyclobutane]-2,5,8,14,17,20,24,27,30,33,55-undecone, PP3093: (1S,7S,11R,13S,19S,22S,26S,29S,35S,37Z)-19-cyclopentyl-11-ethoxy-7-[2-[3-methoxy-4-(trifluoromethyl)phenyl]ethyl]-3,3',3',18,21,25,31,34-octamethyl-29-[(1S)-1-methylpropyl]-22-(piperidine-1-carbonyl)spiro[3,6,9,15,18,21,25,28,31,34,41-undecazapentacyclo[24.15.9.2.sup.43,46.1.sup.35,41.0.sup.9,13]tripentaconta-37,43(52),44,46(51)-tetraene-16,1'-cyclobutane]-2,5,8,14,17,20,24,27,30,33,53-undecone, PP3094: (1S,7S,11R,13S,19S,22S,26S,29S,35S,37Z)-19-cyclopentyl-7-[2-[3-methoxy-4-(trifluoromethyl)phenyl]ethyl]-3,3',3',18,21,25,31,34-octamethyl-29-[(1S)-1-methylpropyl]-11-propoxy-22-(pyrrolidine-1-carbonyl)spiro[3,6,9,15,18,21,25,28,31,34,41-undecazapentacyclo[24.15.9.2.sup.43,46.1.sup.35,41.0.sup.9,13]tripentaconta-37,43(52),44,46(51)-tetraene-16,1'-cyclobutane]-2,5,8,14,17,20,24,27,30,33,53-undecone, PP3095: (2S,8S,12R,14S,20S,23S,27S,30S,36S,38Z)-20-cyclopentyl-32-cyclopropyl-12-ethoxy-8-[2-[3-methoxy-4-(trifluoromethyl)phenyl]ethyl]-4,16,19,22,26,35-hexamethyl-30-[(1S)-1-methylpropyl]-23-(piperidine-1-carbonyl)-27-propyl-2-[[4-(trifluoromethyl)phenyl]methyl]spiro[1,4,7,10,16,19,22,26,29,32,35-undecazatricyclo[34.5.1.0.sup.10,14]dotetracont-38-ene-17,1'-cyclobutane]-3,6,9,15,18,21,25,28,31,34,42-undecone, PP3096: (2S,8S,12R,14S,20S,23S,27S,30S,36S,38Z)-20-cyclopentyl-32-cyclopropyl-12-ethoxy-8-[2-[3-methoxy-4-(trifluoromethyl)phenyl]ethyl]-4,16,19,22,26,35-hexamethyl-30-[(1S)-1-methylpropyl]-23-(piperidine-1-carbonyl)-27-propyl-2-(p-tolylmethyl)spiro[[1,4,7,10,16,19,22,26,29,32,35-undecazatricyclo[34.5.1.0.sup.10,14]dotetracont-38-ene-17,1'-cyclobutane]-3,6,9,15,18,21,25,28,31,34,42-undecone, PP3097: (2S,8S,12R,14S,20S,23S,27S,30S,36S,38Z)-20-cyclopentyl-32-cyclopropyl-12-ethoxy-8-[2-[3-methoxy-4-(trifluoromethyl)phenyl]ethyl]-4,16,19,22,26,35-hexamethyl-30-[(1S)-1-methylpropyl]-27-propyl-23-(pyrrolidine-1-carbonyl)-2-[[4-(trifluoromethyl)phenyl]methyl]spiro[1,4,7,10,16,19,22,26,29,32,35-undecazatricyclo[34.5.1.0.sup.10,14]dotetracont-38-ene-17,1'-cyclobutane]-3,6,9,15,18,21,25,28,31,34,42-undecone, PP3098: (2S,8S,12R,14S,20S,23S,27S,30S,36S,38Z)-20-cyclopentyl-32-cyclopropyl-12-ethoxy-8-[2-[3-methoxy-4-(trifluoromethyl)phenyl]ethyl]-4,16,19,22,26,35-hexamethyl-30-[(1S)-1-methylpropyl]-27-propyl-2-(p-tolylmethyl)-23-(pyrrolidine-1-carbonyl)spiro[1,4,7,10,16,19,22,26,29,32,35-undecazatricyclo[34.5.1.0.sup.10,14]dotetracont-38-ene-17,1'-cyclobutane]-3,6,9,15,18,21,25,28,31,34,42-undecone, PP3099: (1S,4S,10S,13S,17S,20S,26S,28R,32S,38S,42Z)-20-cyclopentyl-28-ethoxy-32-[2-[3-methoxy-4-(trifluoromethyl)phenyl]ethyl]-2,14,18,21,24,36-hexamethyl-10-[(1S)-1-methylpropyl]-17-(piperidine-1-carbonyl)-13-propyl-38-[[4-(trifluoromethyl)phenyl]methyl]spiro[2,8,11,14,18,21,24,30,33,36,39-undecazatetracyclo[37.5.1.0.sup.4,8.0.sup.26,30]pentatetracont-42-ene-23,1'-

cyclobutane]-3,9,12,15,19,22,25,31,34,37,45-undecone, PP3100:
(1S,4S,10S,13S,17S,20S,26S,28R,32S,38S,42Z)-20-cyclopentyl-28-ethoxy-32-[2-[3-methoxy-4-(trifluoromethyl)phenyl]ethyl]-2,14,18,21,24,36-hexamethyl-10-[(1 S)-1-methylpropyl]-17-(piperidine-1-carbonyl)-13-propyl-38-(p-tolylmethyl)spiro[2,8,11,14,18,21,24,30,33,36,39-undecazatetracyclo[37.5.1.0.sup.4,8.0.sup.26,30]pentatetracont-42-ene-23,1'-cyclobutane]-3,9,12,15,19,22,25,31,34,37,45-undecone, PP3101:
(1S,4S,10S,13S,17S,20S,26S,28R,32S,38S,42Z)-20-cyclopentyl-28-ethoxy-32-[2-[3-methoxy-4-(trifluoromethyl)phenyl]ethyl]-2,14,18,21,24,36-hexamethyl-10-[(1 S)-1-methylpropyl]-13-propyl-17-(pyrrolidine-1-carbonyl)-38-[[4-(trifluoromethyl)phenyl]methyl]spiro[2,8,11,14,18,21,24,30,33,36,39-undecazatetracyclo[37.5.1.0.sup.4,8.0.sup.26,30]pentatetracont-42-ene-23,1'-cyclobutane]-3,9,12,15,19,22,25,31,34,37,45-undecone, PP3102:
(1S,4S,10S,13S,17S,20S,26S,28R,32S,38S,42Z)-20-cyclopentyl-28-ethoxy-32-[2-[3-methoxy-4-(trifluoromethyl)phenyl]ethyl]-2,14,18,21,24,36-hexamethyl-10-[(1 S)-1-methylpropyl]-13-propyl-38-(p-tolylmethyl)-17-(pyrrolidine-1-carbonyl)spiro[2,8,11,14,18,21,24,30,33,36,39-undecazatetracyclo[37.5.1.0.sup.4,8.0.sup.26,30]pentatetracont-42-ene-23,1'-cyclobutane]-3,9,12,15,19,22,25,31,34,37,45-undecone, PP3103:
(2S,8S,12R,14S,20S,23S,27S,30S,36S,38Z)-20-cyclopentyl-12-ethoxy-8-[2-[3-methoxy-4-(trifluoromethyl)phenyl]ethyl]-4,16,19,22,26,32,35-heptamethyl-30-[(1 S)-1-methylpropyl]-23-(piperidine-1-carbonyl)-27-propyl-2-[[4-(trifluoromethyl)phenyl]methyl]spiro[1,4,7,10,16,19,22,26,29,32,35-undecazatricyclo[34.5.1.0.sup.10,14]dotetracont-38-ene-17,1'-cyclobutane]-3,6,9,15,18,21,25,28,31,34,42-undecone, PP3104:
(2S,8S,12R,14S,20S,23S,27S,30S,36S,38Z)-20-cyclopentyl-12-ethoxy-8-[2-[3-methoxy-4-(trifluoromethyl)phenyl]ethyl]-4,16,19,22,26,32,35-heptamethyl-30-[(1 S)-1-methylpropyl]-23-(piperidine-1-carbonyl)-27-propyl-2-(p-tolylmethyl)spiro[[1,4,7,10,16,19,22,26,29,32,35-undecazatricyclo[34.5.1.0.sup.10,14]dotetracont-38-ene-17,1'-cyclobutane]-3,6,9,15,18,21,25,28,31,34,42-undecone, PP3105:
(2S,8S,12R,14S,20S,23S,27S,30S,36S,38Z)-20-cyclopentyl-12-ethoxy-8-[2-[3-methoxy-4-(trifluoromethyl)phenyl]ethyl]-4,16,19,22,26,32,35-heptamethyl-30-[(1 S)-1-methylpropyl]-27-propyl-23-(pyrrolidine-1-carbonyl)-2-[[4-(trifluoromethyl)phenyl]methyl]spiro[1,4,7,10,16,19,22,26,29,32,35-undecazatricyclo[34.5.1.0.sup.10,14]dotetracont-38-ene-17,1'-cyclobutane]-3,6,9,15,18,21,25,28,31,34,42-undecone, PP3106:
(2S,8S,12R,14S,20S,23S,27S,30S,36S,38Z)-20-cyclopentyl-12-ethoxy-8-[2-[3-methoxy-4-(trifluoromethyl)phenyl]ethyl]-4,16,19,22,26,32,35-heptamethyl-30-[(1 S)-1-methylpropyl]-27-propyl-2-(p-tolylmethyl)-23-(pyrrolidine-1-carbonyl)spiro[1,4,7,10,16,19,22,26,29,32,35-undecazatricyclo[34.5.1.0.sup.10,14]dotetracont-38-ene-17,1'-cyclobutane]-3,6,9,15,18,21,25,28,31,34,42-undecone, PP3110:
(2S,8S,12R,14S,20S,23S,27S,30S,36S,38Z)-20-cyclopentyl-32-cyclopropyl-8-[2-[3,5-difluoro-4-(trifluoromethyl)phenyl]ethyl]-12-ethoxy-4,16,19,22, 26,35-hexamethyl-30-[(1 S)-1-methylpropyl]-23-(piperidine-1-carbonyl)-27-propyl-2-[[4-(trifluoromethyl)phenyl]methyl]spiro[1,4,7,10,16,19,22,26,29,32,35-undecazatricyclo[34.5.1.0.sup.10,14]dotetracont-38-ene-17,1'-cyclobutane]-3,6,9,15,18,21,25,28,31,34,42-undecone, PP3111:
(2S,8S,12R,14S,20S,23S,27S,30S,36S,38Z)-20-cyclopentyl-32-cyclopropyl-8-[2-[3,5-difluoro-4-(trifluoromethyl)phenyl]ethyl]-12-ethoxy-4,16,19,22,26,35-hexamethyl-30-[(1 S)-1-methylpropyl]-23-(piperidine-1-carbonyl)-27-propyl-2-(p-tolylmethyl)spiro[[1,4,7,10,16,19,22,26,29,32,35-undecazatric clo[34.5.1.0.sup.10,14]dotetracont-38-ene-17,1-cyclobutane]-3,6,9,15,18,21,25,28,31,34,42-undecone, PP3112:

(2S,8S,12R,14S,20S,23S,27S,30S,36S,38Z)-20-cyclopentyl-32-cyclopropyl-8-[2-[3,5-difluoro-4-(trifluoromethyl)phenyl]ethyl]-12-ethoxy-4,16,19,22,26,35-hexamethyl-30-[(1 S)-1-methylpropyl]-27-propyl-23-(pyrrolidine-1-carbonyl)-2-[[4-(trifluoromethyl)phenyl]methyl]spiro[1,4,7,10,16,19,22,26,29,32,35-undecazatricyclo[34.5.1.0.sup.10,14]dotetracont-38-ene-17,1'-cyclobutane]-3,6,9,15,18,21,25,28,31,34,42-undecone, PP3113:

(2S,8S,12R,14S,20S,23S,27S,30S,36S,38Z)-20-cyclopentyl-32-cyclopropyl-8-[2-[3,5-difluoro-4-(trifluoromethyl)phenyl]ethyl]-12-ethoxy-4,16,19,22,26,35-hexamethyl-30-[(1 S)-1-methylpropyl]-27-propyl-2-(p-tolylmethyl)-23-(pyrrolidine-1-carbonyl)spiro[1,4,7,10,16,19,22,26,29,32,35-undecazatricyclo[34.5.1.0.sup.10,14]dotetracont-38-ene-17,1-cyclobutane]-3,6,9,15,18,21,25,28,31,34,42-undecone, PP3114:

(1S,4S,10S,13S,17S,20S,26S,28R,32S,38S,42Z)-20-cyclopentyl-32-[2-[3,5-difluoro-4-(trifluoromethyl)phenyl]ethyl]-28-ethoxy-2,14,18,21,24,36-hexamethyl-10-[(1 S)-1-methylpropyl]-17-(piperidine-1-carbonyl)-13-propyl-38-[[4-(trifluoromethyl)phenyl]methyl]spiro[2,8,11,14,18,21,24,30,33,36,39-undecazatetracyclo[37.5.1.0.sup.4,8.0.sup.26,30]pentatetracont-42-ene-23,1'-cyclobutane]-3,9,12,15,19,22,25,31,34,37,45-undecone, PP3115:

(1S,4S,10S,13S,17S,20S,26S,28R,32S,38S,42Z)-20-cyclopentyl-32-[2-[3,5-difluoro-4-(trifluoromethyl)phenyl]ethyl]-28-ethoxy-2,14,18,21,24,36-hexamethyl-10-[(1 S)-1-methylpropyl]-17-(piperidine-1-carbonyl)-13-propyl-38-(p-tolylmethyl)spiro[2,8,11,14,18,21,24,30,33,36,39-undecazatetracyclo[37.5.1.0.sup.4,8.0.sup.26,30]pentatetracont-42-ene-23,1'-cyclobutane]-3,9,12,15,19,22,25,31,34,37,45-undecone, PP3116:

(1S,4S,10S,13S,17S,20S,26S,28R,32S,38S,42Z)-20-cyclopentyl-32-[2-[3,5-difluoro-4-(trifluoromethyl)phenyl]ethyl]-28-ethoxy-2,14,18,21,24,36-hexamethyl-10-[(1 S)-1-methylpropyl]-13-propyl-17-(pyrrolidine-1-carbonyl)-38-[[4-(trifluoromethyl)phenyl]methyl]spiro[2,8,11,14,18,21,24,30,33,36,39-undecazatetracyclo[37.5.1.0.sup.4,8.0.sup.26,30]pentatetracont-42-ene-23,1'-cyclobutane]-3,9,12,15,19,22,25,31,34,37,45-undecone, PP3117:

(1S,4S,10S,13S,17S,20S,26S,28R,32S,38S,42Z)-20-cyclopentyl-32-[2-[3,5-difluoro-4-(trifluoromethyl)phenyl]ethyl]-28-ethoxy-2,14,18,21,24,36-hexamethyl-10-[(1 S)-1-methylpropyl]-13-propyl-38-(p-tolylmethyl)-17-(pyrrolidine-1-carbonyl)spiro[2,8,11,14,18,21,24,30,33,36,39-undecazatetracyclo[37.5.1.0.sup.4,8.0.sup.26,30]pentatetracont-42-ene-23,1'-cyclobutane]-3,9,12,15,19,22,25,31,34,37,45-undecone, PP3118:

(2S,8S,12R,14S,20S,23S,27S,30S,36S,38Z)-20-cyclopentyl-8-[2-[3,5-difluoro-4-(trifluoromethyl)phenyl]ethyl]-12-ethoxy-4,16,19,22,26,32,35-heptamethyl-30-[(1 S)-1-methylpropyl]-23-(piperidine-1-carbonyl)-27-propyl-2-[[4-(trifluoromethyl)phenyl]methyl]spiro[1,4,7,10,16,19,22,26,29,32,35-undecazatricyclo[34.5.1.0.sup.10,14]dotetracont-38-ene-17,1'-cyclobutane]-3,6,9,15,18,21,25,28,31,34,42-undecone, PP3119:

(2S,8S,12R,14S,20S,23S,27S,30S,36S,38Z)-20-cyclopentyl-8-[2-[3,5-difluoro-4-(trifluoromethyl)phenyl]ethyl]-12-ethoxy-4,16,19,22,26,32,35-heptamethyl-30-[(1 S)-1-methylpropyl]-23-(piperidine-1-carbonyl)-27-propyl-2-(p-tolylmethyl)spiro[[1,4,7,10,16,19,22,26,29,32,35-undecazatricyclo[34.5.1.0.sup.10,14]dotetracont-38-ene-17,1-cyclobutane]-3,6,9,15,18,21,25,28,31,34,42-undecone, PP3120:

(2S,8S,12R,14S,20S,23S,27S,30S,36S,38Z)-20-cyclopentyl-8-[2-[3,5-difluoro-4-(trifluoromethyl)phenyl]ethyl]-12-ethoxy-4,16,19,22,26,32,35-heptamethyl-30-[(1 S)-1-methylpropyl]-27-propyl-23-(pyrrolidine-1-carbonyl)-2-[[4-(trifluoromethyl)phenyl]methyl]spiro[1,4,7,10,16,19,22,26,29,32,35-

undecazatricyclo[34.5.1.0.sup.10,14]dotetracont-38-ene-17,1'-cyclobutane]-3,6,9,15,18,21,25,28,31,34,42-undecone, PP3121: (2S,8S,12R,14S,20S,23S,27S,30S,36S,38Z)-20-cyclopentyl-8-[2-[3,5-difluoro-4-(trifluoromethyl)phenyl]ethyl]-12-ethoxy-4,16,19,22,26,32,35-heptamethyl-30-[(1 S)-1-methylpropyl]-27-propyl-2-(p-tolylmethyl)-23-(pyrrolidine-1-carbonyl)spiro[1,4,7,10,16,19,22,26,29,32,35-undecazatric clo[34.5.1.0.sup.10,14]dotetracont-38-ene-17,1-cyclobutane]-3,6,9,15,18,21,25,28,31,34,42-undecone, (1S,7S,11R,13S,19S,22S,26S,29S,35S,37Z,47Z)-19-cyclopentyl-7-[2-[3,5-difluoro-4-(trifluoromethyl)phenyl]ethyl]-11-ethoxy-N-ethyl-N,3,18,21,25,31,34-heptamethyl-29-[(1S)-1-methylpropyl]-2,5,8,14,17,20,24,27,30,33,55-undecaoxo-spiro[3,6,9,15,18,21,25,28,31,34,41-undecazapentacyclo[24.15.11.2.sup.43,46.1.sup.35,41.0.sup.9,13]pentapentaconta-37,43(54),44,46(53),47-pentaene-16,1'-cyclobutane]-22-carboxamide, (1S,7S,11R,13S,19S,22S,26S,29S,35S,37Z,47Z)-19-cyclopentyl-11-ethoxy-N-ethyl-7-[2-[3-methoxy-4-(trifluoromethyl)phenyl]ethyl]-N,3,3',3',18,21,25,31,34-nonamethyl-29-[(1 S)-1-methylpropyl]-2,5,8,14,17,20,24,27,30,33,55-undecaoxo-spiro[3,6,9,15,18,21,25,28,31,34,41-undecazapentacyclo[24.15.11.2.sup.43,46.1.sup.35,41.0.sup.9,13]pentapentaconta-37,43(54),44,46(53),47-pentaene-16,1'-cyclobutane]-22-carboxamide, (1 S,7S,11 R,13 S,19S,22S,26S,29S,35S,37Z,47Z)-19-cyclopentyl-7-[2-[3,5-difluoro-4-(trifluoromethyl)phenyl]ethyl]-11-ethoxy-3,3',3',18,21,25,31,34-octamethyl-29-[(1 S)-1-methylpropyl]-22-(pyrrolidine-1-carbonyl)spiro[3,6,9,15,18,21,25,28,31,34,41-undecazapentacyclo[24.15.10.2.sup.43,46.1.sup.35,41.0.sup.9,13]tetrapentaconta-37,43(53),44,46(52),47-pentaene-16,1'-cyclobutane]-2,5,8,14,17,20,24,27,30,33,54-undecone, (1 S,7S,11R,13S,19S,22S,26S,29S,35S,37Z,47Z)-7-[2-[3,5-difluoro-4-(trifluoromethyl)phenyl]ethyl]-11-ethoxy-19-(1-ethylpropyl)-3,3',3',18,21,25,31,34-octamethyl-29-[(1 S)-1-methylpropyl]-22-(pyrrolidine-1-carbonyl)spiro[3,6,9,15,18,21,25,28,31,34,41-undecazapentacyclo[24.15.10.2.sup.43,46.1.sup.35,41.0.sup.9,13]tetrapentaconta-37,43(53),44,46(52),47-pentaene-16,1'-cyclobutane]-2,5,8,14,17,20,24,27,30,33,54-undecone, (1 S,7S,11R,13 S,19S,22S,26S,29S,35 S,37Z,47Z)-19-cyclopentyl-31-cyclopropyl-7-[2-[3,5-difluoro-4-(trifluoromethyl)phenyl]ethyl]-11-ethoxy-3,3',3',18,21,25 ,34-heptamethyl-29-[(1 S)-1-methylpropyl]-22-(pyrrolidine-1-carbonyl)spiro[3,6,9,15,18,21,25,28,31,34,41-undecazapentacyclo[24.15.10.2.sup.43,46.1.sup.35,41.0.sup.9,13]tetrapentaconta-37,43(53),44,46(52),47-pentaene-16,1'-cyclobutane]-2,5,8,14,17,20,24,27,30,33,54-undecone, (1S,7S,11R,13S,19S,22S,26S,29S,35S,37Z,47Z)-31-cyclopropyl-7-[2-[3,5-difluoro-4-(trifluoromethyl)phenyl]ethyl]-11-ethoxy-19-(1-ethylpropyl)-3,18,21,25,34-pentamethyl-29-[(1 S)-1-methylpropyl]-22-(pyrrolidine-1-carbonyl)spiro[3,6,9,15,18,21,25,28,31,34,41-undecazapentacyclo[24.15.10.2.sup.43,46.1.sup.35,41.0.sup.9,13]tetrapentaconta-37,43(53),44,46(52),47-pentaene-16,1'-cyclobutane]-2,5,8,14,17,20,24,27,30,33,54-undecone, and (1S,7S,11R,13S,19S,22S,26S,29S,35S,37Z,47Z)-7-[2-(3,4-dichlorophenyl)ethyl]-11-ethoxy-19-(1-ethylpropyl)-3,18,21,25,31,34-hexamethyl-29-[(1 S)-1-methylpropyl]-22-(pyrrolidine-1-carbonyl)spiro[3,6,9,15,18,21,25,28,31,34,41-undecazapentacyclo[24.15.11.2.sup.43,46.1.sup.35,41.0.sup.9,13]pentapentaconta-37,43(54),44,46(53),47-pentaene-16,1'-cyclobutane]-2,5,8,14,17,20,24,27,30,33,55-undecone or a salt thereof, or a solvate thereof. These compounds are included in the formula represented by the formula (1).

(156) In an embodiment, the compound of the present invention is preferably a compound represented by formula (2):

(157) ##STR00010## wherein  custom character,  custom character n, m, P.sub.1, R.sub.2, R.sub.3, P.sub.3, P.sub.4, P.sub.6, R.sub.7, R.sub.8, P.sub.8, R.sub.9, P.sub.9, Q.sub.9, R.sub.10, P.sub.10, R.sub.11 and P.sub.11 are the same as defined in formula (1).

(158) For example, compounds PP1827, PP1830, PP2260, PP2574, PP2576, PP2583, PP2957,

PP3033, PP3034, PP3036, PP3037, PP3047, PP3093 and the following compounds can be included in the formula (2). (1 S,7S,11R,13S,19S,22S,26S,29S,35 S,37Z,47Z)-19-cyclopentyl-7-[2-[3,5-difluoro-4-(trifluoromethyl)phenyl]ethyl]-11-ethoxy-N-ethyl-N,3,18,21,25,31,34-heptamethyl-29-[(1 S)-1-methylpropyl]-2,5,8,14,17,20,24,27,30,33,55-undecaoxo-spiro[3,6,9,15,18,21,25,28,31,34,41-undecazapentacyclo[24.15.11.2.sup.43,46.1.sup.35,41.0.sup.9,13]pentapentaconta-37,43(54),44,46(53),47-pentaene-16,1'-cyclobutane]-22-carboxamide, (1 S,7S,11R,13S,19S,22S,26S,29S,35S,37Z,47Z)-19-cyclopentyl-11-ethoxy-N-ethyl-7-[2-[3-methoxy-4-(trifluoromethyl)phenyl]ethyl]-N,3,3',3',18,21,25,31,34-nonamethyl-29-[(1 S)-1-methylpropyl]-2,5,8,14,17,20,24,27,30,33,55-undecaoxo-spiro[3,6,9,15,18,21,25,28,31,34,41-undecazapentacyclo[24.15.11.2.sup.43,46.1.sup.35,41.0.sup.9,13]pentapentaconta-37,43(54),44,46(53),47-pentaene-16,1'-cyclobutane]-22-carboxamide, (1 S,7S,11R,13S,19S,22S,26S,29S,35S,37Z,47Z)-19-cyclopentyl-7-[2-[3,5-difluoro-4-(trifluoromethyl)phenyl]ethyl]-11-ethoxy-3,3',3',18,21,25,31,34-octamethyl-29-[(1 S)-1-methylpropyl]-22-(pyrrolidine-1-carbonyl)spiro[3,6,9,15,18,21,25,28,31,34,41-undecazapentacyclo[24.15.10.2.sup.43,46.1.sup.35,41.0.sup.9,13]tetrapentaconta-37,43(53),44,46(52),47-pentaene-16,1'-cyclobutane]-2,5,8,14,17,20,24,27,30,33,54-undecone, (1 S,7S,11R,13S,19S,22S,26S,29S,35S,37Z,47Z)-7-[2-[3,5-difluoro-4-(trifluoromethyl)phenyl]ethyl]-11-ethoxy-19-(1-ethylpropyl)-3,3',3',18,21,25,31,34-octamethyl-29-[(1 S)-1-methylpropyl]-22-(pyrrolidine-1-carbonyl)spiro[3,6,9,15,18,21,25,28,31,34,41-undecazapentacyclo[24.15.10.2.sup.43,46.1.sup.35,41.0.sup.9,13]tetrapentaconta-37,43(53),44,46(52),47-pentaene-16,1'-cyclobutane]-2,5,8,14,17,20,24,27,30,33,54-undecone, (1 S,7S,11R,13S,19S,22S,26S,29S,35S,37Z,47Z)-19-cyclopentyl-31-cyclopropyl-7-[2-[3,5-difluoro-4-(trifluoromethyl)phenyl]ethyl]-11-ethoxy-3,3',3',18,21,25,34-heptamethyl-29-[(1 S)-1-methylpropyl]-22-(pyrrolidine-1-carbonyl)spiro[3,6,9,15,18,21,25,28,31,34,41-undecazapentacyclo[24.15.10.2.sup.43,46.1.sup.35,41.0.sup.9,13]tetrapentaconta-37,43(53),44,46(52),47-pentaene-16,1'-cyclobutane]-2,5,8,14,17,20,24,27,30,33,54-undecone, (1S,7S,11R,13S,19S,22S,26S,29S,35S,37Z,47Z)-31-cyclopropyl-7-[2-[3,5-difluoro-4-(trifluoromethyl)phenyl]ethyl]-11-ethoxy-19-(1-ethylpropyl)-3,18,21,25,34-pentamethyl-29-[(1 S)-1-methylpropyl]-22-(pyrrolidine-1-carbonyl)spiro[3,6,9,15,18,21,25,28,31,34,41-undecazapentacyclo[24.15.10.2.sup.43,46.1.sup.35,41.0.sup.9,13]tetrapentaconta-37,43(53),44,46(52),47-pentaene-16,1'-cyclobutane]-2,5,8,14,17,20,24,27,30,33,54-undecone, and (1 S,7S,11R,13S,19S,22S,26S,29S,35S,37Z,47Z)-7-[2-(3,4-dichlorophenyl)ethyl]-11-ethoxy-19-(1-ethylpropyl)-3,18,21,25,31,34-hexamethyl-29-[(1 S)-1-methylpropyl]-22-(pyrrolidine-1-carbonyl)spiro[3,6,9,15,18,21,25,28,31,34,41-undecazapentacyclo [24.15.11.243.46.135.41.09.13] pentapentaconta-37,43 (54),44,46 (53),47-pentaene-16,1'-cyclobutane]-2,5,8,14,17,20,24,27,30,33,55-undecone.

(159) In an embodiment, the compound of the present invention is preferably a compound represented by formula (3):

(160) ##STR00011## In formula (3), R.sub.1 is a C1 to C7 alkyl; R.sub.5 is a benzyl optionally substituted with one or more groups selected from the group consisting of a C.sub.1 to C.sub.6 alkyl, a C.sub.1 to C.sub.6 haloalkyl and a C.sub.3 to C.sub.8 cycloalkyl;  custom character,  custom character,

n, m, P.sub.1, R.sub.2, R.sub.3, P.sub.3, P.sub.4, P.sub.6, R.sub.7, R.sub.8, P.sub.9, R.sub.9, P.sub.9, Q.sub.9, R.sub.10, P.sub.10, R.sub.11 and P.sub.11 are the same as defined in formula (1).

(161) For example, compounds PP1574, PP1650, PP2093, PP2316, PP2320, PP2328, PP2687, PP2691, PP3095, PP3096, PP3097, PP3098, PP3099, PP3100, PP3101, PP3102, PP3103, PP3104, PP3105, PP3106, PP3110, PP3111, PP3112, PP3113, PP3114, PP3115, PP3116, PP3117, PP3118, PP3119, PP3120, and PP3121 can be included in formula (3).

(162) In an embodiment, the cyclic compound of the present invention has high selectivity for

KRAS. In an embodiment, the cyclic compound of the present invention selectively inhibits KRAS. Without wishing to be bound by a particular theory, the divalent group that R.sub.4 and P.sub.5 together form interacts with His95 of KRAS, and thereby high selectivity for KRAS can be achieved. While it is known that there are three isotypes of RAS protein, i.e., HRAS, KRAS, and NRAS, His95 exists only in KRAS. Accordingly, a compound that specifically interacts with His95 of KRAS can inhibit KRAS with high selectivity over NRAS and/or HRAS.

(163) In the present invention, the origin of NRAS, HRAS, and KRAS is not particularly limited, and may encompass those derived from a variety of animals such as human, mouse, rat, rabbit, dog, cat, cattle, horse, pig, goat, rhesus monkey, cynomolgus monkey, chimpanzee, and chicken, but human-derived HRAS, KRAS, and NRAS are preferable. The amino acid sequence of human-derived NRAS is shown in SEQ ID NO: 1, the amino acid sequence of human-derived HRAS is shown in SEQ ID NO: 2, and the amino acid sequence of human-derived KRAS is shown in SEQ ID NO: 3.

(164) In an embodiment, the cyclic compound of the present invention has KRAS inhibitory activity that is 3 times or more higher than NRAS inhibitory activity and/or HRAS inhibitory activity. In an embodiment, the cyclic compound of the present invention has KRAS binding activity that is 3 times or more higher than NRAS binding activity and/or HRAS binding activity.

(165) In an embodiment, the cyclic compound of the present invention has KRAS inhibitory activity that is 5 times, 7 times, 10 times, 15 times, or 20 times or more higher than NRAS inhibitory activity and/or HRAS inhibitory activity. In an embodiment, the cyclic compound of the present invention has KRAS binding activity that is 5 times, 7 times, 10 times, 15 times, or 20 times or more higher than NRAS binding activity and/or HRAS binding activity.

(166) In the present invention, the KRAS inhibitory activity relative to the NRAS inhibitory activity and/or the HRAS inhibitory activity can be determined from the ratio of the KRAS inhibitory activity of the cyclic compound of the present invention to the NRAS inhibitory activity and/or the HRAS inhibitory activity of the cyclic compound of the present invention. For example, when this ratio is defined as [IC₅₀ value of cyclic compound of present invention with respect to NRAS and/or HRAS] divided by [IC₅₀ value of cyclic compound of present invention with respect to KRAS], a larger value of this ratio means that the KRAS inhibitory activity is higher than the NRAS inhibitory activity and/or the HRAS inhibitory activity, or in other words, the KRAS selective inhibitory activity of the cyclic compound of the present invention is high. On the other hand, a smaller value of this ratio means that the KRAS inhibitory activity is lower than the NRAS inhibitory activity and/or the HRAS inhibitory activity, or in other words, the KRAS selective inhibitory activity of the cyclic compound of the present invention is low.

(167) In the present invention, the KRAS binding activity relative to the NRAS binding activity and/or the HRAS binding activity can be determined from the ratio of the KRAS binding activity of the cyclic compound of the present invention to the NRAS binding activity and/or the HRAS binding activity of the cyclic compound of the present invention. For example, when this ratio is defined as [KD value with respect to NRAS and HRAS] divided by [KD with respect to KRAS], a larger value means that the KRAS binding activity is higher than the NRAS binding activity and/or the HRAS binding activity, or in other words, the binding selectivity for KRAS over NRAS and/or HRAS is high. On the other hand, a smaller value means that the KRAS binding activity is lower than the NRAS binding activity and/or the HRAS binding activity, or in other words, the binding selectivity for KRAS over NRAS and/or HRAS is low.

(168) In the present invention, the “interaction” means non-covalent interaction as exemplified by electrostatic interaction (including ionic bonding, hydrogen bonding, and dipole interaction), van der Waals interaction (including hydrophobic interaction), and the like. For example, it means CH- π interaction, NH- π interaction, S- π interaction, cation- π interaction, or halogen- π interaction. The interaction in the present invention may or may not be mediated by another molecule such as a water molecule, but is preferably not mediated by another molecule such as a water molecule.

(169) In the present invention, whether the 95th amino acid residue histidine (denoted as His95 or H95) in the human KRAS wild-type protein interacts with the cyclic compound can be determined by the interatomic distance of their non-hydrogen atoms (in the case of bonding via another molecule such as a water molecule, the interatomic distance between their non-hydrogen atoms taking no account of such another molecule). When the interatomic distance is 5.1 angstroms (Å) or less, it can be determined that their non-hydrogen atoms interact with each other. In some embodiments, the interatomic distance between two interacting non-hydrogen atoms may be, for example, 5.1 Å or less, 4.8 Å or less, 4.5 Å or less, 4.3 Å or less, 4.2 Å or less, 4.1 Å or less, 4.0 Å or less, 3.9 Å or less, or 3.7 Å or less. Also, the interatomic distance may be 2.0 Å or more, 2.1 Å or more, or 2.5 Å or more.

(170) In the present invention, the interatomic distance can be measured, for example, through an analysis of the three-dimensional structure of a complex of the human KRAS wild-type protein and the cyclic compound of the present invention. Specifically, a crystal of the complex of the human KRAS wild-type protein and the cyclic compound of the present invention is prepared. The crystal is subjected to X-ray diffractometry to obtain X-ray diffraction intensity data of space groups, unit cells, and the like. The obtained X-ray diffraction intensity data is applied to a program for initial structure or refined structure determination well known to those skilled in the art, such as Coot (Emsley, P. et al., 2010), Phenix (Adams, P. D. et al., 2010), Phaser (J. Appl. Cryst. 40:658-674 (2007)), Refmac5 (Acta Cryst. D67:355-467 (2011)), and ARP/wARP (Cohen, S. X. et al., 2008), and thereby the three-dimensional structure of the complex of the human KRAS wild-type protein and the cyclic compound of the present invention can be determined.

(171) Once the three-dimensional structure of the complex of the human KRAS wild-type protein and the cyclic compound of the present invention is determined, the interatomic distance can be measured by a method well known to those skilled in the art. For example, the interatomic distance can be measured by allowing a software program for use in molecular modeling or molecular simulation, such as Discovery Studio 2020 Client, MOE (Molecular Operating Environment), or Maestro, to read the structural information of the complex of the cyclic compound of the present invention and the human KRAS wild-type protein, and using a function incorporated in the software program (such as the Distance Monitor function in the case of Discovery Studio 2020 Client). The details of conditions and criteria used by the software to determine the presence or absence of interactions can be viewed in a manual, specification, or the like appended to the software (e.g., in the case of Discovery Studio 2020 Client, the details of conditions and criteria for determining the presence or absence of interactions can be viewed by opening the web page of the specification from the Help button, selecting "Receptor-Ligand Interactions tools", then selecting "Theory-Receptor-Ligand Interactions", and further selecting "Non-bond Interactions").

(172) The crystal of the complex of the human KRAS wild-type protein and the cyclic compound of the present invention can also be obtained by a method well known to those skilled in the art. For example, a solution containing the cyclic compound of the present invention is mixed with a solution containing the human KRAS wild-type protein to obtain the complex of the human KRAS wild-type protein and the cyclic compound of the present invention. By subjecting the resulting complex to a crystallization method well known to those skilled in the art such as a vapor diffusion method, a batch method (a bulk batch method or a microbatch method), a dialysis method, or a counter-diffusion method, the crystal of the complex of the human KRAS wild-type protein and the cyclic compound of the present invention can be prepared. Known vapor diffusion methods are a sitting drop method, a hanging drop method, and a sandwich drop method.

(173) The human KRAS wild-type protein can also be obtained by a method known to those skilled in the art. For example, the human KRAS wild-type protein can be prepared by a recombinant polypeptide expressing method in which cells are used, but the method is not limited thereto. In one embodiment, a nucleic acid that encodes the human KRAS wild-type protein of the present invention is inserted into a suitable expression vector, the vector is introduced into a suitable cell,

the transformed cell is cultured, and the expressed protein is isolated and purified. Such a protein can also be expressed as a fusion protein with another protein to facilitate purification. For example, it is possible to use a method of preparing a fusion protein with a maltose binding protein using *Escherichia coli* as a host (vector pMAL series sold by New England BioLabs, USA), a method of preparing a fusion protein with glutathione-S-transferase (GST)(vector pGEX series sold by Amersham Pharmacia Biotech), a method of preparing a protein to which a histidine tag is added (pET series of Novagen), and a method of preparing a protein to which an HAT tag is added. The host cell is not particularly limited as long as it is a cell suitable for expressing a recombinant protein, and, in addition to *E. coli* mentioned above, for example, yeast, various animal and plant cells, insect cells, and the like can be used. Various methods known to those skilled in the art can be used to introduce a vector into a host cell. For example, for introduction into *Escherichia coli*, introduction methods involving calcium ions (Mandel, M., Higa, A. (1970) Journal of Molecular Biology, 53, 158-162, and Hanahan, D. (1983) Journal of Molecular Biology, 166, 557-580) can be used. The protein expressed in the host cell can be purified and recovered from the host cell or its cell culture or culture supernatant by a method known to those skilled in the art. When a protein is expressed as a fusion protein with the above maltose binding protein, HAT tag, or the like, affinity purification and gel filtration chromatography (size exclusion chromatography, SEC) purification can be easily performed. In affinity chromatography purification and SEC purification, AKTAexpress™ apparatus (GE Healthcare), NGC™ Chromatography System (Bio-Rad), BioLogic DuoFlow™ Chromatography System (Bio-Rad), or the like can be used.

(174) Interatomic energy can also be measured by a method well known to those skilled in the art. For example, interatomic energy can be easily calculated by allowing a molecular simulation program well known to those skilled in the art, such as Discovery Studio 2020 Client, MOE (Molecular Operating Environment), or Maestro, to read the three-dimensional structure of a substance to be measured, and, according to the instructions of the program, selecting a force field (such as Amber or CHARMM) to be used in calculation and atoms for which energy calculation is performed. For example, in the case of Discovery Studio 2020 Client, interatomic energy can be calculated using the Calculate Interaction Energy function.

(175) In one non-limiting embodiment, in a complex of the cyclic compound of the present invention and the human KRAS wild-type protein, the cyclic compound of the present invention interacts with His95 in the human KRAS wild-type protein.

(176) Without wishing to be bound by a specific theory, it is considered that the formation of a complex by the cyclic compound of the present invention and the human KRAS wild-type protein in this manner is associated with contribution to the high binding activity of the cyclic compound of the present invention to the human KRAS wild-type protein and, moreover, with the binding selectivity over HRAS and NRAS.

(177) The present invention also relates to a non-natural amino acid for use in the production of the cyclic compound of the present invention. In an embodiment, the non-natural amino acid of the present invention is an N-protected non-natural amino acid for use in the production of the peptide compound using a solid-phase synthesis method, and in another embodiment, the non-natural amino acid of the present invention is a non-natural amino acid having a free amino group obtained by removing the protecting group from the N-protected non-natural amino acid. Examples of the protecting group of the N-protected non-natural amino acid include an Fmoc group, a Boc group, a Cbz group, an Alloc group, a nosyl group, a dinitronosyl group, a t-Bu group, a trityl group, and a cumyl group. Of these, an Fmoc group, a Boc group, a Cbz group, and an Alloc group are preferable, and an Fmoc group is more preferable.

(178) In an embodiment, examples of the N-protected non-natural amino acid having an Fmoc group as a protecting group in the present invention include the following amino acids listed in Table 4 or salts thereof, or solvates thereof. aa004: (2S)-2-[9H-fluoren-9-ylmethoxycarbonyl(methyl)amino]octanoic acid, aa013: (2R)-2-[9H-fluoren-9-

ylmethoxycarbonyl(methyl)amino]-3-methylsulfanyl-propanoic acid, aa019: (2S)-2-[9H-fluoren-9-ylmethoxycarbonyl(methyl)amino]-4,4-difluoro-butanoic acid, aa023: (2S)-2-[9H-fluoren-9-ylmethoxycarbonyl(methyl)amino]-3-(3-thienyl) propanoic acid, aa028: (2S)-3-(3,4-dichlorophenyl)-2-[9H-fluoren-9-ylmethoxycarbonyl(methyl)amino]propanoic acid, aa043: (2S)-5,5-dichloro-2-[9H-fluoren-9-ylmethoxycarbonyl(methyl)amino] pentanoic acid, aa056: (2R)-3-[9H-fluoren-9-ylmethoxycarbonyl(methyl)amino]-2-methyl-propanoic acid, aa098: (2S,3S)-1-(9H-fluoren-9-ylmethoxycarbonyl)-3-methyl-azetidine-2-carboxylic acid, aa099: (2S,3R)-1-(9H-fluoren-9-ylmethoxycarbonyl)-3-methyl-azetidine-2-carboxylic acid, aa 100: (2S)-1-(9H-fluoren-9-ylmethoxycarbonyl)-3,3-dimethyl-azetidine-2-carboxylic acid, aa111: (2R)-3-allyloxy-2-[9H-fluoren-9-ylmethoxycarbonyl(methyl)amino]propanoic acid, aa136: (2S)-2-[9H-fluoren-9-ylmethoxycarbonyl(propyl)amino]-3-[4-(trifluoromethyl)phenyl] propanoic acid, aa174: (2S)-2-[9H-fluoren-9-ylmethoxycarbonyl(propyl)amino]-3-(p-tolyl) propanoic acid, aa210: (2S)-2-cyclopentyl-2-[ethyl(9H-fluoren-9-ylmethoxycarbonyl)amino]acetic acid, aa220: (2S)-4-(4-chloro-3,5-difluoro-phenyl)-2-(9H-fluoren-9-ylmethoxycarbonylamino) butanoic acid, aa229: (2S)-4-(benzothiophen-5-yl)-2-(9H-fluoren-9-ylmethoxycarbonylamino) butanoic acid, aa233: (2S)-2-(9H-fluoren-9-ylmethoxycarbonylamino)-4-[3-methyl-4-(trifluoromethyl)phenyl]butanoic acid, aa235: (2S)-2-(9H-fluoren-9-ylmethoxycarbonylamino)-4-[3-methoxy-4-(trifluoromethyl)phenyl]butanoic acid, aa239: (2S,4S)-1-(9H-fluoren-9-ylmethoxycarbonyl)-4-methyl-pyrrolidine-2-carboxylic acid, aa244: (2S)-2-[9H-fluoren-9-ylmethoxycarbonyl(methyl)amino]-5,5-difluoro-pentanoic acid, aa246: (2S,4R)-1-(9H-fluoren-9-ylmethoxycarbonyl)-4-propoxy-pyrrolidine-2-carboxylic acid, aa250: (2R)-2-[9H-fluoren-9-ylmethoxycarbonyl(methyl)amino]-3-propoxy-propanoic acid, aa264: (2S,3R)-1-(9H-fluoren-9-ylmethoxycarbonyl)-3-tetrahydropyran-2-yloxy-pyrrolidine-2-carboxylic acid, aa265: (2S,3S)-1-(9H-fluoren-9-ylmethoxycarbonyl)-3-tetrahydropyran-2-yloxy-pyrrolidine-2-carboxylic acid, aa268: (1S,2S,5R)-3-(9H-fluoren-9-ylmethoxycarbonyl)-3-azabicyclo[3.1.0] hexane-2-carboxylic acid, aa279: (2S,4R)-1-(9H-fluoren-9-ylmethoxycarbonyl)-4-tetrahydropyran-2-yloxy-pyrrolidine-2-carboxylic acid, aa281: (2S,4R)-1-(9H-fluoren-9-ylmethoxycarbonyl)-4-methoxy-pyrrolidine-2-carboxylic acid, aa331: (2S)-3-ethyl-2-[9H-fluoren-9-ylmethoxycarbonyl(methyl)amino] pentanoic acid, aa389: (2S)-2-[9H-fluoren-9-ylmethoxycarbonyl(propyl)amino]-4-methyl-pentanoic acid, aa391: (2S,3R)-2-(9H-fluoren-9-ylmethoxycarbonylamino)-3-propoxy-butanoic acid, aa397: (2S)-2-[9H-fluoren-9-ylmethoxycarbonyl(propyl)amino]-3-isopentyloxy-propanoic acid, aa398: (2S)-3-(cyclobutoxy)-2-[9H-fluoren-9-ylmethoxycarbonyl(propyl) amino]propanoic acid, aa399: (2S)-4-(7-chloro-1-methyl-indol-5-yl)-2-(9H-fluoren-9-ylmethoxycarbonylamino) butanoic acid, aa400: (2S)-2-(9H-fluoren-9-ylmethoxycarbonylamino)-4-(2-fluoro-3-methyl-benzothiophen-5-yl) butanoic acid, aa401: (2S)-4-(7-chlorobenzothiophen-5-yl)-2-(9H-fluoren-9-ylmethoxycarbonylamino) butanoic acid, aa402: (2S)-2-(9H-fluoren-9-ylmethoxycarbonylamino)-4-(1-methyl indol-6-yl) butanoic acid, aa403: (2S)-4-(1,3-dimethyl indol-6-yl)-2-(9H-fluoren-9-ylmethoxycarbonylamino) butanoic acid, aa404: (2S)-2-(9H-fluoren-9-ylmethoxycarbonylamino)-4-(1,2,3-trimethyl indol-6-yl) butanoic acid, aa405: (2S)-4-(2,3-dimethyl benzothiophen-5-yl)-2-(9H-fluoren-9-ylmethoxycarbonylamino) butanoic acid, aa406: (2S)-4-(3-chloro-4-ethyl-phenyl)-2-(9H-fluoren-9-ylmethoxycarbonylamino) butanoic acid, aa407: (2S)-2-(9H-fluoren-9-ylmethoxycarbonylamino)-4-[4-(methoxymethyl)-3,5-dimethyl-phenyl]butanoic acid, aa408: (2S)-4-(4-chloro-3-methoxy-phenyl)-2-(9H-fluoren-9-ylmethoxycarbonylamino) butanoic acid, aa409: (2S)-4-[4-chloro-3-(trifluoromethyl)phenyl]-2-(9H-fluoren-9-ylmethoxycarbonylamino) butanoic acid, aa410: (2S)-4-[4-(difluoromethyl)-3,5-difluoro-phenyl]-2-(9H-fluoren-9-ylmethoxycarbonylamino) butanoic acid, aa411: (2S)-4-(4-chloro-3,5-dimethyl-phenyl)-2-(9H-fluoren-9-ylmethoxycarbonylamino) butanoic acid, aa414: (2S,4R)-4-(cyclopentoxo)-1-(9H-fluoren-9-ylmethoxycarbonyl) pyrrolidin-2-carboxylic acid, aa415: (2S,4R)-4-(cyclobutoxy)-1-(9H-fluoren-9-ylmethoxycarbonyl) pyrrolidin-2-carboxylic acid, aa423: (3S)-3-cyclobutyl-3-[9H-fluoren-9-ylmethoxycarbonyl(methyl)amino]propanoic acid,

aa424: (3R)-3-[9H-fluoren-9-ylmethoxycarbonyl(methyl)amino] pentanoic acid, aa425: (3S)-3-[9H-fluoren-9-ylmethoxycarbonyl(methyl)amino]-4-methyl-pentanoic acid, aa426: (3S)-3-cyclohexyl-3-[9H-fluoren-9-ylmethoxycarbonyl(methyl)amino]propanoic acid, and aa443: (2S)-2-[but-3-enyl(9H-fluoren-9-ylmethoxycarbonyl)amino]-3-(p-tolyl) propanoic acid.

(General Production Method)

(179) General production methods for the cyclic compound of the present invention, and the oligopeptide compound and the non-natural amino acid for use in the production of these compounds will be described below. Herein, the cyclic compound may be referred to as “cyclic peptide compound”. Herein, the “cyclic moiety” of a peptide compound means a cyclic portion formed by connection of two or more amino acid residues.

(180) Chemical Synthesis Methods for Peptide Compounds

(181) Examples of chemical synthesis methods for the peptide compounds or the cyclic compounds herein include a liquid-phase synthesis method, a solid-phase synthesis method using Fmoc synthesis, Boc synthesis, or the like, and a combination thereof. In Fmoc synthesis, a basic unit is an amino acid in which a main-chain amino group is protected with an Fmoc group, and a side-chain functional group is protected as necessary with a protecting group that is not cleaved by piperidine or such bases, such as a t-Bu group, a THP group, or a Trt group, and a main-chain carboxylic acid is not protected. The basic unit is not particularly limited as long as it is a combination having an Fmoc-protected amino group and a carboxyl group. For example, dipeptide or tripeptide may be a basic unit, and a cyclic structure may be formed between substituents on nitrogen atoms and/or side chains contained in the dipeptide or tripeptide. The basic unit disposed at the N terminus may be a unit other than the Fmoc amino acid. For example, it may be a Boc amino acid or a Tfa group or Ns group, or a carboxylic acid analog which does not have an amino group. The main-chain carboxyl group, or a side-chain carboxyl group of an amino acid that has a carboxyl group in a side chain and in which the main-chain carboxyl group is protected with a suitable protecting group, is supported on a solid phase by a chemical reaction with the functional group of a solid-phase carrier. Subsequently, the Fmoc group is deprotected by a base such as piperidine or DBU, and a newly produced amino group and a subsequently added, basic-unit protected amino acid having a carboxyl group are subjected to a condensation reaction to produce a peptide bond. In the condensation reaction, various combinations such as a combination of DIC and HOBt, a combination of DIC and HOAt, and a combination of HATU and DIPEA are possible as activating agents for the carboxyl group. The desired peptide sequence can be produced by repeating the Fmoc group deprotection and the subsequent peptide bond forming reaction. After the desired sequence is obtained, cleavage from the solid phase and deprotection of the optionally introduced protecting group of the side-chain functional group are conducted. Further, conformational conversion and cyclization of the peptide can be performed before cleaving from the solid phase. Cleaving from the solid phase and deprotection may be performed under the same conditions, e.g., in 90:10 TFA/H₂O, or deprotection may be performed under different conditions as necessary. Cleaving from the solid phase may be achieved using a weak acid such as 1% TFA in some cases, and a protecting group that can be deprotected with a Pd-containing catalyst or the like may be used to utilize the orthogonality of both chemical reactions. During or at the end of these steps, a step such as cyclization can also be performed. For example, a side-chain carboxylic acid and an N-terminal main-chain amino group can be condensed, and a side-chain amino group and a C-terminal main-chain carboxylic acid can be condensed. In addition, an olefin can be introduced into two or more sites of the side chains and/or the substituents of nitrogen atoms, and cyclized by metathesis reaction. Moreover, a double bond produced by cyclization can be reduced to a single bond. Also, a double bond produced by cyclization can be converted to a cyclopropane ring under conditions involving diiodomethane-diethylzinc or the like. These steps of cyclization, reduction, conversion to a cyclopropane ring, and the like may be carried out during the course of synthesizing a basic unit such as dipeptide or tripeptide. In the meantime, reaction orthogonality is

required between the carboxylic acid on the C-terminal side and the side-chain carboxylic acid to be cyclized, between the main-chain amino group or hydroxy group on the N-terminal side and the side-chain amino group to be cyclized, or between the olefins of side chains and/or substituents of nitrogen atoms and the olefins to be cyclized. As described above, the protecting group is selected in consideration of the orthogonality of the protecting group. In addition, by placing a chloroacetyl group at the N-terminus, cyclization can also be performed between the thiol groups of side chains of cysteine residues. The reaction product thus obtained can be purified by a reverse-phase column, a molecular sieve column, or the like. Details of these procedures are described in, for example, the Solid-Phase Synthesis Handbook published by Merck on May 1, 2002. Commercially available resins for solid phase synthesis are usable, and examples include CTC resin, Wang resin, and SASRIN resin.

(182) A general method for synthesizing an amino acid-supported resin for use in peptide synthesis by a peptide synthesizer will be described below.

(183) An Fmoc amino acid can be supported on a resin by the method described in WO2013/100132 or WO2018/225864. Specifically, for example, 2-chlorotrityl chloride resin and a solvent (e.g., dehydrated dichloromethane) are introduced into a filter-equipped reaction vessel to swell the resin. Next, the solvent and the resin are separated, and then a mixture of the resin, a C-terminal free Fmoc amino acid dissolved in a solvent (e.g., dehydrated dichloromethane), a solvent (e.g., dehydrated methanol), and a base (e.g., diisopropylethylamine) is added to the reaction vessel and mixed to support the Fmoc amino acid on the resin. After the resin and the reaction solution are separated, the resin is mixed with a mixture of one or more solvents and a base (e.g., a mixture of dehydrated dichloromethane, dehydrated methanol, and diisopropylethylamine) to wash the resin. After the resin is washed with a solvent (e.g., dichloromethane) multiple times as necessary, the resin and the reaction solution are separated. By drying the resulting resin under reduced pressure overnight, an Fmoc amino acid-supported resin can be obtained.

(184) ##STR00012##

(185) (wherein n represents an integer of 1 to 11; P.sub.1 to P.sub.11, Q.sub.1 to Q.sub.11, and R.sub.1 to R.sub.11 mean P.sub.1 to P.sub.11, Q.sub.1 to Q.sub.11, and R.sub.1 to R.sub.11 as defined herein, respectively; Li and Ln mean Li and Lu as described herein, respectively; L.sub.2 to L.sub.10 are single bonds; and: (circle) means a resin portion.) The above structure shows that in the Fmoc-amino acid, the 2-chlorotrityl group on the resin is bonded to the carboxylic acid of the Fmoc amino acid via an ester bond.

(186) In the production of the compound described herein, when the defined group undergoes undesired chemical conversion under the conditions of the performed method, the compound can be produced by means of, for example, protection and deprotection of a functional group. Selection and introduction/removal procedures of a protecting group can be performed according to, for example, the methods described in Greene's "Protective Groups in Organic Synthesis" (5th Ed., John Wiley & Sons, 2014), which may be suitably used depending on the reaction conditions. Further, the order of reaction steps such as introduction of a substituent can be changed as necessary. For example, the protecting group for an amino group is an Fmoc, Boc, Cbz, or Alloc group. These carbamate groups can be introduced by reacting an amino group with a carbamating agent in the presence of a basic catalyst. Examples of the carbamating agent include Boc₂O, BocOPh, FmocOSu, FmocCl, CbzCl, and AllocCl. Examples of the basic catalyst include lithium carbonate, sodium carbonate, sodium hydrogen carbonate, potassium carbonate, potassium hydrogen carbonate, cesium carbonate, cesium hydrogen carbonate, lithium hydroxide, sodium hydroxide, potassium hydroxide, cesium hydroxide, sodium phosphate, potassium phosphate, N-methylmorpholine, triethylamine, diisopropylethylamine, and N,N-dimethylaminopyridine. A carbamate group which is a protecting group for an amino group can be removed under basic conditions, acidic conditions, hydrogenolysis reaction conditions, or the like.

(187) (Synthesis Methods for Cyclic Compounds by Cyclization of Peptide Compounds)

(188) A method for transforming a linear peptide compound into a cyclic peptide compound can be performed by carrying out a bond forming reaction within the molecule according to, for example, the method described in Comprehensive Organic Transformations, A Guide to Functional Group Preparations, 3rd Edition by R. C. Larock, or March's Advanced Organic Chemistry: Reactions, Mechanisms, and Structure, 7th Edition by M. B. March. After the bond forming reaction, further, a functional group transforming reaction can also be performed. Examples of the bond forming reaction include a C(O)—N bond formed from carboxylic acid and amine; a C—O—C bond, a C(O)—O bond, and a C(S)—O bond using an oxygen atom; a C(O)—S bond, a C(S)—S bond, a C—S—S—C bond, a C—S—C bond, a C—S(O)—C bond, and a C—S(O.sub.2)—C bond using a sulfur atom; and a C—N—C bond, a C=N—C bond, an N—C(O)—N bond, an N—C(S) N bond, and a C(S)—N bond using a nitrogen atom. Furthermore, examples include C—C bond forming reactions catalyzed by a transition metal, such as Suzuki reaction, Heck reaction, Sonogashira reaction, and metathesis reaction. Examples of the functional group transforming reaction further performed after the bond forming reaction include an oxidation reaction and a reduction reaction. A specific example is a reaction for oxidizing a sulfur atom to transform it into a sulfoxide group or a sulfone group. Another example is a reduction reaction for reducing a triple bond or a double bond of carbon-carbon bonds to a double bond or a single bond. While a closed ring structure is formed by a peptide bond when two amino acids are bonded with the amino acid main chain, a covalent bond between two amino acids may be formed by bonding between side chains of two amino acids, bonding between a side chain and a main chain, or the like. A black circle or a black square below indicates an amino acid residue, and connected black circles or black squares represent a peptide chain connected by an amide bond. The number of amino acid residues constituting a peptide chain are not particularly limited, and the number of amino acid residues is not limited to the number of black circles or black squares exemplified below.

(189) (General Preparation Method 1 for Cyclic Compounds)

(190) ##STR00013##

(191) Cyclic moieties of cyclic compounds having linear moieties can be cyclized by activating the N-terminal amino group and the C-terminal side chain carboxyl group (e.g., L=—CH.sub.2—in the case of aspartic acid or its derivative, and L=—CH.sub.2CH.sub.2—in the case of glutamic acid or its derivative) with an activating reagent or converting them to active esters, and then condensing them in the molecule to form a C(O)—N bond.

(192) (General Preparation Method 2 for Cyclic Compounds)

(193) ##STR00014##

(194) Cyclic compounds described in “General preparation method 1 for cyclic compounds” in which the linear moiety is C-Term can be cyclized by activating the N-terminal amino group and the C-terminal side chain carboxyl group (e.g., L=—CH.sub.2—in the case of aspartic acid or its derivative, and L=—CH.sub.2CH.sub.2—in the case of glutamic acid or its derivative) with an activating reagent or converting them to active esters, and then condensing them in the molecule to form a C(O)—N bond.

(195) (General Preparation Method 3 for Cyclic Compounds)

(196) (Method of Cyclizing with Haloalkyl and SH Groups)

(197) ##STR00015##

(198) Cyclic moieties of cyclic compounds having linear moieties can be cyclized by reacting the haloalkyl group of an amino acid residue with the thiol group of an amino acid residue in the molecule to form a C—S—C bond. Cyclic compounds described in “General preparation method 1 for cyclic compounds” in which the linear moiety is C-Term can also be similarly cyclized by reacting the haloalkyl group of an amino acid residue with the thiol group of an amino acid residue in the molecule to form a C—S—C bond. Further, a C—S(O)—C or C—S(O.sub.2)—C bond can also be formed by oxidizing and converting a sulfur atom to a sulfoxide or sulfone.

(199) (Method of cyclizing with vinyl and SH groups)

(200) ##STR00016##

(201) Cyclic moieties of cyclic compounds having linear moieties can be cyclized by reacting the vinyl group of an amino acid residue with the thiol group of an amino acid residue in the molecule to form a C—S—C bond. Cyclic compounds described in “General preparation method 1 for cyclic compounds” in which the linear moiety is C-Term can also be similarly cyclized by reacting the vinyl group of an amino acid residue with the thiol group of an amino acid residue in the molecule to form a C—S—C bond. Further, a C—S(O)—C or C—S(O₂)—C bond can also be formed by oxidizing and converting a sulfur atom to a sulfoxide or sulfone.

(202) (Method of Cyclizing with Ethynyl and SH Groups)

(203) ##STR00017##

(204) Cyclic moieties of cyclic compounds having linear moieties can be cyclized by reacting the ethynyl group of an amino acid residue with the thiol group of an amino acid residue in the molecule to form a C—S—C bond. Cyclic compounds described in “General preparation method 1 for cyclic compounds” in which the linear moiety is C-Term can also be similarly cyclized by reacting the ethynyl group of an amino acid residue with the thiol group of an amino acid residue in the molecule to form a C—S—C bond. Further, a C—S(O)—C or C—S(O_{sub.2})—C bond can also be formed by oxidizing and converting a sulfur atom to a sulfoxide or sulfone. The double bond site can also be reduced and converted to a single bond.

(205) (Method of Cyclizing with Vinyl and Vinyl Groups)

(206) ##STR00018##

(207) Cyclic moieties of cyclic compounds having linear moieties can be cyclized by reacting different vinyl groups of amino acid residues with each other in the molecule to form a C—C bond. Cyclic compounds described in “General preparation method 1 for cyclic compounds” in which the linear moiety is C-Term can also be similarly cyclized by reacting different vinyl groups of amino acid residues with each other in the molecule to form a C—C bond.

(208) (Method of Cyclizing by Forming a Triazole Ring with Azido and Ethynyl Groups)

(209) ##STR00019##

(210) Cyclic moieties of cyclic compounds having linear moieties can be cyclized by reacting the azido group of an amino acid residue with the ethynyl group of an amino acid residue in the molecule to form a triazole ring. Cyclic compounds described in “General preparation method 1 for cyclic compounds” in which the linear moiety is C-Term can also be similarly cyclized by reacting the azido group of an amino acid residue with the ethynyl group of an amino acid residue in the molecule to form a triazole ring.

(211) General preparation methods for peptide compounds by peptide modification are shown below. In the following schemes, P_{sub.n} represents a substituent for a nitrogen atom, R_{sub.n} and Q_{sub.n} each represent an amino acid side chain, a black circle represents an amino acid residue, linked black circles represent a peptide chain linked by amide bonds, and m represents the number of amino acid residues and may be any integer of 1 or more.

(212) (Method of Preparing Peptides Containing N-Alkylamino Acids)

(213) Peptides containing N-alkylamino acids can be synthesized according to the general peptide synthesis method described in the present Examples using an Fmoc-protected N-alkylamino acid as a raw material, or alternatively can be prepared by alkylating the N-terminal nitrogen on a resin as illustrated below. Specifically, the target peptides having an N-alkylamino acid at the N-terminus can be prepared by reacting the nitrogen of the N-terminal Tfa amide (trifluoroacetamide) of a resin-loaded peptide with an alkyl halide under basic conditions, and then treating the peptide with a reducing agent by referring to Organic Letters, 2008, 10, 4815-4818 or the like. Further, cyclic compounds can be prepared by elongating, cleaving from the resin, cyclizing, deprotecting, and purifying the peptide according to the general peptide synthesis method described in the present Examples.

(214) ##STR00020##

(215) The method described in Nature Protocols, 2012, 7, 432-444 which is shown below can also be used as another method of introducing P.sub.n onto the N-terminal nitrogen. Specifically, the target peptides having P.sub.n at the N-terminus can be obtained by converting the N-terminal amine of a resin-loaded peptide to an Ns-substituted form, introducing P.sub.n by Mitsunobu reaction, and then deprotecting the Ns group. Further, cyclic compounds can be prepared by elongating, cleaving from the resin, cyclizing, deprotecting, and purifying the peptide according to the general peptide synthesis method described in the present Examples.

(216) ##STR00021##

(217) Peptides containing glycine with P.sub.n introduced onto the nitrogen atom can be synthesized according to the general peptide synthesis method described in the present Examples using glycine with P.sub.n introduced onto the Fmoc-protected nitrogen atom as a raw material, or alternatively can be prepared by substitution reaction between the N-terminal halogenated carbon and an amine as illustrated below. Specifically, the target peptides having N-terminal glycine with P.sub.n introduced onto the nitrogen atom can be obtained by reacting the N-terminal amine with iodoacetic acid and then reacting it with any primary amine by referring to Organic Letters, 2010, 12, 4928-4931 or the like. Further, cyclic compounds can be prepared by elongating, cleaving from the resin, cyclizing, deprotecting, and purifying the peptide according to the general peptide synthesis method described in the present Examples.

(218) ##STR00022##

(Method of Preparing Peptides Containing an Aryloxy or Heteroaryloxy Group on the Side Chain)

(219) Peptides containing an aryloxy or heteroaryloxy group on a side chain can be prepared according to the general peptide synthesis method described in the present Examples using an Fmoc amino acid having the target aryloxy or heteroaryloxy group on the side chain as a raw material, or alternatively can be prepared using a peptide having an alcohol on the side chain as a precursor by referring to Organic Letters, 2014, 16, 4944-4947, Tetrahedron Letters, 2003, 44, 3863-3865, or the like, as illustrated below. Specifically, peptides having an aryloxy or heteroaryloxy group on the side chain can be prepared by reacting a peptide having an alcohol on the side chain with triarylboroxane-pyridine complex in the presence of copper (II) acetate.

(220) ##STR00023##

(221) Peptides having an ether group excluding an aryloxy or heteroaryloxy group on a side chain can be prepared according to the general peptide synthesis method described in the present Examples using an Fmoc amino acid having the target ether group on the side chain as a raw material, or alternatively can be prepared using a peptide having an alcohol on the side chain as a precursor by referring to the method described in Journal of Medicinal Chemistry, 2011, 54, 4815-4830 or Journal of Medicinal Chemistry, 2014, 57, 159-170, as illustrated below. Specifically, the peptides having the target ether group on the side chain can be prepared by reacting a peptide having an alcohol with an alkyl halide in the presence of silver (I) oxide, or by reacting a peptide having an alcohol with an alkyl halide using an aqueous sodium hydroxide solution as a base in the presence of a phase transfer catalyst such as a tetraalkylammonium salt.

(222) ##STR00024##

(Method of Preparing Peptides Containing an Aryl or Heteroaryl Group on the Side Chain)

(223) Peptides having an aryl or heteroaryl group on a side chain can be prepared according to the general peptide synthesis method described in the present Examples using an Fmoc amino acid having the target aryl or heteroaryl group on the side chain as a raw material, or alternatively can be prepared using a peptide having a carboxylic acid on the side chain as a precursor by referring to the method described in J. Am. Chem. Soc., 2016, 138, 5016-5019 or the like, as illustrated below. Specifically, the peptides having the target aryl or heteroaryl group on the side chain can be prepared by activating a peptide having a carboxylic acid on the side chain with N-hydroxyphthalimide, and reacting it with any aryl halide or heteroaryl halide.

(224) ##STR00025##

(225) A peptide having carboxylic acid in a substituent of a nitrogen atom and/or in a side chain and having aryl halide or heteroaryl halide in a substituent of another nitrogen atom and/or in another side chain within the molecule can be used to produce a peptide compound in which the peptide main chain is crosslinked. Specifically, a crosslinked compound can be produced by activating a peptide having carboxylic acid with N-hydroxyphthalimide and crosslinking it by reaction with aryl halide or heteroaryl halide within the molecule.

(226) ##STR00026##

(227) Alternatively, peptides having an aryl or heteroaryl group on a side chain can also be synthesized by Suzuki coupling using a peptide having a boronic acid on the side chain as a precursor, as illustrated below. Specifically, the target peptides having an aromatic ring on the side chain can be prepared by synthesizing a precursor peptide using an Fmoc amino acid having a boronic acid on the side chain as a raw material, and reacting it with any aryl halide in the presence of a palladium catalyst.

(228) ##STR00027##

(229) A peptide having boronic acid in a substituent of a nitrogen atom and/or in a side chain and having aryl halide in a substituent of another nitrogen atom and/or in another side chain within the molecule can be used to produce a peptide compound in which the peptide main chain is crosslinked. Specifically, a crosslinked compound can be produced by allowing a peptide having boronic acid to react and thereby crosslink with aryl halide within the molecule in the presence of a palladium catalyst.

(230) ##STR00028##

(231) A peptide having olefin in the substituent of a nitrogen atom and/or in a side chain and having aryl halide in the substituent of another nitrogen atom and/or in another side chain within the molecule can be used to produce a peptide compound that is crosslinked by arylene-containing alkylene. Specifically, the crosslinked compound can be produced by conversion to a boron compound by a hydroboration reaction on olefin, and then causing the boron compound to be reacted, and thereby crosslinked, with the intramolecular aryl halide in the presence of a palladium catalyst.

(232) ##STR00029##

(Method of Preparing Peptides Containing an Amide Group on the Side Chain)

(233) Peptides having an amide group on the side chain can be synthesized using an Fmoc amino acid having the target amide group on the side chain as a raw material, or alternatively can be synthesized by amidation of a peptide having a carboxylic acid on the side chain as a precursor, as illustrated below. Specifically, the target peptides having an amide group on the side chain can be obtained by deprotecting a peptide having a protected carboxylic acid on the side chain to synthesize a precursor peptide having a carboxylic acid on the side chain, and condensing it with any amine using a condensing agent such as HATU.

(234) ##STR00030##

(235) A peptide having carboxylic acid in a substituent of a nitrogen atom and/or in a side chain and having an amino group in a substituent of another nitrogen atom and/or in another side chain within the molecule can be used to produce a peptide compound in which the peptide main chain is crosslinked. Specifically, a crosslinked compound can be produced by synthesizing a precursor peptide having carboxylic acid and an amino group by deprotection, and condensing them using a condensing agent such as HATU for crosslinking by intramolecular amidation reaction.

(236) ##STR00031##

(Synthesis of Peptides Containing a Structure that is Highly Substituted and May Contain a Double Bond on the Side Chain)

(237) Peptides having a structure that is highly substituted and may contain a double bond on the side chain can be synthesized using an Fmoc amino acid having the target double bond on the side chain as a raw material, or alternatively can be prepared by functionalization of a terminal olefin.

Specifically, a peptide having a terminal olefin on the side chain can be synthesized according to the general peptide synthesis method described in the present Examples, and the side chain can be further converted to a side chain having a highly substituted olefin by coupling with a substrate having any terminal olefin by olefin metathesis reaction. Further, the side chain can be converted to a corresponding side chain by reducing the olefin by hydrogenation reaction.

(238) ##STR00032##

(239) Peptide compounds with a peptide backbone crosslinked can also be prepared using a peptide having multiple double bonds in substituents of nitrogen atoms and/or in side chains. Specifically, crosslinked compounds can be prepared by synthesizing a peptide having an olefin at two sites of the substituents of nitrogen atoms and/or the side chains according to the general peptide synthesis method described in the present Examples, and further crosslinking the two olefins by olefin metathesis reaction by referring to Nature Protocols, 2011, 6, 761-771. Further, compounds crosslinked with saturated alkylenes can be prepared by reducing the olefins by hydrogenation reaction.

(240) ##STR00033##

(241) A peptide that contains aryl having an olefin-containing substituent in the substituent of a nitrogen atom and/or in a side chain can be used to produce, as the crosslinked compound, a peptide compound that is crosslinked by arylene and a divalent group containing a double bond. Specifically, the crosslinked compound can be produced by synthesizing a peptide having an aryl containing an olefin in the substituent of a nitrogen atom and/or in a side chain and having olefin in the substituent of another nitrogen atom and/or in another side chain within the molecule according to the general peptide synthesis method described in the Examples, and crosslinking the two olefins by an olefin metathesis reaction. Moreover, a compound crosslinked by arylene-containing alkylene can be produced by reducing olefin by a hydrogenation reaction.

(242) ##STR00034##

(Synthesis of Peptides Containing a Triazole on the Side Chain)

(243) Peptides having a triazole on the side chain can be prepared by click reaction with an azido group. Specifically, peptide compounds having an azido group on the side chain can be prepared by synthesizing a peptide having an azido group on the side chain according to the general peptide synthesis method described in the present Examples, and coupling the peptide with any acetylene in the presence of copper (I) iodide by referring to Bioorganic & Medicinal Chemistry Letters, 2009, 19, 4130-4133 or the like.

(244) ##STR00035##

(245) A peptide having an azide group in a substituent of a nitrogen atom and/or in a side chain and having acetylene in a substituent of another nitrogen atom and/or in another side chain within the molecule can be used to produce a peptide compound in which the peptide main chain is crosslinked. Specifically, a crosslinked compound can be produced by allowing a peptide having an azide group to react and thereby crosslink with acetylene within the molecule in the presence of a palladium catalyst.

(246) ##STR00036##

(Synthesis of Peptides Containing an Aryl Group Substituted with an Alkynyl Group on the Side Chain)

(247) Peptides containing an aryl group substituted with an alkynyl group on the side chain can be synthesized by Sonogashira coupling reaction with an aryl halide group. Specifically, the conversion to peptide compounds having an aryl group substituted with an alkynyl group on the side chain can be conducted by synthesizing a peptide having an aryl iodide group on the side chain according to the general peptide synthesis method described in the present Examples, and coupling the peptide with any acetylene in the presence of copper (I) iodide.

(248) ##STR00037##

(249) A peptide having an aryl halide in a substituent of a nitrogen atom and/or in a side chain and

having acetylene in a substituent of another nitrogen atom and/or in another side chain within the molecule can be used to produce a peptide compound in which the peptide main chain is crosslinked. Specifically, a crosslinked compound can be produced by allowing a peptide having aryl iodide to be couple and thus crosslink with acetylene within the molecule in the presence of copper (I) iodide.

(250) ##STR00038##

(General Method for Producing Oligopeptide Compounds)

(251) Provided below is a general method for producing an oligopeptide having a cyclic structure formed between a substituent of a nitrogen atom and a side chain. In the following scheme, PG.sub.1 and PG.sub.1' represent protecting groups of a nitrogen atom, PG.sub.2 and PG.sub.2' represent protecting groups of an oxygen atom, R.sub.n-1, R.sub.n, R.sub.n+1, and Q.sub.n represent side chains of an amino acid, P.sub.n-1, P.sub.n, and P.sub.n+1 represent substituents of a nitrogen atom, and Y.sub.1 and Y.sub.2 each represent hydrogen, halogen, or alkyl. In the method for producing an amino acid provided below, groups other than the intended functional group may undergo chemical reaction. In such a case, introducing a protecting group to the unintended functional groups enables only the desired reaction to proceed. The reactions for attaching and detaching such a protecting group may be performed, for example, by the methods described in Greene's, "Protective Groups in Organic Synthesis" (5th edition, John Wiley & Sons 2014). As for the conversion reaction of functional groups of a compound, Comprehensive Organic Transformations: A Guide to Functional Group Preparations (5th edition) by Larock or March's Advanced Organic Chemistry: Reactions, Mechanisms, and Structure (8th edition) by Smith can be referred to.

(252) An oligopeptide compound with a cyclic structure formed between a substituent of a nitrogen atom of an amino acid and a side chain of another amino acid can be synthesized using the following method. A protected amino acid can be reacted with an alkylating agent having an olefin in the presence of a base to introduce an alkyl group having an olefin. Then, the C-terminus can be elongated with an amino acid by condensation with a C-terminally protected amino acid. The condensation reaction can be carried out using various combinations of carboxyl-activating agents, such as the combination of DIC and HOBt, the combination of DIC and HOAt, and the combination of HATU and DIPEA. Subsequently, the protected nitrogen atom can be deprotected and then elongated with a protected amino acid having an olefin in the side chain. Then, the olefins within the molecule can be cyclized by metathesis reaction. Then, the C-terminal protecting group can be deprotected to produce an oligopeptide compound that has a free C-terminus and a double bond-containing cyclic structure formed between a substituent of a nitrogen atom and a side chain.

(253) ##STR00039##

(254) Also, an oligopeptide compound having a double bond-containing cyclic structure can be used to produce an oligopeptide compound having a cyclic structure in which the double bond is converted to a single bond. Specifically, a C-terminally protected compound having a double bond-containing cyclic structure is reduced by hydrogenation reaction and then the C-terminal protecting group is deprotected, whereby an oligopeptide compound having a free C-terminus and a cyclic structure formed with alkylene between a substituent of a nitrogen atom and a side chain can be produced.

(255) ##STR00040##

(256) Moreover, an oligopeptide compound having a double bond-containing cyclic structure can be used to produce an oligopeptide compound having a crosslinked structure in which the double bond is converted to a cyclopropane ring. Specifically, a C-terminally protected compound having a double bond-containing cyclic structure can be subjected to conditions such as diiodomethane-diethylzinc to convert the double bond to a cyclopropane ring. Then, the C-terminal protecting group is deprotected, whereby an oligopeptide compound having a crosslinked structure in which the double bond is converted to a cyclopropane ring can be produced.

(257) ##STR00041##

(258) An oligopeptide compound having a cyclic structure formed between a substituent of a nitrogen atom of an amino acid and a side chain of another amino acid can also be synthesized by the following method. By allowing an aldehyde to act on a protected amino acid according to the method of Freidinger et al. (J. Org. Chem., 1983, 48 (1), 77-81), an oxazolidinone compound to which a cyclic protecting group is introduced can be obtained. Then, by carrying out a ring-opening reaction using a silicon compound having an olefin according to the method of Nguyen et al. (Synthesis, 2009, 12, 1991), an alkyl group having an olefin can be introduced to the nitrogen atom. The C-terminus can then be elongated with an amino acid by condensation with a C-terminally protected amino acid. Subsequently, the protected nitrogen atom can be deprotected and then elongated with a protected amino acid having an olefin in the side chain. Then, the olefins within the molecule can be cyclized by metathesis reaction. Then, the C-terminal protecting group is deprotected, whereby an oligopeptide compound having a free C-terminal and a cyclic structure containing a double bond between a substituent of a nitrogen atom and a side chain is formed can be produced.

(259) ##STR00042##

(260) An oligopeptide containing an aryl group substituted with an alkenyl group ($y=0$ to 2) in a side chain can be produced by a Suzuki coupling reaction between an aryl halide group and a boron compound having an alkenyl group. Specifically, a peptide having an aryl iodide group in a side chain can be synthesized according to the general oligopeptide synthesis method described in the Examples, and then coupled with a boron compound having an alkenyl group in the presence of a palladium catalyst for conversion to an oligopeptide compound having an aryl group that is substituted with an alkenyl group ($y=0$ to 2) in a side chain.

(261) ##STR00043##

(262) An oligopeptide compound in which a substituent for the nitrogen atom of an amino acid and the side chain of another amino acid form a cyclic structure by an amide bond can be synthesized using the following method. Specifically, a C-terminally protected peptide is synthesized according to the general synthesis method described in the Examples, then deprotected at the N-terminal, and subjected to a reductive amination reaction with an aldehyde compound containing an amino group having a protecting group to introduce an amino group having a protecting group in the N-substituent. This is then condensed with a protected amino acid that contains a carboxy group having a protecting group in a side chain. After the protecting groups for the carboxy group and the amino group are removed, the molecule is condensed intramolecularly and deprotected at the C-terminal, thereby achieving conversion to an oligopeptide compound in which a substituent for the nitrogen atom of an amino acid and the side chain of another amino acid form a cyclic structure by an amide bond.

(263) ##STR00044##

(Method for Cyclizing Peptide Compound on Resin)

(264) Cyclic compounds and oligopeptide compounds having a cyclic structure can be cyclized by, in addition to cyclization according to the method described in the present Examples, metathesis reaction on a resin as shown in the following method. Specifically, as a resin-supported peptide, a peptide having an olefin in two sites of the substituents of nitrogen atoms and/or the side chains is synthesized according to the common peptide synthesis method described in the present Examples, then the two olefins can be cyclized by metathesis reaction to produce a compound having a cyclic structure. Moreover, peptide elongation, cleavage from resin, cyclization, deprotection, and purification can be performed according to the common peptide synthesis method described in the present Examples to produce a cyclic compound and an oligopeptide compound having a cyclic structure.

(265) ##STR00045##

(General Preparation Methods for Non-Natural Amino Acids)

(266) General preparation methods for C-terminal-free non-natural amino acids where the nitrogen atoms of the amino acids are protected are shown below. In the following schemes, PG.sub.1 and PG.sub.1' each represent a protecting group for a nitrogen atom, PG.sub.2 and PG.sub.2' each represent a protecting group for an oxygen atom, PG.sub.3 and PG.sub.4 each represent a protecting group for an amino acid side chain, R.sub.n and Q.sub.n each represent an amino acid side chain, P.sub.n represents a substituent for a nitrogen atom, P' represents C.sub.1-C.sub.5 alkyl, and R, R', R'', and R'' each represent a substituent for a hydrogen or amino group. In the methods of preparing amino acids shown below, a functional group other than the target functional group may cause chemical reaction. In such a case, only the desired reaction can be allowed to proceed by introducing a protecting group onto a non-target functional group. Examples of such protecting group introduction and removal reactions include methods described in Greene's "Protective Groups in Organic Synthesis" (5th ed., John Wiley & Sons 2014). For conversion reactions of compound functional groups, one can refer to Larock's "Comprehensive Organic Transformations: A Guide to Functional Group Preparations" (5th ed.) or Smith's "March's Advanced Organic Chemistry: Reactions, Mechanisms, and Structure" (8th ed.).

(267) Non-natural amino acids having a protecting group (PG.sub.1) introduced onto the amino acid nitrogen atom can be prepared using the following method. The target C-terminal-free non-natural amino acids can be prepared by introducing a protecting group onto an N-terminal-free amino acid available from a commercial supplier and deprotecting it as necessary according to conventional methods.

(268) ##STR00046##

(269) Non-natural amino acids having a protecting group (PG.sub.1') introduced onto the amino acid nitrogen atom can be prepared using the following method. The target C-terminal-free non-natural amino acids can be prepared by deprotecting an amino acid that has a protecting group (PG.sub.1) introduced onto the N-terminus which is available from a commercial supplier, and introducing a protecting group, according to conventional methods.

(270) ##STR00047##

(271) Non-natural amino acids having an aminoalkyl group introduced onto the substituent (P.sub.n) of the amino acid nitrogen atom can be prepared using the following method. A bromoacetic acid ester derivative available from a commercial supplier is reacted with an amino alcohol according to the method of King et al. (Tetrahedron Letters, 2002, 43 (11), 1987-1990), and then a protecting group (PG.sub.1) is introduced onto the nitrogen atom. Next, the hydroxyl group is oxidized according to the method of Dess et al. (J. Org. Chem., 1983, 48 (22), 4155-4156), and the aldehyde group is reductively aminated according to the method of Borch et al. (J. Org. Chem. 1972, 37 (10), 1673-1674) to introduce an amino group. Next, the target C-terminal-free non-natural amino acid can be prepared by deprotecting the protecting group for the oxygen atom.

(272) ##STR00048##

(273) N-substituted amino acids can also be prepared by the following scheme of introducing a substituent (P.sub.n) onto the amino acid nitrogen atom. A bromoacetic acid ester derivative available from a commercial supplier is reacted with an amine (P.sub.nNH.sub.2) in the presence of a base, and then a protecting group (PG.sub.1) is introduced onto the nitrogen atom. Next, the target C-terminal-free non-natural amino acid can be prepared by deprotecting the protecting group for the oxygen atom.

(274) ##STR00049##

(275) Non-natural amino acids having a-CH.sub.2—P' group introduced onto the amino acid nitrogen atom can be prepared according to the following scheme. An oxazolidinone compound having an introduced cyclic protecting group can be obtained by allowing an aldehyde to act on a C-terminal-free amino acid available from a commercial supplier according to the method of Freidinger et al. (J. Org. Chem., 1983, 48 (1), 77-81). Next, the target C-terminal-free non-natural amino acid can be prepared by ring-opening reaction.

(276) ##STR00050##

(277) Non-natural amino acids having a P.sub.n group introduced onto the amino acid nitrogen atom can be produced according to the following scheme. The P.sub.n group can be introduced onto a commercially available C-terminal-free amino acid by allowing an alkylating agent (P.sub.n—X) to act on it in the presence of a base. Then, a C-terminal-free non-natural amino acid can be produced by carrying out deprotection reaction and protecting group-introducing reaction by conventional methods.

(278) ##STR00051##

(279) Non-natural amino acids having an amide group introduced onto the amino acid side chain can be prepared according to the following scheme. An amide group can be introduced onto the side chain by deprotecting a commercially available protected amino acid (n=1 or 2) and allowing an amine (R''R''NH) to act on the resulting carboxylic acid. Next, a C-terminal-free non-natural amino acid can be prepared by deprotecting the C-terminal protecting group.

(280) ##STR00052##

(281) Non-natural amino acids having an amide group introduced onto the amino acid side chain and a-CH.sub.2—P' group introduced onto the amino acid nitrogen atom can be prepared according to the following scheme. An oxazolidinone compound having an introduced cyclic protecting group can be obtained by allowing an aldehyde to act on a commercially available protected amino acid (n=1 or 2) according to the method of Freidinger et al. (J. Org. Chem., 1983, 48 (1), 77-81). Next, an amide compound can be obtained by deprotecting the side-chain protecting group and then allowing an amine (R''R''NH) to act on it. Next, the target C-terminal-free non-natural amino acid can be prepared by ring-opening reaction.

(282) ##STR00053##

(283) Non-natural amino acids having an amino group introduced onto the amino acid side chain can be prepared according to the following scheme. An amide group can be introduced onto the side chain by allowing an amine (R''R''NH) to act on the carboxyl group of a commercially available protected amino acid (n=1 or 2). Next, a C-terminal-free non-natural amino acid can be prepared by conducting reduction reaction according to the method of Reeves et al. (Advanced Synthesis & Catalysis, 2013, 355 (1), 47-52) and then deprotecting the C-terminal protecting group.

(284) ##STR00054##

(285) Non-natural amino acids having an amino group introduced onto the amino acid side chain and a-CH.sub.2—P' group introduced onto the amino acid nitrogen atom can be prepared according to the following scheme. An amide group can be introduced onto the side chain by allowing an amine (R''R''NH) to act on the carboxyl group of an amino acid protected by a cyclic protecting group (n=1 or 2). Next, the target C-terminal-free non-natural amino acid can be prepared by conducting reduction reaction according to the method of Reeves et al. (Advanced Synthesis & Catalysis, 2013, 355 (1), 47-52) and then performing ring-opening reaction.

(286) ##STR00055##

(287) Non-natural amino acids having a fluoroalkyl group introduced onto the amino acid side chain can be prepared according to the following scheme. The carboxyl group of a commercially available protected amino acid (n=1 or 2) can be reduced and converted to an aldehyde group according to a conventional method, and the aldehyde group can be converted to a difluoromethyl group by introducing a fluorine atom according to a conventional method. Next, a C-terminal-free non-natural amino acid can be prepared by deprotecting the C-terminal protecting group.

(288) ##STR00056##

(289) Non-natural amino acids having a halogenated alkyl group introduced onto the amino acid side chain and a-CH.sub.2—P' group introduced onto the amino acid nitrogen atom can be prepared according to the following scheme. The carboxyl group of an amino acid protected by a cyclic protecting group (n=1 or 2) can be reduced and converted to an aldehyde group according to

a conventional method, and the aldehyde group can be converted to a dihalogenated methyl group by introducing a halogen atom according to a conventional method. Next, a C-terminal-free non-natural amino acid can be prepared by ring-opening of the C-terminal cyclic protecting group.

(290) ##STR00057##

(291) C-terminal-free non-natural amino acids having a halogenated alkyl group introduced onto the amino acid side chain and a-CH.sub.2—P' group introduced onto the amino acid nitrogen atom can also be prepared by the method shown below.

(292) ##STR00058##

(293) Non-natural amino acids having an aryl or heteroaryl group (such groups are referred to as "Ar" in the scheme) introduced onto the amino acid side chain can be prepared according to the following scheme. An N-hydroxyphthalimide (NHPI) group can be introduced onto the side chain by allowing NHPI to act on the carboxyl group of a protected amino acid (n=1 or 2). A non-natural amino acid having an aryl or heteroaryl group introduced and possessing an aralkyl or heteroaralkyl group on the side chain can be prepared by allowing an aryl halide or heteroaryl halide to react according to the method of Huihui et al. (J. Am. Chem. Soc., 2016, 138 (15), 5016-5019). Next, a C-terminal-free non-natural amino acid can be prepared by deprotecting the C-terminal protecting group.

(294) ##STR00059##

(295) Non-natural amino acids having an aryl or heteroaryl group (such groups are referred to as "Ar" in the scheme) introduced onto the amino acid side chain and having a-CH.sub.2—P' group introduced onto the amino acid nitrogen atom can be prepared according to the following scheme. An N-hydroxyphthalimide (NHPI) group can be introduced onto the side chain by allowing NHPI to act on the carboxyl group of an amino acid protected by a cyclic protecting group (n=1 or 2). A non-natural amino acid which has an aryl or heteroaryl group introduced, and has an aralkyl or heteroaralkyl group on the side chain, and is protected by a cyclic protecting group, can be prepared by allowing an aryl halide or heteroaryl halide to react according to the method of Huihui et al. (J. Am. Chem. Soc., 2016, 138 (15), 5016-5019). Next, the target C-terminal-free non-natural amino acid can be prepared by ring-opening reaction.

(296) ##STR00060##

(297) Non-natural amino acids having an aryl or heteroaryl group (such groups are referred to as "Ar" in the scheme) introduced onto the amino acid side chain can be prepared according to the following scheme. A non-natural amino acid having an aryl or heteroaryl group introduced and possessing an aralkyl or heteroaralkyl group on the side chain can be prepared by introducing a protecting group onto a commercially available protected amino group (n=0 or 1) and then allowing an aryl halide or heteroaryl halide to react according to the method of He et al. (Org. Lett. 2014, 16 (24), 6488-6491). Next, the target C-terminal-free non-natural amino acid can be prepared by deprotection reaction and protecting group introduction reaction.

(298) ##STR00061##

(299) Non-natural amino acids having a halogen atom introduced onto the aralkyl group on the amino acid side chain can be prepared according to the following scheme. A boronic acid ester can be introduced onto the aralkyl group which may have a substituent (Ra) on the amino acid side chain according to the method of Ishiyama et al. (J. Am. Chem. Soc. 2002, 124 (3), 390-391). A halogen atom can be introduced onto the introduced boron group using N-halosuccinimide according to the method of Lindner et al. (Chem. Eur. J., 2016, 22, 13218-13235). The target C-terminal-free non-natural amino acid can be prepared by appropriately introducing/removing a protecting group onto/from the obtained non-natural amino acid as necessary.

(300) ##STR00062##

(301) Non-natural amino acids having a halogen atom introduced onto the aralkyl group of the amino acid side chain and a-CH.sub.2—P' group introduced onto the amino acid nitrogen atom can be prepared according to the following scheme. A boronic acid ester can be introduced onto the aryl

group of the amino acid having an aralkyl group which may have a substituent (Ra) on the amino acid side chain according to the method of Ishiyama et al. (J. Am. Chem. Soc. 2002, 124 (3), 390-391). A halogen atom can be introduced onto the introduced boryl group using N-halosuccinimide according to the method of Lindner et al. (Chem. Eur. J., 2016, 22, 13218-13235). The target C-terminal-free non-natural amino acid can be prepared by appropriately introducing/removing a protecting group onto/from the obtained non-natural amino acid as necessary.

(302) ##STR00063##

(303) Non-natural amino acids having an optionally substituted alkoxy or aralkoxy (RbO) group on the amino acid side chain can be prepared according to the following scheme. A cyclized compound can be obtained according to the method of Mitsunobu et al. (Synthesis, 1981, 1, 1-28) after introducing a nosyl(Ns) group onto a commercially available serine derivative (n=1 or 2) according to a conventional method. A serine ether compound can be obtained by ring-opening of the cyclized compound with a suitable alcohol (RbOH) in the presence of a Lewis acid such as BF₃·OEt₂. The target C-terminal-free non-natural amino acid can be prepared by appropriately introducing/removing a protecting group onto/from the obtained non-natural amino acid as necessary.

(304) ##STR00064##

(305) Non-natural amino acids having an optionally substituted alkoxy or aralkoxy (RbO) group on the amino acid side chain can be prepared according to the following scheme. A serine ether compound can be obtained by ring-opening of a commercially available cyclic compound (n=1 or 2) with a suitable alcohol (RbOH) in the presence of a Lewis acid such as BF₃·OEt₂ according to a conventional method. The target C-terminal-free non-natural amino acid can be prepared by appropriately introducing/removing a protecting group onto/from the obtained non-natural amino acid as necessary.

(306) ##STR00065##

(307) Non-natural amino acids having an optionally substituted alkoxy or aralkoxy (RbO) group on the amino acid side chain can be prepared according to the following scheme. A serine ether compound can be obtained by allowing an alkylating agent (Rb—X) to act on a commercially available serine derivative (n=1 or 2) in the presence of a suitable base according to the method of Williamson et al. (Liebigs Ann. Chem. 1851, 77, 37-49). When Rb has a further convertible functional group, Rb can be converted to a target functional group by additional functional group conversion. Examples of such additional functional group conversion include multiple bond reduction reaction. Next, the target C-terminal-free non-natural amino acid can be prepared by deprotecting the obtained non-natural amino acid.

(308) ##STR00066##

(309) Non-natural amino acids having an optionally substituted alkoxy or aralkoxy (RbO) group on the amino acid side chain and having a-CH₂—P' group introduced onto the amino acid nitrogen atom can be prepared according to the following scheme. An oxazolidinone compound having a cyclic protecting group introduced can be obtained by allowing an aldehyde to act on a commercially available serine derivative or a serine derivative prepared by the above-described method (n=1 or 2) according to the method of Freidinger et al. (J. Org. Chem., 1983, 48 (1), 77-81). Next, the target C-terminal-free non-natural amino acid can be prepared by ring-opening reaction.

(310) ##STR00067##

(311) Non-natural amino acids having a protected hydroxy group on the amino acid side chain can be prepared according to the following scheme. The target C-terminal-free non-natural amino acid can be prepared by appropriately introducing/removing a protecting group onto/from a commercially available serine derivative or a serine derivative prepared by the above-described method (n=1 or 2).

(312) ##STR00068##

(313) Non-natural amino acids having a protected hydroxy group on the amino acid side chain and

a-CH.sub.2—P' group introduced onto the amino acid nitrogen atom can be prepared according to the following scheme. An oxazolidinone compound having a cyclic protecting group introduced can be obtained by allowing an aldehyde to act on a commercially available serine derivative or a serine derivative prepared by the above-described method (n=1 or 2) according to the method of Freidinger et al. (J. Org. Chem., 1983, 48 (1), 77-81). Next, the target C-terminal-free non-natural amino acid can be prepared by ring-opening reaction and protecting group introduction reaction.

(314) ##STR00069##

(315) Cyclic non-natural amino acids having a substituent (R.sub.c) introduced onto the hydroxyl group of the cyclic amino acid can be prepared according to the following scheme. The hydroxy group of a commercially available cyclic amino acid can be converted to the target —OR.sub.c group by appropriately introducing a functional group. As a reaction of converting the functional group, an ether bond can be produced by allowing an alkylating agent (R.sub.c—X) to react in the presence of a suitable base according to the method of Williamson et al. (Liebigs Ann. Chem. 1851, 77, 37-49). When R.sub.c has a further convertible functional group, R_c can be converted to a target functional group by additional functional group conversion. Next, the target C-terminal-free non-natural amino acid can be prepared by deprotection reaction.

(316) ##STR00070##

(317) Cyclic non-natural amino acids having a protecting group (PG.sub.3) introduced onto the hydroxyl group of the cyclic amino acid can be prepared according to the following scheme. The target C-terminal-free non-natural amino acid can be prepared by appropriately introducing/removing a protecting group onto/from a commercially available cyclic amino acid.

(318) ##STR00071##

(319) Non-natural amino acids having a boronic acid introduced onto the amino acid side chain can be prepared according to the following scheme. A non-natural amino acid having a boronic acid ester introduced can be obtained by allowing an aldehyde to act on a commercially available glycine derivative according to the method of Lee et al. (Bioorg. Med. Chem. Lett., 2009, 19 (17), 4887-5274). Next, the target C-terminal-free non-natural amino acid can be prepared by appropriately introducing or removing a protecting group.

(320) ##STR00072##

(321) (Synthesis 1 of Fmoc non-natural amino acids having a carboxyl group on the side chain) Fmoc non-natural amino acids having a carboxyl group on the side chain can be prepared according to the following scheme. The main chain carboxyl group of a starting material which is available from a commercial supplier and has a side chain carboxyl group protected by PG.sub.3 (n=1 or 2) can be converted to an amide group by condensing it with an amine (R''R''NH) in the presence of a condensing agent such as DIC. Next, the target Fmoc non-natural amino acid having a carboxyl group on the side chain can be prepared by deprotecting PG.sub.3.

(322) ##STR00073##

(Synthesis 2 of Fmoc Non-Natural Amino Acids Having a Carboxyl Group on the Side Chain)

(323) Fmoc non-natural amino acids having a carboxyl group on the side chain and a-CH.sub.2—P' group introduced onto the amino acid nitrogen atom can be prepared according to the following scheme. The main chain carboxyl group of a starting material which is available from a commercial supplier and has a side chain carboxyl group protected by PG.sub.3 (n=1 or 2) can be converted to an amide group by condensing it with an amine (R''R''NH) in the presence of a condensing agent such as DIC. Next, the target Fmoc non-natural amino acid having a carboxyl group on the side chain can be prepared by deprotecting PG.sub.3.

(324) ##STR00074##

(Synthesis of Fmoc Non-Natural Amino Acids Having Vinyl Halide on the Side Chain)

(325) Fmoc non-natural amino acids having vinyl halide in the side chain can be synthesized by the following scheme according to the method of Shendage et al. (Eur. J. Org. Chem., 2005, 719-727). Boc-2-t-butyl-3-methylimidazolidin-4-one available from a commercial supplier is reacted with an

alkylating agent having vinyl halide in the side chain in the presence of a base, and the intended Fmoc non-natural amino acid having vinyl halide in the side chain can be produced by the method described in the literature.

(326) ##STR00075##

(327) Non-natural amino acids containing a thioether group on the side chain ($n=1$ or 2) can be produced according to the following scheme. An amino acid having a protected side chain thiol group was subjected to carboxylic acid amidation, and following deprotection of the thiol group, halogenated acetic acid having a protected carboxylic acid was allowed to react to form a thioether bond. Next, the amino acid having a thioether group on the side chain can be produced by deprotecting the side chain carboxylic acid.

(328) ##STR00076##

(329) Peptides containing a thioether group on the peptide main chain can be produced by using as a raw material the aforementioned amino acid having a thioether group on the side chain, but alternatively, they can also be produced by the method of Roberts et al. in which an N-terminal bromoacetamide is reacted with a cysteine side chain (Tetrahedron Letters, 1998, 39, 8357-8360), or the method of Robey et al. in which an N-terminal chloroacetamide is reacted with a cysteine side chain (Journal of Peptide Research, 2000, 56, 115-120).

(330) The compounds of the present invention and salts thereof, and solvates thereof include all stereoisomers (such as enantiomers and diastereomers (including cis and trans geometric isomers)) of the target compounds obtained through the above-described reaction steps, and racemates and other mixtures of such isomers. For example, the compounds of the present invention may have one or more asymmetric points, and the present invention encompasses racemic mixtures, diastereomeric mixtures, and enantiomers of such compounds.

(331) When the compounds according to the present invention are obtained as free forms, they can be converted to salts that may be formed by such compounds, or hydrates or solvates thereof, according to conventional methods.

(332) When the compounds according to the present invention are obtained as salts, hydrates, or solvates of such compounds, they can be converted to free forms of such compounds according to conventional methods.

(333) <Pharmaceutical Compositions>

(334) The present invention provides pharmaceutical compositions containing a cyclic compound of the present invention.

(335) The pharmaceutical compositions of the present invention can be formulated by introducing a pharmaceutically acceptable carrier, in addition to a compound of the present invention, a salt thereof, or a solvate thereof by conventional methods. Commonly used excipients, binders, lubricants, colorants, correctives, and as necessary, stabilizers, emulsifiers, absorption promoters, surfactants, pH adjusters, preservatives, antioxidants, and the like can be used for formulation, and they are blended with ingredients generally used as raw materials of pharmaceutical formulations, and formulated by conventional methods.

(336) For example, oral formulations are prepared by adding the compound of the present invention or a salt thereof, and an excipient, and as necessary, a binder, a disintegrant, a lubricant, a colorant, a corrective, and the like, and then formulating them into powder, fine granules, granules, tablets, coated tablets, capsules, and the like by a conventional method.

(337) Examples of these ingredients include animal and vegetable oils such as soybean oil, beef tallow, and synthetic glyceride: hydrocarbons such as liquid paraffin, squalane, and solid paraffin; ester oils such as octyldodecyl myristate and isopropyl myristate: higher alcohols such as cetostearyl alcohol and behenyl alcohol; silicone resin; silicone oil; surfactants such as polyoxyethylene fatty acid ester, sorbitan fatty acid ester, glycerol fatty acid ester, polyoxyethylene sorbitan fatty acid ester, polyoxyethylene hydrogenated castor oil, and polyoxyethylene-polyoxypropylene block copolymer; water-soluble polymers such as hydroxyethylcellulose,

polyacrylic acid, carboxyvinyl polymer, polyethylene glycol, polyvinylpyrrolidone, and methylcellulose; lower alcohols such as ethanol and isopropanol; polyhydric alcohols such as glycerol, propylene glycol, dipropylene glycol, and sorbitol; sugars such as glucose and sucrose; inorganic powders such as silicic anhydride, magnesium aluminum silicate, and aluminum silicate; and purified water.

(338) Examples of the excipients include lactose, corn starch, white soft sugar, glucose, mannitol, sorbitol, microcrystalline cellulose, and silicon dioxide.

(339) Examples of the binders include polyvinyl alcohol, polyvinyl ether, methylcellulose, ethylcellulose, acacia, tragacanth, gelatin, shellac, hydroxypropylmethylcellulose, hydroxypropylcellulose, polyvinylpyrrolidone, polypropylene glycol-polyoxyethylene block polymer, and meglumine.

(340) Examples of the disintegrants include starch, agar, gelatin powder, microcrystalline cellulose, calcium carbonate, sodium bicarbonate, calcium citrate, dextrin, pectin, and carboxymethylcellulose calcium.

(341) Examples of the lubricants include magnesium stearate, talc, polyethylene glycol, silica, and hydrogenated vegetable oil.

(342) For colorants, those approved as additives to pharmaceuticals are used. For correctives, cocoa powder, peppermint camphor, empassm, mentha oil, borneol, powdered cinnamon bark, and the like are used.

(343) Obviously, these tablets and granules may be sugar-coated or otherwise coated appropriately as necessary. When liquid formulations such as syrups and injectable formulations are prepared, they are formulated by adding pH adjusters, solubilizers, tonicity adjusting agents, and the like, and as necessary, solubilizing agents, stabilizers, and the like to the compounds according to the present invention or pharmacologically acceptable salts thereof using conventional methods.

(344) For example, the pharmaceutical compositions can be parenterally used in the form of injectable sterile solutions or suspensions with water or other pharmaceutically acceptable liquids. For example, they would be formulated by appropriately combining with pharmacologically acceptable carriers or media, specifically, sterile water, saline, vegetable oils, emulsifiers, suspending agents, surfactants, stabilizers, flavoring agents, excipients, vehicles, preservatives, or binders, and blending in unit dosage forms required in generally approved formulation.

Specifically, carriers may include light anhydrous silicic acid, lactose, microcrystalline cellulose, mannitol, starch, carmellose calcium, carmellose sodium, hydroxypropylcellulose, hydroxypropylmethylcellulose, polyvinyl acetal diethylaminoacetate, polyvinylpyrrolidone, gelatin, medium chain fatty acid triglyceride, polyoxyethylene hydrogenated castor oil 60, white soft sugar, carboxymethylcellulose, corn starch, and inorganic salts. The amount of the active ingredient in such a formulation is designed to provide a suitable dose within an indicated range.

(345) Sterile compositions for injection can be formulated in a conventional formulation manner using a vehicle such as distilled water for injection.

(346) Aqueous solutions for injection include, for example, isotonic solutions containing saline, glucose, and other adjuvants, such as D-sorbitol, D-mannose, D-mannitol, and sodium chloride, and may be used in combination with appropriate solubilizers, for example, alcohols, specifically, ethanol, polyalcohols, e.g., propylene glycol or polyethylene glycol, and nonionic surfactants, e.g., polysorbate 80 (registered trademark) or HCO-50.

(347) Oily liquids include sesame oil and soybean oil, and may be used in combination with benzyl benzoate and benzyl alcohol as solubilizers. They may also be blended with buffering agents such as phosphate buffer and sodium acetate buffer; analgesics such as procaine hydrochloride; stabilizers such as benzyl alcohol and phenol; and antioxidants. Prepared injections are usually packed in suitable ampoules.

(348) The administration method is preferably oral administration, but is not limited thereto. Specific examples of parenteral administration include dosage forms of injection, nasal

administration, pulmonary administration, and transdermal administration. Examples of injection dosage forms include systemic or local administration by intravenous injection, intramuscular injection, intraperitoneal injection, subcutaneous injection, etc.

(349) The administration method can also be selected according to the age and symptom of the patient. The dosage of the pharmaceutical composition containing the peptide compound prepared by the method of the present invention can be selected, for example, in the range of 0.0001 mg to 1000 mg per kg body weight per dose. Alternatively, the dosage can be selected, for example, in the range of 0.001 to 100000 mg/body per patient; however, it is not necessarily limited to such values. The dosage and the administration method vary according to the body weight, the age, the symptom, and the like of the patient, but can be appropriately selected by those skilled in the art.

(350) In an embodiment, a cyclic compound of the present invention or a salt thereof, or a solvate thereof; or a pharmaceutical composition containing a cyclic compound of the present invention or a salt thereof, or a solvate thereof can bind or selectively bind to KRAS in a subject and can be used for selectively inhibiting KRAS.

(351) In an embodiment, a cyclic compound of the present invention or a salt thereof, or a solvate thereof can bind or selectively bind to KRAS in a subject, and can be used for producing a medical drug for selectively inhibiting KRAS.

(352) In an embodiment, the present invention relates to a method for binding or selectively binding a cyclic compound of the present invention or a salt thereof, or a solvate thereof to KRAS in a subject, or a method for selectively inhibiting KRAS in a subject, including a step of administering an effective amount of a cyclic compound of the present invention or a salt thereof, or a solvate thereof to a subject who needs it.

(353) A cyclic compound of the present invention or a salt thereof, or a solvate thereof as mentioned above has high selectivity to KRAS in a subject. For example, a cyclic compound of the present invention has selectivity to KRAS (KRAS selectivity to NRAS and/or KRAS selectivity to HRAS) higher than PP1820: ((3S,9S,12S,17S,20S,23S,27S,30S,36S)-3-[2-[3-chloro-4-(trifluoromethyl)phenyl]ethyl] 30)-cyclopentyl-23-isobutyl-9-(isopentyloxymethyl)-N,N,7,17,18,24,28,31-octamethyl-20-[(1S)-1-methylpropyl]-2,5,8,11,16,19,22,25,29,32,35-undecaoxo-10-propyl-spiro[1,4,7,10,15,18,21,24,28,31,34-undecazatricyclo[34.3.0.0¹², 15] nonatriacontane-33,l'-cyclobutane]-27-carboxamide). In an embodiment, the pharmaceutical composition of the present invention has KRAS inhibitory activity that is 3 times or more higher than NRAS inhibitory activity and HRAS inhibitory activity.

(354) In an embodiment, the pharmaceutical composition of the present invention has KRAS inhibitory activity that is 5 times, 7 times, 10 times, 15 times, or 20 times or more higher than NRAS inhibitory activity and HRAS inhibitory activity.

(355) In an embodiment, the cyclic compound of the present invention, or a salt thereof, or a solvate thereof, or a pharmaceutical composition containing the cyclic compound of the present invention, or a salt thereof, or a solvate thereof, can be used to treat and/or prevent cancer in a subject.

(356) In an embodiment, the cyclic compound of the present invention, or a salt thereof, or a solvate thereof, can be used in the manufacture of a medicament for treating and/or preventing cancer in a subject.

(357) In an embodiment, the present invention relates to a method for treating and/or preventing cancer in a subject, comprising the step of administering an effective amount of the cyclic compound of the present invention, or a salt thereof, or a solvate thereof, to a subject in need thereof.

(358) Specific examples of the cancer include lung cancer.

(359) The term "subject" herein includes mammals, and mammals are preferably humans. All prior art documents cited in the present specification are incorporated herein by reference.

EXAMPLE

(360) The scope of the present invention will now be further described by way of Examples and Reference Examples below, but the present invention is not limited thereto. When production methods are not described, starting materials and reagents were obtained from commercial suppliers or synthesized using known methods. The analytical conditions of LC/MS are provided in Table 1.

(361) TABLE-US-00001 TABLE 1 Analytical conditions Apparatus Column (I.D. × length)(mm) Mobile phase SQDFA05 Acquity Ascentis Express C18 (2.1 × 50) A) 0.1% FA, H.sub.2O UPLC/SQD B) 0.1% FA, MeCN SQDFA50 Acquity Ascentis Express C18 (2.1 × 50) A) 0.1% FA, H.sub.2O UPLC/SQD B) 0.1% FA, MeCN SQDFA50_2 Acquity Ascentis Express C18 (2.1 × 50) A) 0.1% FA, H.sub.2O UPLC/SQD B) 0.1% FA, MeCN SQDFA40 Acquity Ascentis Express C18 (2.1 × 50) A) 0.1% FA, H.sub.2O UPLC/SQD B) 0.1% FA, MeCN SQDFA05|ong Acquity Ascentis Express C18 (2.1 × 50) A) 0.1% FA, H.sub.2O UPLC/SQD B) 0.1% FA, MeCN SQDFA50|ong Acquity Ascentis Express C18 (2.1 × 50) A) 0.1% FA, H.sub.2O UPLC/SQD B) 0.1% FA, MeCN SQDAA05 Acquity Ascentis Express C18 (2.1 × 50) A) 10 mM AcONH.sub.4, H.sub.2O UPLC/SQD B) MeOH SQDAA50 Acquity Ascentis Express C18 (2.1 × 50) A) 10 mM AcONH.sub.4, H.sub.2O UPLC/SQD B) MeOH SQDAA50_2 Acquity Ascentis Express C18 (2.1 × 50) A) 10 mM AcONH.sub.4, H.sub.2O UPLC/SQD B) MeOH SMD method_02 Shimazu Shim-Pack XR-ODS (3.0 × 50) A) 0.05% TFA, H.sub.2O LCMS-2020 B) 0.05% TFA, MeCN SMD method_03 Nexera/2020 Ascentis Express C18 (2.1 × 50) A) 0.1% FA, H.sub.2O B) 0.1% FA, MeCN SMD method_04 Nexera/2020 Ascentis Express C18 (2.1 × 50) A) 0.1% FA, H.sub.2O B) 0.1% FA, MeCN SMD method_05 Nexera/2020 Ascentis Express RP-Amide A) 0.1% FA, H.sub.2O (2.1 × 50) B) 0.1% FA, MeCN SMD method_06 Nexera/2020 Ascentis Express C18 (2.1 × 50) A) 0.1% FA, H.sub.2O B) 0.1% FA, MeCN SMD method_07 Nexera/2020 Ascentis Express RP-Amide A) 0.1% FA, H.sub.2O (2.1 × 50) B) 0.1% FA, MeCN SMD method_08 Nexera/2020 Accucore C18 A) 10 mM NH₄HCO₃ in (2.1 × 50) H₂O B) MeCN SMD method_10 Nexera/2020 Kinetex EVO C18 A) 10 mM NH₄HCO₃ in (2.1 × 50) H₂O B) MeCN SMD method_11 Shimazu Shim-Pack XR-ODS-C18 (3.0 × 50) A) 0.05% TFA, H.sub.2O LCMS-2020 B) 0.05% TFA, MeCN SMD method_12 Shimazu Shim-Pack XR-ODS (3.0 × 50) A) 0.05% TFA, H.sub.2O LCMS-2020 B) 0.05% TFA, MeCN SMD method_13 Shimazu Shim-Pack XR-ODS (3.0 × 50) A) 0.05% TFA, H.sub.2O LCMS-2020 B) 0.05% TFA, MeCN SMD method_14 Shimazu Shim-Pack XR-ODS (3.0 × 50) A) 0.05% TFA, H.sub.2O LCMS-2020 B) 0.05% TFA, MeCN SMD method_15 Shimazu Shim-Pack XR-ODS-C18 (3.0 × 50) A) 0.05% TFA, H.sub.2O LCMS-2020 B) 0.05% TFA, MeCN SMD method_16 Shimazu Shim-Pack XR-ODS-C18 (3.0 × 50) A) 0.05% TFA, H.sub.2O LCMS-2020 B) 0.05% TFA, MeCN SMD method_17 Shimadzu kinetex XB-C18(3.0 × 50) A) 0.1% FA, H₂O LCMS-2020 B) 0.1% FA, MeCN LC-20ADXR SMD method_18 Shimadzu Shim-Pack XR-ODS (3.0 × 50) A) 0.05% TFA, H₂O LCMS-2020 B) 0.05% TFA, MeCN LC-20AD SMD method_19 Shimadzu ACQUITY BEH C18 (2.1 × 50) A) 0.15% FA, H.sub.2O LCMS-2020 B) 0.15% FA, MeCN SMD method_20 Shimazu Halo C18 (3.0 × 30) A) 0.05% TFA, H.sub.2O LCMS-2020 B) 0.05% TFA, MeCN SMD method_21 Shimadzu Ascentis Express C18 (4.6 × 100) A) 0.1% FA, H₂O LCMS-2020 B) 0.1% FA, MeCN LC-20AD SSC-AA-02/03 Nexera Ascentis Express C18 (2.1 × 50) A) 10 mM AcONH.sub.4, H.sub.2O UC/2020 B) MeOH SSC-FA-02/03 Nexera Ascentis Express C18 (2.1 × 50) A) 0.1% FA, H₂O UC/2020 B) 0.1% FA, MeCN SMD method_22 Shimazu Halo C18 (3.0 × 30) A) 0.05% TFA, H.sub.2O LCMS-2020 B) 0.05% TFA, MeCN SMD method_23 Shimazu Halo C18 (3.0 × 30) A) 0.05% TFA, H.sub.2O LCMS-2020 B) 0.05% TFA, MeCN SMD method_24 Shimazu Halo C18 (3.0 × 30) A) 0.05% TFA H.sub.2O LCMS-2020 B) 0.05% TFA, MeCN SMD method_25 Shimazu Halo C18 (3.0 × 30) A) 0.05% TFA, H.sub.2O LCMS-2020 B) 0.05% TFA, MeCN SMD method_26 Shimazu Halo C18 (3.0 × 30) A) 0.1% FA, H.sub.2O LCMS-2020 B) 0.1% FA, MeCN SMD method_27 Shimazu Halo C18 (3.0 × 30) A) 0.05% TFA, H.sub.2O LCMS-2020 B) 0.05% TFA, MeCN SMD method_28 Shimazu Halo C18 (3.0 × 30) A) 0.1% FA, H.sub.2O LCMS-2020 B) 0.1% FA, MeCN SMD

method_29 Shimazu Halo C18 (3.0 × 30) A) 0.05% TFA, H.sub.2O LCMS-2020 B) 0.05% TFA, MeCN SMD method_30 Shimazu Halo C18 (3.0 × 30) A) 0.05% TFA, H.sub.2O LCMS-2020 B) 0.05% TFA, MeCN SMD method_31 Shimazu Halo C18 (3.0 × 30) A) 0.05% TFA, H.sub.2O LCMS-2020 B) 0.05% TFA, MeCN SMD method_32 Shimazu Halo C18 (3.0 × 30) A) 0.05% TFA, H.sub.2O LCMS-2020 B) 0.05% TFA, MeCN SMD method_33 Shimazu Halo C18 (3.0 × 30) A) 0.05% TFA, H.sub.2O LCMS-2020 B) 0.05% TFA, MeCN SMD method_34 Shimazu Halo C18 (3.0 × 30) A) 0.05% TFA, H.sub.2O LCMS-2020 B) 0.05% TFA, MeCN SMD method_35 Shimazu Halo C18 (3.0 × 30) A) 0.1% FA, H.sub.2O LCMS-2020 B) 0.1% FA, MeCN SMD method_36 Shimazu Halo C18 (3.0 × 30) A) 0.1% FA, H.sub.2O LCMS-2020 B) 0.1% FA, MeCN SMD method_37 Shimazu Halo C18 (3.0 × 30) A) 0.1% FA, H.sub.2O LCMS-2020 B) 0.1% FA, MeCN SMD method_38 Shimazu Halo C18 (3.0 × 30) A) 0.1% FA, H.sub.2O LCMS-2020 B) 0.1% FA, MeCN SMD method_39 Shimazu Halo C18 (3.0 × 30) A) 0.1% FA, H.sub.2O LCMS-2020 B) 0.1% FA, MeCN SMD method_40 Shimazu Halo C18 (3.0 × 30) A) 0.1% FA, H.sub.2O LCMS-2020 B) 0.1% FA, MeCN SMD method_41 Shimazu Halo C18 (3.0 × 30) A) 0.05% TFA, H.sub.2O LCMS-2020 B) 0.05% TFA, MeCN Flow Column Analytical rate temperature Wave conditions Gradient (A/B) (mL/Min) (° C.) length SQDFA05 95/5 => 0/100(1.0 Min) => 1.0 35 210-400 nm 0/100(0.4 Min) PDA total SQDFA50 50/50 => 0/100(0.7 Min) => 1.0 35 210-400 nm 0/100(0.7 Min) PDA total SQDFA50_2 50/50 => 0/100(1 Min) => 1.0 35 210-400 nm 0/100(0.4 Min) PDA total SQDFA40 60/40 => 0/100(1 Min) => 1.0 35 210-400 nm 0/100(0.4 Min) PDA total SQDFA05|ong 95/5 => 0/100(4.5 Min) => 1.0 35 210-400 nm 0/100(0.5 Min) PDA total SQDFA50|ong 50/50 => 0/100(4.5 Min) => 1.0 35 210-400 nm 0/100(0.5 Min) PDA total SQDAA05 95/5 => 0/100(1.0 Min) => 1.0 35 210-400 nm 0/100(0.4 Min) PDA total SQDAA50 50/50 => 0/100(0.7 Min) => 1.0 35 210-400 nm 0/100(0.7 Min) PDA total SQDAA50_2 50/50 => 0/100(0.7 Min) => 0.9 35 210-400 nm 0/100(0.7 Min) PDA total SMD method_02 60/40 => 5/95(3 Min) => 1.2 40 190-400 nm 5/95(0.7 Min) PDA total SMD method_03 95/5 => 0/100(4.5 Min) => 1.0 35 210-400 nm 0/100(0.5 Min) PDA total SMD method_04 95/5 => 0/100(1.5 Min) => 1.0 35 210-400 nm 0/100(0.5 Min) PDA total SMD method_05 95/5 => 0/100(1.5 Min) => 1.0 35 210-400 nm 0/100(0.5 Min) PDA total SMD method_06 50/50 => 0/100(1 Min) => 1.0 35 210-400 nm 0/100(1 Min) PDA total SMD method_07 50/50 => 0/100(1 Min) => 1.0 35 210-400 nm 0/100(1 Min) PDA total SMD method_08 95/5 => 5/95(1.5 Min) => 1.0 35 210-400 nm 5/95(0.5 Min) PDA total SMD method_10 95/5 => 5/95(1.5 Min) => 1.0 35 210-400 nm 5/95(0.5 Min) PDA total SMD method_11 95/5 => 5/95(2 Min) => 1.2 40 190-400 nm 5/95(0.7 Min) PDA total SMD method_12 95/5 => 0/100(1.2 Min) => 1.0 40 190-800 nm 0/100(1 Min) PDA total SMD method_13 95/5 => 0/100(2.2 Min) => 1.0 40 190-800 nm 0/100(1 Min) PDA total SMD method_14 95/5 => 0/100(1.1 Min) => 1.2 40 190-400 nm 0/100(0.6 Min) PDA total SMD method_15 70/30 => 5/95(3.8 Min) => 1.2 40 190-400 nm 5/95(0.8 Min) PDA total SMD method_16 95/5 => 5/95(3.0 Min) => 1.2 40 190-400 nm 5/95(0.7 Min) PDA total SMD method_17 90/10 => 0/100(1.2 Min) => 1.5 40 190-400 nm 0/100(0.5 Min) => PDA total 90/10(0.1 Min) SMD method_18 95/5 => 0/100(2.2 Min) => 1.0 40 190-400 nm 0/100(1.0 Min) PDA total 95/5(0.1 Min) SMD method_19 90/10 => 30/70(3.6 Min) => 0.7 45 190-600 nm 30/70(1.0 Min) PDA total SMD method_20 95/5 => 5/95(1.2 Min) => 1.5 40 190-400 nm 5/95(0.6) => PDA total 95/5(0.02 Min) SMD method_21 95/5 => 70/30(15.0 Min) => 1.0 40 190-400 nm 70/30(3.0 Min) PDA total 5/95(2.0 Min) SSC-AA-02/03 70/30 => 0/100(8.75 Min) => 0.5 50 210-400 nm 0/100(1.25 Min) PDA total SSC-FA-02/03 70/30 => 10/90(7.5 Min) => 0.5 50 210-400 nm 0/100(0.01 Min) => PDA total 0/100(2.49 Min) SMD method_22 95/5 => 5/95(0.7 Min) => 1.5 40 190-400 nm 5/95(0.4 Min) PDA total SMD method_23 60/40 => 0/100(2.0 Min) => 1.2 40 190-400 nm 0/100(0.7 Min) PDA total SMD method_24 95/5 => 0/100(1.1 Min) => 1.5 40 190-400 nm 0/100(0.6 Min) PDA total SMD method_25 95/5 => 0/100(1.1 Min) => 1.2 40 190-400 nm 0/100(0.6 Min) PDA total SMD method_26 95/5 => 5/95(1.1 Min) => 1.5 40 190-400 nm 5/95(0.6 Min) PDA total SMD method_27 50/50 => 0/100(2.0 Min) => 1.2 40 190-400 nm 0/100(0.7 Min) PDA total SMD

method_28 95/5 => 5/95(0.7 Min) => 1.2 40 190-400 nm 5/95(0.4 Min) PDA total SMD
method_29 95/5 => 5/95(1.1 Min) => 1.3 40 190-400 nm 5/95(0.6 Min) PDA total SMD
method_30 95/5 => 0/100(0.7 Min) => 1.5 40 190-400 nm 0/100(0.4 Min) PDA total SMD
method_31 95/5 => 5/95(0.7 Min) => 1.2 40 190-400 nm 5/95(0.4 Min) PDA total SMD
method_32 70/30 => 30/70(1.7 Min) => 1.2 40 190-400 nm 0/100(0.6 Min) => 0/100(0.5 Min)
PDA total SMD method_33 95/5 => 0/100(0.7 Min) => 1.3 40 190-400 nm 0/100(0.4 Min) PDA
total SMD method_34 60/40 => 0/100(2.0 Min) => 1.3 40 190-400 nm 0/100(0.7 Min) PDA total
SMD method_35 95/5 => 0/100(0.7 Min) => 1.0 40 190-400 nm 0/100(0.4 Min) PDA total SMD
method_36 95/5 => 5/95(0.7 Min) => 1.0 40 190-400 nm 5/95(0.4 Min) PDA total SMD
method_37 60/40 => 5/95(2.3 Min) => 1.2 40 190-400 nm 5/95(0.5 Min) PDA total SMD
method_38 95/5 => 5/95(1.1 Min) => 1.2 40 190-400 nm 5/95(0.6 Min) PDA total SMD
method_39 95/5 => 5/95(0.7 Min) => 1.5 40 190-400 nm 5/95(0.4 Min) PDA total SMD
method_40 95/5 => 0/100(0.7 Min) => 1.2 40 190-400 nm 0/100(0.4 Min) PDA total SMD
method_41 95/5 => 5/95(1.1 Min) => 1.5 40 190-400 nm 5/95(0.6 Min) PDA total

Example 1 Solid-Phase Synthesis of Peptide Compound

(362) Peptide elongation was performed through the following basic route (also called the basic peptide synthesis method) according to the peptide synthesis method by the Fmoc method described in WO2013/100132 or WO2018/225864. That is to say, it included the five steps of: 1) a peptide elongation reaction by the Fmoc method from the N-terminal of an amino acid in which the Asp side-chain carboxylic acid or the peptide main-chain carboxylic acid was supported on 2-chlorotrityl resin, 2) a process of cleaving a peptide from 2-chlorotrityl resin, 3) amide cyclization by condensation between the Asp side-chain carboxylic acid or the peptide main-chain carboxylic acid resulting from removal from 2-chlorotrityl resin by the cleaving process and the amino group at the peptide chain N-terminal (a triangular unit), 4) optional deprotection of the protecting group of a side-chain functional group contained in the peptide chain, and 5) purification of a compound by preparative HPLC. In the present Examples, peptide compounds were synthesized through this basic route unless specified otherwise.

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1-1. Fmoc-Amino Acid Used in Peptide Synthesis by Peptide Synthesizer

(364) In the peptide synthesis described herein, the Fmoc-amino acids listed in Table 2 to Table 4 were used in synthesis by a peptide synthesizer.

(365) The Fmoc-amino acids listed in Table 2 were synthesized according to the method described in WO2018/225851 or WO2018/225864.

(366) The Fmoc-amino acids listed in Table 3 were purchased from commercial suppliers.

(367) The Fmoc-amino acids listed in Table 4 were synthesized according to the scheme shown below.

(368) TABLE-US-00002 TABLE 2 Compound No. Abbreviation Structural Formula Name aa044 Fmoc-MeSer(THP)-OH  (2S)-2-[9H-fluoren-9-ylmethoxycarbonyl (methyl)amino]-3-tetrahydropyran-2-yl oxy-propanoic acid aa073 Fmoc-Ser(THP)-OH  (2S)-2-(9H-fluoren-9-ylmethoxycarbonylamino)-3-tetrahydropyran-2-yl oxy-propanoic acid aa270 Fmoc-cisPro(4-pip-4-F2)-OH  (2S,4S)-4-(4,4-difluoro-1-piperidyl)-1-(9H-fluoren-9-ylmethoxycarbonyl) pyrrolidine-2-carboxylic acid aa271 Fmoc-Pro(4-pip-4-F2)-OH  (2S,4R)-4-(4,4-difluoro-1-piperidyl)-1-(9H-fluoren-9-ylmethoxycarbonyl) pyrrolidine-2-carboxylic acid

(369) TABLE-US-00003 TABLE 3 Compound No. Abbreviation Structural Formula Name CAS No. aa001 Fmoc-MeAla(cPr)-OH  (2S)-3-cyclopropyl-2-[9H-fluoren-9-ylmethoxycarbonyl(methyl)amino] propanoic acid 2304413-31-6 aa002 Fmoc-MeAbu-OH  (2S)-2-[9H-fluoren-9-ylmethoxycarbonyl(methyl)amino] butanoic acid 1310575-53-1 aa003 Fmoc-MeHnl-OH  (2S)-2-[9H-fluoren-9-ylmethoxycarbonyl(methyl)amino] heptanoic acid aa007 Fmoc-MeLeu-OH 

(2S)-2-[9H-fluoren-9-ylmethoxycarbonyl(methyl)amino]-4-methyl- pentanoic acid 103478-62-2
aa008 Fmoc-MeAla(tBu)-OH  embedded image (2S)-2-[9H-fluoren-9-ylmethoxycarbonyl(methyl)amino]- 4,4-dimethyl-pentanoic acid 1357308-53-2 aa009 Fmoc-MeAla-OH  embedded image (2S)-2-[9H-fluoren-9-ylmethoxycarbonyl(methyl)amino] propanoic acid 84000-07-7 aa011 Fmoc-MeAlgly-OH  embedded image (2S)-2-[9H-fluoren-9-ylmethoxycarbonyl(methyl)amino] pent-4-enoic acid 2606012-88-6 aa012 Fmoc-MeNva-OH  embedded image (2S)-2-[9H-fluoren-9-ylmethoxycarbonyl(methyl)amino] pentanoic acid 252049-05-1 aa014 Fmoc-MeSer(Me)-OH  embedded image (2S)-2-[9H-fluoren-9-ylmethoxycarbonyl(methyl)amino]-3- methoxy-propanoic acid 1569103-64-5 aa015 Fmoc-MeCha-OH  embedded image (2S)-3-cyclohexyl-2-[9H-fluoren-9-ylmethoxycarbonyl(methyl)amino] propanoic acid 148983-03-3 aa016 Fmoc-MeSer(nPr)-OH  embedded image (2S)-2-[9H-fluoren-9-ylmethoxycarbonyl(methyl)amino]- 3-propoxy-propanoic acid 2255321-10-7 aa017 Fmoc-MeNle-OH  embedded image (2S)-2-[9H-fluoren-9-ylmethoxycarbonyl(methyl)amino] hexanoic acid 112883-42-8 aa024 Fmoc-MeAla(2-Thie)-OH  embedded image (2S)-2-[9H-fluoren-9-ylmethoxycarbonyl(methyl)amino]- 3-(2-thienyl)propanoic acid 1332600-71-1 aa025 Fmoc-MeAla(3-Pyr)-OH  embedded image (2S)-2-[9H-fluoren-9-ylmethoxycarbonyl(methyl)amino]- 3-(3-pyridyl)propanoic acid 1979173-93-7 aa026 Fmoc-MePhe-OH  embedded image (2S)-2-[9H-fluoren-9-ylmethoxycarbonyl(methyl)amino]- 3-phenyl-propanoic acid 77128-73-5 aa027 Fmoc-MeHse(Me)-OH  embedded image (2S)-2-[9H-fluoren-9-ylmethoxycarbonyl(methyl)amino]- 4-methoxy-butanoic acid 1979169-11-3 aa032 Fmoc-MePhe(3-Cl)-OH  embedded image (2S)-3-(3-chlorophenyl)-2-[9H-fluoren-9-ylmethoxycarbonyl(methyl)amino] propanoic acid 1446478-28-9 aa033 Fmoc-MePhe(3-F)-OH  embedded image (2S)-2-[9H-fluoren-9-ylmethoxycarbonyl(methyl)amino]-3-(3-fluorophenyl)propanoic acid 1820567-10-9 aa034 Fmoc-MeAbu(4-F3)-OH  embedded image (2S)-2-[9H-fluoren-9-ylmethoxycarbonyl(methyl)amino]- 4,4,4-trifluoro-butanoic acid aa035 Fmoc-MeGln(Me2)-OH  embedded image (2S)-5-(dimethylamino)-2-[9H-fluoren-9-ylmethoxycarbonyl(methyl)amino]-5-oxo- pentanoic acid 2255321-26-5 aa037 Fmoc-MeVal-OH  embedded image (2S)-2-[9H-fluoren-9-ylmethoxycarbonyl(methyl)amino]-3-methyl-butanoic acid 84000-11-3 aa038 Fmoc-EtLeu-OH  embedded image (2S)-2-[ethyl(9H-fluoren-9-ylmethoxycarbonyl)amino]-4-methyl- pentanoic acid aa039 Fmoc-MeAoc(2)-OH  embedded image (2S)-2-[9H-fluoren-9-ylmethoxycarbonyl(methyl)amino]oct-7- enoic acid 1808268-11-2 aa040 Fmoc-MeAhpe(2)-OH  embedded image (2S)-2-[9H-fluoren-9-ylmethoxycarbonyl(methyl)amino]hept-6- enoic acid 856412-24-3 aa041 Fmoc-MeAhxe(2)-OH  embedded image (2S)-2-[9H-fluoren-9-ylmethoxycarbonyl(methyl)amino]hex-5- enoic acid 856412-21-0 aa042 Fmoc-MeHph-OH  embedded image (2S)-2-[9H-fluoren-9-ylmethoxycarbonyl(methyl)amino]-4-phenyl- butanoic acid 1065076-30-3 aa045 Fmoc-MePhe(4-Me)-OH  embedded image (2S)-2-[9H-fluoren-9-ylmethoxycarbonyl(methyl)amino]-3- (p-tolyl)propanoic acid 227616-20-8 aa046 Fmoc-MePhe(3-Me)-OH  embedded image (2S)-2-[9H-fluoren-9-ylmethoxycarbonyl(methyl)amino]-3- (m-tolyl)propanoic acid aa048 Fmoc-MePhe(4-Cl)-OH  embedded image (2S)-3-(4-chlorophenyl)-2-[9H-fluoren-9-ylmethoxycarbonyl(methyl)amino] propanoic acid 1217716-50-1 aa051 Fmoc-MePhe(4-F)-OH  embedded image (2S)-2-[9H-fluoren-9-ylmethoxycarbonyl(methyl)amino]-3-(4-fluorophenyl)propanoic acid 1979176-87-8 aa052 Fmoc-MePhe(2-F)-OH  embedded image (2S)-2-[9H-fluoren-9-ylmethoxycarbonyl(methyl)amino]-3-(2- fluorophenyl)propanoic acid 2109724-64-1 aa053 Fmoc-MeSer(Al)-OH  embedded image (2S)-3-allyloxy-2-[9H-fluoren-9-ylmethoxycarbonyl(methyl)amino] propanoic acid aa054 Fmoc-MeAla(4-Thz)-OH  embedded image (2S)-2-[9H-fluoren-9-ylmethoxycarbonyl(methyl)amino]-3- thiazol-4-yl- propanoic acid 1446478-22-3 aa055 Fmoc-MeAib-OH  embedded image 2-[9H-fluoren-9-ylmethoxycarbonyl(methyl)amino]-2-methyl- propanoic acid 400779-65-9 aa057 Fmoc-Azp(2)-OH  embedded image (2S)-1-(9H-fluoren-9-ylmethoxycarbonyl)azepane-2-carboxylic acid

2322925-11-9 aa058 Fmoc-D-MeLeu-OH  embedded image (2R)-2-[9H-fluoren-9-ylmethoxycarbonyl(methyl)amino]-4-methyl-pentanoic acid 103478-63-3 aa059 Fmoc-Leu-OH  embedded image (2S)-2-(9H-fluoren-9-ylmethoxycarbonylamino)-4-methyl-pentanoic acid 35661-60-0 aa061 Fmoc-Algly-OH  embedded image (2S)-2-(9H-fluoren-9-ylmethoxycarbonylamino)pent-4-enoic acid 146549-21-5 aa062 Fmoc-Ile-OH 0  embedded image (2S,3S)-2-(9H-fluoren-9-ylmethoxycarbonylamino)-3-methyl-pentanoic acid 71989-23-6 aa063 Fmoc-Val-OH  embedded image (2S)-2-(9H-fluoren-9-ylmethoxycarbonylamino)-3-methyl-butanoic acid 68858-20-8 aa064 Fmoc-Gly(cBu)-OH  embedded image (2S)-2-cyclobutyl-2-(9H-fluoren-9-ylmethoxycarbonylamino)acetic acid 1391630-31-1 aa065 Fmoc-Gly(cBu-3-F2)-OH  embedded image (2S)-2-(3,3-difluorocyclobutyl)-2-(9H-fluoren-9-ylmethoxycarbonylamino)acetic acid 2349734-08-1 aa066 Fmoc-allo-OH  embedded image (2S,3R)-2-(9H-fluoren-9-ylmethoxycarbonylamino)-3-methyl-pentanoic acid 251316-98-0 aa067 Fmoc-Nva(3-Et)-OH  embedded image (2S)-3-ethyl-2-(9H-fluoren-9-ylmethoxycarbonylamino)pentanoic acid 1310680-47-7 aa068 Fmoc-D-Ile-OH  embedded image (2R,3R)-2-(9H-fluoren-9-ylmethoxycarbonylamino)-3-methyl-pentanoic acid 143688-83-9 aa069 Fmoc-Gly(cPent)-OH  embedded image (2S)-2-cyclopentyl-2-(9H-fluoren-9-ylmethoxycarbonylamino)acetic acid 220497-61-0 aa070 Fmoc-Abu(4-F3)-OH  embedded image (2S)-2-(9H-fluoren-9-ylmethoxycarbonylamino)-4,4,4-trifluoro-butanoic acid 181128-48-3 aa071 Fmoc-Abu-OH  embedded image (2S)-2-(9H-fluoren-9-ylmethoxycarbonylamino)butanoic acid 135112-27-5 aa074 Fmoc-MeGly-OH 0  embedded image 2-[9H-fluoren-9-ylmethoxycarbonyl(methyl)amino]acetic acid 77128-70-2 aa075 Fmoc-Gly-OH  embedded image 2-(9H-fluoren-9-ylmethoxycarbonylamino)acetic acid 29022-11-5 aa076 Fmoc-Aze(2)-OH  embedded image (2S)-1-(9H-fluoren-9-ylmethoxycarbonyl)azetidine-2-carboxylic acid 136552-06-2 aa077 Fmoc-Aib-OH  embedded image 2-(9H-fluoren-9-ylmethoxycarbonylamino)-2-methyl-propanoic acid 94744-50-0 aa078 Fmoc-Ala-OH  embedded image (2S)-2-(9H-fluoren-9-ylmethoxycarbonylamino)propanoic acid 35661-39-3 aa079 Fmoc-D-Aze(2)-OH  embedded image (2R)-1-(9H-fluoren-9-ylmethoxycarbonyl)azetidine-2-carboxylic acid 374791-02-3 aa080 Fmoc-(Me)Pro-OH  embedded image (2S)-1-(9H-fluoren-9-ylmethoxycarbonyl)-2-methyl-pyrrolidine-2-carboxylic acid 167275-47-0 aa081 Fmoc-D-(Me)Pro-OH  embedded image (2R)-1-(9H-fluoren-9-ylmethoxycarbonyl)-2-methyl-pyrrolidine-2-carboxylic acid 1286768-33-9 aa082 Fmoc-D-Pro-OH  embedded image (2R)-1-(9H-fluoren-9-ylmethoxycarbonyl)pyrrolidine-2-carboxylic acid 101555-62-8 aa083 Fmoc-Tmo(2)-OH  embedded image (3R)-4-(9H-fluoren-9-ylmethoxycarbonyl)thiomorpholine-3-carboxylic acid 959572-96-4 aa084 Fmoc-Mor(3)-OH 0  embedded image (3S)-4-(9H-fluoren-9-ylmethoxycarbonyl)morpholine-3-carboxylic acid 281655-37-6 aa085 Fmoc-Pic(2)-OH  embedded image (2S)-1-(9H-fluoren-9-ylmethoxycarbonyl)piperidine-2-carboxylic acid 86069-86-5 aa086 Fmoc-Hyp(Et)-OH  embedded image (2S,4R)-4-ethoxy-1-(9H-fluoren-9-ylmethoxycarbonyl)pyrrolidine-2-carboxylic acid 1446478-31-4 aa087 Fmoc-Pro-OH  embedded image (2S)-1-(9H-fluoren-9-ylmethoxycarbonyl)pyrrolidine-2-carboxylic acid 71989-31-6 aa088 Fmoc-EtAla-OH  embedded image (2S)-2-[ethyl(9H-fluoren-9-ylmethoxycarbonyl)amino]propanoic acid 84000-09-9 aa090 Fmoc-nPrGly-OH  embedded image 2-[9H-fluoren-9-ylmethoxycarbonyl(propyl)amino]acetic acid 1310680-42-2 aa091 Fmoc-EtGly-OH  embedded image 2-[ethyl(9H-fluoren-9-ylmethoxycarbonyl)amino]acetic acid 162545-29-1 aa092 Fmoc-DfeGly-OH  embedded image 2-[2,2-difluoroethyl(9H-fluoren-9-ylmethoxycarbonyl)amino]acetic acid 2172536-70-6 aa093 Fmoc-MfeGly-OH  embedded image 2-[9H-fluoren-9-ylmethoxycarbonyl(2-fluoroethyl)amino]acetic acid aa094 Fmoc-PraGly-OH  embedded image 2-[9H-fluoren-9-ylmethoxycarbonyl(prop-2-ynyl)amino]acetic acid 1033622-38-6 aa095 Fmoc-AllylGly-OH 0  embedded image 2-[allyl(9H-fluoren-9-ylmethoxycarbonyl)amino]acetic acid 222725-35-1

aa101 Fmoc-1-ACPr-OH  embedded image 1-(9H-fluoren-9-ylmethoxycarbonylamino) cyclopropanecarboxylic acid 126705-22-4 aa102 Fmoc-NCMeGly-OH  embedded image 2-[cyanomethyl(9H-fluoren-9-ylmethoxycarbonyl)amino]acetic acid 2172570-83-9 aa103 Fmoc-cPrGly-OH  embedded image 2-[cyclopropyl(9H-fluoren-9-ylmethoxycarbonyl)amino]acetic acid 1342767-08-1 aa104 Fmoc-bMeAla-OH  embedded image 3-[9H-fluoren-9-ylmethoxycarbonyl(methyl)amino] propanoic acid 172965-84-3 aa106 Fmoc-MeMethagly-OH  embedded image (2S)-2-[9H-fluoren-9-ylmethoxycarbonyl(methyl)amino]-4-methyl-pent-4-enoic acid 145615-72-1 aa113 Fmoc-ButenylGly-OH  embedded image 2-[but-3-enyl(9H-fluoren-9-ylmethoxycarbonyl)amino]acetic acid 227006-55-5 aa123 Fmoc-nBuGly-OH  embedded image 2-[butyl(9H-fluoren-9-ylmethoxycarbonyl)amino]acetic acid 234442-58-1 aa155 Fmoc-AllylPhe-OH  embedded image (2S)-2-[allyl(9H-fluoren-9-ylmethoxycarbonyl)amino]-3-phenyl-propanoic acid 1054548-99-0 aa208 Fmoc-D-MePhe-OH  embedded image (2R)-2-[9H-fluoren-9-ylmethoxycarbonyl(methyl)amino]-3-phenyl-propanoic acid 138775-05-0 aa212 Fmoc-D-MeAla-OH  embedded image (2R)-2-[9H-fluoren-9-ylmethoxycarbonyl(methyl)amino] propanoic acid 138774-92-2 aa221 Fmoc-Hph(4-Me)-OH  embedded image (2S)-2-(9H-fluoren-9-ylmethoxycarbonylamino)-4-(p-tolyl)butanoic acid 1260587-57-2 aa231 Fmoc-Hph(34-Cl2)-OH  embedded image (2S)-4-(3,4-dichlorophenyl)-2-(9H-fluoren-9-ylmethoxycarbonylamino)butanoic acid 1260616-12-3 aa238 Fmoc-Pro(4-F2)-OH  embedded image (2S)-1-(9H-fluoren-9-ylmethoxycarbonyl)-4,4-difluoro-pyrrolidine-2-carboxylic acid 203866-21-1 aa240 Fmoc-Pro(4R-CF3)-OH  embedded image (2S,4R)-1-(9H-fluoren-9-ylmethoxycarbonyl)-4-(trifluoromethyl) pyrrolidine-2-carboxylic acid 2549171-72-2 aa241 Fmoc-Pro(4R-F)-OH  embedded image (2S,4R)-1-(9H-fluoren-9-ylmethoxycarbonyl)-4-fluoro-pyrrolidine-2-carboxylic acid 203866-20-0 aa245 Fmoc-Hyp(iBu)-OH  embedded image (2S,4R)-1-(9H-fluoren-9-ylmethoxycarbonyl)-4-isobutoxy-pyrrolidine-2-carboxylic acid 865353-14-6 aa249 Fmoc-MeSer(iPen)-OH  embedded image (2S)-2-[9H-fluoren-9-ylmethoxycarbonyl(methyl)amino]-3-isopentyloxy-propanoic acid 2255321-12-9 aa262 Fmoc-Pro(4S-F)-OH  embedded image (2S,4S)-1-(9H-fluoren-9-ylmethoxycarbonyl)-4-fluoro-pyrrolidine-2-carboxylic acid 203866-19-7 aa263 Fmoc-cisHyp(Me)-OH  embedded image (2S,4S)-1-(9H-fluoren-9-ylmethoxycarbonyl)-4-methoxy-pyrrolidine-2-carboxylic acid 1190617-97-0 aa274 Fmoc-Pro(4-keto)-OH  embedded image (2S)-1-(9H-fluoren-9-ylmethoxycarbonyl)-4-oxo-pyrrolidine-2-carboxylic acid 223581-83-7 aa275 Fmoc-Pro(4-Me2)-OH  embedded image (2S)-1-(9H-fluoren-9-ylmethoxycarbonyl)-4,4-dimethyl-pyrrolidine-2-carboxylic acid 1380336-01-5 aa277 Fmoc-Pic(2)(4-F2)-OH  embedded image (2S)-1-(9H-fluoren-9-ylmethoxycarbonyl)-4,4-difluoro-piperidine-2-carboxylic acid 1221793-52-7 aa278 Fmoc-Pro(4-cPr)-OH  embedded image (6S)-5-(9H-fluoren-9-ylmethoxycarbonyl)-5-azaspiro[2.4]heptane-6-carboxylic acid 2170726-27-7 aa280 Fmoc-Nle-OH  embedded image (2S)-2-(9H-fluoren-9-ylmethoxycarbonylamino)hexanoic acid 77284-32-3 aa301 Fmoc-cVal-OH  embedded image 1-(9H-fluoren-9-ylmethoxycarbonylamino) cyclobutanecarboxylic acid 885951-77-9 aa302 Fmoc-cLeu-OH  embedded image 1-(9H-fluoren-9-ylmethoxycarbonylamino) cyclopentanecarboxylic acid 117322-30-2 aa303 Fmoc-cVal(3-Me2)-OH  embedded image 1-(9H-fluoren-9-ylmethoxycarbonylamino)-3,3-dimethyl-cyclobutanecarboxylic acid 1936161-54-4 aa305 Fmoc-Athpc-OH  embedded image 4-(9H-fluoren-9-ylmethoxycarbonylamino)tetrahydropyran-4-carboxylic acid 285996-72-7 aa306 Fmoc-cHex-OH  embedded image 1-(9H-fluoren-9-ylmethoxycarbonylamino) cyclohexanecarboxylic acid 162648-54-6 aa307 Fmoc-cVal(3-F2)-OH  embedded image 1-(9H-fluoren-9-ylmethoxycarbonylamino)-3,3-difluoro-cyclobutanecarboxylic acid 1936532-04-5 aa308 Fmoc-bAla(2-Me2)-OH  embedded image 3-(9H-fluoren-9-ylmethoxycarbonylamino)-2,2-dimethyl-propanoic acid 1076197-00-6 aa310 Fmoc-cLeu(34-d)-OH  embedded image 1-(9H-fluoren-9-ylmethoxycarbonylamino)cyclopent-3-ene-1-carboxylic acid 2219369-66-9 aa311 Fmoc-D-Algly-OH  embedded image (2R)-2-(9H-fluoren-9-ylmethoxycarbonylamino)pent-4-

enoic acid 170642-28-1 aa312 Fmoc-cHex(4-F2)-OH  1-(9H-fluoren-9-ylmethoxycarbonylamino)- 4,4-difluoro-cyclohexanecarboxylic acid 1986905-26-3 aa314 Fmoc-D-Ala-OH  (2R)-2-(9H-fluoren-9-ylmethoxycarbonylamino)propanoic acid 79990-15-1 aa316 Fmoc-(Me)Abu-OH  (2S)-2-(9H-fluoren-9-ylmethoxycarbonylamino)- 2-methyl-butanoic acid 857478-30-9 aa318 Fmoc-AoxeC-OH  3-(9H-fluoren-9-ylmethoxycarbonylamino)oxetane-3- carboxylic acid 1380327-56-9 aa330 Fmoc-MeGly(cPent)-OH  (2S)-2-cyclopentyl-2-[9H-fluoren-9-ylmethoxycarbonyl(methyl)amino]acetic acid 187475-29-2 aa336 Fmoc-Melle-OH  (2S,3S)-2-[9H-fluoren-9-ylmethoxycarbonyl(methyl)amino]-3-methyl-pentanoic acid 138775-22-1 aa337 Fmoc-MeChg-OH  (2S)-2-cyclohexyl-2-[9H-fluoren-9-ylmethoxycarbonyl(methyl)amino]acetic acid 925240-97-7 aa338 Fmoc-MeTle-OH  (2S)-2-[9H-fluoren-9-ylmethoxycarbonyl(methyl)amino]-3,3- dimethyl-butanoic acid 1172579-62-2 aa340 Fmoc-Mealle-OH  (2S,3R)-2-[9H-fluoren-9-ylmethoxycarbonyl(methyl)amino]-3-methyl- pentanoic acid 1821797-58-3 aa387 Fmoc-MeGly(cPr)-OH  (S)-2-(((9H-fluoren-9-yl)methoxy)carbonyl(methyl)amino)-2- cyclopropyl-acetic acid 2642726-08-5 aa388 Fmoc-MeAnone(2)-OH  (2S)-2-[9H-fluoren-9-ylmethoxycarbonyl(methyl)amino] non-8-enoic acid 1808268-51-0 aa390 Fmoc-Leu-OH  (2S)-2-[(9H-fluoren-9-ylmethoxy)carbonyl]amino)-4-methyl- pentanoic acid 35661-60-0 aa392 Fmoc-Thr(Et)-OH  (2S,3R)-3-ethoxy-2-(9H-fluoren-9-ylmethoxycarbonyl amino)butanoic acid 2108708-74-1 aa393 Fmoc-Gly(cPr)-OH  (2S)-2-cyclopropyl-2-(9H-fluoren-9-ylmethoxycarbonyl amino)acetic acid 1212257-18-5 aa394 Fmoc-Gly(4-THP)-OH  (2S)-2-(9H-fluoren-9-ylmethoxycarbonyl amino)-2- tetrahydropyran-4-yl- acetic acid 368866-31-3 aa395 Fmoc-Thr(Me)-OH  (2S,3R)-2-(9H-fluoren-9-ylmethoxycarbonyl amino)-3-methoxy- butanoic acid 928063-81-4 aa396 Fmoc-(MeOEt)Gly-OH  2-[9H-fluoren-9-ylmethoxycarbonyl(2- methoxyethyl)amino]acetic acid 1341969-00-3 aa412 Fmoc-Hph(4-Cl)-OH  (2S)-4-(4-chlorophenyl)-2-(9H-fluoren- 9-ylmethoxycarbonyl amino)butanoic acid 1260608-62-5 aa416 Fmoc-Hyp(iPr)-OH  (2S,4R)-1-(9H-fluoren-9-ylmethoxycarbonyl)-4-isopropoxy- pyrrolidine-2- carboxylic acid aa417 Fmoc-Pro(4R-Me)-OH  (2S,4R)-1-(9H-fluoren-9-ylmethoxycarbonyl)-4-methyl- pyrrolidine-2-carboxylic acid 333777-34-7 aa418 Fmoc-Pro(4R-nPr)-OH  (2S,4R)-1-(9H-fluoren-9-ylmethoxycarbonyl)-4-propyl- pyrrolidine-2-carboxylic acid aa419 Fmoc-Pro(4R-Et)-OH  (2S,4R)-4-ethyl- 1-(9H-fluoren-9-ylmethoxycarbonyl)pyrrolidine-2- carboxylic acid aa420 Fmoc-MecVal(3-Me2)-OH  1-[9H-fluoren-9-ylmethoxycarbonyl(methyl)amino]-3,3- dimethyl- cyclobutane-carboxylic acid 2025106-27-6 aa421 Fmoc-MecVal-OH  1-[9H-fluoren-9-ylmethoxycarbonyl(methyl)amino] cyclobutane-carboxylic acid 1700368-07-5 aa422 Fmoc-MecLeu-OH  1-[9H-fluoren-9-ylmethoxycarbonyl(methyl)amino] cyclopentane-carboxylic acid 1694050-88-8 aa427 Fmoc-Hyp-OH  (2S,4R)-1-(9H-fluoren-9-ylmethoxycarbonyl)-4-hydroxy- pyrrolidine-2-carboxylic acid 88050-17-3

(370) TABLE-US-00004 TABLE 4 Compound No. Abbreviation Structural Formula Name aa004 Fmoc-MeAOC(2)-OH  (2S)-2-[9H-fluoren-9-ylmethoxycarbonyl(methyl) amino]octanoic acid aa006 Fmoc-MeAla(cPent)-OH  (2S)-3-cyclopentyl-2-[9H-fluoren-9-ylmethoxycarbonyl(methyl)amino]p ropanoic acid aa010 Fmoc-MeAla(cBu)-OH  (2S)-3-cyclobutyl-2-[9H-fluoren-9-ylmethoxycarbonyl(methyl)amino] propanoic acid aa013 Fmoc-MeCys(Me)-OH  (2R)-2-[9H-fluoren-9-ylmethoxycarbonyl(methyl)amino]- 3-methylsulfanyl-propanoic acid aa018 Fmoc-MePRA-OH  (2S)-2-[9H-fluoren-9-ylmethoxycarbonyl(methyl)amino] pent-4-ynoic acid aa019 Fmoc-MeAbu(4-F2)-OH  (2S)-2-[9H-fluoren-9-

ylmethoxycarbonyl(methyl)amino]- 4,4-difluoro-butanoic acid aa020 Fmoc-MeSer(cPr)-OH
 (2S)-3-(cyclopropoxy)-2-[9H-fluoren-9-ylmethoxycarbonyl(methyl) amino]propanoic acid aa021 Fmoc-MeSer(Tfe)-OH  (2S)-2-[9H-fluoren-9-ylmethoxycarbonyl(methyl)amino]-3- (2,2,2-trifluoroethoxy)propanoic acid aa022 Fmoc-MeSer(Et)-OH  (2S)-3-ethoxy-2-[9H-fluoren-9-ylmethoxycarbonyl(methyl)amino] propanoic acid aa023 Fmoc-MeAla(3-Thie)-OH  (2S)-2-[9H-fluoren-9- ylmethoxycarbonyl(methyl)amino]- 3-(3-thienyl)propanoic acid aa028 Fmoc-MePhe(34-Cl2)-OH 0  (2S)-3-(3,4-dichlorophenyl)-2-[9H-fluoren-9-ylmethoxycarbonyl(methyl) amino]propanoic acid aa029 Fmoc-MePhe(4-CN)-OH  (2S)-3-(4-cyanophenyl)-2-[9H-fluoren- 9-ylmethoxycarbonyl(methyl)amino] propanoic acid aa030 Fmoc-MePhe(3-CN)-OH  (2S)-3-(3-cyanophenyl)-2-[9H-fluoren- 9-ylmethoxycarbonyl(methyl)amino] propanoic acid aa031 Fmoc-MePhe(2-CN)-OH  (2S)-3-(2-cyanophenyl)-2-[9H-fluoren- 9-ylmethoxycarbonyl(methyl)amino] propanoic acid aa244 Fmoc-MeNva(5-F2)-OH  (2S)-2-[9H-fluoren-9-ylmethoxycarbonyl (methyl)amino]-5,5-difluoro-pentanoic acid aa043 Fmoc-MeNva(5-Cl2)-OH  (2S)-5,5-dichloro-2-[9H-fluoren-9- ylmethoxycarbonyl(methyl)amino] pentanoic acid aa047 Fmoc-MePhe(2-Me)-OH  (2S)-2-[9H-fluoren-9- ylmethoxycarbonyl(methyl)amino]- 3-(o-tolyl)propanoic acid aa049 Fmoc-MePhe(2-Cl)-OH  (2S)-3-(2-chlorophenyl)-2-[9H-fluoren- 9-ylmethoxycarbonyl(methyl)amino] propanoic acid aa050 Fmoc-MePhe(34-F2)-OH  (2S)-3-(3,4-difluorophenyl)-2-[9H-fluoren- 9-ylmethoxycarbonyl(methyl)amino]propanoic acid aa056 Fmoc-R-MeAMPA-OH  (2R)-3-[9H-fluoren-9-ylmethoxycarbonyl(methyl) amino]-2-methyl-propanoic acid aa060 Fmoc-MeGly(cBu)-OH 0  (2S)-2-cyclobutyl-2-[9H-fluoren-9-ylmethoxycarbonyl(methyl)amino]acetic acid aa098 Fmoc-Aze(2)(3S-Me)-OH  (2S,3S)-1-(9H-fluoren-9-ylmethoxycarbonyl)-3- methyl-azetidine-2-carboxylic acid aa099 Fmoc-Aze(2)(3R-Me)-OH  (2S,3R)-1-(9H-fluoren-9-ylmethoxycarbonyl)-3- methyl-azetidine-2-carboxylic acid aa100 Fmoc-Aze(2)(3-Me2)-OH  (2S)-1-(9H-fluoren-9-ylmethoxycarbonyl)-3,3- dimethyl-azetidine-2-carboxylic acid aa111 Fmoc-D-MeSer(Al)-OH  (2R)-3-allyloxy-2-[9H-fluoren-9-ylmethoxycarbonyl(methyl)amino]propanoic acid aa136 Fmoc-nPrPhe(4-CF3)-OH  (2S)-2-[9H-fluoren-9-ylmethoxycarbonyl(propyl) amino]-3-[4-(trifluoromethyl)phenyl]propanoic acid aa164 Fmoc-EtPhe(4-CF3)-OH  (2S)-2-[ethyl(9H-fluoren-9-ylmethoxycarbonyl) amino]-3-[4-(trifluoromethyl)phenyl]propanoic acid aa174 Fmoc-nPrPhe(4-Me)-OH  (2S)-2-[9H-fluoren-9-ylmethoxycarbonyl(propyl) amino]-3-(p-tolyl)propanoic acid aa199 Fmoc-MePhe(4-CF3)-OH  (2S)-2-[9H-fluoren-9-ylmethoxycarbonyl(methyl) amino]-3-[4-(trifluoromethyl)phenyl]propanoic acid aa201 Fmoc-EtPhe(4-Me)-OH  (2S)-2-[ethyl(9H-fluoren-9-ylmethoxycarbonyl) amino]-3-(p-tolyl)propanoic acid aa210 Fmoc-EtGly(cPent)-OH 0  (2S)-2-cyclopentyl-2-[ethyl(9H-fluoren-9-ylmethoxycarbonyl)amino]acetic acid aa217 Fmoc-Hph(4-CF3-35-F2)-OH  (2S)-4-[3,5-difluoro-4-(trifluoromethyl)phenyl]-2- (9H-fluoren-9-ylmethoxycarbonylamino)butanoic acid aa218 Fmoc-Hph(4-CF3-3-Cl)-OH  (2S)-4-[3-chloro-4-(trifluoromethyl)phenyl]-2- (9H-fluoren-9-ylmethoxycarbonylamino)butanoic acid aa219 Fmoc-Hph(4-CF3-3-F)-OH  (2S)-2-(9H-fluoren-9-ylmethoxycarbonylamino)-4- [3-fluoro-4-(trifluoromethyl)phenyl]butanoic acid aa220 Fmoc-Hph(4-Cl-35-F2)-OH  (2S)-4-(4-chloro-3,5-difluoro-phenyl)-2-(9H-fluoren-9-ylmethoxycarbonylamino)butanoic acid aa229 Fmoc-Abu(5-Bzt)-OH  (2S)-4-(benzothiophen-5-yl)-2-(9H-fluoren-9- ylmethoxycarbonylamino)butanoic acid aa233 Fmoc-Hph(4-CF3-3-Me)-OH  (2S)-2-(9H-fluoren-9-ylmethoxycarbonylamino)-4- [3-

methyl-4-(trifluoromethyl)phenyl]butanoic acid aa235 Fmoc-Hph(4-CF3-3-OMe)-OH
 embedded image (2S)-2-(9H-fluoren-9-ylmethoxycarbonylamino)-4- [3-methoxy-4-(trifluoromethyl)phenyl]butanoic acid aa239 Fmoc-Pro(4S-Me)-OH  embedded image (2S,4S)-1-(9H-fluoren-9-ylmethoxycarbonyl)-4- methyl-pyrrolidine-2-carboxylic acid aa246 Fmoc-Hyp(nPr)-OH  embedded image (2S,4R)-1-(9H-fluoren-9-ylmethoxycarbonyl)-4- propoxy-pyrrolidine-2-carboxylic acid aa250 Fmoc-D-MeSer(nPr)-OH 0  embedded image (2R)-2-[9H-fluoren-9-ylmethoxycarbonyl(methyl) amino]-3-propoxy-propanoic acid aa264 Fmoc-cisHyp(3)(THP)-OH  embedded image (2S,3R)-1-(9H-fluoren-9-ylmethoxycarbonyl)-3- tetrahydropyran-2-yloxy-pyrrolidine-2-carboxylic acid aa265 Fmoc-Hyp(3)(THP)-OH  embedded image (2S,3S)-1-(9H-fluoren-9-ylmethoxycarbonyl)-3- tetrahydropyran-2-yloxy-pyrrolidine-2-carboxylic acid aa267 Fmoc-cisHyp(THP)-OH  embedded image (2S,4S)-1-(9H-fluoren-9-ylmethoxycarbonyl)-4-tetrahydropyran-2-yloxy-pyrrolidine-2-carboxylic acid aa268 Fmoc-Pro(3S4-Cl)-OH  embedded image (1S,2S,5R)-3-(9H-fluoren-9-ylmethoxycarbonyl)- 3-azabicyclo[3.1.0]hexane-2-carboxylic acid aa279 Fmoc-Hyp(THP)-OH  embedded image (2S,4R)-1-(9H-fluoren- 9-ylmethoxycarbonyl)-4-tetrahydropyran- 2-yloxy-pyrrolidine-2-carboxylic acid aa281 Fmoc-Hyp(Me)-OH  embedded image (2S,4R)-1-(9H-fluoren-9- ylmethoxycarbonyl)-4- methoxy-pyrrolidine-2-carboxylic acid aa331 Fmoc-MeNva(3-Et)-OH  embedded image (2S)-3-ethyl-2-[9H-fluoren- 9-ylmethoxycarbonyl (methyl)amino]pentanoic acid aa389 Fmoc-nPrLeu-OH  embedded image (2S)-2-[9H-fluoren-9- ylmethoxycarbonyl(propyl) amino]-4-methyl-pentanoic acid aa391 Fmoc-Thr(nPr)-OH  embedded image (2S,3R)-2-(9H-fluoren-9-ylmethoxycarbonylamino)-3-propoxy- butanoic acid aa397 Fmoc-nPrSer(iPen)-OH 0  embedded image (2S)-2-[9H-fluoren-9-ylmethoxycarbonyl (propyl)amino]-3-isopentyloxy-propanoic acid aa398 Fmoc-nPrSer(cBu)-OH  embedded image (2S)-3-(cyclobutoxy)-2-[9H-fluoren-9- ylmethoxycarbonyl(propyl)amino] propanoic acid aa399 Fmoc-Abu(1-Me-7-Cl-5-Indo)-OH  embedded image (2S)-4-(7-chloro-1-methyl-indol-5-yl)-2- (9H-fluoren-9-ylmethoxycarbonylamino) butanoic acid aa400 Fmoc-Abu(5-Bzt-2-F-3-Me)-OH  embedded image (2S)-2-(9H-fluoren-9- ylmethoxycarbonylamino)- 4-(2-fluoro-3-methyl-benzothiophen- 5-yl)butanoic acid aa401 Fmoc-Abu(5-Bzt-7-Cl)-OH  embedded image (2S)-4-(7-chlorobenzothiophen-5-yl)-2- (9H-fluoren-9-ylmethoxycarbonylamino) butanoic acid aa402 Fmoc-Abu(1-Me-6-Indo)-OH  embedded image (2S)-2-(9H-fluoren-9-ylmethoxycarbonylamino)-4- (1-methylindol-6-yl)butanoic acid aa403 Fmoc-Abu(13-Me2-6-Indo)-OH  embedded image (2S)-4-(1,3-dimethyl indol-6-yl)-2- (9H-fluoren-9-ylmethoxycarbonylamino) butanoic acid aa404 Fmoc-Abu(123-Me3-6-Indo)-OH  embedded image (2S)-2-(9H-fluoren-9- ylmethoxycarbonylamino)- 4-(1,2,3-trimethyl indol-6-yl)butanoic acid aa405 Fmoc-Abu(5-Bzt-23-Me2)-OH  embedded image (2S)-4-(2,3-dimethyl benzothiophen-5-yl)-2- (9H-fluoren-9-ylmethoxycarbonylamino) butanoic acid aa406 Fmoc-Hph(3-Cl-4-Et)-OH  embedded image (2S)-4-(3-chloro-4-ethyl-phenyl)-2-(9H- fluoren-9-ylmethoxycarbonylamino)butanoic acid aa407 Fmoc-Hph(4-OMe-35-Me2)-OH 0  embedded image (2S)-2-(9H-fluoren-9- ylmethoxycarbonylamino)-4-[4- (methoxymethyl)-3,5-dimethyl-phenyl] butanoic acid aa408 Fmoc-Hph(4-Cl-3-OMe)-OH  embedded image (2S)-4-(4-chloro-3-methoxy-phenyl)-2- (9H-fluoren-9-ylmethoxycarbonylamino) butanoic acid aa409 Fmoc-Hph(4-Cl-3-CF3)-OH  embedded image (2S)-4-[4-chloro-3-(trifluoromethyl)phenyl]- 2-(9H-fluoren-9-ylmethoxycarbonylamino) butanoic acid aa410 Fmoc-Hph(4-CHF2-35-F2)-OH  embedded image (2S)-4-[4-(difluoromethyl)-3,5-difluoro- phenyl]-2-(9H-fluoren-9-yl methoxycarbonylamino)butanoic acid aa411 Fmoc-Hph(4-Cl-35-Me2)-OH  embedded image (2S)-4-(4-chloro-3,5-dimethyl-phenyl)-2- (9H-fluoren-9-ylmethoxycarbonylamino) butanoic acid aa414 Fmoc-Hyp(cPent)-OH  embedded image (2S,4R)-4-(cyclopentoxy)-1-(9H- fluoren-9-ylmethoxycarbonyl)pyrrolidin- 2-carboxylic acid aa415 Fmoc-Hyp(cBu)-OH  embedded image (2S,4R)-4-(cyclobutoxy)-1-(9H-fluoren- 9-ylmethoxycarbonyl)pyrrolidin-2- carboxylic acid aa443 Fmoc-ButenylPhe(4-Me)-OH  embedded image (2S)-2-[but-3-enyl(9H-fluoren-9-

ylmethoxycarbonyl)amino]-3- (p-tolyl)propanoic acid

Synthesis of Fmoc-Amino Acids

Synthesis of Compound aa004

(371) ##STR00278##

(372) Compound aa004-a (3.00 g, 7.86 mmol) was dissolved in toluene (79 mL), paraformaldehyde (708 mg, 23.6 mmol) and CSA (91 mg, 0.393 mmol) were added, and the mixture was stirred at 90° C. for 5 hours. After cooling the mixture to room temperature, paraformaldehyde (236 mg, 7.86 mmol) and CSA (46 mg, 0.197 mmol) were added, and the mixture was further stirred at 90° C. for 30 minutes. After being cooled to room temperature, the mixture was filtered through Celite, and the residue was washed with ethyl acetate (50 mL). The filtrate was washed twice with a saturated aqueous sodium hydrogen carbonate solution (50 mL) and washed with 50% brine (50 mL). This was dried over sodium sulfate and filtered off, and the solvent was distilled off under reduced pressure to give Compound aa004-b as a crude product.

(373) LCMS (ESI) m/z=394 (M+H)+

(374) Retention time: 1.08 min (Analytical condition SQDFA05)

(375) The crude product (7.86 mmol) of Compound aa004-b was dissolved in DCM (39 mL), then a boron trifluoride diethyl ether complex (BF₃.Math.OEt₂)(2.96 mL, 23.6 mmol), TES (1.95 mL, 23.6 mmol), and water (0.142 mL, 7.86 mmol) were added, and the mixture was stirred at room temperature for 1 hour. The reaction solution was washed with a saturated ammonium chloride solution (40 mL), washed with 50% brine (40 mL), then dried over sodium sulfate, and filtered off, and the solvent was distilled off under reduced pressure. The resulting residue was dissolved in acetonitrile (40 mL) and washed twice with n-hexane (40 mL). The solvent was distilled off under reduced pressure to give Compound aa004 (3.04 g, 98%).

(376) LCMS (ESI) m/z=396 (M+H)+

(377) Retention time: 1.00 min (Analytical condition SQDFA05)

(378) Synthesis of Compound aa028

(379) ##STR00279##

(380) Using aa028-a as a starting material, aa028-b (9.61 g, 94%) was obtained as a crude product in the same manner as the synthesis of Compound aa004-b.

(381) Using aa028-b as a starting material, aa028 (8.10 g, 84%) was obtained in the same manner as the synthesis of Compound aa004.

(382) LCMS (ESI) m/z=470 (M+H)+

(383) Retention time: 0.99 min (Analytical condition SQDFA05)

(384) Synthesis of Compound aa018

(385) ##STR00280##

(386) Using Compound aa018-a, ((2S)-2-[9H-fluoren-9-ylmethoxycarbonylamino] pent-4-ynoic acid, Fmoc-PRA-OH)(3 g, 8.95 mmol) as a starting material, Compound aa018-b (2.71 g, 87%) was obtained in the same manner as the synthesis of Compound aa004-b.

(387) LCMS (ESI) m/z=348 (M+H)+

(388) Retention time: 0.87 min (Analytical condition SQDFA05)

(389) Using the resulting Compound aa018-b (989 mg, 2.85 mmol), Compound aa018 (986 mg, 99%) was obtained in the same manner as the synthesis of Compound aa004.

(390) LCMS (ESI) m/z=350 (M+H)+

(391) Retention time: 0.79 min (Analytical condition SQDFA05)

(392) Synthesis of Compound aa049

(393) ##STR00281##

(394) Using Compound aa049-a, ((2S)-3-(2-chlorophenyl)-2-[9H-fluoren-9-ylmethoxycarbonylamino]propanoic acid, Fmoc-Phe (2-C1)—OH)(5 g, 11.85 mmol) as a starting material, Compound aa049-b (4.2 g, 82%) was obtained in the same manner as the synthesis of Compound aa004-b.

(395) LCMS (ESI) $m/z=434$ (M+H)+
(396) Retention time: 1.01 min (Analytical condition SQDFA05)
(397) Using the resulting Compound aa049-b (4.2 g), Compound aa049 (3.32 g, 79%) was obtained in the same manner as the synthesis of Compound aa004.
(398) LCMS (ESI) $m/z=436$ (M+H)+
(399) Retention time: 0.94 min (Analytical condition SQDFA05)
(400) Synthesis of Compound aa199
(401) ##STR00282##
(402) Using Compound aa199-a, ((2S)-2-[9H-fluoren-9-ylmethoxycarbonylamino]-3-[4-(trifluoromethyl)phenyl] propanoic acid, Fmoc-Phe (4—CF₃)—OH)(200 g, 439 mmol) as a starting material, Compound aa199-b (206.8 g) was obtained as a crude product in the same manner as the synthesis of Compound aa004-b.
(403) LCMS (ESI) $m/z=469$ (M+H)+
(404) Retention time: 3.30 min (Analytical condition SMD method_03)
(405) Using the resulting Compound aa 199-b (205 g), Compound aa199 (195 g, 2 steps, 95%) was obtained in the same manner as the synthesis of Compound aa004.
(406) LCMS (ESI) $m/z=470$ (M+H)+
(407) Retention time: 2.96 min (Analytical condition SMD Method_03)
(408) Synthesis of Compound aa013
(409) ##STR00283##
(410) Using aa013-a as a starting material, aa013-b (5.29 g, 102%) was obtained as a crude product in the same manner as the synthesis of Compound aa004-b.
(411) LCMS (ESI) $m/z=370$ (M+H)+
(412) Retention time: 0.89 min (Analytical condition SQDFA05)
(413) Compound aa013-b (5.29 g, 14.3 mmol) was dissolved in DCM (125 mL), TES (22.9 mL, 143 mmol) and TFA (36.4 mL, 473 mmol) were added, and the mixture was stirred at 38° C. for 8 hours. The solvent was distilled off under reduced pressure, and the residue was dissolved in TBME (100 mL) and washed with a 1 M aqueous dipotassium hydrogen phosphate solution (50 mL). The aqueous layer was extracted 3 times with TBME, the organic layers were combined, and the solvent was distilled off under reduced pressure. Acetonitrile/n-hexane=½ (100 mL) was added, and the mixture was extracted with a 5% aqueous potassium hydrogen carbonate solution (100 mL). The aqueous layer was acidified with 6 M hydrochloric acid, and extracted twice with TBME (30 mL). The organic layer was dried over sodium sulfate and filtered off, and the solvent was distilled off under reduced pressure. Acetonitrile (100 mL) was added, the mixture was washed with n-hexane (50 mL), and the solvent was distilled off under reduced pressure to give Compound aa013 (4.06 g, 76%).
(414) LCMS (ESI) $m/z=372$ (M+H)+
(415) Retention time: 0.83 min (Analytical condition SQDFA05)
(416) Synthesis of Compound aa030
(417) ##STR00284##
(418) In a nitrogen atmosphere, paraformaldehyde (172 mg, 5.74 mmol) and TFA (1.326 mL, 17.22 mmol) were added to a toluene solution (5.7 mL) of (S)-2-(((9H-fluoren-9-yl)methoxy)carbonyl)amino)-3-(3-cyanophenyl) propanoic acid (aa030-a, Fmoc-Phe (3—CN)—OH) (789 mg, 1.913 mmol), and the mixture was stirred at room temperature for 5 hours and 30 minutes. The reaction solution was concentrated under reduced pressure, diluted with DCM, then washed with a saturated aqueous sodium hydrogen carbonate solution, dried over anhydrous magnesium sulfate, and filtered. The resulting solution was concentrated under reduced pressure to give Compound aa030-b (859 mg) as a crude product. The compound was used in the next reaction without further purification.
(419) In a nitrogen atmosphere, TES (2.75 mL, 17.22 mmol) and TFA (3.98 mL, 51.7 mmol) were

added to a DCE (10 mL) solution of Compound aa030-b (853 mg) at room temperature, and the mixture was stirred at 60° C. for 5 hours. The reaction solution was cooled to room temperature, the solvent was distilled off under reduced pressure, t-butyl methyl ether/n-hexane (1/1) was added to the resulting crude product, and the mixture was extracted 5 times with a saturated aqueous sodium hydrogen carbonate solution. The pH of the resulting aqueous layer was acidified with concentrated hydrochloric acid, and then the mixture was extracted with ethyl acetate 3 times. The organic layer was washed with saturated brine, dried over anhydrous magnesium sulfate, and filtered, and then the solvent was distilled off under reduced pressure to give Compound aa030 (811 mg, 2 steps, 99%). The resulting Compound aa060 was used in peptide synthesis without further purification.

(420) LCMS (ESI) $m/z=427$ (M+H)+

(421) Retention time: 0.83 min (Analytical condition SQDFA05)

(422) Synthesis of Compound aa029

(423) ##STR00285##

(424) Using Compound aa029-a, ((2S)-3-(4-cyanophenyl)-2-(9H-fluoren-9-ylmethoxycarbonylamino) propanoic acid, Fmoc-Phe (4—CN)—OH)(1 g, 2.425 mmol) as a starting material, Compound aa029 (1.14 g, 2 steps, 108%) was obtained as a crude product in the same manner as the synthesis of Compound aa030. The resulting Compound aa029 was used in peptide synthesis without further purification.

(425) LCMS (ESI) $m/z=427$ (M+H)+

(426) Retention time: 0.82 min (Analytical condition SQDFA05)

(427) Synthesis of Compound aa031

(428) ##STR00286##

(429) Using Compound aa031-a, ((2S)-3-(2-cyanophenyl)-2-(9H-fluoren-9-ylmethoxycarbonylamino) propanoic acid, Fmoc-Phe (2—CN)—OH)(1 g, 2.425 mmol) as a starting material, Compound aa031-b was obtained as a crude product in the same manner as the synthesis of Compound aa030-b. Using the resulting Compound aa031-b, the crude product obtained in the same manner as the synthesis of Compound aa030 was purified by reverse phase column chromatography (0.1% aqueous formic acid solution/0.1% formic acid acetonitrile solution) to give Compound aa031 (529 mg, 2 steps, 51%).

(430) LCMS (ESI) $m/z=427$ (M+H)+

(431) Retention time: 0.85 min (Analytical condition SQDFA05)

(432) Synthesis of Compound aa050

(433) ##STR00287##

(434) Using Compound aa050-a, ((2S)-3-(3,4-difluorophenyl)-2-[9H-fluoren-9-ylmethoxycarbonyl(methyl)amino]propanoic acid, Fmoc-Phe (34-F2)—OH)(105 mg, 0.247 mmol) as a starting material, Compound aa050-b was obtained as a crude product in the same manner as the synthesis of Compound aa004-b. Furthermore, using aa050-b, Compound aa050 (74.4 mg, 2 steps, 69%) was obtained in the same manner as the synthesis of Compound aa030. LCMS (ESI) $m/z=438$ (M+H)+

(435) Retention time: 0.88 min (Analytical condition SQDFA05)

(436) Synthesis of Compound aa019

(437) ##STR00288##

(438) Compound aa019-a (5.00 g, 13.8 mmol) was suspended in DCM (46 mL), and paraformaldehyde (2.08 g, 69 mmol) and magnesium sulfate (4.16 g, 34.6 mmol) were added. A boron trifluoride diethyl ether complex (BF₃·Math.OEt₂)(2.10 mL, 16.6 mmol) was added dropwise thereto, and the mixture was stirred at room temperature for 1 hour. Solids were filtered off, and TES (5.51 mL, 34.6 mmol) and water (0.249 mL, 13.8 mmol) were added. BF₃·Math.OEt₂ (2.63 mL, 20.8 mmol) was added dropwise thereto at 0° C., and the mixture was stirred at room temperature for 1 hour. TES (1.10 mL, 6.92 mmol) was added, the mixture was stirred at room

temperature for 40 minutes, BF₃.OEt₂ (0.877 mL, 6.92 mmol) was added, and the mixture was stirred at room temperature. The reaction solution was washed with a saturated aqueous sodium chloride solution (25 mL) and washed with saturated brine (50 mL). The organic layer was dried over sodium sulfate and filtered off, and the solvent was distilled off under reduced pressure. The resulting solids were pulverized and washed 3 times with n-hexane (50 mL) to give Compound aa019 (4.93 g, 95%).

(439) LCMS (ESI) m/z=376 (M+H)+

(440) Retention time: 0.80 min (Analytical condition SQDFA05)

(441) Synthesis of Compound aa331

(442) ##STR00289##

(443) Using aa331-a as a starting material, aa331 (9.09 g, 86%) was obtained in the same manner as the synthesis of Compound aa019.

(444) LCMS (ESI) m/z=382 (M+H)+

(445) Retention time: 0.94 min (Analytical condition SQDFA05)

(446) Synthesis of Compound aa020

(447) ##STR00290##

(448) (S)-Methyl N-tritylaziridine-2-carboxylate (aa020-a, CAS No. 75154-68-6)(50 g, 146 mmol) was added to a mixed solution of chloroform (145 mL) and methanol (145 mL), then TFA (33 mL, 3 eq) was added dropwise at 0° C. in a nitrogen atmosphere, and the mixture was stirred for 7 hours. DIPEA (127 mL, 5 eq) was added to this reaction solution at 0° C., and further, a 1,4-dioxane (145 mL) solution of Fmoc-C1 (36 g, 139 mmol) was added dropwise, and the mixture was stirred at 0° C. for 90 minutes in a nitrogen atmosphere. The reaction solution was concentrated under reduced pressure, diluted with ethyl acetate, and then washed sequentially with water, an aqueous ammonium chloride solution, an aqueous sodium hydrogen carbonate solution, and saturated brine. After the resulting organic layer was dried with anhydrous sodium sulfate, the residue obtained by distilling off the solvent under reduced pressure was purified by silica gel column chromatography (ethyl acetate/petroleum ether, 0/100 to 10/90) to give Compound aa020-b (1-O-(9H-fluoren-9-ylmethyl).sub.2-O-methyl(2S)-aziridine-1,2-dicarboxylate)(40 g, 85%).

(449) LCMS (ESI) m/z=324 (M+H)+

(450) Retention time: 2.631 min (Analytical condition SMD method_10))

(451) In a nitrogen atmosphere, Compound aa020-b, (1-O-(9H-fluoren-9-ylmethyl).sub.2-O-methyl(2S)-aziridine-1,2-dicarboxylate)(5 g, 15.46 mmol) was dissolved in DCM (30.9 mL), cyclopropanol (1.665 mL, 26.3 mmol) was added, and then a boron trifluoride diethyl ether complex (BF₃.Math.OEt₂)(0).291 mL, 2.319 mmol) was added while the mixture was ice-cooled. After the mixture was reacted for 2 hours while being ice-cooled, water and a saturated aqueous sodium hydrogen carbonate solution were added to the reaction solution to quench the reaction, the aqueous layer was removed by a phase separator, and then the organic layer was concentrated under reduced pressure. The resulting residue was purified by silica gel column chromatography (n-hexane/ethyl acetate=4/1) to give Compound aa020-c (4.6 g, 78%).

(452) LCMS (ESI) m/z=382 (M+H)+

(453) Retention time: 0.89 min (Analytical condition SQDFA05)

(454) Calcium chloride (20.08 g, 181 mmol) was dissolved in water (50.2 mL), lithium hydroxide monohydrate (2.024 g, 48.2 mmol) was added thereto, and the mixture was stirred at room temperature for 5 minutes to prepare Aqueous Solution A. Compound aa020-c (4.6 g, 12.06 mmol) was dissolved in isopropanol (201 mL) and THF (50.2 mL), Aqueous Solution A prepared in advance was added, and the mixture was stirred at room temperature for 5 hours. Then, 1 N hydrochloric acid (72 mL) was added to the reaction solution, and then isopropanol and THF were removed by concentration under reduced pressure. The resulting aqueous layer was diluted with water (50).2 mL), and extracted 3 times with ethyl acetate (total amount 100 mL). The organic layer was washed with water and saturated brine, dried over anhydrous sodium sulfate, and filtered,

and the filtrate was concentrated under reduced pressure. The resulting residue was washed with ethyl acetate/n-hexane (1/2, 20 v/w) to give aa020-d, ((2S)-3-cyclopropyloxy-2-(9H-fluoren-9-ylmethoxycarbonylamino) propanoic acid, Fmoc-Ser (cPr)—OH) (3.5 g, 79%).

(455) LCMS (ESI) $m/z=368$ (M+H)+

(456) Retention time: 0.78 min (Analytical condition SQDFA05)

(457) Paraformaldehyde (1.815 g, 60.5 mmol), magnesium sulfate (2.36 g, 19.63 mmol), and a boron trifluoride diethyl ether complex (BF₃·Math.OEt₂)(1.184 mL, 9.42 mmol) were added in a nitrogen atmosphere to a DCM (87 mL) solution of Compound aa020-d, ((2S)-3-cyclopropyloxy-2-(9H-fluoren-9-ylmethoxycarbonylamino) propanoic acid, Fmoc-Ser (cPr)—OH) (2.89 g, 7.85 mmol), and the mixture was stirred at room temperature for 2 hours. A saturated aqueous sodium hydrogen carbonate solution was added to the reaction solution to separate the organic layer and the aqueous layer. The aqueous layer was extracted twice with DCM, the combined organic layer was washed with saturated brine and dried over sodium sulfate, and the solvent was distilled off under reduced pressure to give Compound aa020-e as a crude product (3.1 g, quant.).

(458) LCMS (ESI) $m/z=380$ (M+H)+

(459) Retention time: 0.93 min (Analytical condition SQDFA05)

(460) Triethylsilane (3.13 mL, 19.64 mmol), water (0.141 mL, 7.85 mmol), and a boron trifluoride diethyl ether complex (BF₃·Math.OEt₂)(2.49 mL, 19.6 mmol) were added to DCM (26.2 mL) solution of the resulting Compound aa020-e (2.98 g, 7.85 mmol) under an ice-cold condition in a nitrogen atmosphere, and the mixture was stirred for 2 hours. A saturated aqueous ammonium chloride solution was added to the reaction solution to separate the organic layer. The organic layer was washed with a saturated aqueous ammonium chloride solution, then washed with saturated brine, and concentrated under reduced pressure to give a crude product. The resulting crude product was dissolved in acetonitrile, washed with n-hexane, and the acetonitrile layer was concentrated under reduced pressure to give Compound aa020 (2.71 g, 90%).

(461) LCMS (ESI) $m/z=382$ (M+H)+

(462) Retention time: 0.83 min (Analytical condition SQDFA05)

(463) Synthesis of Compound aa006

(464) ##STR00291##

(465) Using Compound aa006-a, ((2S)-3-cyclopentyl-2-[9H-fluoren-9-ylmethoxycarbonylamino]propanoic acid, Fmoc-Ala (cPent)-OH)(10 g, 26.4 mmol) as a starting material, Compound aa006-b (10.5 g) was obtained as a crude product in the same manner as the synthesis of Compound aa020-e.

(466) LCMS (ESI) $m/z=392$ (M+H)+

(467) Retention time: 1.05 min (Analytical condition SQDFA05)

(468) The resulting Compound aa006-b (10.5 g) was reacted in the same manner as the synthesis of Compound aa020, and then the resulting crude product was purified by reverse phase column chromatography (0.1% aqueous formic acid solution/0.1% formic acid acetonitrile solution) to give Compound aa006 (10.11 g, 2 steps, 96%).

(469) LCMS (ESI) $m/z=394$ (M+H)+

(470) Retention time: 0.98 min (Analytical condition SQDFA05)

(471) Synthesis of Compound aa010

(472) ##STR00292##

(473) Using Compound aa010-a, ((2S)-3-cyclobutyl-2-[9H-fluoren-9-ylmethoxycarbonylamino]propanoic acid, Fmoc-Ala (cBu)-OH)(3.36 g, 9.19 mmol) as a starting material, Compound aa010-b (3.63 g) was obtained as a crude product in the same manner as the synthesis of Compound aa020-e.

(474) LCMS (ESI) $m/z=378$ (M+H)+

(475) Retention time: 1.01 min (Analytical condition SQDFA05)

(476) The resulting Compound aa010-b (3.63 g) was reacted in the same manner as the synthesis of

Compound aa020, and then the resulting crude product was purified by reverse phase column chromatography (0.1% formic acid-water/0.1% formic acid-acetonitrile) to give Compound aa010 (3.18 g, 2 steps, 91%).

(477) LCMS (ESI) $m/z=380$ (M+H)+

(478) Retention time: 0.94 min (Analytical condition SQDFA05)

(479) Synthesis of Compound aa047

(480) ##STR00293##

(481) Using Compound aa047-a, ((2S)-2-[9H-fluoren-9-ylmethoxycarbonylamino]-3-(2-methylphenyl) propanoic acid, Fmoc-Phe (2-Me)-OH)(2 g, 4.98 mmol) as a starting material, a saturated aqueous sodium hydrogen carbonate solution was added to the reaction solution obtained in the same manner as the synthesis of Compound aa020-e, a silica gel column (2 v/w) was charged with the mixture, and this was eluted with DCM to give Compound aa047-b as a crude product (1.45 g). Using the resulting Compound aa047-b, Compound aa047 (1.21 g, 2 steps, 58%) was obtained in the same manner as the synthesis of Compound aa020.

(482) LCMS (ESI) $m/z=416$ (M+H)+

(483) Retention time: 0.94 min (Analytical condition SQDFA05)

(484) Synthesis of Compound aa060

(485) ##STR00294##

(486) Using Compound aa060-a, ((2S)-2-cyclobutyl-2-[9H-fluoren-9-ylmethoxycarbonylamino]acetic acid, Fmoc-Gly (cBu)-OH)(2.5 g, 7.11 mmol) as a starting material, Compound aa075-b was obtained as a crude product in the same manner as the synthesis of Compound aa020-e.

(487) LCMS (ESI) $m/z=364$ (M+H)+

(488) Retention time: 0.97 min (Analytical condition SQDFA05)

(489) The entire amount of Compound aa060-b obtained as mentioned above was reacted in the same manner as the synthesis of Compound aa020, and then the resulting crude product was purified by reverse phase column chromatography (0.1% aqueous formic acid solution/0.1% formic acid acetonitrile solution) to give Compound aa060 (2.32 g, 2 steps, 89%).

(490) LCMS (ESI) $m/z=366$ (M+H)+

(491) Retention time: 0.88 min (Analytical condition SQDFA05)

(492) Synthesis of Compound aa021

(493) ##STR00295## ##STR00296##

(494) 2-Nitrobenzenesulfonyl chloride (NsCl)(32.37 g, 146 mmol) and L-serine methyl ester hydrochloride (aa021-a, 25 g, 161 mmol, CAS No. 5680-80-8) were dissolved in DCM (874 mL), and DIPEA (51 mL, 292 mmol) was added at 5° C. The mixture was stirred at room temperature for 1 hour, then washed twice with water (440 mL), washed once with saturated brine/water (1/1, 440 mL), and then dried over magnesium sulfate. Magnesium sulfate was filtered off, and then the filtrate was concentrated under reduced pressure to give Compound aa021-b as a crude product (39.4 g, 89%).

(495) LCMS (ESI) $m/z=302.9$ (M-H)-

(496) Retention time: 0.729 min (Analytical condition SMD method_06)

(497) Compound aa021-b (23 g, 75.6 mmol) was dissolved in DCM (598 mL), and triphenylphosphine (31.7 g, 121 mmol) was added at room temperature. After the mixture was cooled to -14° C., DEAD (55.0 mL, 121 mmol) was added over 10 minutes, and then the mixture was stirred at -5° C. for 50 minutes. n-Hexane (300 mL) was added, and the precipitate was filtered off. The filtrate was purified by silica gel column chromatography (n-hexane/DCM=50:50 to 0:100) to give Compound aa021-c (13.3 g, 62%).

(498) LCMS (ESI) $m/z=287$ (M+H)+

(499) Retention time: 0.872 min (Analytical condition SMD method_06)

(500) Compound aa021-c (10.7 g, 37.3 mmol) was dissolved in TFE (75 mL), a boron trifluoride

diethyl ether complex (BF₃·OEt₂) (0.469 mL, 3.73 mmol) was added, and then the mixture was stirred at 70° C. for 30 minutes. TFE was distilled off under reduced pressure, and the resulting crude product was purified by reverse phase column chromatography (0.1% formic acid-containing acetonitrile/0.1% formic acid-containing distilled water) to give Compound aa021-d (12.3 g, 85%).

(501) LCMS (ESI) m/z=387 (M+H)⁺

(502) Retention time: 0.72 min (Analytical condition SQDFA05)

(503) Compound aa021-d (12 g, 31.1 mmol) was dissolved in methanol (47 mL), and an aqueous solution (31 mL) of lithium hydroxide monohydrate (5.21 g, 124 mmol) was added. The mixture was stirred at room temperature for 90 minutes, then formic acid (11.7 mL, 311 mmol) was added, and the mixture was diluted with water (30 mL) and then purified by reverse phase column chromatography (0.1% formic acid-containing acetonitrile/0.1% formic acid-containing distilled water) to give Compound aa021-e (7.90 g, 68%).

(504) LCMS (ESI) m/z=373 (M+H)⁺

(505) Retention time: 0.63 min (Analytical condition SQDFA05)

(506) Compound aa021-e (7.72 g, 20.7 mmol) was dissolved in acetonitrile (104 mL), potassium carbonate (7.17 g, 51.8 mmol) and dodecanethiol (7.44 mL, 31.1 mmol) were added, and then the mixture was stirred at room temperature for 74 hours. This was diluted with water (1 mL), and washed twice with TBME (200 mL). A 1,4-dioxane (150 mL) solution of Fmoc-OSu (3.5 g) was added to the resulting aqueous solution, and the mixture was stirred for 25 minutes. Further, a 1,4-dioxane (10 mL) solution of Fmoc-OSu (700 mg) was added, and the mixture was stirred for 5 minutes. Furthermore, a 1,4-dioxane (5 mL) solution of Fmoc-OSu (350 mg) was added, and the mixture was stirred for 5 minutes. Further, a 1,4-dioxane (5 mL) solution of Fmoc-OSu (350 mg) was added, the mixture was stirred for 5 minutes, then formic acid (3.9 mL) was added, and the solvent was distilled off under reduced pressure. The resulting crude product was purified by reverse phase column chromatography (0.1% formic acid-containing acetonitrile/0.1% formic acid-containing distilled water) to give Compound aa () 1-f, ((2S)-2-(9H-fluoren-9-ylmethoxycarbonylamino)-3-(2,2,2-trifluoroethoxy) propanoic acid, Fmoc-Ser (Tfe)-OH) (5.67 g, 67%).

(507) LCMS (ESI) m/z=410 (M+H)⁺

(508) Retention time: 2.35 min (Analytical condition SQDFA05 long)

(509) Using Compound aa021-f (2.00 g, 4.89 mmol) as a starting material, Compound aa021-g was obtained as a crude product in the same manner as the synthesis of Compound aa020-e.

Furthermore, Compound aa021 (1.80 g, 2 steps, 87%) was obtained in the same manner as the synthesis of Compound aa020.

(510) LCMS (ESI) m/z=424 (M+H)⁺

(511) Retention time: 0.84 min (Analytical condition SQDFA05)

(512) ##STR00297##

(513) In a nitrogen atmosphere, Compound aa020-b (1 g, 3.09 mmol) was dissolved in toluene (6.2 mL), ethanol (0.542 mL, 9.28 mmol) was added, and then BF₃·OEt₂ (0.059 mL, 0.464 mmol) was added dropwise over 5 minutes while being ice-cooled. This was stirred for 2.5 hours while being brought back to room temperature. The reaction was quenched by adding a saturated aqueous NaHCO₃ solution, and the aqueous layer was removed by a phase separator. The organic layer was dried over anhydrous sodium sulfate, and the solvent was distilled off under reduced pressure. Ethyl acetate/hexane (3/1, 24 mL) was added to the resulting residue, and the solvent was distilled off under reduced pressure. Hexane/TBME (85/15, 24 mL) was added to the resulting solids, and the mixture was stirred for 30 minutes, and then the solvent was filtered off to give Compound aa022-b (787 mg, 69%).

(514) LCMS (ESI) m/z=370 (M+H)⁺

(515) Retention time: 0.86 min (Analytical condition SQDFA05)

(516) Calcium chloride (2.25 g, 20.3 mmol) was dissolved in H₂O (5.7 mL), lithium

hydroxide monohydrate (227 mg, 5.41 mmol) was added thereto, and the mixture was stirred at room temperature for 5 minutes to prepare Aqueous Solution A.

(517) Compound aa022-b was dissolved in isopropanol (22.6 mL) and THF (57 mL), Aqueous Solution A prepared in advance was added, and the mixture was stirred at room temperature for 7 hours. Then, 1 N hydrochloric acid (8.1 mL) was added to the reaction solution, and then isopropanol and THF were removed by concentration under reduced pressure. The resulting aqueous layer was diluted with H₂O and extracted 3 times with ethyl acetate. The organic layer was washed with H₂O and saturated brine, dried over anhydrous sodium sulfate, filtered, and then concentrated under reduced pressure. The residue was triturated with ethyl acetate/n-hexane (1/5, 20 v/w) to give Compound aa022-c (3.5 g, 79%).

(518) LCMS (ESI) m/z=356 (M+H)⁺

(519) Retention time: 0.78 min (Analytical condition SQDFA05)

(520) Using aa022-c as a starting material, aa022-d (871 mg, 84%) was obtained as a crude product in the same manner as the synthesis of Compound aa004-b.

(521) LCMS (ESI) m/z=368 (M+H)⁺

(522) Retention time: 0.84 min (Analytical condition SQDFA05)

(523) Using aa022-d as a starting material, aa022 (254 mg, 90%) was obtained in the same manner as the synthesis of Compound aa004.

(524) LCMS (ESI) m/z=370 (M+H)⁺

(525) Retention time: 0.80 min (Analytical condition SQDFA05)

(526) Synthesis of Compound aa210

(527) ##STR00298##

(528) Compound aa210-a (5.00 g, 13.7 mmol) was dissolved in toluene (17 mL), TFA (9.49 mL, 123 mmol) and paraldehyde (5.42 mL, 41.0 mmol) were added, and the mixture was stirred at 45° C. for 24 hours. The mixture was cooled to 0° C., toluene (17 mL), TFA (19.0 mL, 246 mmol), and TES (19.6 mL, 123 mmol) were added, and the mixture was stirred at 50° C. overnight. The solvent was distilled off under reduced pressure, ethyl acetate was added, and the mixture was washed with a saturated aqueous sodium hydrogen carbonate solution and washed with saturated brine. The organic layer was dried over sodium sulfate and filtered off, and the solvent was distilled off under reduced pressure. Purification by reverse phase column chromatography (0.1% formic acid-containing acetonitrile/0.1% formic acid-containing distilled water) gave Compound aa210 (2.17 g, 40%).

(529) LCMS (ESI) m/z=394 (M+H)⁺

(530) Retention time: 0.94 min (Analytical condition SQDFA05)

(531) Synthesis of Compound aa201

(532) ##STR00299##

(533) In a nitrogen atmosphere, Compound aa201-a, ((2S)-2-(9H-fluoren-9-ylmethoxycarbonylamino)-3-(4-methylphenyl) propanoic acid, Fmoc-Phe (4-Me)-OH)(5.62 g, 14.0 mmol, CAS No. 199006-54-7) was suspended in DCE (17.5 mL), then paraldehyde (5.61 mL, 42.0 mmol) and TFA (9.65 mL, 126 mmol) were added, and the mixture was stirred at 60° C. for 6 hours. The resulting reaction solution containing Compound aa201-b was directly used in the next step.

(534) LCMS (ESI) m/z=428 (M+H)⁺

(535) Retention time: 1.03 min (Analytical condition SQDFA05)

(536) DCE (17.5 mL), TFA (19.3 mL, 252 mmol), and TES (20.1 mL, 126 mmol) were added to the resulting reaction solution of Compound aa201-b, and the mixture was stirred at 60° C. for 17 hours. After the mixture was cooled to room temperature and concentrated under reduced pressure, the resulting residue was dissolved in ethyl acetate (40 mL). The organic layer was washed with a saturated aqueous sodium hydrogen carbonate solution (40 mL) and saturated brine (40 mL) and dried over anhydrous sodium sulfate, and then the solvent was distilled off under reduced pressure.

The resulting residue was dissolved in acetonitrile (30 mL) and washed twice with hexane (15 mL), and the solvent was distilled off under reduced pressure. The resulting residue was purified by reverse phase column chromatography (0.1% formic acid-containing acetonitrile/0.1% formic acid-containing distilled water) to give Compound aa201, ((2S)-2-[ethyl(9H-fluoren-9-ylmethoxycarbonyl)amino]-3-(4-methylphenyl) propanoic acid, Fmoc-EtPhe (4-Me)-OH)(4.4 g, 2 steps, 73%).

(537) LCMS (ESI) $m/z=430$ (M+H)+

(538) Retention time: 0.95 min (Analytical condition SQDFA05)

(539) Synthesis of Compound aa164

(540) ##STR00300##

(541) In a nitrogen atmosphere, Compound aa164-a, ((2S)-2-(9H-fluoren-9-ylmethoxycarbonylamino)-3-[4-(trifluoromethyl)phenyl] propanoic acid, Fmoc-Phe (4—CF₃)—OH) (4.04 g, 8.87 mmol, CAS No. 247113-86-6) was suspended in DCE (11.1 mL), then anhydrous magnesium sulfate (4.27 g, 35.4 mmol), paraldehyde (3.55 mL, 26.6 mmol), and TFA (6.11 mL, 80 mmol) were added, and the mixture was stirred at 60° C. for 3 hours. Furthermore, anhydrous magnesium sulfate (2.14 g, 17.7 mmol) was added, and the mixture was stirred at 60° C. for 1 hour. The resulting reaction solution containing Compound aa164-b was directly used in the next reaction.

(542) LCMS (ESI) $m/z=482$ (M+H)+

(543) Retention time: 1.04 min (Analytical condition SQDFA05)

(544) DCE (11.1 mL), TFA (12.2 mL, 159 mmol), and triethylsilane (12.7 mL, 80 mmol) were added to the resulting reaction solution containing Compound aa164-b, and the mixture was stirred at 60° C. for 10 hours. The mixture was cooled to room temperature, magnesium sulfate was filtered off, and the mixture was concentrated under reduced pressure. Since the intended reaction was not complete, the resulting residue was dissolved in DCE (22.2 mL), then TFA (18.3 mL, 239 mmol) and triethylsilane (12.7 mL, 80 mmol) were added, and the mixture was stirred at 60° C. for 8 hours. After the mixture was cooled to room temperature and concentrated under reduced pressure, the resulting residue was dissolved in ethyl acetate (40 mL). The organic layer was washed with a saturated aqueous sodium hydrogen carbonate solution (40 mL) and saturated brine (40 mL) and dried over anhydrous sodium sulfate, and then the solvent was distilled off under reduced pressure. The resulting residue was dissolved in acetonitrile (30 mL) and washed twice with hexane (15 mL), and the solvent was distilled off under reduced pressure. The resulting residue was purified by reverse phase column chromatography (0.1% formic acid-containing water/acetonitrile) to give Compound aa164 (1.90 g, 2 steps, 44%).

(545) LCMS (ESI) $m/z=484$ (M+H)+

(546) Retention time: 0.97 min (Analytical condition SQDFA05)

(547) Synthesis of Compound aa136

(548) ##STR00301##

(549) Compound aa136-a (10.1 g, 22.1 mmol) was dissolved in toluene (28 mL), then propionaldehyde (14.3 mL, 199 mmol), TFA (15.3 mL, 199 mmol), and magnesium sulfate (7.99 g, 66.4 mmol) were added, and the mixture was stirred at 60° C. for 3 hours. Solids were filtered off with a silica gel pad, and the filtrate was washed twice with water (100 mL). Acetonitrile (20 mL) was added thereto, and the mixture was washed with a 1 M aqueous dipotassium hydrogen phosphate solution (30 mL), washed 3 times with a 3.5% aqueous potassium hydrogen carbonate solution (40 mL), and washed with saturated brine (100 mL). The mixture was dried over anhydrous magnesium sulfate and filtered off, and the solvent was distilled off under reduced pressure to give Compound aa136-b as a crude product. The crude product of Compound aa136-b was dissolved in toluene (28 mL), and TES (7.07 mL, 44.3 mmol) was added. The mixture was cooled to 0° C., TiCl₄.sub.4 (4.88 mL, 44.3 mmol) was added, and the mixture was stirred at room temperature for 15 minutes. Water (50 mL) was added dropwise, and the organic layer was washed

with water (100 mL) and washed with a 1 M aqueous dipotassium hydrogen phosphate solution (50 mL). n-Hexane (90 mL) was added, and the mixture was extracted 3 times with a mixed solvent of acetonitrile (15 mL) and a 1% aqueous potassium hydrogen carbonate solution (30 mL).

Furthermore, the mixture was extracted twice with a mixed solvent of acetonitrile (20 mL) and a 1% aqueous potassium hydrogen carbonate solution (30 mL). n-Hexane (90 mL) was added to the organic layer, and the mixture was extracted twice with a mixed solvent of acetonitrile (20 mL) and a 1% aqueous potassium hydrogen carbonate solution (30 mL). The aqueous layers were combined, n-hexane (80 mL) was added, phosphoric acid was added to adjust pH to 3, and then the mixture was extracted 3 times with TBME (150 mL). The organic layer was washed with saturated brine (50 mL), dried over anhydrous magnesium sulfate, and filtered off, and the solvent was distilled off under reduced pressure to give Compound aa136 (5.03 g, 46%).

(550) LCMS (ESI) $m/z=498$ (M+H)+

(551) Retention time: 1.02 min (Analytical condition SQDFA05)

(552) Synthesis of Compound aa174

(553) ##STR00302##

(554) Using aa174-a as a starting material, aa174 (18.6 g, 84%) was obtained in the same manner as the synthesis of Compound aa136.

(555) LCMS (ESI) $m/z=460$ (M+NH.sub.4)+

(556) Retention time: 1.41 min (Analytical condition SMD method_04)

(557) Synthesis of Compound aa264

(558) ##STR00303##

(559) Compound aa264-a (H-cisHyp (3)—OH, 950 mg, 7.24 mmol) was dissolved in water (22 mL), and DIPEA (3.47 mL, 19.9 mmol) was added. A 1,4-dioxane solution (14.5 mL) of Fmoc-OSu (2.44 g, 7.24 mmol) was added thereto. The resulting solids were crushed with a spatula, and irradiated with ultrasonic waves for further pulverization. 1,4-Dioxane (15 mL) was added to and dissolve the solids, and the solution was stirred at room temperature for 40 minutes. Washing was performed twice with n-hexane/CPME (3/1, 10 mL), and potassium hydrogen sulfate (3.70 g, 27.2 mmol) was added. The mixture was extracted 3 times with isopropyl acetate (30 mL), the organic layer was washed with 50% brine (30 mL), dried over anhydrous sodium sulfate, filtered off, and the solvent was distilled off under reduced pressure to give Compound aa264-b (2.39 g, 93%).

(560) LCMS (ESI) $m/z=355$ (M+H)+

(561) Retention time: 0.64 min (Analytical condition SQDFA05)

(562) Compound 264-b (2.39 g, 6.76 mmol) and PPTS (0.170 g, 0.676 mmol) were suspended in DCM (23 mL), then DHP (1.39 mL, 15.2 mmol) was added, and the mixture was stirred at room temperature for 19 hours. PPTS (0.085 g, 0.338 mmol) and DHP (0.741 mL, 8.11 mmol) were added, and the mixture was stirred at room temperature for 3 hours. This was washed with water, and washed with saturated brine. The mixture was dried over anhydrous sodium sulfate and filtered off, and the solvent was distilled off under reduced pressure. This was dissolved in THF (25 mL), a phosphate buffer (pH=8.2, 25 mL) was added, and the mixture was stirred at 50° C. for 11 hours. Ethyl acetate (25 mL) was added to remove the aqueous layer, and the aqueous layer was extracted twice with ethyl acetate. The organic layer was dried over anhydrous sodium sulfate and filtered off, and the solvent was distilled off under reduced pressure. This was dissolved in DCM (30 mL), and n-hexane (30 mL) was added. Only DCM was distilled off under reduced pressure, and further, n-hexane was distilled off under reduced pressure. n-Hexane was added to the resulting solids, and this was irradiated with ultrasonic waves, and n-hexane was removed by decantation, and this was dried under reduced pressure to give Compound aa264 as a sodium salt. This was dissolved in isopropyl acetate (50 mL), a 0.05 M aqueous phosphoric acid solution (pH=2, 90 mL) was added, the mixture was stirred at room temperature for 10 minutes, and then the aqueous layer was removed. The aqueous layer was extracted with isopropyl acetate, the organic layer was dried over anhydrous sodium sulfate and filtered off, and the solvent was distilled off under reduced pressure.

The resulting crude product was purified by reverse phase chromatography (0.1% formic acid-containing acetonitrile/0.1% formic acid-containing distilled water) to give Compound aa264 (2.33 g, 79%).

(563) LCMS (ESI) $m/z=456$ ($M+NH_{3.3}$)⁺

(564) Retention time: 0.81 min (Analytical condition SQDFA05)

(565) Synthesis of Compound aa265

(566) ##STR00304##

(567) Using aa265-a as a starting material, aa265 (1.38 g, 78%) was obtained in the same manner as the synthesis of Compound aa264.

(568) LCMS (ESI) $m/z=439$ ($M+H$)⁺

(569) Retention time: 0.85 min (Analytical condition SQDFA05)

(570) Synthesis of Compound aa267

(571) ##STR00305##

(572) Using aa267-a as a starting material, aa267 (9.04 g, quant) was obtained in the same manner as the synthesis of Compound aa264.

(573) LCMS (ESI) $m/z=460$ ($M+Na$)⁺

(574) Retention time: 0.81 min (Analytical condition SQDFA05)

(575) Synthesis of Compound aa279

(576) ##STR00306##

(577) Using aa279-a as a starting material, aa279 (5.15 g, 83%) was obtained in the same manner as the synthesis of Compound aa264.

(578) LCMS (ESI) $m/z=460$ ($M+Na$)⁺

(579) Retention time: 0.85 min (Analytical condition SQDFA05)

(580) Synthesis of Compound aa244

(581) ##STR00307## ##STR00308##

(582) A toluene (500 mL) mixed solution of Compound aa244-a, ((2S)-2-(phenylmethoxycarbonylamino) pentanedioic acid)(50 g, 177.8 mmol), paraformaldehyde (11.87 g), and p-toluenesulfonic acid (1.84 g, 10.69 mmol) was stirred at 120° C. for 16 hours. The reaction solution was cooled to room temperature, diluted with ethyl acetate, washed with saturated brine, and dried over anhydrous sodium sulfate, and the solvent was distilled off under reduced pressure to give Compound aa244-b as a crude product (52 g, 96%). This crude product was used in the next reaction without purification.

(583) After a thionyl chloride (50 mL) solution of Compound aa244-b (4 g, 13.64 mmol) was stirred at 85° C. for 1 hour, the solvent was distilled off under reduced pressure. The residue was dissolved in THF (20 mL), and cooled to -78° C. in a nitrogen atmosphere. A THF (20 mL) solution of lithium tri-tert-butoxyaluminum hydride (2.76 g, 10.87 mmol) was added dropwise thereto over 2.5 hours. After the mixture was stirred at -78° C. for 3 hours, water was added to the reaction solution, and the precipitate was removed by filtration. The filtrate was extracted with ethyl acetate, the organic layer was washed with saturated brine and dried over anhydrous sodium sulfate, and the solvent was distilled off under reduced pressure. The resulting residue was purified by silica gel column chromatography (ethyl acetate/petroleum ether) to give Compound aa244-c (2 g).

(584) ¹H-NMR (400-MHz, CDCl₃) δ 7.41-7.32 (m, 5H), 5.49 (br.s, 1H), 5.27-5.11 (m, 3H), 4.38-4.33 (m, 1H), 2.58-2.19 (m, 4H)

(585) In a nitrogen atmosphere, (diethylamino) sulfur trifluoride (DAST)(780) mg, 4.84 mmol) was added to a DCM (20 mL) solution of Compound aa244-c (450) mg, 1.62 mmol) at 0° C., and the mixture was stirred at room temperature for 16 hours. After water was added to the reaction solution, the reaction solution was diluted with DCM, and washed successively with an aqueous sodium hydrogen carbonate solution and saturated brine. The organic layer was dried over anhydrous sodium sulfate, and the solvent was distilled off under reduced pressure. The resulting

residue was purified by silica gel column chromatography (ethyl acetate/petroleum ether) to give Compound aa244-d (0.35 g, 72%). The compound was mixed with another similarly synthesized lot, and subjected to the next reaction.

(586) ¹H-NMR (400-MHz, CDCl₃) δ 7.42-7.35 (m, 5H), 5.97-5.58 (m, 2H), 5.25-5.16 (m, 3H), 4.50-4.35 (m, 1H), 2.14-1.68 (m, 4H)

(587) ¹⁹F-NMR (400 MHz, CDCl₃) δ -116.562

(588) Compound aa244-d (1 g, 3.34 mmol) and TES (12.63 g, 109 mmol) were dissolved in TFA/DCM (10/10 mL), stirred at room temperature for 4 days, and then the solvent was distilled off under reduced pressure. The residue was diluted with an aqueous sodium hydrogen carbonate solution, washed with ether, and adjusted to pH 3 with 2 N hydrochloric acid. Extraction was performed with DCM, the organic layer was washed with saturated brine, dried over anhydrous sodium sulfate, and the solvent was distilled off under reduced pressure to give Compound aa244-e (0.6 g) as a crude product. This crude product was used in the next reaction without purification.

(589) A methanol (10) mL mixed solution of Compound aa244-e (0.6 g) and palladium carbon (10%, 60 mg) was stirred for 16 hours in a hydrogen atmosphere of about 3 atm. Palladium carbon was removed by filtration, and the solvent was distilled off from the filtrate under reduced pressure to give Compound aa244-f (0.23 g) as a crude product. This crude product was mixed with another similarly synthesized lot without purification, and subjected to the next reaction.

(590) Fmoc-OSu (0.9 g, 1.5 eq) was added to a 1,4-dioxane/water (5/5 mL) mixed solution of Compound aa244-f (0.3 g, 1.79 mmol) and potassium carbonate (745 mg, 5.4 mmol), and the mixture was stirred for 2 hours. The reaction solution was diluted with water, washed with diethyl ether, and adjusted to pH 3 with 2 N hydrochloric acid. The reaction solution was extracted 3 times with ethyl acetate, the combined organic layer was washed with saturated brine and then dried over anhydrous sodium sulfate, and the solvent was distilled off under reduced pressure. The resulting residue was purified by reverse phase column chromatography (0.05% TFA-containing acetonitrile/0.05% TFA-containing distilled water) to give Compound aa244 (0.2 g, 29%). Another similarly synthesized lot was also used in the peptide synthesis in the present Examples.

(591) Retention time: 3.153 min (Analytical condition SMD method_19)

(592) ¹H-NMR (300-MHz, DMSO-d₆) δ 12.94 (br.s, 1H), 7.92-7.88 (d, J=7.2 Hz, 2H), 7.66-7.61 (m, 2H), 7.44-7.31 (m, 4H), 6.35-5.75 (m, 1H), 4.49-4.26 (m, 4H), 2.72 (s, 3H), 1.78-1.65 (m, 4H)

(593) ¹⁹F-NMR (300 MHz, DMSO-d₆) δ -115.730

(594) Synthesis of Compound aa043

(595) ##STR00309##

(596) In a nitrogen atmosphere, aa244-c (18 g, 64.98 mmol) was dissolved in dichloromethane (50 mL), and a carbon tetrachloride (550 mL) solution of phosphorus pentachloride (27 g, 130 mmol) was slowly added dropwise at 0° C. The reaction solution was stirred at room temperature for 16 hours, and then dichloromethane was added for dilution. The resulting solution was washed with water, the aqueous phase was extracted twice with dichloromethane, and then the organic phases were combined and washed with an aqueous sodium hydrogen carbonate solution. The resulting solution was dried over anhydrous sodium sulfate, and then the solvent was distilled off under reduced pressure. The resulting crude product was purified by normal phase column chromatography (hexane/ethyl acetate) to give aa043-a (7.2 g, 33%).

(597) aa043-a (7.2 g, 21.75 mmol) was dissolved in dichloromethane (150 mL), then triethylsilane (83.2 g, 718 mmol) and trifluoroacetic acid (24.8 g, 218 mmol) were added, and the mixture was stirred at room temperature for 4 days. The reaction solution was concentrated under reduced pressure, the resulting residue was dissolved in an aqueous sodium hydrogen carbonate solution, and the aqueous phase was washed with diethyl ether. Subsequently, the resulting solution was adjusted to pH 3 with 2 N hydrochloric acid. The mixed solution was extracted twice with dichloromethane, the organic phase was washed with saturated brine and dried over anhydrous

sodium sulfate, and then the solvent was distilled off under reduced pressure. The resulting crude product aa043-b (5.5 g) was directly used in the next step.

(598) A solution obtained by diluting an acetic acid solution (10 mL) of 33% hydrogen bromide with acetic acid (10 mL) was added to the crude product aa043-b (1.1 g), and the mixture was stirred at room temperature for 2 hours. The reaction mixture was concentrated under reduced pressure, and the residue was diluted with water. The resulting solution was washed with diethyl ether, and the solvent was distilled off under reduced pressure to give a crude product (1.0 g) of aa043-c. Crude product aa043-b was directly used in the next reaction. LCMS (ESI) $m/z=200$ (M+H)+

(599) Retention time: 0.88 min (Analytical condition SMD method_12)

(600) Crude product aa043-c (1.00 g) and potassium carbonate (1.48 g, 10.7 mmol) were dissolved in water (10 mL), and a 1,4-dioxane solution (10 mL) of Fmoc-OSu (1.8 g, 5.36 mmol) was added. After being stirred at room temperature for 2 hours, the reaction solution was diluted with water. The resulting solution was washed with diethyl ether (30 mL), and 2 N hydrochloric acid was added to adjust pH to 3. Subsequently, the mixed solution was extracted 3 times with ethyl acetate, the organic phase was washed with saturated brine and dried over anhydrous sodium sulfate, and the solvent was distilled off under reduced pressure. The resulting concentrate was purified by reverse phase column chromatography (water-acetonitrile) to give aa043 (0.5 g, 27%, 3 steps).

(601) LCMS (ESI) $m/z=422$ (M+H)+

(602) Retention time: 2.3 min (Analytical condition SMD method_13)

(603) Synthesis of Compound aa056

(604) ##STR00310##

(605) In a nitrogen atmosphere, aa056-a (90 g, 443 mmol) and methyl iodide (314 g, 2.21 mol) were dissolved in tetrahydrofuran (3.15 μ L). The mixed solution was cooled to 0° C., and sodium hydride (60%, 77.56 g, 2.21 mol) was added while stirring the mixed solution. The reaction solution was stirred at 25° C. for 3 hours, and then added to ice water. The resulting solution was washed 3 times with t-butyl methyl ether/n-hexane (1/3). Then, 1 N hydrochloric acid was added to the aqueous phase to adjust pH to 2-3, and the mixture was extracted 3 times with ethyl acetate. The resulting organic phase was washed with saturated brine, dried over anhydrous sodium sulfate, and concentrated under reduced pressure to give aa056-b (94 g) as a crude product. The resulting crude product was directly used in the next reaction. LCMS (ESI) $m/z=240$ (M+Na)+

(606) Retention time: 1.1 min (Analytical condition SMD method_14)

(607) The crude product (94 g) of aa056-b was dissolved in dichloromethane (600 mL). This solution was cooled to 0° C., and 4 N HCl/1,4-dioxane (660 mL) was added. The reaction solution was stirred at 25° C. for 3 hours and concentrated under reduced pressure to give a crude product (90 g) of aa056-c.

(608) LCMS (ESI) $m/z=118$ (M+H)+

(609) Retention time: 0.25 min (Analytical condition SMD method_14)

(610) The crude product (90 g) of aa056-c was dissolved in water (560 mL), and potassium carbonate was added to adjust pH to 7. Subsequently, potassium carbonate (149 g, 1.08 mol), Fmoc-OSu (131 g, 389 mmol), and 1,4-dioxane (560 mL) were added to the reaction solution, and the mixture was stirred at 25° C. for 3 hours. Next, the reaction solution was washed with t-butyl methyl ether/n-hexane (1/3), then 1 N hydrochloric acid was added to adjust pH to 2 to 3, and solids were collected by filtration. The resulting solids were washed with water and dried under reduced pressure at 50° C. for 16 hours to give aa056 (130 g, 86% (3 steps)).

(611) LCMS (ESI) $m/z=362$ (M+Na)+

(612) Retention time: 2.0 min (Analytical condition SMD method_15)

(613) Synthesis of Compound aa246

(614) ##STR00311##

(615) In a nitrogen atmosphere, aa246-a (30 g, 113 mmol) was dissolved in tetrahydrofuran (300

mL), and sodium hydride (in oil, 60%, 5.97 g, 249 mmol) was added at 0° C. The reaction solution was stirred at room temperature for 4 hours and then cooled to 0° C., and 3-bromo-1-propene (15.73 g, 130 mmol) was added dropwise. Then, the reaction solution was stirred at room temperature for 2 hours, and ice water was added to quench the reaction. The pH of the resulting mixture was adjusted to 2 with concentrated hydrochloric acid, and then the mixture was extracted with ethyl acetate twice. The organic phase was washed with saturated brine, and dried over anhydrous sodium sulfate. The organic phase was concentrated under reduced pressure to give a crude product (27.3 g) of aa246-b.

(616) LCMS (ESI) $m/z=306$ (M+H)+

(617) Retention time: 2.0 min (Analytical condition SMD method_16)

(618) The crude product (26 g) of aa246-b was dissolved in methanol (260 mL), and a 2 M ammonia-methanol solution (63.9 mL) and 10% palladium carbon (3.3 g) were added. The reaction solution was stirred at room temperature for 16 hours in a hydrogen atmosphere and then filtered. The resulting filtrate was concentrated under reduced pressure to give a crude product (14.2 g) of aa246-c.

(619) LCMS (ESI) $m/z=174$ (M+H)+

(620) Retention time: 0.84 min (Analytical condition SMD method_16)

(621) The crude product (10 g) of aa246-c was dissolved in 1,4-dioxane (100 mL) and water (100 mL), and potassium carbonate (23.9 g, 172 mmol) was added. Fmoc-OSu (17.4 g, 51.7 mmol) was added to the reaction solution, and the mixture was stirred at room temperature for 16 hours and then filtered. The resulting filtrate was washed 5 times with t-butyl methyl ether/hexane (1/3), and the aqueous phase was adjusted to pH 2 with concentrated hydrochloric acid. The resulting solution was extracted twice with ethyl acetate. Subsequently, the organic phase was washed with saturated brine, dried over anhydrous sodium sulfate, and concentrated under reduced pressure to give aa246 (15.3 g, 58%, 3 steps).

(622) LCMS (ESI) $m/z=396$ (M+H)+

(623) Retention time: 2.0 min (Analytical condition SMD method_11)

(624) Synthesis of Compound aa281

(625) ##STR00312##

(626) In a nitrogen atmosphere, aa246-a (10 g, 43.2 mmol) was dissolved in tetrahydrofuran (150 mL) and cooled to -10° C., and then sodium hydride (60%, 3.8 g, 95.1 mmol) was added. The reaction solution was stirred at room temperature for 1 hour. Subsequently, methyl iodide (7.4 g, 51.9 mmol) was added dropwise at room temperature. The reaction solution was heated to 50° C., and stirred at 50° C. for 16 hours. Then, a saturated aqueous sodium hydrogen carbonate solution was added to the reaction solution. The aqueous phase was washed with diethyl ether (150 mL), and 1 N hydrochloric acid was added to adjust pH to 3. The resulting mixture was extracted twice with ethyl acetate (200 mL). The organic phase was washed with saturated brine and a 5% aqueous sodium thiosulfate solution, dried over anhydrous sodium sulfate, and then concentrated under reduced pressure to give a crude product (11.1 g) of aa281-a.

(627) LCMS (ESI) $m/z=268$ (M+Na)+

(628) Retention time: 0.75 min (Analytical condition SQDFA05)

(629) The crude product (11.1 g) of aa281-a was dissolved in dichloromethane (110 mL), and 4 N HCl/1,4-dioxane (56.6 mL, 227 mL) was added. The reaction solution was stirred at room temperature for 2 hours, and then the solvent was distilled off under reduced pressure. The resulting residue was suspended in diethyl ether (110 mL), and solids were collected by filtration to give aa281-b (7.4 g, 94%, 2 steps).

(630) LCMS (ESI) $m/z=146$ (M+H)+

(631) Retention time: 0.14 min (Analytical condition SQDFA05)

(632) aa281-b (7.4 g, 40.7 mmol) was dissolved in water (70 mL), and potassium carbonate was added to adjust pH to 7. Then, 1,4-dioxane (70 mL), potassium carbonate (11.3 g, 81.5 mmol), and

Fmoc-OSu (12.4 g, 36.7 mmol) were added thereto at room temperature. After the reaction solution was stirred at room temperature for 2 hours, the aqueous phase was washed 3 times with t-butyl methyl ether-n-hexane (1/3, 70 mL). Subsequently, 1 N hydrochloric acid was added to the aqueous phase to adjust pH to 3, and the mixture was extracted with t-butyl methyl ether (140 mL). The organic phase was washed with saturated brine (70 mL), and dried over anhydrous sodium sulfate. The resulting organic phase was concentrated under reduced pressure to give aa281 (15 g, 98%).

(633) LCMS (ESI) $m/z=368$ (M+H)+

(634) Retention time: 1.79 min (Analytical condition SMD method_11)

(635) Synthesis of Compound aa250

(636) ##STR00313## ##STR00314##

(637) In a nitrogen atmosphere, aa250-a (233 g, 1.14 mol) was dissolved in DMF (1500 mL), and NaH (in oil, 60%, 100 g, 4.17 mol) was added while ice-cooling the mixture. The reaction solution was stirred at room temperature for 3 hours and cooled to 0° C., and a DMF (500 mL) solution of 3-bromo-1-propene (130 g, 1.07 mol) was added dropwise. Subsequently, the reaction solution was stirred at room temperature for 16 hours, and ice water was added to quench the reaction. The resulting solution was adjusted to pH 2 with concentrated hydrochloric acid. The mixed solution was extracted twice with ethyl acetate, the organic phase was washed with saturated brine, dried over anhydrous sodium sulfate, and concentrated under reduced pressure. The resulting residue was purified by silica gel column chromatography (ethyl acetate-n-hexane) to give aa250-b (200 g, 72%).

(638) LCMS (ESI) $m/z=246$ (M+H)+

(639) Retention time: 1.2 min (Analytical condition SMD method_14)

(640) aa250-b (300 g, 1.22 mol) was dissolved in methanol (2 μ L) and dissolved in a 2 M ammonia-methanol solution (918 mL) and methanol (2000 mL), and 10% palladium carbon (60 g) was added. The reaction solution was stirred at room temperature for 16 hours in a hydrogen atmosphere at 10 atm. The reaction solution was filtered and then concentrated under reduced pressure to give a crude product (290 g) of aa250-c.

(641) LCMS (ESI) $m/z=248$ (M+H)+

(642) Retention time: 1.2 min (Analytical condition SMD method_14)

(643) The crude product (100 g) of aa250-c was dissolved in 1,4-dioxane (1500 mL), and 12 N hydrochloric acid (200 mL) was added. The reaction solution was stirred at room temperature for 4 hours, and then the volatile organic solvent was distilled off under reduced pressure to give an aqueous solution of aa250-d. The resulting aqueous solution was adjusted to pH 7 with potassium carbonate, and then 1,4-dioxane (500 mL), Fmoc-OSu (115 g, 0.34 mmol), and potassium carbonate (104.5 g, 0.76 mol) were added. The reaction solution was stirred at room temperature for 16 hours and then filtered. The filtrate was washed 5 times with t-butyl methyl ether/hexane (1/3), and then concentrated hydrochloric acid was added to adjust pH to 1. The resulting solution was extracted twice with ethyl acetate, the organic phase was dried over anhydrous sodium sulfate, and concentrated under reduced pressure to give aa250-e (90 g, 58%, 3 steps).

(644) LCMS (ESI) $m/z=370$ (M+H)+

(645) Retention time: 1.3 min (Analytical condition SMD method_14)

(646) aa250-e (110) g, 298 mmol) was dissolved in toluene (1000 mL), and p-toluenesulfonic acid (3 g, 17.4 mmol) and paraformaldehyde (27 g, 893 mmol) were added. The reaction solution was stirred at 110° C. for 16 hours. The reaction solution was cooled, washed 4 times with an aqueous sodium hydrogen carbonate solution, and dried over anhydrous sodium sulfate. The resulting organic phase was concentrated under reduced pressure to give a crude product (90) g) of aa250-f.

(647) LCMS (ESI) $m/z=382$ (M+H)+

(648) Retention time: 2.9 min (Analytical condition SMD method_16)

(649) The crude product (60 g) of aa250-f was dissolved in dichloromethane (700 mL), then triethylsilane (59 g, 507 mmol) and trifluoroacetic acid (700 mL, 1.41 mol) were added, and the

mixture was stirred at room temperature for 48 hours. Then, the reaction solution was concentrated, and the resulting concentrate was dissolved in water (50 mL). The resulting aqueous solution was neutralized with a saturated aqueous potassium carbonate solution to adjust pH to 8. Subsequently, the mixture was washed 4 times with n-hexane, the resulting solution was adjusted to pH 2 with concentrated hydrochloric acid, and extracted twice with t-butyl methyl ether. The organic phase was dried over anhydrous sodium sulfate and concentrated under reduced pressure to give aa250) (50 g, 66%, 2 steps).

(650) LCMS (ESI) $m/z=384$ (M+H)+

(651) Retention time: 2.0) min (Analytical condition SMD method_11)

(652) ##STR00315##

(653) Fmoc-Glu-OtBu ((4S)-4-(9H-fluoren-9-ylmethoxycarbonylamino)-5-[(2-methylpropan-2-yl)oxy]-5-oxopentanoic acid, CAS No. 84793-7-7)(175 g, 411 mmol), N-hydroxyphthalimide (67.1 g, 411 mmol), WSCI.Math.HCl (78.85 g, 411 mmol), and DMAP (2.51 g, 20.6 mmol) were added to DMF (2 μ L), and the mixture was stirred at room temperature for 16 hours. A 1 N hydrochloric acid solution was added to the reaction solution, and the reaction mixture was extracted with TBME. The organic layers were combined, sequentially washed with water, saturated aqueous sodium hydrogen carbonate solution/water (1/1), and saturated brine, and dried over anhydrous sodium sulfate, and then the solvent was distilled off under reduced pressure. The resulting residue was purified by silica gel column chromatography (ethyl acetate/petroleum ether, 0/100 to 30/70) to give Compound aa220-a (1-O-tert-butyl 5-O-(1,3-dioxoisindol-2-yl)(2S)-2-(9H-fluoren-9-ylmethoxycarbonylamino) pentanedioate)(105 g, 45%).

(654) LCMS (ESI) $m/z=593.5$ (M+Na)+

(655) Retention time: 2.374 min (Analytical condition SMD method_09) .sup.1H-NMR (300-MHz, DMSO-d.sub.6) δ 8.00-7.91 (m, 4H), 7.85 (dd, J=23.0, 7.6 Hz, 3H), 7.75 (d, J=7.4 Hz, 2H), 7.75-7.32 (m, 4H), 4.60-4.31 (m, 2H), 4.30-4.02 (m, 2H), 3.01-2.74 (m, 2H), 2.26-1.86 (m, 2H), 1.41 (s, 9H)

(656) In a nitrogen atmosphere, N,N-dimethylacetamide (35.1 mL) was added to nickel (II) bromide trihydrate (0.76 g, 2.8 mmol) and 4,4'-di-tert-butyl-2,2'-bipyridine (0.75 g, 2.8 mmol), and the mixture was stirred at room temperature for 10 minutes (sol A). Then, sol A and chlorotrimethylsilane (0.9 mL, 7.0 mmol) were added to an N,N-dimethylacetamide (35.1 mL) solution of aa220-a (8.0 g, 14.0 mmol), 5-bromo-2-chloro-1,3-difluorobenzene (4.5 mL, 35.1 mmol), and zinc (4.6 g, 70.1 mmol), and the mixture was stirred at room temperature for 30 minutes. tert-Butyl methyl ether (250 mL) was added, the mixture was passed through a filter obtained by coating Celite with sodium sulfate, and then a 5 wt % aqueous disodium ethylenediaminetetraacetate solution (250 mL) was added for extraction. The organic layer was washed twice with a 5 wt % aqueous sodium carbonate solution (150 mL), washed with a saturated aqueous ammonium chloride solution (150 mL), dried over anhydrous sodium sulfate, and then the solvent was distilled off under reduced pressure. The resulting concentrate was purified by reverse phase chromatography (0.1% formic acid-containing acetonitrile/0.1% formic acid-containing distilled water) to give aa220-b (4.75 g, 64%).

(657) Retention time: 1.18 min (Analytical condition SQDFA05)

(658) aa220-b (4.75 g, 9.0 mmol) was dissolved in 2,2,2-trifluoroethanol (90.0 mL), chlorotrimethylsilane (3.4 mL, 27.0 mmol) was added at room temperature, the mixture was stirred for 15 minutes, and then the solvent was distilled off under reduced pressure. A 20:1 mixed solution (100 mL) of n-hexane and tert-butyl methyl ether was added to the resulting concentrate, then the mixture was filtered under reduced pressure, and the resulting residue was washed twice with a 40:1 mixed solution (100 mL) of n-hexane and tert-butyl methyl ether and dried under reduced pressure to give aa220 (3.92 g, 92%).

(659) LCMS (ESI) $m/z=470$ (M-H)-

(660) Retention time: 0.99 min (Analytical condition SQDFA05)

(661) Synthesis of Compound aa023
(662) ##STR00316##
(663) Fmoc-MeAsp-OtBu ((3S)-4-tert-butoxy-3-[9H-fluoren-9-ylmethoxycarbonyl(methyl)amino]-4-oxo-butanoic acid, CAS No. 2271413-88-6)(300 g, 729 mmol) and N-hydroxyphthalimide (130.8 g, 801.8 mmol) were added to THF (3 μ L), and further, DCC(181.8 g, 1.1 mol) was added, and the mixture was stirred at room temperature for 3 hours. The solvent was distilled off from the reaction solution under reduced pressure, toluene was added to the residue, and solids were removed by filtration. The solvent was distilled off from the filtrate under reduced pressure, and the resulting residue was purified by silica gel column chromatography (hexane/ethyl acetate) to give Compound aa023-a (272 g, 67%).
(664) LCMS (ESI) m/z=578.7 (M+Na)+
(665) Retention time: 1.728 min (Analytical condition SMD method_08)
(666) Using the resulting aa023-a, aa023-b (12.0 g, 51%) was obtained in the same manner as the synthesis of Compound aa220-b using 3-bromothiophene in place of 5-bromo-2-chloro-1,3-difluorobenzene.
(667) LCMS (ESI) m/z=486 (M+Na)+
(668) Retention time: 1.23 min (Analytical condition SMD method_20)
(669) Using the resulting aa023-b, aa023 (8.8 g, 84%) was obtained under the same reaction conditions as the synthesis of Compound aa220.
(670) LCMS (ESI) m/z=408 (M+H)+
(671) Retention time: 0.86 min (Analytical condition SQDFA05)
(672) Synthesis of Compound aa229
(673) ##STR00317##
(674) Using aa220-a as a starting material, aa229-a (3.83 g, 43%) was obtained in the same manner as the synthesis of Compound aa220-b using 5-bromobenzothiophene in place of 5-bromo-2-chloro-1,3-difluorobenzene.
(675) LCMS (ESI) m/z=515 (M+H)+
(676) Retention time: 1.17 min (Analytical condition SQDFA05)
(677) Using the resulting aa229-a, aa229 (3.05 g, 89%) was obtained under the same reaction conditions as the synthesis of Compound aa220.
(678) LCMS (ESI) m/z=458 (M+H)+
(679) Retention time: 0.96 min (Analytical condition SQDFA05)
(680) Synthesis of Compound aa233
(681) ##STR00318##
(682) Using aa220-a as a starting material, aa233-a (1.08 g, 67%) was obtained in the same manner as the synthesis of Compound aa220-b using 4-bromo-2-methyl-1-(trifluoromethyl)benzene in place of 5-bromo-2-chloro-1,3-difluorobenzene.
(683) LCMS (ESI) m/z=541 (M+H)+
(684) Retention time: 1.20 min (Analytical condition SQDFA05)
(685) Using the resulting aa233-a, aa233 (0.65 g, 67%) was obtained under the same reaction conditions as the synthesis of Compound aa220.
(686) LCMS (ESI) m/z=484 (M+H)+
(687) Retention time: 1.01 min (Analytical condition SQDFA05)
(688) Synthesis of Compound aa235
(689) ##STR00319##
(690) Using aa220-a as a starting material, aa235-a (4.73 g, 69%) was obtained in the same manner as the synthesis of Compound aa220-b using 4-bromo-2-methoxy-1-(trifluoromethyl)benzene in place of 5-bromo-2-chloro-1,3-difluorobenzene.
(691) Retention time: 1.15 min (Analytical condition SQDFA05)
(692) Using the resulting aa235-a, aa235 (3.91 g, 92%) was obtained under the same reaction

conditions as the synthesis of Compound aa220.

(693) LCMS (ESI) $m/z=500$ (M+H)+

(694) Retention time: 0.96 min (Analytical condition SQDFA05)

(695) Synthesis of Compound aa217

(696) ##STR00320##

(697) DIC(138 mL, 1.54 eq) was added dropwise at () in a nitrogen atmosphere to a THF (2 μ L) solution of (4S)-4-[(2-methylpropan-2-yl)oxycarbonylamino]-5-oxo-5-phenylmethoxypentanoic acid (Boc-Glu-OBn, CAS number 30924-93-7)(200 g, 592.82 mmol), N-hydroxyphthalimide (106 g, 649.78 mmol, 1.10 eq), and DMAP (3.6 g, 29.47 mmol, 0.05 eq). The reaction solution was stirred at 25° C. for 16 hours, solid matter was removed by filtration, and the solvent was distilled off from the filtrate under reduced pressure. The residue was diluted with toluene, the resulting solids were removed by filtration, and the solvent was distilled off from the filtrate under reduced pressure. The residue was purified by recrystallization (acetone/heptane) to give Compound aa217-a (1-O-benzyl 5-O-(1,3-dioxoisindol-2-yl)(2S)-2-[(2-methylpropan-2-yl)oxycarbonylamino] pentanedioate)(230 g, 80%).

(698) LCMS (ESI) $m/z=505.2$ (M+Na)+

(699) Retention time: 0.992 min (Analytical condition SMD method_16) Nickel bromide trihydrate (NiBr.sub.2: 3H.sub.2O)(13.5 g, 49.7 mmol, 0.3 eq) and 4,4'-di-tert-butyl-2,2'-bipyridyl(dtbbpy, CAS No. 72914-19-3)(13.3 g, 49.7 mmol, 0.3 eq) were added to DMA (400 mL), and the mixture was stirred at 50° C. for 3 hours in a nitrogen atmosphere to prepare a Ni solution.

(700) A DMA (400 mL) mixed solution of Compound aa217-a, (1-O-benzyl 5-O-(1,3-dioxoisindol-2-yl)(2S)-2-[(2-methylpropan-2-yl)oxycarbonylamino] pentanedioate)(80 g, 166 mmol), zinc powder (54.2 g, 829 mmol, 5 eq), and 4-bromo-1,3-difluoro-2-(trifluoromethyl)benzene (CAS No. 156243-64-0), 86.6 g, 332 mmol, 2 eq) was stirred at room temperature for 1 hour in a nitrogen atmosphere, the Ni solution prepared in advance was added, and the mixture was stirred at room temperature for 16 hours. An aqueous EDTA.Math.2Na solution (800 mL, 10%) was added to the reaction solution, and solids were removed by filtration. The filtrate was extracted with ethyl acetate, the combined organic layer was washed with saturated brine and dried over anhydrous sodium sulfate, and the solvent was distilled off under reduced pressure. The resulting residue was purified by silica gel column chromatography (ethyl acetate/petroleum ether) to give Compound aa217-b (57.2 g, 69%).

(701) LCMS (ESI) $m/z=496$ (M+Na)+

(702) Retention time: 1.544 min (Analytical condition SMD method_15)

(703) A toluene mixture (690 mL) of Compound aa217-b (57.2 g, 121 mmol) was cooled to 0° C., and trifluoromethanesulfonic acid (TfOH)(54.4 g, 362 mmol, 3 eq) was added dropwise. After 1 hour of stirring at room temperature, water (58 mL.) was added. The mixed solution was extracted with water, and the combined aqueous layer was extracted with ethyl acetate. The combined organic layer was washed with water and dried over anhydrous sodium sulfate, and the solvent was distilled off under reduced pressure to give 60 g of a residue. Acetonitrile/water (400/400 mL) was added to the residue, and pH was adjusted to 7 with an aqueous sodium hydroxide solution (48%). Fmoc-OSu (36.6 g, 108.6 mmol, 0.9 eq) was added to this solution, and pH was adjusted to 8.0 with an aqueous sodium hydroxide solution (48%), and then the mixture was stirred at room temperature for 16 hours. The reaction solution was filtered while being washed with acetonitrile/water (1/1) to remove solid components. The filtrate was diluted with acetonitrile and acidified with 6 N hydrochloric acid, and precipitated solids were collected by filtration to give Compound aa217 (52 g, 83%).

(704) LCMS (ESI) $m/z=528$ (M+Na)+

(705) Retention time: 3.538 min (Analytical condition SMD method_14)

(706) .sup.1H-NMR (300-MHz, DMSO-d.sub.6) δ 12.69 (s, 1H), 7.90 (d, J=7.5 Hz, 2H), 7.78-7.54

(m, 3H), 7.48-7.20 (m, 6H), 4.33 (d, J=6.3 Hz, 2H), 4.24 (t, J=6.9 Hz, 1H), 3.97-3.84 (m, 1H), 2.79-2.65 (m, 2H), 2.15-2.00 (m, 1H), 2.00-1.83 (m, 1H)

(707) Synthesis of Compound aa218

(708) ##STR00321##

(709) Nickel bromide trihydrate (NiBr.sub.2: 3H.sub.2O)(4 g, 0.07 eq) and 4,4'-di-tert-butyl-2,2'-bipyridyl(dtbbpy, CAS No. 72914-19-3)(3.9 g, 14.55 mmol, 0.07 eq) were added to DMA (500 mL), and the mixture was stirred at 50° C. for 2 hours in a nitrogen atmosphere to prepare a Ni solution.

(710) The Ni solution prepared in advance was added to a DMA (500 mL) mixed solution of Compound aa217-a (100 g, 207.3 mmol), zinc powder (70 g, 5 eq), and 4-bromo-2-chloro-1-(trifluoromethyl)benzene (CAS No. 467435-07-0), 160 g, 617 mmol, 3 eq), and the mixture was stirred at 25° C. for 16 hours. An aqueous EDTA.Math.2Na solution (10 %) was added to the reaction solution, and the mixture was extracted with ethyl acetate. The combined organic layer was washed with saturated brine and dried over anhydrous sodium sulfate, and the solvent was distilled off under reduced pressure. The resulting residue was purified by silica gel column chromatography (ethyl acetate/petroleum ether) to give Compound aa218-b (75 g, 77%).

(711) LCMS (ESI) m/z=494 (M+Na)+

(712) Retention time: 2.863 min (Analytical condition SMD method_17)

(713) A toluene solution (900 mL) of Compound aa218-b (75 g, 158.93 mmol) was cooled to (), and trifluoromethanesulfonic acid (TfOH)(42 mL, 3.00 eq) was added dropwise. After 1 hour of stirring at room temperature, water (75 mL) was added. The mixed solution was extracted with water, and the combined aqueous layer was extracted with ethyl acetate. The combined organic layer was washed with water and dried over anhydrous sodium sulfate, and the solvent was distilled off under reduced pressure. Acetonitrile/water (900/900 mL) was added to the residue, and pH was adjusted to 7 with an aqueous sodium hydroxide solution (48%). Fmoc-OSu (51.2 g, 151.93 mmol, 0.95 eq) was added to this solution, and the mixture was stirred at room temperature for 16 hours while maintaining pH at 7.8 to 8.0. The reaction solution was filtered, and the filtrate was adjusted to pH 2 with 6 N hydrochloric acid. The precipitated solids were collected and dried at 50° C. to give Compound aa218 (70 g, 87%).

(714) LCMS (ESI) m/z=526 (M+Na)+

(715) Retention time: 2.180 min (Analytical condition SMD method_21)

(716) .sup.1H-NMR (300-MHz, DMSO-d.sub.6) δ 12.70 (s, 1H), 7.91 (d, J=7.5 Hz, 2H), 7.79-7.59 (m, 5H), 7.45-7.28 (m, 5H), 4.40-4.19 (m, 3H), 3.96-3.88 (m, 1H), 2.82-2.60(m, 2H), 2.11-1.77 (m, 2H)

(717) Synthesis of Compound aa219

(718) ##STR00322##

(719) Nickel bromide trihydrate (NiBr.sub.2: 3H.sub.2O)(71.5 g, 263 mmol, 0.3 eq) and 4,4'-di-tert-butyl-2,2'-bipyridyl(dtbbpy, CAS Number 72914-19-3)(70.56 g, 263 mmol, 0.3 eq) were added to DMA (2 μL), and the mixture was stirred at 50° C. for 3 hours in a nitrogen atmosphere to prepare a Ni solution.

(720) A DMA (2 μL) mixed solution of Compound aa220-a (1-O-tert-butyl 5-O-(1,3-dioxoisindol-2-yl)(2S)-2-(9H-fluoren-9-ylmethoxycarbonylamino) pentanedioate)(500 g, 876 mmol), zinc powder (287 g, 4.38 mol, 5 eq), and 4-bromo-2-fluoro-1-(trifluoromethyl)benzene (CAS No. 142808-15-9, 425.87 g, 1.753 mol, 2 eq) was stirred at room temperature for 1 hour in a nitrogen atmosphere. The Ni solution prepared in advance was added to this mixed solution, and the mixture was stirred at room temperature for 16 hours. An aqueous EDTA.Math.2Na solution (4 μL, 10%) was added to the reaction solution, and solids were removed by filtration while being washed with ethyl acetate. The filtrate was washed with saturated brine and then dried over anhydrous sodium sulfate, and the solvent was distilled off under reduced pressure. The resulting residue was purified by silica gel column chromatography (ethyl acetate/petroleum ether) to give Compound aa219-b

(230 g, 43%).

(721) LCMS (ESI) $m/z=566$ (M+Na)+

(722) Retention time: 1.317 min (Analytical condition SMD method_18)

(723) A trifluoroethanol (TFE)(2.3 μ L) mixed solution of the resulting Compound aa219-b (230 g, 423 mmol) and chlorotrimethylsilane (TMSCI)(137.9 g, 1.269 mol) was stirred at room temperature for 1 hour, and the precipitated solids were collected by filtration. The operation of dissolving the resulting solids in TBME and distilling off the solvent under reduced pressure was repeated several times to give the intended Compound aa219 (190 g, 90%).

(724) LCMS (ESI) $m/z=510$ (M+Na)+

(725) Retention time: 1.585 min (Analytical condition SMD method_13)

(726) .sup.1H-NMR (300-MHz, DMSO-d.sub.6) δ 12.69 (s, 1H), 7.90 (d, $J=7.5$ Hz, 2H), 7.81-7.66 (m, 4H), 7.47-7.37 (m, 3H), 7.37-7.29 (m, 2H), 7.25 (d, $J=8.1$ Hz, 1H), 4.44-4.14 (m, 3H), 3.97-3.84 (m, 1H), 2.80-2.63 (m, 2H), 2.12-1.81 (m, 2H)

(727) Synthesis of Compound aa239

(728) ##STR00323##

(729) aa239-a (2 g, 8.72 mmol) was dissolved in 1,4-dioxane (2 mL) at room temperature, and 4 N HCl/dioxane (4.36 mL, 17.45 mmol) was added. The reaction mixture was stirred at room temperature for 17.5 hours, the solvent was distilled off under reduced pressure, and the resulting crude product was dissolved in water and freeze-dried to give aa239-a (1.49 g, quant.).

(730) Water (22.5 mL) and DIPEA (5.89 mL, 33.7 mmol) were added to aa239-a (1.49 g, 9 mmol) at room temperature to dissolve it, 1,4-dioxane (15 mL) was added, then N-(9-fluorenylmethoxycarbonyloxy) succinimide (3.03 g, 9 mmol) was added, and the mixture was stirred at room temperature for 15 minutes. Hexane-CPME (3:1)(10 mL) was added to the reaction mixture, and the separated aqueous phase was washed twice with hexane-CPME (3:1) (10 mL). Potassium hydrogen sulfate (4.59 g, 33.7 mmol) and ethyl acetate were added to the aqueous phase, the separated organic phase (about 60 mL) was washed twice with brine (10 mL) and dried over anhydrous sodium sulfate, then the solvent was distilled off under reduced pressure, and the resulting crude product was purified by reverse phase chromatography (0.1% formic acid-containing acetonitrile/0.1% formic acid-containing distilled water) to give aa239 (2.89 g, 91%).

(731) LCMS (ESI) $m/z=352$ (M+H)+

(732) Retention time: 0.81 min (Analytical condition SQDFA05)

(733) Synthesis of Compound aa098

(734) ##STR00324##

(735) Using aa098-a as a starting material, aa098-b (2.30 g, quant.) was obtained in the same manner as the synthesis of Compound aa239-b.

(736) LCMS (ESI) $m/z=116$ (M+H)+

(737) Retention time: 0.16 min (Analytical condition SQDFA05)(retention time of MS peak is described)

(738) Using the resulting aa098-b, aa098 (2.73 g, 87%) was obtained under the same reaction conditions as the synthesis of Compound aa239.

(739) LCMS (ESI) $m/z=338$ (M+H)+

(740) Retention time: 0.73 min (Analytical condition SQDFA05)

(741) Synthesis of Compound aa111

(742) ##STR00325##

(743) In a nitrogen atmosphere, dehydrated THF (40 mL) was added to sodium hydride (1.47 g, 36.7 mmol: calculated as having 60 wt %) cooled in an ice bath. A dehydrated THF (10 mL) solution of aa111-a (3 g, 12.23 mmol) and methyl iodide (4.59 mL, 73.4 mmol) was added dropwise to the resulting suspension. The reaction mixture was stirred for 14 hours from an ice-cooled temperature to 26° C., and then water (2 mL) was added while the mixture was ice-cooled. The resulting mixture was stirred at room temperature for 10 minutes, and then the solvent was

distilled off under reduced pressure. Hexane-CPME (20:1)(20 mL) and water (20 mL) were added to the concentrated residue for extraction, and the aqueous phase (about 30 mL) was washed twice with hexane-CPME (20:1)(20 mL). Potassium hydrogen sulfate (4.1 g, 30.1 mmol) and ethyl acetate were added to the aqueous phase, the separated organic phase (about 50 mL) was washed twice with brine (20 mL) and dried over anhydrous sodium sulfate, and the solvent was distilled off under reduced pressure to give aa111-b (2.66 g, 84%). LCMS (ESI) $m/z=282$ (M+H)⁺ (744) Retention time: 0.62 min (Analytical condition SQDFA05)

(745) Using aa111-b as a starting material, aa111-c (1.55 g, 77%) was obtained in the same manner as the synthesis of Compound aa239-b.

(746) LCMS (ESI) $m/z=160$ (M+H)⁺

(747) Retention time: 0.17 min (Analytical condition SQDFA05)(retention time of MS peak is described)

(748) Using the resulting aa111-c, aa111 (0.837 g, 28%) was obtained under the same reaction conditions as the synthesis of Compound aa239.

(749) LCMS (ESI) $m/z=383$ (M+H)⁺

(750) Retention time: 0.82 min (Analytical condition SQDFA05)

(751) Synthesis of Compound aa268

(752) ##STR00326##

(753) Water (10 mL) and DIPEA (4.81 mL, 27.5 mmol) were added to aa268-a (1 g, 7.87 mmol) at room temperature to dissolve it, 1,4-dioxane (15 mL) was added, then N-(9-fluorenylmethoxycarbonyloxy) succinimide (2.65 g, 7.87 mmol) was added, and the mixture was stirred at room temperature for 30 minutes. Water (15 mL) and hexane (10 mL) were added to the reaction mixture, and the separated aqueous phase was washed twice with hexane-CPME (3:1)(15 mL). Potassium hydrogen sulfate (3.75 g, 27.5 mmol) and ethyl acetate were added to the aqueous phase, the separated organic phase (about 60 mL) was washed twice with brine (15 mL) and dried over anhydrous sodium sulfate, and the solvent was distilled off under reduced pressure to give aa268 (2.85 g, quant).

(754) LCMS (ESI) $m/z=350$ (M+H)⁺

(755) Retention time: 0.78 min (Analytical condition SQDFA05)

(756) Synthesis of Compound aa099

(757) ##STR00327##

(758) Using aa099-a, aa099 (9.91 g, 75%) was obtained under the same reaction conditions as the synthesis of Compound aa268.

(759) LCMS (ESI) $m/z=338$ (M+H)⁺

(760) Retention time: 0.72 min (Analytical condition SQDFA05)

(761) Synthesis of Compound aa100

(762) ##STR00328##

(763) Using aa100-a, aa100 (2.15 g, 80%) was obtained under the same reaction conditions as the synthesis of Compound aa268.

(764) LCMS (ESI) $m/z=352$ (M+H)⁺

(765) Retention time: 0.76 min (Analytical condition SQDFA05)

(766) Synthesis of Compound aa389

(767) ##STR00329##

(768) Using aa059 (Fmoc-Leu-OH) as a starting material, a crude product of aa389-b was obtained in the same manner as synthesis of Compound aa136-b.

(769) Using the resulting aa389-b, aa389 (7.7 g, 69% in 2 steps) was obtained in the same manner as Compound aa136.

(770) LCMS (ESI) $m/z=396$ (M+H)⁺

(771) Retention time: 0.99 min (Analytical condition SQDFA05)

(772) Synthesis of Compound aa397

(773) ##STR00330##

(774) Using aa397-a (N-(((9H-fluoren-9-yl)methoxy) carbonyl)-O-isopentyl-L-serine, Fmoc-Ser (iPen)-OH, CAS No. 2255321-11-8) as a starting material, aa397-b was obtained in the same manner as synthesis of Compound aa136-b.

(775) Using the resulting aa397-b, aa397 (8.1 g, 73% in 2 steps) was obtained in the same manner as synthesis of Compound aa136.

(776) LCMS (ESI) $m/z=440$ (M+H)+

(777) Retention time: 1.04 min (Analytical condition SQDFA05)

(778) Synthesis of Compound aa398

(779) ##STR00331##

(780) Using aa398-a (N-(((9H-fluoren-9-yl)methoxy) carbonyl)-O-cyclobutyl-L-serine, Fmoc-Ser (cBu)-OH, CAS No. 2642331-49-3) as a starting material, aa398-b was obtained in the same manner as synthesis of Compound aa136-b. Using the resulting aa398-b, aa398 (5.3 g, 48% in 2 steps) was obtained in the same manner as synthesis of Compound aa136. LCMS (ESI) $m/z=424$ (M+H)+

(781) Retention time: 0.95 min (Analytical condition SQDFA05)

(782) Synthesis of Compound aa391

(783) ##STR00332##

(784) aa391-a (O-allyl-N-(tert-butoxycarbonyl)-L-threonine, CAS No. 300831-37-2) (5.00 g, 19.3 mmol) was dissolved in methanol (60 mL), and palladium on carbon (5%, 492 mg) and a 7 M ammonia methanol solution (4.13 mL) were added in a nitrogen atmosphere. The mixture was stirred at room temperature for 1.5 hours in a hydrogen atmosphere. Palladium on carbon was removed by filtration, and the solvent was distilled off from the filtrate under reduced pressure to give aa391-b as a crude product. The crude product was used in the next reaction without purification.

(785) aa391-b was dissolved in TFE (89 mL), TMSCI (6.16 mL, 48.2 mmol) was added, and the mixture was stirred at room temperature for 30 minutes. The solvent was distilled off from the mixture under reduced pressure to give aa391-c as a crude product. The crude product was used in the next reaction without purification.

(786) aa391-c was dissolved in water (54 mL), and sodium hydrogen carbonate (6.48 g, 77 mmol) and 1,4-dioxane (54 mL) were added. Fmoc-OSu (6.18 g, 18.3 mmol) was added to the mixture, and the mixture was stirred at room temperature for 4.5 hours. The reaction solution was washed with TBME (100 mL), and 2 N HCl (108 mL) was added to the aqueous layer, followed by extraction with DCM (40 mL) 3 times. The organic layer was dried over anhydrous sodium sulfate, the solvent was distilled off under reduced pressure, and the resulting residue was purified by reverse phase chromatography (0.1% formic acid-containing acetonitrile/0.1% formic acid-containing distilled water). The resulting crude product was dissolved in TBME (100 mL), extracted with a 3.5% aqueous sodium hydrogen carbonate solution (100 mL), and further extracted with a 3.5% aqueous sodium hydrogen carbonate solution (50 mL). After 2 M phosphoric acid was added to the aqueous layer to adjust the pH to 2, the mixture was extracted with ethyl acetate. The mixture was dried over anhydrous sodium sulfate, and then the solvent was distilled off under reduced pressure to give 391 (4.07 g, 55% in 3 steps).

(787) LCMS (ESI) $m/z=384$ (M+H)+

(788) Retention time: 0.84 min (Analytical condition SQDFA05)

(789) Synthesis of Compound aa399

(790) ##STR00333##

(791) 399-a (5-bromo-7-chloro-1H-indole, CAS No. 180623-89-6, 15.0 g, 65.1 mmol) was added to a THF solution (150 mL) of NaH (2.86 g, 71.6 mmol, 60%) under ice cooling, and the mixture was stirred at room temperature for 30 minutes. Iodomethane (10.2 g, 71.6 mmol) was added dropwise thereto under ice cooling, and the mixture was stirred at room temperature for 16 hours.

Water was added thereto, followed by extraction with ethyl acetate 3 times. The solvent was distilled off from the organic layer under reduced pressure to give aa399-b as a crude product (17.2 g).

(792) 1,4-Dioxane solution (70 mL) of the crude product of aa399-b (6.10 g, 24.9 mmol), sodium iodide (18.7 g, 125 mmol), copper (I) iodide (0.48 g, 2.50 mmol), and trans-N.sub.1,N.sub.2-dimethylcyclohexane-1,2-diamine (0.71 g, 4.99 mmol) was stirred at 130° C. for 16 hours in a nitrogen atmosphere. After being cooled to room temperature, the mixture was diluted with water and extracted twice with ethyl acetate. The organic layer was washed twice with brine, dried over anhydrous sodium sulfate, and filtered off, and the solvent was distilled off under reduced pressure to give aa399-c as a crude product (6.1 g).

(793) Using aa220-a (17.7 g, 31.0 mmol), aa399-d (10.5 g, 31%) was obtained in the same manner as Compound aa219-b except that aa399-c was used in place of 4-bromo-2-fluoro-1-(trifluoromethyl)benzene.

(794) LCMS (ESI) m/z=567 (M+Na)+

(795) Retention time: 0.95 min (Analytical condition SMD method_22)

(796) TMSOTf (8.56 g, 38.5 mmol) was added to an IPAC solution (110 mL) of aa399-d (10.5 g, 19.3 mmol) and HMDS (6.22 g, 38.5 mmol) in a nitrogen atmosphere. The mixture was stirred at room temperature for 16 hours. After a 5% aqueous disodium hydrogen phosphate solution was added, the mixture was extracted 3 times with ethyl acetate. The organic layer was washed with a 2% aqueous phosphoric acid solution and washed with 15% brine. The organic layer was dried over sodium sulfate and filtered off, and the solvent was distilled off under reduced pressure. The resultant was purified by reverse phase chromatography (acetonitrile-water) and then washed with hexane/TBME (5/1) to give aa399 (7 g, 74%).

(797) LCMS (ESI) m/z=489 (M+H)+

(798) Retention time: 1.36 min (Analytical condition SMD method_23)

(799) Synthesis of Compound aa400

(800) ##STR00334##

(801) A DMF solution (500 mL) of aa400-a (4-bromobenzenethiol, CAS No. 106-53-6, 50 g, 264 mmol), potassium carbonate (73.1 g, 529 mmol), and chloroacetone (26.9 g, 291 mmol) was stirred at room temperature for 16 hours in a nitrogen atmosphere. Water was added, solids were filtered off, and the solvent was distilled off from the filtrate under reduced pressure to give aa400-b as a crude product (61.0 g).

(802) A chlorobenzene solution (700 mL) of the crude product of aa400-b (84 g, 342 mmol) and PPA (250 g, 2.17 mmol) was stirred at 140° C. for 16 hours in a nitrogen atmosphere. After this solution was cooled to room temperature, the upper layer was added to ethyl acetate to be diluted, and washed with water, and then the solvent was distilled off under reduced pressure. The resulting residue was purified by silica gel column chromatography (ethyl acetate/petroleum ether) to give aa400-c (42.5 g, 54%).

(803) Retention time: 1.13 min (Analytical condition SMD method_24)

(804) A THF solution (300 mL) of aa400-c (40 g, 176 mmol) was cooled to -78° C., and LDA (2 M THF solution, 96 mL, 194 mmol) was added dropwise in a nitrogen atmosphere. The mixture was stirred at -78° C. for 30 minutes, and N-fluorobenzenesulfonimide (NFSI, 100 g, 317 mmol) was added dropwise at -78° C. After stirring this mixture at room temperature for 16 hours, an aqueous ammonium chloride solution was added, followed by extraction with ethyl acetate twice. The organic layer was washed twice with water, dried over anhydrous sodium sulfate, and filtered off, and the solvent was distilled off under reduced pressure. The resulting residue was purified by silica gel column chromatography (ethyl acetate/petroleum ether) to give aa400-d (20 g, 43%).

(805) Retention time: 1.31 min (Analytical condition SMD method_26)

(806) aa400-e was obtained as a crude product (25.0 g) in the same manner as Compound 399-c except that aa400-d (19 g, 77.5 mmol) was used in place of aa399-b.

(807) Using aa220-a (19.0 g, 33.3 mmol), aa400-f (13.8 g, 73%) was obtained in the same manner as Compound aa219-b except that aa400-e was used in place of 4-bromo-2-fluoro-1-(trifluoromethyl)benzene.

(808) LCMS (ESI) $m/z=568$ (M+Na)+

(809) Retention time: 1.31 min (Analytical condition SMD method_26)

(810) aa400 (8 g, 81%) was obtained in the same manner as Compound 399 except that aa400-f (3.0 g, 5.50 mmol) was used in place of aa399-d, and ethyl acetate was used as a solvent in place of IPAC.

(811) LCMS (ESI) $m/z=490$ (M+H)+

(812) Retention time: 1.31 min (Analytical condition SMD method_27)

(813) Synthesis of Compound aa401

(814) ##STR00335##

(815) Methyl thioglycolate (10.2 g, 96.0 mmol) and potassium carbonate (22.1 g, 160 mmol) were added to an acetone solution (300 mL) of aa401-a (5-bromo-3-chloro-2-fluorobenzaldehyde, CAS No. 1280786-80-2, 19 g, 80 mmol) at room temperature, and the mixture was stirred at 60° C. for 2 hours in a nitrogen atmosphere. After the mixture was cooled to room temperature, solids were filtered off, and the solvent was distilled off under reduced pressure. The resulting residue was dissolved in ethyl acetate, dried over anhydrous sodium sulfate, and filtered off, and the solvent was distilled off under reduced pressure. The resulting residue was purified by silica gel column chromatography (ethyl acetate/petroleum ether) to give aa401-b (16.4 g, 67%).

(816) Retention time: 0.89 min (Analytical condition SMD method_22)

(817) Lithium hydroxide monohydrate (6.43 g, 268 mmol) was added to a THF (102 mL)/methanol (34 mL)/water (34 mL) solution of aa401-b (16.4 g, 53.7 mmol) at room temperature. After stirring this mixture at room temperature for 1 hour, the solvent was distilled off under reduced pressure. The resulting residue was dissolved in water, and 2 M hydrochloric acid was added to adjust the pH to 3, followed by extraction with ethyl acetate 3 times. The organic layer was washed with water, dried over anhydrous sodium sulfate, and filtered off, and the solvent was distilled off under reduced pressure to give aa401-c as a crude product (15.9 g).

(818) Copper (I) oxide (31.2 g, 218 mmol) was added to a DMF solution (160 mL) of the crude product of 401-c (15.9 g, 54.5 mmol), and the mixture was stirred at 140° C. for 16 hours. After solids were filtered off, the filtrate was washed with water, dried over anhydrous sodium sulfate, and filtered off, and the solvent was distilled off under reduced pressure. The resulting residue was purified by silica gel column chromatography (ethyl acetate/petroleum ether) to give aa401-d (11.7 g, 87%).

(819) Retention time: 0.99 min (Analytical condition SMD method_28)

(820) aa401-e was obtained as a crude product (12.5 g) in the same manner as Compound 399-c except that aa401-d (11.7 g, 47.3 mmol) was used in place of aa399-b.

(821) Using aa220-a (12.1 g, 21.2 mmol), aa401-f (8.6 g, 74%) was obtained in the same manner as Compound aa219-b except that aa401-e was used in place of 4-bromo-2-fluoro-1-(trifluoromethyl)benzene.

(822) LCMS (ESI) $m/z=570$ (M+Na)+

(823) Retention time: 0.96 min (Analytical condition SMD method_22)

(824) aa401 (4.4 g, 57%) was obtained in the same manner as Compound 399 except that aa401-f (8.6 g) was used in place of aa399-d, and ethyl acetate was used as a solvent in place of IPAC.

(825) LCMS (ESI) $m/z=492$ (M+H)+

(826) Retention time: 1.33 min (Analytical condition SMD method_29)

(827) Synthesis of Compound aa402

(828) ##STR00336##

(829) aa402-b was obtained as a crude product (23 g) in the same manner as Compound 399-c except that aa402-a (6-bromo-1-methyl-1H-indole, CAS No. 125872-95-9, 20 g, 95.2 mmol) was

used in place of aa399-b.

(830) Using aa220-a (9.00 g, 15.8 mmol), aa402-c (3.6 g, 44%) was obtained in the same manner as Compound aa219-b except that aa402-b was used in place of 4-bromo-2-fluoro-1-(trifluoromethyl)benzene.

(831) LCMS (ESI) $m/z=511$ (M+H)+

(832) Retention time: 0.92 min (Analytical condition SMD method_30)

(833) aa402 (5 g, 70%) was obtained in the same manner as Compound 399 except that aa402-c (8.00 g, 15.7 mmol) was used in place of aa399-d.

(834) LCMS (ESI) $m/z=455$ (M+H)+

(835) Retention time: 1.04 min (Analytical condition SMD method_24)

(836) Synthesis of Compound aa403

(837) ##STR00337##

(838) aa403-b was obtained as a crude product (18.1 g) in the same manner as Compound 399-c except that aa403-a (6-bromo-1,3-dimethyl-1H-indole, CAS No. 1616099-24-1, 15.3 g, 68.3 mmol) was used in place of aa399-b.

(839) Using aa220-a (19.1 g, 33.4 mmol), aa403-c (9.5 g, 54%) was obtained in the same manner as Compound aa219-b except that aa403-b was used in place of 4-bromo-2-fluoro-1-(trifluoromethyl)benzene.

(840) LCMS (ESI) $m/z=525$ (M+H)+

(841) Retention time: 0.97 min (Analytical condition SMD method_31)

(842) A crude product of aa403 was obtained in the same manner as Compound 219 except that aa403-c (9.5 g, 18.1 mmol) was used in place of aa219-b. The crude product was purified by reverse phase chromatography (acetonitrile-water) to give aa403 (4.5 g, 53%).

(843) LCMS (ESI) $m/z=469$ (M+H)+

(844) Retention time: 1.82 min (Analytical condition SMD method_32)

(845) Synthesis of Compound aa404

(846) ##STR00338##

(847) A DMF solution (100 mL) of aa404-a (6-bromo-2-methyl-1H-indole, CAS No. 6127-19-1, 19 g, 90.4 mmol), potassium carbonate (18.9 g, 137 mmol), and dimethyl carbonate (24.4 g, 271 mmol) was stirred at 140° C. for 16 hours in a nitrogen atmosphere. After being cooled to room temperature, the solution was diluted with 1 M hydrochloric acid and extracted twice with ethyl acetate. The organic layer was washed with brine, dried over sodium sulfate, and filtered off, and the solvent was distilled off under reduced pressure. The resulting residue was purified by silica gel column chromatography (ethyl acetate/petroleum ether) to give aa404-b (25 g, quant.).

(848) LCMS (ESI) $m/z=224$ (M+H)+

(849) Retention time: 0.98 min (Analytical condition SMD method_33)

(850) Phosphorus oxychloride (16.6 mL, 179 mmol) was added dropwise to DMF (200 mL) under ice cooling in a nitrogen atmosphere. The mixture was stirred for 10 minutes under ice cooling, and a DMF solution (50 mL) of aa404-b (20 g, 89.2 mmol) was added dropwise over 10 minutes under ice cooling. The mixture was stirred at room temperature for 40 minutes, and treated with an aqueous sodium hydrogen carbonate solution. The mixture was stirred for 10 minutes and then extracted twice with ethyl acetate, the organic layer was washed with brine, dried over sodium sulfate, and filtered off, and the solvent was distilled off under reduced pressure to give aa404-c as a crude product (19 g).

(851) Hydrochloric acid (12 M, 21.5 mL, 377 mmol) was added dropwise to a THF solution (500 mL) of the crude product of aa404-c (19 g, 75.4 mmol) and zinc powder (14.8 g, 226 mmol) under ice cooling in a nitrogen atmosphere. The mixture was stirred at room temperature for 16 hours and then diluted with water. The mixture was extracted twice with ethyl acetate, the organic layer was washed with brine, dried over sodium sulfate, and filtered off, and the solvent was distilled off under reduced pressure to give aa404-d as a crude product (13 g).

(852) aa404-e was obtained as a crude product (16 g) in the same manner as Compound 399-c except that aa404-d (13 g, 54.6 mmol) was used in place of aa399-b.

(853) Using aa220-a (16.5 g, 28.9 mmol), Compound aa404-f (7.9 g, 45%) was obtained in the same manner as Compound aa219-b except that aa404-e was used in place of 4-bromo-2-fluoro-1-(trifluoromethyl)benzene.

(854) LCMS (ESI) $m/z=539$ (M+H)+

(855) Retention time: 1.08 min (Analytical condition SMD method_33)

(856) aa404 (4.5 g, 62%) was obtained in the same manner as Compound 399 except that aa404-f (7.9 g, 14.7 mmol) was used in place of aa399-d, and ethyl acetate was used as a solvent in place of IPAC.

(857) LCMS (ESI) $m/z=483$ (M+H)+

(858) Retention time: 1.35 min (Analytical condition SMD method_34)

(859) Synthesis of Compound aa405

(860) ##STR00339##

(861) 3-Bromo-2-butanone (28.8 g, 190 mmol) was added to an aqueous solution (300 mL) of aa405-a (4-bromobenzenethiol, CAS No. 106-53-6, 30 g, 159 mmol), and the mixture was stirred at 80° C. for 1 hour. The mixture was extracted 3 times with ethyl acetate, the organic layer was dried over anhydrous sodium sulfate and filtered off, and the solvent was distilled off under reduced pressure. The resulting residue was purified by silica gel column chromatography (ethyl acetate/petroleum ether) to give aa405-b (40 g, 85%).

(862) LCMS (ESI) $m/z=259$ (M+H)+

(863) Retention time: 0.71 min (Analytical condition SMD method_35)

(864) aa405-c (13.7 g, 70%) was obtained in the same manner as Compound aa400-c except that aa405-b (20 g, 77.1 mmol) was used in place of aa400-b.

(865) Retention time: 0.88 min (Analytical condition SMD method_36)

(866) aa405-d was obtained as a crude product (15.4 g) in the same manner as Compound 399-c except that aa405-c (13.7 g, 56.8 mmol) was used in place of aa399-b.

(867) Using aa220-a (15.3 g, 26.7 mmol), aa405-e (9.06 g, 61%) was obtained in the same manner as Compound aa219-b except that aa405-d was used in place of 4-bromo-2-fluoro-1-(trifluoromethyl)benzene.

(868) LCMS (ESI) $m/z=564$ (M+Na)+

(869) Retention time: 0.99 min (Analytical condition SMD method_36)

(870) aa405 (5.03 g, 61%) was obtained in the same manner as Compound 399 except that aa405-e (9.06 g, 16.7 mmol) was used in place of aa399-d, and ethyl acetate was used as a solvent in place of IPAC.

(871) LCMS (ESI) $m/z=486$ (M+H)+

(872) Retention time: 1.47 min (Analytical condition SMD method_37)

(873) Synthesis of Compound aa406

(874) ##STR00340##

(875) NaH (5.03 g, 210 mmol) was added in small divided portions to a THF solution (100 mL) of (bromomethyl)triphenylphosphonium bromide (49.9 g, 140 mmol) at 55° C. in a nitrogen atmosphere. The mixture was stirred at 55° C. for 30 minutes and then cooled to room temperature, and a THF solution (50 mL) of aa406-a (2-chloro-4-bromobenzaldehyde, CAS No. 158435-41-7, 15.3 g, 70 mmol) was added dropwise. After stirring this mixture at room temperature for 16 hours, water was added. The mixture was extracted 3 times with ethyl acetate, the organic layer was washed with water, dried over sodium sulfate, and filtered off, and the solvent was distilled off under reduced pressure to give aa406-b as a crude product (43 g).

(876) A methanol (10 mL)/THF (10 mL) solution of aa406-b (2.5 g, 11.5 mmol) and platinum (IV) oxide (0.25 g, 1.10 mmol) was stirred at room temperature for 2 hours in a hydrogen atmosphere. Solids were filtered off, and the solvent was distilled off from the filtrate under reduced pressure.

The resulting residue was purified by silica gel column chromatography (ethyl acetate/petroleum ether) to give aa406-c (1.2 g, 42%).

(877) Retention time: 1.41 min (Analytical condition SMD method_38)

(878) Using aa220-a (17 g, 30.0 mmol), Compound aa406-d (4.8 g, 31%) was obtained in the same manner as Compound aa219-b except that aa406-c was used in place of 4-bromo-2-fluoro-1-(trifluoromethyl)benzene.

(879) LCMS (ESI) $m/z=542$ (M+Na)+

(880) Retention time: 1.53 min (Analytical condition SMD method_38)

(881) aa406 (4.3 g, 95%) was obtained in the same manner as Compound 219 except that aa406-d (5 g, 9.61 mmol) was used in place of aa219-b.

(882) LCMS (ESI) $m/z=486$ (M+Na)+

(883) Retention time: 1.32 min (Analytical condition SMD method_38)

(884) Synthesis of Compound aa407

(885) ##STR00341##

(886) Using aa220-a (21 g, 36.8 mmol), aa407-a (9.84 g, 52%) was obtained in the same manner as Compound aa219-b except that 4-bromo-2,6-dimethylanisole (CAS No. 14804-38-7) was used in place of 4-bromo-2-fluoro-1-(trifluoromethyl)benzene.

(887) LCMS (ESI) $m/z=538$ (M+Na)+

(888) Retention time: 1.44 min (Analytical condition SMD method_38)

(889) aa407 (4.3 g, 95%) was obtained in the same manner as Compound 219 except that aa407-a (9.84 g, 19.1 mmol) was used in place of aa219-b.

(890) LCMS (ESI) $m/z=460$ (M+H)+

(891) Retention time: 1.23 min (Analytical condition SMD method_38)

(892) Synthesis of Compound aa408

(893) ##STR00342##

(894) Using aa220-a (15 g, 26.3 mmol), aa408-a (6.4 g, 46%) was obtained in the same manner as Compound aa219-b except that 5-bromo-2-chloroanisole (CAS No. 16817-43-9) was used in place of 4-bromo-2-fluoro-1-(trifluoromethyl)benzene.

(895) LCMS (ESI) $m/z=544$ (M+Na)+

(896) Retention time: 0.89 min (Analytical condition SMD method_39)

(897) aa408 (3.5 g, 62%) was obtained in the same manner as Compound 219 except that aa408-a (6.4 g, 12.3 mmol) was used in place of aa219-b.

(898) LCMS (ESI) $m/z=488$ (M+Na)+

(899) Retention time: 1.00 min (Analytical condition SMD method_26)

(900) Synthesis of Compound aa409

(901) ##STR00343##

(902) Using aa220-a (5 g, 8.76 mmol), aa409-a (3.8 g, 77%) was obtained in the same manner as Compound aa219-b except that 5-bromo-2-chlorobenzotrifluoride (CAS No. 445-01-2) was used in place of 4-bromo-2-fluoro-1-(trifluoromethyl)benzene.

(903) LCMS (ESI) $m/z=582$ (M+Na)+

(904) Retention time: 0.93 min (Analytical condition SMD method_40)

(905) aa409 (3 g, 88%) was obtained in the same manner as Compound 219 except that aa409-a (3.8 g, 6.79 mmol) was used in place of aa219-b.

(906) LCMS (ESI) $m/z=526$ (M+Na)+

(907) Retention time: 1.08 min (Analytical condition SMD method_26)

(908) Synthesis of Compound aa410

(909) ##STR00344##

(910) Using aa220-a (5 g, 8.76 mmol), aa410-a (3.5 g, 73%) was obtained in the same manner as Compound aa219-b except that 5-bromo-2-(difluoromethyl)-1,3-difluorobenzene

(911) (CAS No. 1221272-77-0) was used in place of 4-bromo-2-fluoro-1-(trifluoromethyl)benzene.

LCMS (ESI) $m/z=566$ (M+Na)+

(912) Retention time: 1.21 min (Analytical condition SMD method_24)

(913) aa410 (1.9 g, 60%) was obtained in the same manner as Compound 219 except that aa410-a (3.5 g, 6.44 mmol) was used in place of aa219-b.

(914) LCMS (ESI) $m/z=488$ (M+H)+

(915) Retention time: 0.99 min (Analytical condition SMD method_41)

(916) Synthesis of Compound aa411

(917) ##STR00345##

(918) An N,N-dimethylacetamide (2.92 mL) solution of a nickel (II) bromide 4,4'-di-tert-butyl-2,2'-bipyridine complex (85 mg, 0.175 mmol) and chlorotrimethylsilane (0.111 mL, 0.876 mmol) were added to an N,N-dimethylacetamide (5.84 mL) solution of aa220-a (1.00 g, 1.75 mmol), 5-bromo-2-chloro-1,3-dimethylbenzene (577 mg, 2.63 mmol), and zinc (573 mg, 8.76 mmol), and the mixture was stirred at room temperature for 23 hours in a nitrogen atmosphere. tert-Butyl methyl ether (10 mL) was added, the mixture was passed through a Celite filter covered with sodium sulfate, and then a 5 wt % aqueous disodium ethylenediaminetetraacetate solution (40 mL) was added for extraction. The organic layer was washed with a 5 wt % aqueous sodium carbonate solution (20 mL), washed with a saturated aqueous ammonium chloride solution (20 mL), washed with brine (20 mL), and dried over anhydrous sodium sulfate, and then the solvent was distilled off under reduced pressure to give aa441-a as a crude product.

(919) The crude product of aa441-a was dissolved in TFE (8.76 mL), TMSCI (0.667 mL, 5.26 mmol) was added, and the mixture was stirred at room temperature for 30 minutes. The solvent was distilled off from the mixture under reduced pressure, the resulting crude product was purified by reverse phase column chromatography (0.1% aqueous formic acid solution/0.1% formic acid acetonitrile solution) to give aa441 (364 mg, 45% in 2 steps).

(920) LCMS (ESI) $m/z=462$ (M-H)-

(921) Retention time: 1.01 min (Analytical condition SQDFA05)

(922) Synthesis of Compound aa414

(923) ##STR00346##

(924) Chlorodimethylsilane (18.5 mL, 170 mmol) was added to an acetonitrile (51.5 mL) solution of aa414-a ((2S,4R)-1-(((9H-fluoren-9-yl)methoxy) carbonyl)-4-hydroxypyrrolidine-2-carboxylic acid, Fmoc-Hyp-OH, CAS No. 88050-17-3)(10.0 g, 28.3 mmol) and cyclopentanone (15.0 mL, 170 mmol), and the mixture was stirred at room temperature for 28 hours. A 2 N aqueous sodium hydroxide solution (85 mL) was slowly added dropwise thereto under ice cooling so as not to exceed 30° C. A 2 N aqueous sodium hydroxide solution (14 mL) was further added to adjust the pH to 11. Phosphoric acid was added thereto to adjust the pH to 7 to 8, and the mixture was washed with TBME/hexane (1/3, 150 mL). Phosphoric acid was added to the aqueous layer to adjust the pH to 3, followed by extraction with TBME (150 mL) and further extraction with TBME (100 mL). After the resulting organic layer was washed with brine and dried over anhydrous sodium sulfate, the solvent was distilled off under reduced pressure to give aa414 (11.7 g, 98%).

(925) LCMS (ESI) $m/z=422$ (M+H)+

(926) Retention time: 0.91 min (Analytical condition SQDFA05)

(927) Synthesis of Compound aa415

(928) ##STR00347##

(929) aa415 (10.8 g, 94%) was obtained in the same manner as Compound 414 except that cyclobutanone was used in place of cyclopentanone.

(930) LCMS (ESI) $m/z=408$ (M+H)+

(931) Retention time: 0.86 min (Analytical condition SQDFA05)

(932) Synthesis of Compound aa443

(933) ##STR00348##

(934) Using Compound aa443-a (N-([(9H-fluoren-9-yl)methoxy]carbonyl)-4-(methyl)-L-

phenylalanine) as a starting material, a crude product of aa443-b (2.39 g, 93%) was obtained in the same manner as synthesis of Compound tp006-b.

(935) LCMS (ESI) $m/z=414.5$ (M+H)+

(936) Retention time: 1.01 min (Analytical condition SQDFA05)

(937) Using the resulting crude product of Compound aa443-b, Compound aa443 (2.23 g, 85%) was obtained in the same manner as synthesis of Compound tp006-c.

(938) LCMS (ESI) $m/z=454$ (M-H)+

(939) Retention time: 0.99 min (Analytical condition SQDFA05)

(940) 1-2. Synthesis of Amino Acid-Supporting Resin Used in Solid-Phase Peptide Synthesis by Peptide Synthesizer

(941) Amino acids listed in Table 5 were synthesized by the method shown below, supported on a resin, and used in peptide synthesis by a peptide synthesizer. Resins on which Compound aa427 to Compound aa438 were supported were synthesized by the following method using resins on which the amino acids shown in Table 5 were supported as raw materials, and used in peptide synthesis with a peptide synthesizer. Amino acids listed in Table 6 were purchased from commercial suppliers, supported on a resin, and used in peptide synthesis by a peptide synthesizer. The reaction for causing Fmoc amino acid to be supported on a resin was carried out according to the method described in WO2013/100132 or WO2018/225864.2-Chlorotrityl chloride resin (100 to 200 mesh, 1% DVB) was purchased from Watanabe Chemical Industries Ltd., and Chem-Impex.

(942) TABLE-US-00005 TABLE 5 Compound No. Abbreviation Structural Formula Name
aa358 Fmoc- MeAsp-mor  (3S)-3-[9H-fluoren-9-ylmethoxycarbonyl(methyl)amino]-4-morpholin-4-yl-4-oxobutanoic acid
aa359 Fmoc- MeAsp- NMe2  (3S)-4-(dimethylamino)-3-[9H-fluoren-9-ylmethoxycarbonyl(methyl)amino]-4-oxobutanoic acid
aa360 Fmoc- MeAsp-pyrro  (3S)-3-[9H-fluoren-9-ylmethoxycarbonyl(methyl)amino]-4-oxo-4-pyrrolidin-1-ylbutanoic acid
aa361 Fmoc- MeAsp-aze  (3S)-4-(azetidin-1-yl)-3-[9H-fluoren-9-ylmethoxycarbonyl(methyl)amino]-4-oxobutanoic acid
aa362 Fmoc- MeAsp-pip  (3S)-3-[9H-fluoren-9-ylmethoxycarbonyl(methyl)amino]-4-oxo-4-piperidin-1-ylbutanoic acid
aa363 Fmoc- MeAsp-mor(26-bicyc)  (3S)-3-[9H-fluoren-9-ylmethoxycarbonyl(methyl)amino]-4-(3-oxa-8-azabicyclo[3.2.1]octan-8-yl)-4-oxobutanoic acid
aa364 Fmoc-MeAsp- pyrro(3-Me2)  (3S)-4-(3,3-dimethylpyrrolidin-1-yl)-3-[9H-fluoren-9-ylmethoxycarbonyl(methyl)amino]-4-oxobutanoic acid
aa365 Fmoc-MeAsp- pip(4-Me)  (3S)-3-[9H-fluoren-9-ylmethoxycarbonyl(methyl)amino]-4-(4-methylpiperidin-1-yl)-4-oxobutanoic acid
aa366 Fmoc-MeAsp- piz(oxe)  (3S)-3-[9H-fluoren-9-ylmethoxycarbonyl(methyl)amino]-4-[4-(oxetan-3-yl)piperazin-1-yl]-4-oxobutanoic acid
aa367 Fmoc- MeAsp-oxz  (3S)-3-[9H-fluoren-9-ylmethoxycarbonyl(methyl)amino]-4-(1,3-oxazolidin-3-yl)-4-oxobutanoic acid
aa368 Fmoc- Asp-NMe2  (3S)-4-(dimethylamino)-3-(9H-fluoren-9-ylmethoxycarbonyl(methyl)amino)-4-oxobutanoic acid
aa369 Fmoc- Asp-mor  (3S)-3-(9H-fluoren-9-ylmethoxy-carbonylamino)-4-morpholin-4-yl-4-oxobutanoic acid
aa373 Fmoc-D- MeAsp-NMe2  (R)-3-(((9H-fluoren-9-yl)methoxy)carbonyl(methyl)amino)-4-(dimethylamino)-4-oxobutanoic acid
aa374 Fmoc- MeGly(cPent)- MeAsp-NMe2  (S)-3-(((S)-2-(((9H-fluoren-9-yl)methoxy)carbonyl(methyl)amino)-2-cyclopentyl-N-methylacetamido)-4-(dimethylamino)-4-oxobutanoic acid
aa375 Fmoc- MeGly(cPent)- MeAsp-mor  (S)-3-(((S)-2-(((9H-fluoren-9-yl)methoxy)carbonyl(methyl)amino)-2-cyclopentyl-N-methylacetamido)-4-morpholino-4-oxobutanoic acid
aa376 Fmoc- MeGly(cPent)- EtAsp-NMe2  (3S)-3-(((2S)-2-cyclopentyl-2-[9H-fluoren-9-ylmethoxycarbonyl(methyl)amino]acetyl)-ethyl-amino)-4-(dimethylamino)-4-oxo-butanoic acid
aa377 Fmoc- MeGly(cPent)- nPrAsp-NMe2  (3S)-3-(((2S)-2-cyclopentyl-2-[9H-fluoren-9-ylmethoxycarbonyl(methyl)amino]acetyl)-propyl-amino)-4-(dimethylamino)-4-

oxo- butanoic acid aa378 Fmoc- MeGly(cPent)- cBuEtAsp- NMe2  (3S)-3-[2-cyclobutylethyl- [(2S)-2-cyclopentyl-2- [9H-fluoren-9- ylmethoxycarbonyl(methyl) amino]acetyl]amino]-4- (dimethylamino)-4-oxo-butanoic acid aa379 Fmoc- MeGly(cPent)- cPentEtAsp- NMe2  (3S)-3-[2-cyclopentylethyl- [(2S)-2-cyclopentyl- 2-[9H-fluoren-9- ylmethoxycarbonyl(methyl) amino]acetyl]amino]-4- (dimethylamino)-4-oxo- butanoic acid aa380 Fmoc- MeGly(cPent)- PhenethylAsp- NMe2  (3S)-3-[[(2S)-2-cyclopentyl- 2-[9H-fluoren-9- ylmethoxycarbonyl(methyl) amino]acetyl]-(2- phenylethyl)amino]-4- (dimethylamino)-4-oxo- butanoic acid aa381 Fmoc- MeGly(cPent)- EtAsp- mor  (3S)-3-[[(2S)-2-cyclopentyl- 2-[9H-fluoren-9- ylmethoxycarbonyl(methyl) amino]acetyl]-ethyl- amino]-4-morpholino-4-oxo- butanoic acid aa382 Fmoc- MeGly(cPent)- nPrAsp- mor  (3S)-3-[[(2S)-2-cyclopentyl-2- [9H-fluoren-9- ylmethoxycarbonyl(methyl) amino]acetyl]-propyl- amino]- 4-morpholino-4-oxo-butanoic acid aa383 Fmoc- MeGly(cPent)- cBuEtAsp- mor  (3S)-3-[2-cyclobutylethyl- [(2S)-2-cyclopentyl-2- [9H-fluoren-9- ylmethoxycarbonyl(methyl) amino]acetyl]amino]-4- morpholino-4-oxo-butanoic acid aa384 Fmoc- MeGly(cPent)- cPentEtAsp- mor  (3S)-3-[2-cyclopentylethyl- [(2S)-2-cyclopentyl- 2-[9H-fluoren-9- ylmethoxycarbonyl(methyl) amino]acetyl]amino]-4- morpholino-4-oxo-butanoic acid aa385 Fmoc- MeGly(cPent)- PhenethylAsp- mor  (3S)-3-[[(2S)-2-cyclopentyl- 2-[9H-fluoren-9- ylmethoxycarbonyl(methyl) amino]acetyl]-(2- phenylethyl)amino]-4- morpholino-4-oxo- butanoic acid aa386 Fmoc- MeAsp(OPis)  (3S)-3-[9H-fluoren-9- ylmethoxycarbonyl(methyl) amino]-4-(1-methyl-1-phenyl- ethoxy)-4-oxo-butanoic acid aa427 Fmoc-Hyp(Et)- MecVal- MeGly(cPent)- MeAsp-NMe2  (3S)-3-[[(2S)-2-cyclopentyl-2- [[1-[[(2S,4R)-4-ethoxy-1-(9H- fluoren-9-ylmethoxycarbonyl) pyrrolidin-2- carbonyl]-methyl- amino]cyclobutan carbonyl]- methyl-amino]acetyl]-methyl- amino]-4-(dimethyl amino)- 4-oxo-butanoic acid aa428 Fmoc- Pro(4S-Me)- MecVal- MeGly(cPent)- MeAsp-NMe2  (3S)-3-[[(2S)-2-cyclopentyl-2- [1-[[(2S,4S)-1-(9H-fluoren-9- ylmethoxycarbonyl)-4-methyl- pyrrolidin-2-carbonyl]-methyl- amino]cyclobutan carbonyl]- methyl-amino]acetyl]-methyl- amino]-4-(dimethyl amino)-4- oxo-butanoic acid aa429 Fmoc- Hyp(Et)- MecVal(3-Me2)- MeGly(cPent)- MeAsp-NMe2  (3S)-3-[[(2S)-2-cyclopentyl-2- [1-[[(2S,4R)-4-ethoxy-1-(9H- fluoren-9-ylmethoxycarbonyl) pyrrolidin-2- carbonyl]-methyl- amino]-3,3-dimethyl-cyclobutan carbonyl]-methyl- amino]acetyl]-methyl- amino]-4- (dimethyl amino)-4-oxo- butanoic acid aa430 Fmoc-Hyp(Et)- MecVal- MeNva(3-Et)- MeAsp-NMe2  (3S)-4-(dimethyl amino)-3- [[(2S)-2-[1-[[(2S,4R)-4-ethoxy-1- (9H-fluoren-9-ylmethoxycarbonyl) pyrrolidin-2-carbonyl]- methyl-amino]cyclobutan carbonyl]- methyl-amino]-3-ethyl- pentanoyl]-methyl-amino]-4-oxo- butanoic acid aa431 Fmoc-Hyp(Et)- MecVal(3-Me2)- MeNva(3-Et)- MeAsp-NMe2  (3S)-4-(dimethyl amino)-3- [[(2S)-2-[1-[[(2S,4R)-4-ethoxy-1- (9H-fluoren-9-ylmethoxycarbonyl) pyrrolidin-2-carbonyl]- methyl-amino]-3,3-dimethyl- cyclobutan carbonyl]-methyl- amino]-3-ethyl-pentanoyl]-methyl- amino]-4-oxo-butanoic acid aa432 Fmoc-Hyp(nPr)- MecVal(3-Me2)- MeNva(3-Et)- MeAsp-NMe2  (3S)-4-(dimethyl amino)-3- [(2S)-3-ethyl-2-[1-[[(2S,4R)-1- (9H-fluoren-9-ylmethoxycarbonyl)- 4-propoxy-pyrrolidin-2- carbonyl]-methyl-amino]-3,3- dimethyl-cyclobutan carbonyl]- methyl-amino]pentanoyl]-methyl- amino]-4-oxo-butanoic acid aa433 Fmoc-Hyp(iPr)- MecVal(3-Me2)- MeNva(3-Et)- MeAsp-NMe2  (3S)-4-(dimethyl amino)- 3- [[(2S)-3-ethyl-2-[1-[[(2S,4R)-1- (9H-fluoren-9- ylmethoxycarbonyl)- 4-isopropoxy-pyrrolidin-2- carbonyl]-methyl-amino]-3,3- dimethyl-cyclobutan carbonyl]- methyl-amino]pentanoyl]-methyl- amino]-4-oxo-butanoic acid aa434 Fmoc-Hyp(cBu)- MecVal(3-Me2)- MeNva(3-Et)- MeAsp- NMe2  (3S)-3-[[(2S)-2-[[1-[[(2S,4R)- 4-(cyclobutoxy)-1-(9H-fluoren- 9- ylmethoxycarbonyl)pyrrolidin- 2-carbonyl]-methyl-amino]- 3,3-dimethyl-cyclobutan carbonyl]- methyl-amino]-3-ethyl- pentanoyl]-methyl-amino]-4- (dimethyl amino)-4-oxo- butanoic acid aa435 Fmoc-Hyp(cPent)- MecVal(3-Me2)- MeNva(3-Et)- MeAsp-NMe2  (3S)-3-

[[(2S)-2-[1-[(2S,4R)-4-(cyclopentoxy)-1-(9H-fluoren-9-ylmethoxycarbonyl)pyrrolidin-2-carbonyl]-methyl-amino]-3,3-dimethyl-cyclobutan carbonyl]- methyl-amino]-3-ethyl- pentanoyl]-methyl-amino]-4- (dimethyl amino)-4-oxo- butanoic acid aa436 Fmoc-Pro(4-cPr)- MecVal(3-Me2)- MeNva(3-Et)- MeAsp-NMe2  (3S)-4-(dimethyl amino)-3- [(2S)-3-ethyl-2-[1-[(6S)-5-(9H- fluoren-9-ylmethoxycarbonyl)- 5-azaspiro[2.4]heptan-6- carbonyl]-methyl-amino]-3,3- dimethyl-cyclobutan carbonyl]- methyl-amino]pentanoyl]- methyl-amino]-4-oxo- butanoic acid aa437 Fmoc-Hyp(Et)- MecVal- MeGly(cPent)- MeAsp-pip  (3S)-3-[[[(2S)-2-cyclopentyl-2- [[1-[[[(2S,4R)-4-ethoxy-1-(9H- fluoren-9-ylmethoxycarbonyl) pyrrolidin-2-carbonyl]-methyl- amino]cyclobutan carbonyl]- methyl-amino]acetyl]-methyl-amino]-4-oxo-4-(1-piperidyl) butanoic acid aa438 Fmoc-Hyp(Et)- MecVal- MeGly(cPent)- MeAsp-pyrro  (3S)-3-[[[(2S)-2-cyclopentyl-2- [1-[[[(2S,4R)-4-ethoxy-1-(9H- fluoren-9-ylmethoxycarbonyl) pyrrolidin-2-carbonyl]-methyl- amino]cyclobutan carbonyl]- methyl-amino]acetyl]-methyl- amino]-4-oxo-4-pyrrolidin-1-yl- butanoic acid aa439 Fmoc-MeCys(AcOH)- NMe2  2-[(2R)-3-(dimethyl amino)-2- [9H-fluoren-9-ylmethoxycarbonyl(methyl) amino]- 3-oxo-propyl]sulfanyl- acetic acid aa440 Fmoc-MeGly(cPent)- MeAsp-MeNEt  (3S)-3-[[[(2S)-2-cyclopentyl-2- [9H-fluoren-9-ylmethoxycarbonyl(methyl) amino] acetyl]-methyl-amino]-4- [ethyl(methyl)amino]-4-oxo- butanoic acid aa441 Fmoc- MeGly(cPent)- MeAsp-NEt2  (3S)-3-[[[(2S)-2-cyclopentyl-2- [9H-fluoren-9-ylmethoxycarbonyl(methyl) amino]acetyl]-methyl-amino]-4- (diethyl amino)-4-oxo-butanoic acid aa442 Fmoc- MeGly(cPent)- MeAsp-MeNnPr 0  (3S)-3-[[[(2S)-2-cyclopentyl-2- [9H-fluoren-9-ylmethoxycarbonyl(methyl) amino]acetyl]-methyl-amino]-4- [methyl(propyl)amino]-4-oxo- butanoic acid (943) TABLE-US-00006 TABLE 6 Compound No. Abbreviation Structural Formula Name aa370 Fmoc-D-3-Abu-OH  (3R)-3-(9H-fluoren-9-ylmethoxycarbonylamino)butanoic acid aa371 Fmoc-D-3-MeAbu-OH  (3R)-3-[9H-fluoren-9-ylmethoxycarbonyl(methyl)amino]butanoic acid aa372 Fmoc-D-Pro-(C#CH2)-OH  2-[(2R)-1-(9H-fluoren-9-ylmethoxycarbonyl)pyrrolidin-2-yl]acetic acid aa423 Fmoc-MebAla(3S-cBu)-OH  (3S)-3-cyclobutyl-3-[9H-fluoren-9-ylmethoxycarbonyl(methyl)amino] propanoic acid aa424 Fmoc-MebAla(3R-Et)-OH  (3R)-3-[9H-fluoren-9-ylmethoxycarbonyl(methyl)amino] pentanoic acid aa425 Fmoc-D-MeVal-(C#CH2)-OH  (3S)-3-[9H-fluoren-9-ylmethoxycarbonyl(methyl)amino]-4- methyl-pentanoic acid aa426 Fmoc-MebAla(3S-cHex)-OH  (3S)-3-cyclohexyl-3-[9H-fluoren-9-ylmethoxycarbonyl(methyl)amino] propanoic acid

Synthesis of Resin

Synthesis of Compound aa362-Resin

(944) ##STR00398##

(945) Compound aa362-resin was synthesized by the method described in WO2018/225864.

(946) Herein, when a resin and a compound are bonded, the resin moiety may be indicated as “O” (circle) or “resin”. The 2-chlorotrityl group may be omitted. Further, to clarify the reaction point of the resin moiety, the chemical structure of the reaction moiety may be indicated in connection with “O” (circle) or “resin”. In the above structure of Compound aa362-resin, the 2-chlorotrityl group on the resin is bonded to the side-chain carboxylic acid of MeAsp via an ester bond.

(947) Synthesis of Compound aa358-Resin

(948) ##STR00399##

(949) Fmoc-Asp (OAl)—OH (Compound aa358-a, (2S)-2-(9H-fluoren-9-ylmethoxycarbonylamino)-4-oxo-4-prop-2-enoxybutanoic acid, CAS No. 146982-24-3)(200 g, 506 mmol), p-toluenesulfonic acid (5.7 g, 0.05 eq), and paraformaldehyde (45.6 g, 3 eq) were mixed with toluene (2000 mL), and the mixture was stirred at 110° C. for 16 hours. The solvent was distilled off from the reaction solution under reduced pressure, the residue was dissolved in ethyl

acetate, and the mixture was washed twice with an aqueous sodium hydrogen carbonate solution. The organic layer was dried over sodium sulfate, and the solvent was distilled off under reduced pressure. The residue was purified by silica gel column chromatography (ethyl acetate/petroleum ether, 0/100 to 30/70) to give Compound aa358-b (9H-fluoren-9-ylmethyl(4S)-5-oxo-4-(2-oxo-2-prop-2-enoxyethyl)-1,3-oxazolidine-3-carboxylate)(175 g, 85%). The compound was mixed with another similarly synthesized batch, and used in the following reaction.

(950) LCMS (ESI) $m/z=408$ (M+H)+

(951) Retention time: 1.407 min (Analytical condition SMD method_20)

(952) A DCM (1 μ L) mixed solution of Compound aa358-b (100 g, 245 mmol), zinc bromide (ZnBr.sub.2)(110 g, 496 mmol), and TES (56 g, 481.6 mmol) was stirred at room temperature for 48 hours in a nitrogen atmosphere. Four batches of the same scale of the reaction solution were mixed, and the solvent was distilled off under reduced pressure. The residue was dissolved in TBME and extracted 10 times with 0.5 M phosphate buffer (pH=about 7.5). The aqueous layers were combined, adjusted to pH 2 with 5 N hydrochloric acid, and extracted twice with isopropyl acetate (IPAC). The organic layers were combined, dried over anhydrous sodium sulfate, and the solvent was distilled off under reduced pressure. To remove IPAC, the procedure of adding TBME to the resulting residue and distilling off solvent under reduced pressure was repeated 6 times, and thus Compounds aa358-c ((2S)-2-[9H-fluoren-9-ylmethoxycarbonyl(methyl)amino]-4-oxo-4-prop-2-enoxybutanoic acid) was obtained (270) g, 54%).

(953) LCMS (ESI) $m/z=410$ (M+H)+

(954) Retention time: 1.956 min (Analytical condition SMD method_05)

(955) WSCI.Math.HCl (0.506 g, 2.64 mmol) was dissolved in DMF (4.4 mL), then HOBT (0.356 g, 2.64 mmol), Compound aa358-c ((2S)-2-[9H-fluoren-9-ylmethoxycarbonyl(methyl)amino]-4-oxo-4-prop-2-enoxybutanoic acid), Fmoc-MeAsp (OAl)—OH)(0.9 g, 2.2 mmol) were added, and the mixture was stirred at ($^{\circ}$ C.) for 10 minutes. Morpholine (0.23 mL, 2.64 mmol) was added dropwise to the reaction solution, and the mixture was stirred at 0° C. for 1 hour. Ethyl acetate (9 mL) was added to the reaction solution, the mixture was washed with 0.5 N hydrochloric acid, water, a saturated aqueous sodium hydrogen carbonate solution/water (1/1), and saturated brine/water (1/1), and dried over anhydrous sodium sulfate, and the solvent was distilled off under reduced pressure to give Compound aa358-d as a crude product (522 mg, 50%).

(956) LCMS (ESI) $m/z=479$ (M+H)+

(957) Retention time: 0.87 min (Analytical condition SQDFA05)

(958) Tetrakis(triphenylphosphine) palladium (0)(10.8 mg, 0.0093 mmol) was added to a DCM (1.86 mL) solution of Compound aa358-d (446 mg, 0.932 mmol), and further, phenylsilane (0.081 mL, 0.653 mmol) was added dropwise, and the mixture was stirred at room temperature for 30 minutes. The reaction solution was diluted with TBME, and a 5% aqueous sodium hydrogen carbonate solution was added. The organic layer was removed, phosphoric acid (548 mg) was added to the aqueous layer, and the mixture was extracted with TBME. The organic layer was washed with saturated brine, dried over anhydrous sodium sulfate, and the solvent was distilled off under reduced pressure to give Compound aa358 (321 mg, 79%).

(959) LCMS (ESI) $m/z=439$ (M+H)+

(960) Retention time: 0.69 min (Analytical condition SQDFA05)

(961) The reaction for causing the Fmoc amino acid to be supported on a resin was performed according to the method described in WO2013/100132 or WO2018/225864. 2-Chlorotriyl chloride resin (1.60 mmol/g, 100-200 mesh, 1% DVB, 1 g, 1.6 mmol) and dehydrated dichloromethane (13.3 mL) were placed in a filter-equipped reaction vessel, and shaken at room temperature for 10 minutes. After dichloromethane was removed by applying nitrogen pressure, a mixed solution obtained by adding dehydrated methanol (0.259 mL, 6.4 mmol) and DIPEA (0.671 mL, 3.84 mmol) to a dehydrated dichloromethane (13.3 mL) solution of Compound aa358 (317 mg, 0.723 mmol) was added to the reaction vessel, and the reaction vessel was shaken for 30 minutes. After

the reaction solution was removed by applying nitrogen pressure, a mixed solution obtained by adding dehydrated methanol (1.99 mL) and DIPEA (0.671 mL) to dehydrated dichloromethane (13.3 mL) were added to the reaction vessel, and the reaction vessel was shaken for 1 hour and 30 minutes. After the reaction solution was removed by applying nitrogen pressure, dichloromethane was added to the reaction vessel, the reaction vessel was shaken for 5 minutes, and then the reaction solution was removed by applying nitrogen pressure. This resin washing operation with dichloromethane was repeated 2 more times, and the resulting resin was dried under reduced pressure overnight to give Compound aa358-resin (1.22 g, 0.37 mmol/g).

(962) The amount of amino acid supported on the resin was calculated as follows. The resulting Compound aa358-resin (10.4 mg) was placed in the reaction vessel, DMF (2 mL) was added, and the reaction vessel was shaken at room temperature for 1 hour. Then, DBU (40 μ L) was added, and the reaction vessel was shaken at 30° C. for 30 minutes. Then, DMF (8 mL) was added to the reaction mixture, and 1 mL of the solution was diluted with DMF (11.5 mL). The absorbance (294 nm) of the resulting diluted solution was measured (measured with Shimadzu UV-1600PC (cell length 1.0 cm)), and thus the calculated amount of supported Compound aa358-resin was 0.370 mmol/g.

(963) Another similarly synthesized lot having a different supported amount was also used in the peptide synthesis.

(964) Synthesis of Compound aa359-Resin

(965) ##STR00400##

(966) Using Compound aa358-c ((2S)-2-[9H-fluoren-9-ylmethoxycarbonyl(methyl)amino]-4-oxo-4-prop-2-enoxybutanoic acid) (1.5 g, 3.66 mmol) as a starting material, Compound aa359-a (1.47 g, 92%) was obtained in the same manner as the synthesis of Compound aa358-d using a THE solution (2 mol/L) of dimethylamine in place of morpholine.

(967) LCMS (ESI) $m/z=437$ (M+H)+

(968) Retention time: 0.88 min (Analytical condition SQDFA05)

(969) Using the resulting Compound aa359-a (1.4 g, 3.21 mmol), Compound aa359 (1.30 g, quant.) was obtained in the same manner as the synthesis of Compound aa358.

(970) LCMS (ESI) $m/z=397$ (M+H)+

(971) Retention time: 0.70 min (Analytical condition SQDFA05)

(972) Using the resulting Compound aa359 (1.21 g, 3.05 mmol), Compound aa359-Resin (4.45 g, supported amount 0.318 mmol/g) was obtained in the same manner as the synthesis of Compound aa358-resin.

(973) Another similarly synthesized lot having a different supported amount was also used in the peptide synthesis in the present Example.

(974) Synthesis of Compound aa360-Resin

(975) ##STR00401##

(976) Using Compound aa358-c ((2S)-2-[9H-fluoren-9-ylmethoxycarbonyl(methyl)amino]-4-oxo-4-prop-2-enoxybutanoic acid) (30 g, 73.3 mmol) as a starting material, Compound aa360-a (30.7 g, 91%) was obtained in the same manner as the synthesis of Compound aa358-d using pyrrolidine in place of morpholine.

(977) LCMS (ESI) $m/z=463$ (M+H)+

(978) Retention time: 0.88 min (Analytical condition SQDFA05)

(979) Using the resulting Compound aa360-a (30.7 g, 66.4 mmol), Compound aa360 (26.2 g, 93%) was obtained in the same manner as the synthesis of Compound aa358.

(980) LCMS (ESI) $m/z=423$ (M+H)+

(981) Retention time: 0.71 min (Analytical condition SQDFA05)

(982) Using the resulting Compound aa360 (24.6 g, 58.2 mmol), Compound aa360-resin was obtained in the same manner as the synthesis of Compound aa358-resin (84.1 g, supported amount 0.5216 mmol/g).

(983) Another similarly synthesized lot having a different supported amount was also used in the peptide synthesis in the present Example.

(984) Synthesis of Compound aa361-Resin

(985) ##STR00402##

(986) Using Compound aa358-c ((2S)-2-[9H-fluoren-9-ylmethoxycarbonyl(methyl)amino]-4-oxo-4-prop-2-enoxybutanoic acid)(1.5 g, 3.66 mmol) as a starting material, Compound aa361-a (1.41 g, 86%) was obtained in the same manner as the synthesis of Compound aa358-a using azetidine in place of morpholine.

(987) LCMS (ESI) $m/z=449$ (M+H)+

(988) Retention time: 0.86 min (Analytical condition SQDFA05)

(989) Using the resulting Compound aa361-a (1.4 g, 3.12 mmol), Compound aa361 (1.14 g, 89%) was obtained in the same manner as the synthesis of Compound aa358.

(990) LCMS (ESI) $m/z=409$ (M+H)+

(991) Retention time: 0.69 min (Analytical condition SQDFA05)

(992) Using the resulting Compound aa361 (1.05 g, 2.57 mmol), Compound aa361-Resin (3.64 g, supported amount 0.2984 mmol/g) was obtained in the same manner as the synthesis of Compound aa358-resin.

(993) Another similarly synthesized lot having a different supported amount was also used in the peptide synthesis in the present Example.

(994) Synthesis of Compound aa363-Resin

(995) ##STR00403##

(996) Using Compound aa358-c ((2S)-2-[9H-fluoren-9-ylmethoxycarbonyl(methyl)amino]-4-oxo-4-prop-2-enoxybutanoic acid)(5 g, 12.21 mmol) as a starting material, Compound aa363-a (5.8 g, 94%) was obtained in the same manner as the synthesis of Compound aa358-d using (1R,5S)-3-oxa-8-azabicyclo[3.2.1]octane hydrochloride and 1 eq of DIPEA based on amine, in place of morpholine.

(997) LCMS (ESI) $m/z=505$ (M+H)+

(998) Retention time: 0.87 min (Analytical condition SQDFA05)

(999) Using the resulting Compound aa363-a (5.8 g, 11.49 mmol), Compound aa363 (5.1 g, 96%) was obtained in the same manner as the synthesis of Compound aa358.

(1000) LCMS (ESI) $m/z=465$ (M+H)+

(1001) Retention time: 0.69 min (Analytical condition SQDFA05)

(1002) Using the resulting Compound aa363 (5.1 g, 10.98 mmol), Compound aa363-Resin (18.3 g, supported amount 0.419 mmol/g) was obtained in the same manner as the synthesis of Compound aa358-resin.

(1003) Another similarly synthesized lot having a different supported amount was also used in the peptide synthesis in the present Example.

(1004) Synthesis of Compound aa364-Resin

(1005) ##STR00404##

(1006) Using Compound aa358-c ((2S)-2-[9H-fluoren-9-ylmethoxycarbonyl(methyl)amino]-4-oxo-4-prop-2-enoxybutanoic acid)(0.5 g, 1.221 mmol) as a starting material, Compound aa364-a (359.6 mg, 60%) was obtained in the same manner as the synthesis of Compound aa358-d using 3,3-dimethylpyrrolidine in place of morpholine.

(1007) LCMS (ESI) $m/z=491$ (M+H)+

(1008) Retention time: 0.98 min (Analytical condition SQDFA05)

(1009) Using the resulting Compound aa364-a (338 mg, 0.689 mmol), Compound aa364 (226.2 mg, 73%) was obtained in the same manner as the synthesis of Compound aa358. LCMS (ESI) $m/z=451$ (M+H)+

(1010) Retention time: 0.84 min (Analytical condition SQDFA05)

(1011) Using the resulting Compound aa364 (226 mg, 0.502 mmol), Compound aa364-Resin (731

mg, supported amount 0.455 mmol/g) was obtained in the same manner as the synthesis of Compound aa358-resin.

(1012) Synthesis of Compound aa365-Resin

(1013) ##STR00405##

(1014) Using Compound aa358-c ((2S)-2-[9H-fluoren-9-ylmethoxycarbonyl(methyl)amino]-4-oxo-4-prop-2-enoxybutanoic acid)(0.9 g, 2.198 mmol) as a starting material, a crude material of Compound aa365-a was obtained in the same manner as the synthesis of Compound aa358-d using 4-methylpiperidine in place of morpholine. The resulting crude product was purified by reverse phase column chromatography (0.1% formic acid-containing water-acetonitrile), and further, dissolved in 20% ethyl acetate-hexane, then washed twice with a saturated aqueous sodium hydrogen carbonate solution and once with a saturated aqueous sodium chloride solution, and dried over anhydrous sodium sulfate, and the solvent was distilled off under reduced pressure to give Compound aa365-a (0.587 g, 54%).

(1015) LCMS (ESI) m/z=491 (M+H)+

(1016) Retention time: 1.02 min (Analytical condition SQDFA05)

(1017) Using the resulting Compound aa365-a (535 mg, 1.09 mmol), Compound aa365 (376.6 mg, 77%) was obtained in the same manner as the synthesis of Compound aa358.

(1018) LCMS (ESI) m/z=451 (M+H)+

(1019) Retention time: 0.85 min (Analytical condition SQDFA05) Using the resulting Compound aa365 (364 mg, 0.808 mmol), Compound aa365-Resin (1.2 g, supported amount 0.403 mmol/g) was obtained in the same manner as the synthesis of Compound aa358-resin.

(1020) Synthesis of Compound aa366-Resin

(1021) ##STR00406##

(1022) Using Compound aa358-c ((2S)-2-[9H-fluoren-9-ylmethoxycarbonyl(methyl)amino]-4-oxo-4-prop-2-enoxybutanoic acid)(0.4 g, 0.977 mmol) as a starting material, Compound aa366-a (466 mg, 89%) was obtained in the same manner as the synthesis of Compound aa358-d using 1-(oxetan-3-yl) piperazine in place of morpholine.

(1023) LCMS (ESI) m/z=534 (M+H)+

(1024) Retention time: 0.64 min (Analytical condition SQDFA05)

(1025) A crude product obtained in the same manner as the synthesis of Compound aa358 using the resulting Compound aa366-a (466 mg, 0.873 mmol), was purified by reverse phase column chromatography (0.1% formic acid-containing acetonitrile/0.1% formic acid-containing distilled water) to give Compound aa366 (385 mg, 89%).

(1026) LCMS (ESI) m/z=494 (M+H)+

(1027) Retention time: 0.50 min (Analytical condition SQDFA05)

(1028) Using the resulting Compound aa366 (385 mg, 0.78 mmol), Compound aa366-Resin (1.49 g, supported amount 0.266 mmol/g) was obtained in the same manner as the synthesis of Compound aa358-resin.

(1029) Synthesis of Compound aa367-Resin

(1030) ##STR00407##

(1031) Using Compound aa358-c ((2S)-2-[9H-fluoren-9-ylmethoxycarbonyl(methyl)amino]-4-oxo-4-prop-2-enoxybutanoic acid)(725 mg, 1.77 mmol) as a starting material, Compound aa367-a (164 mg, 20%) was obtained in the same manner as the synthesis of Compound aa358-d. LCMS (ESI) m/z=465 (M+H)+

(1032) Retention time: 0.86 min (Analytical condition SQDFA05)

(1033) Using the resulting Compound aa367-a (146.4 mg, 0.315 mmol), Compound aa367 (111 mg, 83%) was obtained in the same manner as the synthesis of Compound aa358. Another similarly synthesized lot was also added, and the following reaction was carried out. LCMS (ESI) m/z=425 (M+H)+

(1034) Retention time: 0.69 min (Analytical condition SQDFA05)

(1035) Using the resulting Compound aa367 (0.25 g, 0.589 mmol), Compound aa367-Resin (1.06 g, supported amount 0.283 mmol/g) was obtained in the same manner as the synthesis of Compound aa358-resin.

(1036) Synthesis of Compound aa368-Resin

(1037) ##STR00408##

(1038) In a nitrogen atmosphere, dimethylamine hydrochloride (0.991 g, 12.15 mmol), WSCI.Math.HCl (2.8 g, 14.58 mmol), and DIPEA (2.117 mL, 12.15 mmol) were successively added at 0° C. to a DMF (24.3 mL) solution of Fmoc-Asp (OtBu)-OH (Compound aa368-a), 4-tert-butyl N-[(9H-fluoren-9-ylmethoxy) carbonyl]-L-aspartate, CAS No. 71989-14-5 (5 g, 12.15 mmol), and HOBt monohydrate (2.047 g, 13.37 mmol), and the mixture was stirred at 0° C. for 50 minutes. Ethyl acetate/hexane (1/1, 100 mL) and saturated brine/water (1/1, 50 mL) were added to the reaction solution, and the organic layer was separated. The resulting organic layer was successively washed with a saturated aqueous ammonium chloride solution, water, and saturated brine, and dried over anhydrous sodium sulfate, and then the solvent was distilled off under reduced pressure. The resulting residue was purified by silica gel column chromatography (hexane/ethyl acetate) to give Compound aa368-b (5.02 g, 94%).

(1039) LCMS (ESI) $m/z=439$ (M+H)+

(1040) Retention time: 1.03 min (Analytical condition SQDAA05)

(1041) The operation of adding toluene (150 mL) to Compound aa368-b (5 g, 11.4 mmol) and distilling off the solvent under reduced pressure was performed three times. The residue was dissolved in DCM (5.06 mL), TFA (10.13 mL, 137 mmol) was added dropwise at 0° C. in a nitrogen atmosphere, and the mixture was stirred at room temperature for 1 hour. The reaction solution was cooled to 0° C., diethyl ether (10.1 mL) was added, and an 8 N aqueous sodium hydroxide solution (17.1 mL) was added dropwise. Furthermore, a saturated aqueous sodium dihydrogen phosphate solution (7.6 mL) and water (5 mL) were added. This solution was extracted with ethyl acetate, the resulting organic layer was washed with a saturated aqueous sodium dihydrogen phosphate solution/water (1/1), dried over anhydrous sodium sulfate, and then the solvent was distilled off under reduced pressure to give Compound aa368 (2.63 g, 60%). The compound was used in the next reaction without further purification.

(1042) LCMS (ESI) $m/z=383$ (M+H)+

(1043) Retention time: 0.80 min (Analytical condition SQDAA05)

(1044) Using the resulting Compound aa368 (2.367 g, 6.19 mmol), Compound aa368-Resin (9.07 g, supported amount 0.399 mmol/g) was obtained in the same manner as the synthesis of Compound aa358-resin.

(1045) Another similarly synthesized lot having a different supported amount was also used in the peptide synthesis in the present Example.

(1046) Synthesis of Compound aa369-Resin

(1047) ##STR00409##

(1048) Using Compound aa368-a (1 g, 2.43 mmol) as a starting material, Compound aa369-b (713 mg, 61%) was obtained in the same manner as the synthesis of Compound aa368-b using morpholine in place of dimethylamine hydrochloride and DIPEA.

(1049) LCMS (ESI) $m/z=481$ (M+H)+

(1050) Retention time: 0.88 min (Analytical condition SQDFA05)

(1051) Using the resulting Compound aa369-b (400 mg, 0.832 mmol), Compound aa369 (353.4 mg, 100%) was obtained in the same manner as the synthesis of Compound aa368, provided that triethylamine was used in place of an 8 N aqueous sodium hydroxide solution for work-up.

(1052) LCMS (ESI) $m/z=425$ (M+H)+

(1053) Retention time: 0.66 min (Analytical condition SQDFA05)

(1054) Using the resulting Compound aa369 (326 mg, 0.768 mmol), Compound aa369-Resin (1.21 g, supported amount 0.415 mmol/g) was obtained in the same manner as the synthesis of

Compound aa358-resin.

(1055) Another similarly synthesized lot having a different supported amount was also used in the peptide synthesis in the present Example.

(1056) Synthesis of Compound aa370-Resin

(1057) ##STR00410##

(1058) Using Compound aa370 (7.1 g, 21.82 mmol) and 2-chlorotriptyl chloride resin (1.6 mmol/g, 100-200 mesh, 1% DVB, 27.25 g, 43.6 mmol) purchased from a commercial supplier, Compound aa370-resin (33.44 g, supported amount 0.598 mmol/g) was obtained in the same manner as the synthesis of Compound aa358-resin.

(1059) Another similarly synthesized lot having a different supported amount was also used in the peptide synthesis in the present Example.

(1060) Synthesis of Compound aa371-Resin

(1061) ##STR00411##

(1062) Using Compound aa371 (11.5 g, 33.9 mmol) and 2-chlorotriptyl chloride resin (1.69 mmol/g, 100-200 mesh, 1% DVB, 50 g, 84.5 mmol) purchased from a commercial supplier, Compound aa371-resin (58.95 g, supported amount 0.536 mmol/g) was obtained in the same manner as the synthesis of Compound aa358-resin.

(1063) Another similarly synthesized lot having a different supported amount was also used in the peptide synthesis in the present Example.

(1064) Synthesis of Compound aa372-Resin

(1065) ##STR00412##

(1066) Using Compound aa372 (1 g, 2.85 mmol) and 2-chlorotriptyl chloride resin (1.6 mmol/g, 100-200 mesh, 1% DVB, 7 g, 21.39 mmol) purchased from a commercial supplier, Compound aa372-resin (7.19 g, supported amount 0.377 mmol/g) was obtained in the same manner as the synthesis of Compound aa358-resin.

(1067) Synthesis of Compound aa386

(1068) ##STR00413##

(1069) A DCM solution of 2-phenylpropan-2-yl 2,2,2-trichloroacetoimidate (2.74 g, 9.77 mmol) was added to a DCM (15 mL) solution of Compound aa358-c (2 g, 4.88 mmol), and the mixture was stirred at room temperature for 2 hours. The solvent was distilled off from the reaction solution under reduced pressure, and the resulting residue was purified by reverse phase column chromatography to give Compound aa386-a (2.10 g, 81%).

(1070) LCMS (ESI) $m/z=550.3$ (M+Na)+

(1071) Retention time: 0.76 min (Analytical condition SQDAA50)

(1072) In a nitrogen atmosphere, tetrakis(triphenylphosphine) palladium (0) (0.046 g, 0.04 mmol) and phenylsilane (0.343 mL, 2.79 mmol) were added to a DCM (15 mL) solution of Compound aa386-a (2.1 g, 3.98 mmol) at room temperature, and the mixture was stirred for 1 hour. The reaction solution was diluted with TBME and a 5% aqueous sodium hydrogen carbonate solution, the solvent was distilled off under reduced pressure, and the resulting residue was purified by reverse phase column chromatography to give Compound aa386 (1.67 g, 86%). LCMS (ESI) $m/z=510$ (M+Na)+

(1073) Retention time: 0.94 min (Analytical condition SQDFA05)

(1074) Using the resulting Compound aa386 (1.17 g, 2.4 mmol), Compound aa386-resin (3.6 g, 0.377 mmol/g) was obtained in the same manner as the synthesis of Compound aa358-resin.

(1075) Synthesis of Compound aa373-Resin

(1076) ##STR00414##

(1077) In a nitrogen atmosphere, a toluene (500 mL) suspension of aa373-a (20 g, 50.58 mmol), paraformaldehyde (4.56 g, 151.74 mmol), and p-toluenesulfonic acid (0.09 g, 0.5 mmol) was stirred at 105° C. for 16 hours. After the reaction mixture was cooled to room temperature, the solvent was distilled off under reduced pressure. The resulting residue was dissolved in tert-butyl

methyl ether, washed with an aqueous sodium carbonate solution, and the organic phase was dried over sodium sulfate. The solvent was distilled off under reduced pressure to give aa373-b (20 g, 90%).

(1078) LCMS (ESI) $m/z=408$ (M+H)+

(1079) Retention time: 1.05 min (Analytical condition SMD method_02)

(1080) In a nitrogen atmosphere, zinc bromide (11.06 g, 49.09 mmol) was added to a DCM (200 mL) solution of aa373-b (20 g, 49.09 mmol) and triethylsilane (11.42 g, 98.18 mmol). The reaction mixture was stirred at room temperature for 48 hours, and the solvent was distilled off under reduced pressure. The resulting residue was dissolved in an aqueous potassium carbonate solution and washed twice with hexane. The aqueous phase was adjusted to pH 2 with hydrochloric acid, and then extracted with ethyl acetate 3 times. After the organic phase was dried over sodium sulfate, the solvent was distilled off under reduced pressure to give aa373-c (15 g, 93%).

(1081) LCMS (ESI) $m/z=410$ (M+H)+

(1082) Retention time: 0.98 min (Analytical condition SMD method_02)

(1083) HOBt (5.7 g, 42.18 mmol) was added to a DMF (110 mL)-DCM (32 mL) solution of aa373-c (15.7 g, 38.35 mmol), and then WSCDI (8.82 g, 46.01 mmol) was added. The reaction mixture was stirred at room temperature for 15 minutes, and then cooled in an ice bath. Dimethylamine (1.90 g, 42.18 mmol) was added dropwise over 5 minutes. The reaction mixture was stirred while being ice-cooled until the reaction was complete, and then ethyl acetate was added. The resulting mixture was washed with 1 N hydrochloric acid, water, an aqueous sodium hydrogen carbonate solution, and brine. After the organic phase was dried over sodium sulfate, the solvent was distilled off under reduced pressure to give aa373-d (13 g, 78%).

(1084) LCMS (ESI) $m/z=437$ (M+H)+

(1085) Retention time: 1.86 min (Analytical condition SMD method_02)

(1086) In a nitrogen atmosphere, phenylsilane (0.343 mL, 2.72 mmol) was added dropwise to a DCM (7.8 mL) solution of aa373-d (1.70 g, 3.88 mmol) and tetrakis(triphenylphosphine) palladium (()) (0.045 g, 0.039 mmol). The reaction mixture was stirred at room temperature for 30 minutes, and then diluted with tert-butyl methyl ether (17 mL). An aqueous sodium hydrogen carbonate solution (17 mL) was added to the resulting mixture to quench the reaction. The separated organic phase was extracted with water (8.5 mL), and DCM (17 mL) was added to the combined aqueous phase. Phosphoric acid (1.36 mL, 23.31 mmol) was added dropwise to the resulting mixture. After the organic phase was separated, the aqueous phase was extracted with DCM (17 mL), and the combined organic phase was washed with brine (17 mL). After the organic phase was dried over sodium sulfate, the solvent was distilled off under reduced pressure to give aa373 (1.31 g, 85%).

(1087) LCMS (ESI) $m/z=397$ (M+H)+

(1088) Retention time: 0.71 min (Analytical condition SQDFA05)

(1089) 2-Chlorotriyl chloride resin (1.25 mmol/g, 5.3 g, 6.62 mmol) and DCM (37 mL) were placed in a filter-equipped reaction vessel (60 mL), and left to stand still at room temperature for 45 minutes. After DCM was removed by applying nitrogen pressure, a DCM (37 mL) solution of Compound aa373 (1.31 g, 3.31 mmol), DIPEA (2.77 mL, 15.9 mmol), and methanol (1.07 mL, 26.5 mmol) was added to the reaction vessel, and the reaction vessel was shaken at room temperature for 60 minutes. After the reaction solution was removed by applying nitrogen pressure, a DCM (37 mL) solution of DIPEA (2.77 mL, 15.9 mmol) and methanol (7.5 mL, 185 mmol) was added to the reaction vessel, and the reaction vessel was shaken at room temperature for 90 minutes. After the reaction solution was removed by applying nitrogen pressure, DCM (37 mL) was added, mixed, and then discharged by applying nitrogen pressure. This resin washing operation using DCM was repeated 5 times in total, and the resulting resin was dried under reduced pressure overnight to give aa373-resin (6.07 g).

(1090) The amount of supported amino acid on the resin was calculated as follows.

(1091) The resulting Compound aa373-resin (10 mg) was placed in a reaction vessel, DMF (4 mL)

was added, and the reaction vessel was shaken at room temperature for 30 minutes. Then, DBU (40 μ L) was added, and the reaction vessel was shaken at 30° C. for 15 minutes. Then, DMF was added to the reaction mixture to 10 mL, and 80 μ L of this solution was diluted with DMF so as to have a liquid volume of 1 mL. The resulting diluted solution was analyzed by LC/MS (injection volume: 5 μ L), and the amount of supported aa373 was calculated to be 0.404 mmol/g from the UV area value of dibenzofulvene (294 nm: 4211.62; 304 nm: 3791.08). Using a calibration curve created based on the UV area values of dibenzofulvene at wavelengths of 294 nm and 304 nm on each measurement day utilizing, as a standard substance, a mixed solution of Fmoc-Gly-OH (purchased from a commercial supplier) having a known concentration and DBU, the average value of the supported amounts calculated for each wavelength was regarded as the amount of supported resin.

(1092) Synthesis of Compound aa375-Resin

(1093) ##STR00415##

(1094) In a nitrogen atmosphere, DBU (3.15 mL, 20.90 mmol) was added dropwise to a dehydrated DMF (50 mL) solution of aa358-d (10 g, 20.90 mmol) at room temperature, and the mixture was stirred for 10 minutes. Pyridine hydrochloride (2.66 g, 22.99 mmol) was added to the reaction mixture at room temperature, and the mixture was stirred for 10 minutes. A separately prepared active ester solution (as described below) was added to the reaction mixture at room temperature, and then DIPEA (4.01 mL, 22.99 mmol) was added dropwise. After the resulting reaction mixture was stirred at room temperature for 4 hours and 15 minutes, ethyl acetate (100 mL), hexane (20 mL), and 1 N hydrochloric acid (100 mL) were added. The resulting mixture was extracted with ethyl acetate-hexane (5:1)(total amount of organic phase of about 300 mL), the organic phase was washed with 1 N hydrochloric acid (100 mL), water (100 mL), an aqueous sodium hydrogen carbonate solution (100 mL $\times$ 2), and brine (100 mL), dried over sodium sulfate, and the solvent was distilled off under reduced pressure. The resulting crude product was purified by normal phase silica gel chromatography (hexane/ethyl acetate). This was dissolved in TBME (100 mL) and washed with an aqueous potassium carbonate solution (1 w/v %, 50 mL). The organic phase was washed with saturated brine/water (1/1, 50 mL) and dried over sodium sulfate, and the solvent was distilled off under reduced pressure to give aa375-b (9.98 g, 81%).

(1095) The active ester solution was prepared as follows.

(1096) In a nitrogen atmosphere, dehydrated DMF (55 mL) was added to a mixture of (α S)- α -[(9H-fluoren-9-ylmethoxy) carbonyl]methylamino]cyclopentaneacetic acid (7.53 g, 19.85 mmol), WSCI.Math.HCl (5.21 g, 27.2 mmol), and Oxyma (3.56 g, 25.08 mmol) at room temperature and stirred for 40 minutes, and the resulting reaction mixture was used as an active ester solution.

(1097) LCMS (ESI) m/z=618 (M+H)+

(1098) Retention time: 0.96 min (Analytical condition SQDFA05)

(1099) In a nitrogen atmosphere, dehydrated DCM (15 mL) was added to a mixture of aa375-b (9.90 g, 16.03 mmol) and tetrakis(triphenylphosphine) palladium (())(0.185 g, 0.16 mmol), and the mixture was stirred at room temperature. Phenylsilane (1.38 mL, 11.22 mmol) was added dropwise to the resulting solution. The resulting reaction mixture was stirred for 30 minutes, TBME (100 mL) was added, and then an aqueous sodium hydrogen carbonate solution (obtained by diluting a saturated aqueous solution by 2-fold)(100 mL) was slowly added dropwise to quench the reaction. The resulting mixture was extracted twice with water (total amount of aqueous phase of about 150 mL), and then DCM (100 mL) and phosphoric acid (5.6 mL) were added to the aqueous phase. The resulting mixture was extracted twice with DCM (total amount of organic phase of about 200 mL), the organic phase was washed with brine (80 mL $\times$ 2) and dried over sodium sulfate, the solvent was distilled off under reduced pressure, and the resulting aa375 (9.57 g, quant.) was used in the next reaction.

(1100) LCMS (ESI) m/z=578 (M+H)+

(1101) Retention time: 0.79 min (Analytical condition SQDFA05)

(1102) 2-Chlorotrityl chloride resin (1.36 mmol/g, 20.5 g, 15.07 mmol) and DCM (140 mL) were

placed in a filter-equipped reaction vessel, and left to stand still at room temperature for 1 hour. After DCM was removed by applying nitrogen pressure, a DCM (140 mL) solution of Compound aa375 (9.50 g, 16.45 mmol), methanol (5.32 mL, 132 mmol), and DIPEA (13.8 mL, 79 mmol) was added to the reaction vessel, and the reaction vessel was shaken at 25° C. at 60 rpm for 60 minutes. After the reaction solution was removed by applying nitrogen pressure, a DCM (140 mL) solution of methanol (20 mL, 493 mmol) and DIPEA (13.8 mL, 79 mmol) was added to the reaction vessel, and the reaction vessel was shaken at 25° C. at 60 rpm for 60 minutes. After the reaction solution was removed by applying nitrogen pressure, DCM (140 mL) was added, mixed, and then discharged by applying nitrogen pressure. This resin washing operation with DCM was repeated 5 times in total, and the resulting resin was dried under reduced pressure for 1 day to give aa375-resin (26.89 g).

(1103) The amount of supported amino acid on the resin was calculated as follows.

(1104) The resulting compound aa375-resin (10 mg) was placed in a reaction vessel, DMF (2 mL) was added, and the mixture was left to stand still at room temperature for 1 hour. Then, DBU (40 µL) was added, and the reaction vessel was shaken at 25° C. for 30 minutes. Then, DMF (8 mL) was added to the reaction mixture, and 1 mL of this solution was diluted with DMF (11.5 mL). The absorbance (294 nm) of the resulting diluted solution was measured (measured with Shimadzu UV-1600PC (cell length 1.0 cm)), and thus the amount of supported Compound aa375 was calculated to be 0.415 mmol/g.

(1105) Synthesis of Compound aa374-Resin

(1106) ##STR00416##

(1107) Using aa359-a as a starting material, aa074-b (5.0 g, 43%) was obtained in the same manner as the synthesis of Compound aa0375-b.

(1108) LCMS (ESI) $m/z=576.5$ (M+H)⁺

(1109) Retention time: 1.02 min (Analytical condition SQDFA05)

(1110) Using the resulting aa374-b, aa074 (0.786 g, 92%) was obtained under the same reaction conditions as the synthesis of Compound aa375.

(1111) LCMS (ESI) $m/z=536$ (M+H)⁺

(1112) Retention time: 0.85 min (Analytical condition SQDFA05) Using the resulting aa374, aa374-resin (2.72 g) was obtained under the same reaction conditions as the synthesis of Compound aa375-resin. The supported amount calculated in the same manner as aa375-resin was 0.345 mmol/g.

(1113) Synthesis of Compound aa376-Resin

(1114) ##STR00417## ##STR00418##

(1115) Compound aa359-resin (10 g, 0.362 mmol/g, 3.62 mmol) and DCM (100 mL) were added to a filter-equipped reaction vessel to swell the resin. After DCM was removed, the resin was washed 4 times with DMF (100 mL). A DMF solution of DBU (2% v/v, 60 mL) was added, the mixture was shaken for 5 minutes at room temperature, and then the reaction solution was removed. The resin was washed 4 times with NMP (100 mL), and washed with a DCM solution (100 mL) of triethylamine hydrochloride (1.57 g, 11.4 mmol). The resin was washed 4 times with DCM (100 mL) and 4 times with THF (100 mL) to give the resin-supported Compound aa376-a.

(1116) A THE solution (100 mL) of nosyl chloride (3.66 g, 16.5 mmol) and 2,4,6-cholidine (6.60 mL, 49.6 mmol) were added thereto, and the mixture was shaken at room temperature for 3 hours. Then, the reaction solution was removed, and the resin was washed 4 times with THE (100 mL) and washed 4 times with DCM (100 mL) to give the resin-supported Compound aa376-b. The amino acid was cleaved from the resin by TFE/DCM (1/1) using a small amount of the resin-supported Compound aa376-b, and the structure was verified by LC/MS. LCMS (ESI) $m/z=347$ (M+H)⁺

(1117) Retention time: 0.45 min (Analytical condition SQDFA05)

(1118) Then, 6 g of this Compound 376-b-resin was weighed out, and swelled by adding DCM (60

mL). After DCM was removed, the resin was washed 3 times with THF (42 mL). A THF solution (20 mL) of DIAD (2.11 mL, 10.9 mmol) was added to a THF solution (20 mL) of triphenylphosphine (2.85 g, 10.9 mmol), and the mixture was shaken for 15 minutes. Ethanol (1.27 mL, 21.7 mmol) was added thereto, and the mixture was left to stand still for 5 minutes. This was added to the resin, the mixture was shaken at room temperature for 1 hour, and then the reaction solution was removed. The resin was washed 4 times with THF (42 mL) and washed 4 times with DCM (42 mL) to give the resin-supported Compound aa376-c. The amino acid was cleaved from the resin by TFE/DCM (1/1) using a small amount of the resin-supported Compound aa376-c, and the structure was verified by LC/MS.

(1119) LCMS (ESI) $m/z=375$ (M+H)+

(1120) Retention time: 0.56 min (Analytical condition SQDFA05)

(1121) DCM (60 mL) was added to swell this Compound aa376-c-resin. After DCM was removed, the resin was washed 3 times with NMP (42 mL). An NMP solution of DBU (1.64 mL, 10.9 mmol) and an NMP solution (20 mL) of 1-dodecanethiol (4.93 mL, 21.7 mmol) were added, and the mixture was shaken at 40° C. for 1 hour. Then, the reaction solution was removed, and the resin was washed 4 times with NMP (42 mL) and washed 4 times with DCM (42 mL). This was washed 8 times with DMF (42 mL) and washed 4 times with DCM (42 mL) to give the resin-supported Compound aa376-d. The amino acid was cleaved from the resin by TFE/DCM (1/1) using a small amount of the resin-supported Compound aa376-d, and the structure was verified by LC/MS.

(1122) LCMS (ESI) $m/z=189$ (M+H)+

(1123) Retention time: 0.14 min (Analytical condition SQDFA05)

(1124) Then, 5 g of this Compound 376-d-resin was weighed out, and swelled by adding DCM (50 mL). After DCM was removed, the resin was washed 4 times with NMP (50 mL). Compound aa330 (Fmoc-MeGly (cPent)-OH, 2.75 g, 7.24 mmol) was dissolved in DCM (72 mL), thionyl chloride (1.32 mL, 18.1 mmol) and DMF (0.056 mL, 0.724 mmol) were added thereto, and the mixture was stirred at room temperature for 2 hours. After the solvent was distilled off from this under reduced pressure, NMP (70 mL) and 2,4,6-cholidine (2.4 mL, 18.1 mmol) were added. This was added to the resin, and the mixture was shaken at 40° C. for 1.5 hours. Then, the reaction solution was removed, and the resin was washed 4 times with NMP (50 mL) and washed 4 times with DCM (50 mL) to give the resin-supported Compound aa376. The amino acid was cleaved from the resin by TFE/DCM (1/1) using a small amount of the resin-supported Compound aa376, and the structure was verified by LC/MS.

(1125) LCMS (ESI) $m/z=551$ (M+H)+

(1126) Retention time: 0.86 min (Analytical condition SQDFA05)

(1127) Synthesis of Compounds aa377-resin, aa378-resin, aa379-resin, and aa380-Resin

(1128) ##STR00419## ##STR00420## ##STR00421##

(1129) In place of ethanol, the alcohols shown in Table 7 below were reacted with aa376-b-resin in the same manner as aa376-c-resin, and thus resin-supported aa377-c to aa380-c were obtained. Using them, resin-supported aa377-d to aa380-d were obtained in the same manner as aa376-d-resin. Using them, resin-supported aa377 to aa380 were synthesized in the same manner as aa376-resin. The amino acid was cleaved from the resin by TFE/DCM (1/1) using a small amount of the resin-supported Compounds aa377 to aa380, and the structure was verified by LC/MS.

(1130) TABLE-US-00007 TABLE 7 LCMS Retention time (ESI) (Analytical m/z condition Intended product Alcohol (M + H).sup.+ SQDFA05) Compound aa377-resin 1-propanol 565 0.92 Min Compound aa378-resin 2-Cyclobutylethanol 605 1.03 Min Compound aa379-resin 2-Cyclopentylethanol 619 1.07 Min Compound aa380-resin 2-Phenylethanol 627 1.01 Min Synthesis of Compounds aa381-resin, aa382-resin, aa383-resin, aa384-resin, and aa385-Resin

(1131) ##STR00422## ##STR00423## ##STR00424##

(1132) Using aa358-resin in place of aa359-resin, resin-supported aa381-a was obtained in the same manner as aa376-a-resin. Using this compound, resin-supported aa381-b was obtained in the

same manner as aa376-b-resin. Using this compound, resin-supported aa381-c to aa385-c were obtained in the same manner as aa376-c-resin or using alcohols shown in Table 8 below to react with the compound in place of ethanol. Using these compounds, resin-supported aa381-d to aa385-d were obtained in the same manner as aa376-d-resin. Using these compounds, resin-supported aa381 to aa385 were synthesized in the same manner as aa376-resin. The amino acid was cleaved from the resin by TFE/DCM (1/1) using a small amount of resin-supported Compounds aa381 to aa385, and the structure was verified by LC/MS.

(1133) TABLE-US-00008 TABLE 8 LCMS Retention time (ESI) (Analytical m/z condition Intended product Alcohol (M + H).sup.+ SQDFA05) Compound aa381-resin ethanol 593 0.85 Compound aa382-resin 1-propanol 607 0.91 Compound aa383-resin 2-Cyclobutylethanol 647 1.02 Compound aa384-resin 2-Cyclopentylethanol 661 1.06 Compound aa385-resin 2-Phynylethanol 669 1.00

Synthesis of Compound aa427-Resin

(1134) ##STR00425##

(1135) aa374-resin (0.420 mmol/g, 10 g) was placed in a filter-equipped reaction vessel, dichloromethane (100 mL) was added, and the mixture was shaken at room temperature for 5 minutes to swell the resin. After dichloromethane was removed with a filter, the resin was washed 4 times with DMF (100 mL). Subsequently, a 2% DBU/DMF solution (de-Fmoc solution: 60 mL) was added to the resin, and the mixture was shaken at room temperature for 10 minutes to carry out an Fmoc-removal reaction. After the de-Fmoc solution was removed, the resin was washed 6 times with DMF (100 mL). An Fmoc-MecVal-OH (Compound aa421) elongation reaction was carried out on the resulting resin. The elongation reaction was carried out by adding a solution obtained by mixing DIC (2.63 mL, 16.8 mmol) with a DMF solution (40 mL) of Fmoc-MecVal-OH (Compound aa421, 2.95 g, 8.40 mmol) and oxyma (597 mg, 4.20 mmol) to the resin, and shaking the mixture at 30° C. for 86 hours. After the liquid phase of the elongation reaction was removed with a filter, the resin was washed 6 times with DMF (100 mL) and 6 times with dichloromethane (100 mL) and dried to give aa427-a-resin.

(1136) Dichloromethane (100 mL) was added to the resulting aa427-a-resin, and the mixture was shaken at room temperature for 5 minutes to swell the resin. After dichloromethane was removed with a filter, the resin was washed 4 times with DMF (100 mL). Subsequently, a 2% DBU/DMF solution (de-Fmoc solution: 60 mL) was added to the resin, and the mixture was shaken at room temperature for 10 minutes to carry out an Fmoc-removal reaction. After the de-Fmoc solution was removed, the resin was washed 6 times with DMF (100 mL) and 6 times with dichloromethane (100 mL). An Fmoc-Hyp (Et)-OH (Compound aa086) elongation reaction was carried out on the resulting resin. The elongation reaction was carried out by, first, adding thionyl chloride (3.07 mL, 42.0 mmol) and DMF (0.131 mL, 1.68 mmol) to a DCM solution (84 mL) of Fmoc-Hyp (Et)-OH (Compound aa086, 6.41 g, 16.8 mmol), and stirring the mixture at room temperature for 4 hours in a nitrogen atmosphere. The solvent and thionyl chloride were distilled off under reduced pressure to give an acid chloride of Fmoc-Hyp (Et)-OH. This acid chloride was dissolved in DCM (100 mL) and mixed with collidine (5.59 mL, 42.0 mmol), the resulting solution was added to the resin, and the mixture was shaken at 40° C. for 13 hours. After the liquid phase of the elongation reaction was removed with a filter, the resin was washed 6 times with DMF (100 mL) and 6 times with dichloromethane (100 mL) and dried to give aa427-resin. Amino acid was cleaved from the resin by TFE/DCM (1/1) using a small amount of resin-supported Compound aa427, and the structure was verified by LC/MS.

(1137) LCMS (ESI) m/z=789 (M+H)+

(1138) Retention time: 0.84 min (Analytical condition SQDFA05)

(1139) Synthesis of Compound aa428-Resin

(1140) ##STR00426##

(1141) Using aa427-a-resin, aa428-resin was obtained in the same manner as Compound aa427-

resin except that Fmoc-Pro (4S-Me)-OH (Compound aa239) was used in place of Fmoc-Hyp (Et)-OH. Amino acid was cleaved from the resin by TFE/DCM (1/1) using a small amount of resin-supported Compound aa428, and the structure was verified by LC/MS.

(1142) LCMS (ESI) $m/z=759$ (M+H)⁺

(1143) Retention time: 0.85 min (Analytical condition SQDFA05)

(1144) Synthesis of Compound aa429-Resin

(1145) ##STR00427##

(1146) Using aa374-resin, aa429-a-resin was obtained in the same manner as Compound aa427-a-resin except that Fmoc-MecVal (3-Me₂)-OH (Compound aa420) was used in place of Fmoc-MecVal-OH.

(1147) Using the resulting aa429-a-resin, aa429-resin was obtained in the same manner as Compound aa427-resin. Amino acid was cleaved from the resin by TFE/DCM (1/1) using a small amount of resin-supported Compound aa429, and the structure was verified by LC/MS. LCMS (ESI) $m/z=817$ (M+H)⁺

(1148) Retention time: 0.94 min (Analytical condition SQDFA05)

(1149) Synthesis of Compound aa430-Resin

(1150) ##STR00428## ##STR00429##

(1151) aa359-resin (0.320 mmol/g, 5 g) was placed in a filter-equipped reaction vessel, dichloromethane (50 mL) was added, and the mixture was shaken at room temperature for 5 minutes to swell the resin. After dichloromethane was removed with a filter, the resin was washed 4 times with DMF (50 mL). Subsequently, a 2% DBU/DMF solution (de-Fmoc solution: 30 mL) was added to the resin, and the mixture was shaken at room temperature for 10 minutes to carry out an Fmoc-removal reaction. After the de-Fmoc solution was removed, the resin was washed 6 times with DMF (50 mL). An Fmoc-MeNva (3-Et)-OH (Compound aa331) elongation reaction was carried out on the resulting resin. The elongation reaction was carried out by adding to the resin a solution obtained by mixing an NMP solution (15 mL) of Fmoc-MeNva (3-Et)-OH (Compound aa331, 2.44 g, 6.40 mmol) and HOObt (653 mg, 4.00 mmol) with a DMF solution (15 mL) of DIC (1.50 mL, 9.60 mmol), shaking the mixture at 50° C. for 14 hours, removing the liquid phase of the elongation reaction with a filter, again adding to the resin a solution obtained by mixing an NMP solution (15 mL) of Fmoc-MeNva (3-Et)-OH (Compound aa331, 2.44 g, 6.40 mmol) and HOObt (653 mg, 4.00 mmol) with a DMF solution (15 mL) of DIC (1.50 mL, 9.60 mmol), and shaking the mixture at 50° C. for 14 hours. After the liquid phase of the elongation reaction was removed with a filter, the resin was washed 4 times with DMF (100 mL) and 4 times with dichloromethane (100 mL) and dried to give aa430-a-resin.

(1152) Using the resulting aa430-a-resin, aa430-b-resin was obtained in the same manner as Compound aa427-resin except that Fmoc-MecVal-OH was used in place of Fmoc-Hyp (Et)-OH.

(1153) Using the resulting aa430-b-resin, aa430-resin was obtained in the same manner as Compound aa427-resin. Amino acid was cleaved from the resin by TFE/DCM (1/1) using a small amount of resin-supported Compound aa430, and the structure was verified by LC/MS. LCMS (ESI) $m/z=791$ (M+H)⁺

(1154) Retention time: 0.88 min (Analytical condition SQDFA05)

(1155) Synthesis of Compound aa431-Resin

(1156) ##STR00430##

(1157) Using aa430-a-resin, aa431-a-resin was obtained in the same manner as Compound aa427-a-resin except that Fmoc-MecVal (3-Me₂)-OH (Compound aa420) was used in place of Fmoc-MecVal-OH.

(1158) Using the resulting aa431-a-resin, aa431-resin was obtained in the same manner as Compound aa427-resin. Amino acid was cleaved from the resin by TFE/DCM (1/1) using a small amount of resin-supported Compound aa431, and the structure was verified by LC/MS. LCMS (ESI) $m/z=819$ (M+H)⁺

(1159) Retention time: 0.98 min (Analytical condition SQDFA05)
(1160) Synthesis of Compound aa432-Resin
(1161) ##STR00431##
(1162) Using aa431-a-resin, aa432-resin was obtained in the same manner as Compound aa427-resin except that Fmoc-Hyp (nPr)—OH (Compound aa246) was used in place of Fmoc-Hyp (Et)-OH. Amino acid was cleaved from the resin by TFE/DCM (1/1) using a small amount of resin-supported Compound aa432, and the structure was verified by LC/MS.
(1163) LCMS (ESI) m/z=833 (M+H)+
(1164) Retention time: 1.03 min (Analytical condition SQDFA05)
(1165) Synthesis of Compound aa433-Resin
(1166) ##STR00432##
(1167) Using aa431-a-resin, aa433-resin was obtained in the same manner as Compound aa427-resin except that Fmoc-Hyp (iPr)—OH (Compound aa416) was used in place of Fmoc-Hyp (Et)-OH. Amino acid was cleaved from the resin by TFE/DCM (1/1) using a small amount of resin-supported Compound aa433, and the structure was verified by LC/MS.
(1168) LCMS (ESI) m/z=833 (M+H)+
(1169) Retention time: 1.01 min (Analytical condition SQDFA05)
(1170) Synthesis of Compound aa434-Resin
(1171) ##STR00433##
(1172) Using aa431-a-resin, aa434-resin was obtained in the same manner as Compound aa427-resin except that Fmoc-Hyp (cBu)-OH (Compound aa415) was used in place of Fmoc-Hyp (Et)-OH. Amino acid was cleaved from the resin by TFE/DCM (1/1) using a small amount of resin-supported Compound aa434, and the structure was verified by LC/MS.
(1173) LCMS (ESI) m/z=845 (M+H)+
(1174) Retention time: 1.04 min (Analytical condition SQDFA05)
(1175) Synthesis of Compound aa435-Resin
(1176) ##STR00434##
(1177) Using aa431-a-resin, aa435-resin was obtained in the same manner as Compound aa427-resin except that Fmoc-Hyp (cPent)-OH (Compound aa414) was used in place of Fmoc-Hyp (Et)-OH. Amino acid was cleaved from the resin by TFE/DCM (1/1) using a small amount of resin-supported Compound aa435, and the structure was verified by LC/MS.
(1178) LCMS (ESI) m/z=859 (M+H)+
(1179) Retention time: 1.08 min (Analytical condition SQDFA05)
(1180) Synthesis of Compound aa436-Resin
(1181) ##STR00435##
(1182) Using aa431-a-resin, aa436-resin was obtained in the same manner as Compound aa427-resin except that Fmoc-Pro (4-cPr)—OH (Compound aa278) was used in place of Fmoc-Hyp (Et)-OH. Amino acid was cleaved from the resin by TFE/DCM (1/1) using a small amount of resin-supported Compound aa436, and the structure was verified by LC/MS.
(1183) LCMS (ESI) m/z=801 (M+H)+
(1184) Retention time: 0.99 min (Analytical condition SQDFA05)
(1185) Synthesis of Compound aa437-Resin
(1186) ##STR00436##
(1187) Using aa362-resin as a starting material, aa437-a-resin was obtained in the same manner as Compound aa430-a-resin except that Fmoc-MeGly (cPent)-OH (Compound aa330) was used in place of Fmoc-MeNva (3-Et)-OH.
(1188) Using the resulting aa437-a-resin, aa437-b-resin was obtained in the same manner as Compound aa427-a-resin.
(1189) Using the resulting aa437-b-resin, aa437-resin was obtained in the same manner as Compound aa427-resin. Amino acid was cleaved from the resin by TFE/DCM (1/1) using a small

amount of resin-supported Compound aa437, and the structure was verified by LC/MS. LCMS (ESI) $m/z=829$ (M+H)⁺

(1190) Retention time: 0.93 min (Analytical condition SQDFA05)

(1191) Synthesis of Compound aa438-Resin

(1192) ##STR00437##

(1193) Using aa360-resin as a starting material, aa438-a-resin was obtained in the same manner as Compound aa430-a-resin except that Fmoc-MeGly (cPent)-OH (Compound aa330) was used in place of Fmoc-MeNva (3-Et)-OH.

(1194) Using the resulting aa438-a-resin, aa438-b-resin was obtained in the same manner as Compound aa427-a-resin.

(1195) Using the resulting aa438-b-resin, aa438-resin was obtained in the same manner as Compound aa427-resin. Amino acid was cleaved from the resin by TFE/DCM (1/1) using a small amount of resin-supported Compound aa438, and the structure was verified by LC/MS. LCMS

(ESI) $m/z=815$ (M+H)⁺

(1196) Retention time: 0.87 min (Analytical condition SQDFA05)

(1197) Synthesis of Compound aa439-Resin

(1198) ##STR00438## ##STR00439##

(1199) A toluene (200 mL) mixture of (2R)-3-(tert-butylidisulfanyl)-2-[9H-fluoren-9-ylmethoxycarbonyl(methyl)amino]propanonic acid (20.0 g, 46.3 mmol), (+)-camphorsulfonic acid (752.6 mg, 3.2 mmol, 0.07 eq), and paraformaldehyde (13.9 g, 463.4 mmol, 10.0 eq) was stirred at room temperature for 1 hour in a nitrogen atmosphere. The reaction solution was diluted with ethyl acetate, washed 3 times with an aqueous sodium hydrogen carbonate solution, and then washed with brine. The resulting organic layer was dried over anhydrous sodium sulfate, and the solvent was distilled off under reduced pressure to give Compound aa439-b (23 g) as a crude product.

(1200) LCMS (ESI) $m/z=466.1$ (M+Na)⁺

(1201) Retention time: 1.560 min (Analytical condition SMD method_20)

(1202) Trifluoroacetic acid (120 mL) was added to a mixed solution of Compound aa439-b (23 g) obtained above and triethylsilane (60.2 g, 518 mmol) in dichloromethane (120 mL) at room temperature in a nitrogen atmosphere, and the mixture was stirred for 40 hours. The solvent was distilled off from the reaction solution under reduced pressure, the resulting residue was purified by reverse phase column chromatography (C18, acetonitrile/water) to give Compound aa439-c (13 g, 63% in 2 steps).

(1203) LCMS (ESI) $m/z=446.1$ (M+H)⁺

(1204) Retention time: 0.852 min (Analytical condition SMD method_30))

(1205) Compound aa439-c (13.0 g, 29.2 mmol), WSCI.Math.HCl (6.49 g, 33.84 mmol, 1.2 eq), and 3,4-dihydro-3-hydroxy-4-oxo-1,2,3-benzotriazine (HOOBt)(5.23 g, 32.09 mmol, 1.1 eq) were added to a mixed solution of DMF (26 mL) and DCM (90 mL) at room temperature in a nitrogen atmosphere, and the mixture was stirred for 5 minutes. The resulting reaction solution was cooled to 0° C.), dimethylamine (2 mol/L of THE solution, 15.60 mL, 31.2 mmol, 1.07 eq) was added dropwise, and the mixture was stirred at () C for 1 hour. The reaction solution was diluted with ethyl acetate, washed twice with hydrochloric acid (1 mol/L, 130 mL), washed once with water (130 mL), washed twice with an aqueous sodium hydrogen carbonate solution (130 mL), and washed once with brine (130 mL). The resulting organic layer was dried over anhydrous sodium sulfate, and the solvent was distilled off under reduced pressure. The resulting residue was purified by normal phase chromatography (petroleum ether/ethyl acetate) to give Compound aa439-d (12.1 g, 88%).

(1206) LCMS (ESI) $m/z=495.2$ (M+Na)⁺

(1207) Retention time: 1.546 min (Analytical condition SMD method_20)

(1208) Tri-n-butylphosphine (6.0 g, 29.7 mmol, 1.2 eq) was added dropwise to a mixed solution of Compound aa439-d (11.7 g, 24.7 mmol), ethanol (100 mL), DCM (150 mL), and water (25 mL) at

room temperature in a nitrogen atmosphere, and the mixture was stirred at room temperature for 3 hours. The reaction solution was extracted with dichloromethane (DCM), the resulting organic layer was washed with brine and dried over anhydrous sodium sulfate, and then the solvent was distilled off under reduced pressure. The resulting residue was purified by normal phase chromatography (petroleum ether/ethyl acetate), and the resulting mixture was purified by reverse phase column chromatography (C18, acetonitrile/water) to give Compound aa439-e (3.03 g, 32%). (1209) LCMS (ESI) $m/z=407.2$ (M+Na)+

(1210) Retention time: 1.326 min (Analytical condition SMD method_20))

(1211) A mixed solution of Compound aa439-e (2.80 g, 7.29 mmol) and tert-butyl bromoacetate (2.10 g, 10.77 mmol, 1.5 eq) in DMF (40 mL) was stirred at room temperature for 5 minutes, cesium carbonate (2.80 g, 8.59 mmol, 1.2 eq) was added thereto, and the mixture was stirred for 1 hour. The reaction solution was diluted with ethyl acetate, and washed with water. The resulting organic layer was dried over anhydrous sodium sulfate, and the solvent was distilled off under reduced pressure to give Compound aa439-f (3.1 g) as a crude product. The resulting crude product was used in the next reaction without further purification.

(1212) LCMS (ESI) $m/z=499.3$ (M+H)+

(1213) Retention time: 1.467 min (Analytical condition SMD method_20)

(1214) Compound aa439-f (3.1 g) obtained above was dissolved in a mixed solution of trifluoroacetic acid (TFA)(30 mL) and dichloromethane (DCM)(30 mL), and stirred at room temperature for 2 hours in a nitrogen atmosphere. The solvent was distilled off from the reaction solution under reduced pressure, the resulting residue was purified by reverse phase column chromatography (C18, acetonitrile/water), and the resulting mixture was further purified by reverse phase high performance column chromatography (acetonitrile/water, containing TFA), and the resulting eluate was extracted with DCM. The resulting organic layer was washed successively with water and hydrochloric acid (1 mol/L), and the solvent was distilled off under reduced pressure to give Compound aa439 (1.79 g, 55% in 2 steps).

(1215) LCMS (ESI) $m/z=443.2$ (M+H)+

(1216) Retention time: 1.383 min (Analytical condition SMD method_31)

(1217) Using Compound aa439 (0.46 g, 1.04 mmol), Compound aa439-resin (1.54 g, supported amount 0.336 mmol/g) was obtained in the same manner as synthesis of Compound aa358-resin.

(1218) Synthesis of Compound aa440-Resin

(1219) ##STR00440##

(1220) After DIPEA (26.3 mL, 148 mmol) was added to a DMF solution (211 mL) of aa358-c (30.3 g, 74.0 mmol) at 0° C., HATU (30.9 g, 81.0 mmol) and N-methylethylamine (6.97 mL, 81.0 mmol) were added. After being stirred at 0° C. for 30 minutes, the mixture was diluted with toluene (300 mL), and 1 N hydrochloric acid (600 mL) was added. The organic layer was separated, washed with water (300 mL), washed with a 50% aqueous sodium hydrogen carbonate solution (300 mL), and washed with half brine (300 mL). The organic layer was dried over anhydrous sodium sulfate and filtered off, and the solvent was distilled off under reduced pressure to give Compound aa440-a as a crude product (36.6 g). The crude product was dissolved in toluene (211 mL), and DBU (11.0 mL, 74 mmol) was added at 0° C. After stirring at room temperature for 5 minutes, 1 N hydrochloric acid (333 mL) was added to separate the aqueous layer. After the aqueous layer was washed with toluene (333 mL), a 50% aqueous sodium hydroxide solution (7.5 mL) was added to adjust the pH to 8 to 9. The mixture was extracted 5 times with ethyl acetate (200 mL), and the organic layer was washed with half brine (333 mL). The organic layer was dried over anhydrous sodium sulfate and filtered off, and the solvent was distilled off under reduced pressure to give aa440-b as a crude product (12.6 g).

(1221) Oxyma (9.41 g, 66.2 mmol) was added to a DMF solution (158 mL) of Fmoc-MeGly (cPent)-OH (Compound aa330, 21 g, 55.2 mmol) at 0° C. WSCI.Math.HCl (14.8 g, 77 mmol) was added thereto, and the mixture was stirred at room temperature for 30 minutes. A DMF solution (31

mL) of aa440-b (12.6 g, 55.2 mmol) and DIPEA (10.8 mL, 60.7 mmol) were added thereto, and the mixture was stirred at room temperature for 15 hours. Toluene (250 mL) and 0.5 N hydrochloric acid (250 mL) were added, and the organic layer was washed with a 50% aqueous sodium hydrogen carbonate solution (250 mL) and washed with half brine (250 mL). The organic layer was dried over anhydrous sodium sulfate and filtered off, the solvent was distilled off under reduced pressure, and the resulting residue was purified by silica gel chromatography (hexane-ethyl acetate) to give aa440-c (25.6 g, 59% in 3 steps).

(1222) LCMS (ESI) $m/z=612$ (M+Na)+

(1223) Retention time: 3.27 min (Analytical condition SMD method_03)

(1224) Phenylsilane (3.69 mL, 30.0 mmol) was added to a DCM solution (86 mL) of aa440-c (25.3 g, 42.9 mmol). Tetrakis(triphenylphosphine) palladium (0) (496 mg, 0.429 mmol) was added thereto, and the mixture was stirred at room temperature for 30 minutes. The mixture was diluted with TBME (250 mL), and extracted with a 50% aqueous sodium hydrogen carbonate solution (250 mL). The organic layer was further extracted with a 50% aqueous sodium hydrogen carbonate solution (125 mL), and phosphoric acid (17.9 mL, 323 mmol) was added to the combined aqueous layer to adjust the pH to 2 to 3. The mixture was extracted with DCM (250 mL), the organic layer was washed with half brine (250 mL), and the solvent was distilled off under reduced pressure to give aa440 as a crude product (23.6 g).

(1225) LCMS (ESI) $m/z=572$ (M+Na)+

(1226) Retention time: 2.64 min (Analytical condition SMD method_03)

(1227) Using the resulting aa440 (23.5 g, 42.8 mmol), aa440-resin (79.2 g) was obtained under the same reaction conditions as synthesis of Compound aa373-resin. The supported amount calculated in the same manner as aa373-resin was 0.333 mmol/g.

(1228) Synthesis of Compound aa441-Resin

(1229) ##STR00441##

(1230) HOBt (7.30 g, 54.0 mmol) was added to a DMF solution (89 mL) of WSCI.Math.HCl (11.3 g, 58.9 mmol) at 0° C. in a nitrogen atmosphere, and the mixture was stirred for 5 minutes. A DMF/DCM solution (1/1, 74 mL) of aa358-c (20.1 g, 49.1 mmol) was added thereto, and the mixture was stirred at 0° C. for 30 minutes. After diethylamine (5.60 mL, 54.0 mmol) was added thereto, the mixture was stirred at room temperature for 2 hours. The mixture was diluted with ethyl acetate (200 mL), washed with 1 N hydrochloric acid (160 mL), and washed with water (200 mL). The mixture was further washed twice with a 50% aqueous sodium hydrogen carbonate solution (200 mL) and washed with half brine (200 mL). The organic layer was dried over anhydrous sodium sulfate and filtered off, and the solvent was distilled off under reduced pressure to give aa441-a as a crude product (23.3 g).

(1231) DBU (6.61 mL, 44.2 mmol) was added dropwise to a DMF solution (147 mL) of aa441-a (20.5 g, 44.2 mmol) in a nitrogen atmosphere, and the mixture was stirred at room temperature for 5 minutes. Pyridine hydrochloride (5.62 g, 48.6 mmol) was added thereto, and the mixture was stirred at room temperature for 10 minutes. To this mixture was added a solution that is obtained by adding WSCI.Math.HCl (11.9 g, 61.9 mmol) to a DMF solution (147 mL) of Fmoc-MeGly (cPent)-OH (Compound aa330, 15.9 g, 42.0 mmol) and oxyma (7.54 g, 53.0 mmol). The resulting mixture was stirred at room temperature for 30 minutes. Moreover, DIPEA (8.47 mL, 48.6 mL) was added, and then the mixture was stirred at room temperature for 17 hours. The mixture was diluted with ethyl acetate (200 mL), and washed twice with 1 N hydrochloric acid (200 mL). The aqueous layer was extracted with ethyl acetate (200 mL), and the combined organic layer was washed with water (200 mL), washed twice with a 50% aqueous sodium hydrogen carbonate solution (200 mL), and washed with half brine (200 mL). The organic layer was dried over magnesium sulfate and filtered off, the solvent was distilled off under reduced pressure, and the resulting residue was purified twice by silica gel chromatography (hexane-ethyl acetate) to give aa441-b (13.7 g, 52%).

(1232) LCMS (ESI) $m/z=626$ (M+Na)+

(1233) Retention time: 1.50 min (Analytical condition SMD method_04)

(1234) aa441 was obtained as a crude product (10.5 g) in the same manner as Compound aa440 except that aa441-b (13.7 g, 22.7 mmol) was used in place of aa440-c.

(1235) LCMS (ESI) m/z=586 (M+Na)+

(1236) Retention time: 1.17 min (Analytical condition SMD method_05)

(1237) Using the resulting Compound aa441 (10.5 g, 18.6 mmol), aa441-resin (36.3 g) was obtained under the same reaction conditions as synthesis of Compound aa373-resin. The supported amount calculated in the same manner as aa373-resin was 0.362 mmol/g.

(1238) Synthesis of Compound aa442-Resin

(1239) ##STR00442##

(1240) aa442-a was obtained as a crude product (23.6 g) in the same manner as Compound aa441-a except that N-methylpropylamine was used in place of diethylamine.

(1241) aa442-b (15.9 g, 53%) was obtained in the same manner as Compound aa441-b except that aa442-a (22.9 g, 49.4 mmol) was used in place of aa441-a.

(1242) LCMS (ESI) m/z=626 (M+Na)+

(1243) Retention time: 1.51 min (Analytical condition SMD method_04)

(1244) aa442 (13.7 g, 92%) was obtained in the same manner as Compound aa441 except that 442-b (15.9 g, 26.4 mmol) was used in place of aa441-b.

(1245) LCMS (ESI) m/z=564 (M+H)+

(1246) Retention time: 1.28 min (Analytical condition SMD method_04)

(1247) Using the resulting aa442 (13.6 g, 24.2 mmol), aa442-resin (47.1 g) was obtained under the same reaction conditions as synthesis of Compound aa373-resin. The supported amount calculated in the same manner as aa373-resin was 0.348 mmol/g.

(1248) 1-3. Solid-Phase Peptide Synthesis by Peptide Synthesizer

(1249) Cyclic peptide compounds (Table 38) were synthesized by the method described in WO2013/100132 or WO2018/225864. Peptides were synthesized by the Fmoc method, which is described in more detail below, using a peptide synthesizer (Multi pep RS and RSi; manufactured by Intavis). The detailed operational procedure was according to the manual appended to the synthesizer. The relationship between the formal name, structure, and abbreviation of each amino acid residues and oligo peptides constituting the cyclic peptides provided in Table 38 can be understood from Tables 2 to 6 above, and Table 9, Table 10, Table 9-1, Table 17-1, and Table 29-1 below.

(1250) A peptide synthesis method (a basic peptide synthesis method) by the Fmoc method is described in detail in 1-3-1 to 1-3-5 below.

(1251) 1-3-1. Peptide Elongation Reaction by Fmoc Method from N-Terminal of Amino Acid

(1252) A peptide compound was synthesized by a solid-phase synthesis method involving an Fmoc-protected amino acid using a peptide synthesizer (Multi pep RS or Multi pep RSi) manufactured by Intavis. The detailed operational procedure was according to the manual appended to the synthesizer.

(1253) The Fmoc-protected amino acid (0.3 to 0.6 mol/L.) constituting the intended peptide, and HOAt, oxyma, or HOObt (0.375 mol/L.) as a carboxyl group activator, were dissolved in NMP or NMP/DMF (1/1) to prepare Solution 1. When the Fmoc-protected amino acid was poorly soluble in the above solvents, DMSO was added so as to have a concentration of 20 to 30% (v/v) to prepare Solution 1. When, for example, Fmoc-Ser (THP)—OH or Fmoc-MeSer (THP)—OH was used as an Fmoc-protected amino acid having a THP-protecting group in a side chain, Solution 1 was prepared using oxyma as a carboxyl group activator, and Molecular Sieves 4A1/8 (Wako Pure Chemical Industries, Ltd.) or Molecular Sieves 4A1/16 (Wako Pure Chemical Industries, Ltd.) was added and then used in peptide synthesis. When the α,α -di-substituted Fmoc amino acid used was, for example, Fmoc-cVal-OH (Compound aa301), Fmoc-cLeu-OH (Compound aa302), Fmoc-cHex-OH (Compound aa306), Fmoc-cVal (3-Me₂)-OH (Compound aa303), Fmoc-cVal (3-F₂)—OH)

(Compound aa307), Fmoc-AoxeC-OH (Compound aa318), Fmoc-Aib-OH (Compound aa315), or Fmoc-(Me) Abu-OH (Compound aa316), oxyma was used as a carboxyl group activator to prepare Solution 1. When the serin derivative used was, for example, Fmoc-MeSer (nPr)-OH (Compound aa016), Fmoc-MeSer (cPr)-OH (Compound aa020), Fmoc-MeSer (Tfe)-OH (Compound aa021), Fmoc-MeSer (Et)-OH (Compound aa022), Fmoc-MeSer (iPen)-OH (Compound aa249), Fmoc-D-MeSer (nPr)-OH (Compound aa250), or Fmoc-MeSer (Me)-OH (Compound aa252), HOAt or HOObt was used as a carboxyl group activator to prepare Solution 1. When the azetidine or a derivative thereof used was, for example, Fmoc-Aze (2)-OH (Compound 076), Fmoc-D-Aze (2)-OH (Compound aa079), Fmoc-Aze (2)(3S-Me)-OH (Compound aa098), Fmoc-Aze (2)(3R-Me)-OH (Compound aa099), or Fmoc-Aze (2)(3-Me₂)-OH (Compound aa100), HOObt was used as a carboxyl group activator to prepare Solution 1. N,N'-diisopropylcarbodiimide (DIC)(10% v/v) and N,N-dimethylformamide (DMF) were mixed to prepare Solution 2.

(1254) 2-Chlorotrityl resin (100 mg), which is bound to the side-chain carboxylic acid moiety of N-terminal Fmoc-protected aspartic acid or the main-chain carboxylic acid moiety of an N-terminal Fmoc-protected amino acid, was added to a solid-phase reaction vessel, and the reaction vessel was placed in a peptide synthesizer. Dichloromethane (DCM)(0.8 mL) was added to this resin (100 mg), and the mixture was left to stand still for 1 hour to swell the resin. The solution was then discharged from the frit. Solution 1 and Solution 2 were placed in the peptide synthesizer, and automatic synthesis by the peptide synthesizer was started.

(1255) A DMF solution (2% v/v, 0.7 mL.) of 1,8-diazabicyclo [5.4.0]-7-undecene (DBU) was added to a solid-phase reaction vessel containing the resin to deprotect the N-terminal Fmoc group at room temperature. Reaction was carried out for 4.5 minutes in the deprotection of the first residue, and for 10 minutes in the deprotection of the second and subsequent residues, and then the solution was discharged from the frit. DMF (0.7 mL.) was added thereto, and after being left to stand still for 5 minutes, the solution was discharged from the frit. By repeating this resin washing step 3 more times, an amino acid bonded to the resin, or the resin in which the Fmoc group at the N-terminal of the peptide was removed to yield an amino group, was obtained. The washing step may be carried out in such a manner that the resin is washed once with DMF (0.7 mL.), then washed once with a toluene solution (0.7 mL.) of DIPEA. HOAt (2 eq based on resin), and further washed 3 times with DMF (0.7 mL.).

(1256) Subsequently, Solution 1 (0.3 mL.) and Solution 2 (0.36 mL.) were mixed in the mixing vial of the synthesizer and then added to the resin that had undergone the above deprotection treatment, and the solid-phase reaction vessel was heated to 40° C. In the case of a poorly elongatable sequence, the temperature was increased to 60° C. as necessary. The condensation reaction of the amino group on the resin and the Fmoc-protected amino acid was carried out for 2.5 hours. In the case of a poorly elongatable sequence, the reaction was carried out for 20 hours as necessary. In the case of α,α -di-substituted Fmoc amino acid, the temperature was increased to 60° C. and the reaction was carried out for 16 hours, as necessary. In the case of Fmoc-protected amino acid of azetidine or a derivative thereof, the temperature was increased to 50° C. and the reaction was carried out for 10 hours, as necessary, then Solution 2 (0.36 mL.) was added again, and the reaction was further carried out at 50° C. for 10 hours. After reaction, the solution was discharged from the frit. When the elongation efficiency was poor, this condensation reaction of the Fmoc-protected amino acid was repeated one or two more times. For example, this corresponds to the condensation reaction of an N-alkylated Fmoc-protected amino acid on Compound aa358-resin (Fmoc-MeAsp (O-Trt (2-C1) resin)-mor) and Compound aa359-resin (Fmoc-MeAsp (O-Trt (2-C1) resin)-NMe₂). In the case of Fmoc-protected amino acids of Mec Val and MecLeu, the mixture was heated to 60° C. and reacted for 6 hours if necessary, then Solution 2 (0.36 mL.) was added again, and the reaction was further carried out at 60° C. for 6 hours. This operation was repeated, so the reaction was carried out for a total of 24 hours. When elongating Fmoc protected proline or its derivative following MecVal and MecLeu, the operation of heating the mixture to 60° C. to be reacted for 6

hours, then adding again Solution 2 (0.36 mL.), further reacting the mixture at 60° C. for 6 hours, and discharging the solution from the frit was carried out 3 times. Then, the resin was washed 3 times with DMF (0.7 mL.). This Fmoc group deprotection reaction and the subsequent Fmoc amino acid condensation reaction were regarded as constituting one cycle, and a peptide was elongated on the resin surface by repeating this cycle. After completion of peptide elongation, a DMF solution (2% v/v, 0.7 mL) of DBU was added to the resin, and a reaction was carried out for 15 minutes to perform a deprotection reaction of the Fmoc group, and then the solution was discharged from the frit. The resulting resin was washed 4 times with DMF (0.7 mL.), and then washed 4 times with DCM (0.7 mL.).

(1257) 1-3-2. Cleaving of Elongated Peptide from Resin

(1258) To the resin obtained by the method described in WO2013/100132 or WO2018/225864 or by the above method was added 2,2,2-trifluoroethanol (TFE)/DCM (1/1, v/v, 2 mL.) containing 0.75% (v/v) DIPEA, and the resin was reacted for 2 hours at room temperature to carry out the reaction of cleaving a peptide chain from the resin. After reaction, the solution in the tube was recovered from the frit. 2,2,2-Trifluoroethanol (TFE)/DCM (1/1, v/v, 1 mL) was added to the remaining resin, and the operation of recovering the solution from the frit was performed twice. The entirety of the resulting cleaved solutions was combined and mixed with DMF (4 mL) or 1,2-dichloroethane (4 mL.), and then the solvent was distilled off under reduced pressure using a high-throughput centrifugal evaporator (HT-12) manufactured by Genevac.

(1259) 1-3-3. Method of Cyclizing Cleaved Peptide

(1260) The residue obtained by the above method was dissolved in a mixed solution of DMF (4 mL) and DCM (4 mL), a DMF solution (0.5 M, 1.5 eq) of (1-cyano-2-ethoxy-2-oxoethylideneaminoxy)dimethylamino-morpholino-carbenium hexafluorophosphate (COMU) or a DMF solution (0.5 M, 1.5 eq) of O-(7-aza-1 H-benzotriazol-1-yl)-N,N,N,N-tetramethyluronium hexafluorophosphate (HATU) and DIPEA (1.8 eq) were added, and the mixture was stirred at room temperature for 2 hours to carry out a condensation cyclization reaction of the N-terminal amino group and the C-terminal carboxyl group. The equivalent number was calculated based on the product of multiplying the amino acid supporting ratio (mmol/g) of the resin used as a raw material by the amount of the resin used (usually 100 mg). Production of the intended cyclic peptide was verified by LC/MS measurement (SQ Detector 2 manufactured by Waters), and then the solvent was distilled off from the reaction solution under reduced pressure by a high-throughput centrifugal evaporator (HT-12) manufactured by Genevac.

(1261) 1-3-4. Deprotection of Protecting Group of Side-Chain Functional Group Contained in Cyclic Peptide

(1262) In the sequences synthesized using Fmoc-protected amino acids having a THP-protected hydroxyl group in a side chain, such as Fmoc-Ser (THP)—OH (Compound aa0) 73), Fmoc-MeSer (THP)—OH (Compound aa044), Fmoc-Hyp (THP)—OH (Compound aa279), Fmoc-Hyp (3)(THP)—OH (Compound aa265), Fmoc-cisHyp (THP)—OH (Compound 267), and Fmoc-cisHyp (3) (THP)—OH (Compound aa264), and in the sequences synthesized using Fmoc-MeAsp (O-Trt (2-C1) resin)-OPis (Compound aa386-resin), 4 mL of a 1,1,1,3,3,3-hexafluoroisopropyl alcohol (HFIP) solution (containing 2% (v/v) triisopropylsilane (TIPS) and 1% (v/v) 1,2-dichloroethane) of tetramethylammonium hydrogen sulfate (0.05 M) was added to the residue obtained above to dissolve the residue, and then the mixture was left to stand at room temperature for 4 hours to deprotect the THP protecting group or Pis protecting group. After completion of the reaction was confirmed with LC/MS (SQ Detector 2 manufactured by Waters), diisopropylethylamine (DIPEA) (70 µL) was added to the reaction solution, and the solvent was distilled off under reduced pressure with a high-throughput centrifugal evaporator (HT-12) manufactured by Genevac. In this deprotection reaction, another fluoroalcohol such as 2,2,2-trifluoroethanol (THE) can be used in place of HFIP.

(1263) 1-3-5. Cyclic Peptide Purification Method

(1264) DMSO or DMF and acetonitrile were added to the residue obtained by the above method, insoluble matter was removed by filter filtration, and then the solution was purified by preparative-HPLC. The purification apparatus used was Waters Auto Purification System, the column used was YMC-Actus Triart C18 (inner diameter 20 mm, length 100 mm) or YMC-Actus Triart Phenyl(inner diameter 20 mm, length 100 mm), and the mobile phase used was an aqueous methanol-ammonium acetate solution (50 mmol/L.) or acetonitrile-water containing 0.1% formic acid. Fractions containing a compound as the main component were selected, the same amount of DMSO as the amount of the solution of each fraction was added, and the solvent was distilled off under reduced pressure with a high-throughput centrifugal evaporator (HT-12) manufactured by Genevac. The resulting residue was dissolved in DMSO, fractions were integrated as necessary, and again the solvent was distilled off under reduced pressure with a high-throughput centrifugal evaporator (HT-12) manufactured by Genevac to give the intended cyclic peptide.

(1265) Synthesis of Compound PP0247

(1266) Compound aa375-resin (0.448 mmol/g, 100 mg) was used as a raw material, and Fmoc-cLeu-OH, Fmoc-Pro-OH, Fmoc-Hph(4—CF₃-3-C1)—OH, Fmoc-MeGly-OH, Fmoc-nPrPhe (4—CF₃)—OH, Fmoc-Ile-OH, and Fmoc-Azp (2)—OH were used as Fmoc-protected amino acids. The above-described peptide elongation reaction by the Fmoc method, cleaving of an elongated peptide from resin, cyclization of a cleaved peptide (using (1-cyano-2-ethoxy-2-oxoethylidencaminooxy)dimethylamino-morpholino-carbenium hexafluorophosphate (COMU) as a cyclization reagent), and purification of a cyclic peptide were performed to give the intended compound PP0247 (12.7 mg, 19%).

(1267) Compound PP0247

(1268) ##STR00443##

(1269) LCMS (ESI) m/z=1517.6 (M+H)+

(1270) Retention time: 7.775 min (Analytical condition SSC-AA-02/03)

(1271) Using the resins, on which amino acid shown in Table 5 and Table 6 was supported, as raw materials, the following compounds were produced according to the above-described Fmoc method in the same manner as the production method of Compound PP247. A deprotection reaction was added as necessary.

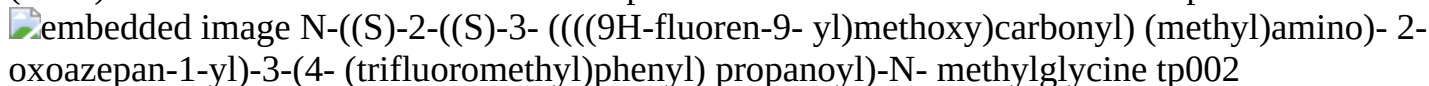
(1272) Compounds PP0001, PP0002, PP0003, PP0004, PP0006, PP0007, PP0011, PP0013, PP0014, PP0015, PP0016, PP0018, PP0019, PP0020, PP0021, PP0022, PP0023, PP0024, PP0025, PP0027, PP0028, PP0029, PP0030, PP0031, PP0032, PP0037, PP0039, PP0041, PP0042, PP0043, PP0044, PP0045, PP0047, PP0048, PP0049, PP0050, PP0051, PP0067, PP0068, PP0070, PP0071, PP0076, PP0077, PP0078, PP0138, PP0141, PP0144, PP0145, PP0146, PP0148, PP0149, PP0150, PP0151, PP0152, PP0153, PP0154, PP0155, PP0156, PP0158, PP0162, PP0165, PP0166, PP0168, PP0169, PP0170, PP0171, PP0172, PP0173, PP0174, PP0175, PP0176, PP0177, PP0178, PP0179, PP0180, PP0181, PP0182, PP0183, PP0184, PP0185, PP0186, PP0187, PP0188, PP0189, PP0190, PP0191, PP0192, PP0193, PP0194, PP0195, PP0196, PP0197, PP0198, PP0199, PP0200, PP0201, PP0202, PP0203, PP0204, PP0205, PP0206, PP0207, PP0208, PP0209, PP0212, PP0217, PP0220, PP0221, PP0222, PP0223, PP0224, PP0225, PP0226, PP0227, PP0228, PP0229, PP0230, PP0231, PP0232, PP0233, PP0234, PP0235, PP0236, PP0237, PP0238, PP0239, PP0240, PP0241, PP0247, PP0248, PP0249, PP0250, PP0251, PP0252, PP0253, PP0254, PP0256, PP0257, PP0258, PP0259, PP0260, PP0261, PP0262, PP0263, PP0264, PP0265, PP0266, PP0267, PP0268, PP0269, PP0270, PP0271, PP0272, PP0273, PP0274, PP0275, PP0276, PP0277, PP0278, PP0279, PP0281, PP0282, PP0283, PP0284, PP0285, PP0286, PP0287, PP0288, PP0289, PP0290, PP0291, PP0292, PP0293, PP0294, PP0295, PP0296, PP0297, PP0298, PP0299, PP0300, PP0301, PP0302, PP0303, PP0304, PP0305, PP0306, PP0307, PP0308, PP0309, PP0310, PP0311, PP0312, PP0313, PP0314, PP0315, PP0320, PP0321, PP0322, PP0323, PP0324, PP0325, PP0326, PP0327, PP0328, PP0329, PP0330, PP0331, PP0332, PP0333, PP0334, PP0335, PP0336, PP0337, PP0338, PP0339, PP0340, PP0341, PP0342, PP0343, PP0344, PP0345, PP0346, PP0347, PP0348, PP0349, PP0350, PP0351, PP0352,

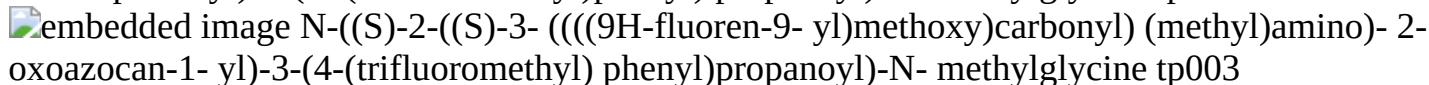
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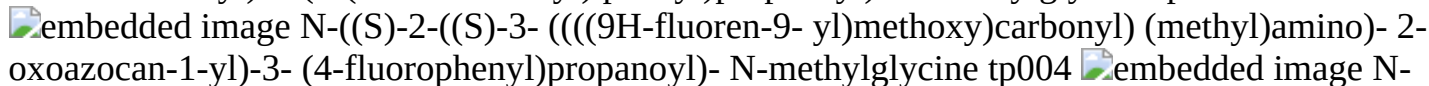
1-4. Cyclic Peptide Synthesis Using Fmoc-Protected Tripeptide as Reagent Synthesis of Fmoc-Protected Tripeptides

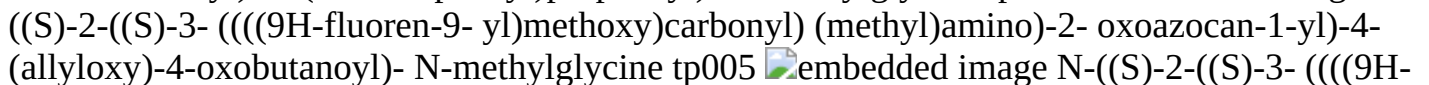
(1273) The tripeptides shown in Table 9 were synthesized by the following method and used in peptide synthesis by a peptide synthesizer.

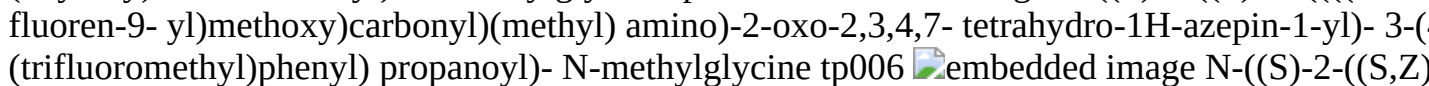
(1274) TABLE-US-00009 TABLE 9 Compound No. Structural Formula Name tp001

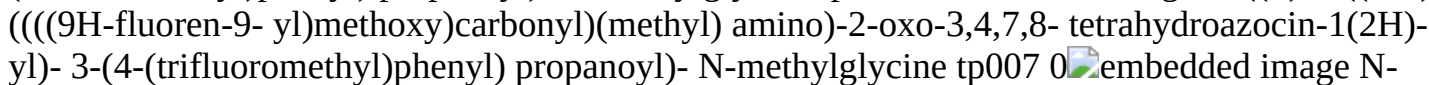
 N-((S)-2-((S)-3-(((9H-fluoren-9-yl)methoxy)carbonyl)(methyl)amino)-2-oxoazepan-1-yl)-3-(4-(trifluoromethyl)phenyl)propanoyl-N-methylglycine tp002

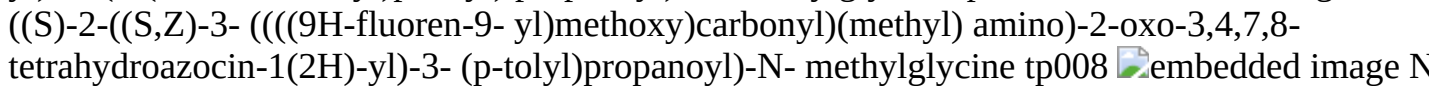
 N-((S)-2-((S)-3-(((9H-fluoren-9-yl)methoxy)carbonyl)(methyl)amino)-2-oxoazocan-1-yl)-3-(4-(trifluoromethyl)phenyl)propanoyl-N-methylglycine tp003

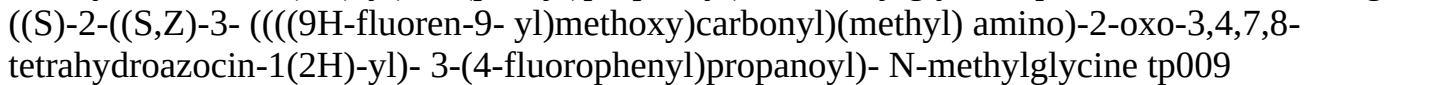
 N-((S)-2-((S)-3-(((9H-fluoren-9-yl)methoxy)carbonyl)(methyl)amino)-2-oxoazocan-1-yl)-3-(4-fluorophenyl)propanoyl-N-methylglycine tp004

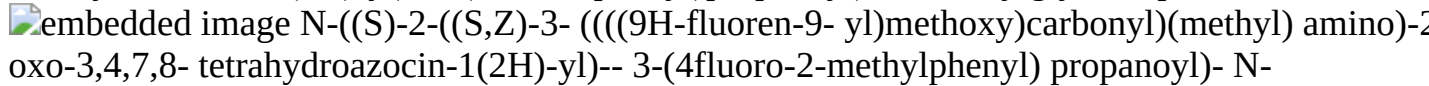
 N-((S)-2-((S)-3-(((9H-fluoren-9-yl)methoxy)carbonyl)(methyl)amino)-2-oxoazocan-1-yl)-4-(allyloxy)-4-oxobutanoyl-N-methylglycine tp005

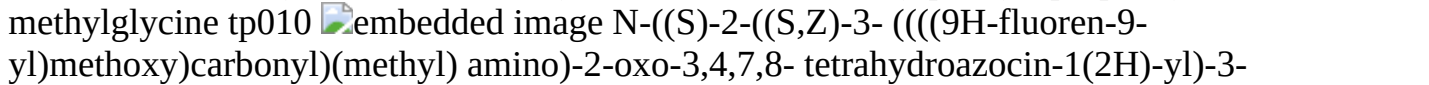
 N-((S)-2-((S)-3-(((9H-fluoren-9-yl)methoxy)carbonyl)(methyl)amino)-2-oxo-2,3,4,7-tetrahydro-1H-azepin-1-yl)-3-(4-(trifluoromethyl)phenyl)propanoyl-N-methylglycine tp006

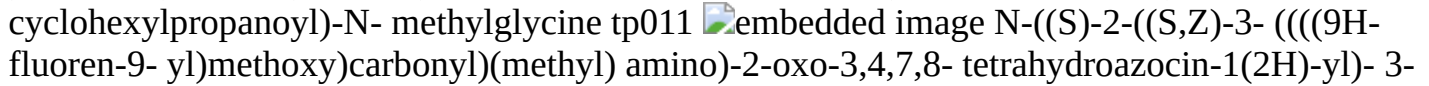
 N-((S)-2-((S,Z)-3-(((9H-fluoren-9-yl)methoxy)carbonyl)(methyl)amino)-2-oxo-3,4,7,8-tetrahydroazocin-1(2H)-yl)-3-(4-(trifluoromethyl)phenyl)propanoyl-N-methylglycine tp007

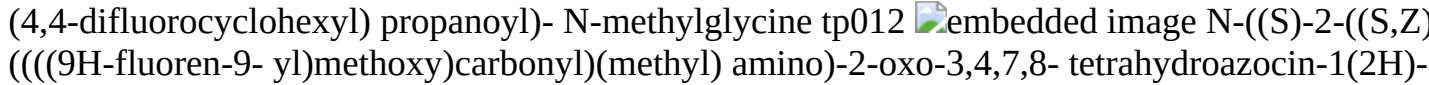
 N-((S)-2-((S,Z)-3-(((9H-fluoren-9-yl)methoxy)carbonyl)(methyl)amino)-2-oxo-3,4,7,8-tetrahydroazocin-1(2H)-yl)-3-(p-tolyl)propanoyl-N-methylglycine tp008

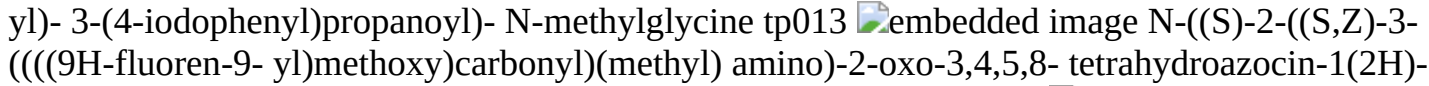
 N-((S)-2-((S,Z)-3-(((9H-fluoren-9-yl)methoxy)carbonyl)(methyl)amino)-2-oxo-3,4,7,8-tetrahydroazocin-1(2H)-yl)-3-(4-fluorophenyl)propanoyl-N-methylglycine tp009

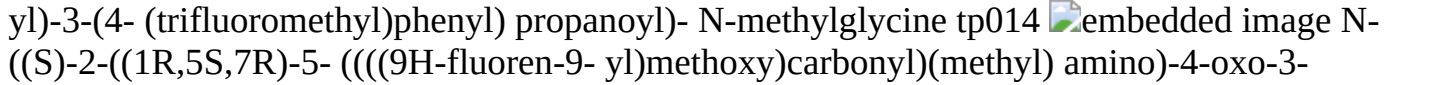
 N-((S)-2-((S,Z)-3-(((9H-fluoren-9-yl)methoxy)carbonyl)(methyl)amino)-2-oxo-3,4,7,8-tetrahydroazocin-1(2H)-yl)-3-(4-fluoro-2-methylphenyl)propanoyl-N-methylglycine tp010

 N-((S)-2-((S,Z)-3-(((9H-fluoren-9-yl)methoxy)carbonyl)(methyl)amino)-2-oxo-3,4,7,8-tetrahydroazocin-1(2H)-yl)-3-cyclohexylpropanoyl-N-methylglycine tp011

 N-((S)-2-((S,Z)-3-(((9H-fluoren-9-yl)methoxy)carbonyl)(methyl)amino)-2-oxo-3,4,7,8-tetrahydroazocin-1(2H)-yl)-3-(4,4-difluorocyclohexyl)propanoyl-N-methylglycine tp012

 N-((S)-2-((S,Z)-3-(((9H-fluoren-9-yl)methoxy)carbonyl)(methyl)amino)-2-oxo-3,4,7,8-tetrahydroazocin-1(2H)-yl)-3-(4-iodophenyl)propanoyl-N-methylglycine tp013

 N-((S)-2-((S,Z)-3-(((9H-fluoren-9-yl)methoxy)carbonyl)(methyl)amino)-2-oxo-3,4,5,8-tetrahydroazocin-1(2H)-yl)-3-(4-(trifluoromethyl)phenyl)propanoyl-N-methylglycine tp014

 N-((S)-2-((1R,5S,7R)-5-(((9H-fluoren-9-yl)methoxy)carbonyl)(methyl)amino)-4-oxo-3-azabicyclo[5.1.0]octan-3-yl)-3-(4-(trifluoromethyl)phenyl)propanoyl-N-methylglycine tp015

 embedded image N-((S)-2-((S)-3- (((9H-fluoren-9- yl)methoxy)carbonyl)(methyl) amino)-6-methyl-2-oxo- 2,3,4,7-tetrahydro-1H- azepin-1-yl)-3-(4- (trifluoromethyl)phenyl) propanoyl)- N-methylglycine tp016  embedded image N-((S)-2-((3S,6R)-3- (((9H-fluoren-9- yl)methoxy)carbonyl)(methyl) amino)-6-methyl-2- oxoazepan-1-yl)-3-(4- (trifluoromethyl)phenyl) propanoyl)- N-methylglycine tp017 0  embedded image N-((S)-2-((3S,5R,6R)-3- (((9H-fluoren-9- yl)methoxy)carbonyl)(methyl) amino)-5,6-dimethyl-2- oxoazepan-1-yl)-3-(4- (trifluoromethyl)phenyl) propanoyl)- N-methylglycine tp018  embedded image N-((S)-2-((3S,5S)-3- (((9H-fluoren-9- yl)methoxy)carbonyl)(methyl) amino)-5,6-dimethyl-2- oxoazepan-1-yl)-3-(4- (trifluoromethyl)phenyl) propanoyl)- N-methylglycine tp019  embedded image N-((S)-2-((S)-3- (((9H-fluoren-9- yl)methoxy)carbonyl) (methyl)amino)-5,6-dimethyl-2- oxo-2,3,4,7-tetrahydro-1H- azepin-1-yl)-3-(4- (trifluoromethyl)phenyl) propanoyl)- N-methylglycine tp020  embedded image N-((S)-2-((3S,5S)-3- (((9H-fluoren-9- yl)methoxy)carbonyl)(methyl) amino)-5-methyl-2- oxoazepan-1-yl)-3-(4- (trifluoromethyl)phenyl) propanoyl)- N-methylglycine tp021  embedded image N-((S)-2-((S)-3- (((9H-fluoren-9- yl)methoxy)carbonyl)(methyl) amino)-5-methyl-2-oxo- 2,3,4,7-tetrahydro-1H- azepin-1-yl)-3-(4- (trifluoromethyl)phenyl) propanoyl)- N-methylglycine tp022  embedded image N-((S)-2-((S,Z)-3- (((9H-fluoren-9- yl)methoxy)carbonyl)(methyl) amino)-2-oxo-3,4,7,8- tetrahydroazocin-1(2H)-yl)-3- (4-vinyl phenyl)propanoyl)-N- methylglycine tp023  embedded image N-((S)-2-((S)-7- (((9H-fluoren-9- yl)methoxy)carbonyl)(methyl) amino)-5,8-dioxo-1,4-diazocan-1- yl)-3-(4-(trifluoromethyl)phenyl) propanoyl)-N-methylglycine tp024  embedded image N-((S)-2-((S,Z)-3- (((9H-fluoren-9- yl)methoxy)carbonyl)(methyl) amino)-2-oxo-3,4,7,8- tetrahydroazocin-1(2H)-yl)- 3-(4-bromophenyl)propanoyl)-N- methylglycine

Synthesis of Compound tp005

(1275) ##STR00468##

(1276) In a nitrogen atmosphere, water (257.0 mL) and a 28% aqueous ammonia solution (94.0 g, 1544.0 mmol) were added to tp005-a (30.0 g, 129.0 mmol), allyl bromide (65.3 mL, 772.0 mmol) was added dropwise, and the mixture was stirred at room temperature for 3 hours. Ethanol (290.0 mL) was added to the reaction solution, and the mixture was stirred at room temperature for 30 minutes. The mixture was filtered, and the resulting residue was washed with water (300 mL) and dried under reduced pressure to give tp005-b (30.4 g, 86%). LCMS (ESI) m/z=274 (M+H)+

(1277) Retention time: 0.64 min (Analytical condition SMD method_04)

(1278) In a nitrogen atmosphere, N-methyl-2-pyrrolidinone (209.0 mL) was added to a reaction vessel containing tp005-b (20.0 g, 73.2 mmol) and tert-butyl methyl glycinate hydrochloride (26.6 g, 146.0 mmol), N,N-diisopropylethylamine (51.1 mL, 293.0 mmol) was added dropwise, 1-[bis(dimethylamino)methylene]-1H-1,2,3-triazolo[4,5-b]pyridinium 3-oxidehexafluorophosphate (55.7 g, 146.0 mmol) was added in several times. After the mixture was stirred at room temperature for 3 hours, ethyl acetate (400 mL), water (200 mL), and a 5 wt % aqueous sodium carbonate solution (200 mL) were added for extraction. The organic layer was washed with a 5 wt % aqueous sodium carbonate solution (400 mL), washed twice with potassium hydrogen sulfate (400 mL), washed with 15 wt % brine (400 mL), and passed through a filter provided with anhydrous sodium sulfate, and the solvent was distilled off under reduced pressure to give tp005-c as a crude product (44.5 g).

(1279) LCMS (ESI) m/z=401 (M+H)+

(1280) Retention time: 0.77 min (Analytical condition SMD method_0) 4)

(1281) 2-Methyltetrahydrofuran (293.0 ml) was added to a reaction vessel containing tp005-c (29.3 g, 73.2 mmol) and (S)-2-(((9H-fluoren-9-yl)methoxy) carbonyl)(methyl)amino) pent-4-enoic acid (33.4 g, 95.0) mmol), N,N-diisopropylethylamine (63.7 mL, 366.0) mmol) was added, and a 50% propylphosphonic anhydride.Math.ethyl acetate solution (93.0 g, 146.0 mmol) and a 50% propylphosphonic anhydride.Math.toluene solution (23.3 g, 36.6 mmol) were added dropwise. After the mixture was stirred at room temperature for 3 days, ethyl acetate (300 mL) and a 5 wt %

aqueous sodium carbonate solution (200 mL) were added for extraction. The organic layer was washed twice with 5 wt % potassium hydrogen sulfate (600 mL), washed twice with a 5 wt % aqueous sodium carbonate solution (600 mL), washed with 15 wt % brine (600 mL), and passed through a filter provided with anhydrous sodium sulfate, and the solvent was distilled off under reduced pressure. The resulting residue was purified by silica gel column chromatography (n-hexane/ethyl acetate) to give tp005-d (39.1 g, 73%). Moreover, ethyl acetate (37.0 mL) was added to tp005-d (23.1 g, 31.5 mmol), the mixture was heated to 50° C. to be dissolved, and then returned to room temperature, and n-hexane (220.0 mL) was gradually added. The resulting precipitate was filtered, washed with a 95:5 mixed solution of n-hexane and ethyl acetate, and dried under reduced pressure to give tp005-d (19.8 g, 85%).

(1282) LCMS (ESI) m/z=734 (M+H)+

(1283) Retention time: 1.68 min (Analytical condition SMD method_0) 4)

(1284) In a nitrogen atmosphere, 1,2-Dichloroethane (467.0 mL) was added to a reaction vessel containing tp005-d (10.3 g, 14.0 mmol) and dichloro [1,3-bis(2-methylphenyl)-2-imidazolidinylidene] (2-isopropoxyphenylmethylene) ruthenium (II)(0).4 g, 0.7 mmol), and the mixture was stirred at 50° C. for 3 hours. The reaction solution was concentrated under reduced pressure, and the resulting residue was purified by silica gel column chromatography (n-hexane/ethyl acetate) to give tp005-e (8.6 g, 87%).

(1285) LCMS (ESI) m/z=706 (M+H)+

(1286) Retention time: 0.70 min (Analytical condition SQDFA50)

(1287) In a nitrogen atmosphere, tp005-e (2.0 g, 2.8 mmol) was dissolved in isopropyl acetate (11.3 mL), then bis(trimethylsilyl)amine (1.5 mL, 7.1 mmol) and trimethylsilyltrifluoromethanesulfonic acid (1.0 mL, 5.7 mmol) were added dropwise at room temperature, and the mixture was stirred for 1 hour. A 1.0 M aqueous disodium hydrogen phosphate solution (15.0 mL) was added, and the mixture was extracted with ethyl acetate (15.0 mL). The organic layer was concentrated under reduced pressure, and the resulting residue was purified by reverse phase chromatography (0.1% formic acid-containing acetonitrile/0.1% formic acid-containing distilled water) to give tp005 (1.2 g, 64%).

(1288) LCMS (ESI) m/z=650 (M+H)+

(1289) Retention time: 0.48 min (Analytical condition SQDFA50)

(1290) Synthesis of Compound tp001

(1291) ##STR00469##

(1292) tp005-e (13.1 g, 18.6 mmol) was dissolved in ethyl acetate (124.0 mL), platinum (IV) oxide (844.0 mg, 3.7 mmol) was added, and the mixture was stirred at room temperature for 3 hours in a hydrogen atmosphere. After the solvent was distilled off under reduced pressure, the resulting residue was purified by silica gel column chromatography (n-hexane/ethyl acetate) to give tp001-a (11.4 g, 87%).

(1293) LCMS (ESI) m/z=708 (M+H)+

(1294) Retention time: 0.73 min (Analytical condition SQDFA50)

(1295) Using tp001-a as a starting material, tp001 (9.6 g, 91%) was obtained in the same manner as synthesis of Compound tp005.

(1296) LCMS (ESI) m/z=652 (M+H)+

(1297) Retention time: 0.46 min (Analytical condition SQDFA50)

(1298) Synthesis of Compound tp002

(1299) ##STR00470##

(1300) Using tp005-c as a starting material, tp002-a (26.3 g, 65%) was obtained in the same manner as synthesis of Compound tp005-d.

(1301) LCMS (ESI) m/z=748 (M+H)+

(1302) Retention time: 1.70 min (Analytical condition SMD method_04)

(1303) Using tp002-a as a starting material, tp002-b (2.9 g, 60%) was obtained in the same manner

as synthesis of Compound tp005-e.
(1304) LCMS (ESI) $m/z=720$ (M+H)+
(1305) Retention time: 1.02 min (Analytical condition SMD method_06)
(1306) Using tp002-b as a starting material, tp002-c (9.1 g, 99%) was obtained in the same manner as synthesis of Compound tp001-a.
(1307) LCMS (ESI) $m/z=722$ (M+H)+
(1308) Retention time: 1.01 min (Analytical condition SMD method_06)
(1309) Using tp002-c as a starting material, tp002 (15.6 g, 90%) was obtained in the same manner as synthesis of Compound tp005.
(1310) LCMS (ESI) $m/z=666$ (M+H)+
(1311) Retention time: 1.30 min (Analytical condition SMD method_04)
(1312) Synthesis of Compound tp003
(1313) ##STR00471##
(1314) Using tp003-a as a starting material, tp003-b (24.6 g, 93%) was obtained in the same manner as synthesis of Compound tp005-c.
(1315) LCMS (ESI) $m/z=533$ (M+H)+
(1316) Retention time: 1.42 min (Analytical condition SMD method_04)
(1317) tp003-b (24.6 g, 46.2 mmol) and toluene (161.0 ml) were added to a reaction vessel, 1,8-diazabicyclo [5.4.0]-7-undecene (7.0 ml, 46.2 mmol) was added, and the mixture was stirred at room temperature for 5 minutes. Ethyl acetate (75 ml) was added, and the organic layer was washed with a 1.0 mmol/L aqueous potassium dihydrogen phosphate solution (150 ml). n-Hexane (225 ml) was added to the organic layer, and the mixture was extracted with a mixed solution of 2 N hydrochloric acid (35 ml) and water (300 ml). The resulting aqueous layer was adjusted to pH 8.01 with a 3.3 mmol/L aqueous potassium phosphate solution. The mixture was extracted with ethyl acetate (200 ml), and the organic layer was dried over magnesium sulfate. After filtration, the solvent was distilled off under reduced pressure to give tp003-c (9.7 g, 68%).
(1318) LCMS (ESI) $m/z=311$ (M+H)+
(1319) Retention time: 0.66 min (Analytical condition SMD method_0) 4)
(1320) tp003-c (9.7 g, 31.2 mmol), N,N-dimethylformamide (208.0 ml), and N,N-diisopropylethylamine (5.7 ml, 32.8 mmol) were added to the reaction vessel, allyl bromide (6.8 ml, 78.0 mmol) was added dropwise, and the mixture was stirred at room temperature for 90 minutes. Ethyl acetate (200 ml) was added, and the mixture was washed with water (200 ml) and then washed with a 1.0 mmol/L aqueous potassium dihydrogen phosphate solution (200 ml). n-Hexane (4 () ml) was added to the organic layer, and the mixture was extracted with a mixed solution of 2 N hydrochloric acid (15.6 ml) and water (200 ml). The resulting aqueous layer was adjusted to pH 8.15 with a 3.3 mmol/L aqueous potassium phosphate solution. The mixture was extracted with ethyl acetate (200 ml), and the organic layer was dried over magnesium sulfate. After filtration, the solvent was distilled off under reduced pressure to give tp003-d (7.6 g, 69%).
(1321) LCMS (ESI) $m/z=351$ (M+H)+
(1322) Retention time: 0.68 min (Analytical condition SMD method_0) 4)
(1323) (S)-2-(((9H-Fluoren-9-yl)methoxy) carbonyl)(methyl)amino) hex-5-enoic acid (26.1 g, 71.3 mmol), dichloromethane (184.0 ml), and N,N-dimethylformamide (0).6 ml, 7.1 mmol) were added to the reaction vessel, thionyl chloride (13.0 ml, 178.0) mmol) was added, the mixture was stirred at room temperature for 1 hour, and the solvent was distilled off under reduced pressure. The resulting residue was dissolved in dichloromethane (81.0 ml), and this was added by cannulation to a reaction vessel containing tp003-d (22.7 g, 64.8 mmol), N,N-diisopropylethylamine (39.5 ml, 227.0) mmol), and dichloromethane (243.0 ml). After the mixture was stirred at room temperature for 3 hours, the solvent was distilled off under reduced pressure. The resulting residue was purified by silica gel column chromatography (n-hexane/ethyl acetate) to give tp003-e (38.3 g, 85%).
(1324) LCMS (ESI) $m/z=698$ (M+H)+

(1325) Retention time: 1.65 min (Analytical condition SMD method_0) 4)

(1326) Using tp003-e as a starting material, tp003-f (17.4 g, 58%) was obtained in the same manner as synthesis of Compound tp005-e.

(1327) LCMS (ESI) m/z=670 (M+H)+

(1328) Retention time: 0.93 min (Analytical condition SMD method_06)

(1329) Using tp003-f as a starting material, tp003-g (17.9 g, 100%) was obtained in the same manner as synthesis of Compound tp001-a.

(1330) LCMS (ESI) m/z=672 (M+H)+

(1331) Retention time: 0.91 min (Analytical condition SMD method_06)

(1332) Using tp003-g as a starting material, tp003 (16.7 g, 82%) was obtained in the same manner as synthesis of Compound tp005.

(1333) LCMS (ESI) m/z=616 (M+H)+

(1334) Retention time: 1.21 min (Analytical condition SMD method_04)

(1335) Synthesis of Compound tp004

(1336) ##STR00472## ##STR00473## ##STR00474##

(1337) A 3.5 wt % aqueous potassium hydrogen carbonate solution (194 ml) and acetonitrile (194 ml) were added to tp004-a (13.0 g, 58.2 mmol), then (2,5-dioxopyrrolidin-1-yl)(9H-fluoren-9-yl)methyl carbonate (19.6 g, 58.2 mmol) was added, and then the mixture was vigorously stirred at room temperature for 1 hour. A 3.5 wt % aqueous potassium hydrogen carbonate solution (195 ml) was added, and the mixture was stirred at room temperature for 30 minutes. The reaction solution was washed with a 1:2 mixed solution (390 ml) of tert-butyl methyl ether and n-hexane, and then adjusted to pH 3 with phosphoric acid. The mixture was extracted with tert-butyl methyl ether (195 ml), and the organic layer was washed with a saturated aqueous sodium chloride solution (130 ml). The organic layer was dried over magnesium sulfate and filtered, and then the solvent was distilled off under reduced pressure to give tp004-b (25.9 g, 100%).

(1338) LCMS (ESI) m/z=446 (M+H)+

(1339) Retention time: 1.22 min (Analytical condition SMD method_04)

(1340) Using tp004-b as a starting material, tp004-c (32.1 g, 96%) was obtained in the same manner as synthesis of Compound tp005-c.

(1341) LCMS (ESI) m/z=573 (M+H)+

(1342) Retention time: 1.43 min (Analytical condition SMD method_04)

(1343) Using tp004-c as a starting material, tp004-d (17.7 g, 90%) was obtained in the same manner as synthesis of Compound tp003-c.

(1344) LCMS (ESI) m/z=351 (M+H)+

(1345) Retention time: 0.95 min (Analytical condition SMD method_08)

(1346) Using tp004-d as a starting material, tp004-e (13.6 g, 69%) was obtained in the same manner as synthesis of Compound tp003-d.

(1347) LCMS (ESI) m/z=391 (M+H)+

(1348) Retention time: 1.15 min (Analytical condition SMD method_08)

(1349) Using tp004-e as a starting material, tp004-f (1.4 g, 76%) was obtained in the same manner as synthesis of Compound tp003-e.

(1350) LCMS (ESI) m/z=760 (M+Na)+

(1351) Retention time: 1.65 min (Analytical condition SMD method_04)

(1352) Using tp004-f as a starting material, tp004-g (6.4 g, 59%) was obtained in the same manner as synthesis of Compound tp005-e.

(1353) LCMS (ESI) m/z=732 (M+Na)+

(1354) Retention time: 1.56 min (Analytical condition SMD method_04)

(1355) tp004-g (6.4 g, 9.0 mmol) was dissolved in methanol (299 ml), 10% palladium carbon (636.0 mg) was added, and the mixture was stirred in a hydrogen atmosphere for 1 hour. After filtration, the solvent was distilled off under reduced pressure. Acetonitrile (128 ml) and a 3.5 wt %

aqueous potassium hydrogen carbonate solution (128 ml) were added, then (2,5-dioxopyrrolidin-1-yl)(9H-fluoren-9-yl)methyl carbonate (5.4 g, 16.0 mmol) was added, and then the mixture was stirred at room temperature for 30 minutes. Moreover, (2,5-dioxopyrrolidin-1-yl)(9H-fluoren-9-yl)methyl carboxylate (5.4 g, 16.0 mmol) was added, and then the mixture was stirred overnight at room temperature. A 3.5 wt % aqueous potassium hydrogen carbonate solution (128 ml) and water (128 ml) were added, and the mixture was washed with a 1:2 mixed solution of ethyl acetate and n-hexane (384 ml) and adjusted to pH 3 with phosphoric acid. The mixture was extracted with ethyl acetate (128 ml), and the organic layer was washed with a saturated aqueous sodium chloride solution (64 ml). The organic layer was dried over magnesium sulfate and filtered, and then the solvent was distilled off under reduced pressure to give tp004-h (12.3 g, 100%).

(1356) LCMS (ESI) $m/z=622$ (M+H)+

(1357) Retention time: 1.28 min (Analytical condition SMD method_0) 4)

(1358) tp004-h (11.4 g, 18.3 mmol) was dissolved in N,N-dimethylformamide (183.0 ml), and after potassium hydrogen carbonate (4.6 g, 45.8 mmol) and then allyl bromide (10.6 ml, 122.0 mmol) were added, the mixture was vigorously stirred at room temperature for 3 hours. Ethyl acetate (200 ml) and n-hexane (200 ml) were added, the organic layer was washed with a 3.5 wt % aqueous potassium hydrogen carbonate solution (200 ml) and then water (200 ml), and the organic layer was dried over anhydrous magnesium sulfate. After filtration, the solvent was distilled off under reduced pressure, and the resulting residue was purified by silica gel column chromatography (n-hexane/ethyl acetate) to give tp004-i (11.4 g, 92%).

(1359) LCMS (ESI) $m/z=684$ (M+Na)+

(1360) Retention time: 1.49 min (Analytical condition SMD method_0) 4)

(1361) Using tp004-i as a starting material, tp004 (5.3 g, 94%) was obtained in the same manner as synthesis of Compound tp005.

(1362) LCMS (ESI) $m/z=628$ (M+Na)+

(1363) Retention time: 1.16 min (Analytical condition SMD method_0) 4)

(1364) Synthesis of Compound tp013

(1365) ##STR00475##

(1366) Using tp002-b as a starting material, tp013 (760.7 mg, 94%) was obtained in the same manner as synthesis of Compound tp005.

(1367) LCMS (ESI) $m/z=664$ (M+H)+

(1368) Retention time: 1.32 min (Analytical condition SMD method_04)

(1369) Synthesis of Compound tp006

(1370) ##STR00476## ##STR00477##

(1371) Magnesium sulfate (6.61 g, 54.9 mmol) and paraformaldehyde (1.98 g, 65.9 mmol) were added at room temperature to a dichloromethane (200 mL) solution of Compound tp006-a (N-([(9H-fluoren-9-yl)methoxy]carbonyl)-4-(trifluoromethyl)-L-phenylalanine)(10.0 g, 22.0 mmol), and then a boron trifluoride diethyl ether complex (2.78 mL, 22.0 mmol) was added dropwise. After the reaction mixture was stirred at room temperature for 1 hour, the reaction solution was filtered through silica gel (24 g) and then concentrated under reduced pressure to give a crude product tp006-b (10.5 g, quant.).

(1372) LCMS (ESI) $m/z=468$ (M+H)+

(1373) Retention time: 1.45 min (Analytical condition SMD method_0) 4)

(1374) The resulting crude product tp006-b (10.3 g, 22.0 mmol) was dissolved in dichloromethane (200 mL), and allyltrimethylsilane (12.2 mL, 77.0 mmol) was added in a nitrogen atmosphere, and a boron trifluoride diethyl ether complex (7.0 mL, 54.9 mmol) was added dropwise. After the reaction mixture was stirred at room temperature for 6 hours, allyltrimethylsilane (12.2 mL, 77.0 mmol) and a boron trifluoride diethyl ether complex (7.0 mL, 54.9 mmol) were added, and the mixture was stirred for 3 days. Water (40 mL) was added to the reaction solution, the mixture was stirred at room temperature for 10 minutes, filtered through Celite, the filtrate was concentrated

under reduced pressure, and the resulting residue was dissolved in toluene (24 mL). The organic layer was washed with water (48 mL), hexane (100 mL) was added, and the mixture was extracted with a mixed solution (2:1, 300 mL) of a 3.5% aqueous potassium hydrogen carbonate solution and acetonitrile. Phosphoric acid was added to the resulting aqueous layer until pH 3, and then the mixture was extracted with ethyl acetate (200 mL). The organic layer was washed with saturated brine (100 mL), dried over anhydrous magnesium sulfate, and then filtered. The resulting solution was concentrated under reduced pressure to give Compound tp006-c (10.3 g, 92%).

(1375) LCMS (ESI) $m/z=510$ (M+H)+

(1376) Retention time: 1.00 min (Analytical condition SMD method_08)

(1377) After HATU (60.4 g, 159 mmol) was added to an NMP (379 mL) solution of Compound tp006-c (67.5 g, 132 mmol) and sarcosine tert-butyl ester hydrochloride (25.3 g, 139 mmol), DIPEA (69.2 mL, 397 mmol) was added dropwise. After the reaction mixture was stirred at room temperature for 1.5 hours, toluene (1.0 μ L, 15 v/w) was added, and the organic layer was washed with water (473 mL, 7 v/w), a 3.5% aqueous potassium hydrogen carbonate solution (473 mL, 7 v/w), and a 1 M aqueous potassium dihydrogen phosphate solution (473 mL, 7 v/w). The organic layer was dried by being passed through an anhydrous magnesium sulfate pad and washed with toluene (67.5 mL, 1 v/w), and the resulting filtrate and washing solution were combined to give Compound tp006-d, which was directly used in the next reaction as a solution.

(1378) LCMS (ESI) $m/z=637$ (M+H)+

(1379) Retention time: 1.66 min (Analytical condition SMD method_10))

(1380) DBU (1.97 mL., 13.2 mmol) was added to a toluene solution of the above Compound tp006-d (132 mmol) at room temperature. The reaction mixture was stirred at room temperature for 17 hours and then washed with a 1 M aqueous potassium dihydrogen phosphate solution (420 mL., 5 v/w). The resulting organic layer was diluted with n-hexane (840 mL, 10) v/w), and then extracted twice with a mixed solution of 2 N hydrochloric acid (99 mL., 198 mmol), water (840 mL., 10 v/w), and acetonitrile (588 mL., 7 v/w). A 3.3 M aqueous potassium phosphate solution was added to the resulting aqueous layer to adjust the pH to 9.5, and the mixture was extracted with ethyl acetate (840 mL., 10) v/w). The organic layer was concentrated under reduced pressure, the resulting residue was dissolved in ethyl acetate (840 mL., 10 v/w), washed with brine (420 mL., 5 v/w), dried over anhydrous magnesium sulfate, and then filtered. The resulting solution was concentrated under reduced pressure to give Compound tp006-e (49.0 g, 90%).

(1381) LCMS (ESI) $m/z=415$ (M+H)+

(1382) Retention time: 1.31 min (Analytical condition SMD method_10)

(1383) DIPEA (72.3 mL., 414 mmol) was added to a dichloromethane (276 mL.) solution of Compound tp006-e (49.0 g, 118 mmol). The resulting mixture was added dropwise to a separately prepared dichloromethane (276 mL.) solution of (9H-fluoren-9-yl)methyl(S)-(1-chloro-1-oxopent-4-en-2-yl)(methyl) carbamate (described below) over 30 minutes. After the resulting reaction mixture was stirred at room temperature for 17 hours, the organic layer was washed with a 5% aqueous sodium dihydrogen phosphate solution (500 mL., 10 v/w), dried over anhydrous magnesium sulfate, and then filtered, and the solution was concentrated under reduced pressure. The resulting crude product was purified by normal phase silica gel chromatography (n-hexane/ethyl acetate) to give Compound tp006-f (45.0 g, 51%).

(1384) (9H-Fluoren-9-yl)methyl(S)-(1-chloro-1-oxopent-4-en-2-yl)(methyl) carbamate was prepared as follows. After thionyl chloride (21.6 mL., 296 mmol) was added to a dichloromethane (296 mL.) solution of (S)-2-(((9H-fluoren-9-yl)methoxy) carbonyl)(methyl)amino) pent-4-enoic acid (41.6 g, 118 mmol), DMF (0.92 mL, 11.8 mmol) was added dropwise. The resulting reaction mixture was stirred at room temperature for 15 minutes and then concentrated under reduced pressure. Toluene (20 mL., 5 v/w) was added to the resulting residue, and the mixture was concentrated under reduced pressure. The resulting crude product was dissolved in dichloromethane (276 mL.) and used in the above-described reaction.

(1385) LCMS (ESI) $m/z=748$ (M+H)+

(1386) Retention time: 1.71 min (Analytical condition SMD method_10)

(1387) A dichloroethane (891 mL) solution of Compound tp006-f (16.7 g, 22.3 mmol), benzoquinone (0.24 g, 2.23 mmol), and a Stewart-Grubbs catalyst (0.51 g, 0.89 mmol) was degassed 5 times, and stirred at 80° C. for 4 hours in a nitrogen atmosphere. The reaction mixture was cooled to room temperature, and then concentrated under reduced pressure, and the resulting crude product was purified by normal phase silica gel column chromatography (hexane/ethyl acetate) to give Compound tp006-g (33.9 g, 39%).

(1388) LCMS (ESI) $m/z=720$ (M+H)+

(1389) Retention time: 1.00 min (Analytical condition SMD method_07)

(1390) After hexamethyldisilazane (24.7 mL, 118 mmol) was added to an ethyl acetate (236 mL) solution of tp006-g (33.9 g, 47.1 mmol) in a nitrogen atmosphere, trimethylsilyl trifluoromethanesulfonate (17.0 mL, 94 mmol) was added dropwise while cooling the mixture in a water bath (25° C.). After the reaction mixture was stirred at room temperature for 1.5 hours, a 5% aqueous disodium hydrogen phosphate solution (340 mL, 10 v/w) was added while cooling the mixture in an ice bath to quench the reaction. Water (170 mL, 5 v/w) and phosphoric acid were added to the resulting mixture to adjust the pH to 3. The separated organic layer was washed with saturated brine (340 mL, 10 v/w), dried over anhydrous magnesium sulfate, and filtered, and the solution was concentrated under reduced pressure. The resulting crude product was purified by reverse phase column chromatography (0.1% formic acid-containing acetonitrile/0.1% formic acid-containing distilled water), and collected fractions were purified by reverse phase column chromatography (acetonitrile/distilled water) to give Compound tp006 (22.4 g, 72%).

(1391) LCMS (ESI) $m/z=664$ (M+H)+

(1392) Retention time: 0.63 min (Analytical condition SMD method_06)

(1393) Synthesis of Compound tp010

(1394) ##STR00478## ##STR00479##

(1395) Using Compound tp010-a ((S)-2-(((9H-fluoren-9-yl)methoxy) carbonyl)amino)-3-cyclohexylpropanoic acid)(14.0 g, 35.6 mmol) as a starting material, a crude product tp010-b (14.4 g, quant.) was obtained in the same manner as synthesis of Compound tp006-b. LCMS (ESI) $m/z=406$ (M+H)+

(1396) Retention time: 0.78 min (Analytical condition SQDAA50_2)

(1397) Using the resulting crude product tp010-b (14.4 g, 35.6 mmol), Compound tp010-c (12.6 g, 79%) was obtained in the same manner as synthesis of Compound tp006-c. LCMS (ESI) $m/z=448$ (M+H)+

(1398) Retention time: 0.71 min (Analytical condition SQDAA50_2)

(1399) Using Compound tp010-c (12.6 g, 28.3 mmol), a toluene solution of Compound tp010-d was obtained in the same manner as synthesis of Compound tp006-d. The resulting solution of Compound tp010-d was directly used in the next reaction.

(1400) LCMS (ESI) $m/z=575$ (M+H)+

(1401) Retention time: 0.88 min (Analytical condition SQDAA50_2)

(1402) Using a toluene solution of the above Compound tp010-d (28.3 mmol), Compound tp010-e (9.0 g, 91%) was obtained in the same manner as synthesis of Compound tp006-e. LCMS (ESI) $m/z=353$ (M+H)+

(1403) Retention time: 0.66 min (Analytical condition SQDAA50_2)

(1404) Using Compound tp010-e (4.5 g, 12.8 mmol), Compound tp010-f (8.6 g, 98%) was obtained in the same manner as synthesis of Compound tp006-f.

(1405) LCMS (ESI) $m/z=708$ (M+Na)+

(1406) Retention time: 0.91 min (Analytical condition SQDAA50_2)

(1407) After nitrogen bubbling in a dichloroethane (469 mL) solution of Compound tp010-f (8.0 g, 11.7 mmol) and a Stewart-Grubbs catalyst (0.17 g, 0.29 mmol), the mixture was stirred at 80° C.

for 16 hours in a nitrogen atmosphere. The reaction mixture was cooled to room temperature and then concentrated under reduced pressure, and the resulting crude product was purified by normal phase silica gel column chromatography (n-hexane/ethyl acetate) to give Compound tp010-g (5.7 g, 73%).

(1408) LCMS (ESI) $m/z=680$ (M+Na)+

(1409) Retention time: 0.84 min (Analytical condition SQDAA50_2)

(1410) Using Compound tp010-g (5.7 g, 8.6 mmol), Compound tp010 (3.8 g, 74%) was obtained in the same manner as synthesis of Compound tp006.

(1411) LCMS (ESI) $m/z=624$ (M+Na)+

(1412) Retention time: 0.69 min (Analytical condition SQDFA40)

(1413) Synthesis of Compound tp011

(1414) ##STR00480## ##STR00481##

(1415) Using Compound tp011-a ((S)-2-(((9H-fluoren-9-yl)methoxy) carbonyl)amino)-3-(4,4-difluorocyclohexyl) propanoic acid)(14.0 g, 32.6 mmol) as a starting material, a crude product tp011-b (14.4 g, quant.) was obtained in the same manner as synthesis of Compound tp006-b.

(1416) LCMS (ESI) $m/z=464$ (M+Na)+

(1417) Retention time: 0.69 min (Analytical condition SQDAA50_2)

(1418) Using the resulting crude product tp011-b (14.4 g, 32.6 mmol), Compound tp011-c (11.6 g, 73%) was obtained in the same manner as synthesis of Compound tp006-c. LCMS (ESI) $m/z=484$ (M+H)+

(1419) Retention time: 0.65 min (Analytical condition SQDAA50_2)

(1420) Using Compound tp011-c (11.6 g, 23.9 mmol), a toluene solution of Compound tp011-d was obtained in the same manner as synthesis of Compound tp006-d. The resulting solution of Compound tp011-d was directly used in the next reaction.

(1421) LCMS (ESI) $m/z=611$ (M+H)+

(1422) Retention time: 0.82 min (Analytical condition SQDAA50_2)

(1423) Using a toluene solution of the above Compound tp011-d (23.9 mmol), Compound tp011-e (8.6 g, 92%) was obtained in the same manner as synthesis of Compound tp-006-e. LCMS (ESI) $m/z=389$ (M+H)+

(1424) Retention time: 0.58 min (Analytical condition SQDAA50_2)

(1425) Using Compound tp011-e (4.3 g, 11.0 mmol), Compound tp011-f (6.9 g, 87%) was obtained in the same manner as synthesis of Compound tp006-f.

(1426) LCMS (ESI) $m/z=744$ (M+Na)+

(1427) Retention time: 0.86 min (Analytical condition SQDAA50_2)

(1428) Using Compound tp011-f (6.9 g, 9.6 mmol), Compound tp011-g (4.8 g, 72%) was obtained in the same manner as synthesis of Compound tp010-g.

(1429) LCMS (ESI) $m/z=716$ (M+Na)+

(1430) Retention time: 0.79 min (Analytical condition SQDAA50_2)

(1431) Using Compound tp011-g (4.8 g, 7.0 mmol), Compound tp011 (2.8 g, 64%) was obtained in the same manner as synthesis of Compound tp006.

(1432) LCMS (ESI) $m/z=660$ (M+Na)+

(1433) Retention time: 0.63 min (Analytical condition SQDFA40)

(1434) Synthesis of Compound tp007

(1435) ##STR00482## ##STR00483## ##STR00484##

(1436) Compound tp007-a ((S)-2-(((9H-fluoren-9-yl)methoxy) carbonyl)amino)-4-methoxy-4-oxobutanoic acid)(30.0 g, 81.0 mmol) as a starting material was reacted in the same manner as synthesis of Compound tp006-b, and the filtrate obtained by Celite-filtering the reaction solution was regarded as Compound tp007-b and directly used in the next reaction.

(1437) LCMS (ESI) $m/z=382$ (M+H)+

(1438) Retention time: 1.22 min (Analytical condition SMD method_04)

(1439) Using a dichloromethane solution of the above Compound tp007-b (81.0 mmol), Compound tp007-c (35.9 g, quant.) was obtained in the same manner as synthesis of Compound tp006-c.

(1440) LCMS (ESI) $m/z=424$ (M+H)+

(1441) Retention time: 0.85 min (Analytical condition SMD method_10)

(1442) Compound tp007-c (34.3 g, 81.0) mmol) was dissolved in tetrahydrofuran (231 mL), reacted in the same manner as synthesis of tp006-d, and then purified by normal phase column chromatography (hexane/ethyl acetate) to give Compound tp007-d (44.4 g, 100%). LCMS (ESI) $m/z=551$ (M+H)+

(1443) Retention time: 1.45 min (Analytical condition SMD method_0) 4)

(1444) Using Compound tp007-d (44.4 g, 81.0 mmol), Compound tp007-e (23.3 g, 88%) was obtained in the same manner as synthesis of tp006-e.

(1445) LCMS (ESI) $m/z=329$ (M+H)+

(1446) Retention time: 0.58 min (Analytical condition SMD method_0) 4)

(1447) Using Compound tp007-e (23.3 g, 71.0) mmol), Compound tp007-f (43.5 g, 93%) was obtained in the same manner as synthesis of tp006-f.

(1448) LCMS (ESI) $m/z=662$ (M+H)+

(1449) Retention time: 1.54 min (Analytical condition SMD method_0) 4)

(1450) After a dichloroethane (996 mL) solution of a Stewart-Grubbs catalyst (0).47 g, 0.82 mmol) was degassed 4 times, a dichloroethane (100) mL) solution of Compound tp007-f (21.8 g, 32.9 mmol) was added dropwise over 1 hour while being stirred at 80° C. in a nitrogen atmosphere. The reaction mixture was stirred at 80° C. for 1 hour and then cooled to room temperature. The reaction mixture was concentrated under reduced pressure, and the resulting crude product was purified by reverse phase column chromatography (0).1% formic acid-containing acetonitrile/() 1% formic acid-containing distilled water) to give Compound tp007-g (20.8 g, 50%).

(1451) LCMS (ESI) $m/z=634$ (M+H)+

(1452) Retention time: 1.39 min (Analytical condition SMD method_0) 4)

(1453) After calcium chloride (61.1 g, 551 mmol) was dissolved in water (153 mL), lithium hydroxide monohydrate (6.16 g, 147 mmol) was added at room temperature. After the mixture was stirred for 5 minutes, isopropanol (612 mL) and a tetrahydrofuran solution (153 mL) of Compound tp007-g (23.3 g, 36.7 mmol) were added and stirred overnight at room temperature. Fmoc-OSu (12.4 g, 36.7 mmol) was added, the mixture was stirred for 2 hours, then water (460) mL, 20) v/w) was added, 2 N hydrochloric acid was added to adjust the pH to 2.5, and the mixture was extracted with ethyl acetate (460) mL, 20 v/w). The organic layer was washed with saturated brine (230) mL, 10) v/w) and concentrated under reduced pressure. Ethyl acetate (115 mL, 5 v/w), acetonitrile (115 mL, 5 v/w), and hexane (230) mL, 10) v/w) were added to and dissolve the resulting residue, and the mixture was extracted with a 3.5% aqueous potassium hydrogen carbonate solution (230) mL, 10) v/w). Phosphoric acid was added to the resulting aqueous layer to adjust the pH to 3, and then the mixture was extracted with ethyl acetate (115 mL, 5 v/w). The organic layer was washed with saturated brine (70) mL, 3 v/w) and concentrated under reduced pressure, and the resulting crude product was purified by reverse phase column chromatography (() 1%-formic acid containing acetonitrile/() 1% formic acid-containing distilled water) to give Compound tp007-h (12.6 g, 56%).

(1454) LCMS (ESI) $m/z=620$ (M+H)+

(1455) Retention time: 0.94 min (Analytical condition SMD method_10))

(1456) 1-(3-Dimethylaminopropyl)-3-ethylcarbodiimide hydrochloride (5.87 g, 30.6 mmol) was added to a dichloromethane (93 ml) suspension of Compound tp007-h (12.6 g, 20.4 mmol) and N-hydroxyphthalimide (3.66 g, 22.4 mmol), and the mixture was stirred at room temperature for 2.5 hours. The reaction mixture was washed with water, then the organic layer was concentrated under reduced pressure, and the resulting crude product was purified by normal phase column chromatography (hexane/ethyl acetate) to give Compound tp007-i (15.1 g, 97%). LCMS (ESI)

m/z=765 (M+H)+

(1457) Retention time: 1.35 min (Analytical condition SMD method_05)

(1458) Nickel (II) bromide trihydrate (2.47 g, 6.46 mmol) and 4,4'-di-tert-butyl-2,2'-bipyridine (1.73 g, 6.46 mmol) were dissolved in DMA (20 mL) to prepare a nickel solution. The nickel solution prepared in advance and chlorotrimethylsilane (0.41 mL, 3.23 mmol) were added to a DMA (110 mL) suspension of Compound tp007-i (4.94 g, 6.46 mmol), zinc powder (6.33 g, 97.0 mmol), and 1-iodo-4-methylbenzene (14.1 g, 64.6 mmol), and the mixture was stirred at room temperature for 2 hours. The resulting reaction mixture was purified by reverse phase column chromatography (() 1% formic acid-containing acetonitrile/0.1% formic acid-containing distilled water) to give Compound tp007-j (2.7 g, 63%).

(1459) LCMS (ESI) m/z=666 (M+H)+

(1460) Retention time: 1.57 min (Analytical condition SMD method_04)

(1461) Using Compound tp007-j (2.6 g, 3.90 mmol), Compound tp007 (1.78 g, 45%) was obtained in the same manner as synthesis of tp006.

(1462) LCMS (ESI) m/z=610 (M+H)+

(1463) Retention time: 1.26 min (Analytical condition SMD method_0) 4)

(1464) Synthesis of Compound tp008

(1465) ##STR00485##

(1466) In a nitrogen atmosphere, nickel (II) bromide trihydrate (1.33 g, 4.89 mmol) and 4,4'-di-tert-butyl-2,2'-bipyridine (1.31 g, 4.89 mmol) were dissolved in DMA (48.9 mL), and the mixture was stirred at room temperature for 30 minutes to prepare a nickel solution. The nickel solution prepared in advance and chlorotrimethylsilane (0.31 mL, 2.45 mmol) were added to a DMA (48.9 mL) suspension of Compound tp007-i (3.74 g, 4.89 mmol), zinc powder (4.80 g, 73.3 mmol), and 1-fluoro-4-iodobenzene (10.2 mL, 88.0 mmol), and the mixture was stirred at room temperature for 1 hour. The resulting reaction mixture was filtered through Celite, and the filtrate was washed with an aqueous ammonium chloride solution (37 mL, 10 v/w). The organic layer was dried over anhydrous magnesium sulfate and then filtered, and the solution was concentrated under reduced pressure. The resulting crude product was purified by normal phase silica gel chromatography (hexane/ethyl acetate) to give Compound tp008-a (2.97 g, 72%). LCMS (ESI) m/z=670 (M+H)+

(1467) Retention time: 1.39 min (Analytical condition SMD method_05)

(1468) Using Compound tp008-a (2.97 g, 4.43 mmol), Compound tp008 (2.36 g, 87%) was obtained in the same manner as synthesis of tp006.

(1469) LCMS (ESI) m/z=614 (M+H)+

(1470) Retention time: 1.22 min (Analytical condition SMD method_04)

(1471) Synthesis of Compound tp009

(1472) ##STR00486##

(1473) In a nitrogen atmosphere, nickel (II) bromide trihydrate (1.76 g, 6.41 mmol) and 4,4'-di-tert-butyl-2,2'-bipyridine (1.72 g, 6.41 mmol) were dissolved in DMA (64.1 mL) to prepare a nickel solution. The nickel solution prepared in advance and chlorotrimethylsilane (0.41 mL, 3.20 mmol) were added to a DMA (64.1 mL) suspension of Compound tp007-i (4.90 g, 6.41 mmol), zinc powder (6.28 g, 96.0 mmol), and 1-bromo-4-fluoro-2-methylbenzene (16.2 mL, 128 mmol), and the mixture was stirred at room temperature for 2 hours. The resulting reaction mixture was filtered through Celite, and a 15% aqueous ammonium chloride solution (150 mL, 30 v/w) was added to the filtrate to separate the organic layer. After the organic layer was washed with an 8% aqueous EDTA disodium solution (150 mL, 30 v/w) and 10% brine (150 mL, 30 v/w), the organic layer was dried over anhydrous magnesium sulfate and then filtered, and the solution was concentrated under reduced pressure. The resulting crude product was purified by normal phase silica gel chromatography (hexane/ethyl acetate) to give Compound tp009-a (2.43 g, 55%).

(1474) LCMS (ESI) m/z=684 (M+H)+

(1475) Retention time: 1.42 min (Analytical condition SMD method_05)

(1476) Using Compound tp009-a (2.42 g, 3.54 mmol), Compound tp009 (1.76 g, 79%) was obtained in the same manner as synthesis of tp006.

(1477) LCMS (ESI) $m/z=628$ (M+H)+

(1478) Retention time: 1.26 min (Analytical condition SMD method_04)

(1479) Synthesis of Compound tp012

(1480) ##STR00487## ##STR00488##

(1481) tp012-a (4.88 g, 9.51 mmol) was dissolved in dichloromethane (95 mL), then anhydrous magnesium sulfate (2.86 g, 23.8 mmol), paraformaldehyde (856 mg, 28.5 mmol), and a boron trifluoride diethyl ether complex (1.2 mL, 9.51 mmol) were added, and the mixture was stirred at room temperature for 3 hours. The reaction solution was filtered through silica gel, silica gel was washed with dichloromethane (100 mL), and the filtrate was concentrated under reduced pressure to give a crude product (4.50 g) of tp012-b.

(1482) LCMS (ESI) $m/z=548$ (M+Na)+

(1483) Retention time: 1.10 min (Analytical condition SQDAA05)

(1484) In a nitrogen atmosphere, the above tp012-b (9.51 mmol) was dissolved in dichloromethane (95 mL), then allyltrimethylsilane (10.6 mL, 66.6 mmol) and a boron trifluoride diethyl ether complex (6.0 mL, 47.6 mmol) were added, and the mixture was stirred at room temperature for 88 hours. The reaction solution was cooled to 0° C., and water (20 mL) was added. The mixed solution was filtered through Celite, and Celite was washed with dichloromethane (100 mL). The resulting filtrate was concentrated under reduced pressure, and the resulting residue was dissolved in toluene (50 mL) and washed with water (24 mL) and a 1 M aqueous dipotassium hydrogen phosphate solution (24 mL). Hexane (100 mL) was added to the organic phase, and the mixture was extracted 5 times with a 3.5% aqueous potassium hydrogen carbonate solution/acetonitrile (2/1, 42 mL). The aqueous phase was washed twice with hexane (40) mL), 2 N hydrochloric acid was added to adjust the pH to 2, and then the mixture was extracted 3 times with ethyl acetate (50 mL). The organic phase was washed with brine (24 mL) and concentrated under reduced pressure to give tp012-c (4.09 g, 76%). LCMS (ESI) $m/z=568$ (M+H)+

(1485) Retention time: 1.03 min (Analytical condition SQDFA05)

(1486) In a nitrogen atmosphere, tp012-c (3.09 g, 5.45 mmol) and sarcosine tert-butyl ester hydrochloride (989 mg, 5.45 mmol) were dissolved in dichloromethane (16 mL), then DIPEA (2.85 mL, 16.34 mmol) and HATU (2.49 g, 6.53 mmol) were added, and the mixture was stirred at room temperature for 1 hour. Subsequently, the reaction solution was successively washed with water (30 mL), a 3.5% aqueous potassium hydrogen carbonate solution (30 mL), and a 1 M aqueous dipotassium hydrogen phosphate solution (30 mL), and the organic phase was dried over anhydrous sodium sulfate. The resulting solution was concentrated under reduced pressure to give a crude product of tp012-d. The resulting tp012-d was directly used in the next step.

(1487) LCMS (ESI) $m/z=695$ (M+H)+

(1488) Retention time: 0.82 min (Analytical condition SQDFA50)

(1489) The crude product (5.45 mmol) of tp012-d was dissolved in toluene (32.7 mL) and acetonitrile (3.63 mL), DBU (0.81 mL) was added, and the mixture was stirred at room temperature for 20 minutes. A 1 M aqueous potassium dihydrogen phosphate solution (30 mL), n-hexane (60) mL), water (40) mL), acetonitrile (20 mL), and 2 N hydrochloric acid (4.2 mL) were added to the reaction solution and shaken, then the mixture separated into three layers. Water (40) mL), acetonitrile (20 mL), and toluene (20 mL) were further added, but the three-layer separation did not disappear, thus the upper-layer organic phase was discarded, and an aqueous 3 M potassium phosphate solution (4 mL) was added to the remaining lower two layers to adjust the pH to 8. The resulting mixed solution was extracted twice with ethyl acetate (30) mL), and the organic phase was washed with saturated brine (20 mL). The organic phase was dried over anhydrous sodium sulfate and concentrated under reduced pressure to give tp012-e (2.74 g, 85%, 2 steps).

(1490) LCMS (ESI) $m/z=473.6$ (M+H)+

(1491) Retention time: 0.57 min (Analytical condition SQDFA05)

(1492) In a nitrogen atmosphere, (S)-2-((((9H-fluoren-9-yl)methoxy) carbonyl)(methyl)amino) pent-4-enoic acid (2.80 g, 7.96 mmol) was dissolved in DCM (20 ml) at room temperature, DMF (0.062 ml, 0.796 mmol) was added, and then thionyl chloride (1.45 ml, 19.9 mmol) was added dropwise. The resulting reaction mixture was stirred for 20 minutes and then concentrated under reduced pressure. Toluene (7 ml) was added to the resulting residue, and the mixture was concentrated under reduced pressure. This operation was performed twice in total, and the resulting crude product was dissolved in DCM (16 mL) and used in the subsequent reaction.

(1493) In a nitrogen atmosphere, a DCM (20 mL) solution of tp012-e (3.42 g, 7.24 mmol) and DIPEA (4.41 mL, 25.3 mmol) was added to the prepared DCM (16 ml) solution of (9H-fluoren-9-yl)methyl(S)-(1-chloro-1-oxopent-4-en-2-yl)(methyl) carbamate. The reaction solution was stirred at room temperature for 3 hours, and then washed with a 1 M aqueous potassium dihydrogen phosphate solution (35 mL). Subsequently, the organic phase was washed with a 3.5% aqueous potassium hydrogen carbonate solution (35 mL) and concentrated under reduced pressure. The resulting crude product was purified by silica gel chromatography (n-hexane/ethyl acetate) to give tp012-f (4.45 g, 76%).

(1494) LCMS (ESI) $m/z=806.5$ (M+H)+

(1495) Retention time: 0.88 min (Analytical condition SQDAA50)

(1496) In a nitrogen atmosphere, tp012-f (2.79 g, 3.46 mmol) was dissolved in 1,2-dichloroethane (115 mL) in a three-neck flask, and a first-generation Grubbs catalyst (60 mg, 0.0715 mmol) was added. While warming to 80° C. and stirring the reaction solution, a 1,2-dichloroethane (15 mL) solution of a first-generation Grubbs catalyst (60 mg, 0.0715 mmol) was added dropwise over 2 hours with a syringe pump. Subsequently, the reaction solution was stirred at 80° C. for 1 hour, cooled to room temperature, and then concentrated under reduced pressure. To complete the reaction, the resulting crude product was again dissolved in 1,2-dichloroethane (115 mL) in a nitrogen atmosphere, a first-generation Grubbs catalyst (240 mg, 0.286 mmol) was added, and the mixture was stirred at 80° C. for 90 minutes. The reaction solution was cooled to room temperature, and then concentrated under reduced pressure. The resulting crude product was purified by silica gel chromatography (n-hexane/ethyl acetate) to give tp012-g (1.57 g, 58%).

(1497) LCMS (ESI) $m/z=778.5$ (M+H)+

(1498) Retention time: 0.74 min (Analytical condition SQDFA50)

(1499) In a nitrogen atmosphere, tp012-g (1.57 g, 2.02 mmol) was dissolved in ethyl acetate (24 mL), hexamethyldisilazane (1.481 mL, 7.07 mmol) was added, and then trimethylsilyl trifluoromethanesulfonate (1.02 mL, 5.65 mmol) was added dropwise. The resulting reaction mixture was stirred for 150 minutes, then the reaction solution was diluted with ethyl acetate (20 mL), and a 1 M aqueous disodium hydrogen phosphate solution (40 ml) was added to quench the reaction. The organic phase was separated, and the solvent was distilled off under reduced pressure. The resulting crude product was purified by reverse phase chromatography (0.1% formic acid-containing acetonitrile/0.1% formic acid-containing distilled water) to give tp012 (1.02 g, 70%).

(1500) LCMS (ESI) $m/z=722$ (M+H)+

(1501) Retention time: 0.47 min (Analytical condition SQDFA50)

(1502) Synthesis of Compound tp014

(1503) ##STR00489##

(1504) In a nitrogen atmosphere, tp005 (0.75 g, 1.06 mmol) was dissolved in 1,4-dioxane (5.3 mL), a 1 M diethylzinc toluene solution (5.31 mL, 5.31 mmol) was added, and then diiodomethane (0.856 mL, 10.63 mmol) was added dropwise. The reaction solution was heated to 60° C., stirred at 60° C. for 4 and a half hours and cooled to room temperature, and a saturated aqueous ammonium chloride solution was added to quench the reaction. The organic phase was separated, the aqueous phase was extracted with ethyl acetate, and the resulting organic phases were combined and concentrated under reduced pressure.

(1505) The same operation was repeated using tp005 (0.75 g, 1.06 mmol), and the resulting crude products were combined and purified by reverse phase chromatography (0.1% formic acid-containing acetonitrile/0.1% formic acid-containing distilled water) to give tp014-a (0.21 g, 14%). At this time, the raw-material tp005 (0.58 g, 39%) and a mixture of tp005 and tp014-a (3/2, 0.18 g, 12%) were also recovered.

(1506) LCMS (ESI) $m/z=720.5$ (M+H)+

(1507) Retention time: 0.74 min (Analytical condition SQDFA50)

(1508) In a nitrogen atmosphere, tp014-a (264 mg, 0.367 mmol) was dissolved in ethyl acetate (2.0 mL), hexamethyldisilazane (0.269 mL, 1.28 mmol) was added, and then trimethylsilyl trifluoromethanesulfonate (0.186 mL, 1.03 mmol) was added dropwise. The resulting reaction mixture was stirred for 1 hour, then the reaction solution was diluted with ethyl acetate (5 mL), and a 1 M aqueous disodium hydrogen phosphate solution (5 mL) was added to quench the reaction. The organic phase was separated, and the solvent was distilled off under reduced pressure. The resulting crude product was purified by reverse phase chromatography (0.1% formic acid-containing acetonitrile/0.1% formic acid-containing distilled water).

(1509) The same operation was repeated using tp014-a (142 mg, 0.197 mmol), hexamethyldisilazane (0.145 mL, 0.690 mmol), and trimethylsilyl trifluoromethanesulfonate (0.100 mL, 0.552 mmol), and, after reverse phase chromatography, the resulting products were combined to give tp014 (274 mg, 73%).

(1510) LCMS (ESI) $m/z=664.5$ (M+H)+

(1511) Retention time: 0.97 min (Analytical condition SQDFA50 long)

(1512) Synthesis of Compound tp015

(1513) ##STR00490## ##STR00491##

(1514) Aqueous ammonia (7.8 mL, 13.2 mmol) was added to a water (17.15 mL) suspension of tp005-a (2 g, 8.58 mmol) at room temperature to dissolve tp005-a, and then 3-bromo-2-methylpropene (5.19 mL, 51.5 mmol) was added dropwise. The reaction mixture was stirred at room temperature for 18 hours, ethanol (6 mL) was added, and the mixture was further stirred at room temperature for 2 hours. The resulting white solids were collected by filtration, washed with water, and the vacuum-dried to give tp015-b (2.61 g, quant.).

(1515) LCMS (ESI) $m/z=289$ (M+H)+

(1516) Retention time: 0.47 min (Analytical condition SQDFA05)

(1517) In a nitrogen atmosphere, dehydrated NMP (10 mL) was added to a mixture of tp015-b (1 g, 3.48 mmol) and sarcosine tert-butyl ester hydrochloride (1.26 g, 6.96 mmol). DIPEA (2.43 mL, 13.92 mmol) was added dropwise to the resulting suspension, the mixture was stirred for about 3 minutes, and then HATU (2.65 g, 6.96 mmol) was added. After the reaction mixture was stirred at room temperature for 4 hours, ethyl acetate (20 mL), water (10 mL), and a 5% aqueous sodium carbonate solution (10 mL) were added. The separated organic phase was washed with a 5% aqueous sodium carbonate solution (20 mL), a 5% aqueous potassium hydrogen sulfate solution (20 mL $\times$ 2), and brine (20 mL), and dried over sodium sulfate. The crude product obtained by distilling off the solvent under reduced pressure was dissolved in TBME (5 mL), then hexane (5 mL) was added to precipitate the product, and the supernatant was removed. The operation of adding hexane (5 mL) and removing the supernatant was performed 2 more times, and vacuum drying was performed to give tp015-c (1.30 g, 90%).

(1518) LCMS (ESI) $m/z=415$ (M+H)+

(1519) Retention time: 0.55 min (Analytical condition SQDFA05)

(1520) In a nitrogen atmosphere, DIPEA (3.79 mL, 21.71 mmol) was added to a dehydrated DCM (20 mL) solution of tp015-c (3.0 g, 7.24 mmol). The resulting mixture was stirred while being cooled in a water bath (25° C.), and a separately prepared DCM (5 mL) solution of (9H-fluoren-9-yl)methyl(S)-(1-chloro-1-oxopent-4-en-2-yl)(methyl) carbamate (described below) was added dropwise over 5 minutes. The resulting reaction mixture was stirred at room temperature for 2.5

hours, and then the solvent was distilled off under reduced pressure. Ethyl acetate (30 ml) and water (12 ml) were added to the resulting residue. The separated organic phase was washed with brine (12 ml×2) and dried over sodium sulfate, and then the solvent was distilled off under reduced pressure. The resulting crude product was purified by normal phase silica gel chromatography (hexane/ethyl acetate) to give tp0) 15-d (4.56 g, 84%).

(1521) (9H-Fluoren-9-yl)methyl(S)-(1-chloro-1-oxopent-4-en-2-yl)(methyl) carbamate was prepared as follows.

(1522) In a nitrogen atmosphere, (S)-2-((((9H-fluoren-9-yl)methoxy) carbonyl)(methyl)amino) pent-4-enoic acid (2.8 g, 7.97 mmol) was dissolved in DCM (15 ml) at room temperature, DMF (0.031 ml, 0.398 mmol) was added, and then thionyl chloride (1.45 ml, 19.92 mmol) was added dropwise. The resulting reaction mixture was stirred for 20 minutes and then concentrated under reduced pressure. Toluene (5 ml) was added to the resulting residue, and the mixture was concentrated under reduced pressure. This operation was performed 3 times in total, and the resulting crude product was dissolved in DCM and used in the above-described reaction.

(1523) LCMS (ESI) m/z=749 (M+H)+

(1524) Retention time: 0.84 min (Analytical condition SQDFA05)

(1525) In a nitrogen atmosphere, tp015-d (1.1 g, 1.47 mmol) and a Stewart-Grubbs catalyst (0.042 g, 0.074 mmol) were dissolved in degassed dehydrated toluene (45 ml), and the mixture was stirred at 75° C. to 80° C. for 45 minutes. After cooling to room temperature, the solvent was distilled off under reduced pressure, and the resulting crude product was purified by normal phase silica gel chromatography (hexane/ethyl acetate) to give tp015-e (1.22 g, 94%). LCMS (ESI) m/z=721 (M+H)+

(1526) Retention time: 0.75 min (Analytical condition SQDFA05)

(1527) In a nitrogen atmosphere, tp015-e (1.0 g, 1.39 mmol) was dissolved in isopropyl acetate (5 ml), hexamethyldisilazane (0.728 ml, 3.47 mmol) was added, and then trimethylsilyl trifluoromethanesulfonate (0.5 ml, 2.78 mmol) was added dropwise. The resulting reaction mixture was stirred for 1 hour, and then 1 M dipotassium hydrogen phosphate (5 ml) was added to quench the reaction. The resulting mixture was extracted with ethyl acetate, the organic phase was washed with brine (10 ml×2) and dried over sodium sulfate, and then the solvent was distilled off under reduced pressure. The resulting crude product was purified by reverse phase chromatography (0.1% formic acid-containing acetonitrile/0.1% formic acid-containing distilled water) to give tp015 (1.0 g, 90%).

(1528) LCMS (ESI) m/z=665 (M+H)+

(1529) Retention time: 0.91 min (Analytical condition SQDFA05)

(1530) Synthesis of Compound tp016

(1531) ##STR00492##

(1532) In a nitrogen atmosphere, 1,1,1,3,3,3-hexafluoropropan-2-ol (100 ml) was added to a mixture of tp015-e (4 g, 5.56 mmol) and platinum oxide (0.379 g, 1.67 mmol). The resulting mixture was depressurized while being stirred, and substituted with hydrogen. This hydrogen substitution operation was performed 3 times in total, and the mixture was stirred at room temperature for 3.5 hours in a hydrogen atmosphere. The reaction mixture was filtered through Celite and washed with ethyl acetate, and the solvent was distilled off under reduced pressure. The resulting crude product was purified by normal phase silica gel chromatography (hexane/ethyl acetate) to give tp016-a (3.05 g, 76%).

(1533) LCMS (ESI) m/z=723 (M+H)+

(1534) Retention time: 2.30 min (Analytical condition SQDFA50 long)

(1535) Using the resulting tp016-a, tp016 (2.61 g, 94%) was obtained under the same reaction conditions as synthesis of Compound tp015.

(1536) LCMS (ESI) m/z=666 (M+H)+

(1537) Retention time: 2.92 min (Analytical condition SQDFA05 long)

(1538) Synthesis of Compound tp019
(1539) ##STR00493##
(1540) Using tp015-c as a starting material, tp0019-a (1.28 g, 70%) was obtained in the same manner as synthesis of Compound tp015-d.
(1541) LCMS (ESI) m/z=763 (M+H)+
(1542) Retention time: 3.13 min (Analytical condition SQDFA50 long)
(1543) Using the resulting tp019-a, tp019-b (0.667 g, 62%) was obtained under the same reaction conditions as synthesis of Compound tp015-e.
(1544) LCMS (ESI) m/z=735 (M+H)+
(1545) Retention time: 3.66 min (Analytical condition SQDFA05 long)
(1546) Using the resulting tp019-b, tp019 (0.54 g, 88%) was obtained under the same reaction conditions as synthesis of Compound tp015.
(1547) LCMS (ESI) m/z=679 (M+H)+
(1548) Retention time: 2.82 min (Analytical condition SQDFA05 long)
(1549) Synthesis of Compound tp021
(1550) ##STR00494##
(1551) Using tp005-c, tp021-c (3.87 g, 72%) was obtained under the same reaction conditions as synthesis of Compound tp015-d.
(1552) LCMS (ESI) m/z=749 (M+H)+
(1553) Retention time: 2.88 min (Analytical condition SQDFA50 long)
(1554) Using the resulting tp021-c, tp021-d (3.35 g, 90%) was obtained under the same reaction conditions as synthesis of Compound tp015-e.
(1555) LCMS (ESI) m/z=720 (M+H)+
(1556) Retention time: 2.19 min (Analytical condition SQDFA50 long)
(1557) Using the resulting tp021-d, tp021 was obtained under the same reaction conditions as synthesis of Compound tp015 (1.13 g, 82%).
(1558) LCMS (ESI) m/z=664 (M+H)+
(1559) Retention time: 2.80 min (Analytical condition SQDFA05 long)
(1560) Synthesis of Compound tp020
(1561) ##STR00495##
(1562) Using tp021-d as a starting material, tp0020-a (1.69 g, quant.) was obtained in the same manner as synthesis of Compound tp016-a.
(1563) LCMS (ESI) m/z=723 (M+H)+
(1564) Retention time: 2.35 min (Analytical condition SQDFA50 long)
(1565) Using the resulting tp020-a, tp020 (1.25 g, 77%) was obtained under the same reaction conditions as synthesis of Compound tp016.
(1566) LCMS (ESI) m/z=666 (M+H)+
(1567) Retention time: 2.91 min (Analytical condition SQDFA05 long)
(1568) Synthesis of Compounds tp017 and tp018
(1569) ##STR00496##
(1570) Using tp019-b as a starting material, a mixture of tp0017-a and tp018-a (2.3:1) was obtained (1.56 g, 58%) in the same manner as synthesis of Compound tp016-a.
(1571) LCMS (ESI) m/z=737 (M+H)+
(1572) Retention time: 2.51 min, 2.59 min (Analytical condition SQDFA50 long)
(1573) Using the resulting mixture of tp0017-a and tp018-a (2.3:1), a mixture of tp017 and tp018 (1.31 g, 91%) was obtained under the same reaction conditions as synthesis of Compound tp016.
(1574) LCMS (ESI) m/z=680 (M+H)+
(1575) Retention time: 3.04 min (Analytical condition SQDFA05 long)
(1576) tp017 and tp018 were not separated.
(1577) Synthesis of Compound tp022

(1578) ##STR00497##

(1579) In a nitrogen atmosphere, tp012-g (20.1 g, 25.8 mmol) was dissolved in 1,4-dioxane (86 mL) and water (43 mL), and then potassium carbonate (10.7 g, 78.0 mmol), [1,1'-bis(diphenylphosphino) ferrocene] dichloropalladium (1.06 g, 1.29 mmol), and vinylboronic acid pinacol ester (5.97 g, 38.8 mmol) were added. The reaction solution was heated to 90° C. and stirred for 4 hours. The reaction solution was cooled to room temperature and left to stand still overnight, then Fmoc-OSu (4.36 g, 12.9 mmol) was added, and the mixture was stirred at room temperature for 1 hour. Ethyl acetate (200 mL) and half brine (200 mL) were added to the reaction solution and subjected to Celite filtration to isolate the organic phase. The organic phase was washed with half brine and dried over magnesium sulfate, and the resulting solution was concentrated under reduced pressure. The resulting crude product was purified by silica gel chromatography (n-hexane-ethyl acetate) to give a crude product of tp022-a (17.43 g). LCMS (ESI) m/z=678 (M+H)+

(1580) Retention time: 0.97 min (Analytical condition SMDF50)

(1581) In a nitrogen atmosphere, the above tp022-a (17.43 g) was dissolved in ethyl acetate (129 mL), then hexamethyldisilazane (13.5 mL, 64.3 mmol) was added, and trimethylsilyl trifluoromethanesulfonate (9.29 mL, 51.4 mmol) was added dropwise over 3 minutes. After the resulting reaction mixture was stirred for 1 hour, a 5% aqueous disodium hydrogen phosphate solution (170 mL) was added to the reaction solution under ice cooling to quench the reaction. After the reaction solution was warmed to room temperature, water (85 mL) was added, and phosphoric acid was added to adjust the pH to 3. The organic phase was separated, and the solvent was distilled off under reduced pressure. The resulting crude product was purified by reverse phase chromatography (0.1% formic acid-containing acetonitrile/0.1% formic acid-containing distilled water) to give tp022 (7.60 g, 48%, 2 steps).

(1582) LCMS (ESI) m/z=622 (M+H)+

(1583) Retention time: 0.60 min (Analytical condition SMDF50)

(1584) Synthesis of Compound tp023

(1585) ##STR00498## ##STR00499## ##STR00500##

(1586) In a nitrogen atmosphere, tp023-a (10.1 g, 22.2 mmol) and sarcosine tert-butyl ester hydrochloride (4.11 g, 22.6 mmol) were dissolved in NMP (55 mL), then DIPEA (11.6 mL, 66.5 mmol) and HATU (10.1 g, 26.6 mmol) were added, and the mixture was stirred at room temperature for 1.5 hours. Subsequently, the reaction solution was diluted with toluene (200 mL) and sequentially washed with water (100 mL), a 3.5% aqueous potassium hydrogen carbonate solution (100 mL), and a 1 M aqueous sodium dihydrogen phosphate solution (100 mL), and the organic phase was dried over anhydrous sodium sulfate. The resulting solution was concentrated under reduced pressure to give a crude product of tp023-b. The resulting tp023-b was directly used in the next step.

(1587) LCMS (ESI) m/z=583 (M+H)+

(1588) Retention time: 1.03 min (Analytical condition SQDF50)

(1589) The above crude product of tp023-b (22.2 mmol) was dissolved in toluene (130 mL), DBU (3.34 mL, 22.2 mmol) was added at 0° C. in a nitrogen atmosphere, and the mixture was stirred at () C for 20 minutes. A 1 M aqueous sodium dihydrogen phosphate solution (100 mL) was added to the reaction solution, and the resulting precipitate was collected by filtration. The resulting precipitate was washed with toluene/n-hexane and water, and dried under reduced pressure to give the desired product. At this time, the organic phase was recovered from the filtrate and the washing solution, and extracted with ().sub.2 N hydrochloric acid/acetonitrile (113 mL/50 mL) and water/acetonitrile (80 mL/40 mL). The aqueous phase was recovered, the pH was adjusted to 11 with a 3 N aqueous phosphoric acid solution, and this was extracted with ethyl acetate. The organic phase was dried over anhydrous sodium sulfate and concentrated under reduced pressure. The solids collected by filtration and the extract were combined to give tp023-c (7.56 g, 95%, 2 steps).

(1590) LCMS (ESI) $m/z=361$ (M+H)+

(1591) Retention time: 0.50 min (Analytical condition SQDFA05)

(1592) In a nitrogen atmosphere, N—BOC-2-aminoacetaldehyde (3.34 g, 21.0 mmol) was dissolved in methanol (210 mL) and ice-cooled to 0° C. 10% palladium carbon (1.67 g, 50 wt %) was added thereto, and then a methanol solution (6 mL) of tp023-c (7.56 g, 21.0 mmol) was added. The reaction solution was stirred at 0° C. for 2.5 hours at 1 atm of a hydrogen atmosphere, then heated to room temperature, and stirred for 1.5 hours. The reaction solution was subjected to Celite filtration, and the Celite was washed with ethyl acetate, followed by concentration under reduced pressure. The resulting crude product was purified by reverse phase chromatography (10 mM aqueous ammonium acetate solution-methanol) to give tp0).sub.23-d (7.03 g, 67%).

(1593) LCMS (ESI) $m/z=505$ (M+H)+

(1594) Retention time: 0.64 min (Analytical condition SQDFA05)

(1595) In a nitrogen atmosphere, Fmoc-MeAsp (OAl)—OH (Compound 358-c, 4.27 g, 10.4 mmol) was dissolved in DCM (26 ml) at room temperature, DMF (0).081 ml, 1.04 mmol) was added, and then thionyl chloride (2.08 ml, 28.7 mmol) was added dropwise. The resulting reaction mixture was stirred for 30 minutes, and then concentrated under reduced pressure. Toluene (30 ml) was added to the resulting residue, and the mixture was concentrated under reduced pressure. This operation was carried out twice in total, and the resulting crude product of Fmoc-MeAsp (OAl)—C1 was dissolved in DCM (10 mL) and used in the subsequent reaction.

(1596) In a nitrogen atmosphere, a DCM (30 mL) solution of tp023-d (3.07 g, 6.10 mmol) and DIPEA (6.05 mL, 34.7 mmol) was added to the prepared DCM (10 ml) solution of Fmoc-MeAsp (OAl)—C1. After the reaction solution was stirred at room temperature for 3 hours, a 3.5% aqueous potassium hydrogen carbonate solution (30 mL) was added. The organic phase was recovered and concentrated under reduced pressure. The resulting crude product was purified by reverse phase chromatography (0).1% formic acid-containing acetonitrile/0.1% formic acid-containing distilled water) to give tp023-e (4.72 g, 87%).

(1597) LCMS (ESI) $m/z=896$ (M+H)+

(1598) Retention time: 1.16 min (Analytical condition SQDFA50)

(1599) In a nitrogen atmosphere, tp023-e (4.72 g, 5.27 mmol) was dissolved in dichloromethane (53 mL), tetrakis(triphenylphosphine) palladium (183 mg, 0.158 mmol) was added, and then phenylsilane (0).454 mL., 3.69 mmol) was slowly added dropwise. The reaction solution was stirred at room temperature for 2.5 hours, and TBME (94 mL) was added. The organic phase was washed with 50% aqueous sodium hydrogen carbonate solution/acetonitrile (47 ml./9.4 mL.), and the organic phase was concentrated under reduced pressure. The resulting crude product was purified by reverse phase chromatography (0).1% formic acid-containing acetonitrile/0.1% formic acid-containing distilled water) to give tp023-f (3.99 g, 88%).

(1600) LCMS (ESI) $m/z=856$ (M+H)+

(1601) Retention time: 1.04 min (Analytical condition SQDFA05)

(1602) tp023-f (1.96 g, 2.29 mmol) was dissolved in 1,4-dioxane (40 mL), a 4 N hydrochloric acid/1,4-dioxane solution (20 mL) was added, and the mixture was stirred at room temperature for 4.5 hours. The reaction solution was concentrated under reduced pressure to give a crude product of tp023-g. The resulting crude product was directly used in the next step.

(1603) HATU (2.66 g, 7.00 mmol) was dissolved in DMF (75 mL), and a DMF solution (100 mL) of the above crude product of tp023-g and DIPEA (1.46 mL, 8.40 mmol) was added dropwise over 100 minutes. Thereafter, a saturated aqueous ammonium chloride solution (200 mL) was added to the reaction solution, and the resulting mixed solution was extracted twice with ethyl acetate (100 mL). The organic phases were combined and washed twice with water (100 mL) and once with brine (100 mL). The resulting organic phase was dried over anhydrous sodium sulfate and concentrated under reduced pressure. The resulting crude product was purified by reverse phase chromatography (0.1% formic acid-containing acetonitrile/0.1% formic acid-containing distilled

water) to give tp023-h (0.56 g, 33%, 2 steps).

(1604) LCMS (ESI) $m/z=738$ (M+H)+

(1605) Retention time: 1.04 min (Analytical condition SQDFA50)

(1606) tp023-h (0.56 g, 0.760 mmol) was dissolved in 2,2,2-trifluoroethanol (10 mL), chlorotrimethylsilane (0.288 mL, 2.28 mmol) was added, and the mixture was stirred at room temperature for 1 hour. Chlorotrimethylsilane (0.15 mL, 1.19 mmol) was added to the reaction solution, then the mixture was stirred for 2.5 hours, chlorotrimethylsilane (0.05 mL, 0.396 mmol) was added, and the mixture was further stirred for 1.5 hours. The reaction solution was concentrated under reduced pressure, and the resulting crude product was purified by reverse phase chromatography (0.1% formic acid-containing acetonitrile/0.1% formic acid-containing distilled water) to give tp023 (0.28 g, 54%).

(1607) LCMS (ESI) $m/z=682$ (M+H)+

(1608) Retention time: 0.80 min (Analytical condition SQDFA50)

(1609) Synthesis of Compound tp024

(1610) ##STR00501## ##STR00502##

(1611) tp024-a (15.0 g, 32.2 mmol) was dissolved in dichloromethane (214 mL), then anhydrous magnesium sulfate (9.68 g, 80.0 mmol), paraformaldehyde (2.90 g, 96.0 mmol), and a boron trifluoride diethyl ether complex (4.08 mL, 32.2 mmol) were added, and the mixture was stirred at room temperature for 3 hours. The reaction solution was filtered through silica gel, the silica gel was washed with dichloromethane (200 mL), and the filtrate was concentrated under reduced pressure. The resulting crude product of tp024-b was directly used in the next reaction.

(1612) Retention time: 1.472 min (Analytical condition SMD method_04)

(1613) In a nitrogen atmosphere, the above crude product of tp0).sub.24-b (32.2 mmol) was dissolved in dichloromethane (322 mL.), then allyltrimethylsilane (35.8 mL., 225 mmol) and a boron trifluoride diethyl ether complex (20.2 mL., 161 mmol) were added, and the mixture was stirred at room temperature for 89 hours. The reaction solution was cooled to 0° C., and water (25 mL.) was added. The mixed solution was subjected to Celite filtration, and the Celite was washed with dichloromethane (100 mL). The resulting filtrate was washed with a 1 N aqueous dipotassium hydrogen phosphate solution (200 mL.) and half brine (100 mL.), dried over sodium sulfate, and concentrated under reduced pressure to give tp024-c (16.4 g, 98%, 2 steps). LCMS (ESI) $m/z=520$ (M+H)+

(1614) Retention time: 1.563 min (Analytical condition SMD method_0) 4)

(1615) In a nitrogen atmosphere, tp024-c (16.4 g, 31.5 mmol) and sarcosine tert-butyl ester hydrochloride (5.72 g, 31.5 mmol) were dissolved in dichloromethane (100 mL), then DIPEA (16.5 mL, 95.0 mmol) and HATU (14.4 g, 37.8 mmol) were added, and the mixture was stirred at room temperature for 1 hour. Subsequently, the reaction solution was sequentially washed with water (100 mL.), a 3.5% aqueous potassium hydrogen carbonate solution (100 mL), and a 1 M aqueous dipotassium hydrogen phosphate solution (100 mL.), and the organic phase was dried over anhydrous sodium sulfate. The resulting solution was concentrated under reduced pressure to give a crude product of tp024-d. The resulting crude product was purified by silica gel chromatography (ethyl acetate-n-hexane) to give tp024-d (13.1 g, 64%).

(1616) LCMS (ESI) $m/z=647$ (M+Na)+

(1617) Retention time: 1.680 min (Analytical condition SMD method_0) 4) tp024-d (13.1 g, 20.2 mmol) was dissolved in toluene (135 mL), DBU (3.02 mL, 20.2 mmol) was added, and the mixture was stirred at room temperature for 20 minutes. The reaction solution was washed with a 1 M aqueous potassium dihydrogen phosphate solution (80 mL.), then n-hexane (140 mL.), water (140 mL.), acetonitrile (70 mL.), and 2 N hydrochloric acid (12 mL.) were added to the organic phase, and the mixture was shaken. The aqueous phase was recovered, and the organic phase was extracted with water/acetonitrile/2 N hydrochloric acid (30 mL./15 mL./3 mL.). The resulting aqueous phases were combined, and 3 N phosphoric acid (12 mL.) was added to adjust the pH to 8. The resulting

mixed solution was extracted twice with ethyl acetate (100 mL, 50 mL.), and the organic phase was washed with brine (50 ml). The organic phase was dried over anhydrous sodium sulfate and concentrated under reduced pressure to give tp024-e (8.20 g, 95%).

(1618) LCMS (ESI) $m/z=425$ (M+H)+

(1619) Retention time: 0.849 min (Analytical condition SMD method_04)

(1620) In a nitrogen atmosphere, (S)-2-((((9H-fluoren-9-yl)methoxy) carbonyl)(methyl)amino) pent-4-enoic acid (7.45 g, 21.2 mmol) was dissolved in DCM (85 ml) at room temperature, DMF (0.164 ml, 2.12 mmol) was added, and then thionyl chloride (3.85 ml, 53.0 mmol) was added dropwise. The resulting reaction mixture was stirred for 20 minutes and then concentrated under reduced pressure. Toluene (20) ml) was added to the resulting residue, and the mixture was concentrated under reduced pressure. This operation was carried out twice in total, and the resulting crude product was dissolved in DCM (23 mL) and used in the subsequent reaction.

(1621) In a nitrogen atmosphere, a DCM (72 mL) solution of tp024-e (8.20 g, 19.3 mmol) and DIPEA (11.8 mL, 67.5 mmol) was added to the prepared DCM (23 ml) solution of (S)-(9H-fluoren-9-yl)methyl(1-chloro-1-oxopent-4-en-2-yl)(methyl) carbamate. The reaction solution was stirred at room temperature for 2 hours, and then washed with a 1 M aqueous potassium dihydrogen phosphate solution (82 mL). Subsequently, the organic phase was washed with a 3.5% aqueous potassium hydrogen carbonate solution (82 mL), and concentrated under reduced pressure. The resulting crude product was purified by silica gel chromatography (n-hexane-ethyl acetate) to give a crude product of tp024-f (14.0 g, 96%).

(1622) LCMS (ESI) $m/z=758$ (M+H)+

(1623) Retention time: 1.737 min (Analytical condition SMD method_0) 4)

(1624) In a nitrogen atmosphere, tp024-f (1.00 g, 1.32 mmol) and 1,4-benzoquinone were dissolved in 1,2-dichloroethane (40 mL) in a three-necked flask, and the reaction solution was heated to 80° C. A 1,2-dichloromethane solution (4 mL) of the first-generation Grubbs catalyst (55 mg, 0.066 mmol) was added dropwise over 1 hour to the reaction solution stirred at 80° C. Subsequently, the reaction solution was stirred at 80° C. for one more hour, cooled to room temperature, and then concentrated under reduced pressure. The resulting crude product was purified by silica gel chromatography (n-hexane-ethyl acetate) to give tp024-g (0.65 g, 68%).

LCMS (ESI) $m/z=730$ (M+H)+

(1625) Retention time: 1.002 min (Analytical condition SMD method_06)

(1626) In a nitrogen atmosphere, tp024-g (1.80 g, 2.46 mmol) was dissolved in ethyl acetate (20) mL), hexamethyldisilazane (1.29 mL, 6.16 mmol) was added, and then trimethylsilyl trifluoromethanesulfonate (0.89 mL, 4.93 mmol) was added dropwise. After the resulting reaction mixture was stirred for 150 minutes, the reaction solution was diluted with n-hexane (50) mL), and extracted once with a 3.5% aqueous potassium hydrogen carbonate solution/acetonitrile (2/1, 40 mL). The pH of the aqueous phase was adjusted to 3 with phosphoric acid, and this was extracted with ethyl acetate (30 mL). The resulting organic phase was dried over sodium sulfate, and the solvent was distilled off under reduced pressure. The resulting crude product was purified by reverse phase chromatography (0.1% formic acid-containing acetonitrile/0.1% formic acid-containing distilled water) to give tp024 (0.92 g, 55%).

(1627) LCMS (ESI) $m/z=674$ (M+H)+

(1628) Retention time: 0.625 min (Analytical condition SMD method_06)

(1629) Synthesis of Compound PP0464

(1630) Using Compound aa375-resin (0.426 mmol/g, 100 mg) as a raw material, Fmoc-cVal-OH, Fmoc-Pro (4-F2)—OH, Fmoc-Hph(4—CF3-35-F2)—OH, Compound tp002, Fmoc-MeGly-OH, Fmoc-Ile-OH, and Fmoc-MeLeu-OH were used. Peptide elongation reaction by the basic Fmoc method described in the present Examples, cleaving of an elongated peptide from resin, cyclization of a cleaved peptide (using (1-cyano-2-ethoxy-2-oxoethylideneaminoxy)dimethylamino-morpholino-carbenium hexafluorophosphate (COMU) as a cyclization reagent), and purification of

a cyclic peptide were performed to give the intended Compound PP0464 (17.2 mg, 26%).

(1631) Compound PP0464

(1632) ##STR00503##

(1633) LCMS (ESI) m/z =1568.2 (M-H)-

(1634) Retention time: 7.979 min (Analytical condition SSC-AA-02/03)

(1635) Using the resins, on which amino acid shown in Table 5 and Table 6 was supported, as raw materials, the following compounds were produced according to the above-described Fmoc method in the same manner as the production method of Compound PP0464. A deprotection reaction was added as necessary.

(1636) Compounds PP0464, PP0465, PP0466, PP0467, PP0468, PP0469, PP0470, PP0471, PP0472, PP0473, PP0474, PP0475, PP0476, PP0477, PP0478, PP0479, PP0480, PP0481, PP0483, PP0484, PP0485, PP0487, PP0488, PP0490, PP0491, PP0492, PP0493, PP0494, PP0495, PP0496, PP0497, PP0498, PP0499, PP0500, PP0501, PP0502, PP0503, PP0504, PP0505, PP0506, PP0507, PP0508, PP0509, PP0510, PP0511, PP0512, PP0513, PP0514, PP0515, PP0516, PP0520, PP0521, PP0522, PP0523, PP0538, PP0539, PP0540, PP0541, PP0542, PP0543, PP0544, PP0545, PP0546, PP0547, PP0548, PP0549, PP0550, PP0551, PP0552, PP0553, PP0554, PP0555, PP0556, PP0557, PP0558, PP0559, PP0560, PP0561, PP0562, PP0563, PP0564, PP0565, PP0566, PP0567, PP0568, PP0569, PP0570, PP0571, PP0572, PP0573, PP0574, PP0575, PP0576, PP0577, PP0578, PP0579, PP0580, PP0581, PP0582, PP0583, PP0584, PP0585, PP0586, PP0587, PP0588, PP0589, PP0590, PP0591, PP0594, PP0595, PP0596, PP0597, PP0598, PP0630, PP0631, PP0632, PP0633, PP0634, PP0635, PP0636, PP0637, PP0638, PP0639, PP0640, PP0641, PP0642, PP0643, PP0644, PP0645, PP0646, PP0647, PP0648, PP0649, PP0650, PP0651, PP0652, PP0653, PP0654, PP0655, PP0656, PP0657, PP0658, PP0659, PP0660, PP0661, PP0662, PP0663, PP0664, PP0665, PP0666, PP0667, PP0668, PP0669, PP0670, PP0671, PP0672, PP0673, PP0674, PP0675, PP0678, PP0679, PP0680, PP0681, PP0682, PP0683, PP0684, PP0685, PP0686, PP0687, PP0688, PP0689, PP0690, PP0693, PP0694, PP0695, PP0696, PP0697, PP0698, PP0699, PP0700, PP0701, PP0702, PP0703, PP0704, PP0705, PP0706, PP0708, PP0709, PP0710, PP0711, PP0712, PP0713, PP0714, PP0715, PP0716, PP0717, PP0718, PP0719, PP0720, PP0721, PP0722, PP0723, PP0724, PP0725, PP0726, PP0727, PP0728, PP0729, PP0730, PP0731, PP0732, PP0733, PP0806, PP0807, PP0808, PP0809, PP0810, PP0811, PP0812, PP0813, PP0814, PP0815, PP0816, PP0817, PP0818, PP0819, PP0820, PP0821, PP0822, PP0823, PP0824, PP0825, PP0826, PP0827, PP0828, PP0829, PP0830, PP0831, PP0832, PP0833, PP0834, PP0835, PP0836, PP0837, PP0838, PP0839, PP0840, PP0841, PP0842, PP0843, PP0844, PP0845, PP0846, PP0847, PP0848, PP0849, PP0850, PP0951, PP0852, PP0853, PP0854, PP0855, PP0856, PP0857, PP0858, PP0859, PP0860, PP0861, PP0862, PP0863, PP0864, PP0865, PP0866, PP0867, PP0868, PP0869, PP0870, PP0871, PP0872, PP0873, PP0926, PP0927, PP0928, PP0929, PP0930, PP0931, PP0932, PP0933, PP0934, PP0935, PP0936, PP0937, PP0938, PP0939, PP0940, PP0941, PP0942, PP0943, PP0944, PP0945, PP0946, PP0947, PP0948, PP0949, PP0950, PP0951, PP0952, PP0953, PP0954, PP0955, PP0956, PP0958, PP0959, PP0960, PP0961, PP0962, PP0963, PP0964, PP0965, PP0966, PP0967, PP0968, PP0969, PP0970, PP0971, PP0972, PP0973, PP0975, PP0976, PP0977, PP0978, PP0979, PP0980, PP0981, PP0982, PP0983, PP0984, PP1017, PP1018, PP1019, PP1020, PP1021, PP1022, PP1024, PP1025, PP1026, PP1027, PP1028, PP1029, PP1030, PP1031, PP1032, PP1033, PP1034, PP1035, PP1036, PP1037, PP1038, PP1039, PP1040, PP1041, PP1042, PP1043, PP1044, PP1045, PP1046, PP1047, PP1048, PP1049, PP1050, PP1051, PP1052, PP1053, PP1054, PP1056, PP1057, PP1058, PP1059, PP1060, PP1063, PP1064, PP1065, PP1086, PP1087, PP1088, PP1089, PP1090, PP1091, PP1092, PP1093, PP1094, PP1095, PP1096, PP1097, PP1098, PP1099, PP1100, PP1101, PP1102, PP1103, PP1104, PP1105, PP1106, PP1107, PP1108, PP1109, PP1110, PP111, PP1112, PP1113, PP1114, PP1115, PP1116, PP1117, PP1118, PP1119, PP1120, PP1121, PP1122, PP1123, PP1124, PP1125, PP1126, PP1127, PP1128, PP1129, PP1130, PP1131, PP1132, PP1133, PP1134, PP1135, PP1136, PP1137, PP1138, PP1139, PP1140, PP1141, PP1142, PP1143, PP1144, PP1145, PP1146, PP1147, PP1148, PP1149, PP1150, PP1155,

[illegible]

PP1821, PP1822, PP1823, PP1825, PP1826, PP1836, PP1837, PP1838, PP1839, PP1840, PP1841, PP1842, PP1844, PP1846, PP1848, PP1849, PP1850, PP1851, PP1852, PP1853, PP1854, PP1855, PP1856, PP1857, PP1859, PP1860, PP1861, PP1862. PP1863, PP1864, PP1865, PP1866, PP1867, PP1868, PP1869, PP1870, PP1871, PP187Z PP1873, PP1874, PP1875, PP1876, PP1877, PP1878, PP1879, PP1880, PP1881, PP188Z PP1883, PP1884, PP1885, PP1886, PP1887, PP1888, PP1889, PP1890, PP1891, PP192, PP1893, PP1894, PP1895, PP1896, PP1897, PP1898, PP1899, PP1900, PP1901, PP1902, PP1903, PP1904, PP1905, PP1906, PP1907, PP1908, PP1909, PP1910, PP1911, PP1913, PP1915, PP1917, PP1919, PP1921, PP1923, PP1925, PP1927, PP1929, PP1931, PP1932, PP1934, PP1935, PP1936, PP1937, PP1938, PP1941, PP1942, PP1943, PP1944, PP1945, PP197Z PP1973, PP1974, PP1975, PP1976, PP1977, PP1979, PP1980, PP1981, PP1983, PP1984, PP1985, PP1987, PP1988, PP1989, PP1990, PP1991, PP1992, PP1993, PP1994, PP1995, PP1996, PP1997, PP1998, PP1999, PP2000, PP2001, PP2002, PP2003, PP2004, PP2005, PP2006, PP2007, PP2008, PP2009, PP2010, PP2011, PP2012, PP2013, PP2014, PP2015, PP2016, PP2017, PP2018, PP2019, PP2020, PP2021, PP22, PP2023, PP2024, PP2025, PP2026, PP2027, PP2028, PP2029, PP2030, PP2031, PP32, PP2033, PP2034, PP2036, PP2037, PP2038, PP2040, PP2041, PP2042, PP2043, PP2044, PP2045, PP2046, PP2047, PP2048, PP2049, PP2050, PP2051, PP2052, PP2053, PP2054, PP2055, PP2056, PP2057, PP2058, PP2059, PP2060, PP2061, PP2062, PP2063, PP2064, PP2065, PP2066, PP2067, PP2068, PP2069, PP2070, PP2071, PP2072, PP2073, PP2074, PP2075, PP2076, PP2077, PP2078, PP2079, PP2080, PP2081, PP202, PP2083, PP2087, PP2091, PP2093, PP2094, PP2095, PP2096, PP2097, PP2098, PP2099, PP2101, PP2102, PP2103, PP2105, PP2106, PP2107, PP2108, PP2109, P2119, PP2120, PP2121, P2122, PP2123, PP2124, PP2125, PP2126, PP2127, PP2128, PP2130, PP2131, 2132, PP2133, PP2135, PP2137, PP2138, PP2139, PP2140, PP2141, P2142, PP2143, PP2144, PP2145, PP2146, PP2147, PP2148, PP2149, PP2150, PP2151, P2152, PP2153, PP2154, PP2155, PP2156, PP2157, PP2158, PP2159, PP2160, PP2161, PP2163, PP2164, PP2165, PP2166, PP2167, PP2168, PP2169, PP2170, PP2171, PP217Z PP2173, PP2174, PP2175, PP2176, PP2178, PP2179, PP2180, PP2181, PP2182, PP2183, PP2184, PP2185, PP2186, PP2187, PP2188, PP2189, PP2190, PP2191, 2192, PP2193, PP2195, PP2196, PP2197, PP2198, PP2199, PP2200, PP2202, PP2203, PP2207, PP2208, PP2209, PP2210, PP2212, PP2214, PP2216, PP2218, PP2219, PP2220, PP2221, PP22Z PP2223, PP2224, PP2225, PP2226, PP2227, PP2229, PP2230, PP2231, PP2232, PP2233, PP2234, PP2235, PP2237, PP2238, PP2242, PP2257, PP2268, PP2269, PP2270, PP2271, PP2272, PP2273, PP2275, PP2385, PP2386, PP2387, PP2388, PP2389, PP2390, PP2391, PP2392, PP2393, PP2394, PP2395, PP2396, PP2397, PP2398, PP2399, PP2400, PP2401, PP2402, PP2403, PP2404, PP2405, PP2406, PP2407, PP2408, PP2409, PP2410, PP2411, PP2412, PP2413, PP2414, PP2415, PP2416, PP2417, PP2418, PP2419, PP2422, PP2423, PP2424, PP2425, PP2426, PP2427, PP2428, PP2429, PP2430, PP2431, PP2432, PP2433, PP2436, PP2437, PP2438, PP2439, PP2440, PP2441, PP2442 PP2443, PP2444, PP2445, PP2446, PP2447, PP2448, PP2449, PP2450, PP2451, PP2452, PP2453, PP2454, PP2455, PP2456, PP2457, PP2458, PP2460, PP2461, PP2462, PP2463, PP2464, PP2465, PP2466, PP2467, PP2468, PP2469, PP2470, PP2471, PP2472, PP2475, PP2477, PP2479, PP2480, PP2481, PP2482, PP2483, PP2484, PP2485, PP2488, PP2489, PP2490, PP2492, PP2494, PP2495, PP2496, PP2497, PP2498, PP2499, PP2500, PP2501, PP2502, PP2504, PP2505, PP2506, PP2507, PP2508, PP2509, PP2510, PP2511, PP2512, PP2513, PP2514, PP2515, PP2516, PP2517, PP2518, PP2519, PP2520, PP2521, PP2522, PP2523, PP2524, PP2525, PP2526, PP2527, PP2528, PP2529, PP2530, PP2531, PP2532, PP2533, PP2534, PP2535, PP2536, PP2537, PP2539, PP2540, PP2541, PP2542, PP2543, PP2545, PP2546, PP2547, PP2548, PP2549, PP2550, PP2551, PP2552, PP2553, PP2554, PP2555, PP2556, PP2557, PP2558, PP2559, PP2560, PP2561, PP2562, PP2563, PP2564, PP2565, PP2566, PP2567, PP2568, PP2569, PP2570, PP2571, PP2572, PP2606, PP2608, PP2610, PP2612, PP2614, PP2615, PP2616, PP2618, PP2619, PP2622, PP2624, PP2626, PP2627, PP2628, PP2630, PP2631, PP2633, PP2634, PP2635, PP2637, PP2638, PP2639, PP2640, PP2641, PP2642, PP2643, PP2644, PP2645, PP2646, PP2648, PP2649, PP2650, PP2651, PP2652, PP2654, PP2655, PP2656, PP2657, PP2659, PP2660, PP2661, PP2662, PP2663,

PP2664, PP2665, PP2666, PP2667, PP2668, PP2669, PP2670, PP2671, PP2706, PP2708, PP2710, PP2712, PP2714, PP2716, PP2718, PP2720, PP2722, PP2724, PP2726, PP2728, PP2730, PP2732, PP2734, PP2736, PP2738, PP2740, PP2742, PP2744, PP2746, PP2748, PP2749, PP2750, PP2751, PP2752, PP2753, PP2754, PP2755, PP2756, PP2757, PP2760, PP2761, PP2762, PP2763, PP2764, PP2765, PP2766, PP2767, PP2768, PP2769, PP2770, PP2771, PP2772, PP2773, PP2776, PP2777, PP2778, PP2779, PP2780, PP2781, PP2782, PP2783, PP2784, PP2785, PP2786, PP2787, PP2788, PP2789, PP2790, PP2791, PP2796, PP2797, PP2798, PP2799, PP2800, PP2801, PP2802, PP2803, PP2804, PP2805, PP2806, PP2807, PP2808, PP2809, PP2811, PP2813, PP2815, PP2817, PP2819, PP2821, PP2822, PP2823, PP2824, PP2825, PP2826, PP2827, PP2828, PP2829, PP2830, PP2831, PP2832, PP2833, PP2834, PP2835, PP2836, PP2837, PP2838, PP2839, PP2840, PP2841, PP2842, PP2843, PP2844, PP2845, PP2846, PP2847, PP2848, PP2888, PP2889, PP2890, PP2891, PP2892, PP2893, PP2894, PP2895, PP2896, PP2897, PP2898, PP2899, PP2900, PP2901, PP2902, PP2904, PP2905, PP2906, PP2907, PP2908, PP2909, PP2910, PP2911, PP2912, PP2913, PP2914, PP2915, PP2916, PP2917, PP2918, PP2920, PP2921, PP2922, PP2923, PP2928, PP2929, PP2931, PP2932, PP2934, PP2935, PP2936, PP2937, PP2938, PP2941, PP2946, PP2947, PP2948, PP2949, PP2950, PP2951, PP2960, PP2961, PP2962, PP2963, PP2964, PP2965, PP2966, PP2967, PP2968, PP2969, PP2970, PP2971, PP2972, PP2973, PP2974, PP2975, PP2976, PP2977, PP2978, PP2979, PP2980, PP2961, PP2962, PP2983, PP2984, PP2985, PP2986, PP2987, PP2988, PP2989, PP2990, PP2991, PP2992, PP2993, PP2994, PP2995, PP2996, PP2997, PP2998, PP2999, PP3000, PP3001, PP3002, PP3003, PP3004, PP3005, PP3006, PP3007, PP3008, PP3009, PP3010, PP3011, PP3012, PP3013, PP3014, PP3015, PP3017, PP3018, PP3019, PP3021, PP3022, PP3026, PP3030, and PP3031.

(1637) The following compounds were produced according to the above-described Fmoc method in the same manner as the production method of Compound PP0464 using aa427-resin to aa438-resin as raw materials:

(1638) Compounds PP2312, PP2313, PP2314, PP2315, PP2316, PP2317, PP2318, PP2319, PP2320, PP2322, PP2323, PP2325, PP2326, PP2327, PP2328, PP2329, PP2330, PP2331, PP2332, PP2333, PP2334, PP2335, PP2336, PP2337, PP2338, PP2339, PP2340, PP2341, PP2342, PP2344, PP2345, PP2347, PP2348, PP2349, PP2350, PP2351, PP2352, PP2353, PP2354, PP2355, PP2356, PP2357, PP2358, PP2359, PP2360, PP2361, PP2362, PP2363, PP2364, PP2365, PP2366, PP2367, PP2368, PP2369, PP2370, PP2371, PP2372, PP2373, PP2374, PP2375, PP2376, PP2377, PP2378, PP2379, PP2380, PP2381, PP2382, PP2383, PP2810, PP2812, PP2814, PP2816, PP2818, PP2820, PP3095, PP3096, PP3097, PP3098, PP3099, PP3100, PP3101, PP3102, PP3103, PP3104, PP3105, PP3106, PP3110, PP3111, PP3112, PP3113, PP3114, PP3115, PP3116, PP3117, PP3118, PP3119, PP3120, and PP3121.

(1639) In a cyclic compound and an oligopeptide compound synthesized with a tripeptide, abbreviations and structures of amino acids contained as partial structures are provided below.

(1640) TABLE-US-00010 TABLE 9-1 Compound Amino acid No. Abbreviation Structural Formula
bb001 Phe(4-CF₃) 04  embedded image
bb002 AllylPhe(4-CF₃) 05  embedded image
bb003 Phe(4-F) 06  embedded image
bb004 AllylPhe(4-F) 07  embedded image
bb005 ButenylPhe(4-CF₃) 08  embedded image
bb006 Cha 09  embedded image
bb007 ButenylCha 0  embedded image
bb008 Cha(4-F₂)  embedded image
bb009 ButenylCha(4-F₂)  embedded image
bb010 ButenylAsp(OMe)  embedded image
bb011 ButenylPhe(4-Me)  embedded image
bb012 ButenylPhe(4-F)  embedded image
bb013 ButenylPhe(4-F-2-Me)  embedded image
bb014 ButenylPhe(4-I)  embedded image
bb015 MethaPhe(4-CF₃)  embedded image

1-5. Cyclic Peptide Synthesis Using Tfa-Protected Amino Acid as Reagent

(1641) According to the general reaction scheme provided below, PP734 to PP0744, PP754 to PP0765, PP771 to PP0776, PP778 to PP0782, PP790 to PP0801, PP0805, and PP0874 to PP0881 were synthesized (• indicates an amino acid unit).

(1642) ##STR00519## ##STR00520##

(1643) Peptide elongation was performed according to the basic route up to the N-terminal before a Tfa-protected amino acid was elongated. Next, Tfa-protected amino acid was elongated through 3 steps of 1) Fmoc deprotection of peptide chain N-terminal, 2) Condensation reaction using Tfa-protected amino acid, and 3) Tfa deprotection. Subsequent peptide elongation, cleaving from resin, and synthesis of a cyclized peptide were performed according to the basic route.

(1644) Synthesis of Compound PP754

(1645) 1) Fmoc Deprotection of Peptide Chain N-Terminal

(1646) Compound aa375-resin (0.276 mmol/g, 1 g) was placed in a filter-equipped reaction vessel, dichloromethane (10 mL) was added, and the mixture was shaken at room temperature for 5 minutes to swell the resin. After dichloromethane was removed with the filter, the resin was washed twice with DMF (7 mL). Subsequently, a 2% DBU/DMF solution (de-Fmoc solution: 7 mL) was added to the resin, and the mixture was shaken at room temperature for 10 minutes to carry out a de-Fmoc reaction. After the de-Fmoc solution was removed, the resin was washed 4 times with DMF (7 mL).

(1647) 2) Condensation Reaction Using Tfa-Protected Amino Acid

(1648) An elongation reaction of Tfa-(Me) Abu-OH (Compound aa317) was performed on the resulting resin. The elongation reaction was performed by adding a mixed solution of a 0.6 M Tfa-(Me) Abu-OH (Compound aa317)/DMF solution (3 mL) and a 10% DIC/DMF solution (3.6 mL) to the resin and shaking the mixture at 60° C. for 24 hours. After the liquid phase of the elongation reaction was removed with a filter, the resin was washed 4 times with DMF (7 mL) and 4 times with dichloromethane (7 mL) and dried to give Compound PP0754-a (Tfa-(Me) Abu-MeGly (cPent)-MeAsp (O-Trt (2-C1)-resin)-mor).

(1649) 3) Tfa Deprotection

(1650) NaBH₄ (0.5 g, 13.2 mmol) was placed in a flask, pumped up, then dissolved in trigrim (6.6 mL) under nitrogen atmosphere to give Solution A. The resulting resin PP754-a (100 mg) was swollen with dichloromethane (1 mL) and then washed twice with tetrahydrofuran (0.7 mL). Tetrahydrofuran (0.5 mL), methanol (0.25 mL), and Solution A (0.25 mL) were added, and the mixture was left to stand still in an open system at room temperature for 2 hours. After the reaction solution was removed, tetrahydrofuran (0.5 mL), methanol (0.25 mL), and Solution A (0.25 mL) were added, and the mixture was left to stand still in an open system at room temperature for 30 minutes. After the reaction solution was removed, the operation of adding methanol (0.7 mL) and washing the mixture for 1 minute was repeated 4 times, and the mixture was further washed with dichloromethane (0.7 mL) 4 times. The amino acid was cleaved from the resin with TFE/DCM (1/1 (v/v)) containing DIPEA (0.045 mmol/L) using a small amount of resin-supported Compound PP754-b, and the structure was verified by LC/MS.

(1651) LCMS (ESI) m/z=455 (M+H)⁺

(1652) Retention time: 0.39 min (Analytical condition SQDFA05)

(1653) Subsequent peptide chain elongation, cleaving from resin, and synthesis of a cyclized peptide were performed according to the basic peptide synthesis method described in the present Examples. LC/MS data is provided in Table 36.

(1654) Using 0.374-resin and aa375-resin as raw materials, PP0734-a to PP0744-a, PP0754-a to PP0765-a, PP0771-a to PP0776-a, PP0778-a to PP0782-a, PP0790-a to PP0801-a, PP0805-a, and PP0874-a to PP0881-a were obtained in the same manner as synthesis of Compound PP0754-a using the Tfa-protected amino acids listed in Table 10 in place of Tfa-(Me)Abu-OH. Then, PP0734-b to PP0744-b, PP0754-b to PP0765-b, PP0771-b to PP0776-b, PP0778-b to PP0782-b, PP0790-b to PP0801-b, PP0805-b, and PP0874-b to PP0881-b were obtained in the same manner as synthesis of PP0754-b. Subsequent peptide chain elongation, cleaving from resin, and synthesis of a cyclized peptide were performed according to the basic peptide synthesis method described in the present Examples to give Compounds PP0734 to PP0744, PP0754 to PP0765, PP0771 to PP0777, PP0778 to PP0782, PP0790 to PP0801, PP0805, and PP0874 to PP0881. LC/MS data is provided in Table 36.

(1655) Synthesis of Tfa-Protected Amino Acids

(1656) The amino acids listed in Table 10 were synthesized by the following method and used in a peptide elongation reaction.

(1657) TABLE-US-00011 TABLE 10 Compound No. Abbreviation Structural Formula Name
aa309 Tfa-D-(Me)Algly-OH  (R)-2-methyl-2-(2,2,2-trifluoroacetamido)pent-4-enoic acid
aa317 Tfa-(Me)Abu-OH  (S)-2-methyl-2-(2,2,2-trifluoroacetamido)butanoic acid
aa319 Tfa-AoxeC-OH  3-(2,2,2-trifluoroacetamido)oxetane-3-carboxylic acid
aa320 Tfa-(Me)Ser(tBu)-OH  (S)-3-(tert-butoxy)-2-methyl-2-(2,2,2-trifluoroacetamido)propanoic acid
aa321 Tfa-(Me)Ser(Al)-OH  (S)-3-(allyloxy)-2-methyl-2-(2,2,2-trifluoroacetamido)propanoic acid
aa322 Tfa-(Me)Ser(Me)-OH  (S)-3-methoxy-2-methyl-2-(2,2,2-trifluoroacetamido)propanoic acid
aa323 Tfa-(Me)Phe-OH  (S)-2-methyl-3-phenyl-2-(2,2,2-trifluoroacetamido)propanoic acid
aa324 Tfa-(Me)Cha-OH  (S)-3-cyclohexyl-2-methyl-2-(2,2,2-trifluoroacetamido)propanoic acid
aa325 Tfa-(Me)Leu-OH  (S)-2,4-dimethyl-2-(2,2,2-trifluoroacetamido)pentanoic acid
aa326 Tfa-(Me)Nva-OH  (S)-2-methyl-2-(2,2,2-trifluoroacetamido)pentanoic acid
aa327 Tfa-(Me)Ile-OH  (2S,3S)-2,3-dimethyl-2-(2,2,2-trifluoroacetamido)pentanoic acid
aa328 Tfa-(Me)Val-OH  (S)-2,3-dimethyl-2-(2,2,2-trifluoroacetamido)butanoic acid
aa329 Tfa-(Me)Gly(cPr)-OH  (S)-2-cyclopropyl-2-(2,2,2-trifluoroacetamido)propanoic acid

Synthesis of Compound aa309

(1658) ##STR00534##

(1659) 4-(3-Phenylpropyl) piperidine (9.0 mL, 42.7 mmol) was added to a dichloromethane (47.4 mL) solution of Compound aa309-a ((R)-2-(((9H-fluoren-9-yl)methoxy) carbonyl)amino)-2-methylpent-4-enoic acid, Fmoc-D-(Me) Algly-OH)(5.0 g, 14.2 mmol), and the mixture was stirred at room temperature for 36 hours in a nitrogen atmosphere. Water (10 mL) and 2 N hydrochloric acid (5 mL) were added to the reaction solution to extract the product, and the aqueous layer was purified by reverse phase column chromatography (0.1% formic acid-water/0.1% formic acid-acetonitrile) to give Compound aa309-b ((R)-2-amino-2-methylpent-4-enoic acid, H-D-(Me) Algly-OH)(1.9 g, quant.), which was used in the next reaction.

(1660) LCMS (ESI) $m/z=130$ (M+H)+

(1661) Retention time: 0.15 min (Analytical condition SQDFA05)(retention time of MS peak is provided)

(1662) After N,N-diisopropylethylamine (7.7 mL, 44.1 mmol) and ethyl 2,2,2-trifluoroacetate (5.3 mL, 44.1 mmol) were added to a methanol (24.5 mL) solution of Compound aa309-b ((R)-2-amino-2-methylpent-4-enoic acid, H-D-(Me) Algly-OH)(1.9 g, 14.7 mmol), and the mixture was stirred at 50° C. for 24 hours. The reaction solution was cooled to room temperature and then concentrated under reduced pressure, and the resulting residue was dissolved in TBME (60 mL) and then washed twice with 1 N hydrochloric acid (60 mL) and once with saturated brine (60 mL). The organic layer was dried over anhydrous sodium sulfate, filtered, and then concentrated under reduced pressure to give a crude product. The resulting crude product was purified by reverse phase column chromatography (0.1% formic acid-water/0.1% formic acid-acetonitrile) to give Compound aa309 (2.5 g, 75%).

(1663) LCMS (ESI) $m/z=224$ (M-H)-

(1664) Retention time: 0.52 min (Analytical condition SQDFA05)

(1665) Synthesis of Compound aa321

(1666) ##STR00535##

(1667) Using Compound aa321-a ((S)-2-(((9H-fluoren-9-yl)methoxy) carbonyl)amino)-3-(allyloxy)-2-methylpropanoic acid, Fmoc-(Me) Ser (Al)—OH)(2.0 g, 5.2 mmol) as a starting material, Compound aa321-b ((S)-3-(allyloxy)-2-amino-2-methylpropanoic acid, H—(Me) Ser (Al)

—OH)(0.87 g, quant.) was obtained in the same manner as synthesis of Compound aa309-b, and was used in the next reaction.

(1668) LCMS (ESI) $m/z=160$ (M+H)+

(1669) Retention time: 0.16 min (Analytical condition SQDFA05)(retention time of MS peak is provided)

(1670) Using Compound aa321-b ((S)-3-(allyloxy)-2-amino-2-methylpropanoic acid, H—(Me) Ser (Al)—OH)(0.84 g, 5.3 mmol), Compound aa321 (1.0 g, 75%) was obtained in the same manner as synthesis of Compound aa309.

(1671) LCMS (ESI) $m/z=256$ (M+H)+

(1672) Retention time: 0.55 min (Analytical condition SQDFA05)

(1673) Synthesis of Compound aa324

(1674) ##STR00536##

(1675) 4-(3-Phenylpropyl) piperidine (4.7 mL, 22.1 mmol) was added to a dichloromethane (18.4 mL) solution of Compound aa324-a ((S)-2-(9H-fluoren-9-ylmethoxycarbonylamino)-3-cyclohexyl-2-methylpropanoic acid, Fmoc-(Me) Cha-OH)(3.0 g, 7.4 mmol), and the mixture was stirred in a nitrogen atmosphere at room temperature for 16 hours. Water (5 mL) and 2 N hydrochloric acid (5 mL) were added to the reaction solution to extract the product, and the aqueous layer was purified by reverse phase column chromatography (0.1% formic acid-water/0.1% formic acid-acetonitrile) to give Compound aa324-b ((S)-2-amino-3-cyclohexyl-2-methylpropanoic acid, H—(Me) Cha-OH)(1.1 g, 81%), which was used in the next reaction.

(1676) LCMS (ESI) $m/z=186$ (M+H)+

(1677) Retention time: 0.32 min (Analytical condition SQDFA05)

(1678) After N,N-diisopropylethylamine (3.1 mL, 18.0 mmol) and ethyl 2,2,2-trifluoroacetate (2.1 mL, 18.0 mmol) were added to a methanol (20.0 mL) solution of Compound aa324-b ((S)-2-amino-3-cyclohexyl-2-methylpropanoic acid, H—(Me) Cha-OH)(1.1 g, 6.0 mmol), and the mixture was stirred at 50° C. for 2 hours. Then, N,N-diisopropylethylamine (3.1 mL, 18.0 mmol) and ethyl 2,2,2-trifluoroacetate (2.1 mL, 18.0 mmol) were added, and the mixture was stirred at 50° C. for 20 hours. The reaction solution was cooled to room temperature and then concentrated under reduced pressure, and the resulting residue was dissolved in TBME (30 mL) and washed twice with 1 N hydrochloric acid (30 mL) and once with saturated brine (30 mL). The organic layer was dried over anhydrous sodium sulfate, filtered, and then concentrated under reduced pressure to give a crude product. The resulting crude product was purified by reverse phase column chromatography (0.1% formic acid-water/0.1% formic acid-acetonitrile) to give Compound aa324 (1.2 g, 72%).

(1679) LCMS (ESI) $m/z=280$ (M-H)–

(1680) Retention time: 0.75 min (Analytical condition SQDFA05)

(1681) Synthesis of Compound aa326

(1682) ##STR00537##

(1683) Using Compound aa326-a ((S)-2-(((9H-fluoren-9-yl)methoxy) carbonyl)amino)-2-methylpentanoic acid, Fmoc-(Me) Nva-OH)(3.0 g, 8.5 mmol) as a starting material, Compound aa326-b ((S)-2-amino-2-methylpentanoic acid, H—(Me) Nva-OH)(1.1 g, 96%) was obtained in the same manner as synthesis of Compound aa309-b, and used in the next reaction.

(1684) LCMS (ESI) $m/z=132$ (M+H)+

(1685) Retention time: 0.13 min (Analytical condition SQDFA05)

(1686) Using Compound aa326-b ((S)-2-amino-2-methylpentanoic acid, H—(Me) Nva-OH)(1.1 g, 8.2 mmol), Compound aa326 (1.2 g, 64%) was obtained in the same manner as synthesis of Compound aa309.

(1687) LCMS (ESI) $m/z=226$ (M-H)–

(1688) Retention time: 0.55 min (Analytical condition SQDFA05)

(1689) Synthesis of Compound aa327

(1690) ##STR00538##

(1691) Using Compound aa327-a ((2S,3S)-2-((((9H-fluoren-9-yl)methoxy) carbonyl)amino)-2,3-dimethylpentanoic acid, Fmoc-(Me) Ile-OH)(3.0 g, 8.2 mmol) as a starting material, Compound aa327-b ((2S,3S)-2-amino-2,3-dimethylpentanoic acid, H—(Me) Ile-OH)(1.1 g, 96%) was obtained in the same manner as synthesis of Compound aa309-b, and used in the next reaction.

(1692) LCMS (ESI) m/z =146 (M+H)+

(1693) Retention time: 0.15 min (Analytical condition SQDFA05)

(1694) Using Compound aa327-b ((2S,3S)-2-amino-2,3-dimethylpentanoic acid, H—(Me) Ile-OH)(1.1 g, 7.9 mmol), Compound aa327 (1.4 g, 72%) was obtained in the same manner as synthesis of Compound aa309.

(1695) LCMS (ESI) m/z =240 (M-H)-

(1696) Retention time: 0.61 min (Analytical condition SQDFA05)

(1697) Synthesis of Compound aa329

(1698) ##STR00539##

(1699) Compound aa329-a ((S)-2-((((9H-fluoren-9-yl)methoxy) carbonyl)amino)-2-cyclopropylpropanoic acid, Fmoc-(Me)Gly (cPr)—OH)(3.0 g, 8.5 mmol) as a starting material, Compound aa329-b ((S)-2-amino-2-cyclopropylpropanoic acid, H—(Me)Gly (cPr)—OH)(1.1 g, 100%) was obtained in the same manner as synthesis of Compound aa309-b, and used in the next reaction.

(1700) LCMS (ESI) m/z =130 (M+H)+

(1701) Retention time: 0.13 min (Analytical condition SQDFA05)

(1702) Using Compound aa329-b ((S)-2-amino-2-cyclopropylpropanoic acid, H—(Me)Gly (cPr)—OH)(1.1 g, 8.5 mmol), Compound aa329 (1.5 g, 76%) was obtained in the same manner as synthesis of Compound aa309.

(1703) LCMS (ESI) m/z =226 (M+H)+

(1704) Retention time: 0.46 min (Analytical condition SQDFA05)

(1705) Synthesis of Compound aa317

(1706) ##STR00540##

(1707) After N,N-diisopropylethylamine (82.7 g, 640 mmol) and ethyl 2,2,2-trifluoroacetate (54.6 g, 384 mmol) were added to a methanol (150 mL) solution of Compound aa317-a ((S)-2-amino-2-methylbutanoic acid, isovaline, H—(Me) Abu-OH)(15.0 g, 128 mmol), the mixture was stirred at 50° C. for 16 hours. The reaction solution was cooled to room temperature and then concentrated under reduced pressure, and the resulting residue was dissolved in TBME and then washed twice with 1 N hydrochloric acid. The organic layer was dried over anhydrous sodium sulfate, filtered, and then concentrated under reduced pressure to give a crude product. The resulting crude product was recrystallized from TBME/hexane (1:7) to give Compound aa317 (12 g, 44%).

(1708) LCMS (ESI) m/z =214 (M+H)+

(1709) Retention time: 0.32 min (Analytical condition SQDFA05)

(1710) Synthesis of Compound aa320

(1711) ##STR00541##

(1712) After N,N-diisopropylethylamine (5.4 mL, 30.8 mmol) and ethyl 2,2,2-trifluoroacetate (3.7 mL, 30.8 mmol) were added to a methanol (17.1 mL) solution of Compound aa320-a ((S)-2-amino-3-(tert-butoxy)-2-methylpropanoic acid, H—(Me) Ser (tBu)-OH)(1.8 g, 10.3 mmol), the mixture was stirred at 50° C. for 5 hours. Then, N,N-diisopropylethylamine (2.7 mL, 15.4 mmol) and ethyl 2,2,2-trifluoroacetate (1.8 mL, 15.4 mmol) were added, and the mixture was stirred at 50° C. for 16 hours. The reaction solution was cooled to room temperature and then concentrated under reduced pressure, and the resulting residue was dissolved in TBME (40 mL) and washed twice with 1 N hydrochloric acid (40 mL) and once with saturated brine (40 mL).

(1713) The organic layer was dried over anhydrous sodium sulfate, filtered, and then concentrated under reduced pressure to give a crude product. The resulting crude product was purified by reverse phase column chromatography (0.1% formic acid-water/0.1% formic acid-acetonitrile) to give

Compound aa320 (2.3 g, 82%).
(1714) LCMS (ESI) m/z =270 (M-H)-
(1715) Retention time: 0.65 min (Analytical condition SQDFA05)
(1716) Synthesis of Compound aa322
(1717) ##STR00542##
(1718) Diisopropylethylamine (5.9 mL, 33.8 mmol) and ethyl 2,2,2-trifluoroacetate (4.0 mL, 33.8 mmol) were added to a methanol (18.8 mL) solution of Compound aa322-a ((S)-2-amino-3-methoxy-2-methylpropanoic acid, H—(Me) Ser (Me)-OH)(1.5 g, 11.3 mmol), and the mixture was stirred at 50° C. for 21 hours. The reaction solution was cooled to room temperature and then concentrated under reduced pressure, and the resulting residue was dissolved in TBME (45 mL) and then washed twice with 1 N hydrochloric acid (45 mL) and once with saturated brine (45 mL). The organic layer was dried over anhydrous sodium sulfate, filtered, and then concentrated under reduced pressure to give a crude product. The resulting crude product was purified by reverse phase column chromatography (0.1% formic acid-water/0.1% formic acid-acetonitrile) to give Compound aa322 (2.1 g, 80%).
(1719) LCMS (ESI) m/z =230 (M+H)+
(1720) Retention time: 0.41 min (Analytical condition SQDFA05)
(1721) Synthesis of Compound aa323
(1722) ##STR00543##
(1723) Using Compound aa323-a ((2S)-2-amino-2-methyl-3-phenylpropanoic acid, H—(Me) Phe-OH)(10.0 g, 55.8 mmol) as a starting material, Compound aa323 (8 g, 52%) was obtained in the same manner as synthesis of Compound aa317.
(1724) LCMS (ESI) m/z =276 (M+H)+
(1725) Retention time: 0.68 min (Analytical condition SQDFA05)
(1726) Synthesis of Compound aa325
(1727) ##STR00544##
(1728) Using Compound aa325-a (2-methylleucine, (S)-2-amino-2,4-dimethylpentanoic acid, H—(Me) Leu-OH)(15.0 g, 103 mmol) as a starting material, Compound aa325 (10 g, 40%) was obtained in the same manner as synthesis of Compound aa317.
(1729) LCMS (ESI) m/z =242 (M+H)+
(1730) Retention time: 0.66 min (Analytical condition SQDFA05)
(1731) Synthesis of Compound aa328
(1732) ##STR00545##
(1733) Using Compound aa328-a ((S)-2-amino-2,3-dimethylbutanoic acid, H—(Me) Val-OH) (2.0 g, 15.3 mmol) as a starting material, Compound aa328 (1.2 g, 34%) was obtained in the same manner as synthesis of Compound aa320.
(1734) LCMS (ESI) m/z =226 (M-H)-
(1735) Retention time: 0.54 min (Analytical condition SQDFA05)
(1736) Synthesis of Compound aa319
(1737) ##STR00546##
(1738) Using Compound aa319-a (3-aminooxetane-3-carboxylic acid, H-AoxeC-OH)(10.0 g, 85.4 mmol) as a starting material, Compound aa319 (9.8 g, 54%) was obtained in the same manner as synthesis of Compound aa317.
(1739) LCMS (ESI) m/z =214 (M+H)+
(1740) Retention time: 0.50 min (Analytical condition SQDFA05) 1-6. Cyclic peptide synthesis by peptide modification 1-6-1. Peptide modification by alkylation of OH
(1741) Synthesis of Compound PP0242
(1742) Compound PP242 was synthesized according to the following scheme.
(1743) ##STR00547##
(1744) Using Compound aa375-resin as a raw material, peptide synthesis by the basic peptide

synthesis method described in the present Examples was performed to give Compound PP0238 (120.3 mg, 0.077 mmol).

(1745) LCMS (ESI) m/z =1562 (M+H)+

(1746) Retention time: 0.99 min (Analytical condition SQDFA05)

(1747) Iodoethane (2.6 μ L, 32.0 μ mol), tetrabutylammonium bromide (3.1 mg, 9.6 μ mol), and a 5 N aqueous sodium hydroxide solution (12.8 μ L, 64.0 μ mol) were added to a dichloromethane (80 μ L) solution of Compound PP238 (5.0 mg, 3.20 μ mol), and the mixture was stirred at room temperature for 7 days in a nitrogen atmosphere. The resulting reaction mixture was purified by reverse phase column chromatography (0.1% formic acid-containing acetonitrile/0.1% formic acid-containing distilled water) to give Compound PP0242 (2.3 mg, 45%). LC/MS data is provided in Table 36.

(1748) Synthesis of Compound PP0243

(1749) Compound PP243 was synthesized according to the following scheme.

(1750) ##STR00548##

(1751) Using Compound aa375-resin as a raw material, peptide synthesis by the basic peptide synthesis method described in the present Examples was performed to give Compound PP0240 (119.2 mg, 0.078 mmol).

(1752) LCMS (ESI) m/z =1536 (M+H)+

(1753) Retention time: 0.97 min (Analytical condition SQDFA05)

(1754) Iodoethane (2.6 μ L, 32.0 μ mol), tetrabutylammonium bromide (3.1 mg, 9.6 μ mol), and a 5 N aqueous sodium hydroxide solution (12.8 μ L, 64.0 μ mol) were added to a dichloromethane (80 μ L) solution of Compound PP240 (4.9 mg, 3.20 μ mol), and the mixture was stirred at room temperature for 7 days in a nitrogen atmosphere. The resulting reaction mixture was purified by reverse phase column chromatography (0.1% formic acid-containing acetonitrile/0.1% formic acid-containing distilled water) to give Compound PP0243 (2.6 mg, 52%). LC/MS data is provided in Table 36.

(1755) Synthesis of Compound PP0245

(1756) Compound PP245 was synthesized according to the following scheme.

(1757) ##STR00549##

(1758) Using Compound aa375-resin as a raw material, peptide synthesis by the basic peptide synthesis method described in the present Examples was performed to give Compound PP0240 (119.2 mg, 0.078 mmol).

(1759) LCMS (ESI) m/z =1536 (M+H)+

(1760) Retention time: 0.97 min (Analytical condition SQDFA05)

(1761) 2,2-Difluoroethyl trifluoromethanesulfonate (4.2 μ L, 32.0 μ mol), tetrabutylammonium bromide (3.1 mg, 9.6 μ mol), and a 5 N aqueous sodium hydroxide solution (12.8 μ L, 64.0 μ mol) were added to a dichloromethane (80 μ L) solution of Compound PP240 (4.9 mg, 3.20 μ mol), and the mixture was stirred at room temperature for 48 hours in a nitrogen atmosphere. The resulting reaction mixture was purified by reverse phase column chromatography (0.1% formic acid-containing acetonitrile/0.1% formic acid-containing distilled water) to give Compound PP0245 (4.2 mg, 82%). LC/MS data is provided in Table 36.

(1762) Synthesis of Compound PP0246

(1763) Compound PP246 was synthesized according to the following scheme.

(1764) ##STR00550##

(1765) Using Compound aa375-resin as a raw material, peptide synthesis by the basic peptide synthesis method described in the present Examples was performed to give Compound PP0241 (109.6 mg, 0.071 mmol).

(1766) LCMS (ESI) m/z =1536 (M+H)+

(1767) Retention time: 0.95 min (Analytical condition SQDFA05)

(1768) 2,2-Difluoroethyl trifluoromethanesulfonate (4.2 μ L, 32.0 μ mol), tetrabutylammonium

bromide (3.1 mg, 9.6 μmol), and a 5 N aqueous sodium hydroxide solution (12.8 μL , 64.0 μmol) were added to a dichloromethane (80 μL) solution of Compound PP241 (4.9 mg, 3.20 μmol), and the mixture was stirred at room temperature for 48 hours in a nitrogen atmosphere. The resulting reaction mixture was purified by reverse phase column chromatography (0.1% formic acid-containing acetonitrile/0.1% formic acid-containing distilled water) to give Compound PP0246 (4.4 mg, 86%). LC/MS data is provided in Table 36.

(1769) Synthesis of Compound PP998

(1770) Compound PP998 was synthesized according to the following scheme.

(1771) ##STR00551##

(1772) Using Compound aa375-resin as a raw material, peptide synthesis by the basic peptide synthesis method described in the present Examples was performed to give Compound PP0868 (76.8 mg, 0.0500 mmol).

(1773) LCMS (ESI) $m/z=1536$ (M+H)⁺

(1774) Retention time: 0.93 min (Analytical condition SQDFA50 long)

(1775) 3,3,3-Trifluoropropyl trifluoromethanesulfonate (5.0 μL , 33.0 μmol), tetrabutylammonium bromide (3.2 mg, 9.77 μmol), and a 5 N aqueous sodium hydroxide solution (13.0 μL , 65.0 μmol) were added to a dichloromethane (65.1 μL) solution of Compound PP0868 (5.0 mg, 3.26 μmol), and the mixture was stirred at room temperature for 9 days in a nitrogen atmosphere.

Dichloromethane (300 μL) and water (300 μL) were added to the resulting reaction mixture, the separated organic layer was concentrated under reduced pressure, and the resulting crude product was purified by reverse phase HPLC(methanol/50 mM aqueous ammonium acetate solution) to give Compound PP0998 (0.31 mg, 6%). LC/MS data is provided in Table 36.

(1776) Synthesis of Compounds PP999 to PP1003 and PP1013 to PP1015 Using Compound PP0868 as a raw material and the alkylating agents shown in Table 11, the following Compounds PP999 to PP1003 and PP1013 to PP1015 were obtained in the same manner as synthesis of PP998. LC/MS data is provided in Table 36.

(1777) TABLE-US-00012 TABLE 11 Raw material Yield Yield Peptide Alkylating agent (mg) (mg) (%) PP0999  5.0 0.94 18 PP1000  5.0 1.56 30 PP1001  5.0 0.16 3 PP1002  5.0 4.07 78 PP1003  5.0 3.32 66 PP1013 — — 1.27 24 PP1014 — — 0.74 14 PP1015 — — 0.20 4 * PP1013 was obtained as a by-product of synthesis of PP0999. * PP1014 was obtained as a by-product of synthesis of PP1000. * PP1015 was obtained as a by-product of synthesis of PP1001.

Synthesis of Compound PP1004

(1778) Compound PP1004 was synthesized according to the following scheme.

(1779) ##STR00557##

(1780) Using Compound aa375-resin as a raw material, peptide synthesis by the basic peptide synthesis method described in the present Examples was performed to give Compound PP0871 (57.5 mg, 0.0371 mmol).

(1781) LCMS (ESI) $m/z=1550$ (M+H)⁺

(1782) Retention time: 1.52 min (Analytical condition SQDFA50 long)

(1783) 1-Iodobutane (3.7 μL , 32.0 μmol), tetrabutylammonium bromide (3.1 mg, 9.68 μmol), and a 5 N aqueous sodium hydroxide solution (12.9 μL , 65.0 μmol) were added to a dichloromethane (64.5 μL) solution of Compound PP0871 (5.0 mg, 3.23 μmol), and the mixture was stirred at room temperature for 7 days in a nitrogen atmosphere. Dichloromethane (300 μL) and water (300 μL) were added to the resulting reaction mixture, the separated organic layer was concentrated under reduced pressure, and the resulting crude product was purified by reverse phase HPLC(methanol/50 mM aqueous ammonium acetate solution) to give Compound PP1004 (0.76 mg, 15%). LC/MS data is provided in Table 36.

(1784) Synthesis of Compounds PP1005 to PP1006 and PP1016

(1785) Using Compound PP0871 as a raw material and the alkylating agents shown in Table 12, the

following Compounds PP1005 to PP1006 and PP1016 were obtained in the same manner as synthesis of PP1004. LC/MS data is provided in Table 36.

(1786) TABLE-US-00013 TABLE 12 Raw material Yield Yield Peptide Alkylating agent (mg) (mg) (%) PP1005  5.0 0.31 6 PP1006  5.0 3.34 64 PP1016  5.0 1.14 22

Synthesis of Compound PP1008

(1787) Compound PP1008 was synthesized according to the following scheme.

(1788) ##STR00561##

(1789) Using Compound aa375-resin as a raw material, peptide synthesis by the basic peptide synthesis method described in the present Examples was performed to give Compound PP0872 (71.7 mg, 0.0463 mmol).

(1790) LCMS (ESI) $m/z=1550$ (M+H)⁺

(1791) Retention time: 1.64 min (Analytical condition SQDFA50 long)

(1792) 3,3,3-Trifluoropropyl trifluoromethanesulfonate (5.0 μ L, 32.0 μ mol), tetrabutylammonium bromide (3.1 mg, 9.68 μ mol), and a 5 N aqueous sodium hydroxide solution (12.9 μ L, 65.0 μ mol) were added to a dichloromethane (64.5 μ L) solution of Compound PP0872 (5.0 mg, 3.23 μ mol), and the mixture was stirred at room temperature for 5 days in a nitrogen atmosphere.

Dichloromethane (300 μ L) and water (300 μ L) were added to the resulting reaction mixture, the separated organic layer was concentrated under reduced pressure, and the resulting crude product was purified by reverse phase HPLC(methanol/50 mM aqueous ammonium acetate solution) to give Compound PP1008 (1.76 mg, 33%). LC/MS data is provided in Table 36.

(1793) Synthesis of Compounds PP1009 to PP1012 and PP1067 to PP1069

(1794) Using Compound PP0872 as a raw material and the alkylating agents shown in Table 13, the following Compounds PP1009 to PP1012 and PP1067 to PP1069 were obtained in the same manner as synthesis of PP1008. LC/MS data is provided in Table 36.

(1795) TABLE-US-00014 TABLE 13 Raw material Yield Yield Peptide Alkylating agent (mg) (mg) (%) PP1009  5.0 1.52 29 PP1010  5.0 1.64 32 PP1011  5.0 0.42 8 PP1012  5.0 1.79 34 PP1067  5.0 3.90 77 PP1068  5.0 1.66 33 PP1069  5.0 2.12 41

Synthesis of Compound PP1070

(1796) Compound PP1070 was synthesized according to the following scheme.

(1797) ##STR00569##

(1798) Using Compound aa375-resin as a raw material, peptide synthesis by the basic peptide synthesis method described in the present Examples was performed to give Compound PP0869 (78.0 mg, 0.0508 mmol).

(1799) LCMS (ESI) $m/z=1536$ (M+H)⁺

(1800) Retention time: 1.01 min (Analytical condition SQDFA50 long)

(1801) 3,3,3-Trifluoropropyl trifluoromethanesulfonate (5.0 μ L, 33.0 μ mol), tetrabutylammonium bromide (3.2 mg, 9.77 μ mol), and a 5 N aqueous sodium hydroxide solution (13.0 μ L, 65.0 μ mol) were added to a dichloromethane (65.1 μ L) solution of Compound PP0869 (5.0 mg, 3.26 μ mol), and the mixture was stirred at room temperature for 5 days in a nitrogen atmosphere.

Dichloromethane (300 μ L) and water (300 μ L) were added to the resulting reaction mixture, the separated organic layer was concentrated under reduced pressure, and the resulting crude product was purified by reverse phase HPLC(methanol/50 mM aqueous ammonium acetate solution) to give Compound PP1070 (0.63 mg, 12%). LC/MS data is provided in Table 36.

(1802) Synthesis of Compounds PP1071 to PP1073 and PP1075 to PP1077

(1803) Using Compound PP0869 as a raw material and the alkylating agents shown in Table 14, the following Compounds PP1071 to PP1077 were obtained in the same manner as synthesis of PP1070. LC/MS data is provided in Table 36.

(1804) TABLE-US-00015 TABLE 14 Raw material Yield Yield Peptide Alkylating agent (mg)

(mg) (%) PP1071 0.47 9 PP1075 1.70 34 PP1076 3.33 65 PP1077 2.28 44
Synthesis of Compound PP1078

(1805) Compound PP1078 was synthesized according to the following scheme.

(1806) ##STR00576##

(1807) Using Compound aa375-resin as a raw material, peptide synthesis by the basic peptide synthesis method described in the present Examples was performed to give Compound PP0870 (53.6 mg, 0.0349 mmol).

(1808) LCMS (ESI) $m/z=1536$ (M+H)+

(1809) Retention time: 0.94 min (Analytical condition SQDFA50 long) Iodomethane (2.0 μ L, 33.0 μ mol), tetrabutylammonium bromide (3.2 mg, 9.77 μ mol), and a 5 N aqueous sodium hydroxide solution (13.0 μ L, 65.0 μ mol) were added to a dichloromethane (65.1 μ L) solution of Compound PP0870 (5.0 mg, 3.26 μ mol), and the mixture was stirred at room temperature for 9 hours in a nitrogen atmosphere. Dichloromethane (300 μ L) and water (300 μ L) were added to the resulting reaction mixture, the separated organic layer was concentrated under reduced pressure, and the resulting crude product was purified by reverse phase HPLC(methanol/50 mM aqueous ammonium acetate solution) to give Compound PP1078 (0.37 mg, 7%). LC/MS data is provided in Table 36.

(1810) Synthesis of Compounds PP1079, PP1080, and PP1084

(1811) Using Compound PP0870 as a raw material and the alkylating agents shown in Table 15, the following Compounds PP1079, PP1080, and PP1084 were obtained in the same manner as synthesis of PP1078. LC/MS data is provided in Table 36.

(1812) TABLE-US-00016 TABLE 15 Raw material Yield Yield Peptide Alkylating agent (mg)

(mg) (%) PP1079 2.35 46 PP1080 2.17 42 PP1084
— 5.0 0.92 17 * PP1084 was obtained as a by-product of synthesis of PP1078.

Synthesis of Compound PP1085

(1813) Compound PP1085 was synthesized according to the following scheme.

(1814) ##STR00579##

(1815) Using Compound aa375-resin as a raw material, peptide synthesis by the basic peptide synthesis method described in the present Examples was performed to give Compound PP0873 (37.3 mg, 0.0241 mmol).

(1816) LCMS (ESI) $m/z=1550$ (M+H)+

(1817) Retention time: 1.50 min (Analytical condition SQDFA50 long)

(1818) Iodomethane (2.0 μ L, 32.0 μ mol), tetrabutylammonium bromide (3.1 mg, 9.68 μ mol), and a 5 N aqueous sodium hydroxide solution (12.9 μ L, 65.0 μ mol) were added to a dichloromethane (64.5 μ L) solution of Compound PP0873 (5.0 mg, 3.23 μ mol), and the mixture was stirred at room temperature for 9 hours in a nitrogen atmosphere. Dichloromethane (300 μ L) and water (300 μ L) were added to the resulting reaction mixture, the separated organic layer was concentrated under reduced pressure, and the resulting crude product was purified by reverse phase HPLC(methanol/50 mM aqueous ammonium acetate solution) to give Compound PP1085 (1.0 mg, 19%). LC/MS data is provided in Table 36.

(1819) Synthesis of Compounds PP1082 and PP1083

(1820) Using Compound PP0873 as a raw material and the alkylating agents shown in Table 17, the following Compounds PP1082 and PP1083 were obtained in the same manner as synthesis of PP1081. LC/MS data is provided in Table 36.

(1821) TABLE-US-00017 TABLE 17 Alkylating Raw material Yield Yield Peptide agent (mg)

(mg) (%) PP1082 1.25 25 PP1083 0.93 18

Synthesis of Compound PP1533

(1822) Compound PP1533 was synthesized according to the following scheme.

(1823) ##STR00580##

(1824) Using Compound aa374-resin as a raw material, peptide synthesis by the basic peptide synthesis method described in the present Examples was performed to give Compound PP1533-a (20.0 mg, 0.013 mmol).

(1825) LCMS (ESI) $m/z=1504$ (M+H)+

(1826) Retention time: 0.91 min (Analytical condition SQDFA05)

(1827) 2,2-Difluoroethyl trifluoromethanesulfonic acid (18.2 μL , 0.14 mmol), tetrabutylammonium bromide (13.3 mg, 0.041 mol), and a 5 N aqueous sodium hydroxide solution (55.1 μL , 0.28 mmol) were added to a dichloromethane (275 μL) solution of Compound PP1533-a (20.7 mg, 0.014 mmol), and the mixture was stirred at room temperature for 3 days in a nitrogen atmosphere.

Dichloromethane (300 μL) and water (300 μL) were added to the resulting reaction mixture, the separated organic layer was concentrated under reduced pressure, and the resulting crude product was purified by reverse phase HPLC(methanol/50 mM aqueous ammonium acetate solution) to give Compound PP1533 (11.5 mg, 54%). LC/MS data is provided in Table 36.

(1828) Synthesis of Compound PP1534

(1829) Compound PP1534 was synthesized according to the following scheme.

(1830) ##STR00581##

(1831) Using Compound aa374-resin as a raw material, PP1534-a (22.6 mg, 0.015 mmol) was obtained in the same manner as synthesis of PP1533-a.

(1832) LCMS (ESI) $m/z=1506$ (M+H)+

(1833) Retention time: 0.93 min (Analytical condition SQDFA05)

(1834) Using Compound PP1534-a (23.4 mg, 0.016 mmol) as a raw material, PP1534 (11.5 mg, 53%) was obtained in the same manner as synthesis of PP1533. LC/MS data is provided in Table 36.

(1835) Synthesis of Compound PP1535

(1836) Compound PP1535 was synthesized according to the following scheme.

(1837) ##STR00582##

(1838) Using Compound aa374-resin as a raw material, PP1535-a (19.5 mg, 0.013 mmol) was obtained in the same manner as synthesis of PP1533-a.

(1839) LCMS (ESI) $m/z=1532$ (M+H)+

(1840) Retention time: 0.96 min (Analytical condition SQDFA05)

(1841) Using Compound PP1535-a (20.3 mg, 0.013 mmol) as a raw material, PP1535 (9.4 mg, 42%) was obtained in the same manner as synthesis of PP1533. LC/MS data is provided in Table 36.

(1842) Synthesis of Compound PP1536

(1843) Compound PP1536 was synthesized according to the following scheme.

(1844) ##STR00583##

(1845) Using Compound aa374-resin as a raw material, PP1536-a (18.0 mg, 0.012 mmol) was obtained in the same manner as synthesis of PP1533-a.

(1846) LCMS (ESI) $m/z=1478$ (M+H)+

(1847) Retention time: 0.94 min (Analytical condition SQDFA05)

(1848) Using Compound PP1536-a (18.4 mg, 0.012 mmol) as a raw material, PP1536 (8.2 mg, 39%) was obtained in the same manner as synthesis of PP1533. LC/MS data is provided in Table 36.

(1849) In a cyclic compound and an oligopeptide compound synthesized by peptide modification by the alkylation of OH, abbreviations and structures of amino acids contained as partial structures are provided below.

(1850) [Table 17-1]

(1851) TABLE-US-00018 TABLE 17-1 Compound Amino acid No. Abbreviation Structural Formula
bb016 cisHyp  bb017 Hyp  bb018 cisHyp(3) 
 bb019 Hyp(3)  bb020 cisHyp(Et)  bb021

cisHyp(Et(2-F2)) 0
bb022 Hyp(Et(2-F2)) 0
bb023 Hyp(Tfp)
bb024 Hyp(nBu)
bb025 Hyp(Me-cPr)
bb026 Hyp(Me-cPent)
bb027 Hyp(3)(nBu)
bb028 Hyp(3)(Tfp)
bb029 Hyp(3)(Me-cPr)
bb030 Hyp(3)(Me-cPent)
bb031 Hyp(3)(Et(2-F2))
bb032 Hyp(3)(Me) 00
bb033 Hyp(3)(Et) 01
bb034 Hyp(3)(nPr) 02
bb035 cisHyp(3)(Me) 03
bb036 cisHyp(3)(Et) 04
bb037 cisHyp(3)(nPr) 05

1-6-2. Peptide Modification by Reductive Coupling Reaction

Synthesis of Compound PP0244

(1852) ##STR00606##

(1853) Using Compound aa375-resin as a raw material, peptide synthesis by the basic peptide synthesis method described in the present Examples was performed to give Compound PP0244-a (456.0 mg, 0.314 mmol).

(1854) LCMS (ESI) $m/z=1452$ (M+H)+

(1855) Retention time: 0.90 min (Analytical condition SQDFA05)

(1856) Compound PP244-a (456.0 mg, 0.314 mmol) and tetrakis(triphenylphosphine) palladium (0)(10.0 mg, 8.65 μ mol) were dissolved in dichloromethane (3.0 mL), and phenylsilane (0.040 mL, 0.325 mmol) was added dropwise in a nitrogen atmosphere. The reaction solution was stirred at room temperature for 2 hours, and then purified by reverse phase chromatography (0.1% formic acid-containing acetonitrile/0.1% formic acid-containing distilled water) to give Compound PP0244-b (80.5 mg, 18%).

(1857) LCMS (ESI) $m/z=1412$ (M+H)+

(1858) Retention time: 0.78 min (Analytical condition SQDFA05)

(1859) Compound PP244-b (5.0 mg, 3.54 μ mol) was placed in a reaction vessel, then a tetrahydrofuran (20 μ L) solution of N-hydroxyphthalimide (0.867 mg, 5.31 μ mol) and a dichloromethane (20 μ L) solution of 1-(3-dimethylaminopropyl)-3-ethylcarbodiimide hydrochloride (0.55 mg, 3.54 μ mol) were added, and the mixture was stirred overnight at room temperature. The solvent was distilled off under reduced pressure to give Compound PP0244-c as a crude product (5.51 mg, 100%).

(1860) LCMS (ESI) $m/z=1557$ (M+H)+

(1861) Retention time: 0.91 min (Analytical condition SQDFA05)

(1862) In a nitrogen atmosphere, nickel (II) bromide trihydrate (0.97 mg, 3.54 μ mol) and 4,4'-di-tert-butyl-2,2'-bipyridine (0.95 mg, 3.54 μ mol) were dissolved in N,N-dimethylacetamide (91 μ L), and this solution was added to an N,N-dimethylacetamide (80 μ L) solution of Compound PP244-c (5.51 mg, 3.54 μ mol), zinc (3.47 mg, 53.1 μ mol), and 5-bromo-1,2,3-trifluorobenzene (4.2 μ L, 35.4 μ mol). After being stirred at room temperature for 19 hours, the mixture was purified by reverse phase chromatography (0.1% formic acid-containing acetonitrile/0.1% formic acid-containing distilled water) to give Compound PP0244 (3.3 mg, 62%). LC/MS data is provided in Table 36.

(1863) Synthesis of Compound PP0882

(1864) ##STR00607##

(1865) Using Compound aa374-resin as a raw material, peptide synthesis by the basic peptide synthesis method described in the present Examples was performed to give Compound PP0882-a (160.3 mg, 0.1086 mmol).

(1866) LCMS (ESI) $m/z=1476$ (M+H)+

(1867) Retention time: 0.93 min (Analytical condition SQDFA05)

(1868) 1,1,1,3,3,3-Hexafluoro-2-propanol (11.66 mL), triisopropylsilane (0.24 mL), 1,2-dichloroethane (0.1 mL), and tetramethylammonium hydrogen sulfate (10.29 mg) were mixed, and the solution thereof (1.0 mL) was added to a reaction vessel containing Compound PP0882-a (5.0 mg, 3.39 μ mol). After the mixture was stirred at room temperature for 1 hour, N,N-

diisopropylethylamine (17.5 μ L) was added. This was combined with a separately synthesized lot and purified by reverse phase chromatography (0.1% formic acid-containing acetonitrile/0.1% formic acid-containing distilled water) to give Compound PP0882-b (123 mg).

(1869) LCMS (ESI) m/z =1358 (M+H)+

(1870) Retention time: 0.74 min (Analytical condition SQDFA05)

(1871) Compound PP0882-b (10.0 mg, 7.37 μ mol), N-hydroxyphthalimide (1.20 mg, 7.37 μ mol), 1-(3-dimethylaminopropyl)-3-ethylcarbodiimide hydrochloride (2.12 mg, 0.011 mmol), and dichloromethane (73.7 μ L) were placed in a reaction vessel, and the mixture was stirred at room temperature for 90 minutes. The reaction solution was diluted with dichloromethane (140 μ L), and washed twice with water (200 μ L). The organic layer was concentrated and dried under reduced pressure to give Compound PP0882-c as a crude product (11.1 mg, 100%).

(1872) LCMS (ESI) m/z =1503 (M+H)+

(1873) Retention time: 0.87 min (Analytical condition SQDFA05)

(1874) Nickel (II) bromide trihydrate (4.02 mg, 14.74 μ mol) and 4,4'-di-tert-butyl-2,2'-bipyridine (3.96 mg, 14.74 μ mol) were dissolved in N,N-dimethylacetamide (37 μ L), and the mixture was stirred at room temperature for 30 minutes in a nitrogen atmosphere. After this solution was added to an N,N-dimethylacetamide (55 μ L) solution of Compound PP0882-c (11.1 mg, 7.37 μ mol), zinc (7.23 mg, 111.0 μ mol), and 4-bromo-2-methyl-1-(trifluoromethyl)benzene (11.1 μ L, 73.7 μ mol), chlorotrimethylsilane (0.94 μ L, 7.37 μ mol) was added, and the mixture was stirred at room temperature for 14 hours. The reaction mixture was diluted with dimethyl sulfoxide (0.8 mL), filtered, and then purified by reverse phase HPLC(methanol/50 mM aqueous ammonium acetate solution) to give Compound PP0882 (4.8 mg, 44%). LC/MS data is provided in Table 36.

(1875) Synthesis of Compounds PP0884 to PP0888 and PP985 to PP0988

(1876) Using Compound PP0882-c (12.17 mg, 8.1 μ mol) as a raw material, PP0884 to PP0888 and PP985 to PP0988 were obtained in the same manner as synthesis of Compound PP0882 using the bromides (10 eq) shown in Table 18. LC/MS data is provided in Table 36.

(1877) TABLE-US-00019 TABLE 18 Intended Yield Yield product Bromide (mg) (%) PP0884 08

 6.19 51 PP0885 09  3.94 33 PP0886 0 

0.45 4 PP0887  2.72 23 PP0888  3.96 34 PP0985

 4.18 34 PP0986  3.78 32 PP0987 

PP0988  4.01 33

Synthesis of Compound PP0833

(1878) ##STR00617##

(1879) Using Compound aa374-resin as a raw material, peptide synthesis by the basic peptide synthesis method described in the present Examples was performed to give Compound PP0883-a (162.4 mg, 0.108 mmol).

(1880) LCMS (ESI) m/z =1490 (M+H)+

(1881) Retention time: 1.03 min (Analytical condition SQDFA05)

(1882) 1,1,1,3,3,3-Hexafluoro-2-propanol (11.66 mL), triisopropylsilane (0).24 mL), 1,2-dichloroethane (0).1 mL), and tetramethylammonium hydrogen sulfate (10).29 mg) were mixed, and the solution thereof (1.0 mL) was added to a reaction vessel containing Compound PP0883-a (5.0) mg, 3.36 μ mol). After the mixture was stirred at room temperature for 1 hour, N,N-diisopropylethylamine (17.5 μ L) was added. This was combined with a separately synthesized lot and purified by reverse phase chromatography (0.1% formic acid-containing acetonitrile/0).1% formic acid-containing distilled water) to give Compound PP0883-b (91 mg).

(1883) LCMS (ESI) m/z =1372 (M+H)+

(1884) Retention time: 0.82 min (Analytical condition SQDFA05)

(1885) Compound PP0883-b (10.0 mg, 7.29 μ mol), N-hydroxyphthalimide (1.189 mg, 7.29 μ mol), 1-(3-dimethylaminopropyl)-3-ethylcarbodiimide hydrochloride (2.10 mg, 10.94 μ mol), and dichloromethane (72.9 μ L) were placed in a reaction vessel, and the mixture was stirred at room

temperature for 90 minutes. The reaction solution was diluted with dichloromethane (140) μL), and washed twice with water (200 μL). The organic layer was concentrated and dried under reduced pressure to give Compound PP0883-c as a crude product (11.1 mg, 100%).

(1886) LCMS (ESI) $m/z=1517$ (M+H)+

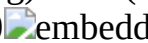
(1887) Retention time: 0.95 min (Analytical condition SQDFA05)

(1888) Nickel (II) bromide trihydrate (3.97 mg, 14.58 μmol) and 4,4'-di-tert-butyl-2,2'-bipyridine (3.91 mg, 14.58 μmol) were dissolved in N,N-dimethylacetamide (37 μL), and the mixture was stirred at room temperature for 30 minutes in a nitrogen atmosphere. After this solution was added to an N,N-dimethylacetamide (55 μL) solution of Compound PP0883-c (11.1 mg, 7.29 μmol), zinc (7.15 mg, 109.0 μmol), and 4-bromo-2-methyl-1-(trifluoromethyl)benzene (11.0) μL , 72.9 μmol), chlorotrimethylsilane (0.93 μL , 7.29 μmol) was added, and the mixture was stirred at room temperature for 14 hours. The reaction mixture was diluted with dimethyl sulfoxide (0).8 mL), filtered, and then purified by reverse phase HPLC(methanol/50) mM aqueous ammonium acetate solution) to give Compound PP0883 (3.7 mg, 34%). LC/MS data is provided in Table 36.

(1889) Synthesis of Compounds PP0889 to PP0893 and PP989

(1890) Using Compound PP0883-c (12.59 mg, 8.3 μmol) as a raw material, PP0889 to PP0893 and PP989 were obtained in the same manner as synthesis of Compound PP0883 using the bromides (10 eq) shown in Table 19. LC/MS data is provided in Table 36.

(1891) TABLE-US-00020 TABLE 19 Intended Yield product Bromide (mg) Yield(%) PP0889

 1.85 15 PP0890  2.03 16 PP0891 0  0.26 2 PP0892  0.27 2 PP0893  0.74 6 PP0989  2.2 17

Synthesis of Compound PP904

(1892) ##STR00624##

(1893) Using Compound aa376-resin as a raw material, peptide synthesis by the basic peptide synthesis method described in the present Examples was performed to give Compound PP0904-a.

(1894) LCMS (ESI) $m/z=1462$ (M+H)+

(1895) Retention time: 0.78 min (Analytical condition SQDAA50)

(1896) The resulting Compound PP904-a was dissolved in dichloromethane (1.0 mL), tetrakis(triphenylphosphine) palladium (0)(31.0 mg, 26.6 μmol) and phenylsilane (58.0 mg, 0.533 mmol) were added, and the mixture was stirred at room temperature for 30 minutes. After the solvent was distilled off under reduced pressure, the resulting crude product was purified by reverse phase HPLC(methanol/50 mM aqueous ammonium acetate solution) to give Compound PP0904-b (22.0 mg, 0.015 mmol, 11%).

(1897) LCMS (ESI) $m/z=1422$ (M+H)+

(1898) Retention time: 0.66 min (Analytical condition SQDAA50)

(1899) Dichloromethane (155 μL) was added to a reaction vessel containing Compound PP904-b (22.0 mg, 0.015 mmol) and N-hydroxyphthalimide (2.5 mg, 0.015 mmol). After the mixture was stirred at room temperature for 1 hour, N,N-diisopropylethylamine (4.1 μL , 0.023 mmol) was added, and the mixture was stirred at room temperature for 1 hour. Then, 1-(3-dimethylaminopropyl)-3-ethylcarbodiimide hydrochloride (4.5 mg, 0.023 mmol) was added, and the mixture was stirred at room temperature for 1 hour. After dilution with dichloromethane, the organic layer was washed with a saturated aqueous ammonium chloride solution and then washed with water. After drying over anhydrous sodium sulfate and filtration, the solvent was distilled off under reduced pressure to give Compound PP0904-c (20.8 mg, 86%).

(1900) LCMS (ESI) $m/z=1567$ (M+H)+

(1901) Retention time: 0.88 min (Analytical condition SQDFA40)

(1902) Nickel (II) bromide trihydrate (1.74 mg, 6.39 μmol) and 4,4'-di-tert-butyl-2,2'-bipyridine (1.71 mg, 6.39 μmol) were dissolved in N,N-dimethylacetamide (75 μL), and this solution was added to an N,N-dimethylacetamide (80 μL) solution of Compound PP904-c (10.0 mg, 6.38 μmol),

zinc (6.3 mg, 0.096 mmol), and 4-bromofluorobenzene (7.0 μ L, 0.064 mmol). After the mixture was stirred at room temperature for 15 hours, chlorotrimethylsilane (0.82 μ L, 6.39 μ mol) was added, and the mixture was stirred for 3 hours. The reaction mixture was diluted with dimethyl sulfoxide, filtered, and then purified by reverse phase HPLC(methanol/50 mM aqueous ammonium acetate solution) to give Compound PP0904 (0.43 mg, 0.292 μ mol). LC/MS data is provided in Table 36.

(1903) Synthesis of Compound PP905

(1904) ##STR00625##

(1905) Using Compound aa381-resin as a raw material, peptide synthesis by the basic peptide synthesis method described in the present Examples was performed to give Compound PP0905-a. Using this as a starting material, PP905-b (27.0 mg, 0.018 mmol, 8%) was obtained in the same manner as synthesis of Compound PP904-b.

(1906) LCMS (ESI) m/z=1464 (M+H)+

(1907) Retention time: 0.66 min (Analytical condition SQDAA50)

(1908) Using PP905-b as a starting material, PP905-c (26.0 mg, 88%) was obtained in the same manner as synthesis of Compound PP904-c.

(1909) LCMS (ESI) m/z=1609 (M+H)+

(1910) Retention time: 0.88 min (Analytical condition SQDFA40)

(1911) Using PP905-c as a starting material, PP905 (1.57 mg, 1.037 μ mol) was obtained in the same manner as synthesis of Compound PP904. LC/MS data is provided in Table 36.

(1912) Synthesis of Compound PP906

(1913) ##STR00626##

(1914) Using Compound aa374-resin as a raw material, peptide synthesis by the basic peptide synthesis method described in the present Examples was performed to give Compound PP0906-a. Using this as a starting material, PP906-b (138.0 mg, 18%) was obtained in the same manner as synthesis of Compound PP904-b.

(1915) LCMS (ESI) m/z=1408 (M+H)+

(1916) Retention time: 0.62 min (Analytical condition SQDAA50)

(1917) Using PP906-b as a starting material, PP906-c (130.3 mg, 100%) was obtained in the same manner as synthesis of Compound PP904-c.

(1918) LCMS (ESI) m/z=1553 (M+H)+

(1919) Retention time: 0.76 min (Analytical condition SQDAA50)

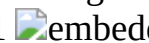
(1920) Using PP906-c as a starting material, PP906 (0.66 mg, 0.447 μ mol) was obtained in the same manner as synthesis of Compound PP904 using 1-bromo-2,4-difluorobenzene in place of 4-bromofluorobenzene. LC/MS data is provided in Table 36.

(1921) Synthesis of Compounds PP907 to PP0915

(1922) Using Compound PP906-c (10 mg, 6.38 μ mol) as a raw material, PP907 to PP0915 were obtained in the same manner as synthesis of Compound PP906 using the bromides (10 eq) shown in Table 20. LC/MS data is provided in Table 36.

(1923) TABLE-US-00021 TABLE 20 Intended Yield Yield product Bromide (mg) (%) PP0907

 0.7 7 PP0908  0.61 6 PP0909  1.23 13

PP0910 0  1.72 18 PP0911  0.72 8 PP0912

 2.03 22 PP0913  0.78 9 PP0914  1.82 19

PP0915  1.72 18

Synthesis of Compound PP916

(1924) ##STR00636##

(1925) Using Compound aa375-resin as a raw material, peptide synthesis by the basic peptide synthesis method described in the present Examples was performed to give Compound PP0916-a. Using this as a starting material, PP916-b (109.0 mg, 14%) was obtained in the same manner as synthesis of Compound PP904-b.

(1926) LCMS (ESI) m/z=1478 (M+H)+

(1927) Retention time: 0.60 min (Analytical condition SQDAA50)

(1928) Using PP916-b as a starting material, PP916-c (95.3 mg, 92%) was obtained in the same manner as synthesis of Compound PP904-c.

(1929) LCMS (ESI) m/z=1623 (M+H)+

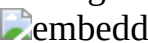
(1930) Retention time: 0.78 min (Analytical condition SQDAA50)

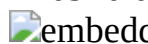
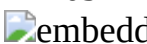
(1931) Using PP916-c as a starting material, PP916 (0.77 mg, 0.498 μ mol) was obtained in the same manner as synthesis of Compound PP904 using 1-bromo-2,4-difluorobenzene in place of 4-bromofluorobenzene. LC/MS data is provided in Table 36.

(1932) Synthesis of Compounds PP917 to PP0924

(1933) Using Compound PP916-c (10 mg, 6.17 μ mol) as a raw material, PP917 to PP0924 were obtained in the same manner as synthesis of Compound PP916 using the bromides (10 eq) shown in Table 21. LC/MS data is provided in Table 36.

(1934) TABLE-US-00022 TABLE 21 Intended Yield Yield product Bromide (mg) (%) PP0917

 0.38 4 PP0918  0.29 3 PP0919  1.02 10
PP0920 0  1.23 12 PP0921  0.81 8 PP0922

 1.02 10 PP0923  0.49 5 PP0924  0.98 10

Synthesis of Compound PP02672

(1935) ##STR00645##

(1936) Using Compound aa374-resin as a raw material, peptide synthesis was carried out by the basic peptide synthesis method described in the present Examples to give Compound PP2672-a. Using this as a starting material, PP2672-b (289 mg, 16%) was obtained in the same manner as the synthesis of Compound PP904-b.

(1937) LCMS (ESI) m/z=1370 (M+H)+

(1938) Retention time: 0.79 min (Analytical condition SQDFA05)

(1939) Using PP2672-b as a starting material, PP2672-c (329 mg, quant.) was obtained in the same manner as the synthesis of Compound PP904-c.

(1940) LCMS (ESI) m/z=1515 (M+H)+

(1941) Retention time: 0.92 min (Analytical condition SQDFA05)

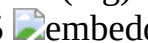
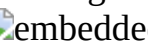
(1942) Using PP2672-c as a starting material, PP2672 (6.2 mg, 41%) was obtained in the same manner as the synthesis of Compound PP904 except that 4-bromo-2-ethyl-1-

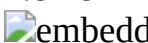
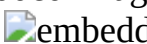
(trifluoromethyl)benzene was used in place of 4-bromofluorobenzene. LC/MS data are provided in Table 36.

(1943) Synthesis of Compounds PP2673 to PP2686

(1944) Using Compound PP02672-c as a raw material, PP2673 to PP2686 were obtained in the same manner as the synthesis of Compound PP02672 using the bromides (10 eq) shown in Table 21-1. LC/MS data are provided in Table 36.

(1945) TABLE-US-00023 TABLE 21-1 Intended product Bromide Yield (mg) Yield (%) PP2673

 3.3 19 PP2674  5.4 31 PP2675  6.4 36
PP2676  7.6 43 PP2677 0  7.2 40 PP2678  7.3 41 PP2679  2.8 28 PP2680  1.1 9.2 PP2681

 2.5 31 PP2682  1.3 17 PP2683  2.4 21
PP2684  2.3 20 PP2685  1.2 11 PP2686  1.2 10

Synthesis of Compound PP02687

(1946) ##STR00660##

(1947) Using Compound aa374-resin as a raw material, peptide synthesis was carried out by the basic peptide synthesis method described in the present Examples to give Compound PP2687-a. Using this as a starting material, PP2687-b (97 mg, 16%) was obtained in the same manner as the synthesis of Compound PP904-b.

(1948) LCMS (ESI) $m/z=1456$ (M+H)+
 (1949) Retention time: 0.91 min (Analytical condition SQDFA05)
 (1950) Using PP2687-b as a starting material, PP2687-c (99 mg, 76%) was obtained in the same manner as the synthesis of Compound PP904-c.
 (1951) LCMS (ESI) $m/z=1601$ (M+H)+
 (1952) Retention time: 1.00 min (Analytical condition SQDFA05)
 (1953) Using PP2687-c as a starting material, PP2687 (2.3 mg, 3.3%) was obtained in the same manner as the synthesis of Compound PP904 except that 1-ethyl-4-iodobenzene was used in place of 4-bromofluorobenzene. LC/MS data are provided in Table 36.
 (1954) Synthesis of Compounds PP2688 to PP2695
 (1955) Using Compound PP02687-c as a raw material, PP2688 to PP2695 were obtained in the same manner as the synthesis of Compound PP02687 using the halides (10 eq) shown in Table 21-2. LC/MS data are provided in Table 36.
 (1956) TABLE-US-00024 TABLE 21-2 Intended product Bromide Yield (mg) Yield (%) PP2688
 2.7 3.8 PP2689  2.7 3.8 PP2690  2.8 3.9
 PP2691  2.6 3.7 PP2692  2.2 3.1 PP2693  2.4 3.4
 PP2694  2.4 3.4 PP2695  0.2 0.3
 Synthesis of Compound PP02696
 (1957) ##STR00669##
 (1958) Using Compound aa440-resin as a raw material, peptide synthesis was carried out by the basic peptide synthesis method described in the present Examples to give Compound PP2696-a. Using this as a starting material, PP2696-b (117 mg, 27%) was obtained in the same manner as the synthesis of Compound PP904-b.
 (1959) LCMS (ESI) $m/z=1442$ (M+H)+
 (1960) Retention time: 0.92 min (Analytical condition SQDFA05)
 (1961) Using PP2696-b as a starting material, PP2696-c (115 mg, 89%) was obtained in the same manner as the synthesis of Compound PP904-c.
 (1962) LCMS (ESI) $m/z=1586$ (M-H)-
 (1963) Retention time: 1.02 min (Analytical condition SQDFA05)
 (1964) Using PP2696-c as a starting material, PP2696 (3.7 mg, 7.4%) was obtained in the same manner as the synthesis of Compound PP904 except that 1-ethyl-4-iodobenzene was used in place of 4-bromofluorobenzene. LC/MS data are provided in Table 36.
 (1965) Synthesis of Compounds PP2697 to PP2704
 (1966) Using Compound PP02696-c as a raw material, PP2697 to PP2704 were obtained in the same manner as the synthesis of Compound PP02696 using the halides (10 eq) shown in Table 21-3. LC/MS data are provided in Table 36.
 (1967) TABLE-US-00025 TABLE 21-3 Intended product Bromide Yield (mg) Yield (%) PP2697 0
 3.8 7.6 PP2698  4.1 8.2 PP2699  4.1 8.0
 PP2700  3.8 7.5 PP2701  3.5 7.0 PP2702  3.9 7.8
 PP2703  3.7 7.3 PP2704  0.2 0.4
 Synthesis of Compound PP02849
 (1968) ##STR00678##
 (1969) Using Compound aa374-resin as a raw material, peptide synthesis was carried out by the basic peptide synthesis method described in the present Examples to give Compound PP2849-a. Using this as a starting material, PP2849-b (345 mg, 22%) was obtained in the same manner as the synthesis of Compound PP904-b.
 (1970) LCMS (ESI) $m/z=1412$ (M+H)+
 (1971) Retention time: 0.80 min (Analytical condition SQDFA05)
 (1972) Using PP2849-b as a starting material, PP2849-c (395 mg, quant. %) was obtained in the same manner as the synthesis of Compound PP904-c.

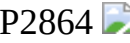
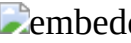
(1973) LCMS (ESI) m/z=1557 (M+H)+

(1974) Retention time: 0.52 min (Analytical condition SQDFA50)

(1975) Using PP2849-c as a starting material, PP2849 (4.5 mg, 39%) was obtained in the same manner as the synthesis of Compound PP904 except that 6-bromo-4-fluoro-2,3-dihydro-1H-indene was used in place of 4-bromofluorobenzene. LC/MS data are provided in Table 36.

(1976) Synthesis of Compounds PP2850 to PP2869

(1977) Using Compound PP02849-c as a raw material, PP2850 to PP2869 were obtained in the same manner as the synthesis of Compound PP02849 using the bromides (10 eq) shown in Table 21-4. LC/MS data are provided in Table 36.

(1978) TABLE-US-00026 TABLE 21-4 Intended product Bromide Yield (mg) Yield (%) PP2850
 3.5 30 PP2851  5.1 44 PP2852  5.1 43
PP2853  5.6 48 PP2854  7.0 61 PP2855  3.4 29 PP2856  5.0 44 PP2857  5.1 44 PP2858
 4.6 40 PP2859  4.3 35 PP2860  7.5 64
PP2861  6.9 58 PP2862  5.8 50 PP2863  3.4 29 PP2864  1.3 11 PP2865  7.8 65 PP2866
 7.6 64 PP2867  7.2 60 PP2868  0.4 3.2
PP2869  4.1 35

Synthesis of Compound PP02870

(1979) ##STR00699##

(1980) Using Compound aa374-resin as a raw material, peptide synthesis was carried out by the basic peptide synthesis method described in the present Examples to give Compound PP2870-a. Using this as a starting material, PP2870-b (289 mg, 16%) was obtained in the same manner as the synthesis of Compound PP904-b.

(1981) LCMS (ESI) m/z=1370 (M+H)+

(1982) Retention time: 0.79 min (Analytical condition SQDFA05)

(1983) Using PP2870-b as a starting material, PP2870-c (329 mg, quant. %) was obtained in the same manner as the synthesis of Compound PP904-c.

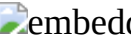
(1984) LCMS (ESI) m/z=1515 (M+H)+

(1985) Retention time: 0.92 min (Analytical condition SQDFA05)

(1986) Using PP2870-c as a starting material, PP2870 (6.2 mg, 41%) was obtained in the same manner as the synthesis of Compound PP904 except that 4-bromo-2-ethyl-1-(trifluoromethyl)benzene was used in place of 4-bromofluorobenzene. LC/MS data are provided in Table 36.

(1987) Synthesis of Compounds PP2871 to PP2886

(1988) Using Compound PP02870-c as a raw material, PP2871 to PP2886 were obtained in the same manner as the synthesis of Compound PP02870 using the bromides (10 eq) shown in Table 21-5. LC/MS data are provided in Table 36.

(1989) TABLE-US-00027 TABLE 21-5 Intended product Bromide Yield (mg) Yield (%) PP2871
00  7.5 51 PP2872 01  6.8 47 PP2873 02  7.4 51 PP2874 03  7.4 50 PP2875 04  8.6 58 PP2876 05
 7.7 52 PP2877 06  6.3 42 PP2878 07  6.4
44 PP2879 08  6.7 46 PP2880 09  6.3 44 PP2881 0
 6.9 46 PP2882  7.8 53 PP2883  8.5 59
PP2884  9.3 64 PP2885  8.5 58 PP2886  7.6 52

1-6-3. Peptide Modification by Amidation of Carboxylic Acid

Synthesis of Compound PP391

(1990) ##STR00716##

(1991) Using Compound aa386-resin as a raw material, peptide synthesis by the basic peptide

synthesis method described in the present Examples was performed to give Compound PP0391-a (5.6 mg, 0.00386 mmol).

(1992) LCMS (ESI) $m/z=1451$ (M+H)+

(1993) Retention time: 0.66 min (Analytical condition SQDFA50)

(1994) 2-Methoxy-N-methylethan-1-amine (0.002 mL, 0.019 mmol) was added to a THF (0.039 mL) solution of Compound PP391-a (5.6 mg, 0.00386 mmol), sodium bicarbonate (1.62 mg, 0.019 mmol), and DEPBT (5.77 mg, 0.019 mmol), the mixture was stirred at room temperature for 1 hour, and then the solvent was distilled off under reduced pressure. The resulting crude product was purified by reverse phase chromatography (0.1% formic acid-containing acetonitrile/0.1% formic acid-containing distilled water) to give PP0391 (5.08 mg, 86%). LC/MS data is provided in Table 36.

(1995) Synthesis of Compounds PP1151 and PP1152

(1996) ##STR00717##

(1997) Using Compound aa386-resin as a raw material, peptide synthesis by the basic peptide synthesis method described in the present Examples was performed to give Compound PP1151-a. Using the resulting Compound PP1151-a (5.0 mg), the intended compounds were obtained in the same manner as synthesis of Compound PP391 using the amines shown in Table 22 in place of 2-methoxy-N-methylethan-1-amine. LC/MS data is provided in Table 36.

(1998) TABLE-US-00028 TABLE 22 Intended product Amine Yield Yield PP1151

 3.2 mg 60% PP1152  2.8 mg 53%

Synthesis of Compounds PP1153 and PP1154

(1999) ##STR00720##

(2000) Using Compound aa386-resin as a raw material, peptide synthesis by the basic peptide synthesis method described in the present Examples was performed to give Compound PP1153-a. Using the resulting Compound PP1153-a (5.1 mg), the intended compounds were obtained in the same manner as synthesis of Compound PP391 using the amines shown in Table 23 in place of 2-methoxy-N-methylethan-1-amine. LC/MS data is provided in Table 36.

(2001) TABLE-US-00029 TABLE 23 Intended product Amine Yield Yield PP1153

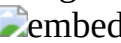
 2.4 mg 44% PP1154  2.5 mg 46%

Synthesis of Compounds PP1300, PP1301, PP1302, PP1303, PP1304, PP1305, PP1306, PP1307, PP1308, PP1309, PP1310, PP1311, PP1312, PP1313, PP1314, PP1315, PP1316, PP1317, PP1318, PP1319, PP1320, PP1331, PP1387, PP1388, and PP1389

(2002) ##STR00723##

(2003) Using Compound aa386-resin as a raw material, peptide synthesis by the basic peptide synthesis method described in the present Examples was performed to give Compound PP1300-a. Using the resulting Compound PP1300-a (0.00574 mmol each), the intended compounds were obtained in the same manner as synthesis of Compound PP391 using the amines shown in Table 24 in place of 2-methoxy-N-methylethan-1-amine. LC/MS data is provided in Table 36.

(2004) TABLE-US-00030 TABLE 24 Intended product R1 R2 Amine Yield Yield PP1300

   3.3 mg 37% PP1301 
  3.1 mg 35% PP1302 0 
  3.5 mg 41% PP1303  
 2.0 mg 23% PP1304   
3.1 mg 35% PP1305  0   2.8 mg 31%
PP1306    2.8 mg 32% PP1307
   2.0 mg 22% PP1308 
 0  3.4 mg 39% PP1309 
  2.4 mg 27% PP1310  
 2.8 mg 32% PP1311   
2.6 mg 28% PP1312 0    2.9 mg 32%

PP1313     4.3 mg 48% PP1314
    4.3 mg 48% PP1315 
0    3.3 mg 36% PP1316 
   3.5 mg 39% PP1317  
  3.0 mg 33% PP1318   0
  3.1 mg 34% PP1319   
3.8 mg 43% PP1320    2.0 mg 22%
PP1331    3.4 mg 38% PP1387 0
   2.8 mg 31% PP1388 
  2.0 mg 22% PP1389  
  2.7 mg 30%

Synthesis of Compounds PP1321, PP1322, PP1323, PP1324, PP1325, PP1326, PP1327, PP1328, and PP1329

(2005) ##STR00799##

(2006) Using Compound aa386-resin as a raw material, peptide synthesis by the basic peptide synthesis method described in the present Examples was performed to give Compound PP1321-a. Using the resulting Compound PP1321-a (0.00574 mmol each), the intended compounds were obtained in the same manner as synthesis of Compound PP391. The intended products and yields are provided in Table 25. LC/MS data is provided in Table 36.

(2007) TABLE-US-00031 TABLE 25 In- tended product R1 R2 Yield Yield PP1321 00
  01   4.2 mg 47% PP1322 02   03
  3.6 mg 42% PP1323 04   05   3.2 mg 37%
PP1324 06   07  3.0 mg 33% PP1325 08   09
  2.6 mg 29% PP1326 0    3.8 mg 43%
PP1327    3.3 mg 37% PP1328 
  2.8 mg 32% PP1329   2.4 mg 27%

Synthesis of Compound PP1330

(2008) ##STR00818##

(2009) Using Compound aa386-resin as a raw material, peptide synthesis by the basic peptide synthesis method described in the present Examples was performed to give Compound PP1330-a. Using the resulting Compound PP1330-a (0.00574 mmol), the intended compound (1.3 mg, 14%) was obtained in the same manner as synthesis of Compound PP391. LC/MS data is provided in Table 36.

(2010) Synthesis of Compounds PP1537 and PP1538

(2011) ##STR00819##

(2012) Using Compound aa386-resin as a raw material, peptide synthesis by the basic peptide synthesis method described in the present Examples was performed to give Compounds PP1537-a and PP1538-a. Using the resulting Compounds PP1537-a and PP1538-a (8.8 mg each), Compounds PP1537 (3.2 mg, 35%) and PP1538 (3.9 mg, 42%) were obtained in the same manner as synthesis of Compound PP391 using N-ethylmethylamine in place of 2-methoxy-N-methylethan-1-amine. LC/MS data is provided in Table 36.

(2013) Synthesis of Compounds PP1552 and PP1553

(2014) ##STR00820##

(2015) Using Compound aa386-resin as a raw material, peptide synthesis by the basic peptide synthesis method described in the present Examples was performed to give Compounds PP1552-a and PP1553-a. Using the resulting Compounds PP1552-a (3.8 mg) and PP1553-a (4.5 mg), Compounds PP1552 (1.7 mg, 44%) and PP1553 (1.9 mg, 41%) were obtained in the same manner as synthesis of Compound PP391 using N-ethylmethylamine in place of 2-methoxy-N-methylethan-1-amine. LC/MS data is provided in Table 36.

(2016) Synthesis of Compounds PP1554 and PP1557

(2017) ##STR00821##

(2018) Using Compound aa386-resin as a raw material, peptide synthesis by the basic peptide synthesis method described in the present Examples was performed to give Compound PP1554-a. Using the resulting Compound PP1554-a (12.6 mg each), Compounds PP1554 (5.9 mg, 44%) and PP1557 (6.7 mg, 51%) were obtained in the same manner as synthesis of Compound PP391 using N-methylhex-5-ene-1-amine or N-methylprop-2-ene-1-amine in place of 2-methoxy-N-methylethan-1-amine. LC/MS data is provided in Table 36.

(2019) 1-6-4. Peptide Synthesis Via N-Alkylation by Mitsunobu Reaction on Resin

(2020) Synthesis of Compound PP55

(2021) PP55 was synthesized according to the following scheme.

(2022) ##STR00822##

(2023) Using Compound aa375-resin as a raw material, peptide synthesis by the basic peptide synthesis method described in the present Examples was performed to give resin-supported Compound PP55-a. Cleaving from the resin was performed with TFE/DCM (1/1) using a small amount of resin-supported Compound PP55-a, and the structure was verified by LC/MS.

(2024) LCMS (ESI) $m/z=1282$ (M+H)+

(2025) Retention time: 1.02 min (Analytical condition SQDFA05)

(2026) Compound PP55-a-resin (100 mg, 0.384 mmol/g, 0.0384 mmol) and DCM (1 mL) were added to a filter-equipped reaction vessel to swell the resin. After DCM was removed, the resin was washed 4 times with DMF (1 mL). A DMF solution (2% v/v, 0.600 mL) of DBU was added, the mixture was shaken for 15 minutes at room temperature, and then the reaction solution was removed. The resin was washed 6 times with NMP (1 mL) to give resin-supported

(2027) Compound PP55-b. Cleaving from the resin was performed with TFE/DCM (1/1) using a small amount of resin-supported Compound PP55-b, and the structure was verified by LC/MS.

(2028) LCMS (ESI) $m/z=1060$ (M+H)+

(2029) Retention time: 0.62 min (Analytical condition SQDFA05)

(2030) An NMP solution (1 mL) of nosyl chloride (34 mg, 0.154 mmol) and 2,4,6-cholidine (0.051 mL, 0.384 mmol) were added thereto, and the mixture was shaken at 40° C. for 1 hour. Then, the reaction solution was removed, and the resin was washed 4 times with NMP (1 mL) and washed 4 times with DCM (1 mL) to give resin-supported Compound PP55-c. Cleaving from the resin was performed with THE/DCM (1/1) using a small amount of resin-supported Compound PP55-c, and the structure was verified by LC/MS.

(2031) LCMS (ESI) $m/z=1245$ (M+H)+

(2032) Retention time: 0.89 min (Analytical condition SQDFA05)

(2033) DCM (1 mL) was added thereto for swelling. After DCM was removed, the resin was washed twice with DCM (1 mL) and then 4 times with THE (1 mL). A THE solution (1 mL) obtained by dissolving triphenylphosphine (50) mg, 0.192 mmol), DIAD (0.038 mL, 0.192 mmol), and 1-butanol (0.035 mL, 0.384 mmol) was added, the mixture was shaken at 40° C. for 30 minutes, and then the reaction solution was removed. The resin was washed 4 times with THF (1 mL) and washed 4 times with DCM (1 mL) to give resin-supported Compound PP55-d. Cleaving from the resin was performed with THE/DCM (1/1) using a small amount of resin-supported Compound PP55-d, and the structure was verified by LC/MS.

(2034) LCMS (ESI) $m/z=1301$ (M+H)+

(2035) Retention time: 1.01 min (Analytical condition SQDFA05)

(2036) DCM (1 mL) was added thereto for swelling. After DCM was removed, the resin was washed 4 times with NMP (1 mL). An NMP solution (0.200 mL) of DBU (0.020 mL, 0.134 mmol) and an NMP solution (0.250 mL) of 1-dodecanethiol (0.063 mL, 0.262 mmol) were added, and the mixture was shaken at room temperature for 1 hour. Then, the reaction solution was removed, and the resin was washed 4 times with NMP (1 mL). An NMP solution (0.200 mL) of DBU (0.020 mL, 0.134 mmol) and an NMP solution (0.250 mL) of 1-dodecanethiol (0.063 mL, 0.262 mmol)

were added, and the mixture was shaken at room temperature for 13 hours. Then, the reaction solution was removed, and the resin was washed 4 times with NMP (1 mL.). An NMP solution (0.200 mL) of DBU (0.020 mL, 0.134 mmol) and an NMP solution (0.250 mL) of 1-dodecanethiol (0.063 mL, 0.262 mmol) were added, and the mixture was shaken at room temperature for 1 hour. Then, the reaction solution was removed, and the resin was washed 4 times with NMP (1 mL) and washed 4 times with DCM (1 mL) to give resin-supported Compound PP55-e. Cleaving from the resin was performed with TFE/DCM (1/1) using a small amount of resin-supported Compound PP55-e, and the structure was verified by LC/MS.

(2037) LCMS (ESI) $m/z=1116$ (M+H)⁺

(2038) Retention time: 0.66 min (Analytical condition SQDFA05)

(2039) Subsequent steps of peptide elongation, cleaving from resin, cyclization, and purification were performed according to the basic route to give Compound PP0055. LC/MS data is provided in Table 36.

(2040) Synthesis of Compound PP52, Compound PP56, Compound PP57, Compound PP59, Compound PP062, Compound PP063, Compound PP064, Compound PP065, Compound PP083, Compound PP084, Compound PP085, Compound PP086, Compound PP087, Compound PP088, Compound PP089, Compound PP90, Compound PP91, Compound PP92, Compound PP93, Compound PP94, Compound PP95, Compound PP96, Compound PP97, Compound PP98, Compound PP99, Compound PP100, Compound PP101, Compound PP102, Compound PP103, Compound PP104, Compound PP105, Compound PP134, Compound PP135, Compound PP139, and Compound PP140

(2041) Using resin-supported Compound PP55-c, the reaction was carried out in the same manner as synthesis of Compound PP55-d using the alcohols shown in Table 26 in place of 1-butanol. Moreover, the Ns group was deprotected in the same manner as synthesis of Compound PP55-e, and subsequent steps of peptide elongation, cleaving from resin, cyclization, and purification were carried out according to the basic route to give the intended compounds. LC/MS data is provided in Table 36.

(2042) TABLE-US-00032 TABLE 26 Intended product Alcohol Compound PP0052 1-Propanol Compound PP0056 Isobutanol Compound PP0057 Cyclopropylmethanol Compound PP0059 1-Propanol Compound PP0062 1-Butanol Compound PP0063 Isobutanol Compound PP0064 Cyclopropylmethanol Compound PP0065 1-Propanol Compound PP0083 Cyclobutylmethanol Compound PP0084 Cyclopentylmethanol Compound PP0085 Cyclohexylmethanol Compound PP0086 3-Methylbutan-1-ol Compound PP0087 4-Methylpentan-1-ol Compound PP0088 2-Propen-1-ol Compound PP0089 3-Buten-1-ol Compound PP0090 2-Propyn-1-ol Compound PP0091 3,3,3-Trifluoro-1-propanol Compound PP0092 2-Methoxyethanol Compound PP0093 3-Methoxy-1-propanol Compound PP0094 2-Propanol Compound PP0095 Cyclobutylmethanol Compound PP0096 Cyclopentylmethanol Compound PP0097 Cyclohexylmethanol Compound PP0098 3-Methylbutan-1-ol Compound PP0099 4-Methylpentan-1-ol Compound PP0100 2-Propen-1-ol Compound PP0101 3-Buten-1-ol Compound PP0102 2-Propyn-1-ol Compound PP0103 3,3,3-Trifluoro-1-propanol Compound PP0104 2-Methoxyethanol Compound PP0105 3-Methoxy-1-propanol Compound PP0134 1-Propanol Compound PP0135 1-Propanol Compound PP0139 2-Propen-1-ol Compound PP0140 3-Buten-1-ol

Synthesis of Compound PP58, Compound PP066, and Compound PP069

(2043) Compound PP58, Compound PP066, and Compound PP069 were synthesized according to the following scheme.

(2044) ##STR00823##

(2045) Using Compound aa375-resin as a raw material, peptide synthesis by the basic peptide synthesis method described in the present Examples was performed to give resin-supported Compound PP58-a. After resin-supported Compound PP58-b was obtained in the same manner as synthesis of Compound PP55-b, resin-supported Compound PP58-c was obtained in the same

manner as synthesis of Compound PP55-c. After resin-supported Compound PP58-d was obtained in the same manner as synthesis of Compound PP55-d using 1-propanol in place of 1-butanol, resin-supported Compound PP58-e was obtained in the same manner as synthesis of Compound PP55-e. Subsequent steps of peptide elongation, cleaving from resin, cyclization, and purification were performed according to the basic route to give the intended compounds. LC/MS data is provided in Table 36.

(2046) Synthesis of Compound PP0053, Compound PP0060, Compound PP0106, Compound PP0107, Compound PP0108, Compound PP0109, Compound PP0110, Compound PP0112, Compound PP0113, Compound PP0114, Compound PP0115, Compound PP0116, Compound PP0117, Compound PP0118, Compound PP0119, Compound PP0120, Compound PP0121, Compound PP0122, Compound PP0123, Compound PP0124, Compound PP0126, Compound PP0127, Compound PP0128, Compound PP0129, Compound PP0130, Compound PP0131, Compound PP0132, and Compound PP0133

(2047) Compound PP0053, Compound PP0060, Compound PP0106, Compound PP0107, Compound PP0108, Compound PP0109, Compound PP0110, Compound PP0112, Compound PP0113, Compound PP0114, Compound PP0115, Compound PP0116, Compound PP0117, Compound PP0118, Compound PP0119, Compound PP0120, Compound PP0121, Compound PP0122, Compound PP0123, Compound PP0124, Compound PP0126, Compound PP0127, Compound PP0128, Compound PP0129, Compound PP0130, Compound PP0131, Compound PP0132, and Compound PP0133 were synthesized according to the following scheme.

(2048) ##STR00824##

(2049) Using Compound aa375-resin as a raw material, peptide synthesis by the basic peptide synthesis method described in the present Examples was performed to give resin-supported Compound PP53-a. After resin-supported Compound PP53-b was obtained in the same manner as synthesis of Compound PP55-b, resin-supported Compound PP53-c was obtained in the same manner as synthesis of Compound PP55-c. After resin-supported Compound PP53-d was obtained in the same manner as synthesis of Compound PP55-d using the alcohols shown in Table 27 in place of 1-butanol, resin-supported Compound PP53-e was obtained in the same manner as synthesis of Compound PP55-e. Subsequent steps of peptide elongation, cleaving from resin, cyclization, and purification were performed according to the basic route to give the intended compounds. LC/MS data is provided in Table 36.

(2050) TABLE-US-00033 TABLE 27 Intended product Alcohol Compound PP0053 1-Propanol Compound PP0060 1-Propanol Compound PP0106 1-Butanol Compound PP0107 Isobutanol Compound PP0108 Cyclopropylmethanol Compound PP0109 Cyclobutylmethanol Compound PP0110 Cyclopentylmethanol Compound PP0112 3-Methylbutan-1-ol Compound PP0113 4-Methylpentan-1-ol Compound PP0114 2-Propen-1-ol Compound PP0115 3-Buten-1-ol Compound PP0116 2-Propyn-1-ol Compound PP0117 3,3,3-Trifluoro-1-propanol Compound PP0118 2-Methoxyethanol Compound PP0119 3-Methoxy-1-propanol Compound PP0120 1-Butanol Compound PP0121 Isobutanol Compound PP0122 Cyclopropylmethanol Compound PP0123 Cyclobutylmethanol Compound PP0124 Cyclopentylmethanol Compound PP0126 3-Methylbutan-1-ol Compound PP0127 4-Methylpentan-1-ol Compound PP0128 2-Propen-1-ol Compound PP0129 3-Buten-1-ol Compound PP0130 2-Propyn-1-ol Compound PP0131 3,3,3-Trifluoro-1-propanol Compound PP0132 2-Methoxyethanol Compound PP0133 3-Methoxy-1-propanol Synthesis of Compound PP54 and Compound PP061

(2051) Compound PP54 and Compound PP061 were synthesized according to the following scheme.

(2052) ##STR00825##

(2053) Using Compound aa375-resin as a raw material, peptide synthesis by the basic peptide synthesis method described in the present Examples was performed to give resin-supported Compound PP54-a. After resin-supported Compound PP54-b was obtained in the same manner as

synthesis of Compound PP55-b, resin-supported Compound PP54-c was obtained in the same manner as synthesis of Compound PP55-c. After resin-supported Compound PP54-d was obtained in the same manner as synthesis of Compound PP55-d using 1-propanol in place of 1-butanol, resin-supported Compound PP54-e was obtained in the same manner as synthesis of Compound PP55-e. Subsequent steps of peptide elongation, cleaving from resin, cyclization, and purification were performed according to the basic route to give the intended compounds. LC/MS data is provided in Table 36.

(2054) Synthesis of Compound PP136 and Compound PP137

(2055) Compound PP136 and Compound PP137 were synthesized according to the following scheme.

(2056) ##STR00826##

(2057) Using Compound aa375-resin as a raw material, peptide synthesis by the basic peptide synthesis method described in the present Examples was performed to give resin-supported

(2058) Compound PP136-a. After resin-supported Compound PP136-b was obtained in the same manner as synthesis of Compound PP55-b, resin-supported Compound PP136-c was obtained in the same manner as synthesis of Compound PP55-c. After resin-supported Compound PP136-d was obtained in the same manner as synthesis of Compound PP55-d using 1-propanol in place of 1-butanol, resin-supported Compound PP136-e was obtained in the same manner as synthesis of Compound PP55-e. Subsequent steps of peptide elongation, cleaving from resin, cyclization, and purification were performed according to the basic route to give the intended compounds. LC/MS data is provided in Table 36.

(2059) Synthesis of Compound PP0210, Compound PP0211, Compound PP0215, and Compound PP216

(2060) Compound PP0210, Compound PP0211, Compound PP0215, and Compound PP0216 were synthesized according to the following scheme.

(2061) ##STR00827##

(2062) Using Compound aa375-resin as a raw material, peptide synthesis by the basic peptide synthesis method described in the present Examples was performed to give resin-supported Compound PP0210-a. After resin-supported Compound PP0210-b was obtained in the same manner as synthesis of Compound PP55-b, resin-supported Compound PP0210-c was obtained in the same manner as synthesis of Compound PP55-c using THE in place of NMP as a reaction solvent. After resin-supported Compound PP0210-d was obtained in the same manner as synthesis of Compound PP55-d using the alcohols shown in Table 28 in place of 1-butanol, resin-supported Compound PP0210-e was obtained in the same manner as synthesis of Compound PP55-e. Subsequent steps of peptide elongation, cleaving from resin, cyclization, and purification were performed according to the basic route to give the intended compounds. LC/MS data is provided in Table 36.

(2063) TABLE-US-00034 TABLE 28 Intended product Alcohol Compound PP0210 Ethanol Compound PP0211 1-Propanol Compound PP0215 Ethanol Compound PP0216 1-Propanol Synthesis of Compound PP0213, Compound PP0214, Compound PP0218, and Compound PP219
(2064) Compound PP0213, Compound PP0214, Compound PP0218, and Compound PP0219 were synthesized according to the following scheme.

(2065) ##STR00828## ##STR00829##

(2066) Using Compound aa375-resin as a raw material, peptide synthesis by the basic peptide synthesis method described in the present Examples was performed to give resin-supported Compound PP0213-a. After resin-supported Compound PP0213-b was obtained in the same manner as synthesis of Compound PP55-b, resin-supported Compound PP0213-c was obtained in the same manner as synthesis of Compound PP55-c using THF in place of NMP as a reaction solvent. After resin-supported Compound PP0213-d was obtained in the same manner as synthesis of Compound PP55-d using the alcohols shown in Table 29 in place of 1-butanol, resin-supported

Compound PP0213-e was obtained in the same manner as synthesis of Compound PP55-e. Subsequent steps of peptide elongation, cleaving from resin, cyclization, and purification were performed according to the basic route to give the intended compounds. LC/MS data is provided in Table 36.

(2067) TABLE-US-00035 TABLE 29 Intended product Alcohol Compound PP0213 Ethanol Compound PP0214 1-Propanol Compound PP0218 Ethanol Compound PP0219 1-Propanol Synthesis of Compound PP524 and Compound PP525

(2068) Compound PP524 and Compound PP525 were synthesized according to the following scheme.

(2069) ##STR00830##

(2070) DCM (1.5 mL) was added to swell resin-supported PP0053-c (210 mg). After DCM was removed, the resin was washed 4 times with THF. Triphenylphosphine (173 mg, 0.661 mmol) was dissolved in THF (0.600 mL), a mixed solution of DIAD (0.128 mL, 0.661 mmol) and THF (0.572 mL) was added thereto, and the mixture was stirred for 15 minutes. 3-Fluoropropan-1-ol (0.099 mL, 1.32 mmol) was added thereto, and the mixture was left to stand still for 5 minutes. This solution was added to the resin, the mixture was shaken at room temperature for 1 hour, and then the reaction solution was removed. The resin was washed 4 times with THF (1.5 mL) and washed 4 times with DCM (1.5 mL) to give resin-supported Compound PP524-d. Cleaving from the resin was performed with TFE/DCM (1/1) using a small amount of resin-supported Compound PP524-d, and the structure was verified by LC/MS.

(2071) LCMS (ESI) $m/z=1359$ (M+H)⁺

(2072) Retention time: 0.98 min (Analytical condition SQDFA05)

(2073) Using this, resin-supported Compound PP524-e was obtained in the same manner as synthesis of Compound PP55-e. Subsequent steps of peptide elongation, cleaving from resin, cyclization, and purification were performed according to the basic route to give the intended compounds. LC/MS data is provided in Table 36.

(2074) Synthesis of Compound PP526 and Compound PP527

(2075) Compound PP526 and Compound PP527 were synthesized according to the following scheme.

(2076) ##STR00831##

(2077) Using Compound aa375-resin as a raw material, peptide synthesis by the basic peptide synthesis method described in the present Examples was performed to give resin-supported Compound PP526-a. After resin-supported Compound PP526-b was obtained in the same manner as synthesis of Compound PP55-b, resin-supported Compound PP526-c was obtained in the same manner as synthesis of Compound PP55-c using THE in place of NMP as a reaction solvent. After resin-supported Compound PP526-d was obtained in the same manner as synthesis of Compound PP524-d allowing 1-propanol to react in place of 3-fluoropropan-1-ol, resin-supported Compound PP526-e was obtained in the same manner as synthesis of Compound PP55-e. Subsequent steps of peptide elongation, cleaving from resin, cyclization, and purification were performed according to the basic route to give the intended compounds. LC/MS data is provided in Table 36.

(2078) Synthesis of Compound PP599

(2079) Compound PP599 was synthesized according to the following scheme.

(2080) ##STR00832##

(2081) Using resin-supported PP210-c, resin-supported Compound PP599-d was obtained in the same manner as synthesis of Compound PP524-d using 2-propen-1-ol in place of 3-fluoropropan-1-ol, and then resin-supported Compound PP599-e was obtained in the same manner as synthesis of Compound PP55-e. Subsequent steps of peptide elongation, cleaving from resin, cyclization, and purification were performed according to the basic route to give the intended compound. LC/MS data is provided in Table 36.

(2082) Synthesis of Compound PP0605

(2083) Compound PP0605 was synthesized according to the following scheme.

(2084) ##STR00833## ##STR00834## ##STR00835##

(2085) Using Compound aa375-resin as a raw material, peptide synthesis by the basic peptide synthesis method described in the present Examples was performed to give resin-supported Compound PP0605-a. After resin-supported Compound PP0605-b was obtained in the same manner as synthesis of Compound PP55-b, resin-supported Compound PP0605-c was obtained in the same manner as synthesis of Compound PP55-c using THF in place of NMP as a reaction solvent. After resin-supported Compound PP0605-d was obtained in the same manner as synthesis of Compound PP524-d using 2-propen-1-ol in place of 3-fluoropropan-1-ol, resin-supported Compound PP0605-e was obtained in the same manner as synthesis of Compound PP55-e. Subsequent steps of peptide elongation, cleaving from resin, cyclization, and purification were performed according to the basic route to give the intended compound. LC/MS data is provided in Table 36.

(2086) Abbreviations and structures of amino acids contained as partial structures in a cyclic compound and an oligopeptide compound synthesized via N-alkylation by a Mitsunobu reaction on a resin are provided below.

(2087) TABLE-US-00036 TABLE 29-1 Compound Amino acid No. Abbreviation Structural Formula
bb038 Phe(4-Me)  bb039 nBuPhe(4-Me)  bb040
iBuPhe(4-Me)  bb041 cPrMePhe(4-Me)  bb042
cBuMePhe(4-Me)  bb043 cPentMePhe(4-Me)  bb044
cHexMePhe(4-Me)  bb045 iPenPhe(4-Me)  bb046
neoHexPhe(4-Me)  bb047 AllylPhe(4-Me)  bb048 PraPhe(4-Me)
 bb049 TfpPhe(4-Me)  bb050 MeOEtPhe(4-Me)
 bb051 MeOnPrPhe(4-Me)  bb052 nBuPhe(4-CF3)  bb053 iBuPhe(4-CF3)  bb054 cPrMePhe(4-CF3)
 bb055 cBuMePhe(4-CF3)  bb056 cPentMePhe(4-CF3)
 bb057 cHexMePhe(4-CF3)  bb058 iPenPhe(4-CF3)
 bb059 neoHexPhe(4-CF3)  bb060 PraPhe(4-CF3)
 bb061 TfpPhe(4-CF3)  bb062 MeOEtPhe(4-CF3)  0
 bb063 MeOnPrPhe(4-CF3)  bb064 nPrGly(cPent)
 bb065 Phe(4-OCF3)  bb066 nPrPhe(4-OCF3)
 bb067 Phe  bb068 EtPhe  bb069 nPrPhe
 bb070 EtPhe(4-F)  bb071 nPrPhe(4-F)  bb072 MfpPhe(4-CF3)  bb073 nPrCha(4-F2)  bb074 AllylPRA
 bb075 AllylPhe(4-Cl)  bb076 AllylPhe(4-OMe)
 bb077 AllylPhe(4-OCF3) 

1-6-5. Peptide Modification by Olefin Metathesis Reaction

Synthesis of Compound PP0616

(2088) Compound PP0616 was synthesized according to the following scheme.

(2089) ##STR00876##

(2090) Using Compound aa374-resin as a raw material, peptide synthesis by the basic peptide synthesis method described in the present Examples was performed to give Compound PP0616-a (6.5 mg, 4.36 μ mol).

(2091) LCMS (ESI) m/z=1491 (M+H)⁺

(2092) Retention time: 0.82 min (Analytical condition SQDFA50)

(2093) Compound PP0616-a (6.5 mg, 4.36 μ mol) and a Stewart-Grubbs catalyst (2.5 mg, 4.36 μ mol) were dissolved in toluene (1 mL), and the mixture was stirred at 80° C. for 24.5 hours in a nitrogen atmosphere. After cooling to room temperature, the solvent was distilled off under reduced pressure. The resulting crude product was purified by reverse phase HPLC(methanol/50 mM aqueous ammonium acetate solution) to give PP0616 (3.3 mg, 52%).

(2094) LC/MS data is provided in Table 36.

(2095) Synthesis of Compound PP0617

(2096) Compound PP0617 was synthesized according to the following scheme.

(2097) ##STR00877##

(2098) Using Compound aa374-resin as a raw material, peptide synthesis by the basic peptide synthesis method described in the present Examples was performed to give Compound PP0617-a (8.4 mg, 5.44 μ mol). Using the resulting PP0617-a, PP0617 (1.7 mg, 22%) was obtained in the same manner as synthesis of Compound PP0616. LC/MS data is provided in Table 36.

(2099) Synthesis of Compound PP0618

(2100) Compound PP0618 was synthesized according to the following scheme.

(2101) ##STR00878##

(2102) Using Compound aa374-resin as a raw material, peptide synthesis by the basic peptide synthesis method described in the present Examples was performed to give Compound PP0618-a (23.9 mg, 16.6 μ mol). Using the resulting PP0618-a, PP0618 (2.4 mg, 11%) was obtained in the same manner as synthesis of Compound PP0616. LC/MS data is provided in Table 36.

(2103) Synthesis of Compound PP0619

(2104) Compound PP0619 was synthesized according to the following scheme.

(2105) ##STR00879##

(2106) Using Compound aa374-resin as a raw material, peptide synthesis by the basic peptide synthesis method described in the present Examples was performed to give Compound PP0619-a (17.6 mg, 11.4 μ mol). Using the resulting PP0619-a, PP0619 (0.57 mg, 3%) was obtained in the same manner as synthesis of Compound PP0616. LC/MS data is provided in Table 36.

(2107) Synthesis of Compound PP0620

(2108) Compound PP0620 was synthesized according to the following scheme.

(2109) ##STR00880##

(2110) Using Compound aa374-resin as a raw material, peptide synthesis by the basic peptide synthesis method described in the present Examples was performed to give Compound PP0620-a (13.7 mg, 9.3 μ mol). Using the resulting PP0620-a, PP0620 (6.7 mg, 53%) was obtained in the same manner as synthesis of Compound PP0616. LC/MS data is provided in Table 36.

(2111) Synthesis of Compound PP0621

(2112) Compound PP0621 was synthesized according to the following scheme.

(2113) ##STR00881##

(2114) Using Compound aa374-resin as a raw material, peptide synthesis by the basic peptide synthesis method described in the present Examples was performed to give Compound PP0621-a (8.3 mg, 5.4 μ mol). Using the resulting PP0621-a, PP0621 (3.9 mg, 44%) was obtained in the same manner as synthesis of Compound PP0616. LC/MS data is provided in Table 36.

(2115) Synthesis of Compound PP0622

(2116) Compound PP0622 was synthesized according to the following scheme.

(2117) ##STR00882##

(2118) Using Compound aa374-resin as a raw material, peptide synthesis by the basic peptide synthesis method described in the present Examples was performed to give Compound PP0622-a (7.6 mg, 4.9 μ mol). Using the resulting PP0622-a, PP0622 (0.25 mg, 4%) was obtained in the same manner as synthesis of Compound PP0616. LC/MS data is provided in Table 36.

(2119) Synthesis of Compound PP0623

(2120) Compound PP0623 was synthesized according to the following scheme.

(2121) ##STR00883##

(2122) Using Compound aa374-resin as a raw material, peptide synthesis by the basic peptide synthesis method described in the present Examples was performed to give Compound PP0623-a (10.7 mg, 7.1 μ mol). Using the resulting PP0623-a, PP0623 (0.4 mg, 4%) was obtained in the same manner as synthesis of Compound PP0616. LC/MS data is provided in Table 36.

(2123) Synthesis of Compound PP0624

(2124) Compound PP0624 was synthesized according to the following scheme.

(2125) ##STR00884##

(2126) Using Compound aa374-resin as a raw material, peptide synthesis by the basic peptide synthesis method described in the present Examples was performed to give Compound PP0624-a (9.7 mg, 6.2 μ mol). Using the resulting PP0624-a, PP0624 (0.3 mg, 3%) was obtained in the same manner as synthesis of Compound PP0616. LC/MS data is provided in Table 36.

(2127) Synthesis of Compound PP0625

(2128) Compound PP0625 was synthesized according to the following scheme.

(2129) ##STR00885##

(2130) Using Compound aa374-resin as a raw material, peptide synthesis by the basic peptide synthesis method described in the present Examples was performed to give Compound PP0625-a (12.3 mg, 8.2 μ mol). Using the resulting PP0625-a, PP0625 (0.3 mg, 3%) was obtained in the same manner as synthesis of Compound PP0616. LC/MS data is provided in Table 36.

(2131) Synthesis of Compound PP0626

(2132) Compound PP0626 was synthesized according to the following scheme.

(2133) ##STR00886##

(2134) Using Compound aa374-resin as a raw material, peptide synthesis by the basic peptide synthesis method described in the present Examples was performed to give Compound PP0626-a (4.7 mg, 3.0 μ mol). Using the resulting PP0626-a, PP0626 (1.6 mg, 40%) was obtained in the same manner as synthesis of Compound PP0616. LC/MS data is provided in Table 36.

(2135) Synthesis of Compound PP0627

(2136) Compound PP0627 was synthesized according to the following scheme.

(2137) ##STR00887##

(2138) Using Compound aa374-resin as a raw material, peptide synthesis by the basic peptide synthesis method described in the present Examples was performed to give Compound PP0627-a (6.6 mg, 4.4 μ mol). Using the resulting PP0627-a, PP0627 (1.2 mg, 20%) was obtained in the same manner as synthesis of Compound PP0616. LC/MS data is provided in Table 36.

(2139) Synthesis of Compound PP0628

(2140) Compound PP0628 was synthesized according to the following scheme.

(2141) ##STR00888##

(2142) Using Compound aa374-resin as a raw material, peptide synthesis by the basic peptide synthesis method described in the present Examples was performed to give Compound PP0628-a (16.6 mg, 10.7 μ mol). Using the resulting PP0628-a, PP0628 (0.5 mg, 3%) was obtained in the same manner as synthesis of Compound PP0616. LC/MS data is provided in Table 36.

(2143) Synthesis of Compound PP0629

(2144) Compound PP0629 was synthesized according to the following scheme.

(2145) ##STR00889##

(2146) Using Compound aa374-resin as a raw material, peptide synthesis by the basic peptide synthesis method described in the present Examples was performed to give Compound PP0629-a (12.3 mg, 8.2 μ mol). Using the resulting PP0629-a, PP0629 (0.7 mg, 6%) was obtained in the same manner as synthesis of Compound PP0616. LC/MS data is provided in Table 36.

(2147) Synthesis of Compound PP1332

(2148) Compound PP1332 was synthesized according to the following scheme.

(2149) ##STR00890##

(2150) Using Compound aa374-resin as a raw material, peptide synthesis by the basic peptide synthesis method described in the present Examples was performed to give Compound PP1332-a (40.9 mg, 25.6 μ mol). The resulting Compound PP1332-a (40 mg) and 1,3-bis(2,4,6-trimethylphenyl)-2-imidazolidinylidene(3-phenyl-1H-inden-1-ylidene)(4,5-dichloro-1,3-diethyl-1,3-dihydro-2H-imidazol-2-ylidene) ruthenium (II) chloride (22 mg, 0.025 mmol) were dissolved

in toluene (3 mL), and the mixture was stirred at 100° C. for 13.5 hours in a nitrogen atmosphere. After cooling to room temperature, the solvent was distilled off under reduced pressure. The resulting crude product was purified by reverse phase HPLC(methanol/50 mM aqueous ammonium acetate solution) to give PP1332 (11.5 mg, 17%). LC/MS data is provided in Table 36.

(2151) Synthesis of Compound PP1333

(2152) Compound PP1333 was synthesized according to the following scheme.

(2153) ##STR00891##

(2154) Using Compound aa375-resin as a raw material, peptide synthesis by the basic peptide synthesis method described in the present Examples was performed to give Compound PP1333-a (55.1 mg, 34.1 μmol). Using the resulting PP1333-a, PP1333 (8.2 mg, 15%) was obtained in the same manner as synthesis of Compound PP1332. LC/MS data is provided in Table 36.

(2155) Synthesis of Compound PP1334

(2156) Compound PP1334 was synthesized according to the following scheme.

(2157) ##STR00892##

(2158) Using Compound aa374-resin as a raw material, peptide synthesis by the basic peptide synthesis method described in the present Examples was performed to give Compound PP1334-a (76.1 mg, 49.1 μmol). Using the resulting PP1334-a, PP1334 (2.8 mg, 4%) was obtained in the same manner as synthesis of Compound PP1332. LC/MS data is provided in Table 36.

(2159) Synthesis of Compound PP1335

(2160) Compound PP1335 was synthesized according to the following scheme.

(2161) ##STR00893##

(2162) Using Compound aa375-resin as a raw material, peptide synthesis by the basic peptide synthesis method described in the present Examples was performed to give Compound PP1335-a (71.8 mg, 45.5 μmol). Using the resulting PP1335-a, PP1335 (2.5 mg, 4%) was obtained in the same manner as synthesis of Compound PP1332. LC/MS data is provided in Table 36.

(2163) Synthesis of Compound PP0894

(2164) Compound PP0894 was synthesized according to the following scheme.

(2165) ##STR00894##

(2166) Using Compound aa374-resin as a raw material, peptide synthesis by the basic peptide synthesis method described in the present Examples was performed to give Compound PP0894-a (26.5 mg, 18 μmol).

(2167) LCMS (ESI) m/z=1496 (M+H)+

(2168) Retention time: 0.78 min (Analytical condition SQDAA50)

(2169) Compound PP0894-a (26.5 mg, 18 μmol) and a Stewart-Grubbs catalyst (5.0 mg, 8.76 μmol) were dissolved in 1,2-dichloroethane (0.58 mL), and the mixture was stirred at 60° C. for 20 hours in a nitrogen atmosphere. After cooling to room temperature, triethylamine (98 μL, 0.701 mmol) and 2-nitrobenzenesulfonohydrazide (152 mg, 0.701 mmol) were added, and the mixture was stirred at room temperature for 5 days. Then, the reaction solution was concentrated under reduced pressure. The resulting crude product was purified by reverse phase HPLC(methanol/50 mM aqueous ammonium acetate solution) to give PP0894 (4.5 mg, 17%). LC/MS data is provided in Table 36.

(2170) Synthesis of Compound PP0895 and Compound PP900

(2171) Compounds PP0895 and PP900 were synthesized according to the following scheme.

(2172) ##STR00895##

(2173) Using Compound aa374-resin as a raw material, peptide synthesis by the basic peptide synthesis method described in the present Examples was performed to give Compound PP0895-a. The resulting Compound PP0895-a (26.5 mg, 18 μmol) and a Stewart-Grubbs catalyst (5.0 mg, 8.76 μmol) were dissolved in 1,2-dichloroethane (0.58 mL), and the mixture was stirred at 60° C. for 20 hours in a nitrogen atmosphere. One-third of the reaction solution was recovered and concentrated under reduced pressure, and the resulting crude product was purified by reverse phase

HPLC(methanol/50 mM aqueous ammonium acetate solution) to give PP0900 (2.1 mg, 8.1%).

LC/MS data is provided in Table 36.

(2174) After the remaining reaction solution was cooled to room temperature, triethylamine (98 μ L, 0.701 mmol) and 2-nitrobenzenesulfonohydrazide (152 mg, 0.701 mmol) were added, and the mixture was stirred at room temperature for 5 days. Then, the reaction solution was concentrated under reduced pressure. The resulting crude product was purified by reverse phase

HPLC(methanol/50 mM aqueous ammonium acetate solution) to give PP0895 (4.4 mg, 17%).

LC/MS data is provided in Table 36.

(2175) Synthesis of Compound PP0896

(2176) Compound PP0896 was synthesized according to the following scheme.

(2177) ##STR00896##

(2178) Using Compound aa374-resin as a raw material, peptide synthesis by the basic peptide synthesis method described in the present Examples was performed to give Compound PP0896-a.

Using the resulting PP0896-a, PP0896 (4.5 mg, 17%) was obtained in the same manner as synthesis of Compound PP0894. LC/MS data is provided in Table 36.

(2179) Synthesis of Compound PP0898 and Compound PP902

(2180) Compounds PP0898 and PP902 were synthesized according to the following scheme.

(2181) ##STR00897##

(2182) Using, as a raw material, resin-supported PP0874-b that was a synthesis intermediate of Compound PP0874 synthesized using Compound aa375-resin, peptide synthesis by the basic peptide synthesis method described in the present Examples was performed to give Compound PP0898-a. Using the resulting PP0898-a, PP902 (1.1 mg, 4.2%) was obtained in the same manner as synthesis of Compound PP900. Using Compound PP902, PP0898 (2.0 mg, 7.7%) was obtained in the same manner as synthesis of Compound PP0895. LC/MS data is provided in Table 36.

(2183) Synthesis of Compound PP0899 and Compound PP903

(2184) Compounds PP0899 and PP903 were synthesized according to the following scheme.

(2185) ##STR00898##

(2186) Using, as a raw material, resin-supported PP0874-b that was a synthesis intermediate of Compound PP0874 synthesized using Compound aa375-resin, peptide synthesis by the basic peptide synthesis method described in the present Examples was performed to give Compound PP0899-a. Using the resulting PP0899-a, PP903 (1.1 mg, 4%) was obtained in the same manner as synthesis of Compound PP900. Using Compound PP903, PP0899 (3.1 mg, 12%) was obtained in the same manner as synthesis of Compound PP0895. LC/MS data is provided in Table 36.

(2187) Synthesis of Compound PP901

(2188) Compound PP901 was synthesized according to the following scheme.

(2189) ##STR00899##

(2190) Using, as a raw material, resin-supported synthesis intermediate PP0874-b of Compound PP0874 synthesized using Compound aa375-resin, peptide synthesis by the basic peptide synthesis method described in the present Examples was performed to give Compound PP0901-a. The resulting Compound PP0901-a (26.5 mg, 18 μ mol) and a Stewart-Grubbs catalyst (5.0 mg, 8.76 μ mol) were dissolved in 1,2-dichloroethane (0.58 mL), and the mixture was stirred at 60° C. for 20 hours in a nitrogen atmosphere. After being cooled to room temperature, the reaction solution was concentrated under reduced pressure. The resulting crude product was purified by reverse phase HPLC(methanol/50 mM aqueous ammonium acetate solution) to give PP0901 (0.8 mg, 3%).

LC/MS data is provided in Table 36.

(2191) Synthesis of Compound PP925

(2192) Compound PP925 was synthesized according to the following scheme.

(2193) ##STR00900##

(2194) Using Compound aa374-resin as a raw material, peptide synthesis by the basic peptide synthesis method described in the present Examples was performed to give Compound PP0925-a.

Using the resulting PP925-a, PP925 (1.1 mg, 4%) was obtained in the same manner as synthesis of Compound PP0894. LC/MS data is provided in Table 36.

(2195) Synthesis of Compound PP992

(2196) Compound PP992 was synthesized according to the following scheme.

(2197) ##STR00901## ##STR00902##

(2198) Using, as a raw material, resin-supported synthesis intermediate PP0874-b of Compound PP0874 synthesized using Compound aa375-resin, peptide synthesis by the basic peptide synthesis method described in the present Examples was performed to give Compound PP0992-a (10.07 mg, 6.33 μ mol).

(2199) LCMS (ESI) $m/z=1590$ (M+H)+

(2200) Retention time: 1.85 min (Analytical condition SQDFA50 long)

(2201) Compound PP992-a (9.09 mg, 5.72 μ mol) and a Stewart-Grubbs catalyst (1.8 mg, 3.1 μ mol) were dissolved in toluene (0.41 mL), and the mixture was stirred at 70° C. for 5 hours in a nitrogen atmosphere. After cooling to room temperature, the solvent was distilled off under reduced pressure. The resulting crude product was purified by reverse phase HPLC(methanol/50 mM aqueous ammonium acetate solution) to give PP0992-b (3.5 mg, 36%).

(2202) LCMS (ESI) $m/z=1562$ (M+H)+

(2203) Retention time: 0.95 min (Analytical condition SQDFA05)

(2204) Compound PP992-b (3.5 mg, 2.24 μ mol) was dissolved in ethanol (1.5 mL.), palladium hydroxide on carbon (20 wt %, 10 mg, 0.028 mmol) was added, and the mixture was stirred at room temperature for 2 hours in a hydrogen atmosphere. Subsequently, the reaction solution was filtered through Celite, Celite was washed with ethanol, and then the filtrate was concentrated under reduced pressure. The resulting crude product was purified by reverse phase HPLC(methanol/50 mM aqueous ammonium acetate solution) to give PP0992 (1.3 mg, 26%).

(2205) LCMS (ESI) $m/z=1564$ (M+H)+

(2206) Retention time: 1.22 min (Analytical condition SQDFA50 long)

(2207) Synthesis of Compound PP993

(2208) Compound PP993 was synthesized according to the following scheme.

(2209) ##STR00903## ##STR00904##

(2210) Using, as a raw material, resin-supported synthesis intermediate PP0874-b of Compound PP0874 synthesized using Compound aa375-resin, peptide synthesis by the basic peptide synthesis method described in the present Examples was performed to give Compound PP0993-a (10.6 mg). Using the resulting PP993-a, PP993-b was obtained in the same manner as synthesis of Compound PP992-b, and then PP993 (1.79 mg, 18%, 2 steps) was obtained in the same manner as synthesis of Compound PP992. LC/MS data is provided in Table 36.

(2211) Synthesis of Compound PP997

(2212) Compound PP997 was synthesized according to the following scheme.

(2213) ##STR00905## ##STR00906##

(2214) Using, as a raw material, resin-supported synthesis intermediate PP0874-b of Compound PP0874 synthesized using Compound aa375-resin, peptide synthesis by the basic peptide synthesis method described in the present Examples was performed to give Compound PP0997-a (9.5 mg). Using the resulting PP997-a, PP997-b was obtained in the same manner as synthesis of Compound PP992-b. A half of this was reacted in the same manner as Compound PP992, the other half was reacted using 10% palladium on carbon in place of palladium hydroxide, and the resulting compounds were combined to give PP0997 (1.56 mg, 31%, 2 steps). LC/MS data is provided in Table 36.

(2215) Synthesis of Compound PP1828

(2216) Synthesis of PP1828 was carried out according to the following scheme.

(2217) ##STR00907##

(2218) Using Compound aa440-resin (Fmoc-MeGly (cPent)-MeAsp (O-Trt (2-C1) resin)-MeNEt)

as a raw material, peptide synthesis was carried out by the basic peptide synthesis method described in the present Examples to give Compound PP1828-a. Compound PP1828-a (10.9 mg, 7.28 μ mol), 1,4-benzoquinone (3.6 mg, 33 μ mol), and a Stewart-Grubbs catalyst (3.2 mg, 5.6 μ mol) were dissolved in 1,4-dioxane (2.0 mL), and the mixture was stirred at 80° C. for 1.5 hours in a nitrogen atmosphere. After being cooled to room temperature, the reaction solution was purified by reverse phase HPLC(methanol/50 mM aqueous ammonium acetate solution) to give PP1828 (3.08 mg, 5.9%).

(2219) LCMS (ESI) m/z =1498 (M+H)+

(2220) Retention time: 3.09 min (Analytical condition SQDFA05 long)

(2221) Synthesis of Compound PP1827

(2222) Synthesis of PP1827 was carried out according to the following scheme.

(2223) ##STR00908##

(2224) Using Compound aa440-resin (Fmoc-MeGly (cPent)-MeAsp (O-Trt (2-C1) resin)-MeNEt) as a raw material, peptide synthesis was carried out by the basic peptide synthesis method described in the present Examples to give Compound PP1827-a. The resulting compound was divided into 3 containers, and subjected to the following experiments. Compound PP1827-a (2.6 mg, 1.7 μ mol), 1,4-benzoquinone (1.1 mg, 10.2 μ mol), and a Stewart-Grubbs catalyst (1.0 mg, 1.7 μ mol) were dissolved in 1,4-dioxane (0.66 mL), and the mixture was stirred at 80° C. for 2 hours in a nitrogen atmosphere and cooled to room temperature. Similarly, the entirety of a reaction solution obtained by stirring Compound PP1827-a (2.7 mg), the Stewart-Grubbs catalyst (1.0 mg, 1.7 μ mol), and 1,4-dioxane (0.66 mL) which are other than 1,4-benzoquinone at 80° C. for 2 hours and the entirety of a reaction solution obtained by stirring PP1827-a (1.6 mg), a Hoveyda-Grubbs first-generation catalyst (1.0 mg, 1.7 μ mol), and 1,4-dioxane (0.66 mL) at 80° C. for 2 hours were mixed, and concentrated under reduced pressure. The resulting crude product was purified by reverse phase HPLC(methanol/50 mM aqueous ammonium acetate solution) to give PP1827 (2.1 mg, 3.9%).

(2225) LCMS (ESI) m/z =1504 (M+H)+

(2226) Retention time: 3.17 min (Analytical condition SQDFA05 long)

(2227) Using any of Compounds aa359-resin, aa360-resin, aa362-resin, aa374-resin, aa429-resin, and aa440-resin as a raw material, peptide synthesis was carried out by the basic peptide synthesis method described in the present Examples to give a cyclic peptide as an intermediate. Using the resulting intermediate peptide, compounds shown in Table 29-2 were produced in the same manner as the synthesis of Compound PP1828 by reacting the side chain moiety of any of MeAlgly, MeAhxe (2), MeAhpe (2), MeAocte (2), and MeAnone (2) with the styrene moiety of ButenylPhe (4—CH=CH.sub.2). Purification was carried out by reverse phase HPLC(methanol/50 mM aqueous ammonium acetate solution or 0.1% formic acid-containing acetonitrile/0.1% formic acid-containing distilled water). The yield (amount/percentage) is provided in Table 29-3. LC/MS data are provided in Table 36.

(2228) ##STR00909##

(2229) TABLE-US-00037

TABLE 29-2	ID	core 1	core 2	core 3	core 4	core 5	core 6	core 7	PP1829
MeAocte(2)	Ile	MeGly	MeAlgly	ButenylPhe(4-CH=CH ₂)	MeGly	Hph(4-CF ₃ -35-F ₂)			PP1830
MeAocte(2)	Ile	MeGly	MeAlgly	ButenylPhe(4-CH=CH ₂)	MeGly	Hph(4-CF ₃ -3-OMe)			PP1831
MeAocte(2)	Ile	MeGly	MeAlgly	ButenylPhe(4-CH=CH ₂)	MeGly	Hph(4-CF ₃ -35-F ₂)			PP1832
MeAocte(2)	Ile	MeGly	MeAlgly	ButenylPhe(4-CH=CH ₂)	MeGly	Hph(4-CF ₃ -3-OMe)			PP1833
MeAhpe(2)	Ile	MeGly	MeAlgly	ButenylPhe(4-CH=CH ₂)	MeGly	Hph(4-CF ₃ -35-F ₂)			PP1834
MeAhpe(2)	Ile	MeGly	MeAlgly	ButenylPhe(4-CH=CH ₂)	MeGly	Hph(4-CF ₃ -3-OMe)			PP1835
MeAhpe(2)	Ile	MeGly	MeAlgly	ButenylPhe(4-CH=CH ₂)	MeGly	Hph(4-CF ₃ -35-F ₂)			PP1827
MeAocte(2)	Ile	MeGly	MeAlgly	ButenylPhe(4-CH=CH ₂)	MeGly	Hph(4-CF ₃ -35-F ₂)			PP1828
MeAocte(2)	Ile	MeGly	MeAlgly	ButenylPhe(4-CH=CH ₂)	MeGly	Hph(4-CF ₃ -3-OMe)			PP2573
MeAhpe(2)	Ile	MeGly	MeAlgly	ButenylPhe(4-CH=CH ₂)	MeGly	Hph(4-CF ₃ -35-F ₂)			PP2574

[illegible]

ButenylPhe(4-CH=CH2) MeGly Hph(34-Cl2) PP3054 MeAhpe(2) Ile MeGly MeAlgly
ButenylPhe(4-CH=CH2) MeGly Hph(34-Cl2) PP3055 MeAhpe(2) Ile MeGly MeAlgly
ButenylPhe(4-CH=CH2) MeGly Hph(34-Cl2) PP3056 MeAlgly Ile MeGly MeAlgly
ButenylPhe(4-CH=CH2) MeGly Hph(34-Cl2) PP3057 MeAnone(2) Ile MeGly MeAlgly
ButenylPhe(4-CH=CH2) MeGly Hph(34-Cl2) PP3058 MeAhpe(2) Ile MeGly MeAlgly
ButenylPhe(4-CH=CH2) MeGly Hph(34-Cl2) PP3059 MeAhpe(2) Ile MeGly MeAlgly
ButenylPhe(4-CH=CH2) MeGly Hph(34-Cl2) PP3060 MeAhpe(2) Ile MeGly MeAlgly
ButenylPhe(4-CH=CH2) MeGly Hph(4-CF3-35-F2) PP3061 MeAhpe(2) Ile cPrGly MeAlgly
ButenylPhe(4-CH=CH2) MeGly Hph(34-Cl2) PP3062 MeAocte(2) Ile MeGly MeAlgly
ButenylPhe(4-CH=CH2) MeGly Hph(34-Cl2) ID core 8 core 9 core 10 core 11 C term PP1829
Hyp(Et) cVal(3-Me2) MeGly(cPent) MeAsp2 MeNEt PP1830 Hyp(Et) cVal(3-Me2) MeGly(cPent)
MeAsp2 MeNEt PP1831 Hyp(Et) cVal(3-Me2) MeGly(cPent) MeAsp2 NMe2 PP1832 Hyp(Et)
cVal(3-Me2) MeGly(cPent) MeAsp2 NMe2 PP1833 Hyp(Et) cVal MeGly(cPent) MeAsp2 MeNEt
PP1834 Hyp(Et) cVal MeGly(cPent) MeAsp2 MeNEt PP1835 Hyp(Et) cVal(3-Me2) MeGly(cPent)
MeAsp2 MeNEt PP1827 Hyp(Et) cVal MeGly(cPent) MeAsp2 MeNEt PP1828 Hyp(Et) cVal
MeGly(cPent) MeAsp2 MeNEt PP2573 Hyp(Et) cVal(3-Me2) MeGly(cPent) MeAsp2 pip PP2574
Hyp(Et) cVal(3-Me2) MeGly(cPent) MeAsp2 pyrro PP2575 Hyp(Et) cVal(3-Me2) MeNva(3-Et)
MeAsp2 NMe2 PP2576 Hyp(Et) cVal(3-Me2) MeNva(3-Et) MeAsp2 pyrro PP2577 Hyp(Et)
cVal(3-Me2) MeGly(cPent) MeAsp2 MeNEt PP2578 Hyp(Et) cVal(3-Me2) MeGly(cPent) MeAsp2
pip PP2579 Hyp(Et) cVal(3-Me2) MeGly(cPent) MeAsp2 pyrro PP2580 Hyp(Et) cVal(3-Me2)
MeNva(3-Et) MeAsp2 NMe2 PP2581 Hyp(Et) cVal(3-Me2) MeNva(3-Et) MeAsp2 pyrro PP2583
Hyp(Et) cVal(3-Me2) MeGly(cPent) MeAsp2 pyrro PP2585 Hyp(Et) cVal(3-Me2) MeGly(cPent)
MeAsp2 pyrro PP2586 Hyp(Et) cVal(3-Me2) MeNva(3-Et) MeAsp2 NMe2 PP2588 Hyp(Et)
cVal(3-Me2) MeGly(cPent) MeAsp2 pyrro PP2589 Hyp(Et) cVal(3-Me2) MeNva(3-Et) MeAsp2
NMe2 PP2590 Hyp(Et) cVal(3-Me2) MeNva(3-Et) MeAsp2 pip PP2591 Hyp(Et) cVal(3-Me2)
MeNva(3-Et) MeAsp2 pyrro PP2592 Hyp(Et) cVal(3-Me2) MeGly(cPent) MeAsp2 MeNEt
PP2593 Hyp(Et) cVal(3-Me2) MeGly(cPent) MeAsp2 pyrro PP2594 Hyp(iPr) cVal(3-Me2)
MeNva(3-Et) MeAsp2 NMe2 PP2595 Hyp(iPr) cVal(3-Me2) MeGly(cPent) MeAsp2 MeNEt
PP2596 Hyp(iPr) cVal(3-Me2) MeGly(cPent) MeAsp2 pyrro PP2597 Hyp(Et) cVal(3-Me2)
MeGly(cPent) MeAsp2 pip PP2598 Hyp(Et) cVal(3-Me2) MeGly(cPent) MeAsp2 MeNEt PP2600
Pro cVal(3-Me2) MeNva(3-Et) MeAsp2 NMe2 PP2601 Pro cVal(3-Me2) MeGly(cPent) MeAsp2
MeNEt PP2602 Pro cVal(3-Me2) MeGly(cPent) MeAsp2 pyrro PP2603 Pro cVal(3-Me2)
MeNva(3-Et) MeAsp2 NMe2 PP2604 Pro cVal(3-Me2) MeNva(3-Et) MeAsp2 pyrro PP2605
Hyp(Et) cVal MeNva(3-Et) MeAsp2 NMe2 PP2952 Hyp(Et) cVal(3-Me2) MeNva(3-Et) MeAsp2
pip PP2953 Hyp(Et) cVal(3-Me2) MeNva(3-Et) MeAsp2 pip PP2954 Hyp(Et) cVal MeGly(cPent)
MeAsp2 pip PP2955 Hyp(Et) cVal(3-Me2) MeNva(3-Et) MeAsp2 NMe2 PP2956 Hyp(Et) cVal
MeNva(3-Et) MeAsp2 pip PP2957 Hyp(Et) cVal MeNva(3-Et) MeAsp2 pyrro PP2958 Hyp(Et)
cVal MeGly(cPent) MeAsp2 pip PP2959 Hyp(Et) cVal(3-Me2) MeNva(3-Et) MeAsp2 NMe2
PP3039 Hyp(Et) cVal MeNva(3-Et) MeAsp2 pip PP3040 Hyp(Et) cVal MeNva(3-Et) MeAsp2
pyrro PP3041 Hyp(Et) cVal(3-Me2) MeGly(cPent) MeAsp2 pyrro PP3042 Hyp(Et) cVal(3-Me2)
MeNva(3-Et) MeAsp2 pip PP3043 Hyp(Et) cVal(3-Me2) MeNva(3-Et) MeAsp2 pyrro PP3044
Hyp(Et) cVal(3-Me2) MeNva(3-Et) MeAsp2 NMe2 PP3045 Hyp(Et) cVal(3-Me2) MeNva(3-Et)
MeAsp2 NMe2 PP3046 Hyp(Et) cVal MeNva(3-Et) MeAsp2 pip PP3047 Hyp(Et) cVal MeNva(3-
Et) MeAsp2 pyrro PP3048 Hyp(Et) cVal(3-Me2) MeNva(3-Et) MeAsp2 NMe2 PP3049 Hyp(Et)
cVal MeGly(cPent) MeAsp2 pyrro PP3050 Hyp(Et) cVal(3-Me2) MeNva(3-Et) MeAsp2 NMe2
PP3051 Hyp(Et) cVal MeNva(3-Et) MeAsp2 pip PP3052 Hyp(Et) cVal MeNva(3-Et) MeAsp2
pyrro PP3053 Hyp(cBu) cVal(3-Me2) MeGly(cPent) MeAsp2 pyrro PP3054 Hyp(cPent) cVal(3-
Me2) MeGly(cPent) MeAsp2 pyrro PP3055 Pro cVal(3-Me2) MeNva(3-Et) MeAsp2 pip PP3056
Hyp(Et) cVal(3-Me2) MeGly(cPent) MeAsp2 MeNEt PP3057 Hyp(Et) cVal(3-Me2) MeGly(cPent)
MeAsp2 NMe2 PP3058 Hyp(cBu) cVal(3-Me2) MeGly(cPent) MeAsp2 MeNEt PP3059 Hyp(Et)

MecVal(3-Me2) MeGly(cPent) MeAsp2 NMe2 PP3061 Hyp(Et) MecVal(3-Me2) MeGly(cPent) MeAsp2 NMe2 PP3062 Hyp(Et) MecVal(3-Me2) MeGly(cPent) MeAsp2 NMe2

(2230) TABLE-US-00038 TABLE 29-3 ID yield (mg) yield (%) PP1829 2.52 4.7 PP1830 2.28 4.3 PP1831 3.09 4.4 PP1832 3.98 5.7 PP1833 3.35 6.4 PP1834 3.23 6.2 PP1835 3.14 5.9 PP1827 2.05 3.9 PP1828 3.08 5.9 PP2573 3.02 22.8 PP2574 3.58 21.1 PP2575 2.56 15.7 PP2576 2.59 16.4 PP2577 4.75 22.3 PP2578 2.28 24.3 PP2579 2.53 27.1 PP2580 2.52 21.9 PP2581 2.73 19.3 PP2583 4.13 23.8 PP2585 4.08 25.8 PP2586 4.11 28.6 PP2588 4.19 27.0 PP2589 4.29 29.0 PP2590 4.31 27.2 PP2591 5.01 27.5 PP2592 4.22 19.0 PP2593 3.09 20.0 PP2594 2.54 19.9 PP2595 4.04 21.0 PP2596 3.5 22.6 PP2597 3.97 29.8 PP2598 0.34 1.4 PP2600 2.37 15.6 PP2601 4.56 18.8 PP2602 3.8 19.6 PP2603 2.53 17.9 PP2604 3.15 16.3 PP2605 4.19 22.2 PP2952 6.42 31.1 PP2953 3.93 21.0 PP2954 9.73 43.1 PP2955 6.03 34.1 PP2956 7.18 40.6 PP2957 7.05 31.2 PP2958 8.50 45.6 PP2959 5.12 34.8 PP3039 3.32 21.3 PP3040 7.15 37.1 PP3041 4.79 16.4 PP3042 2.94 13.6 PP3043 3.39 14.2 PP3044 4.32 21.3 PP3045 4.16 24.4 PP3046 5.42 33.0 PP3047 6.08 27.7 PP3048 2.17 10.6 PP3049 5.46 19.9 PP3050 2.02 12.3 PP3051 3.63 20.9 PP3052 3.14 14.6 PP3053 8.21 27.7 PP3054 8.66 26.6 PP3055 3.91 20.4 PP3056 4.48 15.6 PP3057 4.44 14.7 PP3058 8.67 27.3 PP3059 5.34 41.3 PP3060 4.82 44.4 PP3061 6.7 48.5 PP3062 9.35 36.8

(2231) Concerning the double bonds contained in the crosslinked ring of R.sub.1 to R.sub.5 in PP1827, PP2583, PP2952, PP2954, PP2957, PP2958, PP3040, PP3046, PP3047, PP3049, PP3053, PP3054, PP3056, PP3057, PP3058, PP3059, PP3060, PP3061, and PP3062, the coupling constant between the protons bonded to the respective carbon atoms on the double bonds was observed at 15.0 to 16.4 Hz, and thus the double bonds were all identified as having an E structure. Further, concerning PP1828, PP1829, PP1830, PP1831, PP1832, PP1833, PP1834, PP1835, PP2573, PP2574, PP2575, PP2576, PP2577, PP2578, PP2579, PP2580, PP2581, PP2586, PP2588, PP2589, PP2590, PP2591, PP2592, PP2593, PP2594, PP2595, PP2596, PP2597, PP2598, PP2600, PP2601, PP2602, PP2603, PP2604, PP2605, PP2953, PP2955, PP2956, PP2959, PP3039, PP3044, PP3045, PP3048, PP3050, PP3051, PP3052, and PP3055, which have a similar ring structure, the double bonds are inferred as having an E structure.

(2232) 1-6-6. Peptide Synthesis Via N-Alkylation by Mitsunobu Reaction on Resin and Olefin Metathesis Reaction on Resin

(2233) Synthesis of Compounds PP528 and PP529

(2234) Compounds PP528 and PP529 were synthesized according to the following scheme.

(2235) ##STR00910## ##STR00911## ##STR00912##

(2236) Using, as a raw material, synthesis intermediate PP599-e of Compound PP599 synthesized using Compound aa375-resin, peptide elongation was performed according to the basic route to give resin-supported PP0529-f.

(2237) Compound PP529-f-resin (100 mg, 0.45 mmol/g, 0.045 mmol) and DCM were added to a filter-equipped reaction vessel to swell the resin, DCM was removed, and then the resin was washed 4 times with DCE. A DCE (1.5 mL) solution of a Stewart-Grubbs catalyst (5.13 mg, 0.009 mmol) was added thereto, and the mixture was shaken at 50° C. for 21 hours. Then, the reaction solution was removed, and the resin was washed 4 times with DCE and then washed 4 times with DCM to give resin-supported Compound PP529-g. Cleaving from the resin was performed with TFE/DCM (1/1) using a small amount of resin-supported Compound PP529-g, and the structure was verified by LC/MS.

(2238) LCMS (ESI) m/z=1702 (M+H)+

(2239) Retention time: 3.74 min (Analytical condition SQDFA50_2)

(2240) Subsequent steps of cleaving from resin, cyclization, and purification were performed according to the basic route to give Compound PP0528 (6.9 mg). LC/MS data is provided in Table 36. Moreover, a separately synthesized Compound PP528 (132 mg, 0.09 mmol) at a crude product stage was dissolved in ethyl acetate (1 mL), platinum (IV) oxide (10.2 mg, 0.045 mmol) was

added, and the mixture was stirred at room temperature for 15 hours in a hydrogen atmosphere. Platinum (IV) oxide (5.1 mg, 0.0225 mmol) was added, the mixture was stirred at room temperature in a hydrogen atmosphere for 5 more hours, then platinum (IV) oxide was filtered off, and the solvent was distilled off under reduced pressure. This was purified by reverse phase chromatography (methanol/10 mM aqueous ammonium acetate solution) to give PP0529 (9.8 mg). LC/MS data is provided in Table 36.

(2241) Synthesis of Compounds PP530 to PP0537

(2242) Compounds PP530 to PP0537 were synthesized according to the following scheme.

(2243) ##STR00913## ##STR00914## ##STR00915## ##STR00916## ##STR00917##

(2244) Using Compound aa375-resin as a raw material, peptide synthesis by the basic peptide synthesis method described in the present Examples was performed to give resin-supported Compound PP530-a, Compound PP532-a, Compound PP534-a, and Compound PP536-a.

(2245) After resin-supported Compound PP530-b, Compound PP532-b, Compound PP534-b, and Compound PP536-b were obtained in the same manner as synthesis of Compound PP55-b, resin-supported Compound PP530-c, Compound PP532-c, Compound PP534-c, and Compound PP536-c were obtained in the same manner as synthesis of Compound PP55-c using THE in place of NMP as a reaction solvent. After resin-supported Compound PP530-d, Compound PP532-d, Compound PP534-d, and Compound PP536-d were obtained in the same manner as Compound PP524-d by reacting 2-propen-1-ol in place of 3-fluoropropan-1-ol, resin-supported Compound PP530-e, Compound PP532-e, Compound PP534-e, and Compound PP536-e were obtained in the same manner as synthesis of Compound PP55-e. Subsequent peptide elongation was performed according to the basic route to give resin-supported Compound PP530-f, Compound PP532-f, Compound PP534-f, and Compound PP536-f. In the same manner as Compound PP529-g, resin-supported Compound PP530-g, Compound PP532-g, Compound PP534-g, and Compound PP536-g were obtained.

(2246) Subsequent steps of cleaving from resin, cyclization, and purification were performed according to the basic route to give Compound PP0531, Compound PP533, Compound PP535, and Compound PP537. LC/MS data is provided in Table 36. Moreover, using separately synthesized Compound PP531, Compound PP533, Compound PP535, and Compound PP537 that were at a crude product stage, Compound PP530, Compound PP532, Compound PP534, and Compound PP536 were obtained in the same manner as Compound PP529. LC/MS data is provided in Table 36.

(2247) 1-6-7. Peptide Synthesis Via N-Alkylation by Mitsunobu Reaction on Resin and Olefin Metathesis Reaction

(2248) Synthesis of Compounds PP0606 and PP0611

(2249) Compounds PP0606 and PP0611 were synthesized according to the following scheme.

(2250) ##STR00918## ##STR00919##

(2251) PP599 (25 mg, 0.017 mmol) obtained using Compound aa375-resin as a raw material and a Stewart-Grubbs catalyst (2.4 mg, 0.0042 mmol) were dissolved in DCE (1.5 mL), and after a degassing operation, the mixture was stirred at 50° C. for 16 hours in a nitrogen atmosphere. A Stewart-Grubbs catalyst (2.4 mg, 0.0042 mmol) was added, and after a degassing operation, the mixture was stirred for 20 more hours at 50° C. in a nitrogen atmosphere, and then the solvent was distilled off under reduced pressure. This was purified by reverse phase chromatography (0.1% formic acid-containing acetonitrile/0.1% formic acid-containing distilled water) to give Compound PP0606 (4.1 mg, 17%). LC/MS data is provided in Table 36. Compound PP0606 (3.6 mg, 0.0024 mmol) was dissolved in ethyl acetate (1 mL), platinum (IV) oxide (0.28 mg, 0.0012 mmol) was added, and the mixture was stirred at room temperature for 5 hours in a hydrogen atmosphere. Platinum (IV) oxide was filtered off, and the solvent was distilled off under reduced pressure. The resulting crude product was purified by reverse phase chromatography (0.1% formic acid-containing acetonitrile/0.1% formic acid-containing distilled water) to give Compound PP0611 (1.9

mg, 53%). LC/MS data is provided in Table 36.

(2252) Synthesis of Compounds PP0607 to 0610 and PP0612 to 0615

(2253) Compounds PP0607 to 0610 and PP0612 to 0615 were synthesized according to the following scheme.

(2254) ##STR00920## ##STR00921## ##STR00922## ##STR00923##

(2255) Using Compound aa375-resin as a raw material, peptide synthesis by the basic peptide synthesis method described in the present Examples was performed to give resin-supported Compound PP0612-a, Compound PP0613-a, Compound PP0614-a, and Compound PP0615-a.

(2256) After resin-supported Compound PP0612-b, Compound PP0613-b, Compound PP0614-b, and Compound PP0615-b were obtained in the same manner as synthesis of Compound PP55-b, resin-supported Compound PP0612-c, Compound PP0613-c, Compound PP0614-c, and Compound PP0615-c were obtained in the same manner as synthesis of Compound PP55-c using THF in place of NMP as a reaction solvent. After resin-supported Compound PP0612-d, Compound PP0613-d, Compound PP0614-d, and Compound PP0615-d were obtained in the same manner as synthesis of Compound PP524-d by reacting 2-propen-1-ol in place of 3-fluoropropan-1-ol, resin-supported Compound PP0612-e, Compound PP0613-e, Compound PP0614-e, and Compound PP0615-e were obtained in the same manner as synthesis of Compound PP55-e.

(2257) Subsequent steps of peptide elongation, cleaving from resin, cyclization, and purification were performed according to the basic route to give Compound PP0612-f, Compound PP0613-f, Compound PP0614-f, and Compound PP0615-f. Compound PP0607, Compound PP0608, Compound PP0609, and Compound PP0610 were obtained in the same manner as synthesis of Compound PP0606. Yields in amount and yields in percent are provided in Table 30. LC/MS data is provided in Table 36.

(2258) TABLE-US-00039
TABLE 30 Raw material Intended product R Yield Yield Compound
0612-f Compound 0607 F 6.9 mg 28% Compound 0613-f Compound 0608 Cl 7.2 mg 29%
Compound 0614-f Compound 0609 OMe 6.9 mg 28% Compound 0615-f Compound 0610
OCF.sub.3 5.3 mg 20%

(2259) Moreover, Compound PP0612, Compound PP0613, Compound PP0614, and Compound PP0615 were obtained in the same manner as synthesis of Compound PP0611. Yields in amount and yields in percent are provided in Table 31. LC/MS data is provided in Table 36.

(2260) TABLE-US-00040
TABLE 31 Raw material Intended product R Yield Yield Compound
0607 Compound 0612 F 6.4 mg 100% Compound 0608 Compound 0613 Cl 6.6 mg 98%
Compound 0609 Compound 0614 OMe 5.8 mg 91% Compound 0610 Compound 0615
OCF.sub.3 2.6 mg 54%

Synthesis of Compound PP316, Compound PP317, Compound PP0460, and Compound PP0461

(2261) Compound PP316, Compound PP317, Compound PP0460, and Compound PP0461 were synthesized according to the following scheme.

(2262) ##STR00924## ##STR00925##

(2263) Using PP139 and PP140 obtained from Compound aa375-resin as a raw material, Compound PP0460 and Compound PP0461 were obtained in the same manner as synthesis of Compound PP0606. Yields in amount and yields in percent are provided in Table 32. LC/MS data is provided in Table 36.

(2264) TABLE-US-00041
TABLE 32 Raw material Intended product n Yield Yield Compound
0139 Compound 0460 1 80 mg 58% Compound 0140 Compound 0461 2 32 mg 24%

(2265) Moreover, Compound PP316 and Compound PP317 were obtained in the same manner as synthesis of Compound PP0611. Yields in amount and yields in percent are provided in Table 33. LC/MS data is provided in Table 36.

(2266) TABLE-US-00042
TABLE 33 Raw material Intended product n Yield Yield Compound
0460 Compound 0316 1 60 mg 80% Compound 0461 Compound 0317 2 18 mg 64%

Synthesis of Compound PP318, Compound PP319, Compound PP0462, and Compound PP0463

(2267) Compound PP318, Compound PP319, Compound PP0462, and Compound PP0463 were synthesized according to the following scheme.

(2268) ##STR00926## ##STR00927## ##STR00928## ##STR00929##

(2269) Using Compound aa375-resin as a raw material, peptide synthesis by the basic peptide synthesis method described in the present Examples was performed to give resin-supported Compound PP318-a.

(2270) After resin-supported Compound PP318-b was obtained in the same manner as synthesis of Compound PP55-b, resin-supported Compound PP318-c was obtained in the same manner as synthesis of Compound PP55-c using THE in place of NMP as a reaction solvent. After resin-supported Compound PP318-d was obtained in the same manner as synthesis of Compound PP524-d by reacting 2-propen-1-ol in place of 3-fluoropropan-1-ol, resin-supported Compound PP318-e was obtained in the same manner as synthesis of Compound PP55-e.

(2271) Subsequent steps of peptide elongation, cleaving from resin, cyclization, and purification were performed according to the basic route to give Compound PP0318-f and Compound PP319-f. Compound PP0462 and Compound PP0463 were obtained in the same manner as synthesis of Compound PP0606. Yields in amount and yields in percent are provided in Table 34. LC/MS data is provided in Table 36.

(2272) TABLE-US-00043

TABLE 34	Raw material	Intended product	n	Yield	Yield	Compound
0318-f	Compound 0462	1	59 mg	76%	Compound 0319-f	Compound 0463
		2	18 mg	23%		

(2273) Moreover, Compound PP318 and Compound PP319 were obtained in the same manner as synthesis of Compound PP0611. Yields in amount and yields in percent are provided in Table 35. LC/MS data is provided in Table 36.

(2274) TABLE-US-00044

TABLE 35	Raw material	Intended product	n	Yield	Yield	Compound
0462	Compound 0318	1	46 mg	81%	Compound 0463	Compound 0319
		2	10 mg	65%		

1-6-8. Peptide Modification by Hydroboration Reaction and Suzuki Coupling
Synthesis of Compound PP02117

(2275) ##STR00930##

(2276) Using Compound aa440-resin (Fmoc-MeGly (cPent)-MeAsp (O-Trt (2-C1) resin)-MeNEt) as a raw material, peptide synthesis was carried out by the basic peptide synthesis method described in the present Examples to give Compound PP02117-a. A THF solution of 9-BBN (0.5 M, 240 μ L, 120 μ mol) was added to Compound PP02117-a (12.4 mg, 7.67 μ mol), and the mixture was stirred for 1.5 hours in a nitrogen atmosphere. The resulting reaction solution was diluted with THF (500 μ L), then water (4 μ L, 222 μ mol), cataCXium A Pd G4 (CAS No. 2230788-67-5, 2.7 mg, 3.64 μ mol), and 2-tert-butyl-1,1,3,3-tetramethylguanidine (131 μ L, 110 μ mol) were added, and the mixture was stirred at 60° C. for 1 hour in a nitrogen atmosphere. After being cooled to room temperature, the reaction solution was halved, either a 2% aqueous N-acetylcysteine solution (100 μ L) or a 10% aqueous dithiothreitol solution (100 μ L) was added, the mixture was stirred for 1 hour, the reaction solution was purified twice by reverse phase HPLC(methanol/50 mM aqueous ammonium acetate solution, then 0.1% formic acid-containing acetonitrile/0.1% formic acid-containing distilled water), and the fractions were combined to give PP2117 (3.2 mg, 6.1%).

(2277) LCMS (ESI) m/z=1492 (M+H)+

(2278) Retention time: 0.56 min (Analytical condition SQDFA50)

(2279) Synthesis of Compound PP02118

(2280) Synthesis of PP2118 was carried out according to the following scheme.

(2281) ##STR00931##

(2282) Using Compound aa440-resin (Fmoc-MeGly (cPent)-MeAsp (O-Trt (2-C1) resin)-MeNEt) as a raw material, peptide synthesis was carried out by the basic peptide synthesis method described in the present Examples to give Compound PP02118-a. A THF solution of 9-BBN (0.5 M, 240 μ L, 120 μ mol) was added to Compound PP02118-a (12.1 mg, 7.70 μ mol), and the mixture was stirred for 1.5 hours in a nitrogen atmosphere. The resulting reaction solution was diluted with

THF (500 μ L), then water (4 μ L, 222 μ mol), cataCXium Pd G4 (CAS No. 2230788-67-5, 2.7 mg, 3.64 μ mol), and 2-tert-butyl-1,1,3,3-tetramethylguanidine (131 μ L, 110 μ mol) were added, and the mixture was stirred at 60° C. for 1 hour in a nitrogen atmosphere. After being cooled to room temperature, the reaction solution was halved, either a 2% aqueous N-acetylcysteine solution (100 μ L) or a 10% aqueous dithiothreitol solution (100 μ L) was added, the mixture was stirred for 1 hour, the reaction solution was purified twice by reverse phase HPLC(methanol/50 mM aqueous ammonium acetate solution, then 0.1% formic acid-containing acetonitrile/0.1% formic acid-containing distilled water), and the fractions were combined to give PP2118 (3.2 mg, 6.1%).

(2283) LCMS (ESI) m/z =1486 (M+H)+

(2284) Retention time: 0.53 min (Analytical condition SQDFA50)

(2285) Synthesis of Compound PP02259 Synthesis of PP2259 was carried out according to the following scheme.

(2286) ##STR00932##

(2287) Using Compound aa440-resin (Fmoc-MeGly (cPent)-MeAsp (O-Trt (2-C1) resin)-MeNEt) as a raw material, peptide synthesis was carried out by the basic peptide synthesis method described in the present Examples to give Compound PP02259-a. The resulting compound was azeotroped twice with toluene (1 mL). A THF solution of 9-BBN (0.1 M, 137 μ L, 137 μ mol, 2 eq) was added to Compound PP02259-a (11.2 mg, 6.83 μ mol) azeotroped with toluene, and the mixture was stirred for 1 hour in a nitrogen atmosphere. The resulting reaction solution was diluted with THF (500 μ L), then water (4 μ L, 222 μ mol, 30 eq), [1,1'-bis(di-tert-butylphosphino) ferrocene] dichloropalladium (2.3 mg, 3.5 μ mol, 0.5 eq), and 2-tert-butyl-1,1,3,3-tetramethylguanidine (13 μ L, 110 μ mol, 15 eq) were added, and the mixture was stirred at 60° C. for 1 hour in a nitrogen atmosphere. After the mixture was cooled to room temperature, a 2% aqueous N-acetylcysteine solution (200 μ L) was added, the mixture was stirred for 1 hour, then the reaction solution was purified by reverse phase HPLC(0.1% formic acid-containing acetonitrile/0.1% formic acid-containing distilled water) to give PP2259 (4.2 mg, 7.9%).

(2288) LCMS (ESI) m/z =1514 (M+H)+

(2289) Retention time: 0.62 min (Analytical condition SQDFA50)

(2290) Synthesis of Compound PP02260

(2291) Synthesis of PP2260 was carried out according to the following scheme.

(2292) ##STR00933##

(2293) Using Compound aa440-resin (Fmoc-MeGly (cPent)-MeAsp (O-Trt (2-C1) resin)-MeNEt) as a raw material, peptide synthesis was carried out by the basic peptide synthesis method described in the present Examples to give Compound PP02260-a. The resulting compound was azeotroped twice with toluene (1 mL). A THF solution of 9-BBN (0.1 M, 143 μ L, 143 μ mol, 2 eq) was added to Compound PP02260-a (11.8 mg, 7.14 μ mol) azeotroped with toluene, and the mixture was stirred for 1 hour in a nitrogen atmosphere. The resulting reaction solution was diluted with THF (500 μ L), then water (4 μ L, 222 μ mol, 30 eq), cataCXium Pd G4 (CAS No. 2230788-67-5, 2.7 mg, 3.6 μ mol, 0.5 eq), and 2-tert-butyl-1,1,3,3-tetramethylguanidine (13 μ L, 110 μ mol, 15 eq) were added, and the mixture was stirred at 60° C. for 1 hour in a nitrogen atmosphere. After the mixture was cooled to room temperature, a 10% aqueous dithiothreitol solution (200 μ L) was added, the mixture was stirred for 1 hour, the reaction solution was purified by reverse phase HPLC(0.1% formic acid-containing acetonitrile/0.1% formic acid-containing distilled water) to give PP2260 (4.1 mg, 7.6%).

(2294) LCMS (ESI) m/z =1528 (M+H)+

(2295) Retention time: 0.67 min (Analytical condition SQDFA50)

(2296) Using any of Compounds aa359-resin, aa360-resin, aa362-resin, aa374-resin, aa429-resin, and aa440-resin as a raw material, peptide synthesis was carried out by the basic peptide synthesis method described in the present Examples to give a cyclic peptide as an intermediate. Using the resulting intermediate peptide, compounds shown in Table 35-1 were produced in the same manner

as the synthesis of Compound PP02259 by reacting the side chain moiety of any of MeAlgly, MeAhxe (2), MeAhpe (2), MeAocte (2), and MeAnone (2) with the iodide moiety of ButenylPhe (4-I). Purification was carried out by reverse phase HPLC purification (methanol/50 mM aqueous ammonium acetate solution or 0.1% formic acid-containing acetonitrile/0.1% formic acid-containing distilled water, or both if necessary). The intended product and the yield (amount/percentage) are provided in Table 35-2. LC/MS data are provided in Table 36.

(2297) ##STR00934##

(2298) TABLE-US-00045 TABLE 35-1 ID 1 2 3 4 5 6 7 PP2117 MeAhpe(2) Ile MeGly MeAlgly ButenylPhe(4-CH=CH2) MeGly Hph(4-CF3-35-F2) PP2118 MeAhpe(2) Ile MeGly MeAlgly ButenylPhe(4-CH=CH2) MeGly Hph(4-CF3-3-OMe) PP2259 MeAhpe(2) Ile MeGly MeAlgly ButenylPhe(4-CH=CH2) MeGly Hph(4-CF3-3-OMe) PP2260 MeAocte(2) Ile MeGly MeAlgly ButenylPhe(4-CH=CH2) MeGly Hph(4-CF3-3-OMe) PP2261 MeAhpe(2) Ile MeGly MeAlgly ButenylPhe(4-CH=CH2) MeGly Hph(4-CF3-35-F2) PP2262 MeAhpe(2) Ile MeGly MeAlgly ButenylPhe(4-CH=CH2) MeGly Hph(4-CF3-3-OMe) PP2263 MeAocte(2) Ile MeGly MeAlgly ButenylPhe(4-CH=CH2) MeGly Hph(4-CF3-3-OMe) PP2264 MeAocte(2) Ile MeGly MeAlgly ButenylPhe(4-CH=CH2) MeGly Hph(4-CF3-3-OMe) PP2265 MeAnone(2) Ile MeGly MeAlgly ButenylPhe(4-CH=CH2) MeGly Hph(4-CF3-35-F2) PP2266 MeAnone(2) Ile MeGly MeAlgly ButenylPhe(4-CH=CH2) MeGly Hph(4-CF3-3-OMe) PP2267 MeAhpe(2) Ile MeGly MeAlgly ButenylPhe(4-CH=CH2) MeGly Hph(4-CF3-3-OMe) PP3032 MeAocte(2) Ile MeGly MeAlgly ButenylPhe(4-CH=CH2) MeGly Hph(4-CF3-3-OMe) PP3033 MeAocte(2) Ile MeGly MeAlgly ButenylPhe(4-CH=CH2) MeGly Hph(4-CF3-3-OMe) PP3034 MeAocte(2) Ile MeGly MeAlgly ButenylPhe(4-CH=CH2) MeGly Hph(4-CF3-3-OMe) PP3035 MeAhpe(2) Ile MeGly MeAlgly ButenylPhe(4-CH=CH2) MeGly Hph(4-CF3-3-OMe) PP3036 MeAhpe(2) Ile MeGly MeAlgly ButenylPhe(4-CH=CH2) MeGly Hph(4-CF3-3-OMe) PP3037 MeAhpe(2) Ile MeGly MeAlgly ButenylPhe(4-CH=CH2) MeGly Hph(4-CF3-3-OMe) PP3038 MeAhxe(2) Ile MeGly MeAlgly ButenylPhe(4-CH=CH2) MeGly Hph(4-CF3-3-OMe) PP3063 MeAlgly Ile MeGly MeAlgly ButenylPhe(4-CH=CH2) MeGly Hph(4-CF3-3-OMe) PP3064 MeAlgly Ile MeGly MeAlgly ButenylPhe(4-CH=CH2) MeGly Hph(4-CF3-3-OMe) PP3065 MeAlgly Ile MeGly MeAlgly ButenylPhe(4-CH=CH2) MeGly Hph(4-CF3-3-OMe) PP3066 MeAocte(2) Ile MeGly MeAlgly ButenylPhe(4-CH=CH2) MeGly Hph(4-CF3-3-OMe) PP3067 MeAocte(2) Ile MeGly MeAlgly ButenylPhe(4-CH=CH2) MeGly Hph(4-CF3-3-OMe) PP3068 MeAocte(2) Ile MeGly MeAlgly ButenylPhe(4-CH=CH2) MeGly Hph(4-CF3-3-OMe) PP3069 MeAocte(2) Ile MeGly MeAlgly ButenylPhe(4-CH=CH2) MeGly Hph(4-CF3-3-OMe) PP3070 MeAocte(2) Ile cPrGly MeAlgly ButenylPhe(4-CH=CH2) MeGly Hph(4-CF3-3-OMe) PP3071 MeAocte(2) Ile MeGly MeAlgly ButenylPhe(4-CH=CH2) MeGly Hph(34-Cl2) PP3072 MeAhpe(2) Ile MeGly MeAlgly ButenylPhe(4-CH=CH2) MeGly Hph(4-CF3-3-OMe) PP3073 MeAhpe(2) Ile MeGly MeAlgly ButenylPhe(4-CH=CH2) MeGly Hph(4-CF3-3-OMe) PP3074 MeAhpe(2) Ile MeGly MeAlgly ButenylPhe(4-CH=CH2) MeGly Hph(4-CF3-3-OMe) PP3075 MeAhpe(2) Ile cPrGly MeAlgly ButenylPhe(4-CH=CH2) MeGly Hph(4-CF3-3-OMe) PP3076 MeAhpe(2) Ile MeGly MeAlgly ButenylPhe(4-CH=CH2) MeGly Hph(34-Cl2) PP3077 MeAhxe(2) Ile MeGly MeAlgly ButenylPhe(4-CH=CH2) MeGly Hph(4-CF3-3-OMe) PP3078 MeAhxe(2) Ile MeGly MeAlgly ButenylPhe(4-CH=CH2) MeGly Hph(4-CF3-3-OMe) PP3079 MeAhxe(2) Ile MeGly MeAlgly ButenylPhe(4-CH=CH2) MeGly Hph(4-CF3-3-OMe) PP3080 MeAhxe(2) Ile MeGly MeAlgly ButenylPhe(4-CH=CH2) MeGly Hph(4-CF3-3-OMe) PP3081 MeAhxe(2) Ile MeGly MeAlgly ButenylPhe(4-CH=CH2) MeGly Hph(4-CF3-3-OMe) PP3082 MeAhxe(2) Ile cPrGly MeAlgly ButenylPhe(4-CH=CH2) MeGly Hph(4-CF3-3-OMe) PP3083 MeAlgly Ile MeGly MeAlgly ButenylPhe(4-CH=CH2) MeGly Hph(4-CF3-3-OMe) PP3084 MeAlgly Ile MeGly MeAlgly ButenylPhe(4-CH=CH2) MeGly Hph(4-CF3-3-OMe) PP3085 MeAlgly Ile MeGly MeAlgly ButenylPhe(4-CH=CH2) MeGly Hph(4-CF3-3-OMe) PP3086 MeAlgly Ile MeGly MeAlgly ButenylPhe(4-CH=CH2) MeGly Hph(4-CF3-3-OMe) PP3087 MeAlgly Ile MeGly MeAlgly

ButenylPhe(4-CH=CH2) MeGly Hph(4-CF3-3-OMe) PP3088 MeAlgly Ile cPrGly MeAlgly
 ButenylPhe(4-CH=CH2) MeGly Hph(4-CF3-3-OMe) PP3089 MeAlgly Ile MeGly MeAlgly
 ButenylPhe(4-CH=CH2) MeGly Hph(34-Cl2) PP3090 MeAocte(2) Ile MeGly MeAlgly
 ButenylPhe(4-CH=CH2) MeGly Hph(4-CF3-3-OMe) PP3091 MeAhpe(2) Ile MeGly MeAlgly
 ButenylPhe(4-CH=CH2) MeGly Hph(4-CF3-3-OMe) PP3092 MeAhxe(2) Ile MeGly MeAlgly
 ButenylPhe(4-CH=CH2) MeGly Hph(34-Cl2) PP3093 MeAhxe(2) Ile MeGly MeAlgly
 ButenylPhe(4-CH=CH2) MeGly Hph(4-CF3-3-OMe) PP3094 MeAhxe(2) Ile MeGly MeAlgly
 ButenylPhe(4-CH=CH2) MeGly Hph(4-CF3-3-OMe) ID 8 9 10 11 Cterm PP2117 Hyp(Et) cVal
 MeGly(cPent) MeAsp2 MeNEt PP2118 Hyp(Et) cVal MeGly(cPent) MeAsp2 MeNEt PP2259
 Hyp(Et) cVal(3-Me2) MeGly(cPent) MeAsp2 MeNEt PP2260 Hyp(Et) cVal(3-Me2) MeGly(cPent)
 MeAsp2 MeNEt PP2261 Hyp(Et) cVal MeGly(cPent) MeAsp2 NMe2 PP2262 Hyp(Et) cVal
 MeGly(cPent) MeAsp2 NMe2 PP2263 Hyp(Et) cVal MeGly(cPent) MeAsp2 MeNEt PP2264
 Hyp(Et) cVal MeGly(cPent) MeAsp2 NMe2 PP2265 Hyp(Et) cVal MeGly(cPent) MeAsp2 MeNEt
 PP2266 Hyp(Et) cVal(3-Me2) MeGly(cPent) MeAsp2 MeNEt PP2267 Hyp(Et) cVal(3-Me2)
 MeGly(cPent) MeAsp2 NMe2 PP3032 Hyp(Et) cVal(3-Me2) MeNva(3-Et) MeAsp2 NMe2
 PP3033 Hyp(Et) cVal MeGly(cPent) MeAsp2 pip PP3034 Hyp(Et) cVal(3-Me2) MeGly(cPent)
 MeAsp2 pyrro PP3035 Hyp(Et) cVal(3-Me2) MeNva(3-Et) MeAsp2 NMe2 PP3036 Hyp(Et)
 cVal(3-Me2) MeGly(cPent) MeAsp2 pip PP3037 Hyp(Et) cVal(3-Me2) MeGly(cPent) MeAsp2
 pyrro PP3038 Hyp(nPr) cVal(3-Me2) MeNva(3-Et) MeAsp2 NMe2 PP3063 Hyp(nPr) cVal(3-Me2)
 MeNva(3-Et) MeAsp2 NMe2 PP3064 Hyp(nPr) cVal(3-Me2) MeGly(cPent) MeAsp2 pip PP3065
 Hyp(nPr) cVal(3-Me2) MeGly(cPent) MeAsp2 pyrro PP3066 Hyp(Et) cVal(3-Me2) MeGly(cPent)
 MeAsp2 NMe2 PP3067 Hyp(iPr) cVal(3-Me2) MeGly(cPent) MeAsp2 NMe2 PP3068 Hyp(cBu)
 cVal(3-Me2) MeGly(cPent) MeAsp2 NMe2 PP3069 Pro(4-cPr) cVal(3-Me2) MeGly(cPent)
 MeAsp2 NMe2 PP3070 Hyp(Et) cVal(3-Me2) MeGly(cPent) MeAsp2 NMe2 PP3071 Hyp(Et)
 cVal(3-Me2) MeGly(cPent) MeAsp2 NMe2 PP3072 Hyp(iPr) cVal(3-Me2) MeGly(cPent) MeAsp2
 NMe2 PP3073 Hyp(cBu) cVal(3-Me2) MeGly(cPent) MeAsp2 NMe2 PP3074 Pro(4-cPr) cVal(3-
 Me2) MeGly(cPent) MeAsp2 NMe2 PP3075 Hyp(Et) cVal(3-Me2) MeGly(cPent) MeAsp2 NMe2
 PP3076 Hyp(Et) cVal(3-Me2) MeGly(cPent) MeAsp2 NMe2 PP3077 Hyp(nPr) cVal(3-Me2)
 MeGly(cPent) MeAsp2 NMe2 PP3078 Hyp(iPr) cVal(3-Me2) MeGly(cPent) MeAsp2 MeNEt
 PP3079 Hyp(cBu) cVal(3-Me2) MeGly(cPent) MeAsp2 MeNEt PP3080 Hyp(cPent) cVal(3-Me2)
 MeGly(cPent) MeAsp2 NMe2 PP3081 Pro(4-cPr) cVal(3-Me2) MeGly(cPent) MeAsp2 MeNEt
 PP3082 Hyp(nPr) cVal(3-Me2) MeGly(cPent) MeAsp2 NMe2 PP3083 Hyp(nPr) cVal(3-Me2)
 MeGly(cPent) MeAsp2 MeNEt PP3084 Hyp(iPr) cVal(3-Me2) MeGly(cPent) MeAsp2 MeNEt
 PP3085 Hyp(cBu) cVal(3-Me2) MeGly(cPent) MeAsp2 MeNEt PP3086 Hyp(cPent) cVal(3-Me2)
 MeGly(cPent) MeAsp2 MeNEt PP3087 Pro(4-cPr) cVal(3-Me2) MeGly(cPent) MeAsp2 MeNEt
 PP3088 Hyp(nPr) cVal(3-Me2) MeGly(cPent) MeAsp2 MeNEt PP3089 Hyp(nPr) cVal(3-Me2)
 MeGly(cPent) MeAsp2 MeNEt PP3090 Hyp(Et) MecVal(3-Me2) MeGly(cPent) MeAsp2 NMe2
 PP3091 Hyp(Et) MecVal(3-Me2) MeGly(cPent) MeAsp2 NMe2 PP3092 Hyp(nPr) cVal(3-Me2)
 MeGly(cPent) MeAsp2 NMe2 PP3093 Hyp(Et) cVal(3-Me2) MeGly(cPent) MeAsp2 pip PP3094
 Hyp(nPr) cVal(3-Me2) MeGly(cPent) MeAsp2 pyrro
 (2299) TABLE-US-00046 TABLE 35-2 ID yield (mg) yield (%) PP2117 3.2 6.1 PP2118 3.2 6.1
 PP2259 4.2 7.9 PP2260 4.1 7.6 PP2261 2.5 3.7 PP2262 2.9 4.3 PP2263 2.1 4.0 PP2264 2.7 4.0
 PP2265 2.4 4.5 PP2266 3.5 6.5 PP2267 2.8 4.1 PP3032 2.15 11.4 PP3033 2.33 10.0 PP3034 3.70
 15.4 PP3035 1.38 7.6 PP3036 2.12 9.3 PP3037 1.96 9.8 PP3038 2.38 12.9 PP3063 1.19 0.9 PP3064
 1.87 1.2 PP3065 1.71 1.2 PP3066 2.81 2.6 PP3067 1.4 1.3 PP3068 0.89 0.8 PP3069 0.88 0.8
 PP3070 3.74 3.4 PP3071 2.85 2.7 PP3072 5.48 5.1 PP3073 2.4 2.2 PP3074 2.76 2.6 PP3075 2.26
 2.1 PP3076 1.99 1.9 PP3077 4.53 4.3 PP3078 2.22 2.2 PP3079 2.58 2.5 PP3080 2.02 1.9 PP3081
 3.42 3.5 PP3082 2.56 2.4 PP3083 1.7 1.7 PP3084 1.6 1.6 PP3085 1.2 1.2 PP3086 1.98 1.9 PP3087
 0.9 0.9 PP3088 1.55 1.5 PP3089 1.48 1.5 PP3090 2.18 1.6 PP3091 1.53 1.1 PP3092 1.51 1.5
 PP3093 0.9 0.6 PP3094 0.4 0.3

1-6-9. Synthesis of Cyclic Peptide Having Thioether Group in Main Chain Structure

Synthesis of Compound PP3108 and Compound PP3109

(2300) In the synthesis of Compound PP3108 and Compound PP3109, Compound aa439-Resin (0.336 mmol/g, 100 mg) was used as a raw material, and Fmoc-MeGly (cPent)-OH, Fmoc-cLeu-OH, Fmoc-Pro-OH, Fmoc-Hph(4—CF₃-35-F₂)—OH, Fmoc-MeGly-OH, tp005, tp006, Fmoc-Ile-OH, and Fmoc-MeLeu-OH were used. A peptide elongation reaction by the basic Fmoc method described in the present Examples, cleaving of an elongated peptide from resin, cyclization of the cleaved peptide (using (1-cyano-2-ethoxy-2-oxoethylideneaminoxy)dimethylamino-morpholino-carbenium hexafluorophosphate (COMU) as a cyclization reagent), and purification of a cyclic peptide were carried out to give the intended Compound PP3108 (4.9 mg, 9.8%) and Compound PP3109 (5.6 mg, 11%).

(2301) Compound PP3108

(2302) ##STR00935##

(2303) LCMS (ESI) m/z=1534.2 (M-HI)

(2304) Retention time: 5.989 min (Analytical condition SSC-FA-02)

(2305) Compound PP3109

(2306) ##STR00936##

(2307) LCMS (ESI) m/z=1548.0 (M-H)-

(2308) Retention time: 6.083 min (Analytical condition SSC-FA-02)

(2309) In the measurement conditions in Table 36, FA denotes SSC-FA-02/03, and AA denotes SSC-AA-02/03.

(2310) TABLE-US-00047 TABLE 36 MS Compound Analytical Retention Found MS No.

condition	time (min)	(m/z)	polarity
PP0001	FA 5.285	1462.1	(M - H)-
PP0002	FA 5.776	1530.1	(M - H)-
PP0003	FA 5.504	1441.6	(M + H)+
PP0004	FA 5.528	1476.2	(M - H)-
PP0006	AA 7.936	1489.9	(M - H)-
PP0007	FA 5.081	1435.6	(M + H)+
PP0011	FA 5.204	1449.6	(M + H)+
PP0013	FA 5.120	1491.9	(M - H)-
PP0014	AA 7.951	1521.7	(M + H)+
PP0015	FA 5.405	1451.7	(M + H)+
PP0016	FA 5.643	1465.7	(M + H)+
PP0018	FA 5.321	1437.6	(M + H)+
PP0019	FA 5.552	1451.6	(M + H)+
PP0020	AA 7.809	1464.1	(M - H)-
PP0021	FA 6.133	1478.0	(M - H)-
PP0022	FA 5.495	1450.1	(M - H)-
PP0023	FA 5.713	1505.5	(M + H)+
PP0024	FA 5.419	1413.6	(M - H)-
PP0025	FA 5.555	1450.0	(M - H)-
PP0027	FA 5.724	1464.1	(M - H)-
PP0028	FA 5.736	1464.1	(M - H)-
PP0029	AA 7.771	1464.1	(M - H)-
PP0030	FA 5.965	1477.8	(M - H)-
PP0031	FA 4.945	1407.6	(M - H)-
PP0032	FA 5.312	1477.5	(M + H)+
PP0037	FA 5.348	1467.6	(M + H)+
PP0039	FA 5.147	1423.6	(M + H)+
PP0041	FA 4.941	1484.8	(M + NH ₄)+
PP0042	FA 5.616	1493.8	(M - H)-
PP0043	FA 5.561	1438.0	(M - H)-
PP0044	FA 5.277	1425.6	(M + H)+
PP0045	FA 5.445	1438.1	(M - H)-
PP0047	FA 6.448	1437.8	(M - H)-
PP0048	FA 6.757	1452.1	(M - H)-
PP0049	FA 6.228	1413.7	(M + H)+
PP0050	FA 6.381	1444.8	(M + NH ₄)+
PP0051	FA 5.737	1475.8	(M - H)-
PP0052	FA 5.991	1491.9	(M + H)+
PP0053	FA 6.075	1545.8	(M + H)+
PP0054	FA 6.159	1454.1	(M - H)-
PP0055	FA 6.308	1505.7	(M + H)+
PP0056	FA 6.341	1504.2	(M - H)-
PP0057	FA 6.084	1502.1	(M - H)-
PP0058	FA 6.216	1504.2	(M - H)-
PP0059	FA 5.764	1464.1	(M - H)-
PP0060	FA 5.987	1518.0	(M - H)-
PP0061	FA 5.720	1429.6	(M + H)+
PP0062	AA 7.943	1478.0	(M - H)-
PP0063	FA 5.964	1478.0	(M - H)-
PP0064	AA 7.836	1499.7	(M + Na)+
PP0065	FA 6.012	1477.8	(M - H)-
PP0066	AA 7.939	1479.7	(M + H)+
PP0067	FA 6.075	1576.2	(M - H)-
PP0068	AA 7.907	1550.1	(M - H)-
PP0069	FA 6.248	1492.0	(M - H)-
PP0070	FA 6.247	1515.8	(M - H)-
PP0071	AA 7.843	1494.5	(M + NH ₄)+
PP0076	FA 5.551	1489.9	(M - H)-
PP0077	AA 8.029	1569.8	(M - H)-
PP0078	FA 5.900	1548.7	(M + NH ₄)+
PP0083	AA 8.248	1516.2	(M - H)-
PP0084	FA 6.924	1529.8	(M - H)-
PP0085	FA 7.169	1544.2	(M - H)-
PP0086	FA 6.733	1518.0	(M - H)-
PP0087	FA 6.975	1532.3	(M - H)-
PP0088	FA 5.955	1489.7	(M + H)+
PP0089	FA 6.247	1502.1	(M - H)-
PP0090	FA 5.647	1487.7	(M + H)+
PP0091	FA 6.411	1544.1	(M - H)-
PP0092	FA 5.812	1506.1	(M - H)-
PP0093	FA 5.817	1520.2	(M - H)-
PP0094	AA 7.721	1488.0	(M + Na)+
PP0095	FA 6.296	1491.6	(M + H)+
PP0096	FA		

PP0098 FA 6.475 1492.2 (M - H)-
PP0099 FA 6.683 1506.0 (M - H)- PP0100 FA 5.744 1463.7 (M + H)+ PP0101 FA 6.005 1476.0 (M - H)- PP0102 FA 5.665 1459.7 (M - H)- PP0103 FA 6.069 1518.1 (M - H)- PP0104 FA 5.677 1480.2 (M - H)- PP0105 FA 5.668 1495.7 (M + H)+ PP0106 FA 6.533 1558.1 (M - H)- PP0107 FA 6.641 1559.6 (M + H)+ PP0108 FA 6.367 1556.1 (M - H)- PP0109 FA 6.769 1570.2 (M - H)- PP0110 FA 7.056 1583.7 (M - H)- PP0112 FA 6.804 1572.2 (M - H)- PP0113 FA 7.039 1586.1 (M - H)- PP0114 AA 7.827 1544.1 (M + H)+ PP0115 FA 6.373 1558.1 (M + H)- PP0116 FA 5.773 1540.2 (M - H)- PP0117 FA 6.544 1597.7 (M - H)- PP0118 FA 5.996 1562.1 (M + H)+ PP0119 AA 7.784 1575.8 (M + H)+ PP0120 FA 6.407 1531.8 (M - H)- PP0121 FA 6.313 1532.1 (M - H)- PP0122 FA 6.216 1532. (M + H)- PP0123 FA 6.496 1546.1 (M + H)+ PP0124 FA 6.692 1558.2 (M - H)- PP0126 FA 6.635 1547.7 (M + H)- PP0127 FA 6.82 1560.1 (M - H)- PP0128 FA 5.949 1534.7 (M + NH4)+ PP0129 FA 6.199 1529.9 (M - H)- PP0130 FA 5.883 1513.9 (M - H)- PP0131 FA 6.232 1573.6 (M + H)- PP0132 FA 5.921 1533.9 (M - H)- PP0133 FA 5.917 1547.7 (M - H)- PP0134 FA 5.900 1505.9 (M - H)- PP0135 FA 5.693 1481.7 (M + H)+ PP0136 AA 8.043 1560.2 (M - H)- PP0137 FA 6.304 1534.2 (M - H)- PP0138 FA 6.353 1504.2 (M - H)- PP0139 FA 6.477 1502.2 (M - H)- PP0140 FA 6.779 1516.2 (M - H)- PP0141 FA 6.465 1557.8 (M - H)- PP0144 FA 6.033 1505.9 (M - H)- PP0145 FA 6.329 1532.1 (M - H)- PP0146 FA 6.016 1505.6 (M + H)+ PP0148 FA 6.403 1532.2 (M - H)- PP0149 FA 6.144 1545.7 (M + H)+ PP0150 FA 6.085 1477.7 (M - H)- PP0151 AA 8.072 1581.6 (M + Na)+ PP0152 FA 6.677 1558.0 (M - H)- PP0153 FA 6.077 1542.0 (M + Na)+ PP0154 AA 7.933 1530.1 (M - H)- PP0155 FA 5.944 1506.1 (M + H)+ PP0156 FA 5.841 1532.1 (M + H)+ PP0158 FA 6.103 1530.1 (M - H)- PP0162 FA 6.537 1558.0 (M - H)- PP0165 FA 6.257 1544.0 (M - H)- PP0166 FA 6.159 1555.7 (M - H)- PP0168 FA 6.028 1542.0 (M - H)- PP0169 FA 6.089 1475.9 (M - H)- PP0170 FA 6.245 1520.8 (M + NH4)+ PP0171 FA 6.748 1534.7 (M + NH4)+ PP0172 FA 6.028 1488.1 (M - H)- PP0173 AA 7.937 1543.7 (M - H)- PP0174 FA 5.992 1522.8 (M + NH4)+ PP0175 FA 5.095 1574.7 (M + H)+ PP0176 FA 7.011 1531.7 (M + H)+ PP0177 FA 6.711 1530.0 (M - H)- PP0178 FA 6.243 1502.0 (M - H)- PP0179 FA 6.432 1527.9 (M - H)- PP0180 FA 6.949 1542.0 (M - H)- PP0181 FA 6.175 1513.7 (M - H)- PP0182 FA 6.381 1570.2 (M - H)- PP0183 FA 6.085 1530.1 (M - H)- PP0184 AA 7.676 1599.0 (M - H)- PP0185 AA 8.483 1557.7 (M + H)+ PP0186 FA 6.943 1556.0 (M - H)- PP0187 FA 5.819 1543.7 (M - H)- PP0188 FA 5.935 1515.9 (M - H)- PP0189 FA 5.872 1503.9 (M - H)- PP0190 FA 5.705 1491.6 (M + H)+ PP0191 FA 5.912 1570.1 (M - H)- PP0192 FA 6.140 1543.9 (M + H)+ PP0193 FA 5.992 1529.9 (M - H)- PP0194 FA 5.827 1516.1 (M - H)- PP0195 FA 6.177 1553.7 (M - H)- PP0196 FA 6.276 1581.6 (M + H)+ PP0197 AA 7.636 1527.8 (M + Na)+ PP0198 FA 6.024 1529.9 (M - H)- PP0199 FA 5.963 1581.8 (M + H)+ PP0200 FA 6.084 1551.9 (M - H)- PP0201 FA 6.023 1542.1 (M + H)+ PP0202 FA 5.849 1527.6 (M + H)+ PP0203 FA 6.061 1606.1 (M - H)- PP0204 FA 6.271 1578.1 (M - H)- PP0205 FA 6.149 1567.6 (M + H)+ PP0206 FA 5.999 1553.6 (M + H)+ PP0207 FA 6.007 1541.9 (M + H)+ PP0208 FA 6.131 1566.0 (M - H)- PP0209 FA 7.039 1620.2 (M - H)- PP0210 FA 5.409 1435.6 (M - H)- PP0211 FA 5.681 1451.6 (M + H)+ PP0212 FA 5.231 1439.9 (M - H)- PP0213 FA 5.441 1454.0 (M - H)- PP0214 FA 5.708 1467.9 (M - H)- PP0215 FA 5.587 1463.6 (M + H)+ PP0216 FA 5.899 1477.6 (M + H)+ PP0217 FA 5.273 1466.1 (M - H)- PP0218 FA 5.584 1480.2 (M - H)- PP0219 FA 5.880 1493.7 (M - H)- PP0220 FA 5.635 1503.6 (M + H)+ PP0221 FA 5.637 1504.2 (M - H)- PP0222 FA 6.216 1545.7 (M + H)+ PP0223 FA 5.836 1497.6 (M + H)+ PP0224 FA 5.973 1515.7 (M - H)- PP0225 FA 5.960 1518.0 (M - H)- PP0226 FA 6.484 1559.8 (M + H)+ PP0227 AA 7.968 1534.0 (M + Na)+ PP0228 FA 6.308 1589.9 (M + H)+ PP0229 FA 5.677 1491.6 (M + H)+ PP0230 FA 5.788 1493.6 (M + H)+ PP0231 FA 6.372 1531.7 (M - H)- PP0232 FA 6.023 1484.0 (M - H)- PP0233 FA 6.161 1564.1 (M + H)+ PP0234 FA 5.388 1477.6 (M + H)+ PP0235 FA 5.511 1479.8 (M + H)+ PP0236 FA 6.101 1517.7 (M - H)- PP0237 AA 7.677 1493.7 (M + Na)+ PP0238 FA 5.729 1560.2 (M - H)- PP0239 FA 5.512 1562.1 (M + H)+ PP0240 FA 5.587 1533.7 (M - H)- PP0241 FA 5.384 1534.1 (M - H)- PP0242 FA

6.333 1588.2 (M - H)- PP0243 AA 8.097 1563.7 (M + H)+ PP0244 AA 7.768 1496.2 (M - H)-
PP0245 FA 6.284 1599.7 (M + H)+ PP0246 FA 5.897 1598.2 (M - H)- PP0247 AA 7.775 1517.6
(M + H)+ PP0248 FA 5.639 1489.9 (M - H)- PP0249 FA 5.896 1545.6 (M + H)+ PP0250 FA
5.859 1517.6 (M + H)+ PP0251 FA 5.996 1529.7 (M - H)- PP0252 FA 6.333 1544.2 (M - H)-
PP0253 FA 6.479 1547.9 (M + H)+ PP0254 FA 6.025 1543.0 (M - H)- PP0256 FA 6.420 1559.7
(M + H)+ PP0257 FA 6.197 1544.1 (M - H)- PP0258 FA 6.204 1545.6 (M + H)+ PP0259 AA
7.921 1545.9 (M + H)+ PP0260 FA 5.899 1505.6 (M + H)+ PP0261 AA 8.028 1546.1 (M + H)+
PP0262 FA 5.775 1547.7 (M - H)- PP0263 FA 6.343 1532.1 (M - H)- PP0264 FA 6.232 1569.6
(M + H)+ PP0265 FA 5.860 1533.7 (M - H)- PP0266 FA 5.788 1510.6 (M + NH4)+ PP0267 FA
6.143 1522.1 (M + H)+ PP0268 FA 5.909 1542.0 (M - H)- PP0269 FA 5.763 1516.1 (M - H)-
PP0270 FA 5.991 1571.6 (M + H)+ PP0271 FA 5.983 1543.6 (M + H)+ PP0272 FA 6.127 1556.2
(M - H)- PP0273 FA 6.020 1556.0 (M - H)- PP0274 FA 6.303 1558.2 (M - H)- PP0275 FA
5.851 1556.6 (M + H)+ PP0276 FA 6.281 1541.9 (M - H)- PP0277 FA 6.431 1570.0 (M - H)-
PP0278 FA 6.189 1555.9 (M - H)- PP0279 FA 6.105 1557.7 (M + H)+ PP0281 FA 6.055 1529.7
(M - H)- PP0282 FA 6.295 1570.0 (M - H)- PP0283 FA 5.912 1574.0 (M - H)- PP0284 FA
6.445 1559.6 (M + H)+ PP0285 FA 6.371 1617.4 (M + Na)+ PP0286 FA 6.084 1561.6 (M + H)+
PP0287 FA 6.023 1518.0 (M - H)- PP0288 FA 6.363 1546.0 (M - H)- PP0289 FA 6.003 1529.9
(M - H)- PP0290 FA 5.833 1522.7 (M + NH4)+ PP0291 FA 6.085 1557.9 (M - H)- PP0292 FA
6.072 1529.7 (M - H)- PP0293 FA 6.247 1543.7 (M - H)- PP0294 FA 6.168 1518.1 (M - H)-
PP0295 FA 6.548 1558.2 (M - H)- PP0296 FA 6.023 1562.2 (M - H)- PP0297 FA 6.588 1545.8
(M - H)- PP0298 FA 6.540 1582.0 (M - H)- PP0299 FA 6.165 1548.0 (M - H)- PP0300 FA
6.057 1506.1 (M - H)- PP0301 FA 6.372 1536.1 (M + H)+ PP0302 FA 5.845 1529.6 (M + H)+
PP0303 FA 5.631 1501.9 (M - H)- PP0304 FA 5.873 1557.6 (M + H)+ PP0305 FA 5.871 1528.1
(M - H)- PP0306 FA 6.072 1541.7 (M - H)- PP0307 FA 5.880 1515.7 (M - H)- PP0308 FA
6.325 1557.6 (M + H)+ PP0309 FA 5.909 1561.6 (M + H)+ PP0310 FA 6.143 1573.8 (M - H)-
PP0311 FA 6.369 1544.1 (M - H)- PP0312 FA 6.325 1582.1 (M + H)+ PP0313 FA 5.927 1545.7
(M - H)- PP0314 FA 5.848 1504.1 (M - H)- PP0315 FA 6.211 1533.6 (M + H)+ PP0316 FA
5.553 1477.7 (M + H)+ PP0317 FA 6.133 1491.7 (M + H)+ PP0318 FA 5.684 1530.2 (M - H)-
PP0319 FA 6.311 1544.0 (M - H)- PP0320 FA 5.767 1491.6 (M + H)+ PP0321 FA 6.108 1539.7
(M - H)- PP0322 FA 5.963 1527.6 (M + H)+ PP0323 FA 5.955 1547.7 (M - H)- PP0324 FA
5.823 1533.7 (M - H)- PP0325 FA 5.807 1491.6 (M + H)+ PP0326 FA 5.679 1477.6 (M + H)+
PP0327 AA 7.799 1525.9 (M - H)- PP0328 FA 5.872 1513.6 (M + H)+ PP0329 FA 5.845 1534.2
(M - H)- PP0330 FA 5.759 1521.6 (M + H)+ PP0331 FA 6.181 1519.6 (M + H)+ PP0332 FA
6.035 1504.1 (M - H)- PP0333 AA 7.952 1555.6 (M + H)+ PP0334 FA 6.233 1541.6 (M + H)+
PP0335 FA 6.189 1563.7 (M + H)+ PP0336 FA 6.084 1547.9 (M - H)- PP0337 FA 5.947 1517.6
(M + H)+ PP0338 FA 5.788 1503.6 (M + H)+ PP0339 FA 6.181 1553.6 (M + H)+ PP0340 FA
6.023 1539.6 (M + H)+ PP0341 FA 6.036 1559.9 (M - H)- PP0342 FA 5.881 1545.9 (M - H)-
PP0343 FA 5.985 1529.9 (M - H)- PP0344 FA 5.895 1515.7 (M - H)- PP0345 FA 6.120 1565.6
(M - H)- PP0346 FA 6.015 1552.0 (M - H)- PP0347 FA 6.121 1575.9 (M + H)+ PP0348 FA
6.027 1560.0 (M - H)- PP0349 FA 5.729 1515.7 (M - H)- PP0350 FA 5.572 1501.7 (M - H)-
PP0351 FA 5.844 1553.6 (M + H)+ PP0352 FA 5.717 1539.6 (M + H)+ PP0353 FA 5.808 1561.7
(M + H)+ PP0354 AA 7.747 1546.0 (M - H)- PP0355 FA 5.937 1462.1 (M - H)- PP0356 FA
5.800 1449.6 (M + H)+ PP0357 FA 6.123 1499.6 (M + H)+ PP0358 AA 7.689 1485.6 (M + H)+
PP0359 FA 6.004 1506.1 (M - H)- PP0360 FA 5.879 1493.6 (M + H)+ PP0361 FA 5.819 1449.6
(M + H)+ PP0362 FA 5.713 1433.6 (M - H)- PP0363 FA 6.000 1485.6 (M + H)+ PP0364 FA
5.904 1469.5 (M - H)- PP0365 FA 5.869 1491.7 (M - H)- PP0366 FA 5.791 1477.7 (M - H)-
PP0367 FA 6.216 1477.6 (M + H)+ PP0368 FA 6.093 1463.6 (M + H)+ PP0369 FA 6.405 1513.6
(M + H)+ PP0370 FA 6.281 1497.9 (M - H)- PP0371 FA 6.247 1521.7 (M + H)+ PP0372 FA
6.155 1505.7 (M - H)- PP0373 FA 6.012 1475.6 (M + H)+ PP0374 FA 5.865 1461.6 (M + H)+
PP0375 FA 6.232 1509.9 (M - H)- PP0376 FA 6.075 1497.5 (M + H)+ PP0377 FA 6.127 1517.7

(M - H)- PP0378 FA 5.972 1505.6 (M + H)+ PP0379 FA 6.072 1489.6 (M + H)+ PP0380 FA 5.999 1475.6 (M + H)+ PP0381 FA 6.213 1523.9 (M - H)- PP0382 FA 6.112 1511.6 (M + H)+ PP0383 FA 6.219 1532.0 (M - H)- PP0384 FA 6.128 1518.2 (M - H)- PP0385 FA 5.785 1475.6 (M + H)+ PP0386 FA 5.652 1461.6 (M + H)+ PP0387 FA 5.897 1511.5 (M + H)+ PP0388 FA 5.796 1497.5 (M + H)+ PP0389 FA 5.887 1518.1 (M - H)- PP0390 FA 5.784 1505.8 (M + H)+ PP0391 FA 6.095 1519.8 (M - H)- PP0392 FA 5.848 1549.6 (M + H)+ PP0393 FA 5.679 1491.6 (M + H)+ PP0394 FA 5.975 1539.7 (M - H)- PP0395 FA 5.813 1527.6 (M + H)+ PP0396 FA 5.923 1550.1 (M + H)+ PP0397 FA 5.773 1534.0 (M - H)- PP0398 FA 6.064 1536.9 (M + NH4)+ PP0399 FA 6.321 1568.0 (M - H)- PP0400 FA 6.177 1554.2 (M - H)- PP0401 FA 6.113 1562.2 (M - H)- PP0402 FA 5.827 1516.0 (M - H)- PP0403 FA 6.164 1565.7 (M - H)- PP0404 FA 6.005 1553.7 (M + H)+ PP0405 FA 5.977 1561.9 (M + H)+ PP0406 FA 6.013 1574.2 (M - H)- PP0407 FA 5.537 1534.7 (M + NH4)+ PP0408 FA 5.857 1565.7 (M - H)- PP0409 FA 5.667 1551.7 (M - H)- PP0410 FA 5.876 1575.7 (M + H)+ PP0411 FA 5.677 1560.2 (M - H)- PP0412 FA 5.824 1462.1 (M - H)- PP0413 FA 6.116 1511.9 (M - H)- PP0414 FA 5.952 1497.9 (M - H)- PP0415 FA 6.063 1520.2 (M - H)- PP0416 FA 5.915 1505.9 (M - H)- PP0417 FA 5.884 1461.6 (M - H)- PP0418 FA 5.733 1449.6 (M + H)+ PP0419 FA 6.029 1497.6 (M - H)- PP0420 FA 5.875 1483.6 (M - H)- PP0421 FA 5.967 1505.9 (M - H)- PP0422 FA 5.833 1492.0 (M - H)- PP0423 FA 6.260 1508.6 (M + NH4)+ PP0424 FA 6.132 1476.1 (M - H)- PP0425 FA 6.372 1526.0 (M - H)- PP0426 FA 6.255 1512.0 (M - H)- PP0427 FA 6.320 1535.6 (M + H)+ PP0428 AA 7.909 1519.7 (M - H)- PP0429 FA 6.067 1487.9 (M - H)- PP0430 FA 5.929 1473.9 (M - H)- PP0431 FA 6.240 1524.1 (M - H)- PP0432 FA 6.104 1510.0 (M - H)- PP0433 FA 6.215 1532.0 (M - H)- PP0434 FA 6.076 1537.2 (M + NH4)+ PP0435 FA 6.024 1487.9 (M - H)- PP0436 FA 6.268 1537.9 (M - H)- PP0437 FA 6.144 1523.9 (M - H)- PP0438 FA 6.263 1546.2 (M - H)- PP0439 FA 6.137 1534.2 (M + H)+ PP0440 FA 5.816 1488.0 (M - H)- PP0441 FA 5.639 1473.9 (M - H)- PP0442 AA 7.711 1524.1 (M - H)- PP0443 FA 5.765 1509.6 (M - H)- PP0444 FA 5.948 1532.0 (M - H)- PP0445 AA 7.707 1519.6 (M + H)+ PP0446 FA 6.288 1547.7 (M + H)+ PP0447 FA 6.169 1534.1 (M + H)+ PP0448 AA 8.045 1531.7 (M - H)- PP0449 FA 6.089 1517.7 (M - H)- PP0450 FA 6.512 1561.7 (M + H)+ PP0451 FA 6.408 1547.7 (M + H)+ PP0452 FA 6.423 1558.2 (M - H)- PP0453 FA 6.251 1543.7 (M - H)- PP0454 FA 6.301 1573.6 (M + H)+ PP0455 FA 6.227 1558.0 (M - H)- PP0456 FA 6.077 1558.1 (M - H)- PP0457 FA 6.007 1543.7 (M - H)- PP0460 FA 5.595 1473.7 (M - H)- PP0461 FA 5.591 1489.6 (M + H)+ PP0462 FA 5.761 1529.6 (M + H)+ PP0463 FA 5.769 1543.7 (M + H)+ PP0464 AA 7.979 1568.2 (M - H)- PP0465 FA 6.677 1540.2 (M - H)- PP0466 FA 6.488 1526.2 (M - H)- PP0467 AA 8.103 1577.9 (M + H)+ PP0468 FA 6.623 1549.7 (M + H)+ PP0469 FA 6.449 1536.3 (M + H)+ PP0470 FA 5.811 1448.1 (M - H)- PP0471 FA 5.992 1525.6 (M - H)- PP0472 FA 6.269 1498.0 (M - H)- PP0473 FA 6.093 1484.1 (M - H)- PP0474 FA 5.993 1535.6 (M + H)+ PP0475 FA 6.201 1506.2 (M - H)- PP0476 FA 6.076 1492.2 (M - H)- PP0477 FA 6.547 1561.7 (M + H)+ PP0478 AA 8.063 1533.6 (M + H)+ PP0479 FA 6.223 1532.2 (M - H)- PP0480 FA 6.377 1533.9 (M + H)+ PP0481 FA 6.199 1545.7 (M - H)- PP0483 FA 6.439 1561.7 (M + H)+ PP0484 FA 6.183 1559.7 (M + H)+ PP0485 FA 6.228 1571.8 (M - H)- PP0487 FA 5.892 1504.2 (M - H)- PP0488 FA 6.040 1518.2 (M - H)- PP0490 AA 8.129 1546.2 (M - H)- PP0491 FA 6.712 1582.3 (M - H)- PP0492 FA 6.520 1589.8 (M - H)- PP0493 FA 6.228 1561.6 (M + H)+ PP0494 FA 6.028 1563.7 (M + H)+ PP0495 FA 6.496 1576.3 (M - H)- PP0496 FA 6.079 1518.2 (M - H)- PP0497 AA 7.961 1531.7 (M - H)- PP0498 FA 6.605 1561.7 (M + H)+ PP0499 FA 6.375 1568.2 (M - H)- PP0500 AA 8.120 1533.8 (M - H)- PP0501 FA 5.909 1552.7 (M + NH4)+ PP0502 AA 7.992 1562.2 (M - H)- PP0503 FA 6.119 1521.7 (M + H)+ PP0504 FA 6.072 1532.2 (M - H)- PP0505 FA 6.565 1559.8 (M - H)- PP0506 FA 6.084 1560.2 (M - H)- PP0507 AA 8.101 1504.2 (M - H)- PP0508 AA 8.101 1516.2 (M - H)- PP0509 AA 8.252 1530.2 (M - H)- PP0510 FA 7.147 1543.8 (M - H)- PP0511 FA 6.587 1573.7 (M + H)+ PP0512 FA 6.255 1489.7 (M - H)- PP0513 FA 6.556 1561.7 (M + H)+ PP0514 AA 8.157 1518.2 (M - H)- PP0515 FA 6.679 1549.7 (M + H)+

PP0516 FA 6.165 1532.2 (M - H)- PP0520 FA 6.217 1572.2 (M - H)- PP0521 AA 8.087 1544.2 (M - H)- PP0522 FA 6.173 1544.2 (M - H)- PP0523 FA 6.575 1561.7 (M + H)+ PP0524 FA 5.801 1536.0 (M - H)- PP0525 FA 5.887 1563.7 (M + H)+ PP0526 FA 5.900 1493.9 (M + H)+ PP0527 FA 5.988 1517.8 (M - H)- PP0528 FA 5.440 1460.1 (M - H)- PP0529 FA 5.444 1462.0 (M - H)- PP0530 FA 5.424 1479.7 (M - H)- PP0531 FA 5.407 1479.6 (M + H)+ PP0532 FA 5.639 1497.6 (M + H)+ PP0533 FA 5.695 1493.8 (M - H)- PP0534 FA 5.400 1493.7 (M + H)+ PP0535 FA 5.329 1490.1 (M - H)- PP0536 FA 6.032 1547.6 (M + H)+ PP0537 FA 6.037 1544.0 (M - H)- PP0538 FA 6.023 1546.2 (M - H)- PP0539 FA 5.872 1546.2 (M - H)- PP0540 FA 6.047 1518.2 (M - H)- PP0541 FA 5.669 1518.2 (M - H)- PP0542 FA 5.748 1518.1 (M - H)- PP0543 FA 5.829 1532.2 (M - H)- PP0544 FA 6.107 1546.2 (M - H)- PP0545 AA 7.820 1546.2 (M - H)- PP0546 FA 5.759 1563.2 (M + NH4)+ PP0547 AA 7.824 1557.8 (M - H)- PP0548 FA 6.387 1572.3 (M - H)- PP0549 FA 5.207 1490.2 (M - H)- PP0550 FA 5.363 1504.1 (M - H)- PP0551 AA 7.964 1561.7 (M + H)+ PP0552 FA 5.711 1532.2 (M - H)- PP0553 FA 5.872 1568.2 (M - H)- PP0554 AA 7.859 1575.8 (M - H)- PP0555 FA 5.569 1546.2 (M - H)- PP0556 FA 5.329 1549.6 (M + H)+ PP0557 FA 5.835 1562.2 (M - H)- PP0558 FA 5.372 1566.7 (M + NH4)+ PP0559 FA 5.304 1504.2 (M - H)- PP0560 FA 5.473 1518.2 (M - H)- PP0561 AA 7.955 1546.2 (M - H)- PP0562 FA 5.728 1554.2 (M - H)- PP0563 FA 5.175 1520.2 (M - H)- PP0564 FA 5.371 1549.6 (M + H)+ PP0565 FA 5.459 1506.2 (M - H)- PP0566 FA 5.483 1517.7 (M - H)- PP0567 FA 5.981 1545.7 (M - H)- PP0568 FA 5.647 1546.2 (M - H)- PP0569 FA 5.791 1490.1 (M - H)- PP0570 FA 5.705 1502.2 (M - H)- PP0571 FA 5.968 1517.7 (M + H)+ PP0572 FA 6.503 1530.2 (M - H)- PP0573 AA 7.929 1558.2 (M - H)- PP0574 FA 5.569 1476.2 (M - H)- PP0575 FA 5.659 1554.2 (M - H)- PP0576 FA 5.968 1526.2 (M - H)- PP0577 FA 5.755 1512.1 (M - H)- PP0578 FA 5.640 1562.3 (M - H)- PP0579 FA 5.931 1534.2 (M - H)- PP0580 FA 5.724 1520.2 (M - H)- PP0581 FA 5.087 1434.1 (M - H)- PP0582 FA 5.196 1512.1 (M - H)- PP0583 FA 5.503 1484.1 (M - H)- PP0584 FA 5.305 1469.6 (M - H)- PP0585 FA 5.171 1519.7 (M - H)- PP0586 FA 5.437 1492.1 (M - H)- PP0587 FA 5.249 1478.1 (M - H)- PP0588 FA 5.940 1547.7 (M + H)+ PP0589 AA 7.872 1533.7 (M - H)- PP0590 AA 7.776 1518.2 (M - H)- PP0591 FA 5.945 1532.2 (M - H)- PP0594 FA 5.489 1558.2 (M - H)- PP0595 FA 5.647 1548.6 (M + NH4)+ PP0596 FA 5.632 1530.2 (M - H)- PP0597 FA 5.939 1546.2 (M - H)- PP0598 FA 5.708 1520.2 (M - H)- PP0599 FA 6.553 1502.0 (M - H)- PP0605 FA 5.912 1450.1 (M - H)- PP0606 AA 7.968 1473.7 (M - H)- PP0607 FA 5.943 1492.0 (M - H)- PP0608 FA 6.260 1507.7 (M - H)- PP0609 FA 5.848 1504.1 (M - H)- PP0610 FA 6.572 1559.8 (M + H)+ PP0611 FA 6.048 1476.0 (M - H)- PP0612 FA 6.055 1494.0 (M - H)- PP0613 FA 6.373 1511.6 (M + H)+ PP0614 FA 5.959 1506.2 (M - H)- PP0615 FA 6.671 1562.1 (M + H)+ PP0616 AA 8.116 1459.9 (M - H)- PP0617 AA 8.148 1513.7 (M - H)- PP0618 AA 7.713 1446.0 (M - H)- PP0619 FA 5.937 1501.7 (M + H)+ PP0620 FA 6.131 1446.1 (M - H)- PP0621 FA 6.251 1501.6 (M + H)+ PP0622 FA 6.568 1531.6 (M + H)+ PP0626 AA 7.993 1516.1 (M - H)- PP0627 FA 6.203 1462.0 (M - H)- PP0628 FA 6.401 1515.9 (M - H)- PP0629 AA 7.929 1461.9 (M - H)- PP0630 AA 8.139 1545.9 (M + H)+ PP0631 FA 6.224 1527.5 (M + Na)+ PP0632 FA 6.328 1575.7 (M + H)+ PP0633 FA 6.609 1521.6 (M + H)+ PP0634 AA 8.296 1535.7 (M + H)+ PP0635 AA 8.659 1573.9 (M - H)- PP0636 AA 8.760 1588.3 (M - H)- PP0637 FA 7.508 1597.7 (M + H)+ PP0638 AA 8.284 1592.3 (M + H)+ PP0639 AA 8.431 1604.3 (M - H)- PP0640 FA 7.527 1644.3 (M - H)- PP0641 AA 8.893 1682.3 (M + Na)+ PP0642 FA 7.443 1667.7 (M + H)+ PP0643 AA 7.596 1507.9 (M + H)+ PP0644 FA 6.017 1520.2 (M - H)- PP0645 AA 8.245 1560.2 (M - H)- PP0646 FA 7.035 1592.9 (M + NH4)+ PP0647 AA 8.219 1583.9 (M + H)+ PP0648 AA 7.820 1575.9 (M - H)- PP0649 AA 8.016 1591.9 (M + H)+ PP0650 FA 6.928 1632.3 (M + H)+ PP0651 FA 7.180 1646.0 (M + H)+ PP0652 AA 8.435 1671.3 (M + NH4)+ PP0653 AA 8.127 1582.7 (M + H)+ PP0654 FA 6.003 1447.7 (M - H)- PP0655 FA 6.387 1477.7 (M + H)+ PP0656 FA 4.625 1566.9 (M - H)- PP0657 FA 5.311 1435.7 (M + H)+ PP0658 AA 7.512 1461.7 (M - H)- PP0659 FA 4.784 1581.2 (M - H)- PP0660 FA 5.501 1447.7 (M - H)- PP0661 FA 5.920 1476.2 (M - H)- PP0662 FA

6.243 1562.2 (M - H)- PP0663 FA 6.387 1507.7 (M + H)+ PP0664 FA 6.475 1521.6 (M + H)+
PP0665 AA 8.059 1519.7 (M + H)+ PP0666 FA 6.351 1517.6 (M + H)+ PP0667 AA 8.061 1532.2
(M - H)- PP0668 FA 6.653 1548.2 (M + H)+ PP0669 FA 6.699 1549.9 (M + H)+ PP0670 FA
6.783 1560.2 (M - H)- PP0671 FA 6.963 1574.3 (M - H)- PP0672 FA 6.635 1568.2 (M - H)-
PP0673 FA 4.837 1571.2 (M + H)+ PP0674 FA 6.227 1523.6 (M + H)+ PP0675 FA 6.201 1536.1
(M - H)- PP0678 FA 6.327 1548.0 (M - H)- PP0679 FA 6.401 1561.9 (M - H)- PP0680 AA
8.087 1562.2 (M + H)+ PP0681 FA 6.251 1558.2 (M - H)- PP0682 FA 6.377 1592.9 (M + NH4)+
PP0683 FA 6.559 1589.7 (M + H)+ PP0684 FA 6.600 1591.7 (M + H)+ PP0685 FA 6.700 1604.2
(M + H)+ PP0686 FA 6.872 1616.3 (M - H)- PP0687 FA 6.531 1611.7 (M + H)+ PP0688 FA
4.775 1612.7 (M + H)+ PP0689 FA 6.136 1564.2 (M - H)- PP0690 FA 6.104 1578.2 (M - H)-
PP0693 AA 7.461 1493.7 (M + H)+ PP0694 FA 5.691 1506.2 (M - H)- PP0695 AA 7.520 1506.2
(M + H)+ PP0696 AA 7.404 1501.8 (M - H)- PP0697 FA 5.644 1517.8 (M - H)- PP0698 FA
5.889 1533.6 (M + H)+ PP0699 FA 6.079 1535.7 (M + H)+ PP0700 FA 6.083 1547.9 (M + H)+
PP0701 FA 6.292 1562.2 (M + H)+ PP0702 FA 5.835 1554.2 (M - H)- PP0703 FA 3.977 1555.2
(M - H)- PP0704 FA 5.360 1508.2 (M - H)- PP0705 FA 5.359 1523.6 (M + H)+ PP0706 AA
7.369 1493.6 (M + H)+ PP0708 FA 5.425 1535.6 (M + H)+ PP0709 FA 5.571 1549.7 (M + H)+
PP0710 FA 5.543 1548.2 (M + H)+ PP0711 FA 5.401 1544.1 (M - H)- PP0712 FA 5.567 1560.2
(M - H)- PP0713 FA 5.812 1575.7 (M + H)+ PP0714 FA 6.004 1577.8 (M + H)+ PP0715 FA
6.035 1590.0 (M + H)+ PP0716 FA 6.248 1603.8 (M + H)+ PP0717 FA 5.776 1597.7 (M + H)+
PP0718 AA 7.244 1597.2 (M - H)- PP0719 FA 5.303 1549.8 (M - H)- PP0720 FA 5.309 1565.7
(M + H)+ PP0721 FA 5.220 1535.7 (M + H)+ PP0722 AA 7.623 1552.8 (M + NH4)+ PP0723 AA
8.175 1521.8 (M + H)+ PP0724 FA 6.259 1522.7 (M + NH4)+ PP0725 FA 6.580 1557.7 (M + H)+
PP0726 AA 8.377 1604.3 (M - H)- PP0727 FA 6.560 1590.2 (M + H)+ PP0728 FA 6.849 1641.6
(M + H)+ PP0729 FA 6.168 1492.2 (M - H)- PP0730 FA 5.391 1479.5 (M + H)+ PP0731 FA
5.317 1581.2 (M - H)- PP0732 AA 7.544 1568.8 (M + H)+ PP0733 FA 4.717 1582.7 (M + H)+
PP0734 FA 6.195 1479.7 (M + H)+ PP0735 FA 6.653 1481.7 (M + H)+ PP0736 FA 6.919 1493.8
(M - H)- PP0737 AA 7.889 1480.2 (M - H)- PP0738 FA 6.653 1495.7 (M + H)+ PP0739 FA
7.215 1535.7 (M + H)+ PP0740 AA 8.051 1552.2 (M + Na)+ PP0741 FA 5.955 1481.7 (M - H)-
PP0742 FA 6.300 1509.7 (M + H)+ PP0743 FA 6.732 1525.7 (M + H)+ PP0744 FA 5.749 1466.2
(M - H)- PP0745 FA 6.235 1500.1 (M - H)- PP0746 FA 6.107 1492.1 (M - H)- PP0747 FA
6.400 1492.2 (M - H)- PP0748 FA 6.081 1512.2 (M - H)- PP0749 FA 6.653 1544.2 (M - H)-
PP0750 FA 6.459 1543.9 (M + H)+ PP0751 FA 6.447 1550.2 (M - H)- PP0752 FA 6.635 1587.1
(M + NH4)+ PP0753 FA 6.112 1520.2 (M - H)- PP0754 FA 6.075 1509.8 (M + H)+ PP0755 FA
6.152 1522.2 (M + H)+ PP0756 FA 6.615 1540.7 (M + NH4)+ PP0757 AA 8.203 1536.2 (M - H)-
PP0758 FA 6.361 1523.7 (M + H)+ PP0759 FA 6.611 1555.3 (M + NH4)+ PP0760 AA 8.443
1576.2 (M - H)- PP0761 FA 6.501 1571.7 (M + H)+ PP0762 FA 5.869 1543.2 (M + NH4)+
PP0763 FA 6.224 1551.7 (M + H)+ PP0764 FA 6.683 1584.9 (M + NH4)+ PP0765 FA 5.636
1507.7 (M - H)- PP0766 FA 6.139 1542.1 (M - H)- PP0767 AA 7.645 1533.8 (M - H)- PP0768
FA 6.324 1535.7 (M + H)+ PP0769 AA 7.692 1555.6 (M + H)+ PP0770 FA 6.197 1533.6 (M + H)+
PP0771 AA 7.875 1521.7 (M + H)+ PP0772 FA 6.147 1532.2 (M - H)- PP0773 FA 6.605 1535.8
(M + H)+ PP0774 AA 8.264 1548.0 (M - H)- PP0775 AA 8.021 1535.8 (M + H)+ PP0776 AA
8.173 1547.8 (M - H)- PP0778 AA 8.213 1582.2 (M - H)- PP0779 AA 7.795 1537.7 (M + H)+
PP0780 FA 6.335 1563.7 (M + H)+ PP0781 FA 6.735 1580.3 (M + H)+ PP0782 FA 5.808 1519.8
(M - H)- PP0783 FA 6.280 1555.6 (M + H)+ PP0784 FA 6.172 1547.6 (M + H)+ PP0785 FA
6.397 1547.8 (M + H)+ PP0786 AA 7.799 1589.9 (M + Na)+ PP0787 FA 5.911 1567.6 (M + H)+
PP0788 AA 7.784 1501.6 (M + Na)+ PP0789 FA 5.840 1533.8 (M + H)+ PP0790 FA 5.888 1520.2
(M - H)- PP0791 FA 5.892 1533.7 (M + H)+ PP0792 FA 6.351 1535.8 (M + H)+ PP0793 FA
6.659 1548.2 (M - H)- PP0794 AA 7.957 1533.9 (M - H)- PP0795 FA 6.427 1549.7 (M + H)+
PP0796 FA 6.999 1589.7 (M + H)+ PP0797 FA 6.392 1583.7 (M + H)+ PP0798 FA 5.785 1537.8
(M + H)+ PP0799 AA 8.004 1563.8 (M + H)+ PP0800 FA 6.584 1579.8 (M + H)+ PP0801 AA

7.373 1521.6 (M + H)- PP0802 FA 5.895 1555.6 (M + H)+ PP0803 FA 5.839 1546.2 (M - H)-
PP0804 FA 6.088 1547.8 (M + H)+ PP0805 FA 6.051 1465.8 (M - H)- PP0806 AA 7.783 1561.9
(M - H)- PP0807 FA 5.681 1597.7 (M + H)+ PP0808 FA 5.832 1532.2 (M - H)- PP0809 FA
5.907 1587.7 (M + H)+ PP0810 FA 5.607 1530.2 (M - H)- PP0811 AA 7.841 1565.9 (M + H)+
PP0812 FA 5.688 1550.0 (M - H)- PP0813 FA 5.551 1557.7 (M - H)- PP0814 AA 7.716 1551.7
(M + H)+ PP0815 FA 6.467 1576.2 (M - H)- PP0816 FA 6.365 1610.0 (M - H)- PP0817 FA
6.440 1547.8 (M + H)+ PP0818 FA 6.677 1601.8 (M + H)+ PP0819 FA 6.377 1543.8 (M - H)-
PP0820 AA 8.303 1579.7 (M + H)+ PP0821 AA 8.156 1563.9 (M - H)- PP0822 FA 6.043 1573.6
(M + H)+ PP0823 FA 6.313 1564.2 (M - H)- PP0824 FA 5.619 1505.7 (M + H)+ PP0825 FA
5.912 1574.2 (M + H)+ PP0826 FA 6.151 1575.7 (M + H)+ PP0827 FA 6.003 1505.6 (M + H)+
PP0828 AA 8.135 1609.9 (M - H)- PP0829 FA 6.285 1575.7 (M + H)+ PP0830 AA 8.035 1504.2
(M - H)- PP0831 FA 6.447 1611.6 (M + H)+ PP0832 FA 5.865 1560.2 (M - H)- PP0833 FA
5.912 1539.2 (M + NH4)+ PP0834 FA 6.159 1556.1 (M - H)- PP0835 FA 5.616 1536.1 (M - H)-
PP0836 FA 5.961 1550.2 (M - H)- PP0837 FA 5.941 1550.2 (M - H)- PP0838 FA 6.355 1577.8
(M - H)- PP0839 FA 6.600 1592.2 (M - H)- PP0840 AA 7.792 1529.9 (M + Na)+ PP0841 AA
8.069 1579.7 (M + H)+ PP0842 FA 6.173 1576.2 (M - H)- PP0843 FA 6.319 1600.2 (M - H)-
PP0844 AA 8.259 1592.2 (M - H)- PP0845 FA 6.047 1536.2 (M + H)+ PP0846 FA 6.313 1570.1
(M - H)- PP0847 FA 5.747 1551.8 (M + H)+ PP0848 AA 7.863 1546.2 (M - H)- PP0849 FA
6.176 1562.2 (M - H)- PP0850 AA 7.787 1473.9 (M - H)- PP0851 FA 6.221 1596.7 (M + NH4)+
PP0852 FA 6.313 1591.7 (M + H)+ PP0853 FA 6.447 1633.6 (M + H)+ PP0854 FA 6.255 1585.9
(M + H)+ PP0855 AA 7.997 1618.6 (M + H)+ PP0856 FA 6.527 1629.8 (M + H)+ PP0857 AA
8.320 1643.9 (M - H)- PP0858 FA 6.072 1549.7 (M - H)- PP0859 FA 6.156 1508.1 (M - H)-
PP0860 FA 5.947 1524.7 (M + NH4)+ PP0861 FA 6.225 1543.7 (M + H)+ PP0862 FA 6.185
1583.1 (M + NH4)+ PP0863 FA 5.887 1577.6 (M + H)+ PP0864 FA 6.013 1535.8 (M + H)+
PP0865 FA 5.637 1532.1 (M - H)- PP0866 FA 5.788 1569.7 (M + H)+ PP0867 FA 6.007 1608.8
(M + NH4)+ PP0868 FA 5.119 1535.7 (M + H)+ PP0869 FA 5.223 1535.7 (M + H)+ PP0870 FA
5.128 1535.8 (M + H)+ PP0871 FA 5.860 1566.7 (M + NH4)+ PP0872 FA 6.025 1548.2 (M - H)-
PP0873 FA 5.873 1549.7 (M + H)+ PP0874 FA 5.993 1561.7 (M + H)+ PP0875 FA 6.084 1575.7
(M + H)+ PP0876 FA 6.256 1589.7 (M + H)+ PP0877 FA 6.465 1603.7 (M + H)+ PP0878 FA
6.713 1574.2 (M - H)- PP0879 FA 6.731 1606.7 (M + NH4)+ PP0880 AA 8.416 1604.3 (M + H)+
PP0881 FA 7.051 1617.8 (M + H)+ PP0882 FA 5.689 1470.2 (M - H)- PP0883 FA 6.501 1484.2
(M - H)- PP0884 FA 5.379 1485.8 (M - H)- PP0885 FA 5.504 1474.1 (M - H)- PP0886 FA
5.732 1528.1 (M - H)- PP0887 FA 5.264 1445.6 (M + H)+ PP0888 AA 7.311 1444.2 (M - H)-
PP0889 FA 6.176 1499.9 (M - H)- PP0890 FA 6.303 1506.8 (M + NH4)+ PP0891 FA 6.512
1544.1 (M + H)+ PP0892 FA 6.071 1476.8 (M + NH4)+ PP0893 AA 7.871 1457.8 (M - H)-
PP0894 FA 5.287 1468.2 (M - H)- PP0895 FA 5.477 1481.9 (M - H)- PP0896 FA 5.767 1496.2
(M - H)- PP0898 AA 7.876 1565.7 (M + H)+ PP0899 FA 5.665 1553.7 (M + H)+ PP0900 AA
7.244 1480.2 (M - H)- PP0901 AA 7.671 1547.8 (M - H)- PP0902 FA 6.028 1563.7 (M + H)+
PP0903 AA 7.604 1549.8 (M - H)- PP0904 AA 7.955 1470.2 (M - H)- PP0905 FA 6.173 1513.9
(M + H)+ PP0906 FA 6.232 1474.2 (M - H)- PP0907 FA 6.139 1473.7 (M - H)- PP0908 FA
6.420 1489.7 (M - H)- PP0909 FA 6.411 1491.6 (M + H)+ PP0910 FA 6.383 1494.1 (M - H)-
PP0911 FA 6.157 1470.2 (M - H)- PP0912 AA 8.052 1471.7 (M + H)+ PP0913 FA 5.441 1446.7
(M + H)+ PP0914 AA 7.984 1485.9 (M - H)- PP0915 FA 5.989 1487.9 (M + H)+ PP0916 FA
6.309 1545.7 (M + H)+ PP0917 FA 6.216 1543.8 (M - H)- PP0918 FA 6.515 1560.1 (M - H)-
PP0919 FA 6.487 1560.2 (M - H)- PP0920 FA 6.473 1565.7 (M + H)+ PP0921 FA 6.487 1563.8
(M - H)- PP0922 AA 8.229 1540.3 (M - H)- PP0923 AA 7.612 1515.2 (M - H)- PP0924 FA
6.428 1557.7 (M + H)+ PP0925 AA 7.756 1509.9 (M + H)+ PP0926 FA 6.457 1629.7 (M + H)+
PP0927 FA 6.529 1646.8 (M + NH4)+ PP0928 AA 8.200 1647.7 (M + H)+ PP0929 FA 6.521
1645.7 (M + H)+ PP0930 FA 6.739 1646.2 (M + H)+ PP0931 FA 6.520 1626.0 (M + H)+ PP0932
FA 6.656 1642.7 (M + NH4)+ PP0933 FA 6.693 1625.7 (M + H)+ PP0934 FA 6.569 1604.7 (M +

NH4)+ PP0935 FA 6.603 1586.2 (M - H)- PP0936 FA 6.624 1586.2 (M - H)- PP0937 AA 8.172 1623.0 (M + NH4)+ PP0938 FA 6.639 1602.2 (M - H)- PP0939 FA 6.769 1604.1 (M + H)+ PP0940 FA 6.829 1603.7 (M + H)+ PP0941 FA 6.637 1582.2 (M - H)- PP0942 FA 6.760 1584.2 (M + H)+ PP0943 AA 8.311 1584.3 (M + H)+ PP0944 FA 6.528 1563.7 (M + H)+ PP0945 FA 6.029 1564.2 (M - H)- PP0946 FA 6.293 1579.7 (M + H)+ PP0947 FA 6.225 1594.3 (M + H)+ PP0948 FA 6.453 1630.7 (M + NH4)+ PP0949 FA 6.676 1662.7 (M + NH4)+ PP0950 AA 8.312 1626.0 (M + H)+ PP0951 AA 8.281 1644.0 (M + H)+ PP0952 FA 6.803 1676.7 (M + NH4)+ PP0953 FA 6.944 1638.3 (M - H)- PP0954 FA 6.603 1581.9 (M - H)- PP0955 FA 5.980 1564.2 (M + H)+ PP0956 FA 6.696 1604.0 (M + H)+ PP0958 FA 6.545 1578.3 (M - H)- PP0959 FA 5.652 1467.7 (M + H)+ PP0960 FA 5.573 1483.6 (M + H)+ PP0961 AA 7.997 1530.1 (M + H)+ PP0962 FA 6.109 1546.2 (M - H)- PP0963 FA 6.305 1563.6 (M + H)+ PP0964 FA 5.927 1497.7 (M + H)+ PP0965 FA 6.171 1537.9 (M - H)- PP0966 FA 6.251 1531.7 (M + H)+ PP0967 FA 5.807 1481.6 (M + H)+ PP0968 FA 5.993 1523.9 (M - H)- PP0969 FA 6.053 1516.2 (M - H)- PP0970 FA 6.203 1508.1 (M - H)- PP0971 FA 6.093 1452.2 (M - H)- PP0972 FA 6.095 1482.2 (M - H)- PP0973 FA 6.173 1474.2 (M - H)- PP0975 FA 7.624 1533.9 (M - H)- PP0976 FA 7.545 1564.3 (M - H)- PP0977 FA 7.576 1556.1 (M - H)- PP0978 FA 5.631 1502.1 (M - H)- PP0979 AA 7.675 1578.9 (M + NH4)+ PP0980 FA 5.704 1518.2 (M - H)- PP0981 FA 6.955 1629.8 (M + H)+ PP0982 FA 7.051 1586.3 (M - H)- PP0983 FA 5.533 1489.7 (M - H)- PP0984 AA 7.593 1516.1 (M - H)- PP0985 AA 7.892 1524.0 (M - H)- PP0986 AA 6.993 1471.7 (M - H)- PP0987 FA 5.475 1455.7 (M - H)- PP0988 AA 7.971 1489.7 (M - H)- PP0989 FA 6.839 1539.6 (M + H)+ PP0992 FA 5.424 1563.9 (M + H)+ PP0993 FA 5.696 1577.7 (M + H)+ PP0997 FA 6.353 1591.7 (M + H)+ PP0998 FA 6.000 1629.7 (M - H)- PP0999 FA 6.283 1590.3 (M - H)- PP1000 AA 7.893 1590.2 (M + H)+ PP1001 FA 6.631 1617.8 (M + H)+ PP1002 FA 5.665 1597.9 (M - H)- PP1003 AA 7.601 1548.2 (M - H)- PP1004 FA 6.909 1604.3 (M - H)- PP1005 AA 8.339 1626.3 (M + Na)+ PP1006 FA 6.405 1630.7 (M + NH4)+ PP1008 AA 8.244 1645.7 (M + H)+ PP1009 FA 6.955 1604.3 (M - H)- PP1010 FA 6.699 1621.0 (M + NH4)+ PP1011 FA 7.265 1631.8 (M + H)+ PP1012 FA 6.489 1631.0 (M + NH4)+ PP1013 FA 6.729 1608.8 (M + NH4)+ PP1014 AA 8.123 1611.9 (M + Na)+ PP1015 FA 7.051 1617.8 (M + H)+ PP1016 FA 6.219 1561.8 (M - H)- PP1017 FA 6.248 1547.8 (M - H)- PP1018 FA 6.057 1535.7 (M + H)+ PP1019 AA 7.920 1546.2 (M - H)- PP1020 FA 5.771 1507.7 (M + H)+ PP1021 FA 5.808 1518.2 (M - H)- PP1022 FA 5.924 1520.2 (M - H)- PP1024 AA 8.096 1590.3 (M - H)- PP1025 FA 5.959 1576.2 (M - H)- PP1026 FA 6.064 1606.7 (M + NH4)+ PP1027 FA 5.675 1547.7 (M - H)- PP1028 FA 5.691 1561.7 (M + H)+ PP1029 AA 7.875 1561.8 (M - H)- PP1030 FA 6.152 1608.7 (M + NH4)+ PP1031 FA 6.360 1546.2 (M - H)- PP1032 AA 8.017 1555.8 (M + Na)+ PP1033 FA 6.181 1545.6 (M + H)+ PP1034 FA 5.912 1505.8 (M + H)+ PP1035 FA 5.881 1517.6 (M + H)+ PP1036 FA 6.043 1520.2 (M + H)+ PP1037 FA 6.359 1547.6 (M + H)+ PP1038 FA 6.264 1588.3 (M - H)- PP1039 FA 6.065 1592.7 (M + NH4)+ PP1040 FA 6.065 1586.2 (M - H)- PP1041 AA 7.861 1546.2 (M - H)- PP1042 FA 5.777 1557.9 (M - H)- PP1043 FA 5.941 1560.2 (M - H)- PP1044 FA 6.265 1589.7 (M + H)+ PP1045 FA 6.608 1561.7 (M + H)+ PP1046 FA 6.440 1547.7 (M + H)+ PP1047 FA 6.531 1559.9 (M + H)+ PP1048 FA 6.189 1519.9 (M + H)+ PP1049 FA 6.319 1531.9 (M + H)+ PP1050 FA 6.288 1532.2 (M - H)- PP1051 FA 6.620 1560.2 (M - H)- PP1052 FA 5.807 1518.2 (M - H)- PP1053 FA 6.072 1532.2 (M - H)- PP1054 FA 5.976 1518.2 (M - H)- PP1056 AA 7.648 1530.2 (M - H)- PP1057 FA 5.992 1575.7 (M + H)+ PP1058 FA 6.261 1545.9 (M + H)+ PP1059 FA 5.929 1545.8 (M - H)- PP1060 FA 6.044 1563.6 (M + H)+ PP1063 FA 5.780 1544.2 (M - H)- PP1064 AA 7.671 1517.8 (M + H)+ PP1065 FA 5.823 1504.2 (M - H)- PP1067 FA 6.312 1563.6 (M + H)+ PP1068 FA 6.479 1577.7 (M + H)+ PP1069 FA 6.716 1591.8 (M + H)+ PP1070 FA 5.967 1631.9 (M + H)+ PP1071 FA 6.284 1590.3 (M - H)- PP1072 AA 7.909 1589.8 (M + H)+ PP1073 FA 6.641 1618.3 (M + H)+ PP1075 FA 5.537 1549.6 (M + H)+ PP1076 FA 5.732 1563.7 (M + H)+ PP1077 FA 6.011 1576.2 (M - H)- PP1078 AA 7.361 1548.2 (M - H)- PP1079 AA 7.759 1562.2 (M - H)- PP1080 FA 6.051 1577.8 (M + H)+ PP1082 FA

6.401 1577.7 (M + H)- PP1083 FA 6.653 1609.3 (M + NH4)+ PP1084 FA 5.477 1549.8 (M + H)+ PP1085 FA 6.173 1563.7 (M + H)+ PP1086 FA 5.815 1519.7 (M + H)+ PP1087 FA 5.608 1491.7 (M - H)- PP1088 FA 5.253 1478.1 (M - H)- PP1089 AA 7.491 1492.2 (M - H)- PP1090 FA 5.711 1531.8 (M + H)+ PP1091 FA 5.492 1516.2 (M - H)- PP1092 AA 7.515 1546.2 (M - H)- PP1093 FA 5.676 1573.7 (M + H)+ PP1094 FA 5.495 1489.7 (M - H)- PP1095 FA 6.157 1448.7 (M + H)+ PP1096 FA 6.688 1490.8 (M + H)+ PP1097 FA 6.003 1459.2 (M - H)- PP1098 FA 5.993 1432.7 (M - H)- PP1099 FA 6.355 1432.7 (M + H)+ PP1100 FA 6.528 1475.1 (M - H)- PP1101 FA 6.099 1416.7 (M - H)- PP1102 FA 6.528 1445.2 (M - H)- PP1103 FA 5.424 1481.7 (M - H)- PP1104 FA 5.535 1505.7 (M + H)+ PP1105 FA 6.016 1547.7 (M + H)+ PP1106 AA 7.469 1474.1 (M - H)- PP1107 FA 5.843 1533.7 (M + H)+ PP1108 AA 7.376 1489.7 (M - H)- PP1109 FA 5.671 1519.6 (M + H)+ PP1110 AA 7.577 1517.9 (M - H)- PP1111 AA 7.600 1529.8 (M - H)- PP1112 FA 6.104 1577.0 (M + NH4)+ PP1113 FA 5.385 1521.6 (M + H)+ PP1114 FA 5.647 1516.2 (M - H)- PP1115 FA 5.529 1515.7 (M + H)+ PP1116 AA 7.688 1565.9 (M - H)- PP1117 FA 6.079 1547.7 (M + H)+ PP1118 FA 5.919 1562.7 (M + NH4)+ PP1119 FA 5.551 1536.2 (M - H)- PP1120 AA 7.349 1539.7 (M - H)- PP1121 FA 5.601 1554.2 (M - H)- PP1122 FA 5.167 1577.0 (M + H)+ PP1123 AA 7.657 1571.8 (M - H)- PP1124 AA 7.379 1558.2 (M - H)- PP1125 FA 6.317 1573.8 (M + H)+ PP1126 AA 7.655 1584.2 (M - H)- PP1127 AA 7.815 1600.2 (M - H)- PP1128 FA 5.533 1489.7 (M - H)- PP1129 FA 5.568 1497.6 (M + H)+ PP1130 FA 5.319 1489.7 (M - H)- PP1131 FA 5.267 1478.1 (M - H)- PP1132 FA 5.405 1479.6 (M + H)+ PP1133 AA 7.509 1527.7 (M - H)- PP1134 FA 5.625 1475.7 (M + H)+ PP1135 FA 5.355 1461.8 (M + H)+ PP1136 FA 5.765 1474.2 (M - H)- PP1137 FA 5.643 1492.1 (M - H)- PP1138 AA 7.303 1476.2 (M - H)- PP1139 FA 5.311 1445.8 (M - H)- PP1140 FA 5.249 1433.6 (M - H)- PP1141 FA 5.629 1462.2 (M - H)- PP1142 FA 5.868 1476.1 (M - H)- PP1143 FA 5.436 1447.6 (M - H)- PP1144 AA 7.804 1505.9 (M - H)- PP1145 AA 7.487 1478.1 (M - H)- PP1146 AA 7.409 1447.7 (M - H)- PP1147 FA 6.083 1507.6 (M + H)+ PP1148 FA 6.739 1534.2 (M - H)- PP1149 FA 6.693 1530.3 (M - H)- PP1150 AA 7.776 1489.7 (M - H)- PP1151 FA 6.608 1523.7 (M + H)+ PP1152 FA 6.169 1521.6 (M + H)+ PP1153 AA 8.340 1552.3 (M - H)- PP1154 FA 6.147 1550.1 (M - H)- PP1155 FA 5.957 1544.7 (M + H)+ PP1156 AA 7.688 1544.7 (M + H)+ PP1157 FA 5.939 1544.7 (M + H)+ PP1158 FA 6.536 1587.6 (M + H)+ PP1159 FA 5.964 1577.7 (M + H)+ PP1160 FA 6.277 1591.7 (M + H)+ PP1161 AA 8.323 1592.3 (M + H)+ PP1162 FA 6.900 1623.3 (M + NH4)+ PP1163 AA 8.179 1616.2 (M - H)- PP1164 FA 6.441 1617.7 (M + H)+ PP1165 AA 8.039 1637.3 (M + H)+ PP1166 FA 6.287 1637.3 (M + H)+ PP1167 AA 7.977 1637.2 (M + H)+ PP1168 FA 6.872 1680.1 (M + H)+ PP1169 FA 6.585 1575.6 (M + H)+ PP1170 FA 6.533 1574.2 (M - H)- PP1171 FA 6.425 1594.8 (M + H)+ PP1172 FA 6.393 1594.7 (M + H)+ PP1173 FA 6.416 1594.7 (M + H)+ PP1174 FA 6.972 1637.7 (M + H)+ PP1175 FA 5.511 1483.7 (M - H)- PP1176 FA 5.735 1498.2 (M - H)- PP1177 AA 7.697 1535.8 (M + H)+ PP1178 FA 5.811 1525.7 (M + H)+ PP1179 FA 5.547 1514.2 (M - H)- PP1180 AA 7.696 1528.2 (M - H)- PP1181 FA 5.991 1560.2 (M - H)- PP1182 FA 4.067 1563.3 (M + H)+ PP1183 FA 6.015 1565.9 (M - H)- PP1184 AA 7.915 1565.8 (M - H)- PP1185 FA 5.825 1603.8 (M + NH4)+ PP1186 AA 7.707 1587.2 (M + H)+ PP1187 FA 5.829 1586.8 (M + H)+ PP1188 FA 6.431 1629.7 (M + H)+ PP1189 AA 7.653 1441.7 (M - H)- PP1190 AA 7.765 1479.7 (M + Na)+ PP1191 AA 7.675 1491.7 (M - H)- PP1192 AA 7.845 1482.2 (M - H)- PP1193 FA 5.637 1472.2 (M - H)- PP1194 FA 5.600 1486.2 (M - H)- PP1195 AA 7.931 1518.2 (M - H)- PP1196 FA 4.143 1518.8 (M - H)- PP1197 FA 6.119 1524.2 (M - H)- PP1198 FA 6.065 1523.8 (M - H)- PP1199 FA 5.584 1493.7 (M + H)+ PP1200 FA 5.756 1524.8 (M + NH4)+ PP1201 AA 7.749 1535.7 (M + H)+ PP1202 FA 6.536 1563.9 (M + H)+ PP1203 FA 5.868 1532.2 (M - H)- PP1204 FA 6.251 1561.7 (M + H)+ PP1205 AA 7.549 1522.2 (M - H)- PP1206 FA 5.335 1487.7 (M + H)+ PP1207 AA 7.448 1499.8 (M - H)- PP1208 FA 5.652 1515.8 (M + H)+ PP1209 AA 7.608 1528.3 (M - H)- PP1210 FA 6.345 1558.3 (M + H)+ PP1211 FA 5.637 1544.8 (M + NH4)+ PP1212 FA 6.049 1555.7 (M + H)+ PP1213 FA 5.344 1516.2 (M - H)- PP1214 FA 5.631 1457.6 (M + H)+ PP1215 FA 5.824 1471.6 (M + H)+ PP1216 AA 7.828 1486.1

(M + H)+ PP1217 FA 6.039 1497.8 (M - H)- PP1218 FA 6.607 1527.7 (M + H)+ PP1219 AA 7.847 1496.0 (M - H)- PP1220 AA 8.073 1524.2 (M - H)- PP1221 FA 5.649 1486.0 (M - H)- PP1222 FA 5.239 1445.7 (M + H)+ PP1223 AA 7.503 1457.9 (M - H)- PP1224 FA 5.563 1472.2 (M - H)- PP1225 FA 5.691 1488.0 (M + H)+ PP1226 AA 8.040 1514.3 (M - H)- PP1227 AA 7.605 1484.2 (M - H)- PP1228 AA 7.867 1512.3 (M - H)- PP1229 AA 7.435 1474.2 (M - H)- PP1230 FA 5.683 1473.7 (M + H)+ PP1231 FA 5.797 1487.7 (M + H)+ PP1232 AA 7.791 1500.2 (M - H)- PP1233 FA 5.783 1498.2 (M - H)- PP1234 AA 7.499 1505.8 (M + H)+ PP1235 AA 7.947 1559.7 (M + H)+ PP1242 FA 6.381 1561.7 (M + H)+ PP1243 FA 5.303 1498.2 (M - H)- PP1244 FA 5.468 1512.2 (M - H)- PP1245 FA 5.629 1545.0 (M + NH4)+ PP1246 AA 7.908 1555.8 (M + H)+ PP1247 FA 5.460 1525.7 (M + H)+ PP1248 FA 5.913 1570.7 (M + NH4)+ PP1249 FA 5.547 1468.0 (M - H)- PP1250 AA 7.691 1482.1 (M - H)- PP1251 FA 5.847 1497.7 (M + H)+ PP1252 FA 6.417 1524.1 (M - H)- PP1253 AA 7.715 1496.0 (M + H)+ PP1254 FA 6.123 1523.6 (M + H)+ PP1255 AA 7.335 1456.2 (M - H)- PP1256 FA 5.347 1471.7 (M + H)+ PP1257 AA 7.549 1484.2 (M - H)- PP1258 FA 5.872 1519.8 (M - H)- PP1259 FA 6.092 1513.7 (M + H)+ PP1260 FA 5.344 1482.2 (M - H)- PP1261 FA 5.805 1511.8 (M + H)+ PP1262 FA 5.427 1471.7 (M + H)+ PP1263 FA 5.732 1497.9 (M - H)- PP1264 FA 6.008 1524.2 (M - H)- PP1265 AA 7.596 1481.9 (M - H)- PP1266 FA 5.564 1533.7 (M - H)- PP1267 FA 5.824 1589.7 (M + H)+ PP1268 FA 5.659 1547.7 (M + H)+ PP1269 FA 5.432 1502.2 (M - H)- PP1270 FA 5.327 1490.2 (M - H)- PP1271 FA 5.203 1445.7 (M - H)- PP1272 AA 7.575 1487.8 (M - H)- PP1273 FA 5.551 1475.7 (M + H)+ PP1274 AA 7.539 1540.2 (M - H)- PP1275 FA 6.077 1532.2 (M - H)- PP1276 FA 5.689 1519.9 (M + H)+ PP1277 FA 5.968 1532.2 (M - H)- PP1278 FA 5.320 1505.8 (M - H)- PP1279 FA 5.421 1534.2 (M - H)- PP1280 FA 5.469 1492.2 (M - H)- PP1281 FA 5.535 1546.1 (M + H)+ PP1282 FA 5.740 1575.7 (M + H)+ PP1283 FA 5.593 1573.7 (M + H)+ PP1284 AA 7.412 1532.2 (M - H)- PP1285 FA 5.439 1583.6 (M + H)+ PP1286 FA 5.663 1549.7 (M + H)+ PP1287 FA 5.864 1575.8 (M - H)- PP1288 AA 7.624 1560.2 (M - H)- PP1289 FA 6.219 1545.7 (M + H)+ PP1290 FA 6.519 1574.0 (M + H)+ PP1291 FA 6.359 1571.7 (M + H)+ PP1292 FA 6.045 1530.2 (M - H)- PP1293 FA 6.211 1580.2 (M - H)- PP1294 FA 6.377 1558.2 (M - H)- PP1295 FA 6.000 1559.7 (M + H)+ PP1296 FA 5.868 1557.7 (M + H)+ PP1297 AA 7.569 1516.2 (M - H)- PP1298 FA 5.740 1565.8 (M - H)- PP1299 FA 5.865 1562.8 (M + NH4)+ PP1300 FA 6.151 1547.7 (M + H)+ PP1301 FA 5.992 1544.2 (M - H)- PP1302 FA 5.676 1504.1 (M - H)- PP1303 FA 5.844 1554.2 (M - H)- PP1304 FA 5.995 1532.2 (M - H)- PP1305 FA 6.144 1532.2 (M - H)- PP1306 FA 6.459 1561.7 (M + H)+ PP1307 FA 6.293 1559.9 (M + H)+ PP1308 FA 5.967 1519.7 (M + H)+ PP1309 FA 6.141 1569.6 (M + H)+ PP1310 FA 6.301 1546.2 (M - H)- PP1311 FA 5.915 1602.3 (M + H)+ PP1312 FA 5.753 1617.0 (M + NH4)+ PP1313 FA 5.436 1558.2 (M - H)- PP1314 FA 5.607 1590.8 (M + NH4)+ PP1315 AA 7.737 1585.9 (M - H)- PP1316 FA 5.688 1549.7 (M + H)+ PP1317 FA 5.995 1577.7 (M + H)+ PP1318 FA 5.843 1574.2 (M - H)- PP1319 AA 7.475 1533.9 (M - H)- PP1320 FA 5.704 1584.2 (M - H)- PP1321 FA 5.451 1543.7 (M + H)+ PP1322 FA 5.337 1499.8 (M - H)- PP1323 FA 5.713 1513.6 (M + H)+ PP1324 FA 5.793 1570.3 (M - H)- PP1325 FA 5.675 1528.2 (M - H)- PP1326 AA 7.889 1542.1 (M + H)+ PP1327 FA 5.620 1568.3 (M - H)- PP1328 FA 5.512 1527.7 (M + H)+ PP1329 FA 5.864 1556.7 (M + NH4)+ PP1330 FA 5.904 1551.7 (M + H)+ PP1331 FA 5.848 1563.7 (M + H)+ PP1333 FA 6.700 1589.8 (M + H)+ PP1334 FA 6.580 1519.7 (M + H)+ PP1335 AA 8.217 1560.2 (M - H)- PP1336 FA 5.760 1474.1 (M - H)- PP1337 FA 5.693 1516.2 (M - H)- PP1338 FA 5.500 1502.1 (M - H)- PP1339 FA 6.060 1542.2 (M - H)- PP1340 FA 5.845 1483.8 (M - H)- PP1341 AA 7.875 1540.2 (M - H)- PP1342 FA 6.224 1513.9 (M - H)- PP1343 FA 5.909 1542.2 (M - H)- PP1344 FA 5.701 1528.2 (M - H)- PP1345 FA 6.132 1554.2 (M - H)- PP1346 FA 5.961 1500.2 (M - H)- PP1347 FA 6.155 1533.8 (M - H)- PP1348 FA 6.073 1595.0 (M + NH4)+ PP1349 FA 6.011 1490.2 (M - H)- PP1350 FA 5.919 1533.7 (M + H)+ PP1351 FA 5.971 1575.7 (M + H)+ PP1352 FA 6.032 1535.7 (M + H)+ PP1353 FA 5.951 1594.8 (M + NH4)+ PP1354 FA 5.827 1561.9 (M - H)- PP1355 FA 6.349 1603.8 (M + H)+ PP1356 FA 6.199 1561.8

(M + H)+ PP1357 AA 7.967 1574.2 (M + H)+ PP1358 FA 6.139 1532.1 (M - H)- PP1359 FA 6.060 1592.7 (M + NH4)+ PP1360 FA 5.929 1560.1 (M - H)- PP1361 FA 6.429 1601.7 (M + H)+ PP1362 FA 6.301 1558.1 (M - H)- PP1363 FA 5.805 1532.2 (M + H)+ PP1364 FA 5.861 1490.2 (M - H)- PP1365 FA 5.781 1531.8 (M - H)- PP1366 FA 5.615 1519.6 (M + H)+ PP1367 AA 8.011 1557.9 (M - H)- PP1368 AA 7.852 1515.9 (M - H)- PP1369 FA 5.765 1560.2 (M - H)- PP1370 FA 5.812 1519.9 (M - H)- PP1371 FA 5.745 1563.7 (M + H)+ PP1372 FA 5.612 1547.9 (M - H)- PP1373 FA 6.120 1589.7 (M + H)+ PP1374 FA 5.983 1564.7 (M + NH4)+ PP1375 FA 5.852 1557.8 (M - H)- PP1376 FA 5.928 1517.8 (M - H)- PP1377 AA 7.845 1561.8 (M + H)+ PP1378 AA 7.735 1564.9 (M + NH4)+ PP1379 FA 6.200 1587.7 (M + H)+ PP1380 FA 6.089 1544.2 (M - H)- PP1381 FA 5.613 1517.7 (M + H)+ PP1382 FA 5.653 1476.1 (M - H)- PP1383 FA 5.583 1520.2 (M + H)+ PP1384 FA 5.400 1504.2 (M - H)- PP1385 FA 5.967 1545.7 (M + H)+ PP1386 FA 6.115 1535.1 (M + NH4)+ PP1387 FA 6.464 1561.9 (M + H)+ PP1388 FA 6.303 1559.6 (M + H)+ PP1389 FA 6.311 1547.7 (M + H)+ PP1390 FA 5.417 1461.7 (M + H)+ PP1391 FA 5.585 1487.7 (M + H)+ PP1392 FA 5.889 1532.2 (M - H)- PP1393 AA 7.932 1600.0 (M - H)- PP1394 AA 7.883 1546.3 (M - H)- PP1395 AA 7.717 1548.2 (M - H)- PP1396 AA 7.880 1559.7 (M + H)+ PP1397 FA 6.065 1545.9 (M - H)- PP1398 AA 8.033 1577.8 (M + H)+ PP1399 FA 6.152 1587.8 (M - H)- PP1400 FA 6.135 1522.3 (M - H)- PP1401 FA 6.012 1535.8 (M + H)+ PP1402 FA 6.027 1539.7 (M + H)+ PP1403 FA 6.073 1526.2 (M + H)+ PP1404 FA 6.577 1480.7 (M - H)- PP1405 FA 6.612 1468.6 (M + H)+ PP1406 AA 7.607 1516.2 (M - H)- PP1407 FA 5.520 1477.7 (M + H)+ PP1408 AA 7.584 1536.7 (M + NH4)+ PP1409 FA 5.711 1502.2 (M - H)- PP1410 FA 6.169 1418.9 (M - H)- PP1411 FA 5.968 1506.3 (M + H)+ PP1412 AA 7.989 1530.3 (M - H)- PP1413 AA 8.175 1448.7 (M + H)+ PP1414 AA 7.632 1477.9 (M - H)- PP1415 AA 7.667 1522.2 (M + H)+ PP1416 FA 5.856 1504.3 (M - H)- PP1417 FA 6.381 1420.8 (M - H)- PP1418 FA 6.243 1614.3 (M - H)- PP1419 FA 6.464 1598.3 (M - H)- PP1420 FA 6.109 1561.8 (M + H)+ PP1421 FA 6.329 1544.3 (M - H)- PP1422 FA 5.837 1585.9 (M - H)- PP1423 FA 6.085 1588.7 (M + NH4)+ PP1424 AA 7.751 1551.3 (M + NH4)+ PP1425 FA 5.940 1517.7 (M + H)+ PP1426 FA 4.965 1432.2 (M - H)- PP1427 FA 5.144 1434.2 (M - H)- PP1428 AA 8.004 1469.9 (M - H)- PP1429 FA 5.699 1506.3 (M - H)- PP1430 FA 6.507 1500.0 (M + H)+ PP1431 FA 6.131 1535.9 (M + H)+ PP1432 FA 6.237 1472.3 (M - H)- PP1433 FA 5.861 1508.3 (M - H)- PP1434 AA 7.965 1457.8 (M + H)+ PP1435 FA 6.085 1471.8 (M + H)+ PP1436 FA 5.544 1493.7 (M + H)+ PP1437 FA 5.663 1506.3 (M - H)- PP1438 FA 6.015 1487.8 (M + H)+ PP1439 FA 6.131 1500.3 (M - H)- PP1440 FA 5.564 1522.0 (M - H)- PP1441 FA 5.676 1535.9 (M - H)- PP1442 FA 5.488 1505.7 (M + H)+ PP1443 FA 5.885 1467.8 (M - H)- PP1444 AA 7.355 1491.8 (M + H)+ PP1445 AA 7.763 1455.8 (M + H)+ PP1446 AA 7.881 1469.7 (M + H)+ PP1447 FA 5.481 1505.8 (M + H)+ PP1448 FA 5.311 1470.1 (M - H)- PP1449 FA 5.375 1510.2 (M - H)- PP1450 FA 5.488 1491.7 (M - H)- PP1451 FA 5.473 1501.7 (M - H)- PP1452 AA 7.363 1478.2 (M - H)- PP1453 FA 5.677 1506.3 (M - H)- PP1454 FA 5.431 1491.8 (M + H)+ PP1455 FA 5.756 1518.3 (M - H)- PP1456 FA 5.275 1479.9 (M + H)+ PP1457 AA 7.371 1447.7 (M - H)- PP1458 FA 5.947 1533.8 (M + H)+ PP1459 FA 5.857 1521.7 (M + H)+ PP1460 FA 5.896 1521.7 (M + H)+ PP1461 FA 5.057 1433.7 (M - H)- PP1462 FA 5.712 1435.7 (M + H)+ PP1463 AA 7.776 1475.7 (M + H)+ PP1464 FA 5.871 1455.9 (M - H)- PP1465 FA 5.880 1465.8 (M - H)- PP1466 FA 5.697 1441.8 (M - H)- PP1467 FA 6.031 1471.7 (M + H)+ PP1468 AA 7.808 1454.2 (M - H)- PP1469 FA 6.119 1482.3 (M - H)- PP1470 FA 5.681 1442.2 (M - H)- PP1471 FA 5.720 1411.8 (M - H)- PP1472 FA 6.307 1497.8 (M + H)+ PP1473 AA 8.117 1483.9 (M - H)- PP1474 FA 6.255 1485.8 (M + H)+ PP1475 FA 5.435 1397.8 (M - H)- PP1476 FA 5.559 1457.8 (M - H)- PP1477 FA 5.440 1460.2 (M - H)- PP1478 FA 5.333 1444.2 (M - H)- PP1479 FA 5.215 1445.7 (M - H)- PP1480 AA 7.805 1460.0 (M - H)- PP1481 FA 5.632 1462.2 (M - H)- PP1482 FA 5.499 1447.7 (M + H)+ PP1483 FA 5.383 1448.2 (M - H)- PP1484 FA 5.401 1443.7 (M + H)+ PP1485 FA 5.524 1439.5 (M - H)- PP1486 AA 7.860 1470.2 (M - H)- PP1487 FA 5.971 1467.8 (M - H)- PP1488 FA 5.497 1483.7 (M + H)+ PP1489 FA 6.191 1436.9 (M - H)-

PP1490 FA 5.980 1424.7 (M + H)+ PP1491 AA 8.220 1465.2 (M - H)- PP1492 AA 8.020 1452.7 (M + H)+ PP1493 FA 6.399 1440.7 (M + H)+ PP1494 FA 6.183 1425.1 (M - H)- PP1495 AA 7.635 1535.9 (M - H)- PP1496 AA 7.413 1522.2 (M - H)- PP1497 FA 5.436 1505.6 (M + H)+ PP1498 FA 5.876 1555.6 (M + H)+ PP1499 FA 5.815 1500.2 (M - H)- PP1500 FA 5.736 1522.1 (M - H)- PP1501 FA 5.937 1532.2 (M - H)- PP1502 AA 7.665 1531.8 (M - H)- PP1503 FA 5.743 1544.2 (M - H)- PP1504 AA 7.969 1582.2 (M - H)- PP1505 FA 6.292 1582.2 (M - H)- PP1506 FA 6.220 1595.7 (M + H)+ PP1507 AA 7.760 1531.8 (M - H)- PP1508 FA 5.968 1533.6 (M + H)+ PP1509 FA 5.904 1544.2 (M - H)- PP1510 FA 5.981 1525.9 (M - H)- PP1511 AA 7.813 1515.7 (M + H)+ PP1512 FA 5.919 1549.6 (M + H)+ PP1513 FA 5.925 1535.8 (M - H)- PP1514 FA 5.755 1497.7 (M + H)+ PP1515 AA 7.488 1481.8 (M - H)- PP1516 AA 7.971 1524.3 (M - H)- PP1517 AA 7.741 1511.7 (M + H)+ PP1518 FA 5.907 1498.2 (M - H)- PP1519 AA 7.551 1485.7 (M + H)+ PP1520 FA 6.412 1438.8 (M - H)- PP1521 FA 6.200 1424.7 (M - H)- PP1522 FA 6.879 1466.8 (M - H)- PP1523 AA 8.104 1454.7 (M + H)+ PP1524 FA 6.605 1440.9 (M - H)- PP1525 AA 7.932 1428.7 (M + H)+ PP1526 FA 5.655 1538.2 (M - H)- PP1527 FA 5.416 1524.2 (M - H)- PP1528 AA 7.601 1493.8 (M - H)- PP1529 FA 5.335 1499.2 (M + NH4)+ PP1530 FA 6.001 1541.0 (M + NH4)+ PP1531 FA 5.759 1507.9 (M - H)- PP1532 FA 5.747 1497.7 (M + H)+ PP1533 AA 7.528 1565.9 (M - H)- PP1534 FA 5.857 1569.7 (M + H)+ PP1535 FA 6.119 1595.7 (M + H)+ PP1536 FA 5.976 1541.7 (M + H)+ PP1537 FA 6.085 1514.2 (M - H)- PP1538 FA 6.028 1537.6 (M + H)+ PP1552 AA 7.925 1456.2 (M - H)- PP1553 FA 5.576 1492.2 (M - H)- PP1554 AA 8.196 1583.7 (M + H)+ PP1557 AA 7.747 1539.9 (M - H)- PP1558 AA 7.908 1565.3 (M + NH.sub.4)+ PP1559 AA 7.691 1526.2 (M - H)- PP1560 FA 5.992 1540.2 (M - H)- PP1561 FA 6.204 1572.9 (M + NH.sub.4)+ PP1562 FA 6.160 1573.0 (M + NH.sub.4)+ PP1563 FA 5.988 1552.4 (M - H)- PP1564 FA 6.257 1585.3 (M + NH.sub.4)+ PP1565 FA 6.461 1582.3 (M + H)+ PP1566 AA 7.821 1552.7 (M + NH.sub.4)+ PP1568 AA 7.964 1582.2 (M + Na)+ PP1569 AA 7.920 1564.2 (M + H)+ PP1570 FA 6.217 1546.3 (M - H)- PP1571 FA 6.527 1560.0 (M - H)- PP1572 FA 6.585 1591.7 (M + H)+ PP1573 FA 6.211 1566.3 (M - H)- PP1574 FA 6.333 1561.9 (M + H)+ PP1575 FA 6.881 1606.3 (M + H)+ PP1576 AA 8.111 1561.8 (M + H)+ PP1577 AA 8.172 1537.8 (M + H)+ PP1578 AA 8.343 1552.3 (M + H)+ PP1579 AA 8.029 1546.2 (M - H)- PP1580 FA 6.640 1579.2 (M + NH.sub.4)+ PP1582 AA 8.784 1559.2 (M - H)- PP1583 FA 7.421 1533.0 (M - H)- PP1584 AA 8.520 1531.2 (M - H)- PP1586 AA 8.591 1568.7 (M + H)+ PP1587 AA 8.632 1506.7 (M + H)+ PP1588 AA 8.492 1522.2 (M + NH.sub.4)+ PP1589 AA 8.289 1478.7 (M + H)+ PP1590 AA 8.180 1476.7 (M + H)+ PP1591 AA 8.352 1514.7 (M + H)+ PP1592 AA 8.265 1512.6 (M + H)+ PP1593 AA 8.721 1532.7 (M + H)+ PP1594 AA 8.617 1531.2 (M + H)+ PP1595 AA 8.395 1504.7 (M + H)+ PP1596 AA 8.313 1502.7 (M + H)+ PP1598 AA 8.400 1538.7 (M + H)+ PP1599 AA 8.156 1496.7 (M + H)+ PP1600 AA 8.020 1470.7 (M + H)+ PP1601 AA 8.405 1414.7 (M + H)+ PP1603 AA 8.695 1442.7 (M + H)+ PP1605 AA 8.229 1490.7 (M + H)+ PP1606 AA 8.119 1489.1 (M + H)+ PP1607 AA 7.893 1462.7 (M + H)+ PP1608 FA 6.600 1498.6 (M + H)+ PP1609 AA 7.920 1496.6 (M + H)+ PP1610 AA 8.288 1516.7 (M + H)+ PP1611 AA 8.208 1514.6 (M + H)+ PP1612 AA 7.989 1488.7 (M + H)+ PP1613 AA 7.927 1486.7 (M + H)+ PP1614 AA 8.115 1524.6 (M + H)+ PP1615 AA 8.041 1522.7 (M + H)+ PP1616 AA 7.749 1480.7 (M + H)+ PP1617 AA 7.637 1454.7 (M + H)+ PP1620 AA 8.259 1426.7 (M + H)+ PP1622 AA 8.452 1492.7 (M + H)+ PP1623 AA 8.305 1490.7 (M + H)+ PP1624 AA 8.091 1464.6 (M + H)+ PP1625 AA 7.991 1462.6 (M + H)+ PP1626 FA 6.973 1500.6 (M + H)+ PP1627 AA 8.080 1498.7 (M + H)+ PP1628 AA 8.536 1518.7 (M + H)+ PP1629 FA 7.181 1516.7 (M + H)+ PP1630 AA 8.204 1491.2 (M + H)+ PP1631 AA 8.123 1488.7 (M + H)+ PP1632 AA 8.295 1526.7 (M + H)+ PP1633 AA 8.229 1523.2 (M - H)- PP1634 FA 7.851 1545.2 (M + H)+ PP1635 AA 8.684 1542.7 (M + H)+ PP1636 AA 8.468 1516.7 (M + H)+ PP1637 AA 8.403 1514.7 (M + H)+ PP1639 AA 8.500 1549.2 (M - H)- PP1640 FA 5.499 1606.8 (M + NH.sub.4)+ PP1641 FA 5.891 1517.7 (M - H)- PP1643 AA 7.899 1583.7 (M + H)+ PP1644 FA 6.128 1585.7 (M - H)- PP1645 FA 6.215 1546.4 (M - H)- PP1646 FA 6.489 1561.8 (M + H)+ PP1648 FA 6.481 1574.2 (M + H)+

PP1649 FA 6.412 1569.0 (M + NH.sub.4)+ PP1650 FA 6.355 1560.0 (M - H)- PP1651 FA 6.459 1572.3 (M - H)- PP1653 AA 7.856 1540.3 (M - H)- PP1654 FA 6.403 1572.9 (M + NH.sub.4)+ PP1655 FA 6.093 1578.3 (M + H)+ PP1656 FA 6.216 1588.8 (M + NH.sub.4)+ PP1657 AA 7.909 1571.8 (M + H)+ PP1658 FA 6.184 1539.7 (M + H)+ PP1660 AA 8.020 1547.2 (M + NH.sub.4)+ PP1661 FA 6.581 1530.3 (M - H)- PP1662 FA 6.501 1530.2 (M - H)- PP1663 FA 6.189 1521.6 (M + H)+ PP1665 FA 6.529 1560.2 (M - H)- PP1666 FA 5.971 1519.6 (M + H)+ PP1667 FA 6.224 1558.2 (M - H)- PP1668 AA 8.079 1575.9 (M - H)- PP1670 FA 6.651 1561.9 (M + H)+ PP1671 FA 6.157 1563.7 (M + H)+ PP1672 AA 8.133 1578.2 (M + H)+ PP1673 FA 6.879 1606.3 (M + H)+ PP1674 FA 5.632 1512.4 (M - H)- PP1675 FA 5.928 1581.7 (M + H)+ PP1676 FA 6.260 1573.3 (M + NH.sub.4)+ PP1677 FA 5.779 1554.8 (M + NH.sub.4)+ PP1678 AA 7.967 1585.3 (M + NH.sub.4)+ PP1679 FA 5.915 1558.3 (M + H)+ PP1682 AA 7.832 1558.2 (M + H)+ PP1683 FA 6.364 1517.8 (M + H)+ PP1684 AA 8.164 1542.1 (M - H)- PP1687 FA 6.327 1571.7 (M + H)+ PP1688 AA 7.523 1562.1 (M + H)+ PP1689 AA 7.737 1521.7 (M + H)+ PP1691 AA 7.749 1531.7 (M + H)+ PP1692 AA 7.945 1545.7 (M + H)+ PP1693 AA 7.861 1534.0 (M - H)- PP1694 FA 6.349 1549.7 (M + H)+ PP1696 AA 7.957 1551.7 (M + H)+ PP1697 AA 8.031 1587.9 (M + Na)+ PP1698 AA 8.215 1578.3 (M - H)- PP1699 AA 7.909 1532.0 (M - H)- PP1700 AA 7.791 1570.2 (M + H)+ PP1701 AA 7.597 1572.2 (M - H)- PP1702 AA 7.816 1588.2 (M + H)+ PP1703 FA 6.120 1542.2 (M - H)- PP1704 AA 8.124 1560.3 (M - H)- PP1705 FA 6.528 1573.8 (M + H)+ PP1706 AA 7.849 1567.2 (M + NH.sub.4)+ PP1707 FA 5.896 1510.3 (M - H)- PP1709 AA 7.623 1560.2 (M - H)- PP1710 FA 6.144 1578.3 (M - H)- PP1711 FA 6.217 1540.2 (M + H)+ PP1713 FA 6.231 1551.8 (M + H)+ PP1714 AA 7.972 1554.2 (M + H)+ PP1715 FA 6.363 1553.8 (M + H)+ PP1716 AA 8.016 1566.2 (M + H)+ PP1717 AA 7.724 1541.7 (M + H)+ PP1718 AA 7.992 1569.7 (M + H)+ PP1721 AA 7.981 1554.2 (M + H)+ PP1722 FA 5.993 1543.8 (M + H)+ PP1727 FA 6.205 1516.2 (M - H)- PP1728 FA 6.307 1529.7 (M + H)+ PP1729 FA 6.356 1533.7 (M + H)+ PP1731 FA 6.260 1555.8 (M + H)+ PP1732 AA 7.944 1569.8 (M + H)+ PP1733 FA 6.392 1606.2 (M + H)+ PP1734 FA 6.473 1624.2 (M + H)+ PP1735 FA 6.259 1567.7 (M - H)- PP1736 FA 6.561 1583.7 (M + H)+ PP1737 FA 6.391 1580.2 (M + H)+ PP1738 FA 6.256 1581.7 (M + H)+ PP1739 FA 6.551 1595.7 (M + H)+ PP1740 FA 6.164 1569.7 (M + H)+ PP1741 FA 6.064 1570.3 (M + H)+ PP1742 FA 6.128 1598.7 (M + NH.sub.4)+ PP1743 FA 6.196 1603.2 (M + NH.sub.4)+ PP1744 AA 8.092 1614.3 (M + H)+ PP1746 AA 7.887 1549.1 (M + NH.sub.4)+ PP1747 AA 7.759 1566.2 (M - H)- PP1748 FA 6.205 1544.3 (M - H)- PP1749 FA 6.196 1558.1 (M + H)+ PP1750 FA 6.392 1558.2 (M - H)- PP1751 AA 8.076 1560.1 (M + H)+ PP1752 FA 6.437 1572.1 (M + H)+ PP1753 AA 7.813 1569.3 (M + Na)+ PP1754 FA 6.291 1576.1 (M + H)+ PP1757 AA 8.087 1564.1 (M + H)+ PP1758 FA 6.044 1511.9 (M - H)- PP1759 FA 6.033 1526.1 (M + H)+ PP1760 FA 6.241 1527.7 (M + H)+ PP1761 FA 6.177 1526.2 (M - H)- PP1762 AA 8.099 1540.2 (M + H)+ PP1763 AA 8.125 1544.2 (M + H)+ PP1764 AA 8.045 1561.2 (M + NH.sub.4)+ PP1765 AA 7.795 1548.3 (M + H)+ PP1767 FA 6.097 1548.1 (M + H)+ PP1768 FA 6.196 1546.3 (M - H)- PP1769 FA 6.077 1546.4 (M - H)- PP1771 FA 6.479 1561.7 (M + H)+ PP1772 FA 6.388 1561.9 (M + H)+ PP1773 AA 8.093 1560.3 (M - H)- PP1776 FA 5.999 1490.2 (M - H)- PP1777 AA 7.727 1490.3 (M - H)- PP1779 FA 5.569 1520.1 (M + H)+ PP1780 FA 5.317 1474.2 (M - H)- PP1781 AA 7.419 1464.2 (M - H)- PP1782 FA 5.153 1420.3 (M - H)- PP1783 FA 5.467 1477.7 (M + H)+ PP1784 FA 5.213 1432.3 (M - H)- PP1785 FA 5.668 1490.3 (M - H)- PP1786 FA 5.432 1447.7 (M + H)+ PP1788 AA 7.464 1446.3 (M - H)- PP1789 FA 5.693 1461.9 (M + H)+ PP1790 AA 7.716 1492.0 (M - H)- PP1791 AA 7.531 1450.1 (M + H)+ PP1792 AA 7.692 1462.4 (M - H)- PP1793 AA 7.583 1478.4 (M - H)- PP1794 FA 5.392 1434.3 (M - H)- PP1795 FA 5.648 1489.7 (M - H)- PP1796 AA 7.409 1465.2 (M + NH.sub.4)+ PP1797 FA 5.919 1518.2 (M + H)+ PP1798 FA 5.683 1471.9 (M - H)- PP1799 FA 5.660 1449.7 (M + H)+ PP1800 AA 7.576 1460.3 (M - H)- PP1801 FA 5.945 1486.3 (M - H)- PP1802 FA 5.719 1461.8 (M + H)+ PP1803 AA 7.785 1474.4 (M - H)- PP1804 FA 5.693 1531.9 (M - H)- PP1805 FA 5.436 1488.2 (M - H)- PP1806 AA 7.547 1478.3 (M - H)- PP1807 FA 5.288 1435.7 (M + H)+

PP1808 FA 5.581 1490.1 (M - H)- PP1809 AA 7.423 1465.2 (M + NH.sub.4)+ PP1810 FA 5.740 1505.9 (M + H)+ PP1811 FA 5.504 1460.2 (M - H)- PP1812 AA 7.540 1448.2 (M - H)- PP1813 FA 5.588 1460.2 (M - H)- PP1814 AA 7.717 1474.2 (M - H)- PP1815 FA 6.153 1517.6 (M - H)- PP1816 AA 7.645 1429.6 (M - H)- PP1817 FA 6.024 1447.6 (M + H)+ PP1818 AA 7.941 1458.2 (M + H)+ PP1819 FA 6.377 1473.4 (M + H)+ PP1820 FA 6.493 1432.2 (M + H)+ PP1821 FA 5.871 1461.6 (M - H)- PP1822 FA 5.683 1417.7 (M - H)- PP1823 FA 6.043 1475.7 (M - H)- PP1825 FA 6.259 1561.3 (M + H)+ PP1826 FA 6.137 1545.9 (M - H)- PP1827 FA 5.697 1503.8 (M + H)+ PP1828 FA 5.503 1497.8 (M + H)+ PP1829 FA 6.296 1531.8 (M + H)+ PP1830 FA 6.123 1542.5 (M + NH.sub.4)+ PP1831 FA 6.029 1517.8 (M + H)+ PP1832 FA 5.848 1529.0 (M + NH.sub.4)+ PP1833 FA 5.473 1506.4 (M + NH.sub.4)+ PP1834 FA 5.283 1501.3 (M + NH.sub.4)+ PP1835 FA 6.069 1516.0 (M - H)- PP1836 AA 7.579 1444.2 (M + H)+ PP1837 FA 6.201 1498.1 (M + H)+ PP1838 FA 5.519 1442.4 (M - H)- PP1839 AA 7.921 1470.3 (M - H)- PP1840 FA 5.577 1444.3 (M + H)+ PP1841 FA 6.097 1471.7 (M + H)+ PP1842 FA 5.516 1442.4 (M - H)- PP1844 FA 5.939 1446.2 (M - H)- PP1846 FA 5.844 1447.6 (M + H)+ PP1848 AA 7.795 1447.6 (M + H)+ PP1849 FA 6.016 1471.7 (M + H)+ PP1850 FA 5.839 1447.6 (M + H)+ PP1851 AA 7.664 1501.7 (M + H)+ PP1852 FA 5.765 1447.6 (M + H)+ PP1853 FA 5.360 1428.4 (M - H)- PP1854 AA 7.684 1469.4 (M + Na)+ PP1855 AA 7.369 1429.7 (M + H)+ PP1856 FA 5.852 1447.8 (M - H)- PP1857 FA 5.292 1447.3 (M + NH.sub.4)+ PP1859 FA 5.227 1430.2 (M + H)+ PP1860 FA 5.799 1449.7 (M + H)+ PP1861 FA 5.284 1440.4 (M - H)- PP1862 AA 7.695 1450.3 (M + H)+ PP1863 FA 5.225 1440.4 (M - H)- PP1864 AA 7.623 1467.3 (M + NH.sub.4)+ PP1865 FA 5.469 1442.3 (M - H)- PP1866 FA 5.619 1448.4 (M - H)- PP1867 FA 5.412 1442.4 (M - H)- PP1868 FA 5.879 1475.3 (M + NH.sub.4)+ PP1869 AA 7.411 1458.3 (M - H)- PP1870 FA 5.792 1456.4 (M - H)- PP1871 FA 5.211 1458.4 (M - H)- PP1872 FA 5.808 1486.4 (M - H)- PP1873 AA 7.563 1474.3 (M + H)+ PP1874 FA 6.220 1460.2 (M - H)- PP1875 FA 5.693 1478.4 (M - H)- PP1876 FA 6.124 1460.3 (M - H)- PP1877 FA 5.609 1480.3 (M + H)+ PP1878 FA 5.859 1495.1 (M + NH.sub.4)+ PP1879 FA 5.501 1484.3 (M - H)- PP1880 FA 6.141 1491.7 (M + H)+ PP1881 AA 7.921 1524.4 (M + Na)+ PP1882 FA 6.128 1462.4 (M - H)- PP1883 FA 6.449 1506.2 (M + H)+ PP1884 AA 7.880 1462.3 (M - H)- PP1885 FA 6.072 1470.5 (M - H)- PP1886 FA 6.052 1492.4 (M - H)- PP1887 FA 5.995 1502.4 (M + H)+ PP1888 FA 5.823 1457.7 (M + H)+ PP1889 FA 6.092 1471.7 (M + H)+ PP1890 FA 5.737 1456.4 (M - H)- PP1891 AA 7.837 1470.3 (M - H)- PP1892 AA 7.688 1486.0 (M - H)- PP1893 FA 6.419 1474.3 (M - H)- PP1894 AA 8.072 1476.1 (M + H)+ PP1895 FA 5.815 1476.3 (M - H)- PP1896 FA 6.023 1490.3 (M - H)- PP1897 FA 5.575 1454.3 (M - H)- PP1898 FA 5.821 1468.4 (M - H)- PP1899 FA 6.092 1482.4 (M - H)- PP1900 FA 5.520 1456.4 (M + H)+ PP1901 AA 7.687 1470.2 (M + H)+ PP1902 AA 7.857 1482.3 (M - H)- PP1903 AA 7.709 1500.4 (M + H)+ PP1904 FA 5.759 1456.4 (M - H)- PP1905 FA 6.000 1488.8 (M + NH.sub.4)+ PP1906 FA 6.280 1484.4 (M - H)- PP1907 FA 5.696 1456.3 (M - H)- PP1908 AA 7.796 1489.4 (M + NH.sub.4)+ PP1909 FA 6.209 1503.2 (M + NH.sub.4)+ PP1910 AA 7.641 1505.4 (M + NH.sub.4)+ PP1911 FA 6.275 1497.9 (M + H)+ PP1913 AA 7.848 1565.2 (M + NH.sub.4)+ PP1914 FA 5.860 1493.6 (M + H)+ PP1915 AA 8.093 1560.4 (M + H)+ PP1916 FA 5.925 1492.2 (M - H)- PP1917 FA 6.399 1590.4 (M - H)- PP1918 FA 6.004 1506.2 (M - H)- PP1919 AA 8.085 1590.0 (M - H)- PP1920 FA 6.113 1536.2 (M - H)- PP1921 AA 8.113 1590.4 (M - H)- PP1922 AA 7.480 1506.2 (M - H)- PP1923 FA 6.164 1616.4 (M - H)- PP1925 FA 6.241 1560.4 (M + H)+ PP1926 FA 6.055 1506.3 (M - H)- PP1927 AA 8.071 1570.3 (M + Na)+ PP1928 FA 6.119 1537.2 (M + NH.sub.4)+ PP1929 AA 8.004 1542.4 (M - H)- PP1930 FA 6.532 1503.6 (M + H)+ PP1931 FA 6.131 1542.4 (M - H)- PP1932 FA 6.161 1509.9 (M - H)- PP1934 FA 6.071 1509.1 (M + NH.sub.4)+ PP1935 FA 6.017 1491.6 (M + H)+ PP1936 FA 6.197 1521.6 (M + H)+ PP1937 FA 6.011 1535.9 (M + H)+ PP1938 FA 6.128 1562.3 (M - H)- PP1941 FA 6.237 1533.8 (M - H)- PP1942 FA 6.345 1546.3 (M - H)- PP1943 FA 6.035 1558.3 (M - H)- PP1944 AA 7.965 1560.4 (M + H)+ PP1945 FA 6.639 1572.3 (M - H)- PP1946 FA 6.001 1518.2 (M - H)- PP1947 FA 6.237 1518.2 (M - H)- PP1948 FA 6.028 1520.1 (M + H)+ PP1949 FA

6.159 1519.6 (M + H)- PP1950 FA 6.037 1517.8 (M - H)- PP1952 FA 6.071 1520.1 (M + H)-
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(M + H)+ PP1956 FA 6.005 1476.2 (M - H)- PP1957 FA 6.160 1502.2 (M - H)- PP1958 FA
6.016 1506.2 (M - H)- PP1959 FA 6.028 1520.1 (M + H)+ PP1960 FA 6.096 1464.2 (M - H)-
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(M - H)- PP1965 FA 6.484 1562.2 (M - H)- PP1967 FA 6.059 1518.2 (M - H)- PP1968 FA
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(M - H)- PP1974 FA 6.568 1559.9 (M - H)- PP1975 FA 6.212 1586.3 (M + H)+ PP1976 FA
6.583 1529.8 (M - H)- PP1977 FA 6.235 1553.9 (M - H)- PP1979 FA 5.961 1540.1 (M - H)-
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(M - H)- PP1984 FA 6.657 1541.3 (M - H)- PP1985 FA 6.323 1565.9 (M - H)- PP1987 FA
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(M - H)- PP1993 FA 6.243 1539.9 (M - H)- PP1994 FA 6.103 1544.0 (M - H)- PP1995 FA
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(M - H)- PP2017 FA 6.175 1569.8 (M - H)- PP2018 FA 5.735 1555.7 (M - H)- PP2019 FA
5.725 1525.9 (M - H)- PP2020 AA 7.812 1540.0 (M - H)- PP2021 FA 5.557 1524.1 (M - H)-
PP2022 FA 5.809 1537.9 (M - H)- PP2023 FA 6.128 1552.1 (M - H)- PP2024 FA 6.025 1531.8
(M - H)- PP2025 FA 6.019 1525.9 (M - H)- PP2026 FA 5.721 1511.8 (M - H)- PP2027 FA
5.983 1526.1 (M - H)- PP2028 FA 6.287 1539.7 (M - H)- PP2029 FA 5.825 1525.8 (M - H)-
PP2030 AA 8.088 1596.1 (M - H)- PP2031 FA 6.628 1589.9 (M - H)- PP2032 FA 6.325 1575.8
(M - H)- PP2033 FA 6.448 1590.4 (M - H)- PP2034 AA 8.095 1560.1 (M - H)- PP2036 FA
6.291 1557.8 (M - H)- PP2037 FA 6.587 1572.1 (M - H)- PP2038 FA 6.544 1574.0 (M + H)+
PP2040 FA 6.715 1566.0 (M - H)- PP2041 AA 8.279 1560.1 (M - H)- PP2042 FA 6.479 1546.0
(M - H)- PP2043 FA 6.725 1559.8 (M - H)- PP2044 FA 6.200 1506.3 (M - H)- PP2045 FA
6.480 1558.3 (M - H)- PP2046 AA 7.969 1556.3 (M - H)- PP2047 FA 5.607 1464.3 (M - H)-
PP2048 FA 5.901 1516.2 (M - H)- PP2049 FA 5.717 1515.6 (M + H)+ PP2050 AA 8.169 1551.0
(M + NH.sub.4)+ PP2051 AA 7.737 1550.3 (M - H)- PP2052 FA 6.320 1530.3 (M - H)- PP2053
AA 7.919 1566.4 (M + H)+ PP2054 FA 6.223 1533.3 (M + NH.sub.4)+ PP2055 FA 6.137 1539.2
(M + NH.sub.4)+ PP2056 AA 7.997 1526.4 (M - H)- PP2057 FA 6.435 1552.7 (M + NH.sub.4)+
PP2058 AA 8.039 1526.4 (M - H)- PP2059 FA 5.801 1536.4 (M - H)- PP2060 FA 6.108 1524.3
(M - H)- PP2061 FA 5.639 1492.3 (M - H)- PP2062 FA 6.319 1533.9 (M - H)- PP2063 AA
7.685 1506.3 (M - H)- PP2064 AA 7.968 1494.4 (M - H)- PP2065 FA 5.737 1496.2 (M - H)-
PP2066 FA 6.035 1530.3 (M + H)+ PP2067 FA 6.044 1510.4 (M - H)- PP2068 FA 5.948 1489.7
(M + H)+ PP2069 FA 5.559 1495.6 (M + H)+ PP2070 FA 5.851 1508.4 (M - H)- PP2071 FA
5.831 1508.4 (M - H)- PP2072 FA 6.136 1522.3 (M - H)- PP2073 FA 6.179 1558.2 (M + H)+
PP2074 FA 6.485 1572.4 (M + H)+ PP2075 FA 6.400 1527.7 (M + H)+ PP2076 FA 6.689 1540.3
(M - H)- PP2077 AA 7.863 1530.6 (M + NH.sub.4)+ PP2078 FA 5.973 1521.1 (M + NH.sub.4)+
PP2079 FA 6.275 1516.2 (M - H)- PP2080 AA 7.699 1500.3 (M - H)- PP2081 FA 6.093 1515.6
(M + H)+ PP2082 FA 6.048 1514.2 (M - H)- PP2083 AA 8.031 1528.2 (M - H)- PP2087 AA
7.925 1557.6 (M + H)+ PP2091 FA 6.004 1569.2 (M + NH.sub.4)+ PP2093 FA 6.337 1520.2 (M +
H)+ PP2094 FA 6.387 1518.4 (M - H)- PP2095 FA 6.199 1516.3 (M - H)- PP2096 AA 7.893

1502.4 (M + H)+ PP2097 FA 6.113 1513.3 (M + NH.sub.4)+ PP2098 FA 6.149 1512.3 (M - H)-
PP2099 FA 5.964 1510.3 (M - H)- PP2101 FA 6.448 1574.4 (M + H)+ PP2102 FA 6.528 1572.4
(M - H)- PP2103 FA 6.379 1570.3 (M - H)- PP2105 FA 6.224 1568.2 (M + H)+ PP2106 FA
6.301 1566.4 (M - H)- PP2107 FA 6.129 1566.3 (M + H)+ PP2108 AA 8.140 1522.4 (M + H)+
PP2109 FA 6.515 1532.3 (M - H)- PP2117 FA 5.437 1492.0 (M + H)+ PP2118 FA 5.233 1484.0
(M - H)- PP2119 FA 6.471 1560.0 (M - H)- PP2120 FA 6.548 1484.1 (M - H)- PP2121 FA
6.183 1550.0 (M - H)- PP2122 AA 8.148 1457.8 (M - H)- PP2123 FA 6.328 1576.0 (M - H)-
PP2124 FA 6.539 1558.0 (M - H)- PP2125 FA 6.475 1548.2 (M - H)- PP2126 FA 6.260 1548.0
(M - H)- PP2127 FA 6.515 1562.2 (M - H)- PP2128 FA 6.421 1573.8 (M - H)- PP2130 FA
6.559 1546.0 (M - H)- PP2131 FA 6.344 1546.2 (M - H)- PP2132 FA 6.617 1560.1 (M - H)-
PP2133 FA 6.643 1575.6 (M + H)+ PP2135 FA 6.527 1548.2 (M - H)- PP2137 FA 6.264 1506.0
(M - H)- PP2138 FA 6.737 1572.1 (M - H)- PP2139 FA 6.128 1522.1 (M - H)- PP2140 FA
6.619 1546.1 (M - H)- PP2141 AA 8.064 1508.1 (M - H)- PP2142 FA 6.356 1506.4 (M + H)+
PP2143 FA 5.965 1467.6 (M + H)+ PP2144 FA 6.235 1520.1 (M - H)- PP2145 FA 6.123 1492.1
(M - H)- PP2146 FA 6.371 1492.3 (M - H)- PP2147 FA 6.179 1480.1 (M - H)- PP2148 FA
6.455 1506.0 (M - H)- PP2149 FA 6.333 1507.5 (M + H)+ PP2150 FA 6.052 1465.5 (M + H)+
PP2151 FA 6.477 1520.1 (M - H)- PP2152 FA 6.217 1492.3 (M + H)+ PP2153 FA 6.348 1494.3
(M - H)- PP2154 FA 6.268 1478.0 (M - H)- PP2155 FA 6.196 1554.2 (M - H)- PP2156 FA
6.569 1518.1 (M - H)- PP2157 FA 5.897 1544.2 (M - H)- PP2158 FA 6.436 1491.6 (M - H)-
PP2159 FA 6.051 1570.1 (M - H)- PP2160 FA 6.208 1542.0 (M - H)- PP2161 FA 6.305 1556.3
(M - H)- PP2163 FA 6.059 1542.0 (M + H)+ PP2164 FA 6.420 1568.1 (M - H)- PP2165 FA
6.240 1541.6 (M - H)- PP2166 FA 6.016 1500.1 (M - H)- PP2167 FA 5.856 1516.1 (M - H)-
PP2168 FA 6.141 1502.3 (M - H)- PP2169 FA 5.837 1486.3 (M - H)- PP2170 FA 5.887 1473.6
(M - H)- PP2171 FA 6.041 1500.2 (M - H)- PP2172 FA 6.257 1514.2 (M - H)- PP2173 FA
6.057 1488.3 (M - H)- PP2174 AA 8.243 1539.9 (M - H)- PP2175 FA 6.547 1513.9 (M + H)+
PP2176 FA 6.596 1525.8 (M - H)- PP2178 FA 6.428 1511.4 (M + H)+ PP2179 FA 6.492 1498.1
(M - H)- PP2180 FA 6.719 1538.1 (M - H)- PP2181 AA 8.216 1512.1 (M - H)- PP2182 FA
6.329 1469.8 (M - H)- PP2183 FA 6.216 1487.5 (M + H)+ PP2184 FA 6.344 1459.5 (M + H)+
PP2185 FA 6.421 1471.9 (M - H)- PP2186 FA 6.023 1429.8 (M - H)- PP2187 FA 6.201 1455.9
(M - H)- PP2188 FA 6.257 1443.8 (M - H)- PP2189 FA 5.912 1461.5 (M - H)- PP2190 FA 6.111
1489.7 (M - H)- PP2191 FA 6.179 1476.1 (M - H)- PP2192 FA 6.049 1516.2 (M - H)- PP2193
AA 8.176 1490.3 (M - H)- PP2195 FA 5.952 1490.0 (M - H)- PP2196 FA 6.337 1504.0 (M - H)-
PP2197 FA 5.652 1486.1 (M - H)- PP2198 FA 6.151 1502.1 (M - H)- PP2199 SQD 1.050 1493.9
(M + H)+ PP2200 FA 6.428 1492.1 (M - H)- PP2202 FA 6.291 1491.8 (M - H)- PP2203 FA
6.717 1520.3 (M - H)- PP2207 FA 6.516 1505.8 (M - H)- PP2208 FA 6.365 1476.0 (M - H)-
PP2209 FA 6.325 1504.1 (M - H)- PP2210 FA 6.120 1459.7 (M - H)- PP2212 FA 6.379 1474.0
(M - H)- PP2214 FA 6.671 1488.1 (M - H)- PP2216 AA 7.876 1462.2 (M - H)- PP2218 FA
6.152 1493.5 (M + H)+ PP2219 FA 6.275 1492.0 (M - H)- PP2220 FA 5.949 1462.2 (M - H)-
PP2221 FA 6.096 1489.8 (M - H)- PP2222 FA 6.188 1480.0 (M + H)+ PP2223 FA 6.203 1476.2
(M - H)- PP2224 FA 6.073 1496.6 (M + NH.sub.4)+ PP2225 FA 5.904 1478.0 (M - H)- PP2226
AA 7.797 1503.9 (M - H)- PP2227 FA 6.089 1462.0 (M - H)- PP2229 FA 6.440 1506.2 (M - H)-
PP2230 AA 8.119 1491.6 (M - H)- PP2231 FA 6.232 1475.7 (M - H)- PP2232 FA 6.200 1492.0
(M - H)- PP2233 FA 5.851 1448.0 (M - H)- PP2234 FA 6.285 1490.0 (M - H)- PP2235 FA
6.537 1504.0 (M - H)- PP2237 FA 5.869 1445.6 (M - H)- PP2238 FA 6.157 1460.0 (M - H)-
PP2242 FA 5.483 1458.2 (M - H)- PP2257 FA 6.225 1490.0 (M - H)- PP2259 FA 5.783 1511.9
(M - H)- PP2260 AA 7.825 1526.2 (M - H)- PP2261 AA 7.179 1476.2 (M - H)- PP2262 FA
4.957 1470.2 (M - H)- PP2263 FA 5.505 1497.5 (M - H)- PP2264 FA 5.225 1483.8 (M - H)-
PP2268 FA 6.137 1515.9 (M - H)- PP2269 FA 5.912 1512.3 (M - H)- PP2270 FA 6.419 1513.8
(M - H)- PP2271 FA 6.111 1521.0 (M + NH.sub.4)+ PP2272 FA 6.377 1515.6 (M - H)- PP2273
FA 6.372 1516.0 (M - H)- PP2275 AA 8.187 1543.8 (M - H)- PP2312 FA 5.669 1525.8 (M - H)-

PP2313 FA 6.263 1552.1 (M – H)– PP2314 FA 5.736 1556.2 (M – H)– PP2315 FA 6.072 1570.3 (M – H)– PP2316 FA 6.085 1552.1 (M – H)– PP2317 FA 6.079 1554.1 (M – H)– PP2318 FA 6.163 1582.1 (M – H)– PP2319 FA 5.867 1569.7 (M – H)– PP2320 FA 5.915 1551.5 (M – H)– PP2322 FA 5.975 1582.2 (M – H)– PP2323 FA 5.815 1504.3 (M – H)– PP2325 FA 6.177 1500.1 (M – H)– PP2326 FA 5.604 1502.1 (M – H)– PP2327 FA 6.269 1530.2 (M – H)– PP2328 FA 5.995 1498.1 (M – H)– PP2329 AA 7.744 1500.1 (M – H)– PP2330 FA 6.083 1527.7 (M – H)– PP2331 FA 6.067 1529.9 (M – H)– PP2332 FA 5.804 1498.2 (M – H)– PP2333 FA 6.072 1465.7 (M – H)– PP2334 FA 5.879 1530.3 (M + H)+ PP2335 FA 5.727 1468.1 (M – H)– PP2336 FA 5.760 1463.7 (M – H)– PP2337 FA 6.515 1492.3 (M – H)– PP2338 FA 5.619 1480.3 (M – H)– PP2339 FA 6.144 1494.3 (M – H)– PP2340 FA 6.223 1490.3 (M – H)– PP2341 FA 6.401 1492.3 (M – H)– PP2342 FA 6.079 1506.3 (M – H)– PP2344 FA 6.100 1490.1 (M – H)– PP2345 FA 5.720 1502.1 (M – H)– PP2347 FA 5.556 1518.1 (M – H)– PP2348 AA 7.396 1500.2 (M – H)– PP2349 FA 6.140 1528.3 (M – H)– PP2350 FA 5.516 1530.3 (M – H)– PP2351 FA 5.944 1544.1 (M – H)– PP2352 FA 5.843 1526.3 (M – H)– PP2353 FA 5.949 1528.3 (M – H)– PP2354 FA 5.933 1556.1 (M – H)– PP2355 FA 5.833 1544.3 (M – H)– PP2356 FA 5.648 1526.3 (M – H)– PP2357 FA 5.793 1556.3 (M – H)– PP2358 FA 5.732 1496.2 (M – H)– PP2359 FA 5.541 1511.6 (M – H)– PP2360 FA 6.183 1522.1 (M – H)– PP2361 FA 5.952 1538.0 (M – H)– PP2362 FA 5.984 1521.6 (M – H)– PP2363 FA 5.741 1538.0 (M – H)– PP2364 FA 5.613 1442.4 (M – H)– PP2365 FA 5.701 1472.2 (M – H)– PP2366 FA 6.071 1467.6 (M – H)– PP2367 FA 6.167 1498.1 (M – H)– PP2368 FA 5.865 1467.6 (M – H)– PP2369 FA 5.947 1498.1 (M – H)– PP2370 FA 5.848 1435.6 (M + H)+ PP2371 FA 5.727 1449.9 (M – H)– PP2372 FA 6.309 1461.6 (M + H)+ PP2373 FA 6.181 1476.3 (M – H)– PP2374 FA 6.204 1460.2 (M – H)– PP2375 FA 6.143 1476.1 (M – H)– PP2376 FA 5.533 1471.4 (M + H)+ PP2377 FA 5.659 1500.1 (M – H)– PP2378 FA 5.952 1495.6 (M – H)– PP2379 FA 6.071 1526.1 (M – H)– PP2380 FA 5.785 1496.2 (M – H)– PP2381 FA 5.957 1526.3 (M – H)– PP2382 FA 5.837 1527.6 (M – H)– PP2383 FA 5.655 1543.6 (M – H)– PP2385 FA 5.567 1458.2 (M – H)– PP2386 FA 6.048 1498.1 (M – H)– PP2387 FA 5.480 1459.5 (M + H)+ PP2388 AA 7.869 1512.3 (M – H)– PP2389 FA 5.765 1491.6 (M – H)– PP2390 FA 5.676 1514.3 (M – H)– PP2391 FA 5.680 1494.0 (M – H)– PP2392 FA 5.712 1502.2 (M – H)– PP2393 FA 5.713 1472.0 (M – H)– PP2394 FA 6.085 1476.1 (M – H)– PP2395 FA 5.945 1505.7 (M – H)– PP2396 FA 5.956 1508.2 (M – H)– PP2397 FA 5.856 1508.1 (M – H)– PP2398 FA 6.004 1516.3 (M – H)– PP2399 FA 6.469 1489.9 (M – H)– PP2400 FA 6.347 1520.1 (M – H)– PP2401 FA 6.233 1519.9 (M – H)– PP2402 AA 8.067 1520.1 (M – H)– PP2403 FA 6.264 1522.0 (M – H)– PP2404 FA 6.144 1522.3 (M – H)– PP2405 FA 6.316 1500.1 (M – H)– PP2406 FA 6.209 1530.3 (M – H)– PP2407 FA 6.561 1506.3 (M – H)– PP2408 FA 6.448 1536.3 (M – H)– PP2409 FA 6.293 1496.1 (M – H)– PP2410 FA 6.555 1510.1 (M – H)– PP2411 FA 6.768 1516.3 (M – H)– PP2412 FA 5.867 1486.3 (M – H)– PP2413 AA 7.915 1500.3 (M – H)– PP2414 FA 6.355 1506.3 (M – H)– PP2415 FA 6.167 1500.0 (M – H)– PP2416 FA 6.440 1514.1 (M – H)– PP2417 FA 6.657 1520.3 (M – H)– PP2418 FA 6.000 1512.0 (M – H)– PP2419 FA 6.255 1526.4 (M – H)– PP2422 FA 6.067 1508.3 (M – H)– PP2423 FA 6.105 1516.1 (M – H)– PP2424 FA 6.343 1522.2 (M – H)– PP2425 FA 6.133 1516.1 (M – H)– PP2426 FA 6.381 1521.7 (M – H)– PP2427 FA 6.175 1512.2 (M – H)– PP2428 FA 6.288 1548.1 (M – H)– PP2429 FA 6.461 1526.3 (M – H)– PP2430 FA 6.649 1532.1 (M – H)– PP2431 FA 6.228 1542.3 (M – H)– PP2432 FA 6.343 1578.1 (M – H)– PP2433 FA 5.944 1528.3 (M – H)– PP2436 FA 6.401 1578.1 (M – H)– PP2437 FA 6.081 1478.2 (M – H)– PP2438 FA 5.696 1518.3 (M – H)– PP2439 FA 5.992 1478.0 (M – H)– PP2440 FA 5.971 1524.0 (M – H)– PP2441 FA 6.119 1486.3 (M – H)– PP2442 FA 5.893 1488.0 (M – H)– PP2443 AA 7.855 1486.1 (M – H)– PP2444 FA 6.069 1522.0 (M – H)– PP2445 FA 5.901 1516.1 (M – H)– PP2446 FA 5.843 1532.3 (M – H)– PP2447 FA 6.027 1486.3 (M – H)– PP2448 FA 6.199 1494.0 (M – H)– PP2449 FA 5.901 1516.1 (M – H)– PP2450 FA 6.088 1537.7 (M – H)– PP2451 FA 6.357 1490.0 (M – H)– PP2452 FA 6.120 1482.3 (M – H)– PP2453 FA 6.344 1490.0 (M – H)– PP2454 AA 7.987 1515.6 (M + Na)+ PP2455 FA

6.375 1492.2 (M - H)- PP2456 FA 6.444 1488.2 (M - H)- PP2457 FA 6.268 1492.2 (M - H)-
PP2458 FA 6.033 1521.1 (M + NH.sub.4)+ PP2460 FA 6.188 1537.6 (M + H)+ PP2461 FA 6.292
1500.1 (M - H)- PP2462 FA 6.317 1508.0 (M - H)- PP2463 FA 6.180 1530.3 (M - H)- PP2464
FA 6.079 1546.1 (M - H)- PP2465 FA 6.641 1504.3 (M - H)- PP2466 FA 6.329 1452.1 (M - H)-
PP2467 FA 6.623 1504.0 (M - H)- PP2468 FA 6.528 1486.1 (M - H)- PP2469 FA 6.496 1534.1
(M - H)- PP2470 FA 6.261 1496.3 (M - H)- PP2471 FA 6.527 1506.0 (M - H)- PP2472 FA
6.329 1553.5 (M + H)+ PP2475 FA 6.420 1506.1 (M - H)- PP2477 FA 6.051 1516.1 (M - H)-
PP2479 FA 6.560 1502.2 (M - H)- PP2480 FA 6.015 1528.1 (M - H)- PP2481 FA 6.349 1522.3
(M - H)- PP2482 FA 6.031 1516.1 (M - H)- PP2483 FA 6.284 1516.1 (M - H)- PP2484 FA
6.419 1550.3 (M - H)- PP2485 FA 6.277 1534.3 (M - H)- PP2488 FA 6.636 1499.9 (M - H)-
PP2489 FA 6.291 1522.3 (M - H)- PP2490 FA 6.557 1522.2 (M - H)- PP2492 FA 6.443 1520.1
(M - H)- PP2494 FA 6.144 1498.0 (M - H)- PP2495 FA 6.365 1532.1 (M - H)- PP2496 FA
6.445 1492.2 (M - H)- PP2497 FA 6.173 1486.2 (M - H)- PP2498 FA 6.389 1520.0 (M - H)-
PP2499 FA 6.653 1538.7 (M + NH.sub.4)+ PP2500 FA 6.413 1504.2 (M - H)- PP2501 FA 6.792
1516.0 (M - H)- PP2502 FA 5.905 1461.5 (M - H)- PP2504 FA 5.835 1472.0 (M - H)- PP2505
FA 5.751 1472.3 (M - H)- PP2506 FA 6.181 1476.0 (M - H)- PP2507 FA 6.049 1505.8 (M - H)-
PP2508 FA 6.423 1518.1 (M - H)- PP2509 FA 6.675 1501.4 (M + H)+ PP2510 FA 6.232 1512.3
(M - H)- PP2511 FA 6.667 1525.7 (M - H)- PP2512 FA 6.260 1535.2 (M + NH.sub.4)+ PP2513
FA 6.736 1490.1 (M - H)- PP2514 FA 6.031 1510.1 (M - H)- PP2515 FA 7.073 1528.3 (M - H)-
PP2516 FA 6.308 1506.0 (M - H)- PP2517 AA 8.683 1493.5 (M + H)+ PP2518 FA 6.089 1500.3
(M - H)- PP2519 FA 6.536 1560.1 (M - H)- PP2520 FA 6.271 1536.0 (M - H)- PP2521 FA
6.595 1523.8 (M - H)- PP2522 FA 6.052 1530.3 (M - H)- PP2523 FA 6.567 1564.3 (M - H)-
PP2524 FA 6.511 1516.0 (M - H)- PP2525 AA 8.288 1527.9 (M - H)- PP2526 FA 6.335 1514.1
(M - H)- PP2527 FA 6.607 1556.1 (M - H)- PP2528 FA 6.383 1504.0 (M - H)- PP2529 FA
6.775 1569.6 (M - H)- PP2530 FA 6.353 1535.6 (M + H)+ PP2531 AA 7.967 1494.2 (M - H)-
PP2532 FA 5.975 1490.0 (M - H)- PP2533 AA 8.221 1551.4 (M + Na)+ PP2534 FA 5.771 1483.6
(M - H)- PP2535 FA 6.479 1507.9 (M - H)- PP2536 FA 5.785 1488.2 (M - H)- PP2537 FA
6.576 1473.5 (M + H)+ PP2539 FA 5.839 1478.0 (M - H)- PP2540 FA 5.611 1472.2 (M - H)-
PP2541 FA 5.761 1508.0 (M - H)- PP2542 FA 5.539 1504.0 (M + H)+ PP2543 FA 6.053 1488.2
(M - H)- PP2545 FA 5.907 1476.0 (M - H)- PP2546 FA 5.848 1505.9 (M - H)- PP2547 FA
5.577 1522.4 (M + H)+ PP2548 FA 5.375 1515.5 (M + H)+ PP2549 FA 5.660 1517.8 (M - H)-
PP2550 FA 5.645 1520.3 (M - H)- PP2551 FA 5.425 1514.1 (M - H)- PP2552 FA 5.731 1520.0
(M + H)+ PP2553 FA 6.060 1550.0 (M + H)+ PP2554 FA 5.857 1542.3 (M - H)- PP2555 AA
8.049 1545.9 (M - H)- PP2556 FA 6.125 1548.0 (M - H)- PP2557 FA 5.904 1542.4 (M - H)-
PP2558 FA 6.207 1546.1 (M - H)- PP2559 FA 6.451 1483.5 (M + H)+ PP2560 FA 6.704 1521.7
(M + H)+ PP2561 FA 6.779 1503.3 (M + NH.sub.4)+ PP2562 AA 8.208 1512.2 (M - H)- PP2563
FA 6.537 1477.4 (M + H)+ PP2564 FA 6.908 1514.1 (M - H)- PP2565 AA 8.583 1477.9 (M - H)-
PP2566 FA 6.360 1545.7 (M - H)- PP2567 FA 6.395 1528.9 (M + NH.sub.4)+ PP2568 FA 6.393
1550.3 (M - H)- PP2569 FA 6.500 1515.4 (M + H)+ PP2570 FA 6.608 1531.9 (M - H)- PP2571
FA 6.656 1497.4 (M + H)+ PP2572 FA 6.575 1536.3 (M - H)- PP2573 FA 6.411 1560.5 (M +
NH.sub.4)+ PP2574 FA 5.873 1528.2 (M - H)- PP2575 FA 5.852 1504.0 (M - H)- PP2576 FA
5.976 1530.3 (M - H)- PP2577 FA 6.103 1481.4 (M + H)+ PP2578 FA 6.511 1507.4 (M + H)+
PP2579 AA 7.833 1510.8 (M + NH.sub.4)+ PP2580 FA 5.911 1486.5 (M + NH.sub.4)+ PP2581 FA
6.063 1493.8 (M - H)- PP2583 FA 6.408 1554.0 (M - H)- PP2586 FA 5.509 1490.2 (M - H)-
PP2588 FA 5.575 1496.5 (M + NH.sub.4)+ PP2589 FA 5.575 1455.5 (M + H)+ PP2590 FA 6.331
1512.6 (M + NH.sub.4)+ PP2591 AA 7.741 1498.9 (M + NH.sub.4)+ PP2592 FA 6.320 1512.4 (M
+ NH.sub.4)+ PP2593 FA 6.159 1507.9 (M + H)+ PP2594 FA 6.072 1537.2 (M + NH.sub.4)+
PP2595 FA 6.320 1512.9 (M + NH.sub.4)+ PP2596 AA 7.975 1505.9 (M - H)- PP2597 FA 6.147
1493.5 (M + H)+ PP2598 FA 6.247 1492.2 (M - H)- PP2600 FA 5.611 1460.0 (M - H)- PP2601
FA 5.824 1436.1 (M - H)- PP2602 FA 5.676 1447.5 (M - H)- PP2603 FA 5.659 1425.4 (M + H)+

PP2604 FA 5.825 1449.8 (M - H)- PP2605 FA 6.024 1481.4 (M + H)+ PP2606 FA 6.780 1590.0 (M - H)- PP2608 FA 6.968 1604.3 (M - H)- PP2610 FA 6.911 1555.4 (M + H)+ PP2612 FA 7.109 1569.5 (M + H)+ PP2614 FA 6.735 1544.2 (M - H)- PP2615 FA 6.716 1576.3 (M - H)- PP2616 AA 8.399 1507.9 (M - H)- PP2618 FA 6.788 1538.1 (M - H)- PP2619 FA 6.907 1562.3 (M - H)- PP2622 FA 6.337 1576.0 (M + H)+ PP2624 FA 6.415 1539.3 (M + H)+ PP2626 FA 6.607 1560.0 (M - H)- PP2627 FA 6.901 1589.9 (M - H)- PP2628 FA 6.701 1525.4 (M + H)+ PP2630 FA 7.033 1552.1 (M - H)- PP2631 FA 6.556 1584.3 (M - H)- PP2633 FA 6.759 1598.3 (M - H)- PP2634 AA 8.184 1588.4 (M - H)- PP2635 FA 6.367 1539.9 (M - H)- PP2637 FA 6.407 1539.0 (M + NH.sub.4)+ PP2638 AA 8.244 1483.5 (M - H)- PP2639 FA 6.561 1554.1 (M - H)- PP2640 FA 6.580 1535.4 (M + H)+ PP2641 AA 8.371 1521.8 (M + Na)+ PP2642 AA 8.316 1539.8 (M - H)- PP2643 FA 6.620 1521.3 (M + H)+ PP2644 AA 8.369 1483.8 (M - H)- PP2645 AA 8.343 1566.0 (M - H)- PP2646 AA 8.264 1546.0 (M - H)- PP2648 FA 6.859 1555.4 (M + H)+ PP2649 FA 6.539 1537.5 (M + H)+ PP2650 FA 6.669 1501.7 (M + H)+ PP2651 FA 6.955 1581.8 (M + H)+ PP2652 FA 6.641 1561.9 (M - H)- PP2654 FA 6.395 1576.3 (M - H)- PP2655 FA 6.532 1588.3 (M - H)- PP2656 FA 6.759 1602.3 (M - H)- PP2657 FA 6.544 1588.2 (M - H)- PP2659 FA 5.972 1508.2 (M - H)- PP2660 FA 6.180 1523.6 (M + H)+ PP2661 FA 6.263 1526.2 (M - H)- PP2662 AA 8.237 1540.0 (M - H)- PP2663 FA 6.691 1602.3 (M - H)- PP2664 FA 6.813 1614.3 (M - H)- PP2665 FA 7.031 1628.3 (M - H)- PP2666 FA 6.851 1614.3 (M - H)- PP2667 FA 7.024 1628.4 (M - H)- PP2668 FA 6.327 1534.2 (M - H)- PP2669 FA 6.523 1548.2 (M - H)- PP2670 FA 6.591 1553.4 (M + H)+ PP2671 FA 6.792 1567.6 (M + H)+ PP2672 FA 6.280 1496.3 (M - H)- PP2673 FA 5.659 1482.4 (M - H)- PP2674 FA 6.103 1476.1 (M - H)- PP2675 FA 5.731 1478.1 (M - H)- PP2676 FA 5.663 1482.2 (M - H)- PP2677 FA 5.977 1497.8 (M - H)- PP2678 FA 5.521 1498.1 (M - H)- PP2679 FA 6.307 1537.7 (M + H)+ PP2681 FA 6.192 1509.7 (M + H)+ PP2683 FA 5.773 1459.7 (M + H)+ PP2684 FA 6.069 1476.2 (M + H)+ PP2685 FA 6.003 1454.4 (M - H)- PP2686 FA 5.817 1447.7 (M + H)+ PP2687 FA 6.243 1514.3 (M - H)- PP2688 FA 6.092 1518.2 (M - H)- PP2689 FA 6.059 1518.2 (M - H)- PP2690 FA 6.076 1550.3 (M - H)- PP2691 FA 6.247 1526.1 (M - H)- PP2692 FA 6.492 1528.4 (M - H)- PP2693 FA 5.804 1518.2 (M + H)+ PP2694 FA 5.616 1528.4 (M - H)- PP2695 FA 6.063 1581.3 (M + NH.sub.4)+ PP2696 FA 6.275 1500.3 (M - H)- PP2697 FA 6.117 1504.2 (M - H)- PP2698 FA 6.124 1503.7 (M - H)- PP2699 FA 6.099 1535.6 (M - H)- PP2700 FA 6.255 1512.3 (M - H)- PP2701 AA 8.196 1514.3 (M - H)- PP2702 FA 5.807 1502.1 (M - H)- PP2703 FA 5.624 1532.5 (M + NH.sub.4)+ PP2704 FA 6.109 1548.4 (M - H)- PP2706 FA 6.205 1494.3 (M - H)- PP2708 FA 6.340 1524.1 (M - H)- PP2710 FA 5.971 1488.1 (M - H)- PP2712 FA 6.123 1518.1 (M - H)- PP2714 FA 6.240 1460.3 (M + H)+ PP2716 FA 6.389 1487.7 (M - H)- PP2718 FA 6.499 1520.1 (M - H)- PP2720 FA 6.384 1535.7 (M - H)- PP2722 FA 6.277 1514.1 (M - H)- PP2724 FA 6.161 1530.3 (M - H)- PP2726 FA 6.551 1485.5 (M + H)+ PP2728 FA 6.440 1500.2 (M - H)- PP2730 FA 6.377 1548.2 (M - H)- PP2732 AA 8.056 1564.3 (M - H)- PP2734 FA 6.159 1542.2 (M - H)- PP2736 FA 6.053 1558.3 (M - H)- PP2738 AA 8.199 1512.0 (M - H)- PP2740 FA 6.332 1527.5 (M - H)- PP2742 FA 6.635 1574.3 (M - H)- PP2744 FA 6.369 1576.2 (M - H)- PP2746 FA 6.421 1568.1 (M - H)- PP2748 FA 6.137 1569.6 (M - H)- PP2749 FA 6.697 1539.4 (M + H)+ PP2750 AA 8.179 1540.1 (M - H)- PP2751 FA 6.053 1491.6 (M - H)- PP2752 FA 6.201 1521.6 (M - H)- PP2753 FA 5.839 1486.1 (M - H)- PP2754 FA 5.980 1516.3 (M - H)- PP2755 FA 6.071 1456.1 (M - H)- PP2756 FA 6.223 1486.1 (M - H)- PP2757 FA 6.379 1518.2 (M - H)- PP2760 FA 6.049 1527.6 (M - H)- PP2761 AA 8.205 1481.8 (M - H)- PP2762 AA 8.177 1498.0 (M - H)- PP2763 FA 6.123 1562.0 (M - H)- PP2764 FA 6.024 1542.0 (M + H)+ PP2765 FA 5.907 1556.3 (M - H)- PP2766 FA 6.264 1528.5 (M + NH.sub.4)+ PP2767 FA 6.167 1525.5 (M - H)- PP2768 FA 6.520 1572.3 (M - H)- PP2769 FA 6.240 1574.0 (M - H)- PP2770 FA 6.315 1566.3 (M - H)- PP2771 FA 6.020 1568.3 (M - H)- PP2772 FA 6.023 1469.7 (M - H)- PP2773 FA 6.115 1542.1 (M - H)- PP2776 FA 6.231 1546.2 (M - H)- PP2777 FA 6.813 1506.1 (M - H)- PP2778 FA 6.300 1450.0 (M - H)- PP2779 FA 6.611 1452.1 (M - H)- PP2780 FA

5.708 1527.6 (M - H)- PP2781 FA 5.960 1530.0 (M - H)- PP2782 FA 5.519 1474.3 (M - H)-
PP2783 FA 5.760 1476.2 (M - H)- PP2784 FA 6.280 1544.2 (M - H)- PP2785 FA 6.588 1546.0
(M - H)- PP2786 FA 6.111 1489.7 (M - H)- PP2787 FA 6.389 1492.1 (M - H)- PP2788 FA 5.749
1505.7 (M - H)- PP2789 FA 6.021 1508.1 (M - H)- PP2790 AA 7.643 1452.1 (M - H)- PP2791
FA 5.812 1453.7 (M - H)- PP2796 AA 7.867 1491.6 (M - H)- PP2797 FA 5.783 1477.6 (M - H)-
PP2798 FA 5.709 1478.2 (M - H)- PP2799 FA 5.761 1441.8 (M - H)- PP2800 FA 6.216 1505.6
(M - H)- PP2801 FA 6.115 1504.2 (M - H)- PP2802 FA 6.007 1492.3 (M - H)- PP2803 FA
5.932 1498.2 (M - H)- PP2804 FA 5.713 1478.0 (M - H)- PP2805 AA 8.045 1467.5 (M - H)-
PP2806 AA 8.213 1574.1 (M - H)- PP2807 FA 5.979 1532.0 (M - H)- PP2808 FA 6.503 1520.3
(M - H)- PP2809 FA 5.776 1525.8 (M - H)- PP2810 FA 6.349 1473.8 (M - H)- PP2811 AA
7.927 1496.0 (M - H)- PP2812 FA 6.707 1489.4 (M + H)+ PP2813 FA 6.268 1558.1 (M - H)-
PP2814 FA 6.611 1489.5 (M + H)+ PP2815 AA 7.840 1551.7 (M - H)- PP2816 FA 6.765 1502.3
(M + H)+ PP2817 FA 6.267 1540.8 (M + NH.sub.4)+ PP2818 FA 7.097 1532.7 (M + NH.sub.4)+
PP2819 AA 8.221 1535.9 (M - H)- PP2820 FA 6.523 1457.5 (M + H)+ PP2821 AA 8.080 1561.6
(M + Na)+ PP2822 FA 6.369 1528.9 (M + NH.sub.4)+ PP2823 FA 6.396 1544.1 (M - H)- PP2824
FA 6.221 1521.6 (M - H)- PP2825 FA 5.976 1552.3 (M - H)- PP2826 FA 6.267 1556.1 (M - H)-
PP2827 FA 6.435 1520.1 (M - H)- PP2828 FA 6.191 1550.3 (M - H)- PP2829 FA 6.472 1553.9
(M - H)- PP2830 AA 8.208 1540.1 (M - H)- PP2831 FA 6.457 1574.0 (M - H)- PP2832 FA
6.145 1551.8 (M + H)+ PP2833 FA 6.180 1585.8 (M + H)+ PP2834 FA 6.311 1541.5 (M + H)+
PP2835 FA 6.177 1527.4 (M + H)+ PP2836 FA 6.205 1559.8 (M - H)- PP2837 FA 6.029 1509.6
(M - H)- PP2838 FA 6.079 1544.0 (M - H)- PP2839 FA 6.181 1510.1 (M - H)- PP2840 FA
5.932 1540.0 (M - H)- PP2841 AA 7.976 1543.9 (M - H)- PP2842 FA 6.152 1556.0 (M - H)-
PP2843 AA 8.048 1520.0 (M - H)- PP2844 FA 6.335 1554.0 (M - H)- PP2845 AA 7.976 1533.9
(M + Na)+ PP2846 FA 5.877 1540.2 (M - H)- PP2847 FA 6.183 1544.0 (M - H)- PP2848 FA
6.485 1504.0 (M - H)- PP2849 FA 6.236 1500.2 (M - H)- PP2850 FA 5.533 1501.7 (M - H)-
PP2852 FA 6.504 1550.0 (M - H)- PP2853 FA 6.061 1512.3 (M - H)- PP2854 FA 6.109 1500.1
(M - H)- PP2855 FA 6.333 1495.6 (M - H)- PP2856 FA 6.161 1488.3 (M - H)- PP2857 FA
5.945 1519.8 (M - H)- PP2858 FA 6.411 1517.6 (M + H)+ PP2859 AA 8.267 1580.3 (M + H)+
PP2860 FA 5.881 1542.5 (M + NH.sub.4)+ PP2861 FA 6.511 1540.2 (M + H)+ PP2862 FA 6.343
1518.3 (M - H)- PP2863 FA 6.645 1510.3 (M - H)- PP2864 FA 5.904 1523.9 (M - H)- PP2865
FA 6.160 1559.8 (M - H)- PP2866 FA 6.205 1540.0 (M - H)- PP2867 FA 5.716 1540.0 (M - H)-
PP2868 FA 5.692 1540.7 (M + NH.sub.4)+ PP2869 FA 6.197 1500.3 (M - H)- PP2870 FA 6.725
1510.3 (M - H)- PP2871 FA 6.103 1496.1 (M - H)- PP2872 FA 6.544 1509.2 (M + NH.sub.4)+
PP2873 FA 6.159 1492.4 (M - H)- PP2874 FA 6.095 1495.7 (M - H)- PP2875 FA 6.425 1511.6
(M - H)- PP2876 FA 5.951 1512.1 (M - H)- PP2877 FA 6.377 1533.4 (M + H)+ PP2879 FA
6.431 1472.2 (M - H)- PP2880 FA 5.748 1474.2 (M - H)- PP2881 FA 6.707 1522.1 (M - H)-
PP2883 FA 6.267 1484.1 (M - H)- PP2884 FA 6.308 1472.1 (M - H)- PP2885 AA 8.295 1488.1
(M - H)- PP2886 FA 6.524 1468.1 (M - H)- PP2888 AA 8.199 1536.1 (M + H)+ PP2889 FA
6.476 1545.3 (M + NH.sub.4)+ PP2890 FA 6.416 1481.8 (M + H)+ PP2891 AA 8.177 1472.3 (M -
H)- PP2892 FA 6.859 1533.7 (M + H)+ PP2893 FA 6.203 1542.3 (M + NH.sub.4)+ PP2894 FA
6.661 1480.4 (M + H)+ PP2895 FA 6.008 1487.8 (M + NH.sub.4)+ PP2896 FA 6.300 1531.3 (M -
H)- PP2897 FA 5.977 1528.4 (M + NH.sub.4)+ PP2898 FA 6.095 1478.9 (M + H)+ PP2899 FA
5.781 1455.5 (M - H)- PP2900 FA 6.797 1528.3 (M - H)- PP2901 FA 5.625 1496.9 (M + H)+
PP2902 AA 8.317 1474.4 (M - H)- PP2904 FA 6.497 1525.5 (M - H)- PP2905 FA 5.593 1510.3
(M + H)+ PP2906 FA 6.292 1471.5 (M - H)- PP2907 FA 6.101 1518.3 (M - H)- PP2908 FA
6.263 1512.7 (M + H)+ PP2909 FA 5.573 1478.3 (M - H)- PP2910 AA 7.989 1457.5 (M - H)-
PP2911 FA 5.092 1450.3 (M - H)- PP2912 FA 5.909 1515.8 (M + NH.sub.4)+ PP2913 FA 6.317
1558.4 (M - H)- PP2914 FA 5.697 1443.4 (M - H)- PP2915 FA 5.887 1474.3 (M - H)- PP2916
FA 6.348 1521.6 (M + H)+ PP2917 AA 7.725 1522.3 (M - H)- PP2918 FA 6.041 1492.3 (M - H)-
PP2920 AA 7.939 1484.1 (M - H)- PP2921 FA 6.265 1454.0 (M - H)- PP2922 FA 6.144 1466.4

(M - H)- PP2923 AA 7.680 1468.3 (M - H)- PP2928 FA 6.740 1516.3 (M - H)- PP2929 FA 6.316 1518.3 (M - H)- PP2931 FA 6.245 1464.2 (M - H)- PP2932 FA 5.867 1466.3 (M - H)- PP2934 AA 7.883 1514.4 (M + H)+ PP2935 FA 5.977 1514.3 (M - H)- PP2936 FA 5.960 1484.3 (M - H)- PP2937 FA 5.852 1460.4 (M - H)- PP2938 FA 5.505 1462.4 (M - H)- PP2941 FA 6.033 1487.6 (M + H)+ PP2946 FA 6.283 1532.3 (M - H)- PP2947 FA 6.099 1478.1 (M - H)- PP2948 FA 6.524 1530.3 (M - H)- PP2949 FA 6.351 1476.3 (M - H)- PP2950 FA 5.992 1531.1 (M + H)+ PP2951 AA 7.791 1475.3 (M - H)- PP2952 FA 6.541 1546.4 (M + H)+ PP2953 FA 6.660 1526.9 (M + NH.sub.4)+ PP2954 AA 7.919 1540.3 (M - H)- PP2955 FA 6.400 1549.4 (M + NH.sub.4)+ PP2956 FA 6.559 1543.7 (M + H)+ PP2957 FA 6.020 1530.4 (M + H)+ PP2958 FA 6.521 1523.1 (M + NH.sub.4)+ PP2959 FA 6.475 1513.2 (M + NH.sub.4)+ PP2960 FA 6.047 1480.3 (M - H)- PP2961 FA 6.456 1532.2 (M - H)- PP2962 AA 8.156 1441.9 (M - H)- PP2963 FA 6.228 1478.3 (M - H)- PP2964 FA 6.672 1530.1 (M - H)- PP2965 FA 6.563 1530.2 (M - H)- PP2966 AA 7.964 1478.1 (M - H)- PP2967 FA 6.339 1476.2 (M - H)- PP2968 FA 6.335 1526.4 (M - H)- PP2969 FA 6.480 1546.2 (M - H)- PP2970 FA 5.832 1474.3 (M - H)- PP2971 FA 6.519 1558.2 (M - H)- PP2972 AA 8.081 1498.1 (M - H)- PP2973 AA 8.073 1532.2 (M - H)- PP2974 FA 6.131 1445.8 (M + H)+ PP2975 FA 6.192 1558.1 (M - H)- PP2976 FA 6.215 1523.9 (M - H)- PP2977 FA 6.259 1532.2 (M - H)- PP2978 FA 5.941 1470.3 (M - H)- PP2979 FA 6.697 1528.2 (M - H)- PP2980 FA 6.337 1522.0 (M - H)- PP2981 FA 6.735 1540.2 (M - H)- PP2982 FA 6.055 1468.1 (M - H)- PP2983 FA 6.647 1514.1 (M - H)- PP2984 FA 6.287 1556.2 (M - H)- PP2985 FA 6.333 1540.1 (M - H)- PP2986 FA 6.467 1538.2 (M - H)- PP2987 FA 6.447 1514.1 (M - H)- PP2988 FA 6.025 1504.4 (M - H)- PP2989 FA 6.259 1506.3 (M - H)- PP2990 FA 6.219 1486.4 (M - H)- PP2991 FA 6.116 1480.3 (M - H)- PP2992 FA 5.808 1498.4 (M - H)- PP2993 FA 5.924 1506.2 (M - H)- PP2994 FA 6.016 1480.1 (M - H)- PP2995 FA 6.005 1480.3 (M - H)- PP2996 AA 8.119 1570.2 (M - H)- PP2997 FA 6.417 1476.2 (M - H)- PP2998 FA 6.335 1516.2 (M - H)- PP2999 FA 6.447 1488.3 (M - H)- PP3000 FA 5.547 1572.1 (M - H)- PP3001 FA 6.341 1462.2 (M - H)- PP3002 FA 5.379 1518.1 (M - H)- PP3003 FA 6.047 1488.0 (M - H)- PP3004 FA 6.167 1462.2 (M - H)- PP3005 FA 6.041 1500.3 (M - H)- PP3006 FA 5.853 1474.3 (M - H)- PP3007 FA 5.753 1500.0 (M - H)- PP3008 FA 5.793 1474.3 (M - H)- PP3009 FA 6.189 1470.3 (M - H)- PP3010 FA 6.220 1482.3 (M - H)- PP3011 FA 6.059 1456.2 (M - H)- PP3012 FA 5.869 1482.2 (M - H)- PP3013 FA 5.959 1456.3 (M - H)- PP3014 FA 6.759 1572.1 (M - H)- PP3015 FA 6.685 1558.1 (M - H)- PP3017 FA 6.524 1558.1 (M - H)- PP3018 AA 8.265 1518.2 (M - H)- PP3019 FA 6.489 1504.1 (M - H)- PP3021 FA 6.353 1505.4 (M + H)+ PP3022 FA 5.872 1574.1 (M - H)- PP3026 FA 5.624 1520.2 (M - H)- PP3030 FA 6.300 1534.4 (M - H)- PP3031 FA 6.657 1497.5 (M + H)+ PP3032 FA 5.919 1514.4 (M - H)- PP3033 FA 5.880 1524.4 (M - H)- PP3034 FA 5.919 1538.3 (M - H)- PP3035 FA 5.607 1500.4 (M - H)- PP3036 FA 6.156 1539.9 (M + H)+ PP3037 FA 5.600 1523.8 (M - H)- PP3038 FA 5.808 1500.4 (M - H)- PP3039 FA 6.689 1524.9 (M + NH.sub.4)+ PP3040 FA 6.124 1493.4 (M + H)+ PP3044 FA 6.105 1518.1 (M - H)- PP3045 AA 7.909 1482.0 (M - H)- PP3046 FA 6.276 1495.4 (M + H)+ PP3047 FA 5.712 1480.1 (M - H)- PP3048 FA 6.433 1532.4 (M - H)- PP3049 AA 7.837 1491.9 (M - H)- PP3050 FA 6.499 1497.4 (M + H)+ PP3051 FA 6.605 1507.6 (M - H)- PP3052 FA 6.025 1493.9 (M - H)- PP3053 FA 6.445 1517.8 (M - H)- PP3054 FA 6.717 1533.4 (M + H)+ PP3055 FA 6.424 1465.4 (M + H)+ PP3056 FA 5.672 1452.0 (M - H)- PP3057 FA 6.404 1493.9 (M - H)- PP3058 FA 6.616 1524.5 (M + NH.sub.4)+ PP3059 FA 6.168 1498.5 (M + NH.sub.4)+ PP3060 FA 6.128 1516.2 (M - H)- PP3061 AA 8.037 1524.9 (M + NH.sub.4)+ PP3062 AA 7.981 1493.6 (M - H)- PP3063 FA 5.748 1486.0 (M - H)- PP3064 FA 6.260 1524.3 (M - H)- PP3065 FA 5.752 1510.0 (M - H)- PP3066 FA 5.829 1511.9 (M - H)- PP3067 FA 6.045 1526.1 (M - H)- PP3068 FA 6.267 1537.7 (M - H)- PP3069 FA 5.955 1495.5 (M + H)+ PP3070 FA 6.140 1538.2 (M - H)- PP3071 FA 6.035 1483.4 (M + H)+ PP3072 FA 5.720 1512.1 (M - H)- PP3073 FA 5.929 1524.2 (M - H)- PP3074 FA 5.615 1480.2 (M - H)- PP3075 FA 5.944 1523.5 (M - H)- PP3076 FA 5.733 1467.8 (M - H)- PP3077 FA 5.711 1498.3

(M - H)- PP3078 FA 5.872 1512.1 (M - H)- PP3079 FA 6.055 1543.4 (M + NH.sub.4)+ PP3080 FA 6.057 1524.1 (M - H)- PP3081 FA 5.748 1480.3 (M - H)- PP3082 FA 6.144 1525.7 (M + H)+ PP3083 FA 5.876 1498.3 (M - H)- PP3084 FA 5.783 1498.2 (M - H)- PP3085 FA 5.949 1510.0 (M - H)- PP3086 FA 6.189 1524.4 (M - H)- PP3087 FA 5.604 1466.2 (M - H)- PP3088 FA 6.248 1524.1 (M - H)- PP3089 AA 7.901 1469.4 (M + H)+ PP3090 FA 6.193 1526.1 (M - H)- PP3091 FA 5.963 1512.3 (M - H)- PP3092 FA 5.997 1469.5 (M + H)+ PP3093 FA 6.052 1524.1 (M - H)- PP3094 FA 5.849 1524.1 (M - H)- PP3095 FA 6.699 1592.4 (M - H)- PP3096 FA 6.632 1540.4 (M + H)+ PP3097 FA 6.199 1580.4 (M + H)+ PP3098 FA 6.112 1524.4 (M - H)- PP3099 FA 6.524 1594.3 (M + H)+ PP3100 FA 6.423 1538.4 (M - H)- PP3101 FA 6.023 1578.4 (M - H)- PP3102 FA 5.915 1526.2 (M + H)+ PP3103 FA 6.276 1566.4 (M - H)- PP3104 FA 6.141 1512.4 (M - H)- PP3105 FA 5.753 1552.3 (M - H)- PP3106 FA 5.616 1498.3 (M - H)- PP3108 FA 5.989 1534.2 (M - H)- PP3109 FA 6.083 1548.0 (M - H)- PP3110 FA 6.917 1598.4 (M - H)- PP3111 FA 6.932 1546.3 (M + H)+ PP3112 FA 6.408 1584.1 (M - H)- PP3113 FA 6.312 1530.3 (M - H)- PP3114 FA 6.748 1598.4 (M - H)- PP3115 FA 6.651 1544.4 (M - H)- PP3116 FA 6.232 1584.3 (M - H)- PP3117 FA 6.125 1530.3 (M - H)- PP3118 FA 6.501 1572.3 (M - H)- PP3119 FA 6.367 1518.3 (M - H)- PP3120 FA 5.971 1558.2 (M - H)- PP3121 FA 5.831 1504.2 (M - H)-

Example 2 Pharmacological Test

Evaluation of Compound Binding to KRAS, NRAS, and HRAS by Surface Plasmon Resonance (SPR)

(2311) Surface plasmon resonance was used to analyze the binding affinity of compounds for KRAS, NRAS, and HRAS. The apparatus used was Biacore 8K, 8K+, or T200 (GE Healthcare), and the running buffer used was HBS (10 mM HEPES-NaOH, 150 mM NaCl, pH 7.4) containing 1 mM DTT, 10 mM MgCl.sub.2, 0.01% Tween 20, 10 μ M GDP, and 4% DMSO. The dilution series of each compound solution was prepared by, first, preparing a dilution series of a compound solution having a concentration 100-fold greater than the final concentration using DMSO as a solvent, and then diluting the solution 100-fold with a dilution buffer (1 mM DTT, 10 mM MgCl.sub.2, 0.01% Tween 20, 10 μ M GDP, 3.03% DMSO-containing HBS).

(2312) Biotinylated Avi-tag fusion KRAS, NRAS, and HRAS proteins were expressed in *E. coli* BL.21 (DE3) strains, and after cells were homogenized, they were purified using Streptavidin Mutein Matrix and Superdex 75. After purification, GDP-loaded KRAS, NRAS, and HRAS proteins were immobilized on the surface of Sensor Chip CAP (Cytiva) coated with Biotin CAPture Reagent in an immobilization amount of about 100 to 300 RU. Binding of each compound was evaluated by the Single Cycle Kinetics method, and a running buffer or a compound solution was added to the RAS protein non-immobilized surface and the immobilized surface to obtain a binding response. The flow rate was set to 100 μ L/min, and in the compound-adding cycle, compound solutions having different concentrations were intermittently added for 75 seconds each, starting from a low concentration. The dissociation phase was observed for 3500 seconds. In the blank cycle, a running buffer was intermittently added and not the compound solution. The Biotin CAPture Reagent and RAS proteins were immobilized for each cycle, and at the end of each cycle, the sensor chip was regenerated by the regeneration solution appended to the Biotin CAPture Kit (Cytiva). Measurement was carried out at 30° C.

(2313) The resulting sensorgram was subjected to curve fitting based on a 1:1 binding model on Biacore Insight Evaluation Software or T200 Evaluation Software, and thereby the dissociation constant K_D of each compound with respect to KRAS, NRAS, and HRAS was determined. Table 37 shows the K_D value of each test compound with respect to GDP-bound KRAS wild-type protein, and the ratio of the K_D value with respect to GDP-bound NRAS wild-type protein and HRAS wild-type protein to the K_D value with respect to KRAS. A compound having a K_D ratio of 0.1 or more and less than 3 is denoted as C, a compound having 3 or more and less than 10 is denoted as B, a compound having 10 or more and less than 20 is denoted as A, and a compound

having 20 or more is denoted as AA. Blank indicates a compound from which the KD value with respect to NRAS or HRAS was not obtained.

(2314) Measurement of NCI-H441 Cell Proliferation Inhibitory Activity

(2315) A test compound was dispensed into a U-bottom 384-well plate as 40 nL serially diluted dimethyl sulfoxide solutions using a liquid handler Echo (LABCYTE). As for human lung cancer strain NCI-H441 (ATCC), a cell suspension was prepared so as to have 250 cells/40 μ L in an RPMI-1640 medium (Sigma) supplemented with 10% fetal bovine serum (Sigma), a 2.5 g/L D-(+)-glucose solution, 10 mmol/L HEPES, and 1 mmol/L sodium pyruvate. This cell suspension was dispensed in an amount of 40 μ L per well into a plate containing the test compound, and cultured at 37° C. in a 5% carbon dioxide gas incubator. Four days later, 20 μ L of CellTiter-Glo (registered trademark)(Promega) was added to each well and fluorescence was measured. From the growth inhibition ratio attained when the test compound was added to that of the control to which the test compound was not added, the cell proliferation inhibitory activity of the test compound was calculated in terms of a 50% proliferation inhibitory concentration (an IC.sub.50 value). The IC.sub.50 value of each test compound is shown in Table 37.

(2316) TABLE-US-00048 TABLE 37 Compound KRAS NRAS/ HRAS/ NCI-H441 No. KD (M) KRAS KRAS IC.sub.50 (nM) PP0001 1.4E-10 C C 6.3 PP0002 1.8E-10 C C 2.1 PP0003 2.0E-09 C C 41.4 PP0004 2.4E-09 C C 52.9 PP0006 1.5E-09 C C 20.6 PP0007 7.4E-11 C C 4.8 PP0011 3.0E-10 C C 5.8 PP0013 7.3E-11 C C 9.8 PP0014 9.0E-11 C C 2.0 PP0015 2.7E-09 C C 24.5 PP0016 1.6E-09 C C 24.6 PP0018 7.7E-11 C C 7.0 PP0019 3.4E-10 C C 6.7 PP0020 7.7E-10 C C 9.9 PP0021 6.5E-10 C C 9.8 PP0022 1.8E-10 B C 14.4 PP0023 1.4E-10 B C 4.5 PP0024 5.9E-09 B C 101.6 PP0025 2.4E-09 C C 27.5 PP0027 8.0E-10 C C 17.1 PP0028 2.4E-09 C C 54.5 PP0029 7.0E-09 C C 26.4 PP0030 7.7E-09 C C 86.7 PP0031 1.8E-10 B C 11.4 PP0032 3.7E-10 C C 26.9 PP0037 7.6E-09 C C 429.7 PP0039 1.7E-10 C C 20.1 PP0041 1.2E-10 B C 62.7 PP0042 1.2E-10 C C 7.2 PP0043 8.7E-10 C C 20.4 PP0044 3.6E-09 C C 142.2 PP0045 2.3E-09 C C 292.3 PP0047 1.8E-08 C C 229.3 PP0048 6.3E-08 C C 464.1 PP0049 6.3E-09 C C 201.4 PP0050 2.3E-08 B C 499.0 PP0051 9.1E-10 C C 22.1 PP0052 1.2E-10 C B 1.7 PP0053 1.2E-10 C C 4.6 PP0054 4.5E-09 C C 82.0 PP0055 2.7E-10 C C 3.5 PP0056 2.9E-09 C B 49.6 PP0057 2.1E-10 C B 6.2 PP0058 2.0E-09 C C 25.8 PP0059 1.5E-10 B C 6.1 PP0060 9.9E-11 B B 1.5 PP0061 3.8E-09 B C 84.4 PP0062 2.1E-10 C C 3.8 PP0063 2.9E-09 C C 39.3 PP0064 4.2E-10 C C 9.9 PP0065 5.4E-10 C C 4.3 PP0066 1.4E-09 B C 26.1 PP0067 7.2E-11 C C 2.7 PP0068 1.4E-10 C C 9.1 PP0069 6.9E-09 C C 39.7 PP0070 1.6E-10 C B 2.1 PP0071 1.5E-10 3.8 PP0076 6.4E-11 C C 10.9 PP0077 2.0E-10 C B 5.3 PP0078 1.9E-10 C B 4.8 PP0083 1.1E-09 11.3 PP0084 2.7E-09 C B 50.0 PP0085 8.4E-09 C C 108.5 PP0086 2.6E-10 C B 6.4 PP0087 7.5E-10 C C 24.0 PP0088 1.2E-10 4.1 PP0089 1.7E-10 C B 5.5 PP0090 8.3E-11 1.9 PP0091 2.3E-10 C B 4.1 PP0092 1.5E-10 C C 3.7 PP0093 7.6E-11 C B 2.2 PP0094 9.4E-10 C C 10.6 PP0095 7.9E-10 C C 18.2 PP0096 2.8E-09 C C 68.9 PP0097 9.3E-09 C C 240.1 PP0098 2.3E-10 9.4 PP0099 7.0E-10 18.7 PP0100 1.9E-10 B C 24.3 PP0101 1.6E-10 B C 10.4 PP0102 2.0E-10 C C 13.4 PP0103 1.5E-10 B C 8.2 PP0104 2.5E-10 B C 6.7 PP0105 1.4E-10 B C 8.3 PP0106 1.9E-10 C B 12.3 PP0107 1.7E-09 C C 32.2 PP0108 2.0E-10 C C 9.0 PP0109 1.3E-09 C B 21.7 PP0110 4.0E-09 C B 56.5 PP0112 3.6E-10 C B 6.1 PP0113 7.5E-10 C B 7.1 PP0114 1.6E-10 C C 6.1 PP0115 2.1E-10 C C 2.0 PP0116 2.7E-10 C C 6.8 PP0117 1.9E-10 B B 9.2 PP0118 2.1E-10 C C 6.0 PP0119 1.0E-10 C C 1.6 PP0120 1.9E-10 B C 5.6 PP0121 2.1E-09 C C 29.5 PP0122 3.2E-10 C C 12.0 PP0123 8.3E-10 B C 12.5 PP0124 3.5E-09 C C 48.8 PP0126 4.3E-10 C C 9.3 PP0127 9.2E-10 C C 25.3 PP0128 2.5E-10 C C 10.4 PP0129 2.0E-10 B C 8.8 PP0130 2.5E-10 C C 8.8 PP0131 1.4E-10 B C 11.1 PP0132 2.0E-10 C C 14.8 PP0133 8.0E-11 B C 6.4 PP0134 6.6E-10 C B 18.5 PP0135 7.5E-10 B C 26.1 PP0136 1.3E-09 19.6 PP0137 4.7E-10 13.4 PP0138 7.7E-10 C C 20.6 PP0139 2.3E-07 C C 1495.7 PP0140 8.0E-07 C B 712.5 PP0141 5.5E-10 C B 3.5 PP0144 2.4E-10 B C 18.3 PP0145 5.0E-10 B B 8.8 PP0146 4.2E-10 B B 23.9 PP0148 3.4E-09 B B 38.0 PP0149 2.3E-08 C B 317.4 PP0150 1.7E-09 B C 49.1 PP0151 5.0E-10

C C 5.3 PP0152 2.8E-09 C B 37.9 PP0153 3.3E-10 C B 11.6 PP0154 2.3E-10 C B 10.9 PP0155
1.8E-10 B C 2.8 PP0156 1.2E-10 C B 8.1 PP0158 8.5E-10 A B 15.4 PP0162 1.7E-08 B C 165.8
PP0165 2.9E-09 B B 35.0 PP0166 2.0E-09 C B 24.9 PP0168 6.7E-10 C B 26.6 PP0169 3.5E-11
B B 2.3 PP0170 1.2E-10 B B 5.2 PP0171 3.1E-10 B C 6.9 PP0172 1.3E-10 B C 3.8 PP0173
6.7E-10 B C 11.6 PP0174 1.5E-10 B B 5.2 PP0175 3.5E-10 B B 6.6 PP0176 5.5E-10 B C 17.7
PP0177 5.5E-10 B C 34.4 PP0178 7.3E-11 C B 1.5 PP0179 1.1E-10 C B 4.5 PP0180 7.2E-10 C
C 7.5 PP0181 8.1E-11 C B 1.5 PP0182 8.8E-10 C B 6.2 PP0183 1.7E-10 C B 1.6 PP0184
3.2E-10 C B 2.1 PP0185 1.1E-09 C B 32.2 PP0186 1.7E-09 C B 20.5 PP0187 1.9E-10 B C 3.0
PP0188 2.5E-10 B C 3.9 PP0189 2.2E-10 B C 3.0 PP0190 1.3E-10 B B 4.1 PP0191 1.4E-10 C B
3.9 PP0192 1.9E-10 C B 2.6 PP0193 1.7E-10 C C 1.4 PP0194 9.0E-11 C B 1.9 PP0195 9.1E-11
B C 2.7 PP0196 1.5E-10 C C 3.7 PP0197 1.3E-10 B C 3.6 PP0198 1.9E-10 C B 5.7 PP0199
1.3E-10 B C 3.5 PP0200 2.2E-10 C C 2.5 PP0201 6.0E-11 B C 3.4 PP0202 7.6E-11 B C 3.9
PP0203 9.4E-11 C C 2.8 PP0204 2.1E-10 C C 5.4 PP0205 1.1E-10 C C 4.6 PP0206 7.1E-11 C B
2.2 PP0207 6.6E-11 B C 1.9 PP0208 2.6E-10 C C 4.6 PP0209 1.1E-07 C C 568.8 PP0210
5.2E-10 C C 17.1 PP0211 2.7E-10 B C 10.8 PP0212 9.8E-11 C C 26.6 PP0213 2.5E-10 C C 10.3
PP0214 1.3E-10 B C 5.4 PP0215 4.1E-10 C C 4.8 PP0216 2.8E-10 C B 3.0 PP0217 1.8E-10 C C
18.1 PP0218 2.0E-10 C C 3.9 PP0219 1.3E-10 C B 2.2 PP0220 3.5E-10 C C 7.7 PP0221 1.9E-10
C C 2.9 PP0222 5.7E-10 C C 17.1 PP0223 4.0E-10 C C 5.0 PP0224 2.2E-10 C B 2.0 PP0225
2.3E-10 C C 2.4 PP0226 4.7E-10 C B 15.3 PP0227 2.6E-10 C B 4.1 PP0228 3.7E-11 C B 1.8
PP0229 2.5E-10 B B 24.7 PP0230 3.9E-10 B C 15.5 PP0231 5.5E-10 B C 13.1 PP0232 2.8E-10
B B 4.7 PP0233 5.2E-11 B C 1.6 PP0234 4.9E-10 B C 49.9 PP0235 1.5E-10 B C 4.5 PP0236
5.4E-10 B C 20.3 PP0237 2.8E-10 B C 4.0 PP0238 4.1E-11 C C 1.3 PP0239 5.0E-11 C B 6.2
PP0240 3.5E-11 B C 7.9 PP0241 5.2E-11 B C 4.9 PP0242 2.0E-10 C C 5.0 PP0243 3.1E-10 B C
23.2 PP0244 3.5E-09 C B 17.4 PP0245 3.7E-10 22.9 PP0246 5.4E-11 B C 1.8 PP0247 3.8E-10
B C 41.3 PP0248 1.1E-09 B B 84.0 PP0249 6.9E-10 B C 25.5 PP0250 1.2E-10 B C 6.7 PP0251
2.5E-10 8.4 PP0252 3.4E-10 B B 19.4 PP0253 5.5E-10 B B 15.1 PP0254 1.6E-10 B C 5.6
PP0256 9.9E-10 B C 26.6 PP0257 1.4E-09 B B 30.8 PP0258 5.2E-10 B C 11.3 PP0259 4.2E-10
C B 10.2 PP0260 3.9E-10 B C 16.8 PP0261 1.4E-10 B C 4.6 PP0262 4.4E-11 B B 1.9 PP0263
4.7E-10 B C 19.2 PP0264 4.2E-10 B C 16.4 PP0265 2.8E-10 B C 13.3 PP0266 2.0E-10 B C 15.2
PP0267 8.7E-10 B B 38.2 PP0268 3.4E-10 C B 15.4 PP0269 1.4E-09 C B 34.7 PP0270 7.6E-10
C B 8.5 PP0271 2.2E-10 C B 6.5 PP0272 1.7E-10 C B 3.7 PP0273 5.6E-10 C B 12.2 PP0274
4.7E-10 C B 9.1 PP0275 3.0E-10 C C 9.3 PP0276 1.5E-07 752.2 PP0277 5.3E-10 C C 6.7
PP0278 6.2E-10 C B 12.4 PP0279 3.6E-10 15.0 PP0281 4.8E-10 C C 7.8 PP0282 1.1E-10 C C
3.9 PP0283 6.3E-11 C C 1.9 PP0284 6.7E-10 C B 29.0 PP0285 7.5E-10 C B 18.9 PP0286
5.2E-10 C B 35.0 PP0287 2.6E-10 C C 14.5 PP0288 4.2E-10 C B 14.3 PP0289 2.2E-09 B B 76.4
PP0290 4.8E-09 B C 149.0 PP0291 2.2E-09 B B 31.0 PP0292 5.4E-10 B C 36.3 PP0293 8.4E-10
B C 37.6 PP0294 1.7E-09 B C 55.4 PP0295 5.2E-10 B C 33.2 PP0296 1.5E-10 B C 4.3 PP0297
2.3E-09 B C 74.3 PP0298 3.0E-09 B C 41.1 PP0299 1.4E-09 B B 20.2 PP0300 1.4E-09 B C 73.9
PP0301 4.8E-09 B C 88.2 PP0302 1.6E-09 B C 126.2 PP0303 4.0E-09 B B 349.2 PP0304
1.9E-09 B B 112.4 PP0305 9.6E-10 B C 49.6 PP0306 1.2E-09 B B 141.5 PP0307 2.1E-09 B B
204.8 PP0308 1.1E-09 B B 56.0 PP0309 2.9E-10 B B 11.7 PP0310 4.5E-10 B C 13.8 PP0311
3.3E-09 B B 146.6 PP0312 3.4E-09 B B 82.4 PP0313 2.2E-09 B C 56.6 PP0314 1.6E-09 B C
58.1 PP0315 3.8E-09 B B 148.0 PP0316 5.4E-11 C B 2.8 PP0317 8.6E-10 B B 32.2 PP0318
3.4E-11 B B 2.3 PP0319 1.8E-09 B B 23.5 PP0320 9.0E-10 B B 92.3 PP0321 1.0E-09 B B 55.7
PP0322 8.4E-10 B C 74.2 PP0323 2.5E-10 B B 64.0 PP0324 4.1E-10 B B 38.5 PP0325 1.1E-09
B B 163.2 PP0326 1.5E-09 B B 172.5 PP0327 1.9E-09 B B 162.5 PP0328 1.6E-09 B B 242.6
PP0329 7.4E-10 B C 65.8 PP0330 9.5E-10 B C 98.2 PP0331 2.8E-09 B B 84.7 PP0332 3.2E-09
B B 167.1 PP0333 2.4E-09 B C 41.6 PP0334 4.7E-09 B C 129.7 PP0335 1.3E-09 B C 41.7
PP0336 1.6E-09 B B 73.2 PP0337 2.5E-09 B B 333.4 PP0338 3.6E-09 B B 295.9 PP0339
5.0E-09 B B 157.9 PP0340 3.7E-09 B B 177.3 PP0341 1.4E-09 B B 134.9 PP0342 1.7E-09 B B

141.0 PP0343 1.3E-09 C B 47.6 PP0344 1.1E-09 C B 82.9 PP0345 1.3E-09 C B 54.9 PP0346
1.2E-09 C B 69.5 PP0347 3.6E-10 C B 10.9 PP0348 3.8E-10 C B 15.1 PP0349 7.3E-10 C B 75.9
PP0350 1.1E-09 C B 69.8 PP0351 1.1E-09 C B 39.0 PP0352 8.2E-10 C B 72.4 PP0353 4.7E-10
C B 46.7 PP0354 4.4E-10 C B 24.7 PP0355 3.3E-10 B B 48.2 PP0356 4.9E-10 B B 99.4 PP0357
4.0E-10 B B 39.8 PP0358 3.3E-10 B C 34.8 PP0359 1.4E-10 B B 33.1 PP0360 2.5E-10 B C 18.0
PP0361 7.2E-10 B C 69.0 PP0362 1.0E-09 B B 148.6 PP0363 1.0E-09 B C 91.4 PP0364 7.7E-10
B B 103.0 PP0365 2.9E-10 B B 42.6 PP0366 3.8E-10 B B 80.2 PP0367 1.3E-09 B B 99.1
PP0368 1.7E-09 B B 114.8 PP0369 1.9E-09 B C 68.9 PP0370 1.5E-09 B C 63.2 PP0371 4.8E-10
B C 30.9 PP0372 7.9E-10 B B 34.2 PP0373 1.2E-09 B B 152.0 PP0374 1.8E-09 B B 141.9
PP0375 2.3E-09 B B 57.1 PP0376 2.0E-09 B B 114.2 PP0377 7.0E-10 B C 33.4 PP0378 8.5E-10
B B 109.6 PP0379 3.8E-10 C B 4.4 PP0380 5.9E-10 C B 16.4 PP0381 4.9E-10 C B 11.1 PP0382
3.5E-10 C B 12.1 PP0383 2.4E-10 C B 2.6 PP0384 1.8E-10 C B 7.0 PP0385 3.7E-10 15.2
PP0386 6.9E-10 C B 29.6 PP0387 5.4E-10 C B 20.0 PP0388 7.8E-10 C B 11.6 PP0389 1.6E-10
C B 3.0 PP0390 2.0E-10 C B 7.6 PP0391 9.2E-11 B B 1.1 PP0392 6.7E-11 B C 0.9 PP0393
1.1E-10 B B 2.7 PP0394 2.4E-10 C C 2.6 PP0395 6.8E-11 B C 0.8 PP0396 1.3E-10 B C 0.9
PP0397 1.2E-10 C C 0.7 PP0398 4.8E-10 B C 3.6 PP0399 3.3E-10 B C 6.3 PP0400 1.7E-10 B B
7.8 PP0401 1.5E-10 B B 5.1 PP0402 1.2E-09 B C 58.1 PP0403 8.3E-10 B B 19.4 PP0404
3.7E-10 B C 21.7 PP0405 3.1E-10 B C 14.3 PP0406 4.2E-11 C B 2.5 PP0407 1.1E-10 C B 7.6
PP0408 9.4E-11 C B 5.7 PP0409 5.6E-11 C C 4.4 PP0410 5.8E-11 C B 1.4 PP0411 4.6E-11 C C
2.8 PP0412 4.3E-11 B C 1.8 PP0413 4.9E-11 C C 2.0 PP0414 2.9E-11 B C 1.9 PP0415 1.8E-11
B B 1.9 PP0416 2.7E-11 B C 1.1 PP0417 7.4E-11 B C 3.9 PP0418 7.4E-11 B C 4.6 PP0419
6.7E-11 B C 4.2 PP0420 3.4E-11 B C 1.2 PP0421 3.3E-11 B C 2.0 PP0422 1.7E-11 B B 1.6
PP0423 1.8E-10 B C 7.1 PP0424 1.7E-10 B C 9.5 PP0425 9.9E-11 B C 6.5 PP0426 1.2E-10 C C
4.8 PP0427 1.1E-10 B C 2.2 PP0428 6.3E-11 B C 4.6 PP0429 3.9E-10 B C 12.4 PP0430 2.6E-10
B B 27.9 PP0431 3.5E-10 B C 9.9 PP0432 1.7E-10 B B 10.7 PP0433 1.5E-10 B B 11.0 PP0434
1.2E-10 B C 9.7 PP0435 4.7E-11 C C 2.6 PP0436 5.4E-11 C C 1.0 PP0437 3.6E-11 C B 1.2
PP0438 3.0E-11 C C 1.0 PP0439 2.1E-11 0.7 PP0440 4.6E-11 C B 4.4 PP0441 7.2E-11 C C 3.4
PP0442 3.7E-11 C C 0.8 PP0443 2.7E-11 C C 0.9 PP0444 3.1E-11 C C 1.4 PP0445 2.5E-11 C C
1.7 PP0446 1.0E-10 B C 1.6 PP0447 1.8E-10 B C 3.9 PP0448 1.5E-10 B C 2.0 PP0449 3.5E-10
C C 9.2 PP0450 3.8E-10 B C 8.4 PP0451 7.5E-10 11.2 PP0452 7.3E-10 B B 34.1 PP0453
1.2E-09 B C 20.3 PP0454 7.3E-11 C C 2.9 PP0455 1.8E-10 9.9 PP0456 7.0E-11 C C 7.0 PP0457
8.3E-11 C B 12.1 PP0460 1.4E-10 C C 5.4 PP0461 4.9E-10 A A 23.2 PP0462 3.2E-10 C C 15.1
PP0463 5.4E-10 A A 26.3 PP0464 9.2E-10 B B 17.6 PP0465 5.1E-10 B B 18.0 PP0466 2.0E-10
B B 22.6 PP0467 5.9E-10 B B 33.8 PP0468 8.0E-10 B B 37.2 PP0469 7.6E-10 B B 36.1 PP0470
5.5E-10 B B 51.9 PP0471 9.3E-10 B B 37.7 PP0472 7.7E-10 B B 24.4 PP0473 5.6E-10 C C 14.8
PP0474 6.3E-10 B B 19.9 PP0475 4.6E-10 B B 18.7 PP0476 2.0E-10 B B 4.9 PP0477 1.1E-08 A
A 138.6 PP0478 4.0E-07 2751.8 PP0479 2.9E-08 B B 2435.2 PP0480 3.1E-09 A B 47.3 PP0481
8.5E-07 B B >1000 PP0483 3.4E-08 B B 524.9 PP0484 3.2E-09 A B 165.1 PP0485 2.8E-08 A B
1264.5 PP0487 1.7E-09 B B 260.4 PP0488 2.8E-09 A B 54.3 PP0490 2.4E-09 B B 50.5 PP0491
1.1E-09 B B 33.1 PP0492 1.2E-09 B B 31.9 PP0493 2.0E-09 B B 628.4 PP0494 5.1E-10 B B
135.9 PP0495 5.1E-09 A B 837.4 PP0496 4.8E-09 B B 398.7 PP0497 8.5E-10 B B 44.2 PP0498
1.8E-08 B B 147.2 PP0499 1.1E-09 B B 44.9 PP0500 1.8E-08 B B 962.8 PP0501 3.2E-09 B B
715.8 PP0502 1.2E-08 A B 1818.1 PP0503 4.4E-09 A B 282.9 PP0504 1.4E-08 B B 234.9
PP0505 1.0E-08 B B 209.6 PP0506 2.5E-08 B B 705.0 PP0507 9.4E-10 B B 62.1 PP0508
2.9E-09 B B 251.6 PP0509 2.4E-09 B B 108.6 PP0510 1.2E-08 B B 391.2 PP0511 1.6E-08 A B
413.5 PP0512 3.7E-10 B B 29.4 PP0513 6.2E-09 B B 238.3 PP0514 1.1E-08 B B 311.3 PP0515
2.9E-08 B B 632.8 PP0516 1.0E-08 B B 457.0 PP0520 9.9E-10 B B 17.1 PP0521 7.2E-08 B B
585.5 PP0522 2.8E-09 B B 83.0 PP0523 1.2E-08 A B 734.1 PP0524 1.1E-10 B C 4.8 PP0525
1.4E-10 C B 2.1 PP0526 7.4E-11 B C 6.6 PP0527 2.1E-10 C C 5.7 PP0528 3.7E-10 C C 21.8
PP0529 9.5E-11 C B 6.5 PP0530 6.0E-11 C B 7.0 PP0531 1.4E-10 C B 21.2 PP0532 6.6E-11 C

6.9 PP0533 1.7E-10 C B 8.9 PP0534 1.6E-10 B B 16.7 PP0535 5.0E-10 C B 55.4 PP0536
6.3E-10 B B 21.8 PP0537 5.8E-09 C C 196.7 PP0538 6.7E-10 B B 78.0 PP0539 1.7E-10 B B 6.3
PP0540 1.7E-08 B B 765.9 PP0541 4.7E-10 B B 83.3 PP0542 1.1E-10 B B 3.4 PP0543 1.0E-09
B B 41.5 PP0544 1.8E-08 B B 646.9 PP0545 2.4E-10 B B 14.0 PP0546 1.4E-10 B B 5.9 PP0547
3.6E-10 B B 21.5 PP0548 1.5E-08 C B 500.8 PP0549 6.2E-11 C B 13.0 PP0550 5.7E-11 B B 4.9
PP0551 3.8E-08 C B 401.4 PP0552 5.2E-11 B B 2.8 PP0553 4.9E-11 B B 5.6 PP0554 3.6E-11 B
B 1.6 PP0555 6.6E-11 B B 48.5 PP0556 2.4E-11 C B 21.0 PP0557 1.1E-10 B B 14.8 PP0558
6.7E-11 B B 31.4 PP0559 1.6E-10 B B 23.1 PP0560 5.0E-11 B B 5.2 PP0561 3.1E-10 B B 20.9
PP0562 4.0E-11 B B 4.4 PP0563 1.0E-10 B B 29.0 PP0564 1.8E-10 B B 18.4 PP0565 1.6E-10 C
B 11.5 PP0566 2.9E-10 B B 18.4 PP0567 2.1E-10 B B 8.3 PP0568 1.1E-09 B B 79.5 PP0569
3.4E-11 C B 2.8 PP0570 4.6E-11 B B 7.5 PP0571 5.0E-11 B B 5.4 PP0572 1.4E-10 B B 9.0
PP0573 3.4E-10 B B 8.7 PP0574 2.6E-11 C B 3.1 PP0575 2.3E-11 B B 3.2 PP0576 1.9E-11 C B
2.4 PP0577 1.9E-11 C C 2.4 PP0578 2.9E-11 B B 2.2 PP0579 1.7E-11 C B 2.4 PP0580 1.5E-11
C C 1.8 PP0581 2.0E-11 C B 8.0 PP0582 2.0E-11 C B 9.6 PP0583 1.9E-11 C C 10.3 PP0584
7.7E-12 C C 7.0 PP0585 2.2E-11 B B 6.2 PP0586 1.0E-11 C B 5.3 PP0587 1.4E-11 C C 5.1
PP0588 1.1E-10 B B 16.7 PP0589 3.6E-10 B B 24.8 PP0590 2.7E-10 C B 20.9 PP0591 8.7E-08
B B 3887.5 PP0594 7.3E-11 C B 2.5 PP0595 1.0E-09 C B 80.0 PP0596 1.0E-10 B B 8.3 PP0597
3.3E-10 B B 18.2 PP0598 2.9E-10 B B 19.6 PP0599 1.8E-07 C B 2800.0 PP0605 2.8E-07 C C
888.7 PP0606 1.8E-09 C B 266.0 PP0607 9.7E-10 C B 119.0 PP0608 1.7E-09 C B 252.2 PP0609
3.9E-09 B B 468.3 PP0610 4.9E-08 B B 689.4 PP0611 7.3E-10 B B 88.9 PP0612 3.9E-10 B B
43.3 PP0613 6.0E-10 B B 42.4 PP0614 1.1E-09 B B 110.3 PP0615 6.2E-09 B B 216.3 PP0616
4.9E-09 B B 73.8 PP0617 6.9E-09 B B 159.1 PP0618 6.2E-09 C C 79.1 PP0619 3.7E-09 C C
70.7 PP0620 6.9E-10 C C 38.4 PP0621 2.4E-09 C C 83.6 PP0622 1.1E-08 B B 330.2 PP0626
7.1E-09 C C 158.0 PP0627 3.3E-09 C C 98.3 PP0628 2.4E-09 C C 46.3 PP0629 2.4E-09 B C
45.1 PP0630 6.9E-09 C B 388.8 PP0631 5.9E-10 C B 41.4 PP0632 1.1E-09 C B 56.7 PP0633
2.3E-09 B B 116.3 PP0634 4.0E-09 B B 273.8 PP0635 2.7E-09 B B 85.6 PP0636 8.6E-09 A A
728.8 PP0637 2.3E-08 B B 951.6 PP0638 3.4E-09 B B 340.7 PP0639 8.5E-09 B B 630.6 PP0640
2.0E-08 B B >600 PP0641 2.6E-08 A A >1300 PP0642 1.1E-07 B B 2650.9 PP0643 7.4E-11 C B
6.5 PP0644 1.5E-10 C B 20.7 PP0645 2.8E-11 B B 12.0 PP0646 1.4E-10 C B 17.0 PP0647
6.8E-10 C B 45.0 PP0648 2.1E-10 B B 20.8 PP0649 7.5E-10 C B 30.2 PP0650 6.2E-10 C B 45.7
PP0651 6.0E-10 B B 136.3 PP0652 2.5E-09 B B 208.5 PP0653 1.4E-09 B B 165.7 PP0654
2.0E-09 B B 690.0 PP0655 1.9E-09 B B 124.7 PP0656 3.9E-11 C C 26.5 PP0657 7.6E-11 B B
32.9 PP0658 3.4E-11 C B 10.3 PP0659 4.2E-10 B B 184.2 PP0660 2.9E-10 B B 131.3 PP0661
3.4E-10 C B 56.6 PP0662 3.1E-09 B B 1569.0 PP0663 2.8E-10 B B 22.7 PP0664 1.9E-10 B B
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3.8E-10 B B 22.7 PP0669 5.2E-10 B B 20.3 PP0670 2.8E-10 B B 11.3 PP0671 6.7E-10 B B 32.5
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B B 25.5 PP0678 4.7E-10 B B 35.6 PP0679 5.8E-10 B B 40.2 PP0680 4.3E-10 B B 21.2 PP0681
4.4E-10 B B 24.2 PP0682 5.4E-10 B B 22.6 PP0683 6.1E-10 B B 44.7 PP0684 1.1E-09 B B 45.0
PP0685 8.0E-10 B B 52.3 PP0686 1.0E-09 B B 65.7 PP0687 6.4E-10 B B 44.1 PP0688 2.1E-10
A B 187.4 PP0689 3.0E-10 B B 34.4 PP0690 4.6E-10 B B 60.7 PP0693 1.6E-11 C C 3.8 PP0694
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PP0698 1.5E-11 C C 4.1 PP0699 3.6E-11 C C 2.9 PP0700 1.3E-11 C B 2.9 PP0701 1.7E-11 C B
3.8 PP0702 4.4E-11 C C 3.3 PP0703 1.3E-11 C B 26.7 PP0704 1.1E-11 C C 11.4 PP0705
2.7E-11 C C 17.5 PP0706 3.6E-09 B B 5394.0 PP0708 2.6E-11 B B 5.5 PP0709 3.0E-11 C B 1.0
PP0710 2.9E-11 C B 4.9 PP0711 2.5E-11 C B 6.5 PP0712 4.2E-11 C C 7.1 PP0713 3.3E-11 C B
4.2 PP0714 4.7E-11 C B 5.1 PP0715 3.5E-11 C B 4.5 PP0716 5.5E-11 C B 6.3 PP0717 3.2E-11
B B 4.8 PP0718 2.9E-11 C B 42.3 PP0719 2.0E-11 C B 8.5 PP0720 3.6E-11 C B 13.7 PP0721
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127.3 PP0725 5.0E-09 B B 193.9 PP0726 1.1E-09 B B 32.7 PP0727 1.3E-09 B B 85.2 PP0728

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2.7E-10 B C 40.9 PP0736 1.8E-09 B C 129.9 PP0737 5.5E-10 B C 38.2 PP0738 6.4E-10 B B
99.2 PP0739 3.7E-09 B C 296.8 PP0740 4.7E-10 B C 51.2 PP0741 2.8E-10 B B 51.5 PP0742
5.1E-10 B B 73.3 PP0743 6.8E-10 B C 76.3 PP0744 6.0E-11 B B 57.0 PP0745 2.9E-11 B C 3.9
PP0746 4.3E-10 B C 35.4 PP0747 3.6E-10 B C 45.9 PP0748 8.0E-11 B B 10.7 PP0749 5.9E-10
B C 19.8 PP0750 3.7E-10 B C 11.1 PP0751 5.2E-10 B C 25.5 PP0752 5.4E-10 B B 25.8 PP0753
1.4E-10 B C 8.2 PP0754 4.9E-10 B C 40.4 PP0755 5.6E-10 B B 83.8 PP0756 8.7E-10 B B 69.5
PP0757 4.0E-09 B C 480.2 PP0758 1.6E-09 B C 153.1 PP0759 2.7E-09 B B 184.7 PP0760
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83.0 PP0764 3.4E-09 B B 215.3 PP0765 3.1E-10 B C 123.0 PP0766 9.6E-11 B B 8.7 PP0767
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62.2 PP0771 3.3E-09 B B 203.7 PP0772 4.3E-09 B B 434.5 PP0773 4.0E-09 B B 170.1 PP0774
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545.9 PP0779 5.1E-09 B B 364.5 PP0780 7.8E-09 B B 527.3 PP0781 1.7E-08 B B 651.2 PP0782
2.3E-09 B B 425.1 PP0783 8.5E-10 B B 122.1 PP0784 1.7E-08 B B 446.6 PP0785 3.0E-09 B B
324.4 PP0786 6.0E-10 B B 44.8 PP0787 3.4E-10 C B 38.3 PP0788 5.4E-11 B C 7.1 PP0789
1.2E-10 C B 12.5 PP0790 5.6E-10 C B 28.8 PP0791 9.8E-10 C B 86.0 PP0792 9.0E-10 C B
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36.3 PP0800 2.7E-09 C B 186.4 PP0801 4.5E-10 C B 28.7 PP0802 8.2E-11 C B 6.5 PP0803
1.7E-09 C B 52.5 PP0804 1.4E-09 C B 40.9 PP0805 1.6E-10 B C 10.3 PP0806 1.9E-10 B B 16.9
PP0807 5.9E-11 B B 16.7 PP0808 2.7E-11 B B 16.7 PP0809 2.7E-11 B B 8.4 PP0810 4.0E-11 B
B 12.8 PP0811 8.3E-11 B B 9.8 PP0812 2.2E-10 B B 17.9 PP0813 4.0E-10 C B 90.3 PP0814
3.9E-11 B B 19.4 PP0815 3.9E-09 B B 344.7 PP0816 5.0E-10 A B 92.5 PP0817 3.6E-10 A B
111.7 PP0818 1.7E-09 B B 43.7 PP0819 5.6E-10 A B 75.7 PP0820 1.2E-09 A B 60.1 PP0821
2.0E-09 A B 250.2 PP0822 5.7E-09 B B 1988.1 PP0823 8.2E-10 A B 130.5 PP0824 5.5E-11 A A
23.6 PP0825 2.3E-10 A A 26.5 PP0826 1.3E-10 C B 11.5 PP0827 1.1E-10 C C 15.7 PP0828
1.7E-10 C C 7.3 PP0829 8.1E-11 B C 4.4 PP0830 5.8E-11 C C 3.4 PP0831 3.2E-10 C C 3.7
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B B 122.1 PP0836 6.3E-10 B B 359.8 PP0837 1.9E-10 B B 63.4 PP0838 5.3E-10 B B 50.4
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6.0E-10 B B 173.8 PP0843 5.2E-10 B B 38.6 PP0844 1.2E-09 B B 50.2 PP0845 2.2E-09 B B
134.9 PP0846 9.9E-10 B B 22.2 PP0847 2.8E-10 B B 135.6 PP0848 5.7E-10 B B 37.5 PP0849
4.6E-10 B B 35.7 PP0850 1.2E-08 B B 3085.5 PP0851 5.2E-10 B B 37.9 PP0852 4.6E-10 B B
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9.6E-10 B B 66.7 PP0857 1.4E-09 B B 111.2 PP0858 6.6E-10 B B 52.5 PP0859 3.1E-10 B B
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6.0E-10 B B 228.5 PP0864 4.9E-10 B B 185.2 PP0865 1.1E-09 B B 767.1 PP0866 5.3E-10 B B
197.4 PP0867 1.2E-09 B B 205.9 PP0868 5.3E-11 C B 70.8 PP0869 5.6E-11 B B 134.3 PP0870
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B B 154.0 PP0892 3.3E-10 B B 53.6 PP0893 1.3E-09 B B 226.5 PP0894 8.8E-12 C C 3.7
PP0895 7.2E-12 C C 2.2 PP0896 1.8E-11 C C 3.1 PP0898 8.8E-10 B B 119.8 PP0899 9.9E-10 A
B 922.8 PP0900 1.2E-11 C C 4.9 PP0901 3.0E-09 B B 332.0 PP0902 6.3E-10 A B 72.6 PP0903
3.6E-09 B C 596.2 PP0904 8.0E-10 B B 58.7 PP0905 7.9E-10 B B 92.0 PP0906 2.9E-10 B B

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6.2E-10 B B 66.2 PP0911 6.0E-11 B B 10.2 PP0912 3.2E-10 B B 40.5 PP0913 1.1E-09 B B
206.4 PP0914 4.3E-10 B B 40.2 PP0915 1.4E-10 B B 37.6 PP0916 3.4E-10 B B 35.9 PP0917
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75.5 PP0921 2.6E-09 B B 139.0 PP0922 5.0E-10 B B 98.2 PP0923 1.6E-09 B B 445.2 PP0924
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112.9 PP0928 1.0E-09 A B 182.3 PP0929 2.7E-09 A B 316.8 PP0930 1.8E-09 A B 141.9 PP0931
1.9E-09 A B 180.7 PP0932 2.9E-09 B B 158.3 PP0933 1.6E-09 A B 150.1 PP0934 7.2E-10 B B
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63.9 PP0942 9.9E-10 B B 66.1 PP0943 1.1E-09 B B 154.2 PP0944 3.9E-09 A A 344.2 PP0945
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8.5E-09 A A 1080.4 PP0953 2.7E-08 A A 1612.8 PP0954 7.5E-09 A A 438.5 PP0955 1.9E-08 A A
6475.4 PP0956 7.2E-09 A A 690.3 PP0958 7.5E-09 B B 1038.1 PP0959 3.9E-10 B B 156.2
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B B 100.0 PP0964 5.9E-10 B B 56.6 PP0965 5.7E-10 B B 48.8 PP0966 2.3E-10 B B 34.6
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PP0975 8.9E-09 1567.6 PP0976 4.8E-09 B B 1178.5 PP0977 7.7E-09 B A 1115.5 PP0978
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A 797.4 PP0982 9.9E-10 A A A 372.4 PP0983 7.8E-11 A A 19.5 PP0984 2.8E-10 A A 64.6
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B 10.3 PP0989 1.3E-09 B B 134.8 PP0992 4.6E-09 B B 790.3 PP0993 1.4E-09 B B 331.8
PP0997 5.6E-08 A A 2627.1 PP0998 1.2E-10 B B 19.7 PP0999 1.1E-10 C B 7.6 PP1000 3.9E-11
B B 7.5 PP1001 1.0E-10 B B 28.1 PP1002 2.5E-11 B B 2.1 PP1003 7.6E-11 C B 29.1 PP1004
3.7E-09 B B 254.6 PP1005 1.1E-09 A A 274.7 PP1006 3.2E-10 B B 81.9 PP1008 1.1E-09 A B
101.6 PP1009 2.0E-09 B B 195.0 PP1010 1.5E-09 B B 179.1 PP1011 3.7E-09 A B 314.2 PP1012
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110.0 PP1016 2.7E-10 A B 61.2 PP1017 4.8E-11 C B 5.0 PP1018 1.7E-11 C B 6.8 PP1019
5.5E-11 C B 7.4 PP1020 2.2E-11 C B 18.0 PP1021 5.7E-11 C B 14.0 PP1022 3.6E-11 C B 17.5
PP1024 8.8E-11 B B 12.7 PP1025 5.6E-11 C B 10.7 PP1026 1.2E-10 B B 10.4 PP1027 5.6E-11
C B 7.2 PP1028 9.9E-11 B B 17.3 PP1029 7.8E-11 C B 24.3 PP1030 2.6E-10 B B 27.9 PP1031
6.7E-10 C C 86.2 PP1032 3.8E-10 C C 43.9 PP1033 4.3E-10 C C 67.2 PP1034 4.3E-10 C C 96.8
PP1035 5.2E-10 C C 127.6 PP1036 7.5E-10 C C 149.8 PP1037 7.6E-10 C C 296.0 PP1038
3.0E-09 C C 187.8 PP1039 5.9E-10 C C 98.4 PP1040 1.2E-09 C C 18.7 PP1041 5.7E-10 C C
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133.8 PP1049 9.1E-10 C C 358.6 PP1050 9.4E-10 C C 815.0 PP1051 1.8E-09 C C 285.7 PP1052
2.5E-10 A A 58.7 PP1053 2.3E-10 A A 67.8 PP1054 3.9E-09 A B 468.9 PP1056 2.1E-10 A A
75.0 PP1057 3.1E-10 A B 96.9 PP1058 2.9E-09 A B 599.1 PP1059 9.1E-10 A B 200.7 PP1060
2.6E-09 A A 306.8 PP1063 1.2E-09 A A 380.1 PP1064 1.7E-08 A A 1392.2 PP1065 4.5E-10 A B
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PP1075 6.6E-11 B B 21.8 PP1076 6.8E-11 B B 4.6 PP1077 8.5E-11 B B 14.9 PP1078 4.7E-10 B
B 610.2 PP1079 2.3E-10 B B 27.2 PP1080 2.4E-10 B B 39.2 PP1082 3.5E-09 A B 299.6 PP1083
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22.4 PP1087 9.5E-11 A A 40.4 PP1088 5.5E-10 A A 195.0 PP1089 1.2E-10 A A A 57.3 PP1090
8.6E-11 A A A 20.9 PP1091 1.1E-10 A B 17.9 PP1092 1.5E-10 A A 15.8 PP1093 5.8E-10 A A A
78.7 PP1094 8.4E-10 A B 778.0 PP1095 1.8E-10 B B 14.9 PP1096 2.4E-10 B B 38.1 PP1097

1.2E-10 B B 10.2 PP1098 6.8E-10 B B 100.9 PP1099 5.8E-10 B A 68.1 PP1100 1.0E-10 B B
17.6 PP1101 4.6E-10 B B 68.4 PP1102 9.5E-10 B A 83.1 PP1103 3.6E-11 A B 34.8 PP1104
5.9E-11 A A 44.6 PP1105 7.0E-11 A A 12.0 PP1106 8.1E-10 A A 154.6 PP1107 3.8E-11 A A 5.4
PP1108 3.3E-11 A A 28.3 PP1109 1.2E-10 A B 17.0 PP1110 5.6E-11 A B 11.2 PP1111 4.5E-11 A
A 13.0 PP1112 6.0E-11 A B 10.4 PP1113 3.1E-11 A A 21.4 PP1114 5.4E-11 A A 12.2 PP1115
5.4E-11 A A 29.0 PP1116 6.9E-11 A A 19.8 PP1117 7.0E-11 A A 16.5 PP1118 5.6E-11 A B 6.5
PP1119 3.7E-11 A A 25.5 PP1120 4.2E-11 A A 20.6 PP1121 4.9E-11 A A 17.3 PP1122 4.7E-11 A
B 274.9 PP1123 7.2E-11 A B 13.4 PP1124 5.6E-11 A B 12.4 PP1125 7.0E-11 A A 15.0 PP1126
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1.5E-10 A A 1776.6 PP1141 3.2E-10 A A 423.6 PP1142 6.5E-10 A A 321.5 PP1143 1.7E-10 A
A 75.3 PP1144 2.4E-10 A A 155.4 PP1145 2.1E-10 A B 140.0 PP1146 3.2E-10 A B 202.3
PP1147 7.1E-10 A A 888.4 PP1148 1.2E-09 A A 357.1 PP1149 4.3E-10 A A 173.0 PP1150
1.7E-10 A A 42.7 PP1151 6.4E-10 B B 97.1 PP1152 1.3E-09 B B 299.8 PP1153 2.8E-10 B B
15.5 PP1154 4.3E-10 B B 69.6 PP1155 1.5E-10 B B 73.1 PP1156 6.3E-11 B B 33.7 PP1157
6.1E-11 B B 30.3 PP1158 1.8E-10 B B 57.5 PP1159 3.6E-11 C B 13.0 PP1160 3.2E-11 B B 18.4
PP1161 6.0E-10 B B 153.4 PP1162 1.2E-09 B B 198.5 PP1163 5.9E-10 B B 93.3 PP1164
5.2E-10 B B 95.9 PP1165 1.5E-09 A B 295.3 PP1166 5.0E-10 B B 124.5 PP1167 5.0E-10 B B
134.1 PP1168 1.8E-09 A A 337.7 PP1169 3.2E-10 B B 74.0 PP1170 2.3E-10 B B 64.7 PP1171
6.9E-10 B B 117.5 PP1172 1.7E-10 B B 48.3 PP1173 1.8E-10 B B 70.8 PP1174 1.0E-09 B B
85.2 PP1175 1.6E-10 B B 23.5 PP1176 1.3E-10 B B 50.0 PP1177 8.1E-11 B B 32.5 PP1178
1.6E-10 B B 26.3 PP1179 9.1E-11 B B 56.9 PP1180 1.7E-10 B B 46.9 PP1181 1.9E-10 B B 26.3
PP1182 8.0E-11 B B 134.5 PP1183 2.6E-10 B B 27.8 PP1184 1.6E-10 B B 91.9 PP1185 5.0E-10
B B 175.0 PP1186 9.4E-11 B B 54.4 PP1187 1.2E-10 B B 52.3 PP1188 4.3E-10 B B 33.5 PP1189
6.6E-11 B B 16.1 PP1190 9.2E-11 B B 15.5 PP1191 5.6E-11 A B 13.0 PP1192 4.0E-11 B A 10.5
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B B 50.8 PP1197 9.6E-11 B B 14.8 PP1198 1.9E-10 B B 35.2 PP1199 1.7E-10 A A 39.1 PP1200
2.0E-10 A A 19.2 PP1201 1.7E-10 A A 9.7 PP1202 4.0E-10 24.1 PP1203 2.3E-10 A A 20.8
PP1204 2.4E-10 A A 23.4 PP1205 1.6E-10 A A 25.4 PP1206 1.7E-10 A A 109.4 PP1207
2.1E-10 A A 29.9 PP1208 1.7E-10 A A 24.4 PP1209 2.2E-10 A A 15.9 PP1210 4.0E-10 A A
A 44.3 PP1211 2.7E-10 A B 26.8 PP1212 2.8E-10 A A 35.6 PP1213 1.8E-10 A A 40.4 PP1214
3.4E-10 A A 93.4 PP1215 4.2E-10 A A 33.4 PP1216 3.9E-10 A A 19.8 PP1217 4.8E-10 A A
A 17.8 PP1218 8.8E-10 A A 67.9 PP1219 3.5E-10 A A 21.4 PP1220 5.7E-10 A A 22.2
PP1221 4.0E-10 A A 40.6 PP1222 1.2E-09 A A 301.2 PP1223 1.2E-09 A A 117.3 PP1224
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127.6 PP1228 1.6E-09 A A 55.1 PP1229 1.3E-09 A A 144.5 PP1230 7.7E-10 A A 109.7 PP1231
5.6E-10 A A 23.1 PP1232 6.7E-10 A A 59.7 PP1233 6.6E-10 A A 34.7 PP1234 3.8E-09 B C
398.0 PP1235 6.7E-09 B C 386.0 PP1242 7.6E-11 A A 12.8 PP1243 5.8E-11 A A 13.2 PP1244
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A A 21.3 PP1259 4.9E-10 A A 62.0 PP1260 3.7E-10 A A 60.4 PP1261 4.4E-10 A A 76.8 PP1262
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PP1266 4.5E-11 A A 12.2 PP1267 4.6E-11 A A 10.8 PP1268 4.5E-11 A A 12.1 PP1269 9.1E-11 A
B 29.0 PP1270 1.3E-10 A B 53.5 PP1271 5.4E-11 A A 90.7 PP1272 1.1E-10 A A 24.0 PP1273
1.2E-10 A B 16.5 PP1274 3.2E-11 A A 7.5 PP1275 4.7E-11 A A 4.9 PP1276 6.6E-11 A B 20.8
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AA 34.4 PP1281 1.4E-10 AA A 10.2 PP1282 2.0E-10 AA A 12.3 PP1283 2.0E-10 AA A 42.6
PP1284 1.2E-10 AA A 54.8 PP1285 1.5E-10 AA A 24.6 PP1286 6.5E-10 AA A 58.8 PP1287
5.6E-10 AA A 49.7 PP1288 1.8E-10 AA A 22.2 PP1289 1.7E-10 AA A 34.7 PP1290 2.4E-10 AA
A 15.4 PP1291 2.4E-10 A A 16.7 PP1292 1.6E-10 A A 24.6 PP1293 2.2E-10 AA A 4.9 PP1294
2.9E-10 A A 10.0 PP1295 1.2E-10 AA A 6.2 PP1296 1.2E-10 AA A 14.7 PP1297 8.7E-11 AA A
23.1 PP1298 1.3E-10 AA A 11.4 PP1299 1.4E-10 A A 11.0 PP1300 8.2E-11 A A 9.5 PP1301
8.3E-11 A A 4.7 PP1302 5.7E-11 A A 9.2 PP1303 6.8E-11 AA A 4.2 PP1304 7.0E-11 A A 8.8
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3.4E-10 AA A 34.5 PP1316 7.2E-11 A A 13.2 PP1317 9.4E-11 A B 7.6 PP1318 9.1E-11 A A 11.5
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A 56.2 PP1323 2.6E-10 AA A 41.8 PP1324 9.3E-11 A A 4.4 PP1325 5.5E-10 A A 42.7 PP1326
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34.5 PP1330 4.2E-10 AA A 79.6 PP1331 8.0E-11 A A 13.9 PP1333 5.8E-08 C B 979.5 PP1334
1.1E-08 B B 968.3 PP1335 1.8E-08 B B 736.5 PP1336 7.4E-11 B B 10.8 PP1337 1.7E-10 B B
29.8 PP1338 1.4E-10 B B 35.0 PP1339 6.5E-10 A B 42.1 PP1340 8.1E-11 B B 10.9 PP1341
8.1E-10 A B 46.3 PP1342 9.1E-11 B B 5.0 PP1343 2.7E-10 A B 19.3 PP1344 1.4E-10 B B 16.7
PP1345 7.1E-10 A B 34.8 PP1346 8.4E-11 B B 7.2 PP1347 1.7E-11 B B 3.6 PP1348 7.6E-11 B
B 8.7 PP1349 9.0E-11 C B 14.7 PP1350 1.8E-10 B B 20.1 PP1351 1.3E-10 A B 12.8 PP1352
2.4E-11 A B 4.6 PP1353 1.0E-10 A B 3.7 PP1354 5.6E-11 AB 3.6 PP1355 1.6E-10 A B 10.2
PP1356 4.7E-11 B B 3.3 PP1357 1.3E-10 A B 5.4 PP1358 3.0E-11 A B 4.1 PP1359 6.8E-11 A B
3.2 PP1360 5.5E-11 A B 4.9 PP1361 1.5E-10 A B 6.2 PP1362 3.7E-11 A B 3.7 PP1363 3.2E-10
A B 34.7 PP1364 5.0E-11 A B 7.1 PP1365 1.3E-10 A B 4.2 PP1366 1.1E-10 A B 14.0 PP1367
3.4E-10 A B 29.3 PP1368 6.1E-11 A B 2.9 PP1369 2.6E-10 B B 13.4 PP1370 5.7E-11 C C 2.3
PP1371 7.7E-11 A B 5.1 PP1372 7.8E-11 B B 5.0 PP1373 3.5E-10 B B 12.2 PP1374 3.3E-11 B
B 1.9 PP1375 2.3E-10 B B 11.5 PP1376 3.7E-11 B B 3.5 PP1377 8.7E-11 B B 5.8 PP1378
8.1E-11 B C 6.6 PP1379 2.9E-10 A B 10.3 PP1380 4.5E-11 B B 4.1 PP1381 7.2E-10 B B 80.4
PP1382 6.8E-11 B B 15.8 PP1383 1.7E-10 B B 11.2 PP1384 1.1E-10 A B 13.5 PP1385 8.0E-10
B B 19.6 PP1386 8.2E-11 B B 4.2 PP1387 9.8E-11 A B 15.3 PP1388 7.6E-11 A B 8.8 PP1389
6.6E-11 AA A 2.4 PP1390 1.1E-09 A A 305.4 PP1391 1.6E-09 A A 271.1 PP1392 4.7E-11 A A
1.4 PP1393 1.9E-10 A A 18.2 PP1394 2.4E-10 AA A 25.1 PP1395 6.3E-11 A B 3.8 PP1396
1.4E-10 AA A 19.4 PP1397 7.3E-11 A A 3.7 PP1398 3.2E-11 AA A 2.3 PP1399 1.9E-10 A A 18.8
PP1400 4.9E-11 A B 5.1 PP1401 2.6E-10 A A 34.4 PP1402 3.0E-10 A A 82.0 PP1403 6.0E-10
AA A 110.6 PP1404 4.8E-09 A A 498.2 PP1405 8.5E-09 A A 466.1 PP1406 7.2E-11 A B 10.9
PP1407 7.4E-11 A A 18.7 PP1408 2.2E-10 A A 37.6 PP1409 2.1E-10 A A 43.5 PP1410 3.2E-10
B B 51.0 PP1411 8.0E-11 A B 6.0 PP1412 1.9E-10 A A 23.0 PP1413 4.2E-10 B B 62.5 PP1414
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PP1418 2.4E-10 AA A 29.9 PP1419 2.6E-10 AA A 40.4 PP1420 3.5E-10 A A 31.4 PP1421
3.0E-10 A A 27.9 PP1422 2.2E-10 AA A 14.1 PP1423 1.4E-10 AA A 4.4 PP1424 2.3E-10 A A
26.3 PP1425 2.4E-10 A B 20.2 PP1426 1.0E-09 A B 474.8 PP1427 9.9E-10 A B 465.1 PP1428
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2.3E-10 A A 32.9 PP1461 2.7E-10 A A 120.7 PP1462 1.2E-10 A A 27.5 PP1463 1.3E-10 A A 44.5
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A 11.2 PP1468 1.2E-10 A A 9.5 PP1469 1.8E-10 A A 22.2 PP1470 2.2E-10 A A 110.7 PP1471
2.5E-10 A A 48.2 PP1472 1.4E-10 A A 37.1 PP1473 1.9E-10 A A 36.6 PP1474 2.3E-10 A A 48.2
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14.8 PP1482 4.5E-10 A B 76.4 PP1483 5.1E-10 A B 90.3 PP1484 4.5E-10 A A 59.3 PP1485
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9.4E-10 B B 129.2 PP1493 2.3E-09 B B 135.1 PP1494 5.8E-10 B B 142.5 PP1495 5.5E-10 A B
98.7 PP1496 2.6E-10 A B 36.7 PP1497 2.2E-11 A A 10.7 PP1498 6.9E-11 A A 10.3 PP1499
5.4E-11 A B 5.5 PP1500 3.9E-11 A A 6.9 PP1501 2.4E-11 A A 1.5 PP1502 3.8E-11 A A 2.0
PP1503 1.1E-10 A A 3.8 PP1504 3.2E-10 A A 38.0 PP1505 9.2E-11 A A 4.0 PP1506 3.4E-10 A A
A 73.4 PP1507 1.7E-10 A A 16.2 PP1508 1.5E-10 A A 28.1 PP1509 4.0E-10 A A 67.9 PP1510
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8.3 PP1514 6.5E-10 A A 72.7 PP1515 4.2E-10 A A 91.9 PP1516 9.8E-10 A B 45.7 PP1517
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575.0 PP1521 2.6E-09 A B 173.2 PP1522 8.8E-09 A A 566.7 PP1523 4.1E-09 320.2 PP1524
5.7E-09 B B 438.6 PP1525 1.7E-09 A A 131.6 PP1526 1.3E-09 A A 118.0 PP1527 9.5E-10 A A
142.9 PP1528 2.0E-10 A B 45.6 PP1529 8.1E-11 A B 30.2 PP1530 2.1E-10 A B 17.2 PP1531
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24.7 PP1535 1.8E-10 A A 18.9 PP1536 2.0E-10 A B 24.8 PP1537 6.7E-11 A A A 7.9 PP1538
8.6E-11 A B 7.5 PP1552 1.9E-10 A A 15.9 PP1553 1.3E-10 A A 12.1 PP1554 2.7E-09 A A 120.8
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A 7.6 PP1577 1.7E-10 B B 6.7 PP1578 2.0E-10 A A 12.6 PP1579 8.7E-11 A B 4.1 PP1580
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A A 86.6 PP1590 2.9E-10 A B 8.0 PP1591 4.6E-09 A A 85.4 PP1592 7.1E-10 A B 17.2 PP1593
4.3E-09 A A 143.9 PP1594 7.1E-10 A A 48.8 PP1595 2.3E-09 A A 64.6 PP1596 5.6E-10 A B
18.1 PP1598 7.6E-10 A A 36.6 PP1599 3.7E-10 A A 10.2 PP1600 3.2E-10 A B 13.8 PP1601
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12.6 PP1607 5.7E-10 A 39.3 PP1608 6.7E-10 A A 70.1 PP1609 2.0E-10 A A 37.0 PP1610
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B B 41.9 PP1620 6.4E-09 B B 170.3 PP1622 1.3E-09 A A 28.1 PP1623 1.9E-10 A A 4.5 PP1624
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22.9 PP1628 2.5E-09 A A 149.9 PP1629 3.6E-10 B B 20.7 PP1630 1.8E-09 A A 144.5 PP1631
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6.8E-10 A A 44.8 PP1640 1.4E-10 B B 28.3 PP1641 9.8E-11 A A 4.6 PP1643 5.5E-11 A A 1.6
PP1644 3.8E-11 A A 2.7 PP1645 4.0E-11 A A 1.8 PP1646 1.2E-10 A A 2.8 PP1648 1.1E-10 A A
2.1 PP1649 6.4E-11 A B 2.2 PP1650 5.7E-11 A A 1.2 PP1651 7.9E-11 A B 2.3 PP1653 9.7E-11 A
A 1.6 PP1654 9.7E-11 A A 2.5 PP1655 8.3E-11 A A 1.6 PP1656 6.5E-11 A A A 1.5 PP1657
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PP1662 1.4E-10 A B 6.4 PP1663 3.8E-11 A A A 3.6 PP1665 6.6E-11 A A A 3.1 PP1666 7.6E-11 A
A 8.1 PP1667 8.1E-11 A A 2.0 PP1668 1.3E-10 A A 5.3 PP1670 1.0E-10 A A 6.7 PP1671 7.7E-11

AA 3.0 PP1672 8.7E-11 A A 6.0 PP1673 1.6E-10 A A 6.0 PP1674 3.6E-11 A A 5.1 PP1675
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PP1679 6.5E-11 A A 3.1 PP1682 8.2E-11 AA A 2.6 PP1683 9.7E-11 A A 2.5 PP1684 1.5E-10 A A
7.7 PP1687 7.5E-11 A A 2.7 PP1688 2.5E-11 A A 1.5 PP1689 2.3E-11 A B 1.4 PP1691 6.0E-11 A
A 3.1 PP1692 1.1E-10 AA A 4.3 PP1693 2.7E-11 AA A 1.1 PP1694 5.1E-11 A A 4.1 PP1696
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PP1700 1.1E-10 A A 3.6 PP1701 8.0E-11 A A 3.7 PP1702 8.5E-11 A A 4.9 PP1703 1.3E-10 A A
7.2 PP1704 1.4E-10 A A 10.3 PP1705 1.4E-10 AA A 8.0 PP1706 7.5E-11 A A 6.7 PP1707
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A 3.8 PP1717 5.1E-11 A A 2.4 PP1718 6.7E-11 AA A 2.5 PP1721 3.0E-10 A A 10.0 PP1722
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PP1731 7.0E-11 A A 7.4 PP1732 9.9E-11 AA A 6.4 PP1733 1.1E-10 A A 3.8 PP1734 1.7E-10 A
A 4.4 PP1735 8.1E-11 AA A 2.2 PP1736 1.3E-10 A A 3.6 PP1737 1.1E-10 A A 2.7 PP1738
7.1E-11 AA A 5.1 PP1739 5.4E-11 AA AA 5.5 PP1740 5.3E-11 A A 3.6 PP1741 3.9E-11 A B 2.3
PP1742 5.1E-11 A A 1.6 PP1743 7.6E-11 AA A 5.1 PP1744 1.3E-10 A A 5.3 PP1746 5.4E-11 A A
1.9 PP1747 5.8E-11 A A 2.3 PP1748 6.2E-11 A B 3.1 PP1749 3.9E-11 A A 3.3 PP1750 6.9E-11 A
B 3.4 PP1751 4.7E-11 A A 2.2 PP1752 8.5E-11 A A 5.8 PP1753 3.8E-11 A A 5.9 PP1754 6.4E-11
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3.5 PP1784 8.3E-11 A B 7.8 PP1785 5.1E-11 A A 6.9 PP1786 1.4E-10 A B 11.4 PP1787 7.1E-11
A B 9.0 PP1788 6.3E-11 B B 6.2 PP1789 1.3E-10 A A 9.5 PP1790 3.0E-11 A B 1.7 PP1791
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A B 5.8 PP1803 1.0E-10 A B 7.6 PP1804 5.6E-11 A A 3.5 PP1805 7.1E-11 A A 7.0 PP1806
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22.6 PP1814 2.7E-10 A A 26.2 PP1815 3.4E-10 B B 83.0 PP1816 1.2E-09 B C 63.8 PP1817
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40.7 PP1821 6.3E-10 A A 104.7 PP1822 1.1E-09 A B 107.0 PP1823 1.0E-09 A A 55.9 PP1825
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B 10.7 PP1837 2.0E-10 A A 15.6 PP1838 1.0E-10 A A 20.0 PP1839 3.1E-10 A B 24.3 PP1840
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A A 22.2 PP1863 1.3E-10 A A 88.8 PP1864 6.4E-11 A B 10.2 PP1865 8.3E-11 A B 36.0 PP1866
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B 8.5 PP1874 1.3E-10 A A 9.9 PP1875 8.3E-11 A B 7.7 PP1876 2.0E-10 A A 23.0 PP1877
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AA 17.3 PP1885 2.3E-10 AA A 18.3 PP1886 1.2E-10 A B 1.9 PP1887 3.0E-10 A B 19.2 PP1888
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PP1892 7.4E-11 A B 4.8 PP1893 1.7E-10 A B 18.5 PP1894 2.0E-10 A A 15.9 PP1895 1.3E-10 A
B 15.6 PP1896 6.0E-11 A B 5.3 PP1897 1.6E-10 B B 22.6 PP1898 1.3E-10 B B 6.3 PP1899
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C 44.1 PP1923 3.4E-09 A A 68.0 PP1925 2.3E-10 A A 9.7 PP1926 3.6E-10 B B 17.4 PP1927
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B C 74.0 PP1970 6.2E-10 B B 16.8 PP1971 5.0E-09 B B 38.0 PP1972 1.5E-10 A A 13.2 PP1973
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13.4 PP2127 1.8E-10 A A 12.5 PP2128 4.0E-10 A A 14.8 PP2130 6.2E-10 A A 7.5 PP2131
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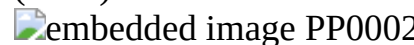
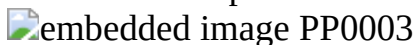
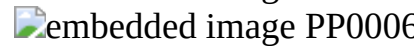
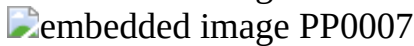
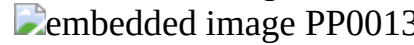
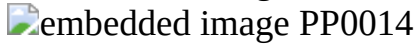
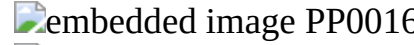
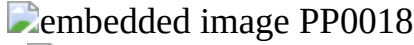
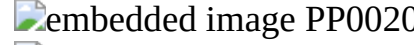
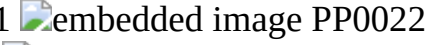
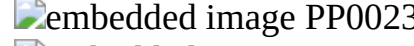
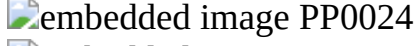
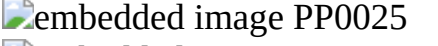
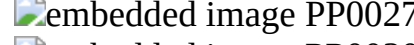
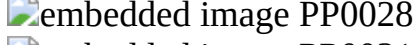
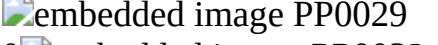
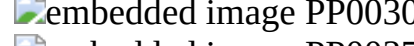
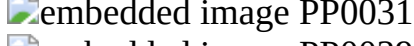
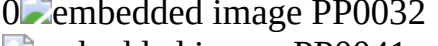
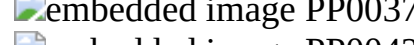
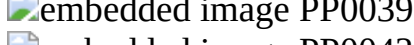
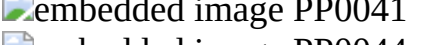
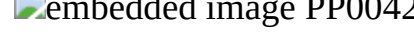
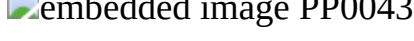
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 B 19.1 PP3019 2.0E-09 AA A 42.6 PP3021 1.1E-09 AA A 76.7 PP3022 5.9E-10 AA A 51.4
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List of Structures of Cyclic Compounds

(2317) TABLE-US-00049 TABLE 38 Compound No. Structural Formula PP0001

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 PP0006  PP0007  PP0011
 PP0013  PP0014  PP0015
 PP0016  PP0018  PP0019
 PP0020 0  PP0021  PP0022
 PP0023  PP0024  PP0025
 PP0027  PP0028  PP0029
 PP0030  PP0031 0  PP0032
 PP0037  PP0039  PP0041
 PP0042  PP0043  PP0044

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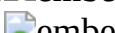
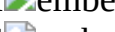
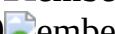
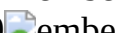
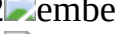
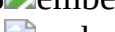
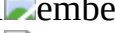
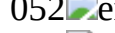
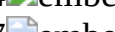
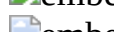
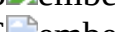
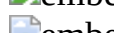
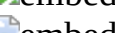
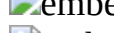
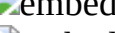
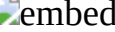
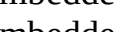
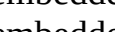
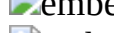
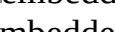
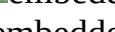
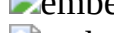
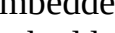
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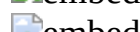
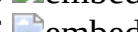
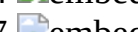
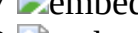
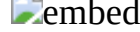
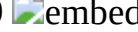
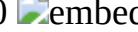
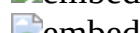
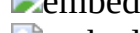
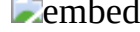
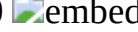
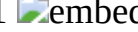
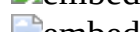
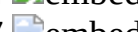
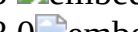
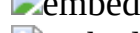
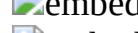
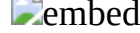
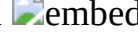
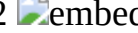
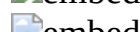
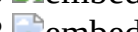
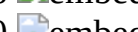
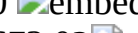
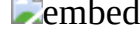
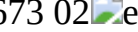
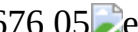
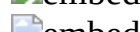
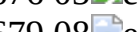
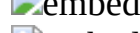
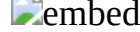
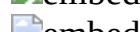
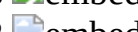
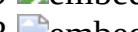
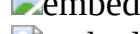
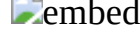
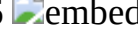
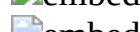
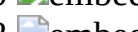
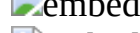
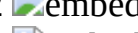
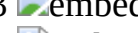
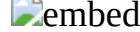
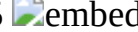
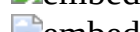
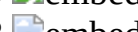
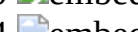
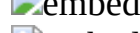
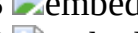
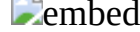
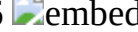
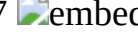
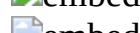
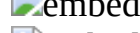
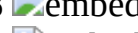
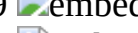
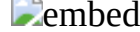
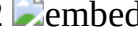
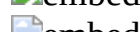
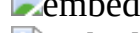
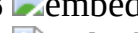
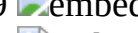
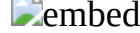
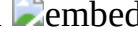
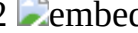
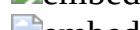
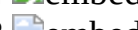
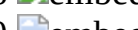
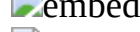
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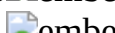
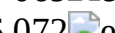
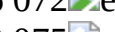
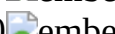
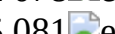
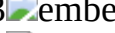
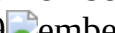
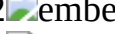
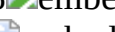
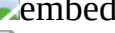
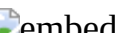
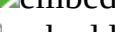
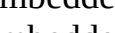
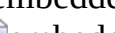
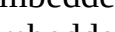
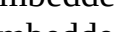
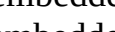
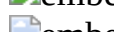
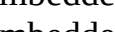
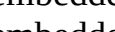
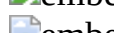
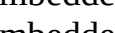
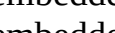
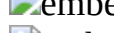
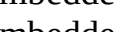
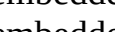
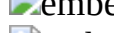
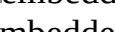
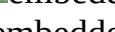
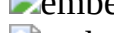
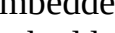
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INDUSTRIAL APPLICABILITY

(2318) The present invention provides cyclic compounds that selectively inhibit KRAS.

Claims

1. A compound, which is (1S,4S,10S,13S,17S,20S,26S,28R,32S,38S,42Z)-20-cyclopentyl-28-ethoxy-32-[2-[3-methoxy-4-(trifluoromethyl)phenyl]ethyl]-N,N,2,14,18,21,24,36-octamethyl-10-[(1S)-1-methylpropyl]-3,9,12,15,19,22,25,31,34,37,45-undeca-oxo-13-propyl-38-[[4-(trifluoromethyl)phenyl]methyl]spiro[2,8,11,14,18,21,24,30,33,36,39-undecazatetracyclo[37.5.1.0.sup.4,8.0.sup.26,30] pentatetracont-42-ene-23,l'-cyclobutane]-17-carboxamide.
2. A pharmaceutical composition comprising the compound of claim 1.
3. A method of inhibiting KRAS in a subject, the method comprising administering to the subject a

- pharmaceutical composition comprising an effective amount of the compound of claim 1.
4. A method of treating or preventing cancer in a subject, the method comprising administering to the subject a pharmaceutical composition comprising an effective amount of the compound of claim 1.
 5. The method of claim 4, wherein the cancer is lung cancer.
 6. The method of claim 4, wherein the subject is human.
 7. A solvate of (1S,4S,10S,13S,17S,20S,26S,28R,32S,38S,42Z)-20-cyclopentyl-28-ethoxy-32-[2-[3-methoxy-4-(trifluoromethyl)phenyl]ethyl]-N,N,2,14,18,21,24,36-octamethyl-10-[(1S)-1-methylpropyl]-3,9,12,15,19,22,25,31,34,37,45-undecaoxo-13-propyl-38-[[4-(trifluoromethyl)phenyl]methyl]spiro[2,8,11,14,18,21,24,30,33,36,39-undecazatetracyclo[37.5.1.0.sup.4,8.0.sup.26,30] pentatetracont-42-ene-23,1'-cyclobutane]-17-carboxamide.
 8. A pharmaceutical composition comprising the solvate of claim 7.
 9. A method of selectively inhibiting KRAS in a subject, the method comprising administering to the subject a pharmaceutical composition comprising an effective amount of the solvate of claim 7.
 10. A method of treating or preventing cancer in a subject, the method comprising administering to the subject a pharmaceutical composition comprising an effective amount of the solvate of claim 7.
 11. The method of claim 10, wherein the cancer is lung cancer.
 12. The method of claim 10, wherein the subject is human.
 13. The solvate according to claim 7, which is a hydrate.
 14. A pharmaceutical composition comprising the solvate of claim 13.
 15. A method of selectively inhibiting KRAS in a subject, the method comprising administering to the subject a pharmaceutical composition comprising an effective amount of the solvate of claim 13.
 16. A method of treating or preventing cancer in a subject, the method comprising administering to the subject a pharmaceutical composition comprising an effective amount of the solvate of claim 13.
 17. The method of claim 16, wherein the cancer is lung cancer.
 18. The method of claim 16, wherein the subject is human.
 19. A salt of (1S,4S,10S,13S,17S,20S,26S,28R,32S,38S,42Z)-20-cyclopentyl-28-ethoxy-32-[2-[3-methoxy-4-(trifluoromethyl)phenyl]ethyl]-N,N,2,14,18,21,24,36-octamethyl-10-[(1S)-1-methylpropyl]-3,9,12,15,19,22,25,31,34,37,45-undecaoxo-13-propyl-38-[[4-(trifluoromethyl)phenyl]methyl]spiro[2,8,11,14,18,21,24,30,33,36,39-undecazatetracyclo[37.5.1.0.sup.4,8.0.sup.26,30] pentatetracont-42-ene-23,1'-cyclobutane]-17-carboxamide, or a solvate thereof.
 20. A pharmaceutical composition comprising the salt of claim 19, or a solvate thereof.
 21. A method of inhibiting KRAS in a subject, the method comprising administering to the subject a pharmaceutical composition comprising an effective amount of the salt of claim 19, or a solvate thereof.
 22. A method of treating or preventing cancer in a subject, the method comprising administering to the subject a pharmaceutical composition comprising an effective amount of the salt of claim 19, or a solvate thereof.
 23. The method of claim 22, wherein the cancer is lung cancer.
 24. The method of claim 22, wherein the subject is human.
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